

G P U POWER DISTRIBUTION
OVERHEAD LINES MATERIAL MANUAL

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SECTION 1

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3452040A	204A	JUMPER PARKING BAR HENLEY
3452050	205	MOUNTING CHANNEL AUTO RECLOSER SINGLE POLE
3452052	205/2	HSAR MOUNTING BRACKET 'H' POLE
3452180	218	COPPER ELBOW FOR RECLOSER CONNECTIONS
3452181	218/1	PMR RECLOSER STRAP FOR MOUNTING SURGE ARRESTERS

SECTION 2

MIN NUMBER	M	DESCRIPTION
3451130	113	EARTH WIRE GUARD AND CABLE COVERS
3451750	175	CLEAT CROSSARM
3451753	175/3	CLEAT CROSSARM
3451760	176A	CABLE BACKBOARD FITTINGS FOR 33kv & 66kv TERMINAL POLES
3451763	176/3	FISH PLATE
3451780	178	33kv & 66kv CABLE CLEATS
3453040	304	TERMINATION SUPPORT BRACKET HEATSHRINK 11kv (H- POLE)
3453080	308	ANTI CLIMBING GUARD SUPPORT H.V. TERMINAL WITH U/G CABLE
3453110	311A	CABLE PROTECTION CASING & LID FOR 33 & 66kv TERMINAL POLES
3453470	347	HEAT SHRINK BREAKOUT FOR CONCENTRIC CABLES
3454530	453	CROSSARM & INSULATOR TAKE-OFF BRACKET FOR 33k.v POLE TERMINATION
3454532	453/2	CROSSARM & INSULATOR TAKE-OFF BRACKET FOR 66k.v POLE TERMINATION

SECTION 3

MIN NUMBER	M	DESCRIPTION
3453520	352	OVERHEAD LINE CONDUCTORS BARE & P.V.C COVERED (ALUMINIUM)
3453530	353	OVERHEAD LINE CONDUCTORS BARE & P.V.C COVERED (AL. ALLOY)
3453540	354	OVERHEAD LINE CONDUCTORS BARE & P.V.C COVERED (A.C.S.R)
3453560	356	OVERHEAD LINE CONDUCTORS P.V.C COVERED HARD DRAWN COPPER
3453570	357	OVERHEAD LINE CONDUCTORS BARE & P.V.C COVERED CADMIUM COPPER
3453940	394	L.V. AERIAL BUNDLED CONDUCTORS (A.B.C)

SECTION 4

MIN NUMBER	M	DESCRIPTION
3453500	350	OVERHEAD LINE CONDUCTORS PVC INSULATED & SHEATHED COPPER CABLES
3453510	351	OVERHEAD LINE CONDUCTORS PVC INSULATED & SHEATHED ALUMINIUM CABLES
3453580	358	OVERHEAD LINE CONDUCTORS PVC INSULATED COPPER & ALUMINIUM FLEXIBLE L/L CABLE
3453590	359	OVERHEAD LINE CONDUCTORS BARE & PVC COVERED BINDING WIRE,& CHAFING TAPE
3453600	360	CLAMP ADAPTOR 125kN

SECTION 5

MIN NUMBER	M	DESCRIPTION
3451410	141	CONDUCTOR GAME GUARDS
3451416	141/6	CLYDESDALE SNAPFAST BIRD DIVERTERS
3453200	320	CONDUCTOR VIBRATION DAMPER
3455000	500	CONDUCTOR SPACER TWIN 175/400mm ² ACSR 200mm SPACING
3455010	501	CONDUCTOR SPACER TWIN 175/400mm ² ACSR 300mm SPACING
3455020	502	CONDUCTOR SPACER TWIN 175/400mm ² ACSR 400mm SPACING

SECTION 6

MIN NUMBER	M	DESCRIPTION
3451220	122	TENSION JOINT AAAC & ACSR
3451230	123	TENSION JOINT HD ALUMINIUM
3451590	159	DEAD END 70 - 175 ACSR
3451610	161	TENSION JOINT 70 - 400 ACSR
3451730	173	RECONDUCTORING TENSION JOINT
3452100	210	DEAD END 400 HD ALUMINIUM
3452200	220	ADAPTOR PLATE
3452330	233	TENSION JOINT COPPER
3452530	253	DEAD END 400 ACSR
3452870	287	SOCKET DEAD END COPPER,AL ALLOY & ACSR
3452880	288	BALL DEAD END AL ALLOY & ACSR
3453331	333/1	DEAD END JUMPER TERMINAL ANGULAR ACSR & COPPER
3453332	333/2	DEAD END JUMPER TERMINAL STRAIGHT ACSR & COPPER
3453350	335	DEAD END DOUBLE JUMPER 100 - 400 ACSR
3453360	336	DEAD END INTEGRAL JUMPER 70 -175 - 400 ACSR
3453380	338	REPLACEMENT BOLTS FOR DISCONNECTABLE DEAD ENDS
3453390	339	TENSION JOINT CADMIUM COPPER
3453400	340	SOCKET DEAD END CADMIUM COPPER
3453410	341	BALL DEAD END CADMIUM COPPER
3454460	446	HARDEX HELICAL TERMINATION
3454610	461	TERMINATING THIMBLE
3455640	564	BOLTED TERMINATING CLAMP 200 AAAC
3455660	566	DEAD END UNITYPE FORAAAC REMOVABLE JUMPER
3455680	568	WEDGE CLAMP ASSEMBLY AAAC

SECTION 7

MIN NUMBER	M	DESCRIPTION
3451950	195	SINGLE HOLE BI-METAL LUG
3452190	219	T JOINT WITH JUMPER LUGS
3452360	236	SINGLE HOLE COPPER LUG
3452370	237	SINGLE HOLE COPPER LUG
3452380	238	TWO HOLE COPPER LUG
3452730	273	CRIMPET FOR COPPER CONDUCTORS
345277	277	LINE TAP LUGS FOR FLEXIBLE 50mm AL
3452850	285	SINGLE HOLE H D ALUMINIUM LUG
3452950	295	LINE TAP LUGS FOR FLEXIBLE 35mm COPPER
3453020	302	CRIMPET COVERS
3453100	310	LUGS FOR AL ALLOY & ACSR
3453230	323	PARALLEL GROVE CLAMP
3453270	327	BI-METAL COMPRESSION TERMINAL PIN TYPE 175/400 ACSR
3453280	328	BI-METAL COMPRESSION TERMINAL STRAIGHT PALM 175/400 ACSR
3453290	329	BI-METAL COMPRESSION TERMINAL 30° PALM 175/400 ACSR
3453980	398	BI-METAL TERMINAL LUG
3453990	399	BI-METAL SPACER KIT

SECTION 8

MIN NUMBER	M	DESCRIPTION
3451200	120	NON-TENSION ALUMINIUM JOINT
3451350	135	NON-TENSION ALUMINIUM TO COPPER JOINT
3451690	169	NON-TENSION CONNECTOR 100mm ACSR
3451700	170	NON-TENSION CONNECTOR 150 & 175mm ACSR
3451710	171	NON-TENSION CONNECTOR 100mm ACSR TO 70mm COPPER
3451720	172	NON-TENSION CONNECTOR COPPER TO ALUMINIUM
3452060	206	NON-TENSION CONNECTOR COPPER
3452430	243	NON-TENSION CONNECTOR 0.4 COPPER EQUIV 400 ALUM
3452630	263	NON-TENSION CONNECTOR 240 COPPER TO 400 SCA
3453300	330	NON-TENSION CONNECTOR 175 & 400mm ACSR
3454840	484	INSULATED TWINSHIELDS

SECTION 9

MIN NUMBER	M	DESCRIPTION
3453210	321	CONDUCTOR REPAIR SLEEVE (COMPRESSION TYPE) 70,175,400mm ² ACSR
3453220	322	CONDUCTOR REPAIR SLEEVE (PREFORMED TYPE)

SECTION 10

MIN NUMBER	M	DESCRIPTION
3451920	192	LIVE LINE TAP FOR COPPER CONDUCTORS UP TO 70mm ²
3451921	192/1	LIVE LINE TAP FOR COPPER
3451980	198	LIVE LINE TAP FOR AAAC & ACSR CONDUCTORS UP TO 100mm ²
3451981	198/1	LIVE LINE TAP FOR ALLOY - ACSR
3452790	279	LIVE LINE TAPPING BAIL

SECTION 11

MIN NUMBER	M	DESCRIPTION
3454480	448	HELICAL FITTINGS - HARDEX INTERMEDIATE
3454630	463	HELICAL FITTINGS - BARE 11kv SIDE TIES
3454640	464	HELICAL FITTINGS - 11kv BARE CONDUCTOR DOUBLE TOP TIES
3454650	465	HELICAL FITTINGS - COPPER 11kv SIDE TIES
3454660	466	HELICAL FITTINGS - PVC L.V. TERMINATIONS/DEAD ENDS
3454670	467	HELICAL FITTINGS - 11kv COPPER CONDUCTOR DOUBLE TOP TIES
3454680	468	HELICAL FITTINGS - PVC SIDE TIES
3454683	468/3	HELICAL FITTINGS - PVC SIDE TIES WITH SPIRAL
3454690	469	HELICAL FITTINGS - SINGLE TOP TIE TIES FOR COPPER
3454700	470	HELICAL FITTINGS - BARE L.V. TERMINATIONS/DEAD ENDS
3454710	471	HELICAL FITTINGS - 33/66kv BARE CONDUCTOR SIDE TIES
3454720	472	HELICAL FITTINGS - BARE L.V. INTERMEDIATE/ANGLE TIES
3454730	473	HELICAL FITTINGS - SINGLE TOP TIE TIES FOR ALUMINIUM
3454740	474	HELICAL FITTINGS - SINGLE TOP TIE TIES FOR PVC
3454743	474/3	HELICAL FITTINGS - SINGLE TOP TIE TIES FOR PVC WITH SPIRAL
3454750	475	HELICAL FITTINGS - 33/66kv BARE CONDUCTOR SINGLE TOP TIES
3454760	476	HELICAL FITTINGS - 33/66kv BARE CONDUCTOR DOUBLE TOP TIES
3454770	477	HELICAL FITTINGS - PVC L.V. INTERMEDIATE/ANGLE TIES
3454780	478	HELICAL FITTINGS CONCENTRIC CABLE DEAD ENDS
3455710	571	HELICAL FITTINGS CONDUCTOR DEAD END GRIPS A.A.A.C
3455720	572	SOCKET THIMBLE H.V HELICAL DEAD ENDS 70 Kn
3455722	572/2	THIMBLE CLEVIS 125 Kn
3455723	572/3	DOUBLE EYE LINK 120kN
3455740	574	HELICAL FITTINGS CONDUCTOR DEAD END GRIPS H.D COPPER

SECTION 12

MIN NUMBER	M	DESCRIPTION
3450041	4/1	SECTION ANGLE CROSSARM SINGLE/H POLES (HEAVY CONSTRUCTION)
3450050	5	ANGLE STRUT
3450480	48	ANGLE CROSSARM
3450491	49/1	ANGLE CROSSARM
3450492	49/2	ANGLE CROSSARM
3450500	50	ANGLE CROSSARM
3450510	51	INTERMEDIATE CROSSARM H POLE 11 & 33 kv (HEAVY CONSTRUCTION)
3450560	* 56	LINE CROSSARM 1M SPACING
3450560	* 56	LINE CROSSARM 1M SPACING MODIFIED
3450561	* 56/1	LINE CROSSARM SINGLE PHASE
3450570	* 57	CROSSARM TERMINAL
3450570	* 57	CROSSARM TERMINAL MODIFIED
3450571	* 57/1	CROSSARM TERMINAL SINGLE PHASE
3450580	58	SECTION ANGLE CROSSARM 1M SPACING
3450581	* 58/1	SECTION ANGLE CROSSARM SINGLE PHASE
3450590	59	TIE STRAP
3450630	63	SQUARE CURVED WASHER
3450730	73	SQUARE TAPERED CURVED WASHER
3450762	76/2	AUTO RECLOSER & TRANSFORMER MOUNTING CROSSARM (H-POLE)
3450820	82	H-POLE TERMINAL CROSSARM / LINE
3450961	96/1	LINE CROSSARM 11 & 33 kv HEAVY
3451000	100	CHANNEL CROSSARM 66kv
3451010	101	BRACED ANGLE CROSSARM
3451032	103/2	BRACED TERMINAL CROSSARM WITH PROVISION FOR SURGE DIVERTERS (66kv HEAVEY CONSTRUCTION)
3451033	103/3	TERMINAL CROSSARM 66 Kv
3451060	106	TERMINATING STRAP
3451070	107	11 & 33kv H-POLE THRO,(PIN) CROSSARM (HEAVY)
3451450	145	LINE CROSSARM (TERMINAL)
3451460	146	LINE TERMINATING STRAP
3451480	148	TERMINAL CROSSARM 11kv HEAVY CONSTRUCTION
3451490	149	BRACED TERMINAL CROSSARM 33kv HEAVY CONSTRUCTION
3451500	150	LINE STRAP
3451501	150/1	LANDING STRAP TERMINAL
3451510	151	LONG SECTION STRAP HEAVY CONSTRUCTION
3451511	151/1	LANDING STRAP SECTION
3451520	152	SHORT SECTION STRAP (HEAVY CONSTRUCTION)
3451530	153	BRACED ANGLE CROSSARM
3451540	154	11 & 33kv PILOT INSULATOR SPINDLE
3451560	156	SINGLE CROSSARM LINE SECTION STRAP
3451860	186	SECTION CROSSARM
3451870	187	SECTION CROSSARM
3451880	188	LINE CROSSARM 66kv DUAL (SUSP)
3451890	189	LINE CROSSARM LONG 66 Kv DUAL (SUSP)
3451910	191	LINE CROSARM 1.2M SPACING SINGLE & H-POLE
3451940	194	SECTION ANGLE CROSSARM 1.2M SPACING SINGLE & H-POLE
3451960	196	SECTION ANGLE CROSSARM 1.4M SPACING SINGLE & H-POLE
3451961	196/1	CROSSARM STRUT FOR 1.4 SPACING SECTION ANGLE & 1.2 SPACING TERMINAL CROSSARMS
3452220	222	DUAL CIRCUIT CROSSARM
3452260	226	ANGLE STRUT
3452270	227	LINE CROSSARM
3452280	228	D; SHAPED TAPERED WASHER
3452460	246	CROSSARM TIE STRAP

SECTION 12

MIN NUMBER	M	DESCRIPTION
3452571	257/1	BRACED ANGLE CROSSARM 66kV UP TO 200mm ² AAAC
3452580	258	SUSPENSION TYPE CHANNEL CROSSARM 66kV
3452590	259	LONG SECTION STRAP 125kN
3452591	259/1	LONG SECTION STRAP (MIN FAILING LOAD 70kN)
3452600	260	SHORT SECTION STRAP 125kN
3452601	260/1	SHORT SECTION STRAP (MIN FAILING LOAD 70kN)
3452610	261	LONG TERMINAL STRAP 125kN
3452620	262	SHORT TERMINAL STRAP 125kN
3452680	268	SECTION STRAP (MIN FAILING LOAD 20kN)
3453150	315	BRASS WASHER
3453170	317	SQUARE FLAT WASHER
3453180	318	CROSSARM STRAP FOR USE WITH M107 400mm H.D.A
3453190	319	ROUND FLAT WASHER
3453620	362	POLE BOLTS 16mm DIA
3453630	363	POLE BOLTS 20mm DIA
3453640	364	ASSEMBLY BOLTS 12mm DIA
3453650	365	ASSEMBLY BOLTS 16mm DIA
3453660	366	ASSEMBLY BOLTS 20mm DIA
3454000	400	TERMINAL CROSSARM 1.2M SPACING SINGLE POLES 100mm ² MAX
3456050	605	TOWER FITTING STEP BOLTS

SECTION 13

MIN NUMBER	M	DESCRIPTION
3450930	93	NON EXTENSIBLE GALVANISED STEEL EARTH ROD
3451240	124	WIRE ROPE GRIP
3451930	193	EARTHING CLAMP
3452250	225	STEEL TOWER EARTH CONDUCTOR TO STEEL BONDS

SECTION 14

MIN NUMBER	M	DESCRIPTION
3450640	64	11kV PIN INSULATOR
3450645	64/5	11kV PIN INSULATOR LONG CREEPAGE
3450850	85	DISC INSULATOR M.F.L 70kN
3451320	132	33kV PIN INSULATOR
3451651	165/1	33kV LINE POST LAPP INSULATOR
3451652	165/2	33kV LINE POST POLYMER INSULATOR
3451680	168	66kV LINE POST LAPP INSULATOR
3452541	254/1	125kN DISC INSULATOR 146 CENTRES
3453010	301	80kN HIGH IMPACT DISC
3453091	309/1	125kN DISC INSULATOR PORCELAIN
3453092	309/2	125kN DISC INSULATOR GLASS
3453130	313	11kV PIN INSULATOR HIGH IMPACT
3454171	417/1	11kV COMPOSITE INSULATOR PIG-TAIL HOOK 20kN LIGHT CONSTRUCTION
3454172	417/2	11kV COMPOSITE INSULATOR SOCKET & BALL 70kN
3454180	418	DISC INSULATOR M.F.L 80kN 132kV SUSPENSION
3454190	419	DISC INSULATOR M.F.L 120kN 132kV SUSPENSION
3454200	420	DISC INSULATOR M.F.L 190kN 132kV SUSPENSION - 132kV TOWER LINES TWIN 400 mm ² A.C.S.R CONDUCTORS
3454240	424	L.V SUPPORT INSULATOR
3454241	424/1	L.V BRACKET INSULATOR
3455400	540	INSULATOR SECURITY CLIP

SECTION 15

MIN NUMBER	M	DESCRIPTION
3450010	1	INSULATOR FITTING SHACKLES M.F.L 70kN
3450250	25	EYEBOLT
3450620	62	11kV INSULATOR PIN 10kN
3450670	67	BALL ENDED HOOK M.F.L
3450710	71	PILOT INSULATOR 11 & 33kV
3450840	84	33kV INSULATOR PIN 10kN
3450860	86	BALL ENDED HOOK M.F.L 70kgf
3450870	87	SOCKET CLEVIS M.F.L 70kN
3450880	88	SOCKET TONGUE M.F.L 70kN
3450890	89	ARCING HORNS
3451040	104	SUSPENSION 'U' LINK
3451050	105	GUSSET PLATE 66KV SUSP
3451080	108	M.S. EYE NUT
3451330	133	BALL ENDED EYE LINK M.F.L 70kN
3451570	157	INSULATOR BRACKET
3451840	184	SOCKET CLEVIS M.F.L 70kN
3451850	185	BALL CLEVIS M.F.L 70kN
3452350	235	CLAMP ADAPTORS M.F.L 70kN
3452490	249	BALL ENDED EYE LINK M.F.L 125kN
3452500	250	D' SHACKLE M.F.L 125kN
3452510	251	SOCKET TONGUE M.F.L 125 kN
3452550	255	ARCING HORNS FOR USE WITH 12700KG EQUIPMENT
3453060	306	CRANKED LINKS M.F.L 125kN
3453070	307	LANDING PLATE M.F.L 125kN
3453160	316	VANDAL SHIELD
3453430	343	EARTHWIRE FITTINGS CLEVIS CONNECTORS 70 & 175mm ² TERMINATIONS
3453970	397	INSULATOR SUPPORT PLATE
3454011	401/1	EXTENSION LINK M.F.L 70kN
3454270	427	U' BOLTS
3454280	428	EARTHWIRE SUSPENSION LINK
3454310	431	BALL ENDED EYE LINK M.F.L 190kN
3454320	432	LANDING PLATE M.F.L 190 kN
3454330	433	SHACKLE M.F.L 190kN
3454340	434	TOWER SHACKLE 400mm ² ACSR CONDUCTORS M.F.L 190kN
3454350	435	SOCKET TONGUE M.F.L 190kN
3454360	436	CRANKED LINKS M.F.L 190kN
3454431	443/1	CRANKED LINKS M.F.L 70kN
3454441	444/1	LANDING PLATE M.F.L 70kN
3455090	509	INSULATOR FITTING / CONDUCTOR TERMINATION CLEVIS PINS
3455100	510	SOCKET CLEVIS M.F.L125kN
3455130	513	SOCKET CLEVIS M.F.L 70kN
3455500	550	INSULATOR ARCING HORN 132kV SUSPENSION LIVE END TYPE 4SC
3455510	551	INSULATOR ARCING HORN 132kV LOW DUTY EARTH END TYPES 4LA & 7LA
3455520	552	INSULATOR ARCING HORN 132kV SUSPENSION EARTH END TYPE 4SA FOR 39" GAP
3455530	553	INSULATOR ARCING HORN 132kV SUSPENSION EARTH END TYPE 4SB FOR 44" GAP
3455540	554	INSULATOR ARCING HORN 132kV SUSPENSION EARTH END TYPE 7SA FOR 39" GAP
3455550	555	INSULATOR ARCING HORN 132kV SUSPENSION EARTH END TYPE 7SB FOR 44" GAP
3455560	556	INSULATOR ARCING HORN 132kV SUSPENSION LIVE END TYPE 4TC
3455570	557	INSULATOR ARCING HORN 132kV SUSPENSION LIVE END TYPE 7TC

SECTION 15

MIN NUMBER	M	DESCRIPTION
3455580	558	INSULATOR ARCING HORN 132KV TENSION EARTH END TYPE 4TB FOR 44" GAP
3455590	559	INSULATOR ARCING HORN 132KV TENSION EARTH END TYPE 4TA FOR 39" GAP
3455600	560	INSULATOR ARCING HORN 132KV TENSION EARTH END TYPE 7TA FOR 39" GAP
3455610	561	INSULATOR ARCING HORN 132KV TENSION EARTH END TYPE 7TB FOR 44" GAP
3455620	562	INSULATOR ARCING HORN 132KV SUSPENSION LIVE END TYPE 3/2/132 SC
3455690	569	INSULATOR BRACKET

SECTION 16

MIN NUMBER	M	DESCRIPTION
3450130	13	INSULATOR - L.V. REEL TYPE
3450140	14	D' IRON & PIN
3450220	22	TIE ROD FOR STRUT POLES
3453730	373	A.B.C - STAND-OFF CLEAT
3453740	374	A.B.C. - ANCHOR CLAMP
3453800	380	A.B.C COMPRESION TYPE END SEAL
3453810	381	A.B.C COMPRESION JOINT INSULATED
3453820	382	A.B.C COPPER LUG (INSULATED BI-METAL)
3453830	383	A.B.C. SERVICE ANCHOR CLAMP
3453840	384	A.B.C. INSULATOR PIERCING CONNECTOR
3453850	385	A.B.C. - SUPPORT HOOK
3453860	386	A.B.C. - EXTENDED SUPPORT HOOK
3453870	387	A.B.C. - NUT HOOK
3453880	388	A.B.C. - SUSPENSION STRAP
3453890	389	A.B.C. - LIVE POT END
3453910	391	A.B.C - SUSPENSION CLAMP
3453920	392	ASSEMBLY A.B.C. SERVICE TERMINATION BOX
3453930	393	A.B.C. INSULATING SLEEVE FOR TENSION & NON TENSION JOINTS

SECTION 17

MIN NUMBER	M	DESCRIPTION
3451390	139	POLE - CIRCUIT IDENTIFICATION PLATE (SYMBOL TYPE)
3452760	276	POLE NUMBER AND NAMEPLATE
3452811	281/1	SAFETY SIGN TRANSMISION TOWERS
3453690	369	FISHING WARNING & PROHIBITION NOTICES
3454621	462/1	SAFETY SIGN
3454900	490	TOWER SAFETY SIGN BRACKET
3455800	580	POLE NUMBERING SYSTEMFITTING RAILS
3455810	581	DIGITS FOR POLE NUMBERING SYSTEM

SECTION 18

MIN NUMBER	M	DESCRIPTION
3454250	425	WOOD POLE PLUGS
3454410	441	FOUNDATION BLOCKS FOR SINGLE & H-POLES
3454420	442	FOUNDATION BLOCKS FOR SINGLE & H-POLES
3454501	450/1	POLE DRILLINGS MP TYPES AS SHOWN
3454502	450/2	POLE DRILLINGS MP TYPES AS SHOWN
3454503	450/3	POLE DRILLINGS MP TYPES AS SHOWN
3454504	450/4	POLE DRILLINGS MP TYPES AS SHOWN

SECTION 19

MIN NUMBER	M	DESCRIPTION
3450370	37	HOUSE SERVICE WALL BRACKET (SHORT)
3450380	38	HOUSE SERVICE WALL BRACKET ABC PIG TAIL
3450390	39	HOUSE SERVICE WALL BRACKET (LONG)
3450430	43	HOUSE SERVICE CORNER BRACKET (DIAGONAL)
3450440	44	EAVES EXTENSION BRACKET
3450520	52	COACH SCREW SERVICE INSULATOR
3452900	290	EXTENSION STRAPS FOR HOUSE SERVICE BRACKETS
3452910	291	SERVICES ENTRY TUBE
3452990	299	INSOLE 'CABLE TIE'
3454520	452	UNDER EAVES SERVICE BREECHES JOINT BOX CONCENTRIC CABLES

SECTION 20

MIN NUMBER	M	DESCRIPTION
3450020	2	TRANSFORMER FIXING STRAP
3450540	54	PILOT CROSSARM
3450751	75/1	TX PLATFORM GUSSET PLATE
3450920	92	TRANSFORMER PLATFORM SINGLE POLE
3451554	155/4	LUCY M.P.C.400 POLE MOUNTED L.V. FUSE UNIT
3452120	212	PILOT BAR ASSEMBLY
3452470	247	CABLE CLEAT

SECTION 21

MIN NUMBER	M	DESCRIPTION
3450721	72/1	SUPPORTING BRACKET FOR FUSES ON SINGLE POLES
3450722	72/2	STAND - OFF SUPPORT BRACKET FOR FUSES ON SINGLE POLES
3451360	136	66kV SURGE DIVERter SUPPORTING BRACKET
3452340	234	33kV SURGE DIVERter SUPPORTING BRACKET
3452342	234/2	33kV SURGE ARRESTER BASE PLATE HEC 36
3452820	282	BUSBAR & CONNECTORS FOR 33kV MOUNTED TERMINATION
3453421	342/1	BOWTHORPE BMP 33kv SURGE ARRESTER TYPE HEC 36
3454601	460/1	11kV SURGE ARRESTER
3454602	460/2	24kV SURGE ARRESTER

SECTION 22

MIN NUMBER		M	DESCRIPTION
3450170		17	TUBULAR STAY ROD
3451150		115	STAY TERMINATING THIMBLE
3451190		119	STAY STRAP
3452700	*	270	PREFORMED STAY GRIPS
3452710	*	271	PREFORMED GRIPS
3453050		305	STAY INSULATORS FOR LV,11kV & 33kV LINES
3454090		409	DUCKBILL STAY ANCHOR (2.2 TONNE)
3454100		410	WOOD STAY BLOCKS
3454450		445	STAY INSULATORS FOR 66kV LINES

SECTION 23

MIN NUMBER	M	DESCRIPTION
3450530	53	CROSSARM - 'H' POLE FUSE & CABLE SUPPORT
3450910	91	OUTRIGGER A.C.D BRACKET
3452452	245/2	POST INS 11KV SWITCH & FUSE GEAR
3453440	344	11KV POLE MOUNTED EXPULSION FUSE UNIT BRUSH TYPE A3463
3453450	345	11KV POLE MOUNTED EXPULSION FUSE UNIT TYPE DJP (GEC)
3454590	459	ASSEMBLY OF SINGLE POLE ANGLE BRACKET TYPE A3468 (HAWKER)
3455050	505	UNIVERSAL EXPULSION FUSE ELEMENTS
3457150	715	CROSSARM MOUNTING STRAP FOR SPRITE ENCLOSURE
3457160	716	MOUNTING BRACKET FOR COOPER KYLE VCS SWITCH

M	ME	ISSUE	SECTION
1		B	15
2			20
4/1			12
5			12
13		A	16
14		A	16
17		A	22
22			16
25			15
37			19
38			19
39			19
43			19
44			19
48			12
49/1			12
49/2			12
50			12
51			12
52			19
53			23
54			20
56			12
56/1			12
57			12
57/1			12
58			12
58/1			12
59			12
62			15
63			12
64			14
64/5			14
67			15
71		B	15
72/1		A	21
72/2			21
73			12
75/1			20
76/2			12
82			12
84			15
85			14
86			15
87			15
88			15

M	ME	ISSUE	SECTION
	89		15
91			23
92		B	20
93		A	13
96/1			12
100			12
101			12
103/2			12
103/3			12
104			15
	105	B	15
106			12
107			12
108			15
113			2
115			22
119			22
120			8
122			6
123			6
124		A	13
132			14
133			15
135			8
136			21
139			17
141			5
141/6			5
145			12
146			12
148			12
149			12
150			12
150/1			12
151			12
151/1			12
152			12
153			12
154		G	12
155/4			20
156			12
157			15
159			6
161			6
165/1			14
165/2			14

M	ME	ISSUE	SECTION
168			14
169			8
170			8
171			8
172			8
173			6
175		A	2
175/3		B	2
176A			2
176/3			2
178			2
184		D	15
185		E	15
186			12
187			12
188			12
189			12
191			12
192			10
192/1			10
	193	C	13
194			12
195			7
196			12
196/1			12
198			10
198/1			10
204		G	1
205			1
205/2			1
206			8
	212		20
210			6
218			1
218/1			1
219			7
220			6
222			12
225			13
226			12
227			12
228			12
233			6
234		E	21
234/2			21
235		A	15

M	ME	ISSUE	SECTION
236			7
237			7
238			7
243			8
245			23
2452			23
246			12
247			20
249			15
250			15
251			15
253			6
254/1			14
	255	B	15
257/1			12
258			12
259			12
259/1			12
260			12
260/1			12
261			12
262			12
263			8
268			12
270		G	22
	271	F	22
273			7
276		D	17
277			7
279			10
281/1		A	17
282			21
285			7
287			6
288			6
290			19
291			19
295			7
299			19
301			14
302			7
304			2
305			22
306		A	15
307			15
308		A	2

M	ME	ISSUE	SECTION
309/1			14
309/2			14
310			7
311A			2
313			14
315			12
316			15
317			12
318			12
319			12
320		B	5
321			9
322			9
323			7
327		A	7
328			7
329			7
330			8
333/1			6
333/2			6
335			6
336			6
338			6
339			6
340			6
341			6
342/1			21
343			15
344			23
345			23
347			2
352		C	3
350		C	4
351			4
353		B	3
354		E	3
356		C	3
357		C	3
358		A	4
359			4
362			12
363			12
364			12
365			12
366			12
369			17

M	ME	ISSUE	SECTION
373			16
374		B	16
380			16
381		A	16
382		A	16
383		B	16
384		C	16
385			16
386			16
387			16
388		A	16
389			16
391		C	16
392			16
393			16
394		A	3
397			15
398			7
399			7
400			12
401/1			15
409			22
410			22
417/1			14
417/2			14
418			14
419			14
420			14
424			14
424/1			14
427		A	15
428			15
431			15
432			15
433			15
434			15
435			15
436			15
443/1			15
444/1			15
445			22
446			6
448			11
452			19
453		B	2
453/2			2

SECTION 1

MIN NUMBER	M	DESCRIPTION
3452040	204	JUMPER PARKING BAR
3452050	205	MOUNTING CHANNEL AUTO RECLOSER SINGLE POLE
3452052	205/2	HSAR MOUNTING BRACKET 'H' POLE
3452180	218	COPPER ELBOW FOR RECLOSER CONNECTIONS
3452181	218/1	PMR RECLOSER STRAP FOR MOUNTING SURGE ARRESTERS

6x25
M.S. Washer welded
in position shown

10 Dia M.S. Rod

M.C.I. Insert
tapping M10 Hole

Insulator

Insulator Resilient Pad

75x12 ϕ
Coachscrew

380

100

50

50

75

SCALE	-
DATE	11.7.01
DRAWN	A.J.E.
TRACED	
CHECKED	K.D.
APPROVED	15.07.01

JUMPER PARKING ROD

ISSUE	DATE
G	11.7.01

REDRAWN ON CAD

AJE/RD

Produced by
Output Services

GPU
POWER UK

DRG. No.

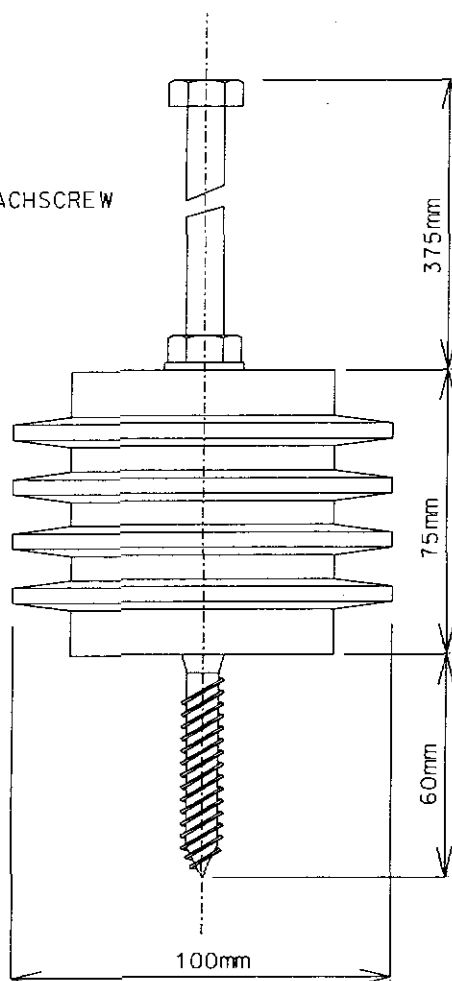
M204

DRG. No.

M204

A4

CYLOALIPHATIC INSULATOR
STAINLESS STEEL BAR & COACHSCREW



ELECTRICAL DATA	
TEST	VALUE
CREEPAGE (mm)	236 (30mm/kV)
TAUT STRING (mm)	141
IMPULSE FLASHOVER (DRY)	+ve 77.7kVp / -VE 146.1kVp
IMPULSE WITHSTAND (DRY)	+ve 74.2kVp / -VE 138.5kVp
AC 1 MIN. FLASHOVER (DRY)	48.8kV
AC 1 MIN. WITHSTAND (DRY)	44.0kV
AC 1 MIN. FLASHOVER (WET)	27.8kV
AC 1 MIN. WITHSTAND (WET)	23.1kV

MECHANICAL DATA (INSULATOR)		
TEST	VALUE	SPECIFICATION
GLASS TRANS. TEMP T _g (DSC)	105-115 C	
FLEXURAL STRENGTH	120-130 N/mm ²	ISO 178
TENSILE STRENGTH	80-95 N/mm ²	ISO R527
IMPACT STRENGTH	9-11kJ/m ²	ISO 179

INSULATOR MARKING:

1. Product Code.
2. Manufacturer's Identity.
3. Year/Week number batch identity.

PRODUCT CODE: 56141-20

Produced by
Primary CAD

SCALE	-
DATE	06.02.03
DRAWN	M.J.B
TRACED	-
CHECKED	KD
APPROVED	13.02.03

CYLOALIPHATIC PARKING BAR INSULATOR

ISSUE	DATE

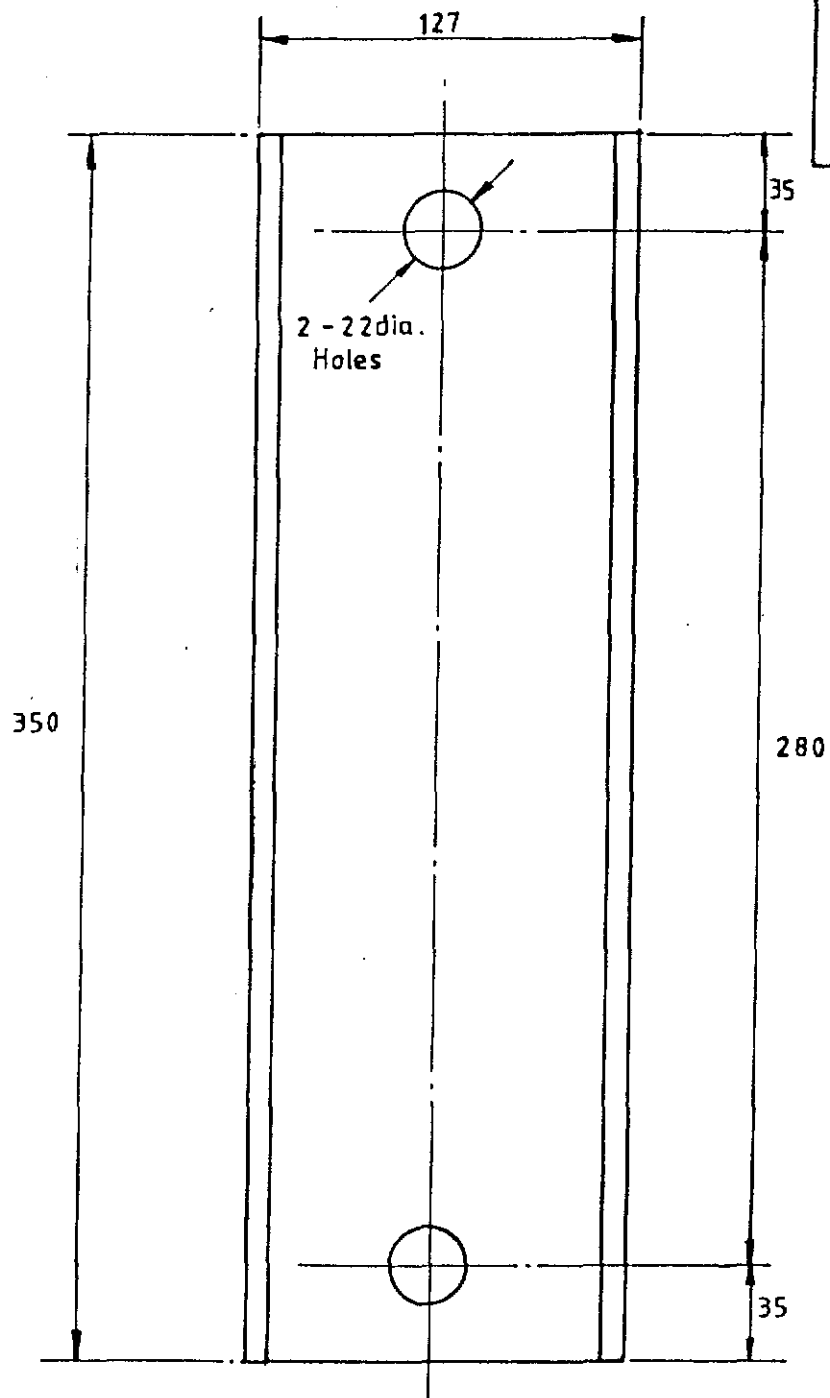
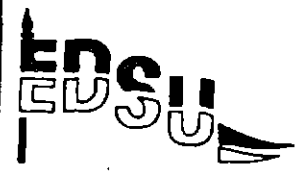


Aquila

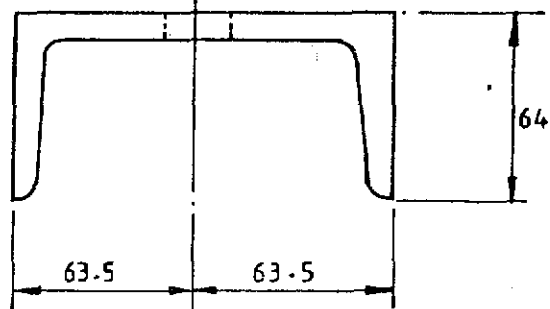
DRG. No.

M204a

DRG. No. M424



127x64x14.9kg./MMS Channel
To be galvanised to B.S.729
Spec. after cutting & drilling.



TITLE

MOUNTING CHANNEL
AUTO RECLOSER SINGLE POLE

SCALE

DATE 17.8.92

DRAWN A.J.B.

TRACED S.G.T.

CHECKED

APPROVED

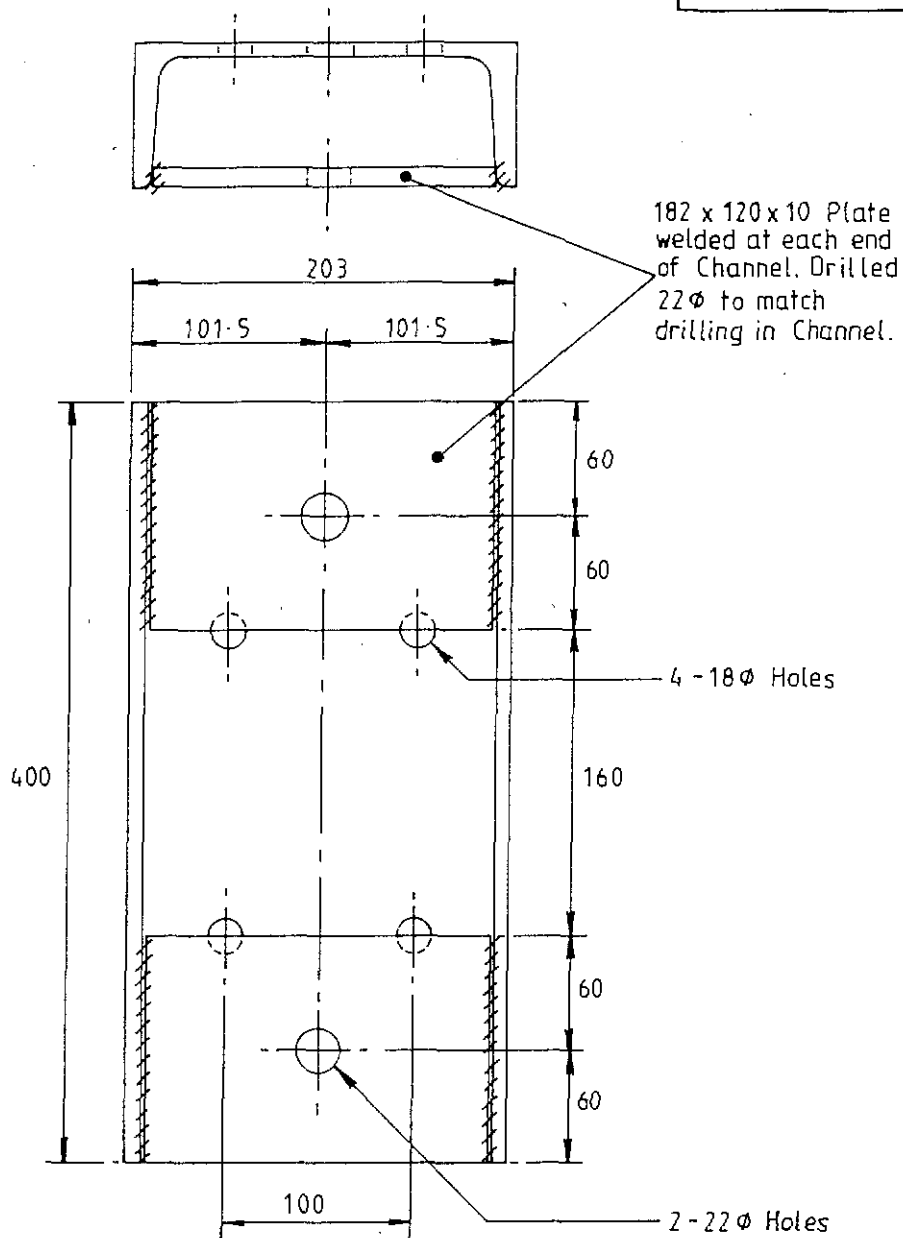
MEB

Midlands Electricity plc

DRG No

M. 205

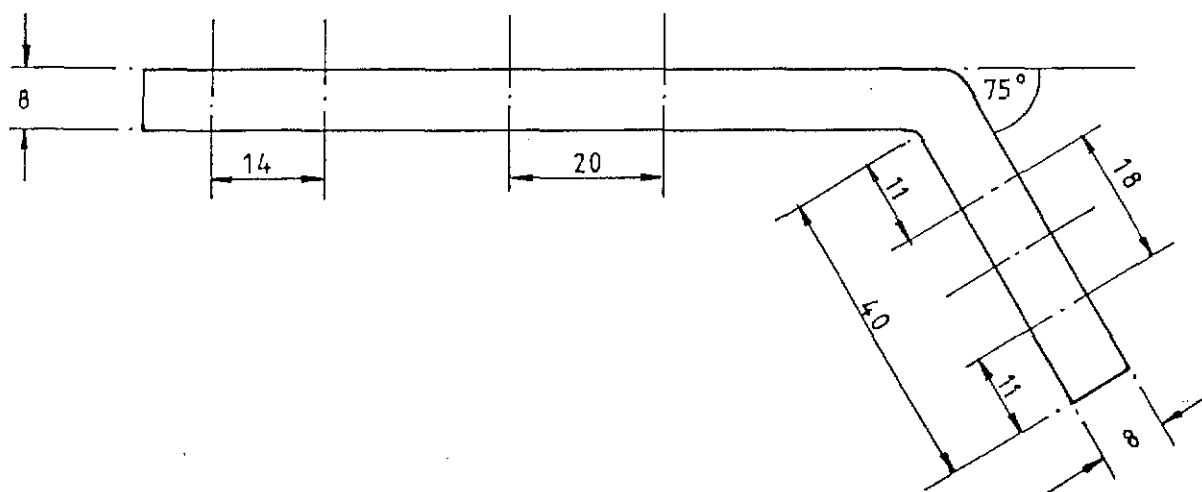
ADDED BY
DATE



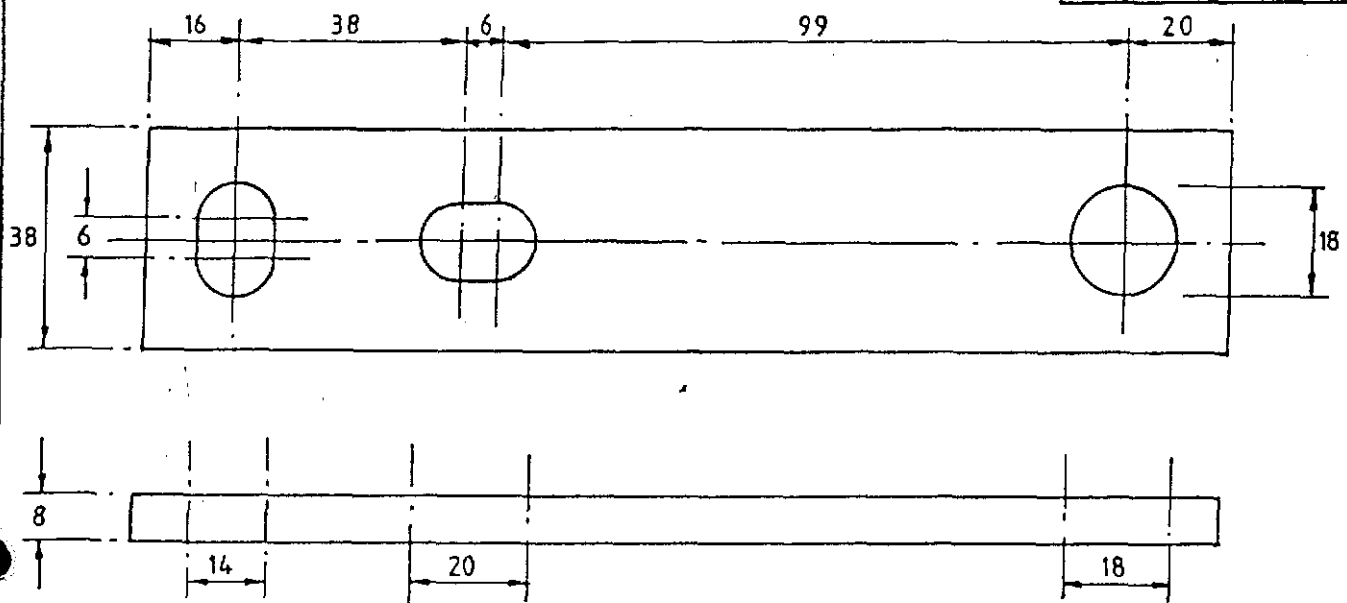
MATERIAL 400 x 203 x 76 x 23.82 kg/m M.S. Channel.
 2 - 182 x 120 x 10 M.S. Plates Welded as shown
 c/w 4 - M16 x 50 Bolts
 Threaded 38mm c/w Nut & Washer

FINISH Galv. to B.S. 729 after Cutting, Drilling & Welding.

ADDED BY DATE	TITLE HSAR MOUNTING BRACKET 'H' POLES	SCALE	—	 Midlands Electricity plc ORG No M205/2
		DATE	20-4-94	
		DRAWN	A.J.B.	
		TRACED	P.J.W.	
		CHECKED	RCJ	
		APPROVED		

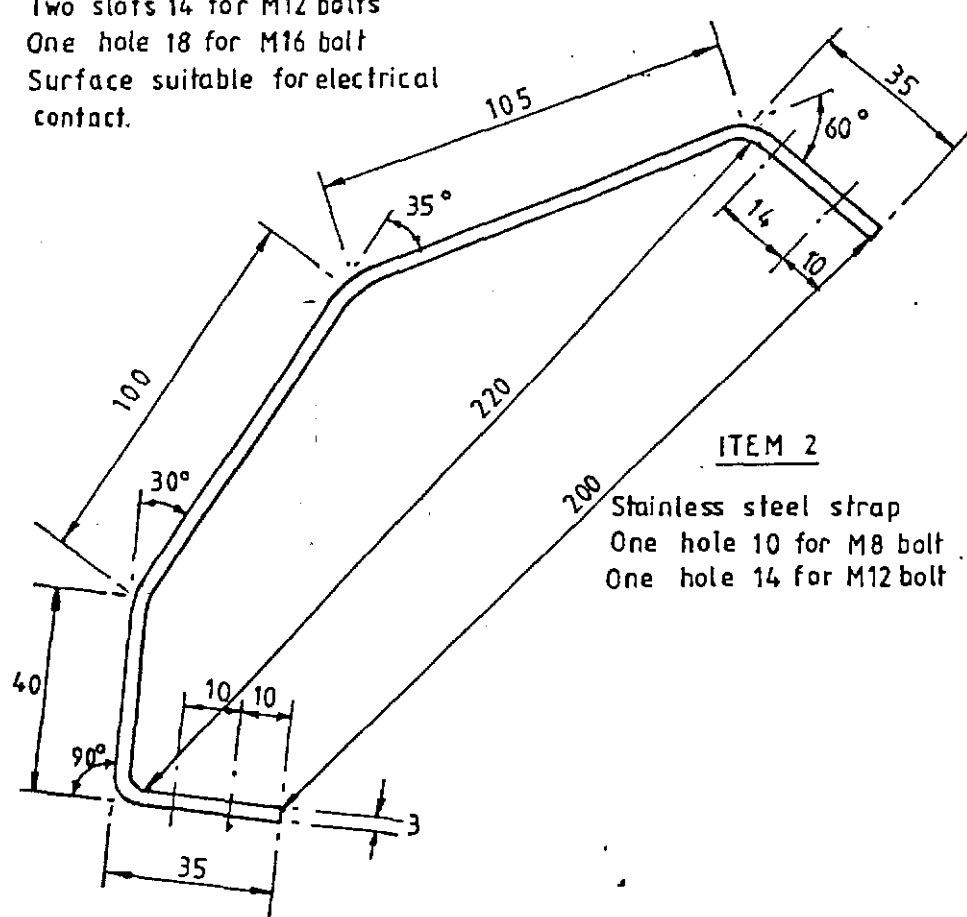


ADDED BY DATE	TITLE COPPER ELBOW FOR RECLOSER CONNECTIONS	SCALE	—	 Midlands Electricity plc
		DATE	11.1.94	
		DRAWN	R.C.F.	
		TRACED	S.G.T.	
		CHECKED		
		APPROVED		
	ESI STANDARD 41-16			DRG No M218



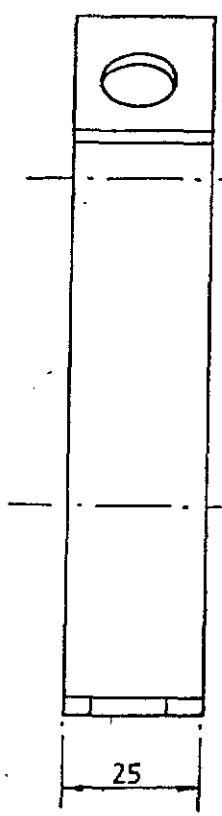
ITEM 1

Copper strap
 Two slots 14 for M12 bolts
 One hole 18 for M16 bolt
 Surface suitable for electrical contact.



ITEM 2

Stainless steel strap
 One hole 10 for M8 bolt
 One hole 14 for M12 bolt



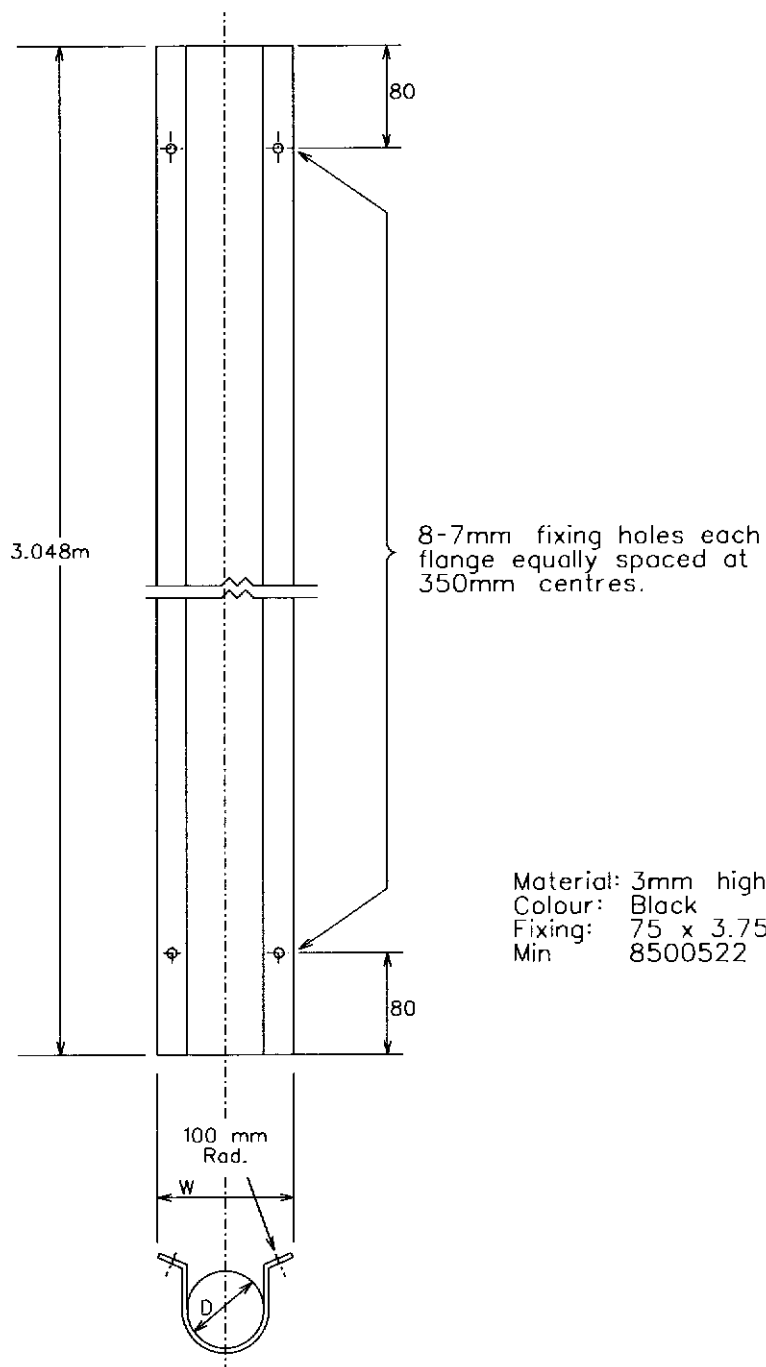
ADD'D DATE	TITLE	SCALE	 Midlands Electricity plc. M218/1	
	PMR RECLOSER STRAP FOR MOUNTING SURGE ARRESTERS	DATE		14.11.95
		DRAWN		RCF
		TRACED		SGT
		CHECKED		RCF
		APPROVED		

AMDO 15628 E

ESI 41-16

SECTION 2

MIN NUMBER	M	DESCRIPTION
3451130	113	EARTH WIRE GUARD AND CABLE COVERS
3451750	175	CLEAT CROSSARM
3451753	175/3	CLEAT CROSSARM
3451760	176A	CABLE BACKBOARD FITTINGS FOR 33kv & 66kv TERMINAL POLES
3451763	176/3	FISH PLATE
3451780	178	33kv & 66kv CABLE CLEATS
3453040	304	TERMINATION SUPPORT BRACKET HEATSHRINK 11kv (H- POLE)
3453080	308	ANTI CLIMBING GUARD SUPPORT H.V. TERMINAL WITH U/G CABLE
3453110	311A	CABLE PROTECTION CASING & LID FOR 33 & 66kv TERMINAL POLES
3453470	347	HEAT SHRINK BREAKOUT FOR CONCENTRIC CABLES
3454530	453	CROSSARM & INSULATOR TAKE-OFF BRACKET FOR 33k.v POLE TERMINATION
3454532	453/2	CROSSARM & INSULATOR TAKE-OFF BRACKET FOR 66k.v POLE TERMINATION



Material: 3mm high impact rigid P.V.C
 Colour: Black
 Fixing: 75 x 3.75 Galvanised Clout Nails
 Min 8500522

Min No	Cable Type & Size			Internal Dia	Manufactures Ref
	L.V Cable	H.V Cable	Earth Cable		
113/1	25 & 35 CNE Single		Earth Wire	25	CG003X10
113/2	25 & 35 CNE3 Phase		Earth Wire	38	CG004X10
113/4	185 Wavecon			64	CG006X10
113/5		95 & 185 XLPE		84	CG007X10
113/7		300 XLPE		100	CG008X10

Produced by
Primary CAD

SCALE	1:22
DATE	20.10.03
DRAWN	M.J.B
TRACED	
CHECKED	17.11.03
APPROVED	KD

EARTH WIRE GUARD AND CABLE COVERS

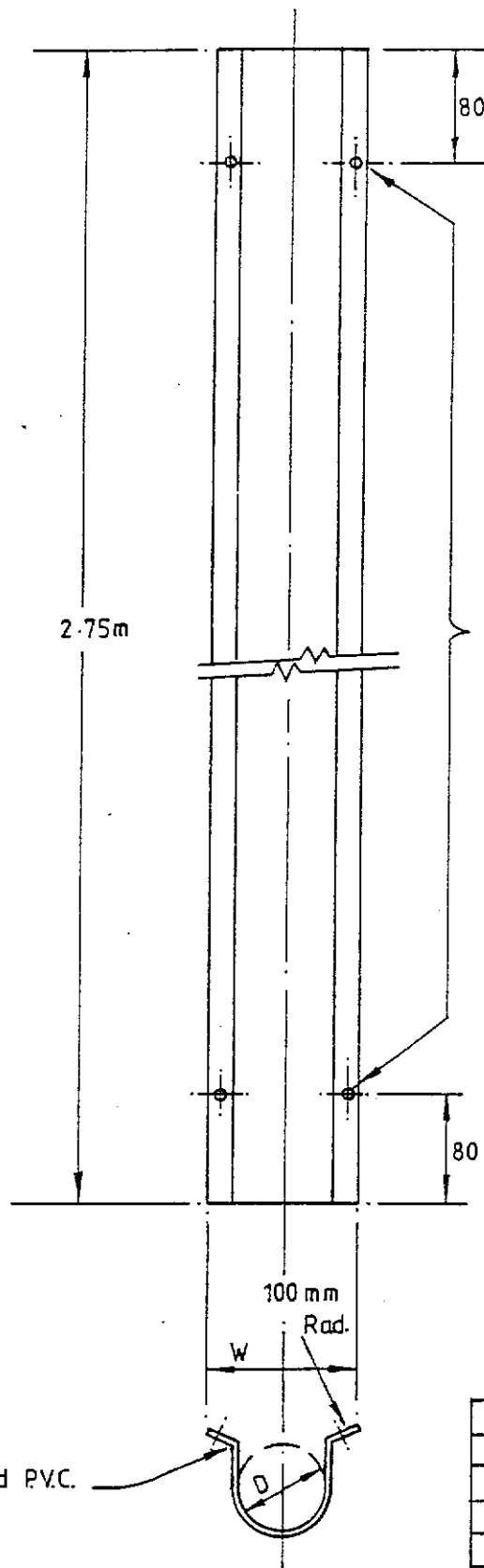
ISSUE DATE



Aquila

ORG. No.

M113



8-7mm fixing holes each flange equally spaced at 350mm centres.

Material : 3mm high impact rigid P.V.C.

Colour : Black

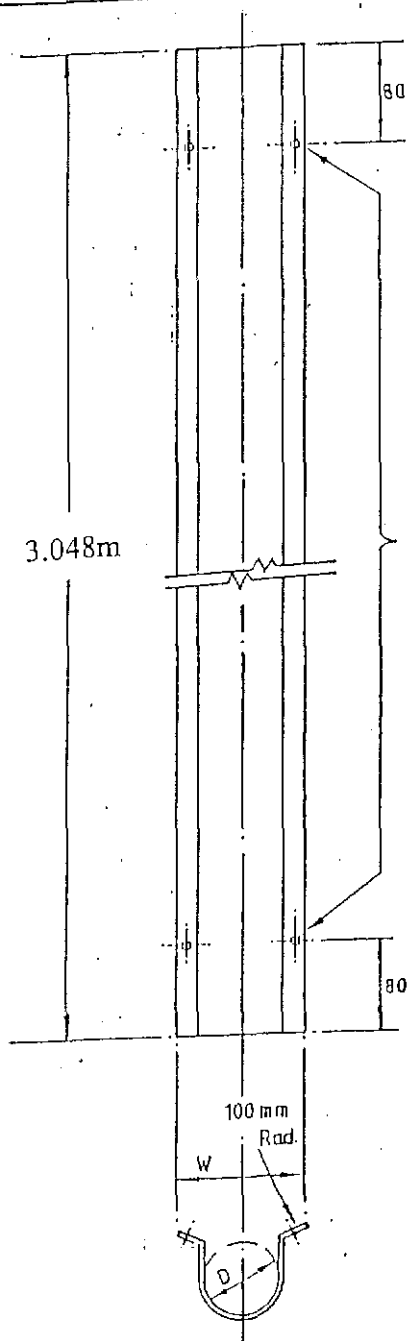
Fixing : Cover to be fixed with 16 - N°14 drive screws.

REF N°	D	W
M 113/0	16	56
M 113/1	25	65
M 113/2	38	78
M 113/3	50	90
M 113/4	64	104

SCALE	N.T.S.
DATE	27.7.81
DRAWN	
TRACED	SGT
CHECKED	
APPROVED	

EARTH WIRE GUARD AND CABLE COVERS.

ISSUE	MIDLANDS ELECTRICITY BOARD
DRG. No.	M 113.
ESI STANDARD	ESI. Drg N°



8-7mm fixing holes each
flange equally spaced at
350mm centres.

Material : 3mm high impact rigid P.V.C

Colour : Black

Fixing : 75 x 3.75 Galvanised Clout Nails
Min 8500522

Min No	Cable Type & Size			Internal Dia	Manufactures Ref
	L.V Cable	H.V Cable	Earth Cable		
113/1	25 & 35 CNE Single		Earth wire	25	CG003X10
113/2	25 & 35 CNE 3 Phase		Earth wire	38	CG004X10
113/4	185 Wavecon			64	CG006X10
113/5		95 & 185 XLPE		84	CG007X10
113/7		300 XLPE		100	CG008X10

TITLE

EARTH WIRE GUARD AND
CABLE COVERS

SCALE

DATE

DRAWN

TRACED

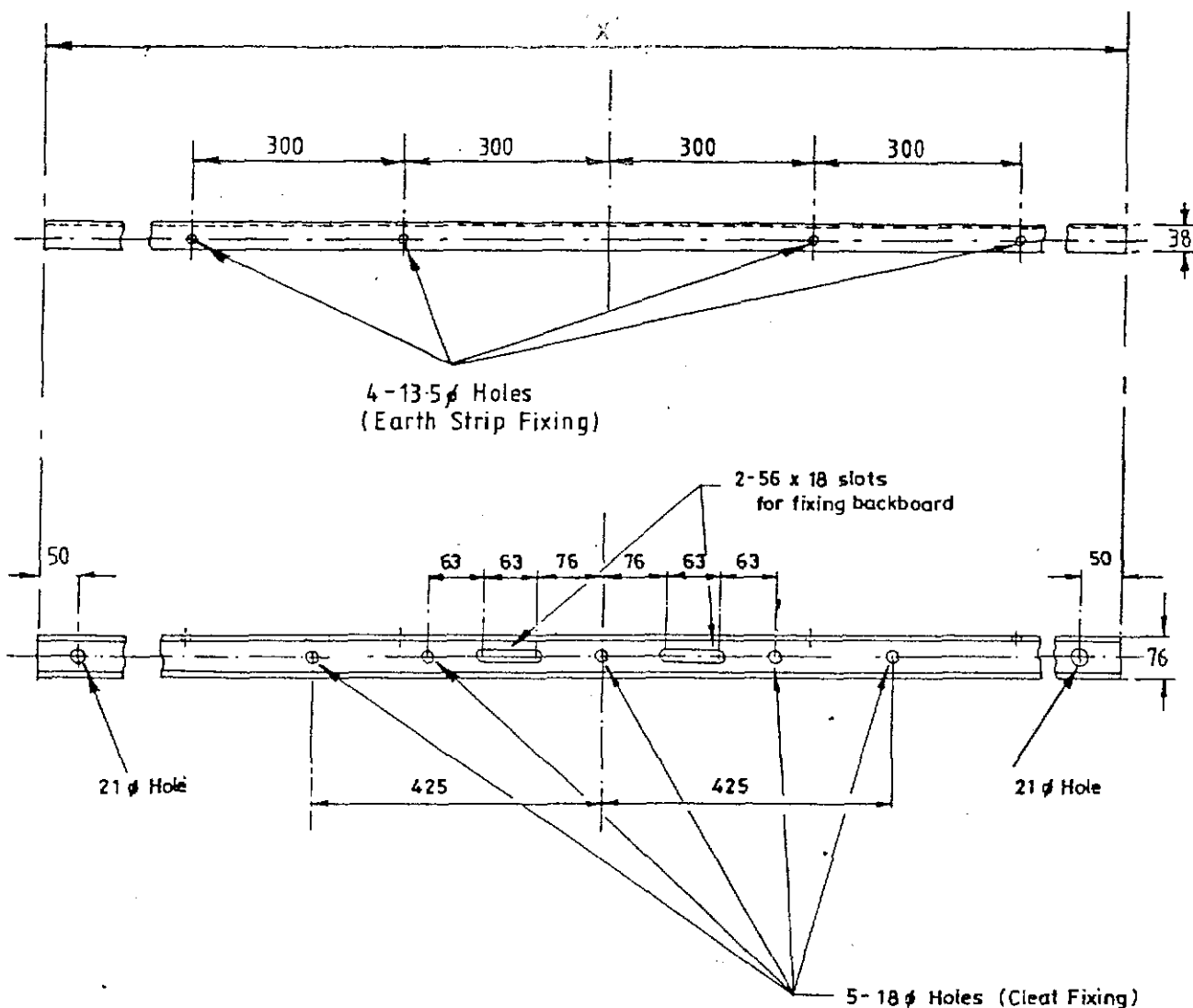
CHECKED

APPROVED

GPU
POWER UK

DRG No

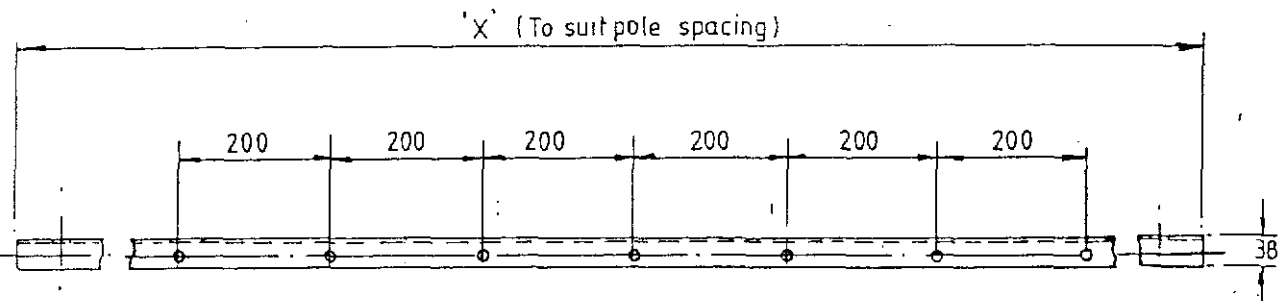
M 113



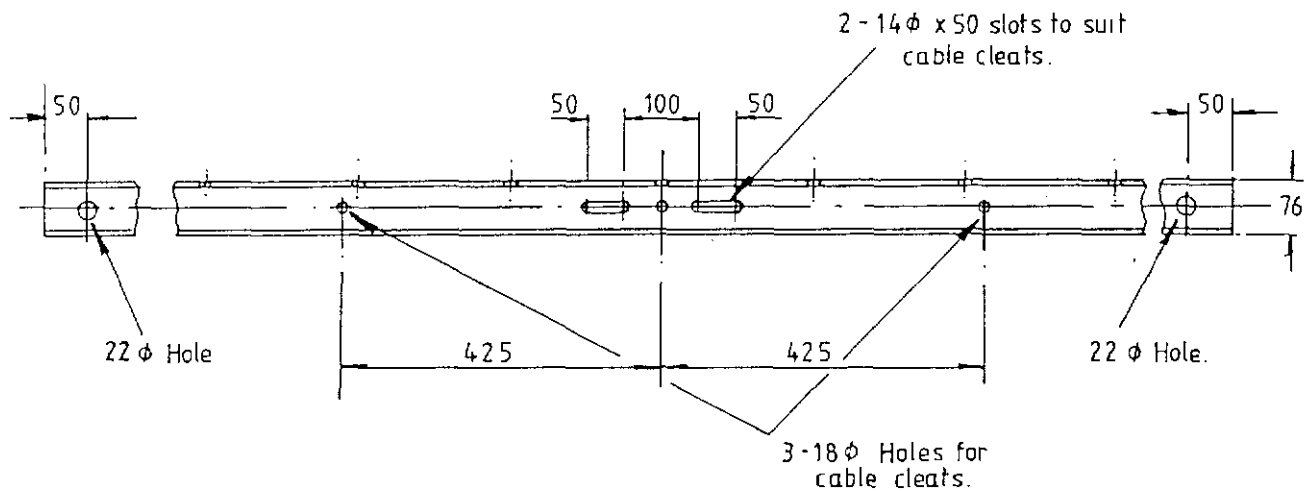
DRG N ^o	POLE SPACING	DIM'N. X.	MATERIAL.	SECTION.	NOTES.
175/1	1-83	1-93	M.S.	76x38x6	GALV TO BS.729 AFTER CUTTING & DRILLING
175/2	2-13	2-24	TO BS 4 QUALITY	CHANNEL	

SCALE	N. T. S.	CLEAT CROSSARM	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	26.10.82		A	
DRAWN	R.A.C.		Channel Changed to 76x38x6	DRG. No. M.175.
TRACED			27-7-87	
CHECKED	J.M.B.			E.S.I. DRG. N ^o
APPROVED				
E.S.I. STANDARD				

(MF)



NOTE:- 12 ϕ Holes to be equi-spaced at 200mm c/s to give max N^o available.



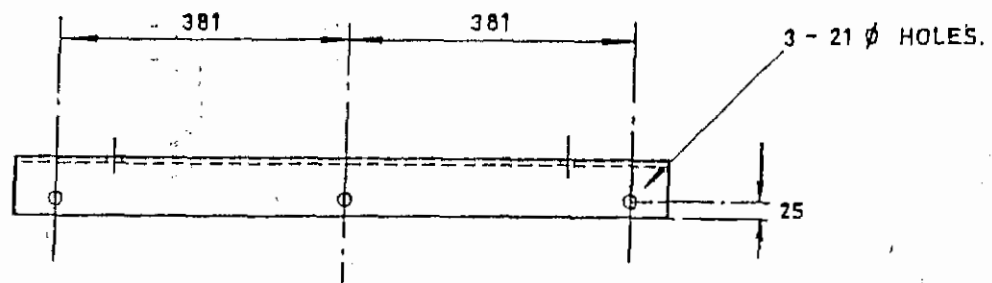
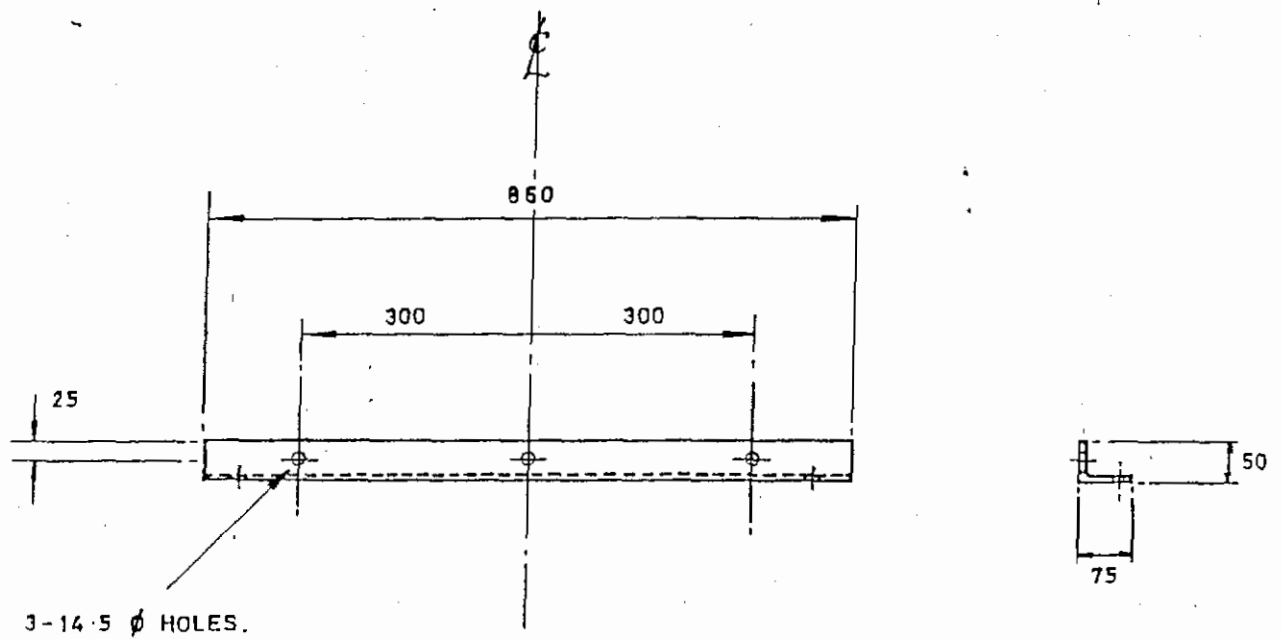
DRG N ^o	POLE SPACING.	DIM'N "X"	MATERIAL.	SECTION	NOTES.
175/3	1.83	1.93	MS.	76 x 38 x 6	GALV. TO BS.729
175/4	2.13	2.23			AFTER CUTTING & DRILLING.

SCALE	1:100
DATE	12-8-87
DRAWN	R.A.C.
TRACED	
CHECKED	
APPROVED	

CLEAT CROSSARM.

ISSUE	B
175/4 dim'n.	"X" corrected
187-94	A.J.B.

MIDLANDS ELECTRICITY BOARD
ORG. No. M.175/3



FINISH:—

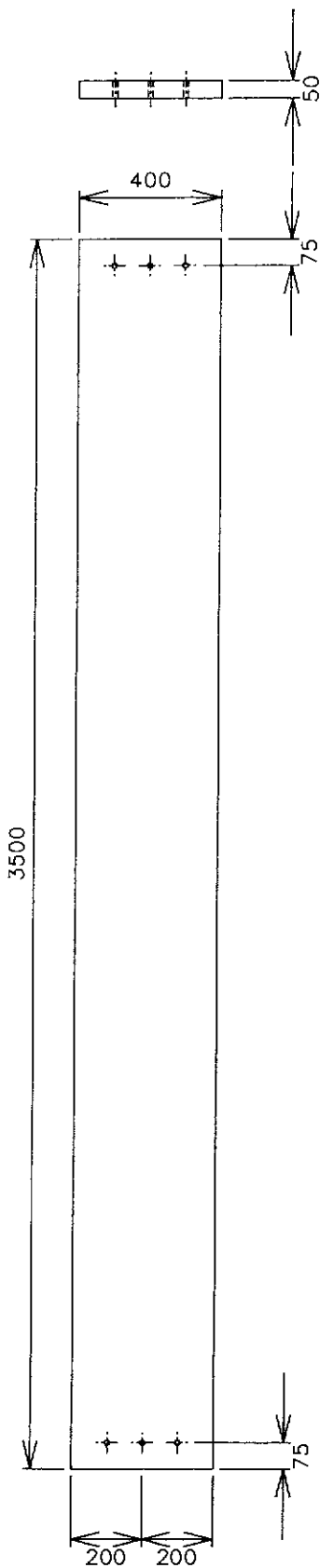
Galv. to B.S. 729 after cutting & drilling.

MATERIAL:—

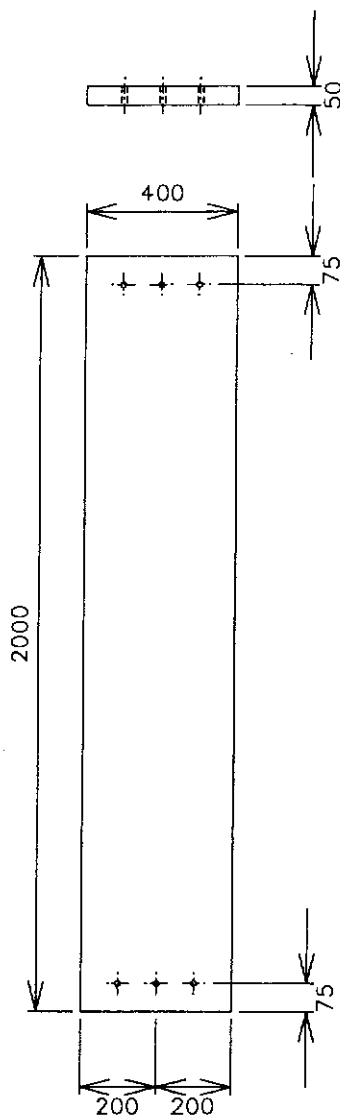
76 X 51 X 8 M.S. Angle to B.S. 4

28 / 33 ton quality.

SCALE	1-10	TERMINATION SUPPORT BRACKET HEATSHRINK 11 kV. TERMINATION (H-POLE)	ISSUE	MIDLANDS
DATE	26-10-82			ELECTRICITY BOARD
DRAWN	R.A.C.			ORG. No.
TRACED	A.E.			M.304
CHECKED	J.M.A.			
APPROVED	J.V.	E.S.I. STANDARD		E.S.I. ORG. No.



MIN No 176/1



MIN No 176/2

CABLE BACKBOARD

Material ELM

Finish Creosoted

Min No	Item	Length	Width
3451761	Backboard	3500	400
3451762	Backboard	2000	400
3451763	Fish Plate	250	300

SCALE	1:20
DATE	09.01.03
DRAWN	ADD
TRACED	
CHECKED	KD
APPROVED	

CABLE BACKBOARD FITTINGS PLATE
FOR 33kV & 66kV TERMINAL POLES

ISSUE	DATE



Aquila

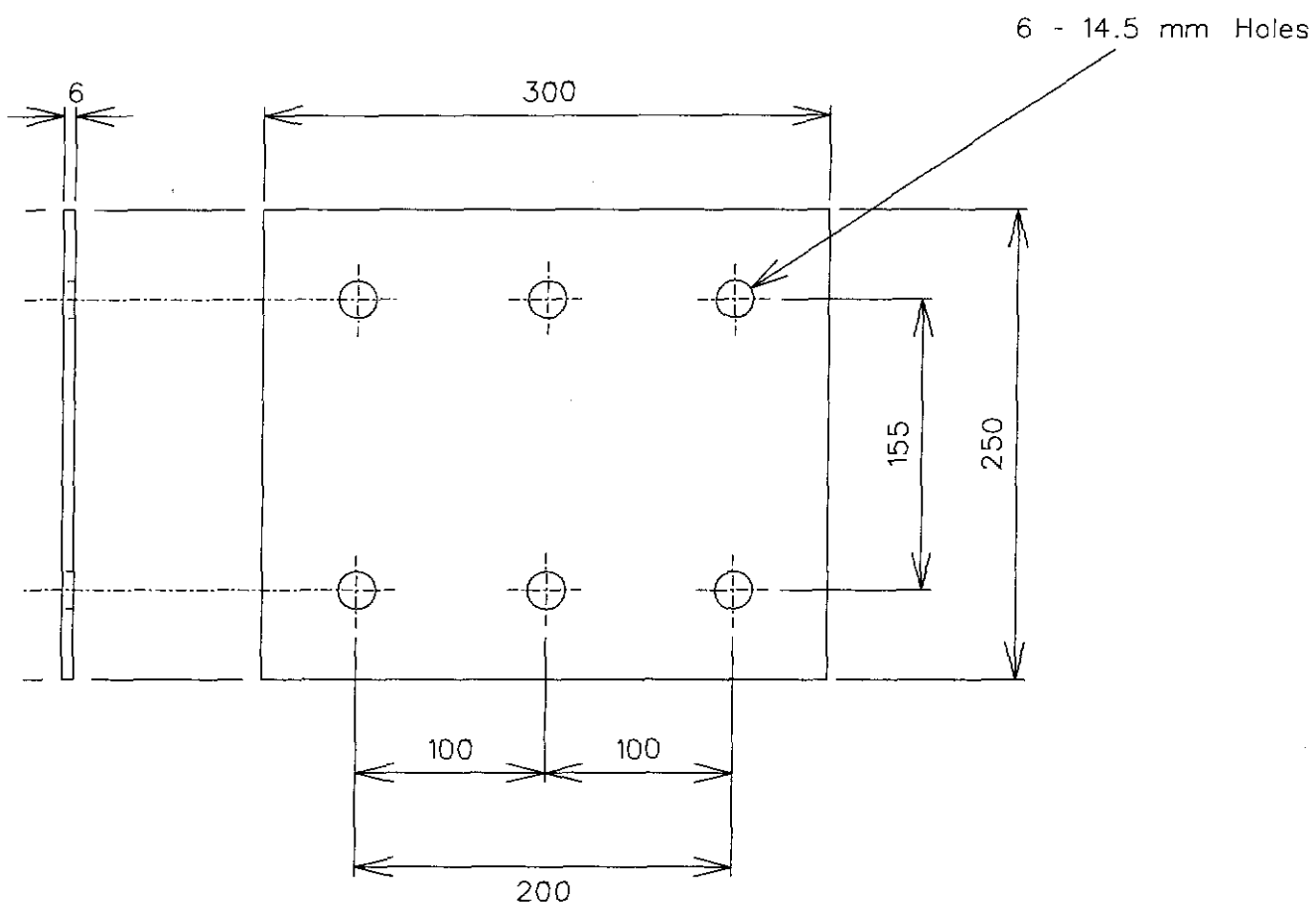
DRG. No.

M176A

DRG. No.

M176A

A4



MIN No 3451763

Produced by
Output Services

SCALE	1:4
DATE	09.01.03
DESIGN	ADD
TRACED	
CHECKED	KD
APPROVED	

CABLE BACKBOARD FISH PLATE
FOR 33kV & 66kV TERMINAL POLES

ISSUE	DATE



Aquila

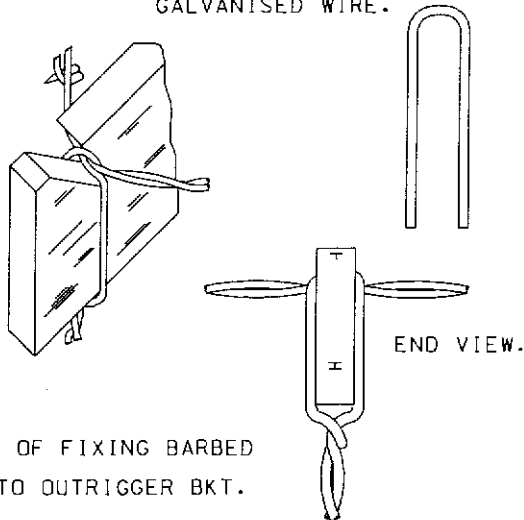
DRG. No.

M176

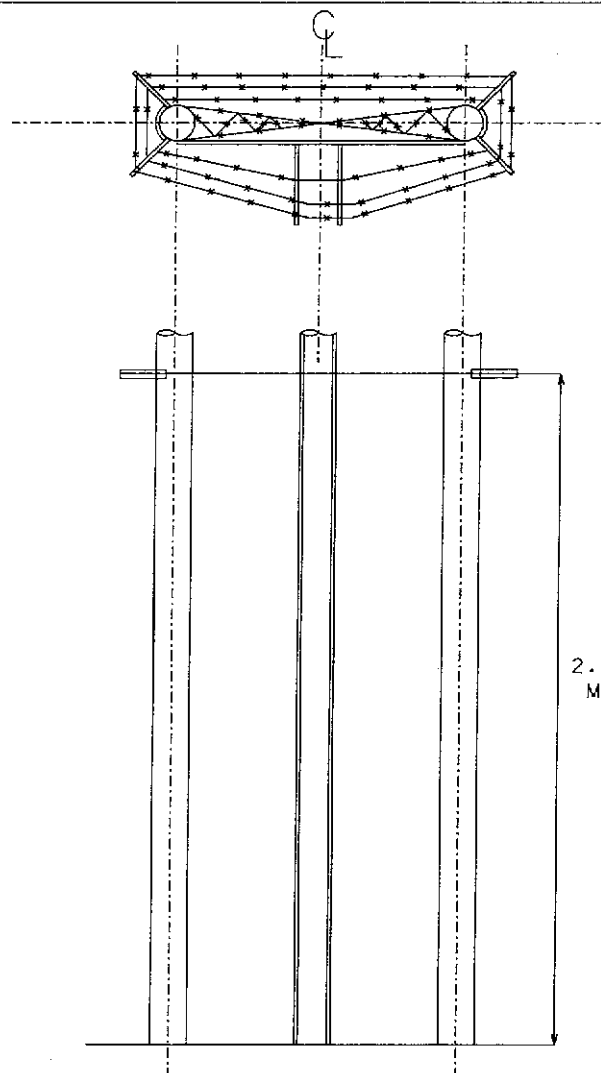
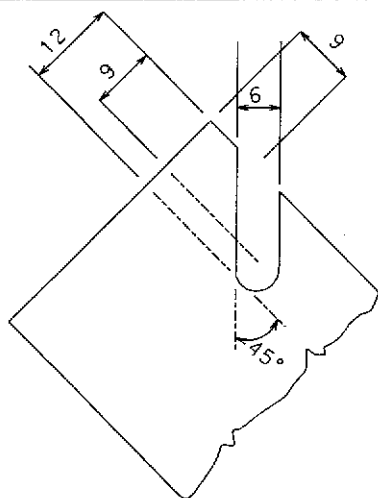
DRG. No.

M176

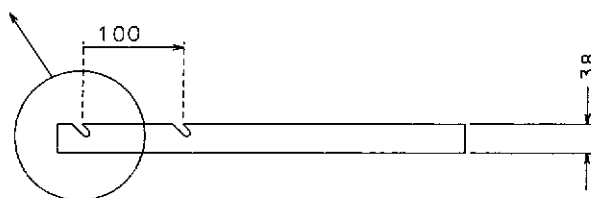
PREFORMED No.12 S.W.G.
GALVANISED WIRE.



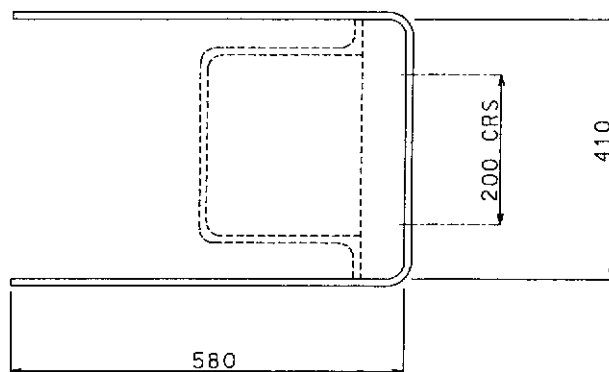
METHOD OF FIXING BARBED
WIRE TO OUTRIGGER BKT.



GENERAL ARRANGEMENT.



15 Ø HOLES.



MATERIAL :-
38x10 M.S. FLAT
28x33 TON QUALITY

FINISH:-
GALV TO B.S. 729 AFTER
CUTTING AND DRILLING.

SCALE	N.T.S
DATE	11.02.03
DRAWN	R.A.C.
TRACED	A.E.
CHECKED	
APPROVED	

ANTI CLIMBING GUARD SUPPORT
FOR H.V. TERMINAL POLE WITH
CABLE TERMINATION

ISSUE	DATE
B	05.06.03
MINOR MODS	
K.D/A00	

DRG. No.



Aquila

M 308

Produced by
Primary CAD

DRG. No. M 308

Cable Protection Case Kit
MIN No 3453110

Material

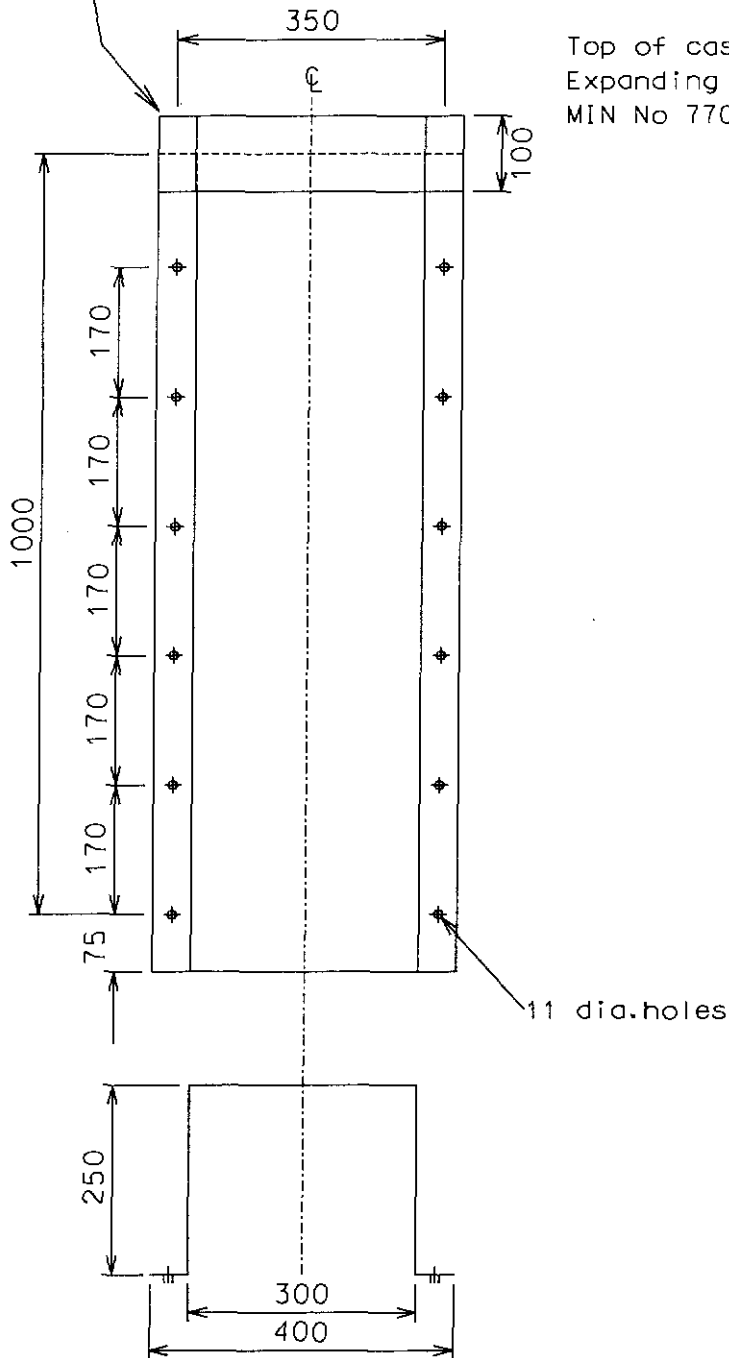
- 3 Cable Casings MIN No 3453114
- 1 Cable Backboard 3.5m MIN No 3451761

If backboards length is required to be longer than 3.5m

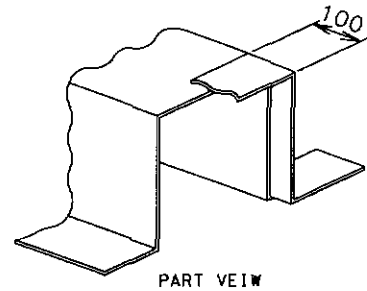
- 1 Cable Backboard 2m MIN No 3451762
- 1 Fish Plate MIN No 3451763

Top of casing to be sealed with Hilti Expanding Foam CF125
MIN No 7701300

For details see part view



Note:-Cable cover to be cut to suit varying cable length



MIN No 311/4

SCALE	1:10
DATE	14.01.03
DRAWN	ADD
TRACED	-
CHECKED	
APPROVED	

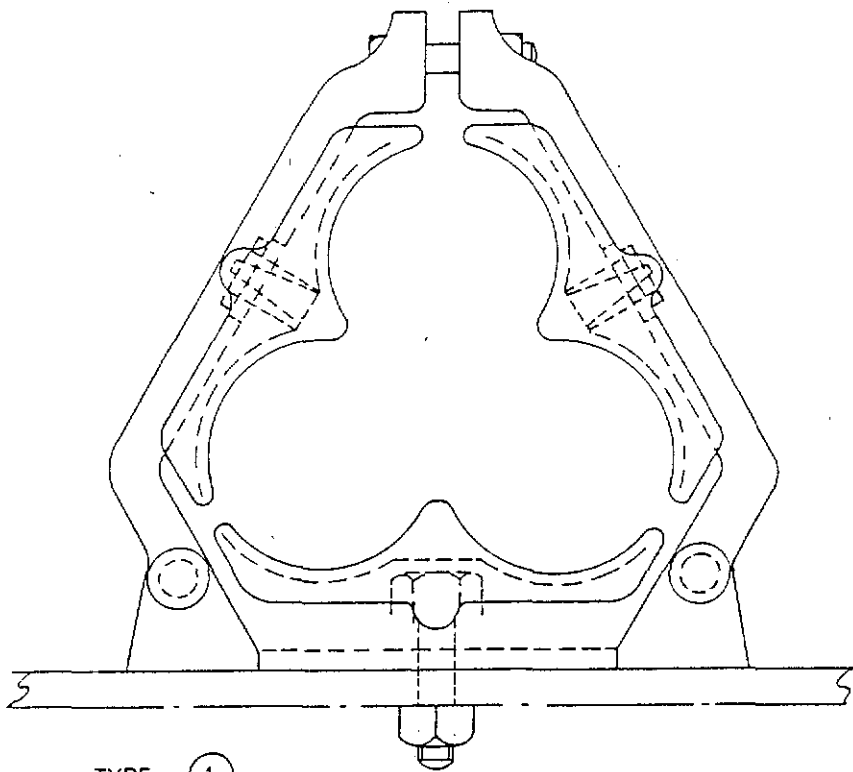
CABLE PROTECTION CASING AND LID
FOR 33KV & 66KV TERMINAL POLE

ISSUE	DATE

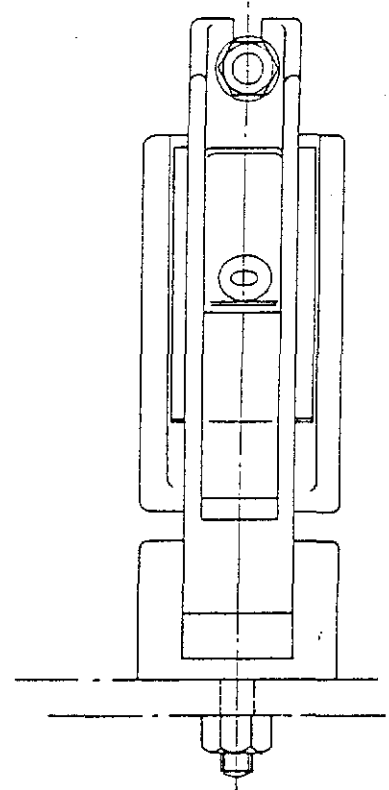
Produced by Primary CAD
 Aquila
DRG. No. M311

DRG. No.

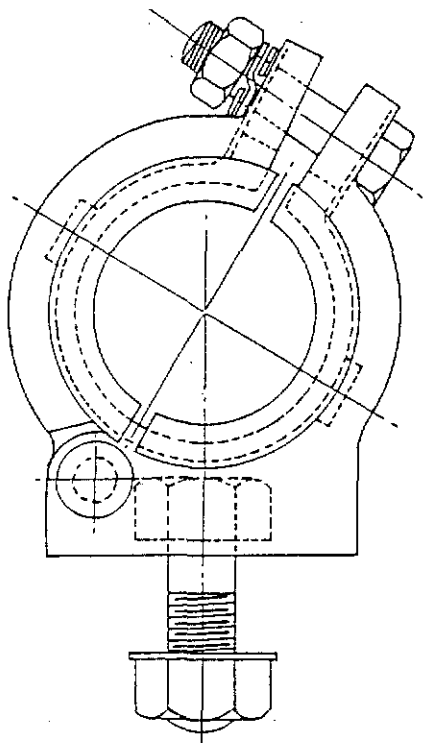
M311



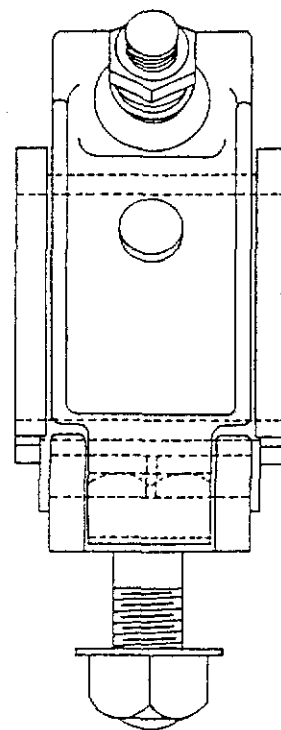
TYPE—①



LOOSE ADAPTOR TYPE TREFOIL CLEAT
SINGLE BOLT FIXING.



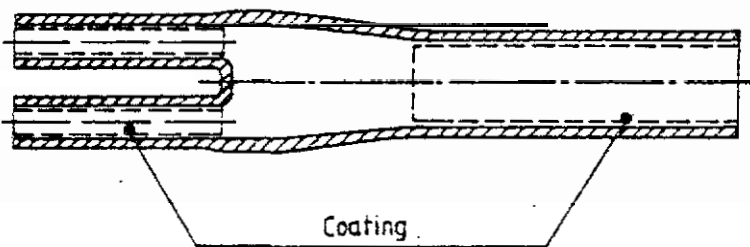
TYPE—②



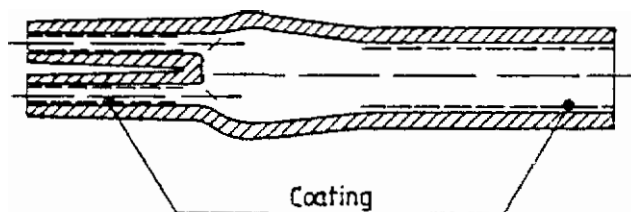
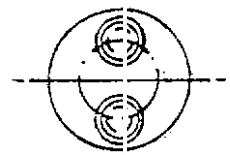
LOOSE ADAPTOR TYPE INSULATED SINGLE WAY
SINGLE BOLT FIXING CLEAT

NOTE: CLEATS TO BE ORDERED TO SUIT O/D OF CABLE.

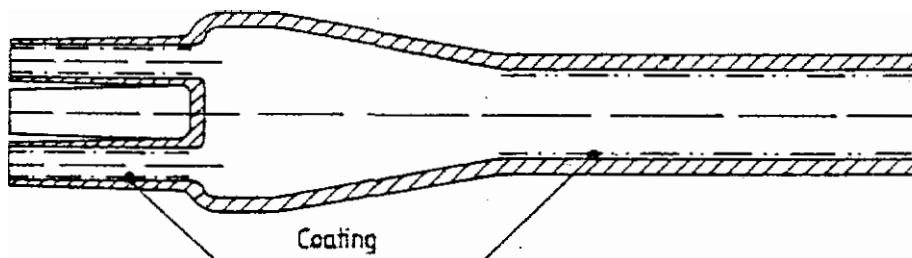
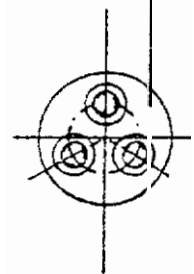
SCALE	M.T.S.	33 kV & 66 kV CABLE CLEATS	ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	26.10.82				
DRAWN	R.A.C.				
TRACED	P.G.				
CHECKED	<i>Amv</i>				
APPROVED	<i>Amv</i>	E.S.I. STANDARD			DRG. No. M178
					E.S.I. DRAWING No.



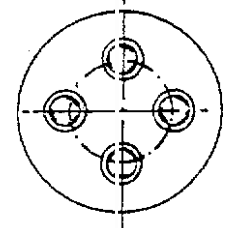
TYPE 1.



TYPE 2.



TYPE 3.

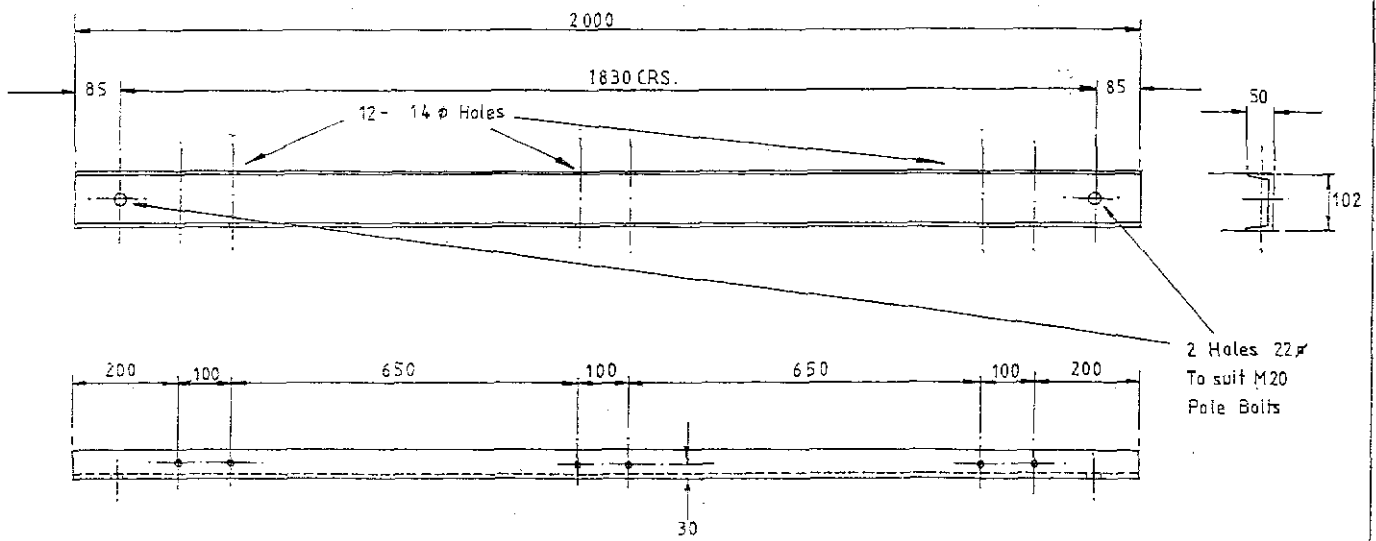


TYPE	CABLE TYPE & SIZE	RAYCHEM REF
347/1	4-25mm ² 1 CORE WITH CONCENTRIC N/E	302 K 333 -3/70
347/2	16-35mm ² 1 CORE WITH SPLIT CON. N+E	402 W 533 -3/70
347/3	16-35mm ² 3 CORE WITH CONCENTRIC N/E	502 K 033 -3/70

DATE	3.10.83
DRAWN	RAG
TRACED	J.W.
CHECKED	J.W.
APPROVED	J.W.

HEAT SHRINK
BREAKOUT FOR
CONCENTRIC CABLES

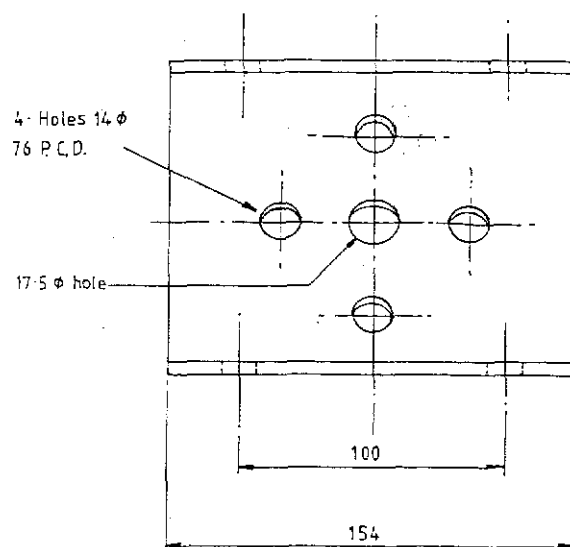
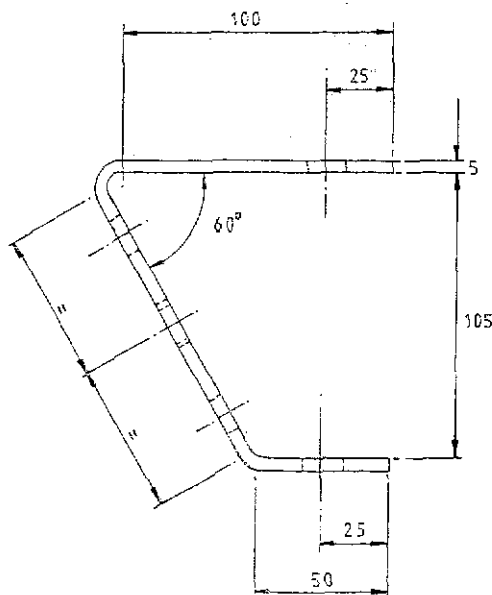
ISSUE	MIDLANDS ELECTRICITY BOARD
DRG. No.	M 347



SCALE: 1:10

CROSSARM:

102 x 50 x 2000 M.S. Channel
Galv. to BS 729 After Cutting
and drilling

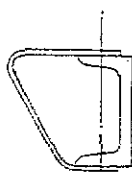


SCALE: 1:2

NOTE: All Holes 14 ϕ unless
otherwise stated.

INSULATOR TAKE OFF BRACKET

5 mm. Thick M.S. Galv to BS 729
after Cutting & Drilling.

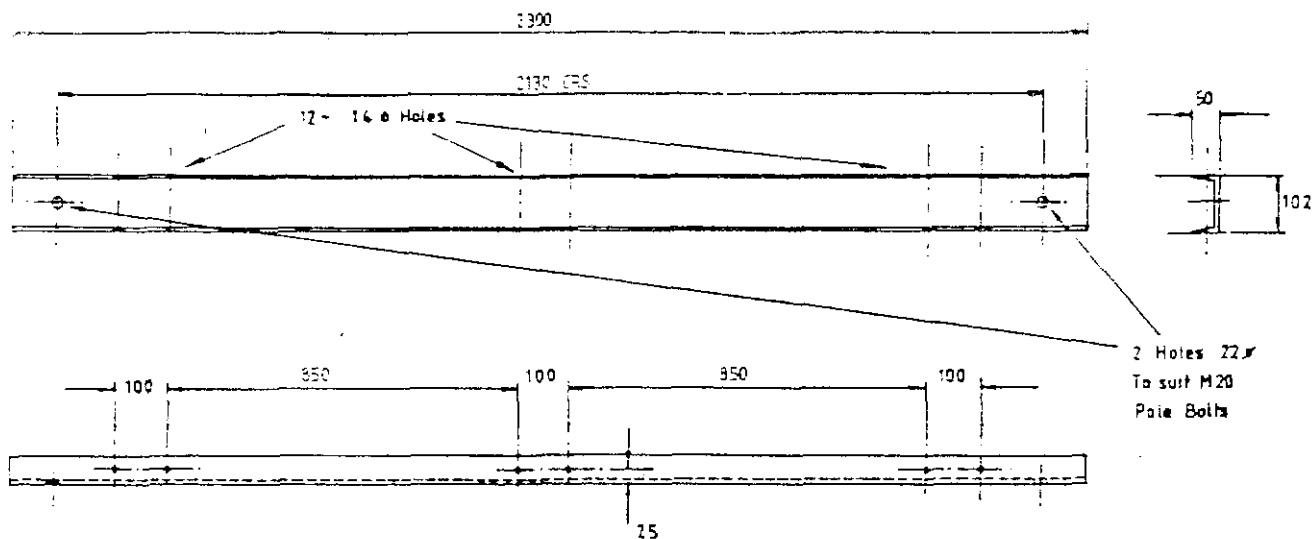


ASSEMBLY

SCALE	1:10 & 1:2
DATE	24-4-87
DRAWN	RAC
TRACED	
CHECKED	
APPROVED	

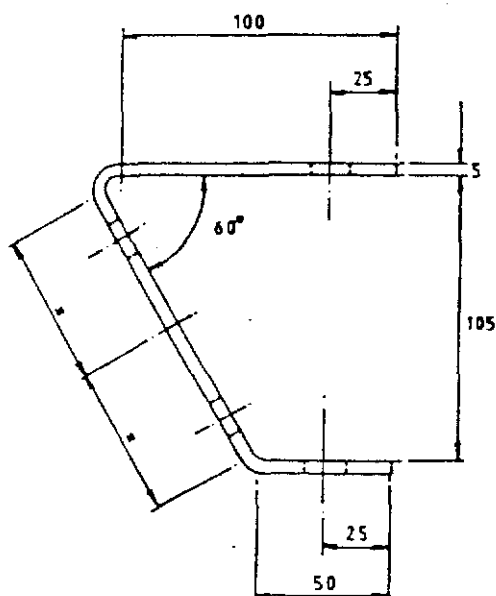
**CROSSARM & INSULATOR TAKE-OFF BRACKET
FOR 33KV. POLE TERMINATION**

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
B	A	
Arm hole posns. made 30mm. 20-4-93 A.J.B.	Bracket centre hole added. 12-3-93 A.J.B.	DRG. No. M453

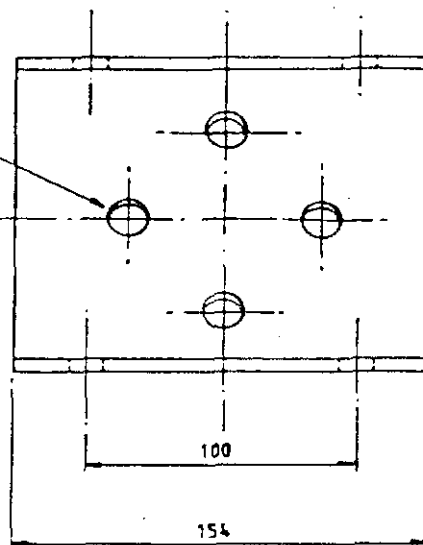


CROSSARM:

102 x 50 x 2000 M.S. Channel
Galv. to BS 729 After Cutting
and drilling



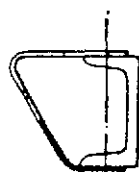
4 Holes 16 ϕ
74 P.C.D.



INSULATOR TAKE OFF BRACKET

5 mm. Thick M.S. Galv. to BS 729
after Cutting & Drilling.

NOTE: All Holes 16 ϕ unless
otherwise stated.



ASSEMBLY

SCALE	N.T.S.
DATE	24.7.90
DRAWN	
TRACED	
CHECKED	
APPROVED	

CROSSARM & INSULATOR TAKE-OFF BRACKET FOR 66 kV. POLE TERMINATION

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		ORG No
		M453/2

SECTION 3

MIN NUMBER	M	DESCRIPTION
3453520	352	OVERHEAD LINE CONDUCTORS BARE & P.V.C COVERED (ALUMINIUM)
3453530	353	OVERHEAD LINE CONDUCTORS BARE & P.V.C COVERED (AL. ALLOY)
3453540	354	OVERHEAD LINE CONDUCTORS BARE & P.V.C COVERED (A.C.S.R)
3453560	356	OVERHEAD LINE CONDUCTORS P.V.C COVERED HARD DRAWN COPPER
3453570	357	OVERHEAD LINE CONDUCTORS BARE & P.V.C COVERED CADMIUM COPPER
3453940	394	L.V. AERIAL BUNDLED CONDUCTORS (A.B.C)
3451416	141/6	CLYDESDALE BFO

NON-INSULATED CONDUCTORS

CONDUCTOR SIZE (mm ²)	STRANDING (mm)	DIAMETER BARE (mm)	WEIGHT (Kg/m)	BREAKING LOAD (Kgf)		RATING (Amperes)		STORES REFERENCE NUMBER
						SUMMER	WINTER	
25	7 / 2.14	6.42	.069	439				3453521
50	7 / 3.10	9.30	.145	847		187	260	3453522
100	7 / 4.39	13.17	.290	1632		284	406	3453523
400	37 / 3.78	26.46	1.145	6436		633	994	3453524

INSULATED CONDUCTORS

CONDUCTOR SIZE (mm ²)	STRANDING (mm)	DIAMETER BARE (mm)	WEIGHT INSULATED (Kg/m)	BREAKING LOAD (Kgf)	DIAMETER OVER INSULATION	RATING (Amperes)		STORES REFERENCE NUMBER
						SUMMER	WINTER	
25	7 / 2.14	6.42	.110	439	8.44			3453525
50	7 / 3.10	9.30	.200	847	11.70	187	260	3453526
100	7 / 4.39	13.17	.360	1632	16.00	284	406	3453527

- Conductor Stranding - 50 mm², 100 mm² & 400 mm² Conductors.
to BS. 215 pt. 1. (1970)
25 mm² Stranding as tabled above.
Material - all sizes to BS 215 pt. 1 (1970)
- P.V.C. Insulation - To BS. 6485 (1971) - colour Black.
Type 8
25 mm², 50 mm² and 100 mm² only.
- Conductor Greasing - To Engineering Recommendation L38/1
Category 1. Greasing P.V.C. Covered
Category 4. Greasing 25 mm², 50 mm² & 100 mm² Bare
Category 3. Greasing 400 mm² Bare

SCALE	-	OVERHEAD LINE CONDUCTORS BARE AND P.V.C. COVERED ALUMINIUM CONDUCTORS	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	16.6.82		C	
DRAWN	W.D.H.		Eng. Rec	
TRACED	J.W.		L38/1 mod.	
CHECKED			9-8-93 AJB	
APPROVED	<i>[Signature]</i>	E.S.I. STANDARD		DRG. No. M. 352
				ESI. DRG. No.

NON-INSULATED CONDUCTORS

CONDUCTOR SIZE (mm ²)	STRANDING (mm)	DIAMETER BARE (mm)	WEIGHT (Kg/m)	BREAKING LOAD (Kg.f)	RATING (Amperes)		STORES REFERENCE NUMBER
					SUMMER	WINTER	
25	7/2-34	7.02	.082	862	127	171	3453531
50	7/3-30	9.90	.164	1714	192	266	3453532
100	7/4-65	13.95	.325	3395			3453533
200 *	37/2-87	20.09	.659	6840			3453534
300	37/3-53	24.71	.997	10350		935	3453539

INSULATED CONDUCTORS

CONDUCTOR SIZE (mm ²)	STRANDING (mm)	DIAMETER BARE (mm)	WEIGHT INSULATED (Kg/m)	BREAKING LOAD (Kg.f)	DIAMETER OVER INSULATION	RATING (Amperes)		STORES REFERENCE NUMBER
						SUMMER	WINTER	
25	7/2-34	7.02	170	862	10.6	127	171	3453535
50	7/3-30	9.90	280	1713	13.9	192	266	3453536
100	7/4-65	13.95	470	3395	18.0			3453537
200 *	37/2-87	20.09	880	6840	24.2			3453538

Conductor Stranding - To B.S. 3242 (1970) or As shown *

P.V.C. Insulation - To B.S. 6485 (1971) Colour-Green
Type 16.

Conductor Greasing - To Engineering Recommendation L38/1

Category 1 Greasing P.V.C. Covered.

Category 4 Greasing 25mm², 50mm² & 100mm² Bare

Category 3 Greasing 200mm² 300mm² Bare

SCALE	—	OVERHEAD LINE CONDUCTORS BARE AND P.V.C. COVERED AL. ALLOY CONDUCTORS.	ISSUE	MEEB
DATE	16.6.82		B	
DRAWN	W.D.H.		Table Eng. Rec. & cat. 2 mods. -8-93 A.J.B.	ORG. No. M.353.
TRACED	S.J.H.			
CHECKED	Ref			
APPROVED		E.S.I. STANDARD		E.S.I. DRG. N°

NON INSULATED CONDUCTORS

CONDUCTOR SIZE (mm ²)	STRANDING (mm.)	DIAMETER BARE (mm.)	WEIGHT (kg/m)	BREAKING LOAD (Kg.f.)		RATING		STORES REFERENCE NUMBER
						SUMMER	WINTER	
70	7/12-79 + 12/12-79	13.95	0.538	6241		-	-	3453541
100	7/11-57 + 6/14-72	14.15	0.394	3335		264	388	3453542
100 *	7/11-80 + 26/2-30	14.60	0.449	4082		264	388	3453543
150	7/12-59 + 30/2-59	18.13	0.726	7057		351	535	3453544
175	7/12-79 + 30/2-79	19.53	0.842	8137		383	590	3453545
400	7/13-18 + 54/3-18	28.62	1.621	13450		617	1000	3453546

INSULATED CONDUCTORS

CONDUCTOR SIZE (mm ²)	STRANDING (mm.)	DIAMETER BARE (mm.)	WEIGHT INSULATED (Kg/m)	BREAKING LOAD (Kg.f.)	DIAMETER OVER INSULATION	RATING		STORES REFERENCE NUMBER
						SUMMER	WINTER	
100	7/11-57 + 6/14-72	14.15	0.550	3335	18.2	264	388	3453548
175	7/12-79 + 30/2-79	19.53	1.050	8137	23.6	383	590	3453549

Conductor Stranding - To B.S. 215 Part 2 (1970)

P.V.C. Insulation - To B.S. 6485 (1971) Colour Green, Type 16.

Conductor Greasing - To Engineering Recommendation L38/1

3453541	Category 4	Greasing
3453542	"	4
3453543	"	3
3453544		
3453545		
3453546		
3453548	"	1
3453549	"	1

* 100 mm.² A.C.S.R.
(7/11-80 + 26/2-30)

— Aluminium stranding (26/2-30 mm.)
material as specified in B.S. 215 Part 2
Steel Strand to B.S. 4565 (1970)
Standard grade steel.

SCALE	—	OVERHEAD LINE CONDUCTORS BARE AND P.V.C. COVERED A.C.S.R. CONDUCTORS	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	15.6.82		E	
DRAWN	W.D.H.		Eng. Rec L38/1 mod.	
TRACED	S.J.H.		9-8-93 A.J.B	DRG. No.
CHECKED				M. 354
APPROVED				E.S.I. DRG. No.
E.S.I. STANDARD				

CONDUCTOR SIZE (m.m. ²)	STRANDING (m.m.)	DIAMETER BARE (m.m.)	WEIGHT INSULATED (Kg./m.)	BREAKING LOAD (Kg.f.)	DIAMETER OVER INSULATION	RATING (Amperes)		STORES REFERENCE NUMBER
						SUMMER	WINTER	
14	7/1.60	4.80	.160	586	6.80	100	140	3453561
35	7/2.50	7.50	.360	1438	9.90	198	269	3453562
70	7/3.55	10.65	.690	2741	13.50	273	379	3453563
16	3/2.65	5.69	.220	672	9.30	121	161	3453565
32	3/3.75	8.05	.390	1296	12.10	184	250	3453566
70	7/3.55	10.65	.750	2741	14.70	273	379	3453567

Conductor Stranding - To B.S. 125 (1970)

P.V.C. Covering - To B.S. 6485 (1971)

3453561	} Colour Black Type 8
3453562	
3453563	
3453565	} Colour Green Type 16
3453566	
3453567	

Conductor Greasing To Engineering Recommendation L 38/1

3453561	} Category 1 Grease.
3453562	
3453563	
3453567	

SCALE	---	OVERHEAD LINE CONDUCTORS P.V.C. COVERED HARD DRAWN COPPER CONDUCTORS	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	14.6.82		C	
DRAWN	W.D.H.		Eng Rec. L38/1 Mod.	DRG. No.
TRACED	S.J.H.		9.8.93 A.J.B.	M.356
CHECKED				
APPROVED	<i>[Signature]</i>	E.S.I. STANDARD		E.S.I. DRG. No.

CONDUCTOR SIZE (mm ²)	STRANDING (mm)	DIAMETER BARE (mm)	WEIGHT BARE (Kg/m)	WEIGHT INSULATED (Kg/m)	BREAKING LOAD (Kgf)	RATING		STORES REFERENCE NUMBER
						SUMMER	WINTER	
22 ·025	7 / 2·00 7 / ·073	6·00	·1984	-	1320			3453571
30 ·04	7 / 2·30 7 / ·093	6·90	·2623	-	1713			3453572
38 ·05	7 / 2·60 7 / ·103	7·80	·3352	-	2212			3453573
55 ·075	7 / 3·20 7 / ·127	9·60	·5077	-	3155			3453574
75 ·10	7 / 3·70 7 / ·146	11·10	·6786	-	4102			3453575
85 ·125	19 / 2·36 19 / ·093	11·80	·7530	-	4643			3453576
115 ·15	19 / 2·77 19 / ·109	13·85	1·0373	-	6396			3453577
135 ·175	19 / 3·00 19 / ·118	15·00	1·2168	-	7502			3453578
22 ·025	7 / 2·00 7 / ·073	6·00	·1984	·270	1320			3453579

Conductor Stranding - 3453571
3453572
3453573
3453574
3453575
3453579

Strand Diameters and Material
to BS 125 (1970)

- 3453576
3453577
3453578

Strand Diameter as tabled Material
to BS 125 (1970)

P.V.C. Insulation - 3453579

To BS 6485 (1971)
colour green Type 16

Conductor Greasing To Engineering Recommendation L 38/1

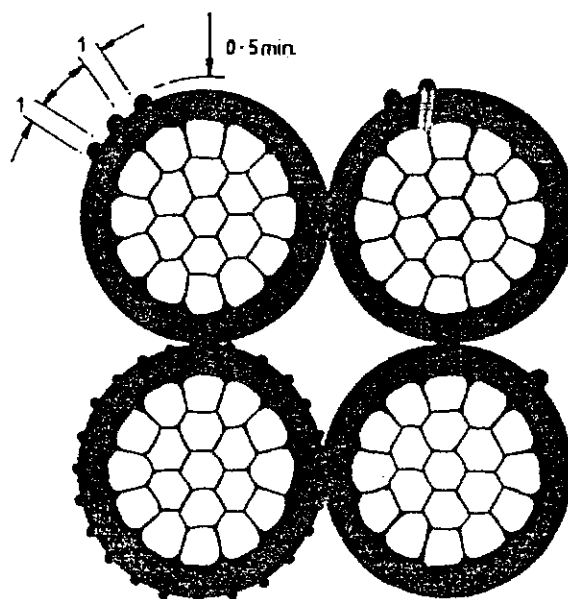
3453579 Category 1 Grease.

SCALE		OVERHEAD LINE CONDUCTORS BARE AND P.V.C. COVERED CADMIUM COPPER CONDUCTORS	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	22.6.82		C	
DRAWN	W.D.H.		Eng. Rec L38/1 mod	DRG No.
TRACED	J.W.		9.8.93 A.J.B.	M 357
CHECKED				
APPROVED		E.S.I. STANDARD		ESI DRG. No.



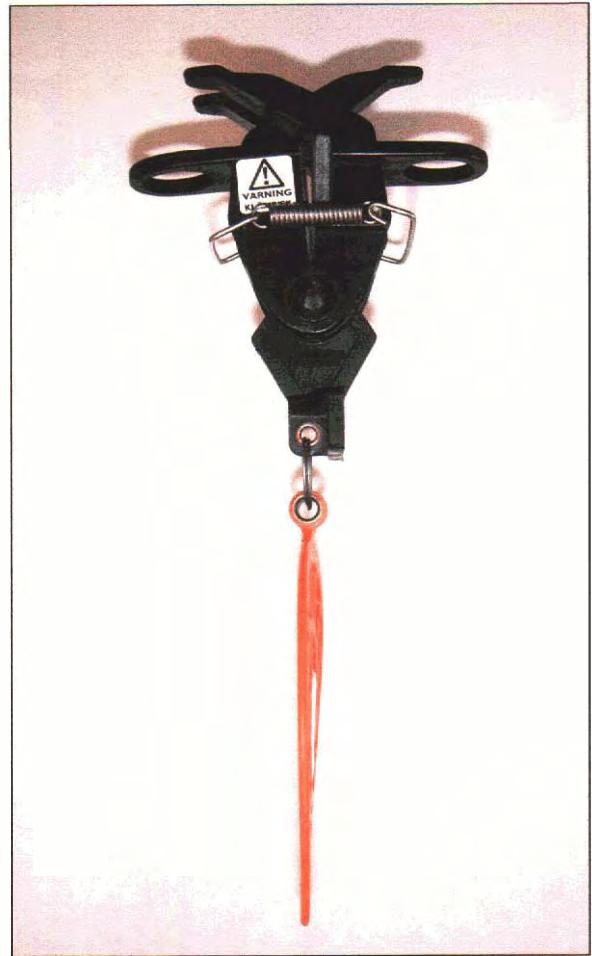
NUMBER OF CORES X CONDUCTOR SIZE	NUMBER OF WIRES IN CORE	DIA OF BARE CORE (mm.)		MAX. DIA. OF INSULATED CORE (mm.) EXCLUDING RIBS	APPROX WEIGHT OF BUNDLE (kg / m)	MAX. CONTINUOUS RATING @ 75°C ON POLES (25°C AMBIENT)	STORES REF No
		MIN.	MAX.				
2 X 35	7	6.6	7.5	10.7	0.26	138	0
4 X 50	19 ± 1	7.7	8.5	12.1	0.70	143	345394/1
4 X 95	19 ± 1	11.0	12.0	15.1	1.30	228	2
2 X 50	19 ± 1	7.7	8.5	12.1	0.35	167	3
2 X 95	19 ± 1	11.0	12.0	16.1	0.65	262	4

4 x 120

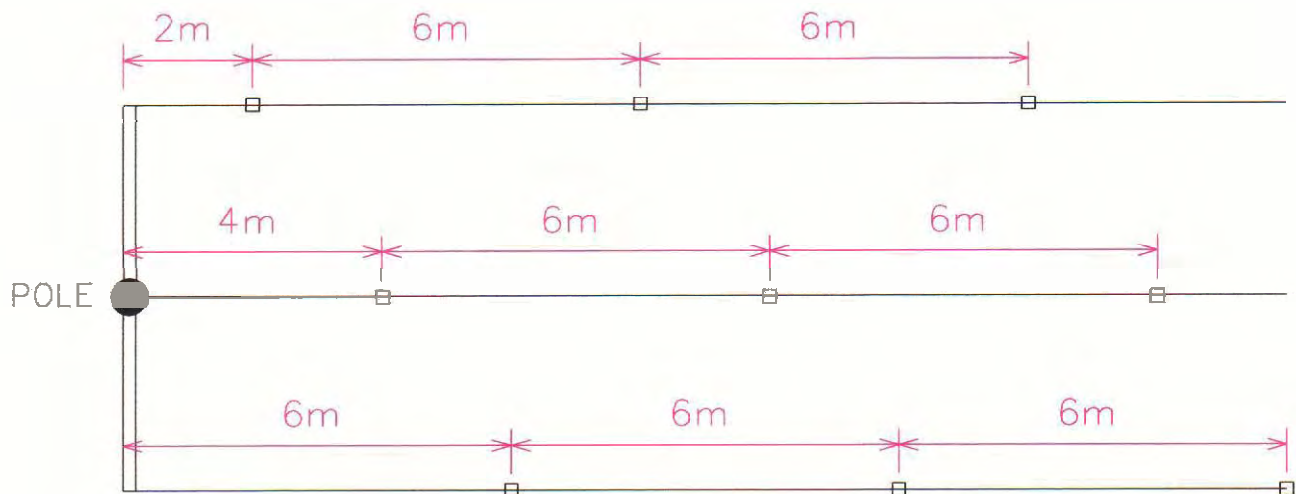


Sequence and Dimensions of
Phase Identification Ribs

SCALE	-	L.V. AERIAL BUNDLED CONDUCTORS (A.B.C)	ISSUE	MEB
DATE	8-7-88		A	
DRAWN	R.A.C.		2 x 50 AND 2 x 95 ADDED AJB 7-5-91	DRG. No. M.394 (MF)
TRACED	S.J.H.			
CHECKED	20			
APPROVED				
E.S.I. STANDARD 43-13		DRG No 431301		



H.V. Horizontal Construction Only



Note: Conductor Spacing (Mid Span) should be 1m minimum for 11kV construction

Use Clydesdale Fixing head for Insulated pole Part No ENG KL70 OP

SCALE	-
DATE	29/05/03
DRAWN	ADD
TRACED	-
CHECKED	11/06/03
APPROVED	KD

CLYDESDALE "SNAPFAST"
BIRD FLIGHT DIVERTERS

ISSUE	DATE



Aquila

DRG. No.

M141/6

11/1/03

11/1/03

A

SECTION 4

MIN NUMBER	M	DESCRIPTION
3453500	350	OVERHEAD LINE CONDUCTORS PVC INSULATED & SHEATHED COPPER CABLES
3453510	351	OVERHEAD LINE CONDUCTORS PVC INSULATED & SHEATHED ALUMINIUM CABLES
3453580	358	OVERHEAD LINE CONDUCTORS PVC INSULATED COPPER & ALUMINIUM FLEXIBLE L/L CABLE
3453590	359	OVERHEAD LINE CONDUCTORS BARE & PVC COVERED BINDING WIRE,& CHAFING TAPE
3453600	360	CLAMP ADAPTOR 125kN

CONDUCTOR SIZE (mm ²)	STRANDING (mm)	DIAMETER (BARE) (mm)	WEIGHT (BARE) (Kg/m)	INSULATION THICKNESS (mm)	DIAMETER OVER INSULATION (mm)	MATERIAL CONDITION	RATING (Amperes)	STORES REFERENCE NUMBER
16	7 / 1.70	5.10	.145	1.0 / 1.0	9.40	PLAIN ANNEALED	75	3453501
25	7 / 2.14	6.48	.229	1.2 / 1.1	11.40	"	110	3453502
35	19 / 1.53	7.65	.317	1.2 / 1.1	12.60	"	145	3453503
50	19 / 1.78	8.90	.429	1.4 / 1.2	14.50	"		3453504
70	19 / 2.14	10.70	.620	1.4 / 1.2	16.30	"	205	3453505
95	19 / 2.52	12.60	.860	1.6 / 1.3	18.80	"	260	3453506
120	37 / 2.03	14.21	1.086	1.6 / 1.4	20.70	"	320	3453507
35	19 / 1.53	7.65	.317	1.2 / 1.1	12.60	"	145	3453508
16	7 / 1.70	5.10	.145	1.0 / 1.0	9.40	"	75	3453509
25	7 / 2.14	6.48	.229	1.2 / 1.1	11.40	"	110	3453500

Conductor Stranding.

Strand Diameters as above,
Material in accordance with
BS.6360 (1975)

P.V.C. Insulation & Sheath.

Insulation and Sheath thickness as above,
Material in accordance with
BS.6004 (1984)

Items	3453503	} Insulation colour - Red. Sheath colour - Brown
	3453504	
	3453505	
	3453506	
	3453507	
	3453508	} Insulation Black - Sheath Brown
Items	3453501	
	3453502	} Insulation colour - Red. Sheath colour - Grey.
Items	3453509	
	3453500	} Insulation colour - Black Sheath colour - Grey

SCALE		OVERHEAD LINE CONDUCTORS. P.V.C. INSULATED AND SHEATHED COPPER CABLES.	ISSUE	MIDLANDS
DATE	15.6.82.		C	ELECTRICITY BOARD
DRAWN	W.D.H.		B.S. N°	DRG. No.
TRACED	J.W.		MODIFIED	M. 350
CHECKED			R.A.C. 3-287	
APPROVED		E.S.I. STANDARD		E.S.I. DRG. N°

CONDUCTOR SIZE (m.m. ²)	STRANDING (m.m.)	DIAMETER BARE (m.m.)	WEIGHT BARE (Kg./m.)	INSULATION THICKNESS (m.m.)	DIAMETER OVER INSULATION (m.m.)	MATERIAL CONDITION	RATING (Amperes)	STORES REFERENCE NUMBER
25	7/2-14	6.42	.0700	1.2 / 1.1	11.4	G1AH68	85	3453511
35	19/1-53	7.65	.0960	1.2 / 1.1	12.6	"	105	3453512
50	19/1-78	8.90	.1310	1.4 / 1.2	14.5	"	130	3453513

Conductor Stranding - Strand Diameters as listed above.
Material in accordance with BS.6004 (1975).

P.V.C. Insulation and - Insulation and Sheath thickness as above
Sheath. Material in accordance with BS6004 (1975).
Insulation Colour-Red.
Sheath Colour-Brown.

SCALE	—	OVERHEAD LINE CONDUCTORS P.V.C. INSULATED AND SHEATHED ALUMINIUM CABLES	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	16.6.82			
DRAWN	W.D.H.			DRG. No. M.351.
TRACED	S.J.H.			
CHECKED				
APPROVED	<i>[Signature]</i>			
E.S.I. STANDARD			E.S.I. DRG. N°	

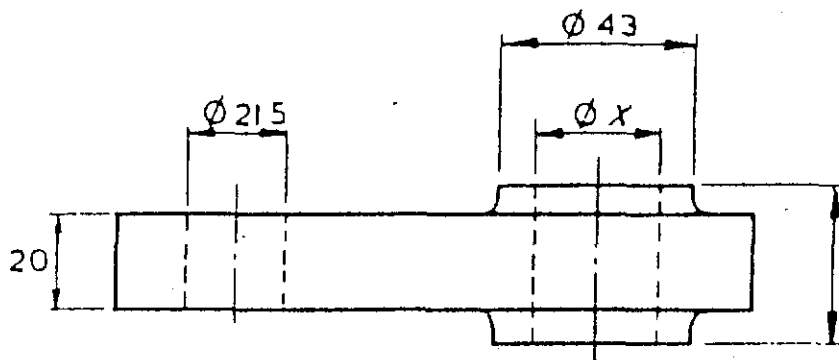
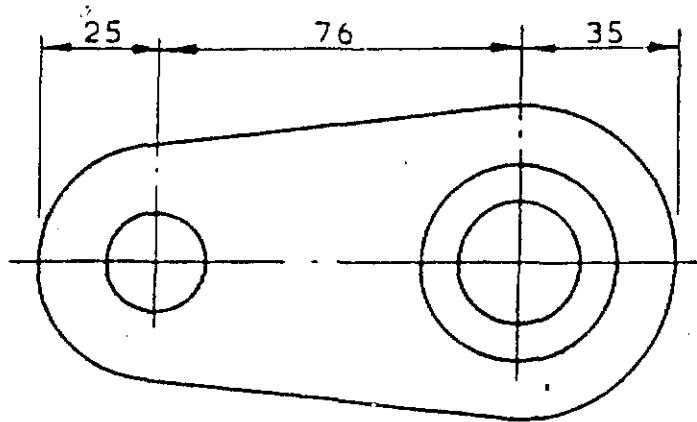
CONDUCTOR SIZE (mm ²)	MATERIAL	STRANDING (mm)	INSULATION THICKNESS (mm)	OVERALL DIAMETER (mm)	WEIGHT (Kg/m)		RATING (amperes)	STORES REFERENCE NUMBER
35	COPPER	1107 / 2	1.0 (BLACK)	9.80	317		190	3453581
50	ALUMINIUM	312 / 45	0.7-1.0 CLEAR	13.00	145		190	3453585

- Conductor Stranding - Type 1 as tabled
Type 5 as tabled with 7 bunches of 45 strands.
- P.V.C. Insulation - Type 1 - PVC Type 4 BS 6746
- Type 5 - PVC Type T.1.2 BS. 6746 or equivalent.

SCALE		OVERHEAD LINE CONDUCTORS P.V.C. INSULATED COPPER AND ALUMINIUM FLEXIBLE L/L CABLE	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M 358 MF
DATE	22. 6. 82.		A	
DRAWN	W. D. H.		Type 5 stranding	
TRACED	J. W.		Revised.	
CHECKED				
APPROVED	#	E.S.I. STANDARD		E.S.I. DRG. N°

DESCRIPTION	DIMENSIONS (mm)	SPECIFICATION	P.V.C. COVERING	STORES REFERENCE NUMBER
BARE COPPER BINDING WIRE	2.5 DIA	BS 4109 Condition 0	-	3453590
P.V.C. COPPER BINDING WIRE	2.5 DIA	BS 4109 Condition 0	BLACK P.V.C.-BS6746 T.11	3453591
BARE ALUM. BINDING WIRE	2.5 DIA	BS 2627 Condition H8	-	3453592
P.V.C. ALUM. BINDING WIRE	2.5 DIA	BS 2627 Condition H8	BLACK P.V.C.-BS6746 T.11	3453593
STIRRUP WIRE				
CHAFING TAPE				
COPPER CHAFING TAPE	4.75 x 1.5	ANNEALED COPPER	-	3453598

SCALE		OVERHEAD LINE CONDUCTORS BARE AND P.V.C. COVERED BINDING WIRE, STIRRUP WIRE, CHAFING TAPE AND BAIL MATERIAL	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	22.6.82			
DRAWN	W.D.H.			DRG. No.
TRACED	J.W.			M359
CHECKED				
APPROVED		E.S.I. STANDARD		E.S.I. DRG. No.



M.N°	BS	X	Y.
360/1	28/27C	29	38

TO BS. 3288 Pt 1x2 Fig.45

SCALE	NTS.	CLAMP ADAPTOR MINIMUM FAILING LOAD 125 kN.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	12.10.82			
DRAWN	R.A.C.			
TRACED				ORG. No
CHECKED				M 360
APPROVED	#			

SECTION 5

MIN NUMBER	M	DESCRIPTION
3451410	141	Conductor Game Guards
3453200	320	Conductor Vibration Damper
3455000	500	Conductor Spacer Twin 175/400mm ² ACSR 200mm Spacing
3455010	501	Conductor Spacer Twin 175/400mm ² ACSR 300mm Spacing
3455020	502	Conductor Spacer Twin 175/400mm ² ACSR 400mm Spacing

MEB. REF.	CONDUCTOR RANGE		LENGTH	MAN. REF NUMBER
	MIN	MAX		
M141/1	4.45	6.34	178	BFD5249001
M141/2	6.35	8.88	216	BFD5249002
M141/3	8.89	11.43	241	BFD5249003
M141/4	11.43	15.21	279	BFD5249004
M141/5	15.22	19.55	330	BFD5249005

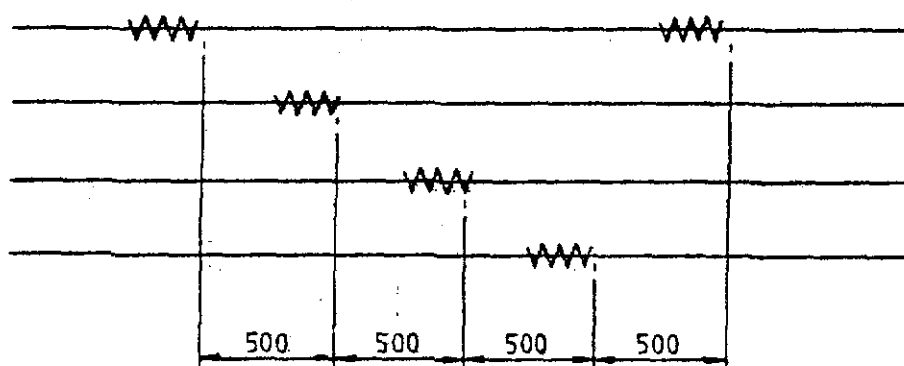
MANUFACTURER : PREFORMED LINE PRODUCTS (GREAT BRITAIN) LTD.
ANDOVER,
HAMPSHIRE.



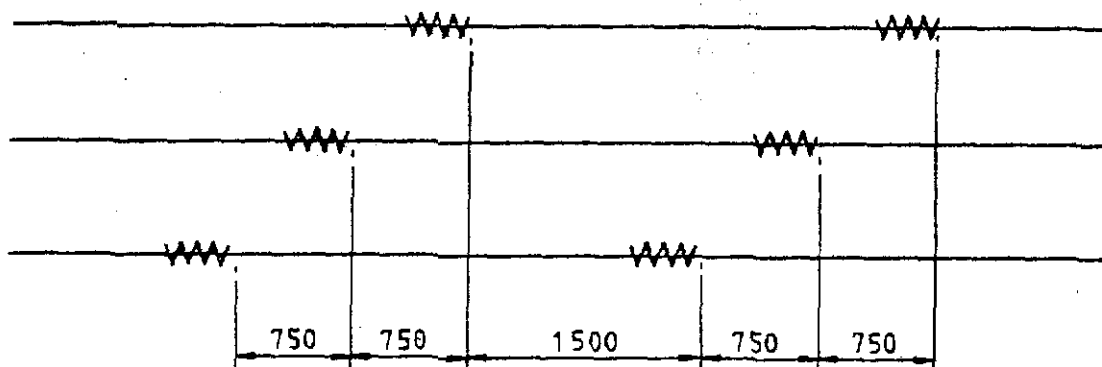
CHECK
100
200
400
600
800
1000

METHOD OF ATTACHMENT

NOTE:- Conductor Spacing (Mid Span) should be 1m. minimum for 11kV. Construction.



M.V. VERTICAL CONSTRUCTION



H.V. HORIZONTAL CONSTRUCTION

SCALE	-
DATE	13.8.84
DRAWN	R.A.C.
TRACED	C.A.R.
CHECKED	
APPROVED	

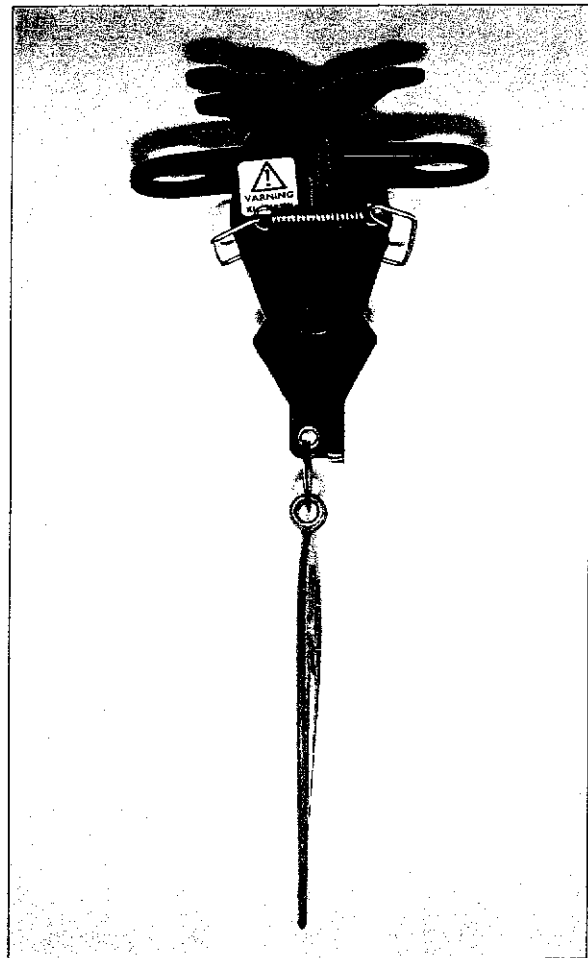
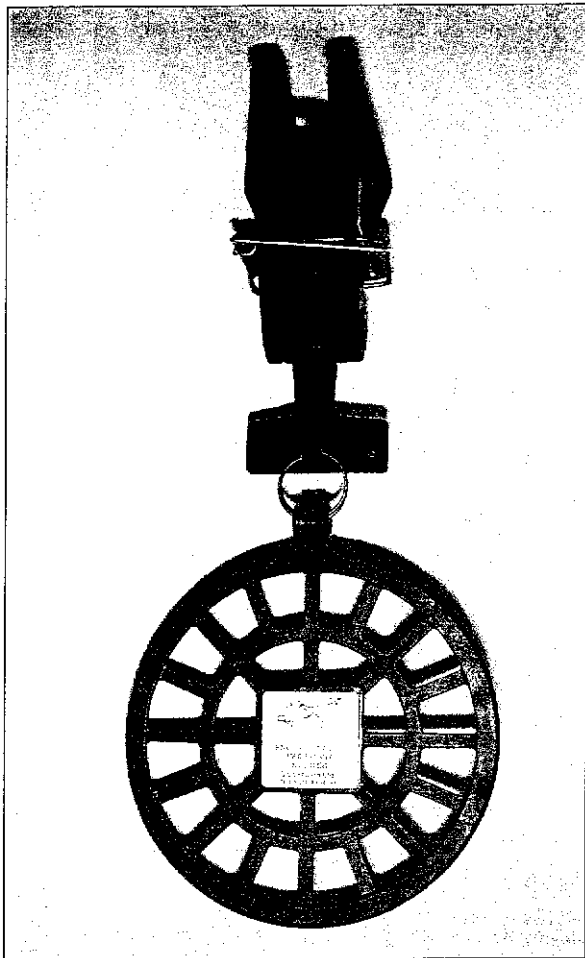
CONDUCTOR GAME GUARDS

ISSUE

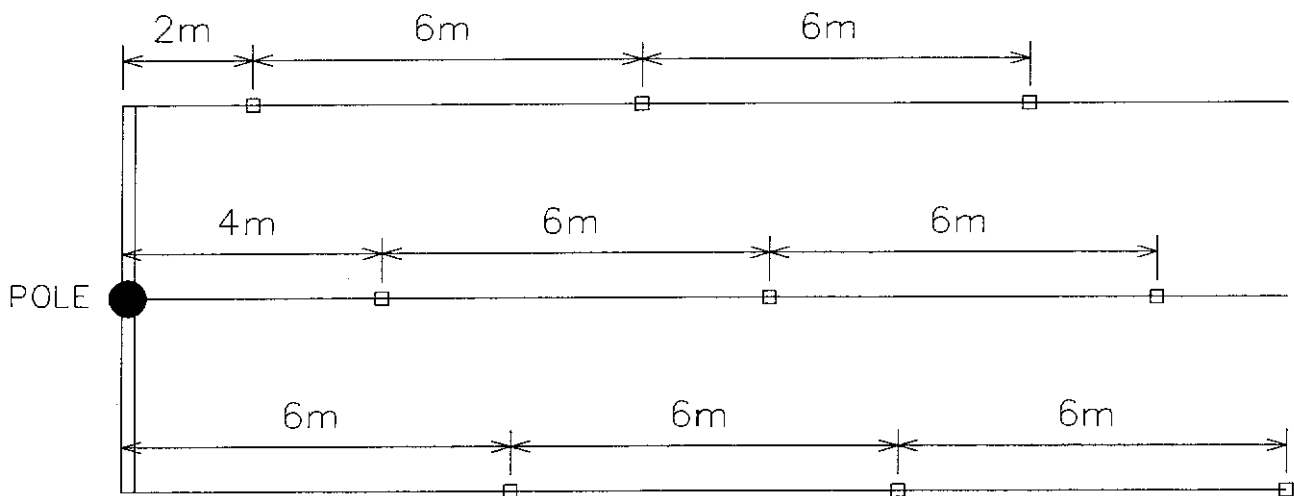
MIDLANDS
ELECTRICITY BOARD

ORG. No.

M. 141.



H.V. Horizontal Construction Only



Note: Conductor Spacing (Mid Span) should be 1m minimum for 11kV construction.

Use Clydesdale Fixing head for Insulated pole Part No ENG KL70 OP

SCALE	-
DATE	29/05/03
DRAWN	ADD
TRACED	-
CHECKED	11/06/03
APPROVED	KD

CLYDESDALE "SNAPFAST"
BIRD FLIGHT DIVERTERS

ISSUE	DATE



Aquila

DRG. No.

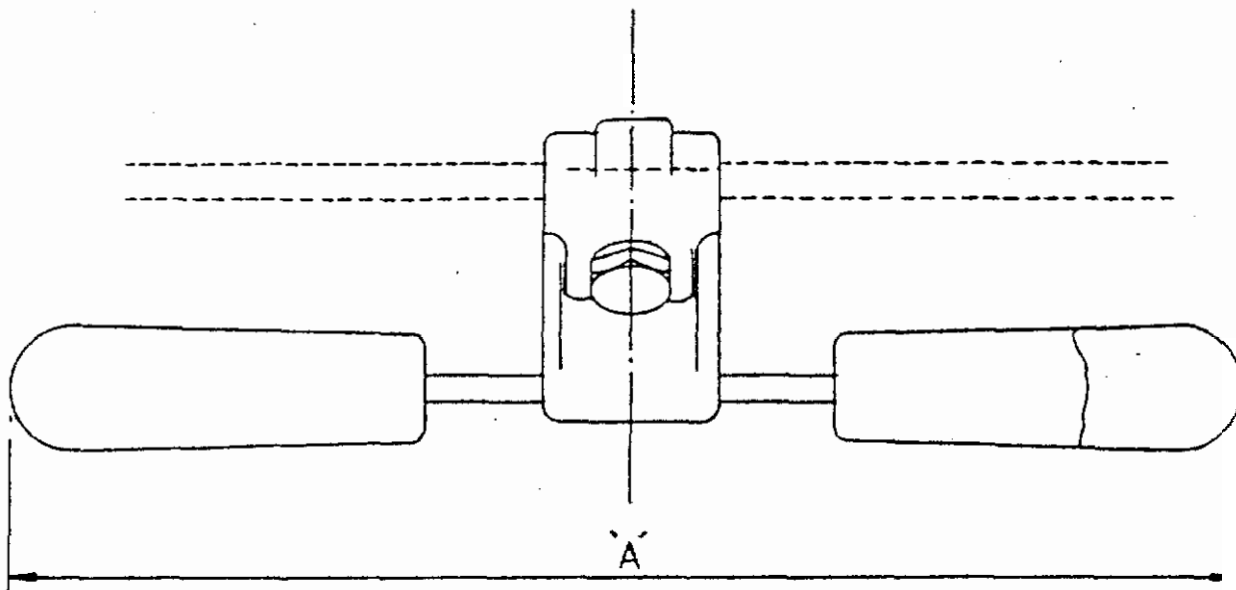
M141/6

Produced by
Primary CAD

DRG. No.

M141/6

A4



BDD 290 6000
 300 "
 500 "

MEB REF	CEGB CCN ^o	CONDUCTOR TYPE x SIZE	DULMISON			BOWTHORPE			BRUSH POWER		
			REF	Amm	Wt.kg.	REF	Amm	Wt.kg.	REF	Amm	Wt.kg.
320/0		70mm ² ACSR	SB14/	454	3.4	ST	477	3.2	A606	432	3.2
		100mm ² ACSR	120P	454	3.4	3201	477	3.2	A606	432	3.2
320/1		150mm ² ACSR	SB23/	556	5.8	ST	504	4.1	A610	514	5.4
		175mm ² ACSR	180P	556	5.8	5001T	504	4.1	A610	514	5.4
320/2		400mm ² ACSR	SB32/	620	7.6	ST	631	7.89	A616	635	8.5
			274P	620	7.6	8351T	631	7.89	A616	635	8.5
320/3		75 CAD CU	DK								
		70 HD CU	5199/	454	3.4	-	-	-	-	-	-
		85 CAD CU	105	454	3.4	-	-	-	-	-	-
320/4		115 CAD CU	DK								
			5199/	454	3.4	-	-	-	-	-	-
			120								
320/5		135 CAD CU	OK								
			5199/	454	3.4	-	-	-	-	-	-
			140								

Dulmison
 DK 60011
 DK 60016
 DK 60000

NOTE:- 320/0 } For use with ACSR conductors
 320/1 } conductor clamp - cast aluminium
 320/2 }
 320/3 } For use with cu x cad cu conductors
 320/4 } conductor clamp - malleable C.I. or gunmetal
 320/5 }

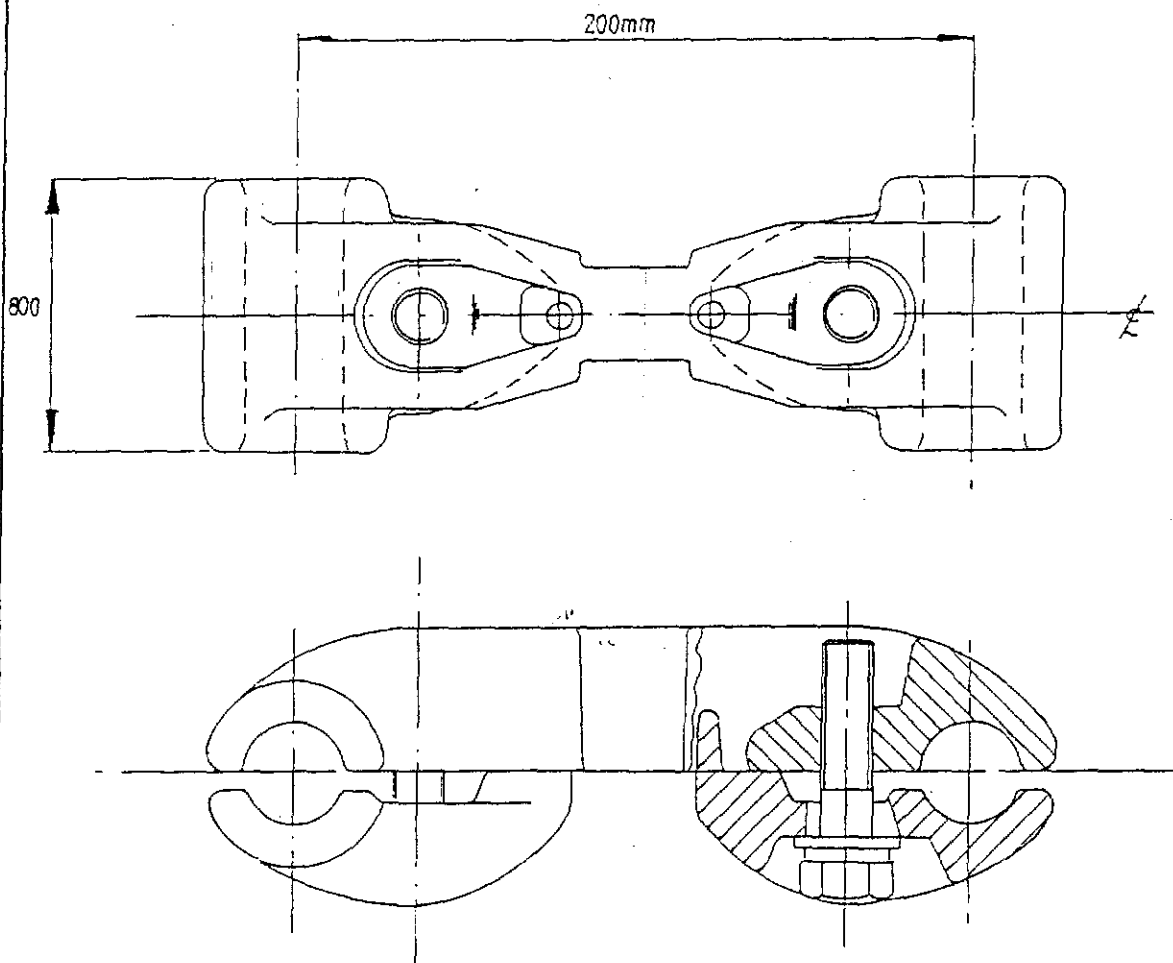
SCALE	N.T.S.
DATE	25.4.79
DRAWN	R.A.C.
CHECKED	
APPROVED	

CONDUCTOR VIBRATION DAMPER

E.S.I. STANDARD

ISSUE	MIDLANDS
B	ELECTRICITY BOARD
Dulmison Ref Corrected. CAR 8-8-84	DRG. No. M 320.
	E.S.I. Org. No.

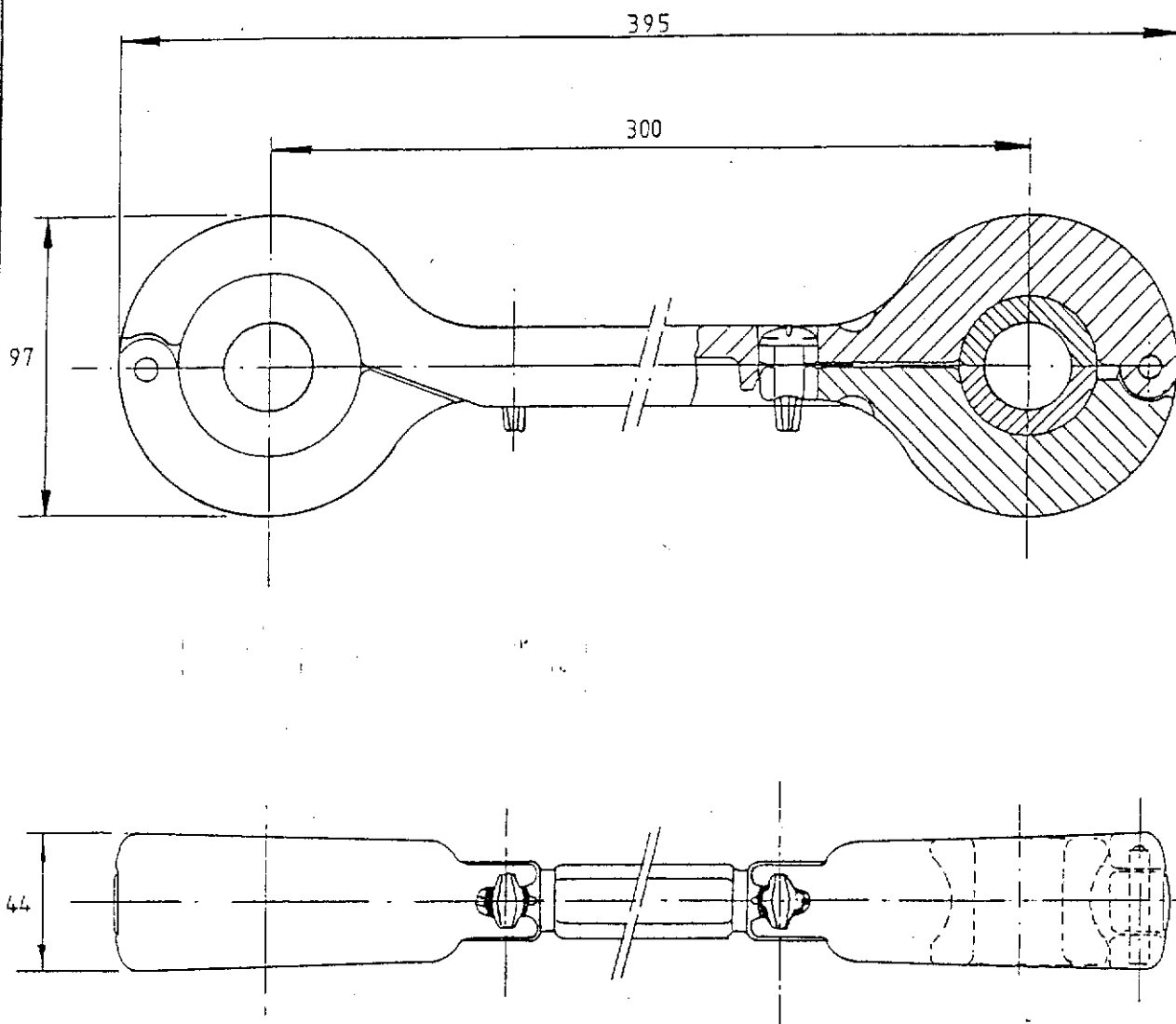
(MF)



NOTE: DULMISON TYPE SHOW

ME.B. REF	CEGB CC N°	CONDUCTOR SIZE	MANUFACTURER	REFERENCE.
M500/0		175 mm ²	DULMISON	2JS352/193HSZ
M500/2	4302577	175 mm ²	BOWTHORPE	LSR1956 203
M500/6		400 mm ²	"	LSR 2862 203
M500/9		400 mm ²	DULMISON	2JS 352/278 HSZ

SCALE	NTS.	CONDUCTOR ² SPACER FOR TWIN 175mm ² ACSR CONDUCTORS. 200mm SPACING.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	5-1-83		A	
DRAWN	RAC.		Man. Ref.	DRG. No.
TRACED			Added.	M 500 MF
CHECKED	#		HPS. 2-4-84	
APPROVED	#		ES1 Drg N°	7828/8
2044G		E.S.I. STANDARD		

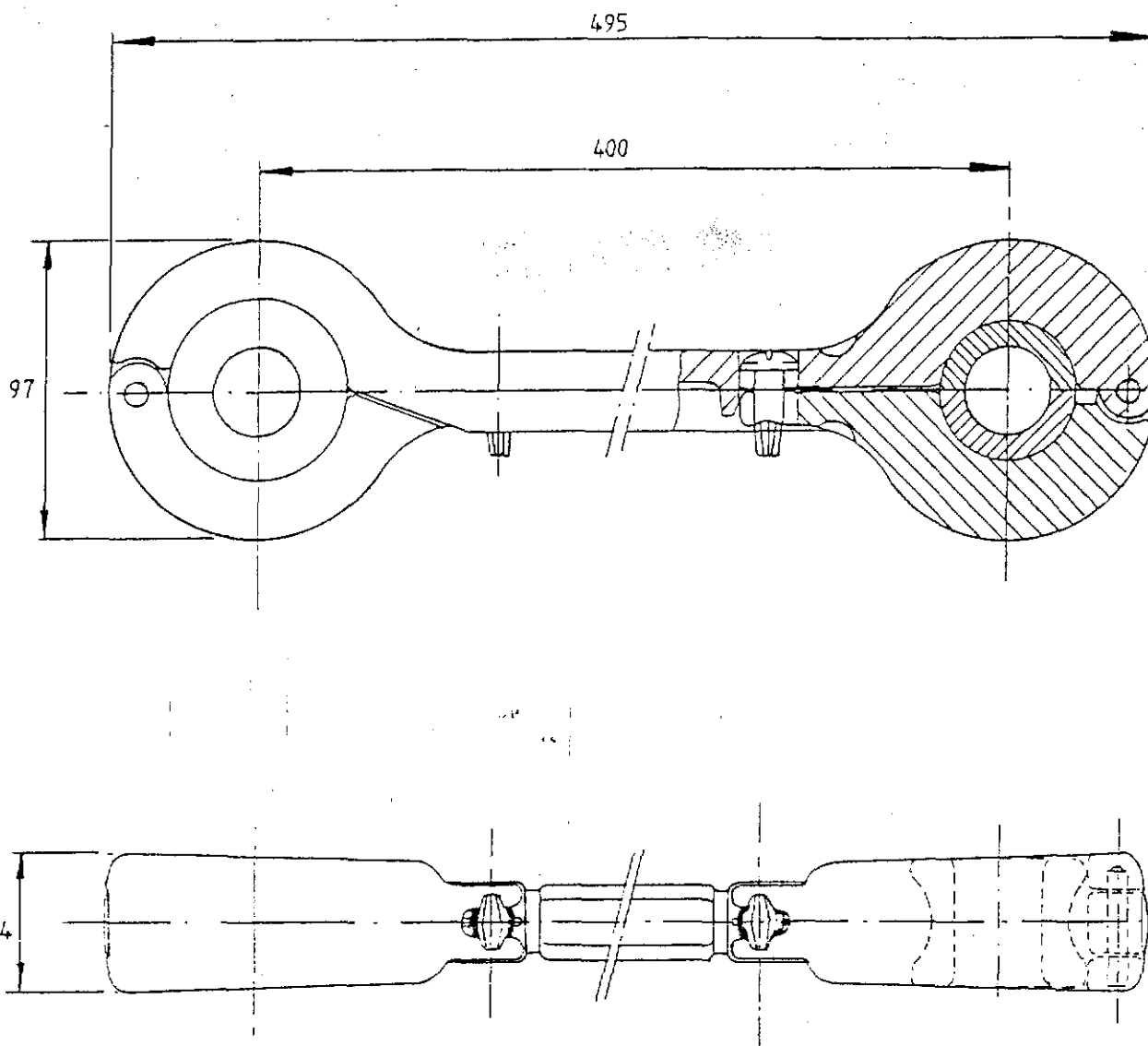


NOTE: DULMISON TYPE SHOWN

M.E.B REF	CEGB CC N°	CONDUCTOR SIZE	MANUFACTURER	REFERENCE
501/0		175 mm ²	DULMISON	2S2 300/1861
501/2	4302623	175 mm ²	BOWTHORPE	LSF 1956 305
501/6		400 mm ²	"	LSF 2862 305
501/9		400 mm ²	DULMISON	2S2 300/2862

~~AD~~ 700 ?
~~700~~

SCALE	V.T.S.	CONDUCTOR₂ SPACER FOR TWIN 175mm ACSR CONDUCTORS 300mm SPACING.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	5-1-83		A	
DRAWN	R.A.C.		Man. Ref. Added.	
TRACED			H.P.S. 2-4-84	
CHECKED				
APPROVED				ORG. No. M.501 MF
2044G	17	E.S.I. STANDARD		E.S.I Org N° 7828/81



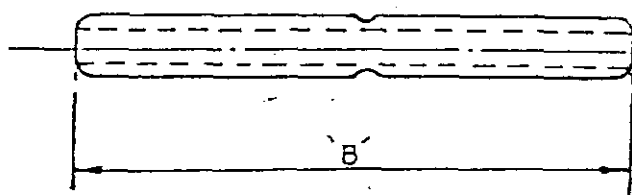
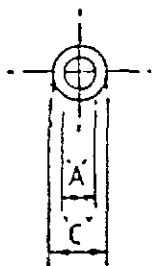
NOTE DULMISON TYPE SHOWN

MEB REF	CEGB CC N°	CONDUCTOR SIZE	MANUFACTURER	REFERENCE
50210		175 mm ²	DULMISON	2S2400/1861
50212	4302672	175 mm ²	BOWTHORPE	LSF1956406
50216		400 mm ²	II	LSF2862406
50219		400 mm ²	DULMISON	2S2400/2862

SCALE	N.I.S.	CONDUCTOR SPACER FOR TWIN 175mm² ACSR CONDUCTORS 400mm SPACING	ISSUE	MIDLANDS
DATE	5-1-83		A	ELECTRICITY BOARD
DRAWN	R.A.C.		Man. Ref.	DRG. No.
TRACED			Added.	M. 502 (MF)
CHECKED	14		CAR. 3/84	
APPROVED				
2044G		ESI STANDARD	ESI. Drg N° 7828/8	

SECTION 6

MIN NUMBER	M	DESCRIPTION
3451220	122	TENSION JOINT AAAC & ACSR
3451230	123	TENSION JOINT HD ALUMINIUM
3451590	159	DEAD END 70 - 175 ACSR
3451610	161	TENSION JOINT 70 - 400 ACSR
3451730	173	RECONDUCTORING TENSION JOINT
3452100	210	DEAD END 400 HD ALUMINIUM
3452200	220	ADAPTOR PLATE
3452330	233	TENSION JOINT COPPER
3452530	253	DEAD END 400 ACSR
3452870	287	SOCKET DEAD END COPPER,AL ALLOY & ACSR
3452880	288	BALL DEAD END AL ALLOY & ACSR
3453331	333/1	DEAD END JUMPER TERMINAL ANGULAR ACSR & COPPER
3453332	333/2	DEAD END JUMPER TERMINAL STRAIGHT ACSR & COPPER
3453350	335	DEAD END DOUBLE JUMPER 100 - 400 ACSR
3453360	336	DEAD END INTEGRAL JUMPER 70 -175 - 400 ACSR
3453380	338	REPLACEMENT BOLTS FOR DISCONNECTABLE DEAD ENDS
3453390	339	TENSION JOINT CADMIUM COPPER
3453400	340	SOCKET DEAD END CADMIUM COPPER
3453410	341	BALL DEAD END CADMIUM COPPER
3454460	446	HARDEX HELICAL TERMINATION
3454610	461	TERMINATING THIMBLE
3455640	564	BOLTED TERMINATING CLAMP 200 AAAC
3455660	566	DEAD END UNITYPE FORAAAC REMOVABLE JUMPER
3455680	568	WEDGE CLAMP ASSEMBLY AAAC



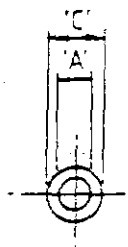
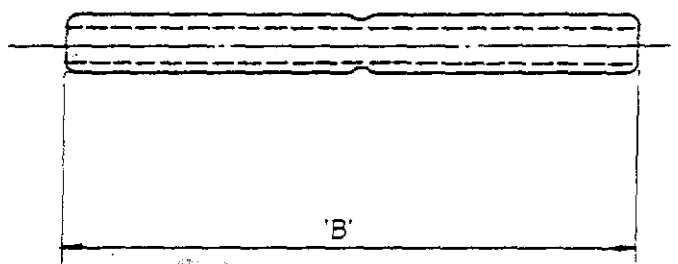
JOINTING INSTRUCTIONS

1. Scratch brush both ends of conductor to be joined.
2. Unlay aluminium strands.
3. Remove surplus grease from the steel core then insert the steel core into the joint and withdraw again.
4. Spread the joint compound evenly over the steel core and wipe any surplus onto the aluminium strands then replace the aluminium strands.
5. Mark each conductor at half joint length from the end and insert both ends together up to this mark.
6. Compress as instructions on joint sleeve.

Note * Tension joint Ref. 122/3 maybe used for 50mm² AAAC conductor.

TYPE	CONDUCTOR TYPE AND SIZE	MANUFACTURERS REF	DIMENSIONS (mm)			COMP'N DIE DETAILS	
			A	B	C	HAND	HYDRAULIC
122/1	25mm ² AL ALLOY 712-34mm ²					-	U 698
122/2 *	50mm ² AAAC * 713-30					-	U 659
122/3	50mm ² ACSR 713-35mm ²					-	U 659
122/4	40mm ² ACSR 713-00					-	U 699
122/5	40mm² AL ALLOY 712-95mm²					-	U 699
122/6	25mm² ACSR 712-36					-	U 698
122/7	100mm ² AAAC 714-65					-	U 654
122/8	200mm ² AAAC 3712-87					-	A - 9
122/9	300mm ² AAAC 3713-53					-	A - 10
122/0	500mm ² AAAC 6113-50					-	A - 12

AMUC 15628 E Addn A - 3/ELM out 1-12-92 Redrawn and Table revised. Added AJB Date 9.3.94 192/0 added	TITLE	SCALE		MEB Midlands Electricity plc M.122
	CONDUCTOR TENSION JOINT	DATE	1-12-92	
	(ONE PIECE)	DRAWN	AJU	
	AAAC & ACSR CONDUCTORS	TRACED	E.L.W.	
		CHECKED	Ref	
	APPROVED			



JOINTING INSTRUCTIONS

1. Scratch brush both ends of conductor to be joined.
2. Unlay aluminium strands.
3. Remove surplus grease from the steel core then insert the steel core into the joint and withdraw again.
4. Spread the joint compound evenly over the steel core and wipe any surplus onto the aluminium strands then replace the aluminium strands.
5. Mark each conductor at half joint length from the end and insert both ends together up to this mark.
6. JOINT
Compress as instructions on Joint Sleeve.

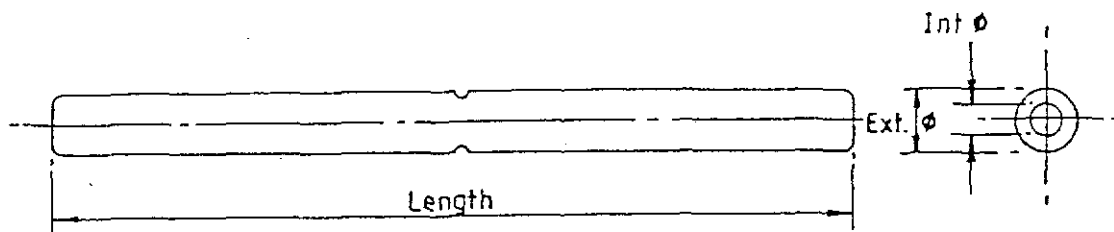
TYPE	CONDUCTOR TYPE AND SIZE	MANUFACTURERS REF BICCON	DIMENSIONS (mm)			COMP'N DIE DETAILS	
			A	B	C	HAND	HYDRAULIC
						MH100(MD6)	G10(Y35)
122 / 1	25 mm ² AL ALLOY 7/2-34 mm ²	SF12L (YDS3L1-BG)	8.12	190.5	16.9	W 698	U 698
122 / 2	50 mm ² AL ALLOY 7/3-30	SF20L (YDS 248L-BG)	11.1	216	20.3	W 659	U 659
122 / 3 *	50 mm ² ACSR * 7/3-35 mm ²	SF20R (YDS 248R-BG)	11.1	438	20.3	W 659	U 659
122 / 4	40 mm ² ACSR 7/3-00	SF18R (YDS1R1-BG)	10.7 9.9	428.6	17.9	W 699	U 699
122 / 5	40 mm ² AL ALLOY 7/2-95 mm ²	SF16L (YDS1L1-BG)	9.9	190.5	17.9	W 699	U 699
122 / 6	25 mm ² ACSR 7/2-36	SF12R (YDS 3R1-BG)	8.12	415.9	16.9	W 698	U 698

- * NOTE 1. Tension Joint Ref. 122 / 3 may be used for 50mm Al. Alloy
2. Biccon new and equivalent old numbers shown

SCALE	-
DATE	10.2.81
DRAWN	R.A.C.
TRACED	
CHECKED	
APPROVED	

CONDUCTOR TENSION JOINT
25 mm² 40 mm² 50 mm² ALAL
& ACSR CONDUCTORS

ISSUE	MIDLANDS
A	ELECTRICITY BOARD
Dimensions Added 28.10.85	DRG. No. M. 122



MEB Ref.	Conductor Size (mm ²)	Manufacturer Ref.		Dimensions			Die/No. of Compressions				No. 60 Ton	No.
				Length	Int. Dia.	Ext. Dia.	M06	No.	Y35/610	No.		
	25 Aluminium											
123/1	7/2-14						W163		U163		N/A	
	50 Aluminium											
123/2	7/3-10						NOT		U243		N/A	
	100 Aluminium											
123/3	7/4-39						TO		U249		N/A	
123/4	95 ABC						BE		U658		N/A	
	400 Aluminium											
123/5	37/3-78						USED		N/A		A-11	

JOINTING INSTRUCTIONS

- 1 Scratch brush both ends of conductor to be jointed.
- 2 Apply a tape marker to each conductor $\frac{1}{2}$ length of joint from the end
- 3 Insert conductors into joint to the tape mark Pushing both ends in together to force grease into interstices.
- 4 Compress overall commencing at centre of joint and work outwards (See instructions on joint Sleeve)

AD000 & J/PW
Date 28-6-94
Table & Note Rev
Retraced.

TITLE

CONDUCTOR TENSION JOINT
HD ALUMINIUM CONDUCTORS

SCALE

DATE

DRAWN

TRACED

CHECKED

APPROVED

4-7-89

A.J.B

P.J.W.

MEB

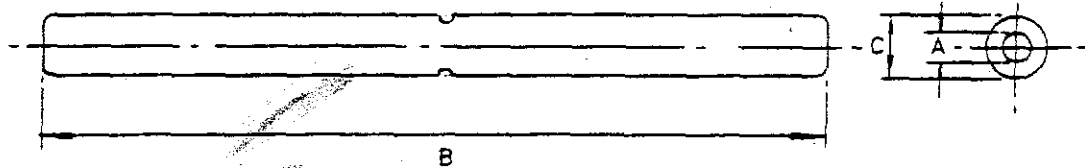
Midlands Electricity plc

ORG No

M.123

ESI STANDARD 43-92

E.S.I. ORG. N°



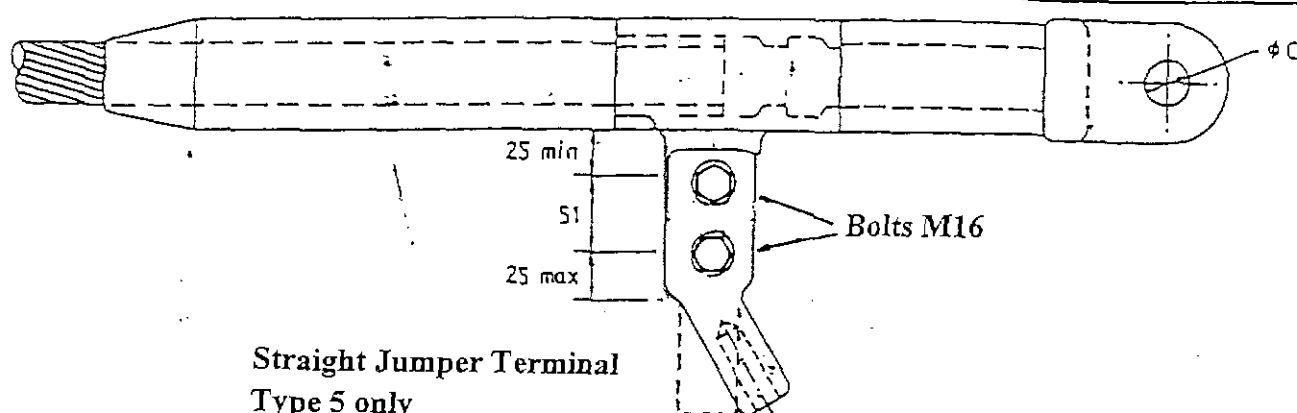
TYPE N°	H.D. ALUMINIUM CONDUCTOR SIZE	MANUFACTURER	CATALOGUE NUMBER	DIMENSIONS			COMPRESSOR DIE DETAILS	
				A	B	C	HAND	HYDRAULIC
123/1	25mm ² 7/2-14mm -025in ² 7/1-087"	BICCON	SF5A YDS2CAS	7.4	98.4	12.7	MH 100 (MD 6)	G 10 (Y 35)
123/2	50mm ² 7/3-10 (.05in ²) 7/1-122"	BICCON	SF9A YDS249AT	9.9	194.1	15.9	W 243	U 243
123/3	100mm ² 7/4-39 (.1in ²) 7/1-173"	BICCON	SF25A YDS279AT	14.3	265.9	21.4		U 249

JOINTING INSTRUCTIONS

1. Scratch brush both ends of conductor to be jointed.
2. Apply a tape marker to each conductor $\frac{1}{2}$ length of joint from the end.
3. Insert conductors into joint to the tape mark. Pushing both ends in together to force grease into interstices.
4. BURNDY JOINT
Compress as instructions on Joint Sleeve

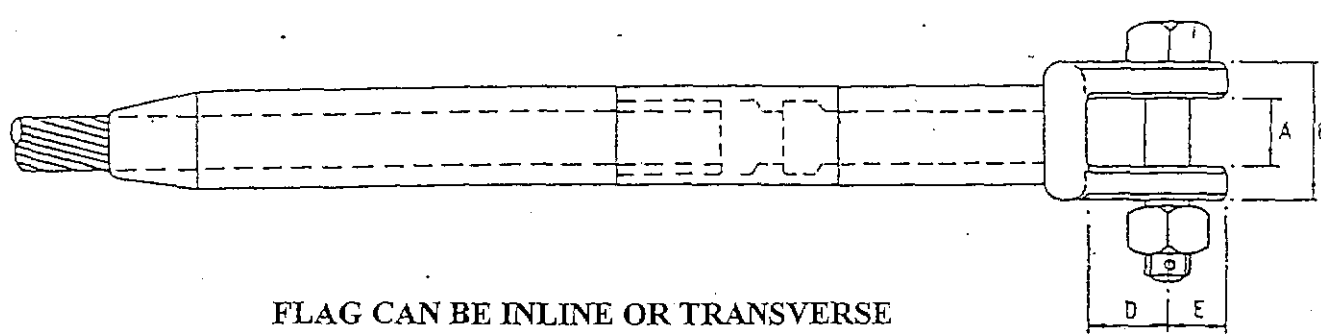
NOTE : Biccon new and equivalent old numbers shown.

SCALE	—	CONDUCTOR TENSION JOINT 25mm ² 50mm ² 100mm ² H D ALUMINIUM CONDUCTORS	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	22-5-73		A	
DRAWN	W.D.H.		Dimension added to 123/1/2/3 R.A.C. OCT.85	
TRACED	C.E.			
CHECKED	<i>[Signature]</i>			
APPROVED	<i>[Signature]</i>			DRG. No. M. 123



**Straight Jumper Terminal
Type 5 only**

Seperate angled and straight jumper
assembly to drg M333

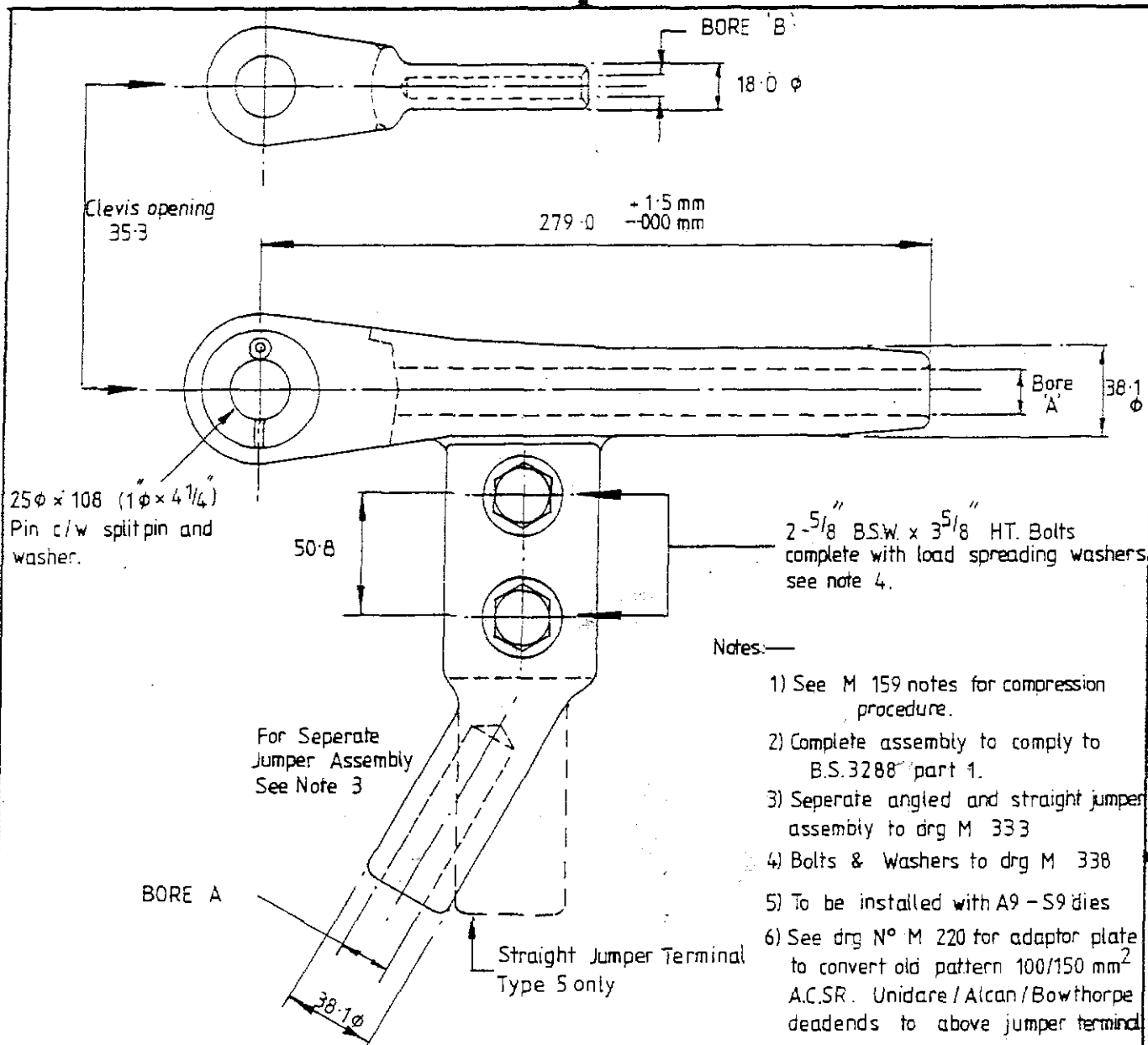


**FLAG CAN BE INLINE OR TRANSVERSE
CAST OR MANUFACTURED**

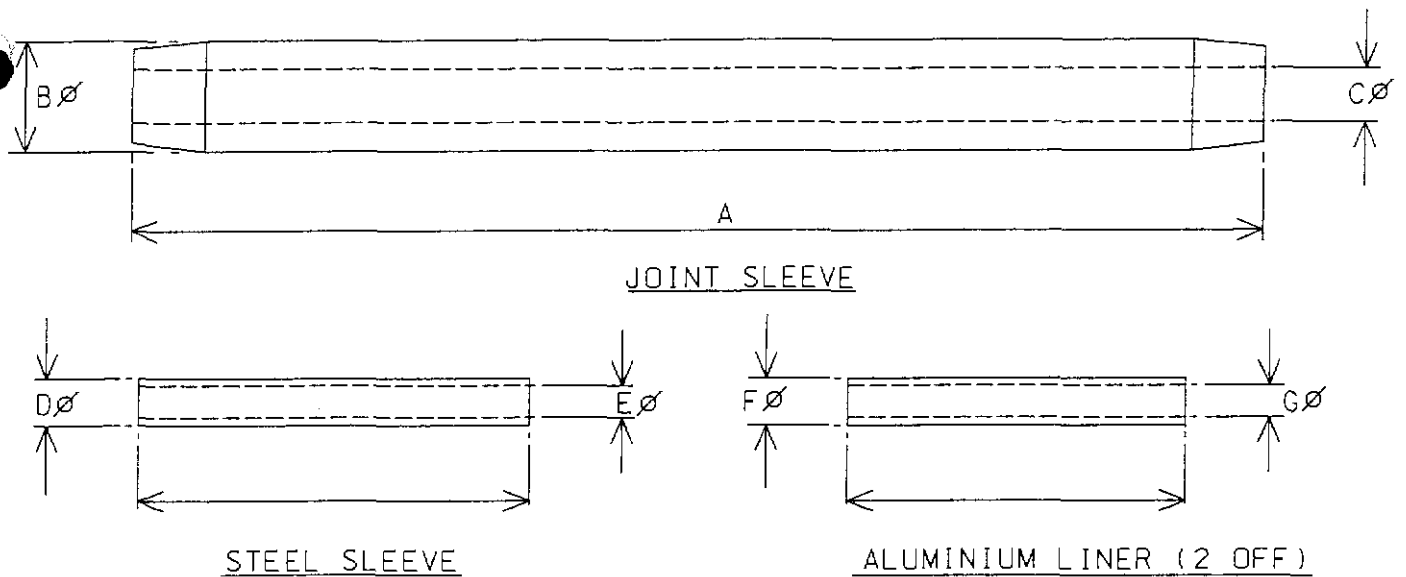
A min	B max	C Dia	D min	E max	BOLT TO BS 3288
33	59	27	35	33	M24 x 100

ME. REF	CONDUCTOR SIZE		MANUFACTURER REFERENCE			BORE DIA		DIE REF	
	SIZE	STRAND				ALUMINA	STEEL B	AL	ST
159/1	150mm ²	30/2-59Al 7/2-59St						A9	S9
159/2	175mm ²	30/2-79Al 7/2-79St						A9	S9
159/3	100mm ²	26/2-27Al 7/1-78 St						A9	S9
159/4	100mm ²	6/4-72Al 7/1-57St						A9	S9
159/5	70mm ²	12/2-79 Al 7/2-79 St						A9	S9

ADD'D BY DATE	TITLE CONDUCTOR DEAD-END (BOLTED JUMPER) 70mm² - 175mm² ACSR CONDUCTORS.	SCALE		 Midlands Electricity plc DRG No M 159
		DATE		
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		



ME. REF	CEGB COMMODITY CODE	CONDUCTOR SIZE		MANUFACTURER REFERENCE			BORE DIA		DIE REF	
		SIZE	STRAND	BICCON	UNIDARE	BOWTHORPE	ALUMA	STEEL 'B'	AL	ST
159/1	-	150mm ²	30/2-59Al 7/2-59St	DT 77- 102 RK8	-	DDA 1813	19.8	9.10	A9	S9
159/2	43-01-115 43-01-727	175mm ²	30/2-79Al 7/2-79St	DT 77- 104 RK8	ZYQ-220	DDA 1953	21.4	9.10	A9	S9
159/3	-	100mm ²	26/2-27Al 7/1-78 St	DT 77- 100 R1K8	-	DDA 1443	17.10	6.35	A9	S9
159/4	-	100mm ²	6/4-72Al 7/1-57St	DT 77- 100 R2K8	-		17.10	5.90	A9	S9
159/5	43-01-130 BODY ONLY	70mm ²	12/2-79 Al 7/2-79 St	DT 77- 100 R K8	ZYQ-91 ZYQ-182A	DDS 1395	17.10	9.10	A9	S9
SCALE		CONDUCTOR DEAD-END (BOLTED JUMPER) 70mm ² - 175mm ² ACSR CONDUCTORS.					ISSUE		MIDLANDS ELECTRICITY BOARD	
DATE							J			
DRAWN							CEGB C.C.		DRG. No. M 159	
TRACED							N° Change			
CHECKED							RAC 10-85			
APPROVED										
2044G		OZALID (UK) LTD								



MEB. REF.	C.E.G.B. COMODITY CODE	CONDUCTOR SIZE (mm)	JOINT SLEEVE			STEEL SLEEVE		ALUMINIUM LINER	
			LENGTH A (mm)	O/D B (mm)	I/D C (mm)	O/D D (mm)	I/D E (mm)	O/D F (mm)	I/D G (mm)
161/0	4303117	175mm ²	572	38	21.8	18	9.1	—	—
161/1	4303109	70mm ²	420	38	22 / 17	18	9.1	21	17
161/2	—	100mm ² (13 STRAND)	420	38	22 / 17	18	6	21	17.5
161/3	—	150mm ²	465	38	24.6	18	9.3	24	19
161/4	—	100mm ² (33 STRAND)	420	38	22 / 17	18	6.5	24	17.3
161/5	4303123	400mm ²	760	48	30.5	19.2	9.9	—	—

MEB. REF.	MANUFACTURER REF N°		DIE REF.
	OHLF		
161/0	84 RR 00		A9-S9
161/1	84 NP 02		A9-S9
161/2	84 NP 06		A9-S9
161/3	84 NP 05		A9-S9
161/4	84 NP 11		A9-S9
161/5	84 RT 00		A11- S10

- NOTES**
1. To comply with BS 3288. Part. 1.
 2. For jointing instruction See M161. Notes.

E.S.I. STANDARD 43.92

Produced by Primary CAD

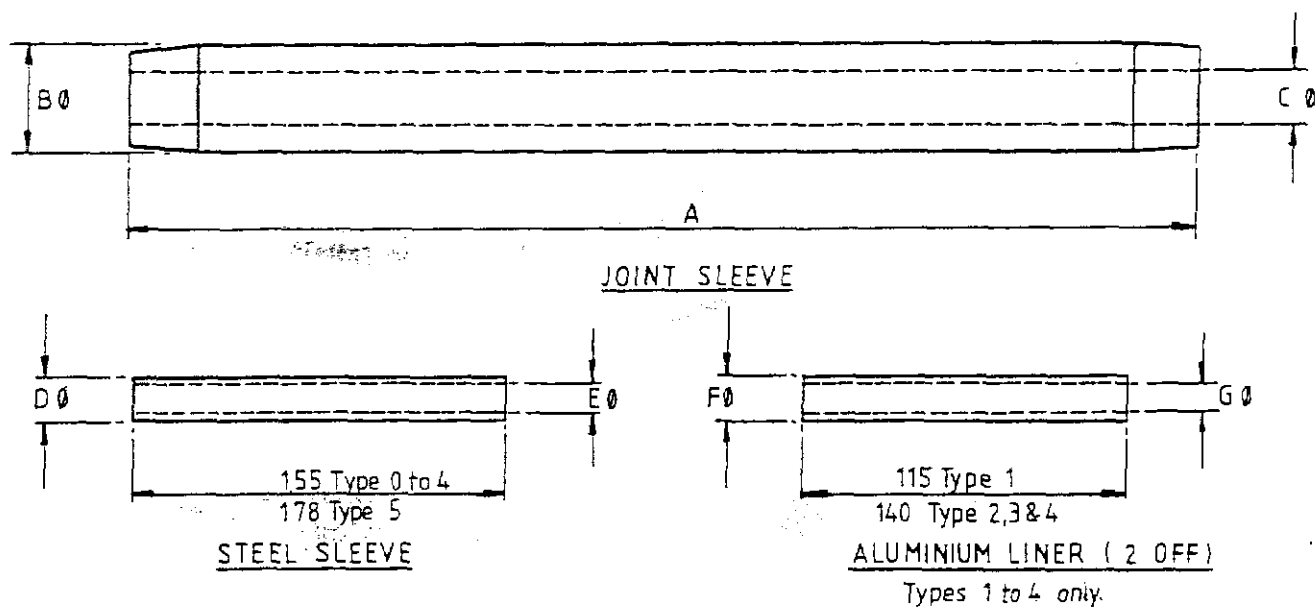
DATE	—
DRAWN	28.7.82
CHECKED	R.A.C.
APPROVED	S.J.H.

CONDUCTOR COMPRESSION JOINT
TENSION
70mm² TO 400mm² A.C.S.R. CONDUCTOR

ISSUE	DATE	ISSUE	DATE
C	12.2.03	B	1.2.84
MOD OHLF ADDED		DIMENSIONS & REF. REVISED	
PA		HPS	

		ORG. No.
M161		

ORG. No. M161

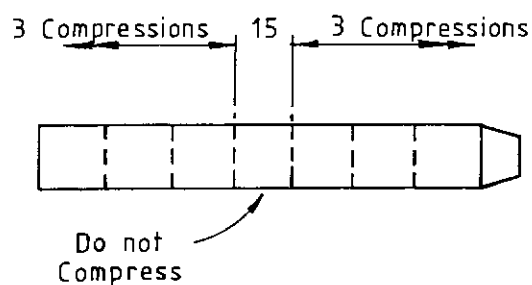
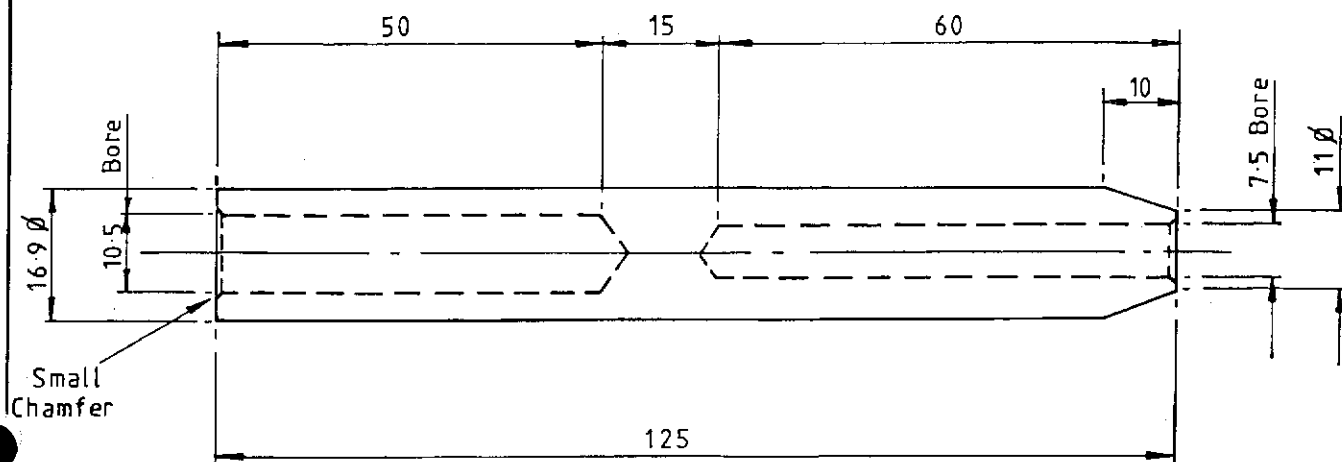


MEB. REF.	C.E.G.B. COMODITY CODE	CONDUCTOR SIZE (m.m.)	JOINT SLEEVE			STEEL SLEEVE		ALUMINIUM LINER	
			LENGTH A (m.m.)	O/D B (m.m.)	I/D C (m.m.)	O/D D (m.m.)	I/D E (m.m.)	O/D F (m.m.)	I/D G (m.m.)
161/0	4303117	175mm ²	570	38.1	21.4	18.00	9.12	—	—
161/1	4303109	70 mm. ²	420	38.1	21.4	18.00	9.12	20.24	16.28
161/2	—	100mm ² (13 STRAND)	465	38.1	24.6	19.05	6.35	24.13	15.47
161/3	—	150 mm. ²	465	38.1	24.6	18.03	9.30	24.13	19.05
161/4	—	100mm ² (33 STRAND)	465	38.1	24.6	18.03	6.35	24.13	17.27
161/5	4303123	400 mm. ²	660	47.6	30.2	19.20	9.93	—	—

MEB. REF.	MANUFACTURER REF N°			DIE REF.
	BICCON LTD.	BOWTHORPE	UNIDARE O.H.L.F.	
161/0	—	MS 1953	RKT 220	A9-S9
161/1	—	MS 1395	—	A9-S9
161/2	SFT 100 R2	—	—	A9-S9
161/3	SFT 102R	—	—	A9-S9
161/4	SFT 100R1	—	—	A9-S9
161/5	—	MS 2862	RKT 241	A11-S10

- NOTES**
1. To comply with BS.3288. Part.1.
 2. For Jointing Instruction See M.161. Notes.

SCALE	—	CONDUCTOR COMPRESSION JOINT TENSION 70mm ² TO 400 mm. ² A.CSR. CONDUCTOR	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M. 161.
DATE	28.7.82		B	
DRAWN	R.A.C.		Dimensions & Ref. Revised.	
TRACED	S.J.H.		HPS. 1-2-84	
CHECKED	—			
APPROVED	—	E.S.I. STANDARD 43-92		E.S.I. DRG. N°

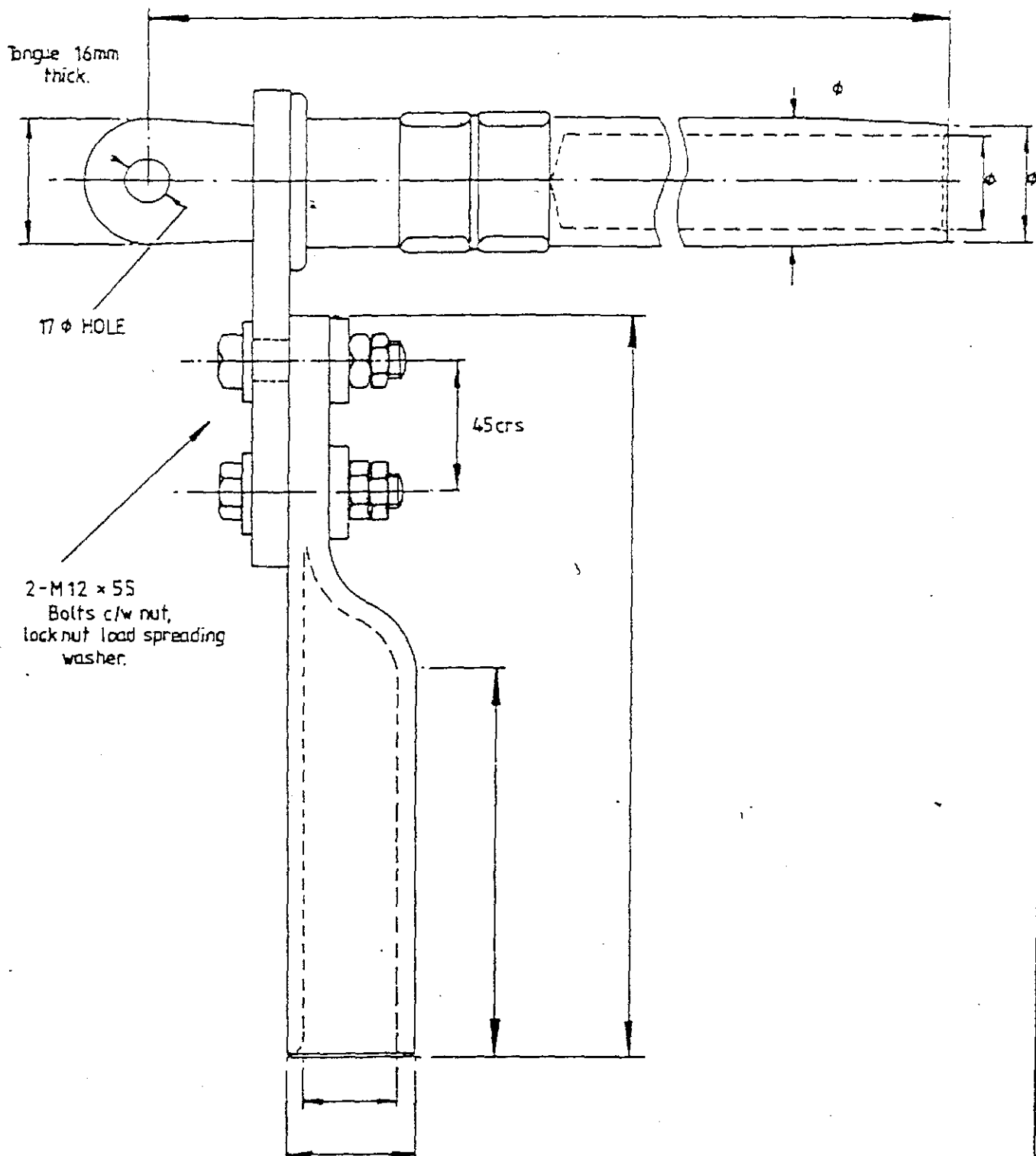


Use U698 die
G10 (Y35) tool

Material :- Aluminium alloy
Fill with grit-loaded grease to
maximise mechanical performance

Mechanical Performance :- To meet the requirements of BS 3288 (Pt 1)
for tension joints.

DATE BY	TITLE O/H LINE RECONDUCTORING TENSION JOINT FOR CONNECTING NEW 50mm ² AAAC (HAZEL) TO OLD 25mm ² AAAC (ALMOND) OR 16mm ² HD COPPER (3/2-65)	SCALE DATE 26-2-91 DRAWN B. B. TRACED G. S. CHECKED APPROVED	MEB Midlands Electricity plc ORG No M173
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MEB REF	ITEM	COMP'N DIE DETAILS
		G10 (Y35)
210 / 0	Dead End Kit C/W Terminal	U 342
210 / 1	Jumper Terminal	U 342

SCALE	NTS	CONDUCTOR TERMINATION COMPR- -SSION DEAD-END 400mm ² H.D. ALUMINIUM CONDUCTOR.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	22-11-82		A	
DRAWN	R.A.C.		Revised	DRG. No. M.210.
TRACED			J.W. 3/84	
CHECKED				E.S.I. Drg No.
APPROVED				
2044G		E.S.I. STANDARD 43-92		

1-14 x 12 CRS.

1-14 ϕ Hole

1-18 ϕ Hole

1-18 x 12 CRS Slot

Material:—
99.5% purity
63.5 x 15.9 aluminium
plate 193 long.

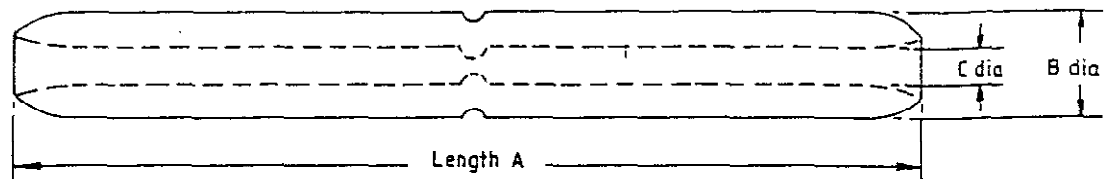
Note:—

Jumper Terminal to be connected
with 2-M338/2 bolts and
load spreading washers

SCALE	FULL
DATE	18-12-80
DRAWN	R.A.
TRACED	
CHECKED	

ADAPTOR PLATE FOR BICC & ALCAN
UNIDARE DEAD-END & TEE
CONNECTIONS

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		ORG No
		M 220 MF

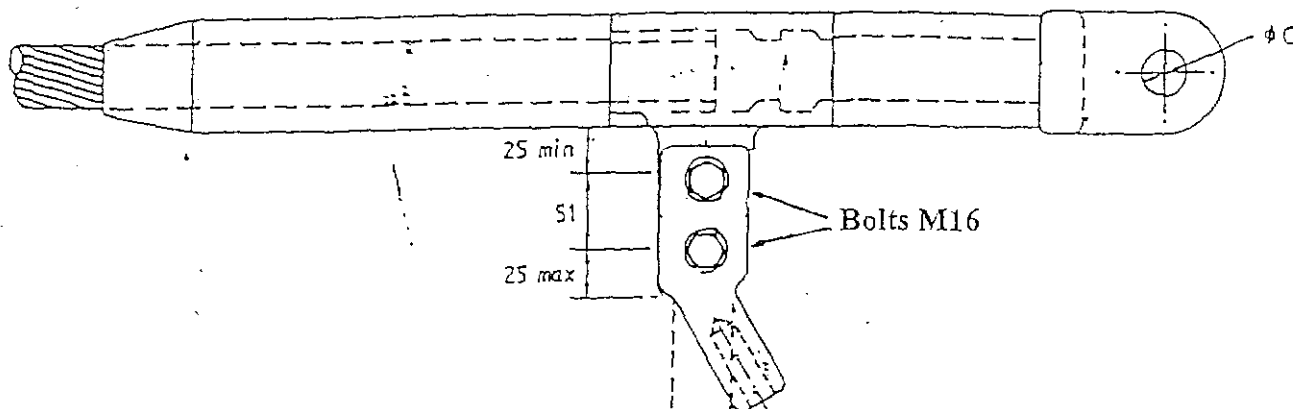


MIN No.	Conductor Stranding	Compression Die		Manufacturers Reference	Dimensions (mm)		
		MH100/MD6	G10/Y35		A	B	C
233/0	14mm (.0225) 7/1.60		U161	BICC SF-1C	68	8	5.5
				Burndy YDS 5C6	66	8.5	5.5
233/1	16mm (.025) 3/2.65	USED	U162	BICC SF-2C	95	10	6
				Burndy YDS 5C3	74	10	6
233/2	32mm (.05) 3/3.75 35mm (.058) 7/2.50		U163	BICC SF-4C	98	13	9
				Burndy YDS 2C5	100	13.5	8.6
				Simel J32U-UK	94	13.3	8.8
233/3	70mm (.10) 7/3.55	BE	U166	BICC SF-7C	157	18	11
				Burndy YDS 258	156	18	11
233/4	100mm (.15) 7/4.30	TO	J168	BICC SF-8C	182	23	14
				Burndy YDS 275	189	22.5	13.5
233/5	125mm (.20) 19/2.90		J169	BICC SF-9C	198	24	15
				Burndy YDS 291	191	24	15.3
233/6	50mm (.075) 7/3.00	NOT	U164	BICC SF-6C	114	14.5	9.5
				Burndy YDS 244	115	14.5	9.5
233/7	150mm (.25) 19/3.20		U170	BICC SF-10C	192	27	17
				Burndy YDS 303	208	27	17

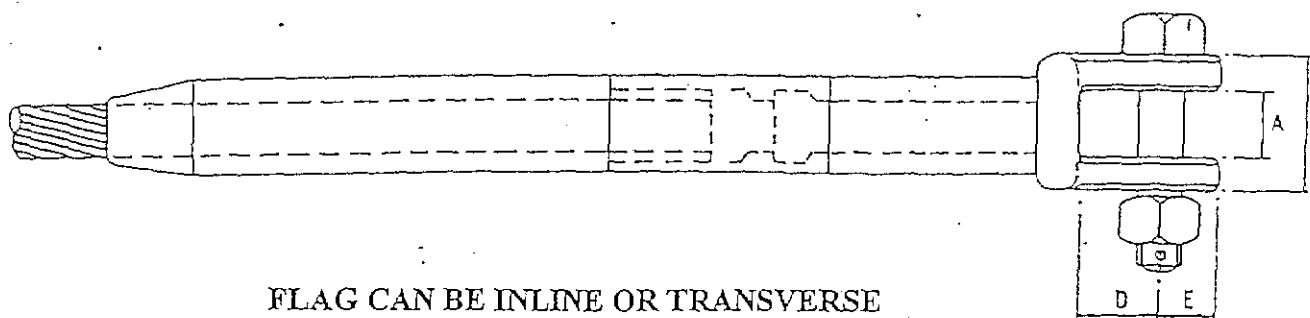
Notes

- Each fitting to have stamped on it the conductor size, stranding and Die Reference
- Joints must meet the requirements of ESI 43-92

C ADDL. A.J.B. DATE 2-7-93 Table Revised.	TITLE CONDUCTOR TENSION JOINT COPPER CONDUCTORS	SCALE	—	 Midlands Electricity plc ORG No M233
		DATE	2-7-93	
		DRAWN	R.C.F.	
		TRACED	P.J.W.	
		CHECKED		
		APPROVED		



Straight Jumper Terminal
Type 2



FLAG CAN BE INLINE OR TRANSVERSE
CAST OR MANUFACTURED

A	B	C	D	E	BOLT TO
min	max	Dia	min	max	BS 3288
41	70	29	48	59	M27 x 110

M. REF	CEGB COMMODITY CODE	CONDUCTOR SIZE		MANUFACTURER REF		DIE REF	
		SIZE	STRAND			ALUMINIUM	STEEL
ZS311	BODY 4301123 TERM 4301135	2 400mm ACSR	54/3-18 Al 7/3-18 st			A 11	S 10
ZS312	BODY 4301123 TERM 4301133	2 400mm ACSR	54/3-18 Al 7/3-18 st			A 11	S 10

TITLE

CONDUCTOR DEAD-END
400mm² ACSR CONDUCTOR

SCALE

DATE

DRAWN

TRACED

CHECKED

APPROVED

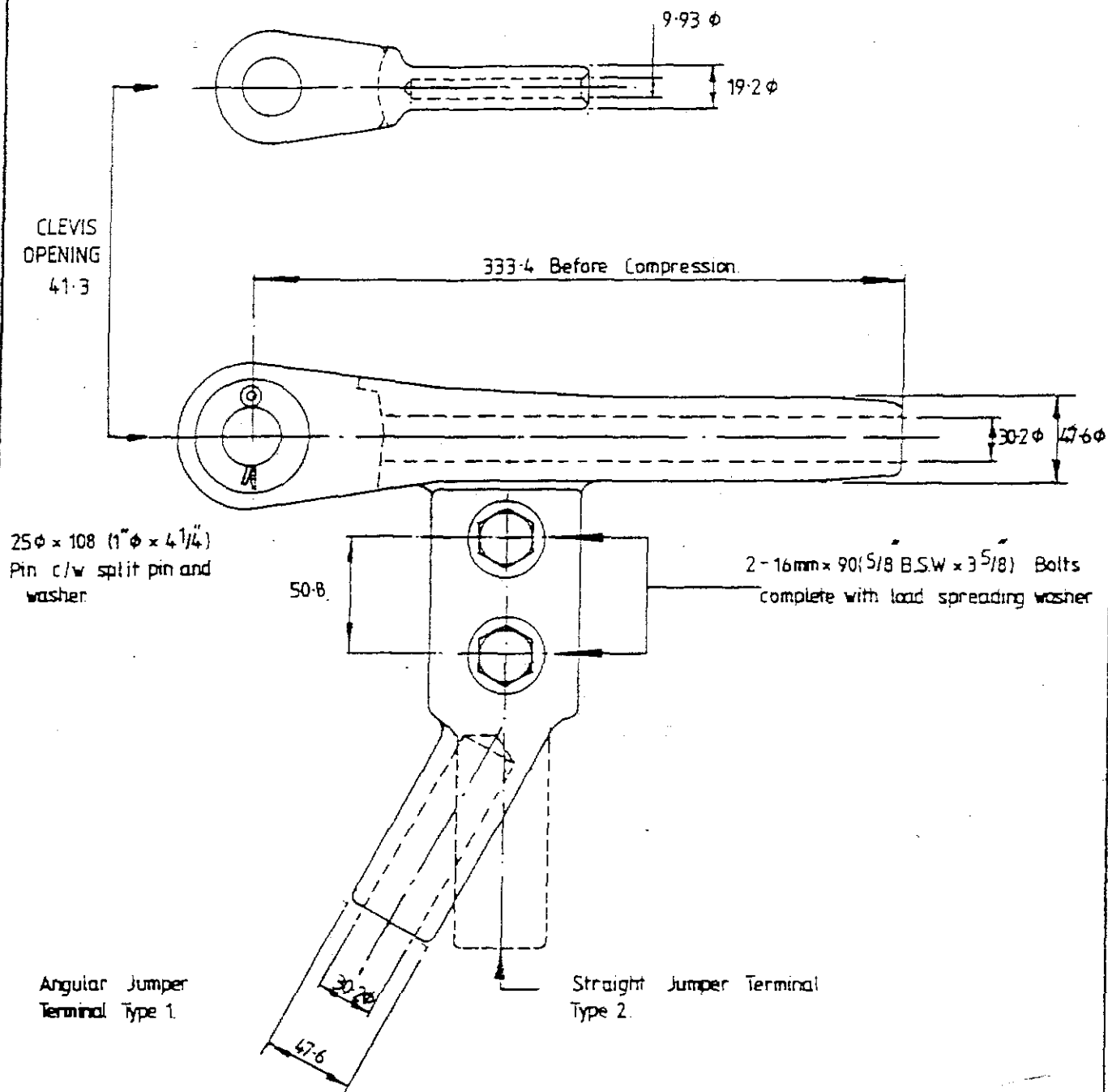
mee

Midlands Electricity plc

JRG No

M. 253

ADDED
DATE



M. REF	CEGB COMMODITY CODE	CONDUCTOR SIZE		MANUFACTURER REF		DIE REF	
		SIZE	STRAND	BOWTHORPE	UNIDARE	ALUMINIUM	STEEL
253/1	BODY 4301123 TERM 4301735	2 400mm ACSR	54/3-18 Al 7/3-18 st	DDA 2862	ZYQ-241	A 11	S.10
253/2	BODY 4301123 TERM 4301783	2 400mm ACSR	54/3-18 Al 7/3-18 st	DDS 2862	ZYQ-241	A 11	S.10

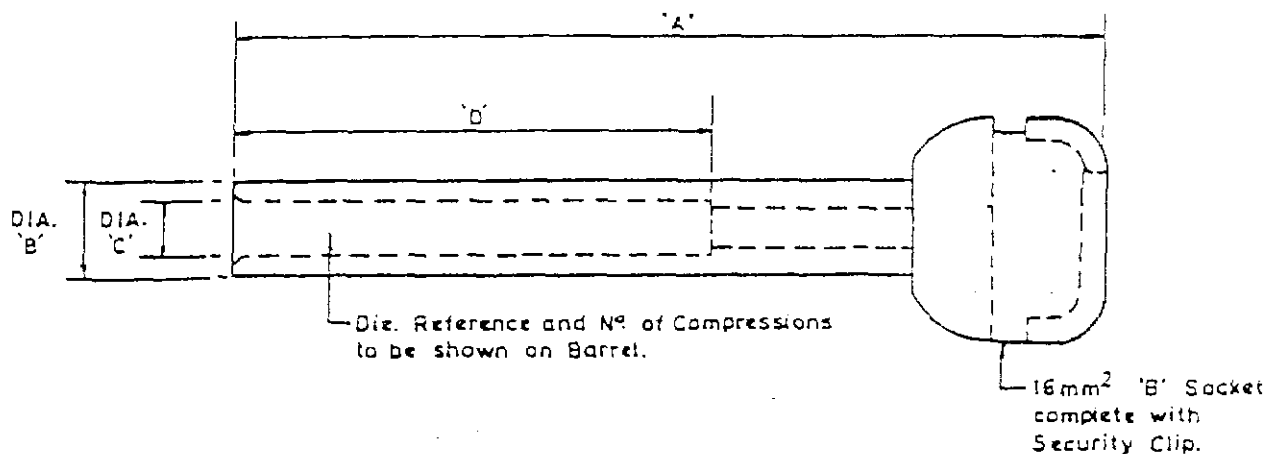
SCALE	N.T.S.
DATE	16.11.82
DRAWN	R.A.C.
TRACED	
CHECKED	
APPROVED	

CONDUCTOR DEAD-END
400mm² ACSR CONDUCTOR

ISSUE	MIDLANDS ELECTRICITY BOARD
A	DRG No
Bowthorpe Ref added. R.A.C. 4/84	M. 253

ESI. STANDARD

ESI. DRG. No

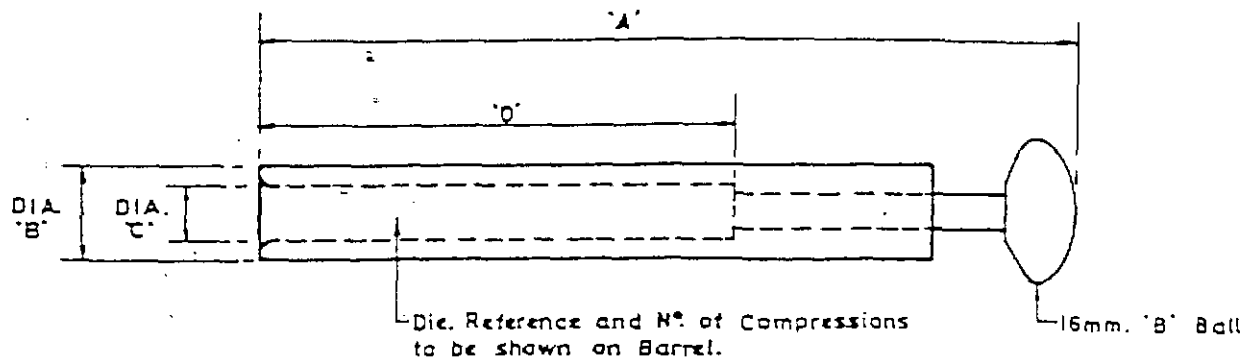


- NOTES:-
1. Joint must meet the requirements of BS3288 Pt 1.
 2. The Joint must be compressed in accordance with the manufacturers instructions impressed on the sleeve.
 3. Ensure that the Joint is filled with 'Penatrox' (AL Joints) or the inside face is treated with Berilium (Cu. Joints) before installation.

ME. REF. NO.	CONDUCTOR		LENGTH A (mm)	DIA. B (mm)	DIA. C (mm)	DEPTH D (mm)	BICCON REF. No.	DIE REF.	
	SIZE	STRANDING						MDO (MDO6)	G10 (Y35)
287/1	16mm ²	3/2.65mm.	138.1	10.16	6.22	34.9	D50-2C (YDW5C3-BG1)	W162	U162 U4CYDMT
	.025in ²	3/1.04 in.							
287/2	32mm ²	3/3.75mm.	147.6	13.36	8.64	49.2	D50-4C (YDW2C5-BG)	W163	U163 U2CYDMT
	.05 in ²	3/1.47 in.							
287/3	70mm ²	7/3.55mm.	177.8	18.03	11.30	77.8	D50-7C (YDW25B-BG1)	—	U166 U26YDT
	.10 in ²	7/1.36 in.							
287/4	50mm ²	7/3.00mm.	154.3	14.5	9.5	54.3	D50-6C (YDW244-BG1)	—	U164.
	.075 in ²	7/1.16 in.							
287/5	25mm ²	7/2.34mm.	308.0	16.89	8.13	203.2	D50-12RL (YDW3RL-BG)	W698	U698 U3RLYRT
	.025 in ²	7/0.92 in.							
287/6	50mm ²	7/3.30mm.	319.1	20.32	11.13	215.9	D50-20RL (YDW248RL-BG)	W659	U659 U26RLYRT
	.05 in ²	7/1.30 in.							
287/5	25mm ²	6/2.36 mm AL + 1/2.36 mm ST	308.0	16.89	8.13	203.2	D50-12RL (YDW3RL-BG1)	W698	U698 U3RLYRT
	.025 in ²	6/0.93" AL + 1/0.93" ST							
287/6.	50mm ²	6/3.35 mm AL + 1/3.35 mm ST	319.1	20.32	11.13	215.9	D50-20RL (YDW248RL-BG1)	W659	U659 U26RLYRT
	.05 in ²	6/1.32" AL + 1/1.32" ST							
287/9	100mm ²	7/1.57 mm ST + 6/4.72 mm AL	396.9	25.91	15.47	266.7	D50-34RL (YDW279RL-BG1)	—	U654 U28RLYRT
	.10 in ²	7/0.62" ST + 6/1.85" AL							

Dimensions are in millimetres
unless otherwise stated.
Biccon New and Equivalent
Old Numbers Shown

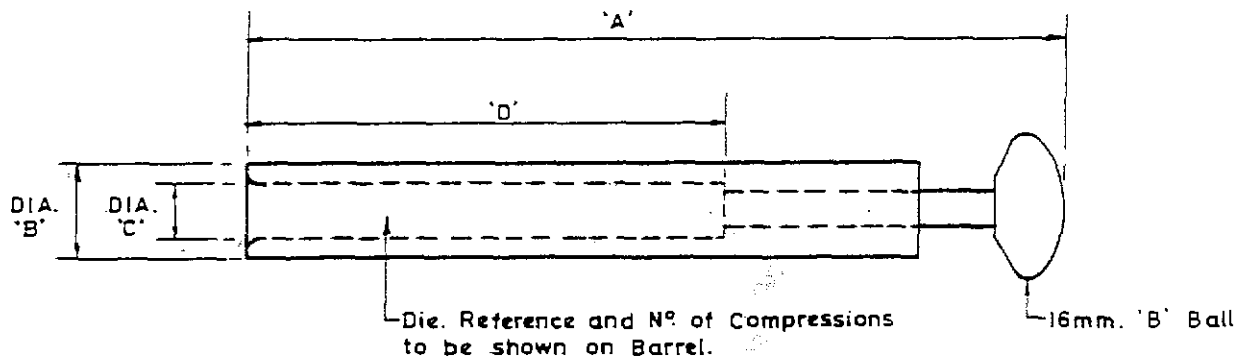
SCALE	N.T.S.	CONDUCTOR TERMINATION COMPRESSION SOCKET DEAD END COPPER, AL ALLOY & ACSRS CONDUCTORS	ISSUE 4-12-85 RAC A	MIDLANDS ELECTRICITY BOARD
DATE	27-11-73			
DRAWN	A.R.B.	Die Ref & Dim. Change M287/ 3/4 & TITLE MOD	ORG. No. M 287	
TRACED	D.F.B.			
CHECKED	<i>[Signature]</i>			
APPROVED	<i>[Signature]</i>			



- NOTES:-
1. Joint must meet the requirements of BS 3288, Pt1.
 2. The Joint must be compressed in accordance with the manufacturers instructions impressed on the sleeve.
 3. Ensure that the joint is filled with 'Penatrox' (AL Joints) or the inside face is treated with Berilium (Cu. Joints) before installation.

M. REF. No	CONDUCTOR		LENGTH	DIA.	DIA.	DIA.		DIE REF.	
	SIZE	STRANDING						MH100/MD6	G10 / Y35
288/5	25mm ²	7 / 2.34mm						W698	U698
	.025in ²	7 / .092in							
288/6	50mm ²	7 / 3.30mm						W659	U659
	.05in ²	7 / .130in							
288/9	100mm ²	7 / 4.65mm						Not To Be Used	U654
288/5	25mm ²	6 / 2.36mm AL + 1 / 2.36mm ST						W698	U698
	.025in ²	6 / .093in AL + 1 / .093in ST							
288/6	50mm ²	6 / 3.35mm AL + 1 / 3.35mm ST						W659	U659
	.05in ²	6 / .132in AL + 1 / .132in ST							
288/9	100mm ²	7 / 1.57mm ST + 6 / 4.72mm AL						Not To Be Used	U654
	.10in ²	7 / .062in ST + 6 / .186in AL							

ADDED BY DATE	TITLE	SCALE		 Midlands Electricity plc M288
	CONDUCTOR TERMINATION COMPRESSION BALL DEAD END ACSR & AAAC	DATE	11/11/99	
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		



- NOTES:-
1. Joint must meet the requirements of B.S.3288 Pt1.
 2. The Joint must be compressed in accordance with the manufacturers instructions impressed on the sleeve.
 3. Ensure that the joint is filled with 'Penatrox' (AL Joints) or the inside face is treated with Berilium (Cu. Joints) before Installation.

M. REFN ^o		CONDUCTOR		LENGTH	DIA.	DIA.	DEPTH	BICCON	DIE REF.	
		SIZE	STRANDING	'A'(mm)	'B'(mm)	'C'(mm)	'D'(mm)	REF. N ^o	MH100(MDS)	G10(Y35)
288/1	H.D. COPPER	16mm ²	3 / 2.65 mm.	100.0	10.16	6.22	34.9	D60 - 2C (YDW5C3-BG)	W 162	U162 U4CYDMT
		.025in ²	3 / .104 in.							
288/2		32mm ²	3 / 3.75 mm.	109.5	13.36	8.64	49.2	D60 - 4C (YDW2C5-BG)	W 163	U163 U2CYDMT
		.05in ²	3 / .147 in.							
288/3	H.D. COPPER	70mm ²	7 / 3.55 mm.	152.4	18.03	11.30	77.8	D60 - 7C (YDW258-BG)	-	U166 U26YDT
		.10in ²	7 / .136 in.							
288/4	H.D. COPPER	50mm ²	7 / 3.00 mm	128.5	14.5	9.5	54.3	D60 - 6C (YDW244-BG)	-	U166.
		.075in ²	7 / .166 in							
288/5	AL ALLOY	25mm ²	7 / 2.34 mm.	285.8	16.89	8.13	203.2	D60-12RL (YDW3RIL-BG)	W 698	U698 U3RLYRT
		.025in ²	7 / .092 in.							
288/6	AL ALLOY	50mm ²	7 / 3.30 mm.	296.9	20.32	11.13	215.9	D60-20RL (YDW248RL-BG)	W 659	U659 U26RLYRT
		.05in ²	7 / .130 in.							
288/5	ACSR (S.C.A.)	25mm ²	6 / 2.36 mm AL + 1 / 2.36 mm ST	285.8	16.89	8.13	203.2	D60-12RL (YDW3RIL-BG)	W 698	U698 U3RLYRT
		.025in ²	6 / .093" AL + 1 / .093" ST							
288/6	ACSR (S.C.A.)	50mm ²	6 / 3.35 mm AL + 1 / 3.35 mm ST	296.9	20.32	11.13	215.9	D60-20RL (YDW248RL-BG)	W 659	U659 U26RLYRT
		.05in ²	6 / 1.32" AL + 1 / 1.32" ST							
288/9	ACSR (S.C.A.)	100mm ²	7 / 1.57 mm ST + 6 / 4.72 mm AL	381.0	25.91	15.47	266.7	D60-34RL (YDW279RL-BG)	-	U654 U28RLYRT
		.10in ²	7 / .062" ST + 6 / .186" AL							

Dimensions are in millimetres unless otherwise stated

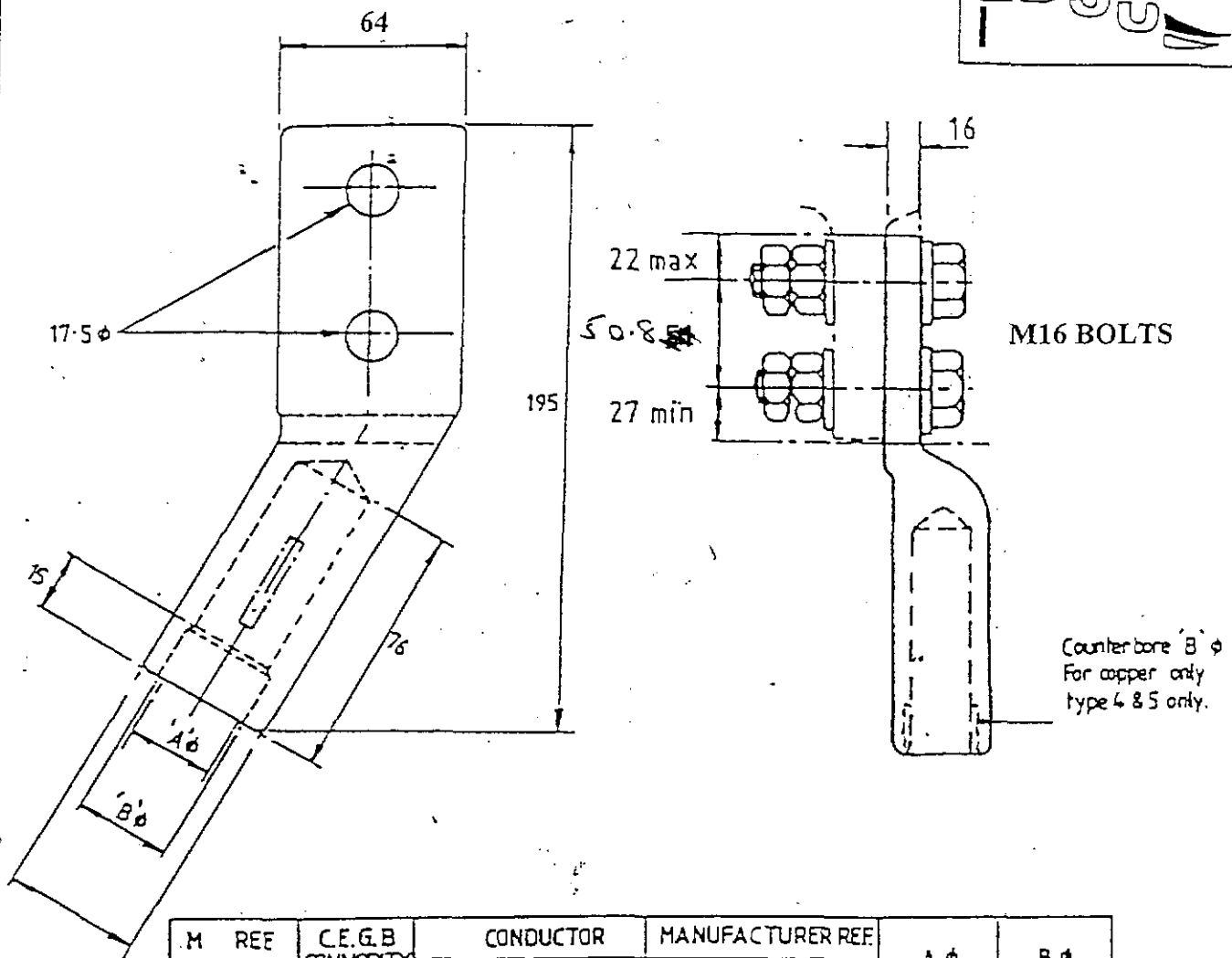
Biccon New and Equivalent Number Shown



SCALE	N. T. S.
DATE	27.11.73
DRAWN	A. R. B.
TRACED	D. F. B.
CHECKED	<i>[Signature]</i>
APPROVED	<i>[Signature]</i>
OZALID 16339	

CONDUCTOR TERMINATION COMPRESSION BALL DEAD END COPPER, AL ALLOY & ACSR CONDUCTORS

ISSUE	MIDLANDS ELECTRICITY BOARD
4.12.85 R.A.C. FA	DRG. No.
Die Ref & Dim. Change M288/ 3/4 & TITLE MODS	M 288

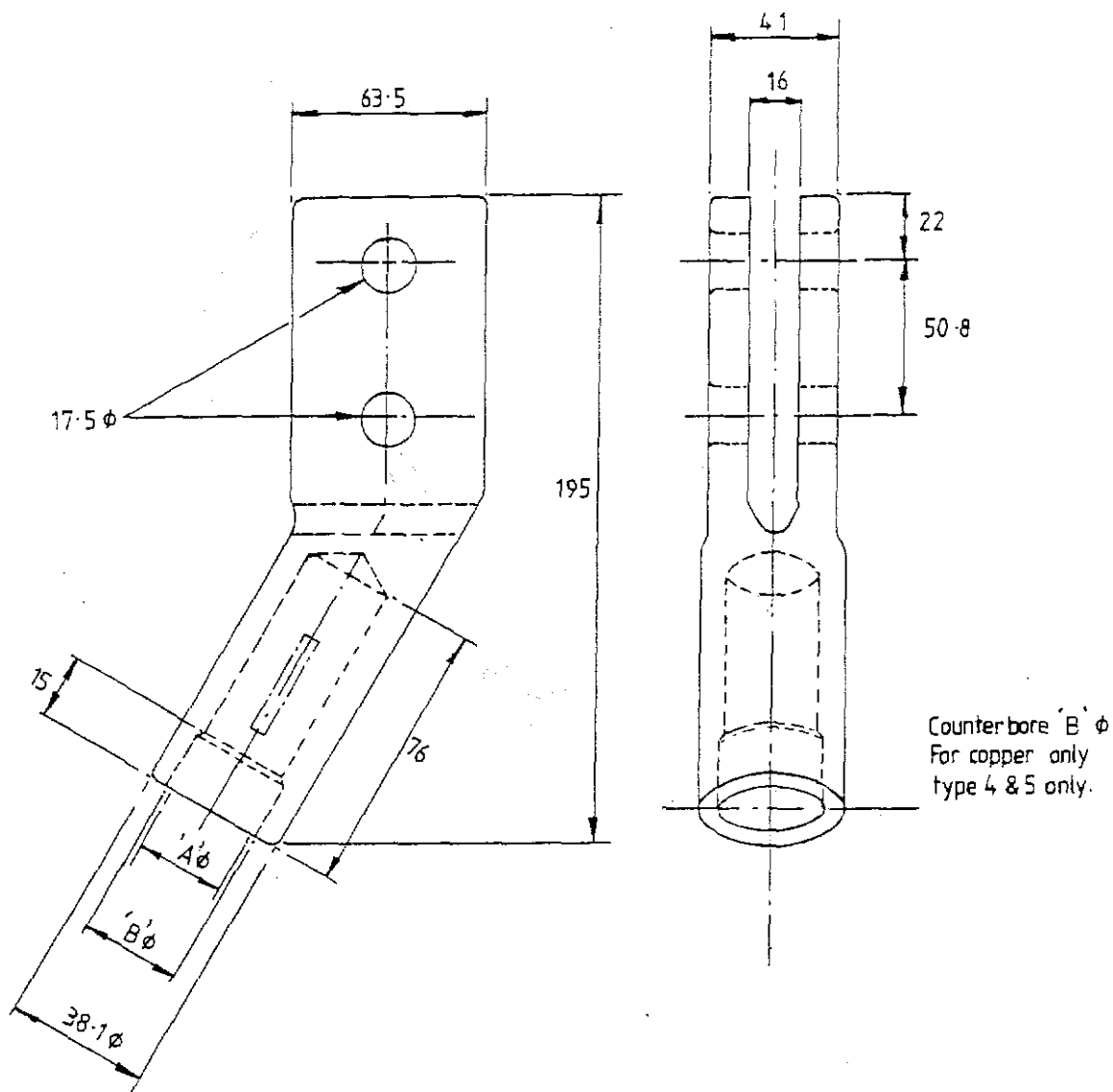


M REF	C.E.G.B COMMODITY CODE	CONDUCTOR		MANUFACTURER REF	A φ	B φ
		SIZE	STRAND			
333/1		150mm ² ACSR	30/2-59A1 7/2-59st			
333/2	43-01-727	175mm ² ACSR	30/2-79A1 7/2-79 st			
333/3		100mm ² ACSR	A1 ACSR SIZES			
333/4		70 mm ² Cu	7/3-55			
333/5		125 mm ² Cu	7/4-90 19/2-92			

NOTES: — See M 159 notes for compression procedure types 1-3 only.
Terminal supplied c/w nuts bolts & washers.

Types 4 & 5 conductor to be greased before insertion.
Terminal to be double wrapped with denso tape after compression.

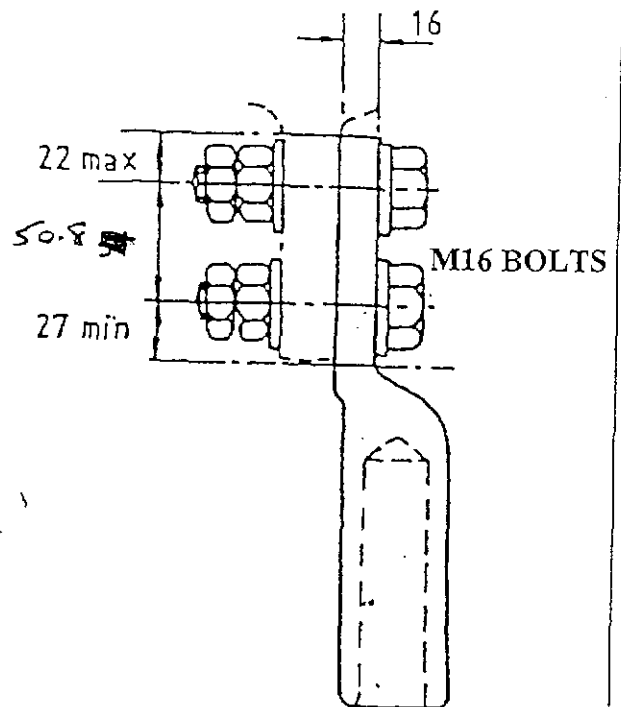
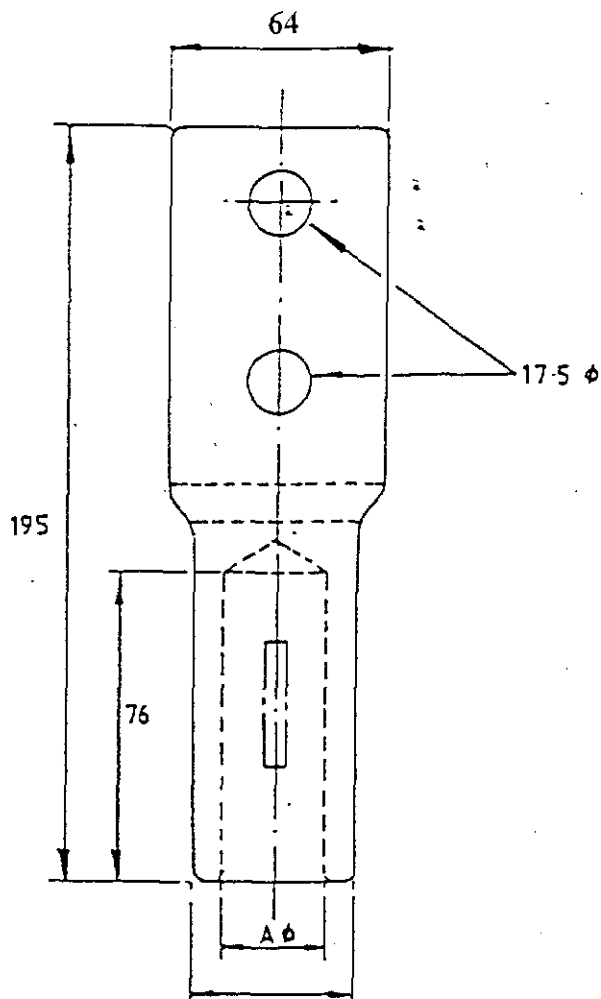
ADDED BY DATE	TITLE	SCALE	MEB Midlands Electricity plc ORG No M 333 st 1.
		GATE	
		DRAWN	
		TRACED	
		CHECKED	
		APPROVED	



M REF	C.E.G.B COMMODITY CODE	CONDUCTOR		MANUFACTURER REF BIC CON	A φ	B φ
		SIZE	STRAND			
333/1	—	150mm ² ACSR	30/2-59Al 7/2-59st	L102R-16DU30K1 (YCAT150R-16U30)	19.83	—
333/2	43-01-727	175mm ² ACSR	30/2-79Al 7/2-79 st	L104R-16DU30K1 (YCAT175R-16U30)	21.8	—
333/3	—	100mm ² ACSR	All ACSR SIZES	L100R-16DU30K1 (YCAT100R-16U30)	17.06	—
333/4	—	70 mm ² Cu	7/3-55	L107CH-16DU30K1 (YCAT070C-16U30)	11.11	14.68
333/5	—	125 mm ² Cu	7/4-90 19/2-92	L109CH-16DU30K1 (YCAT125C-16U30)	15.86	21.43

- NOTES:—
1. See M 159 notes for compression procedure types 1-3 only.
 2. Terminal supplied c/w nuts bolts & washers.
 3. Replacement nuts bolts & washers to drg M .338.
 4. Types 4 & 5 conductor to be greased before insertion.
 5. Terminal to be double wrapped with denso tape after compression.

SCALE	—	CONDUCTOR DEAD-END JUMPER TERMINAL ANGULAR FOR A.C.S.R. & COPPER CONDUCTORS.	ISSUE	MIDLANDS
DATE	28-8-81		B	ELECTRICITY BOARD
DRAWN	R.A.C.		Man. Ref.	DRG. No. M 333 sht 1.
TRACED			Revised.	
CHECKED			RAC 10-85	
APPROVED				

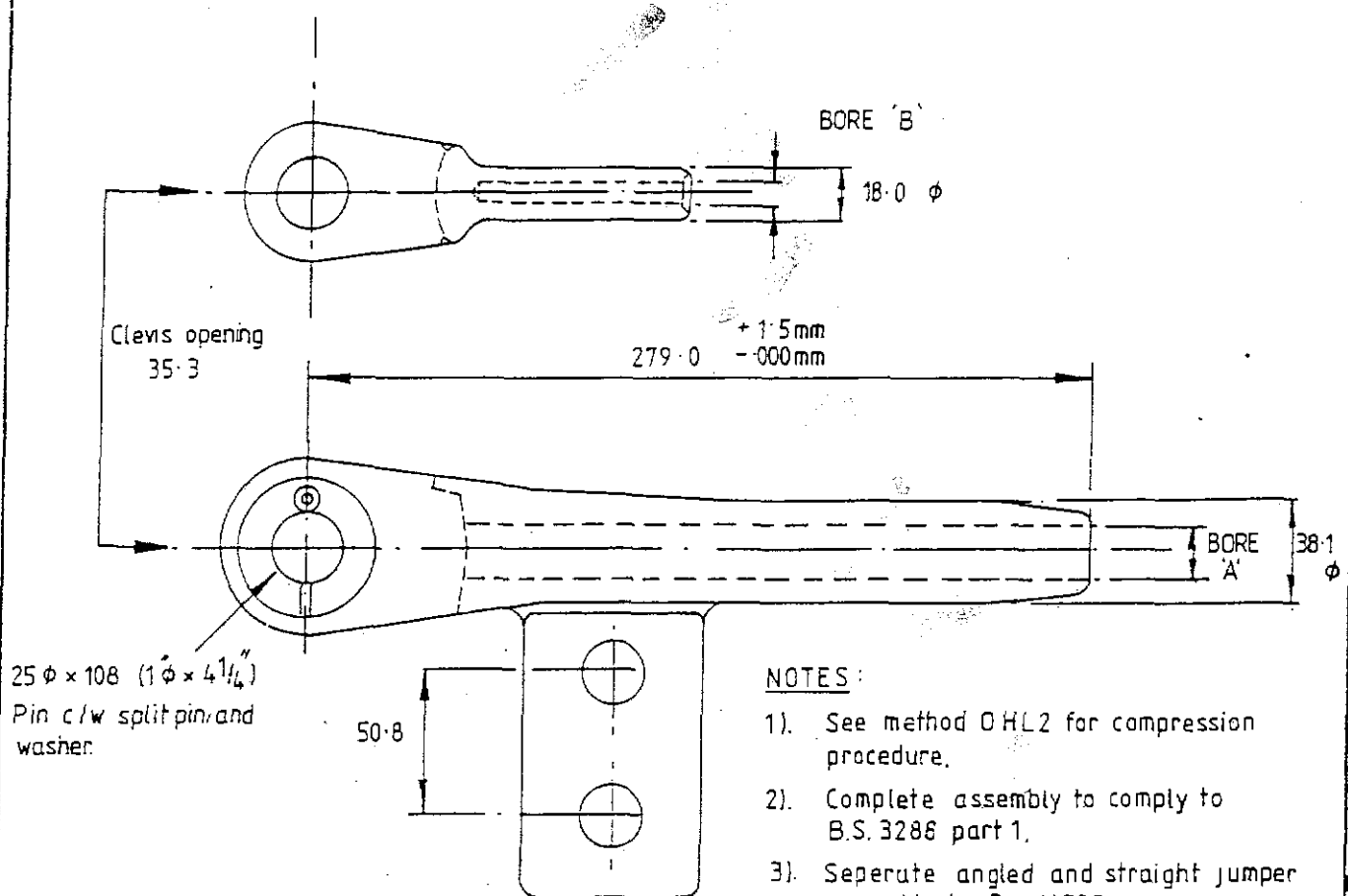


M. REF.	C.E.G.B. COMMODITY CODE	CONDUCTOR SIZE		MANUFACTURERS REF	A ϕ
		SIZE	STRAND		
333/7	43-01-775	175 mm ² ACSR	30/2-79 Al 7/2-79 St		21.8
333/8	-	150 mm ² ACSR	30/2-59 Al 7/2-59 St		19.83
333/9	-	70 mm ² ACSR	12/2-79 Al 7/2-79 St		17.06

NOTES: -

1. See M 159 notes for compression procedure.
2. Terminal supplied with Nuts, Bolts and Washers.

ADDED BY DATE	TITLE	SCALE	 Midlands Electricity plc DRG No M 333 sht 2.
	CONDUCTOR DEAD-ENDS (JUMPER TERMINAL STRAIGHT FOR ACSR CONDUCTORS.	DATE	
		DRAWN	
		TRACED	
		CHECKED	
		APPROVED	



NOTES:

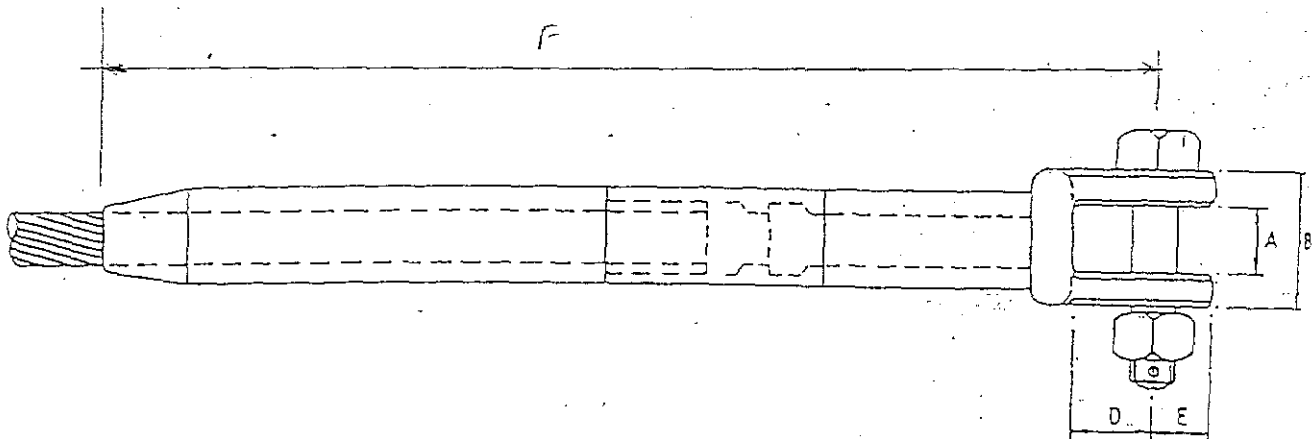
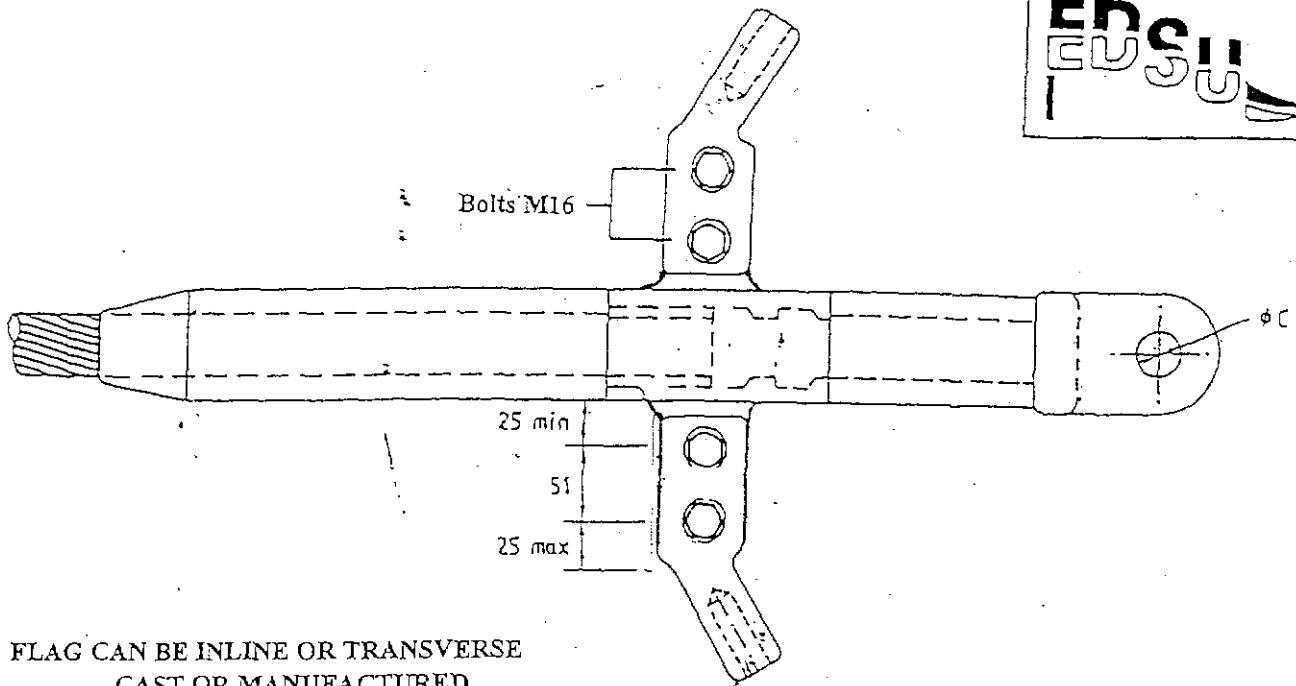
- 1). See method OHL2 for compression procedure.
- 2). Complete assembly to comply to B.S. 3286 part 1.
- 3). Separate angled and straight jumper assembly to Drg. M333
- 4). Bolts and Washers to Drg. M338
- 5). To be installed with A9 - S9 dies
- 6). See Drg. No. M220 for adaptor plate to convert old pattern 100/150 mm.² ACSR. Unidare / Alcan / Bowthorpe deadends to above jumper terminal.
- 7). Clevis pin to Drg. M509.

M. REF.	CEGB COMMODITY CODE	CONDUCTOR SIZE		MANUFACTURER REF		BORE DIA		DIA REF	
		SIZE	STRAND	BICCON	UNIDARE	ALUM A	STEEL 'B'	ALUM	STEEL.
334/1	—	150mm ²	30/2-59Al 7/2-59St	YDWT 150. R1U-C	—	19.8	9.10	A9	S9.
334/2	43-01-115	175mm ²	30/2-79Al 7/2-79St	YDWT 175 R1U-C	—	21.4	9.10	A9	S9.
334/3	—	100mm ²	26/2-27Al 7/1-78St	YDWT 100 XR1U-C	—	17.10	6.35	A9	S9
334/4	—	100mm ²	6/4-72Al 7/1-57St	YDWT 100 R1U-C	—	17.10	5.90	A9	S9
334/5	43-01-130	70mm ²	12/2-79Al 7/2-79St	YDWT 070 R1U-C	—	17.10	9.10	A9	S9

SCALE	N.T.S.
DATE	22-10-81
DRAWN	R.A.C.
TRACED	
CHECKED	
APPROVED	

CONDUCTOR DEAD-END
(TERMINATION ONLY) 70mm² -
175mm² ACSR CONDUCTORS.

ISSUE	MIDLANDS ELECTRICITY BOARD
C	DRG. No.
Bore Dia's corrected R.A.C 19-11-86	M.334.



Size mm	A min	B max	C Dia	D min	E max	BOLT TO BS 3288
100						
150	33	59	27	35	33	M24 x 100
175						
400	41	70	29	48	59	M27 x 110

M. REF.	CONDUCTOR SIZE		MANUFACTURER REF.		DIMENSIONS (mm.)					DIE REF
	SIZE	STRAND		LENGTH	A0	B0	C0	D0	L	
335/1	175mm ²	30/2-79 7/2-79	84GR03	406						A9 S9
335/2	400mm ²	54/3-18 7/3-18	83XT07	462						A11 S10
335/3	100mm ²	26/4-72 7/1-78								A9 S9
335/5	150mm ²	30/2-59 7/2-59								A9 S9

TITLE

CONDUCTOR DEAD-END DOUBLE
JUMPER CONNECTION 100mm², 400mm²
ACSR CONDUCTORS.

SCALE

DATE

DRAWN

TRACED

CHECKED

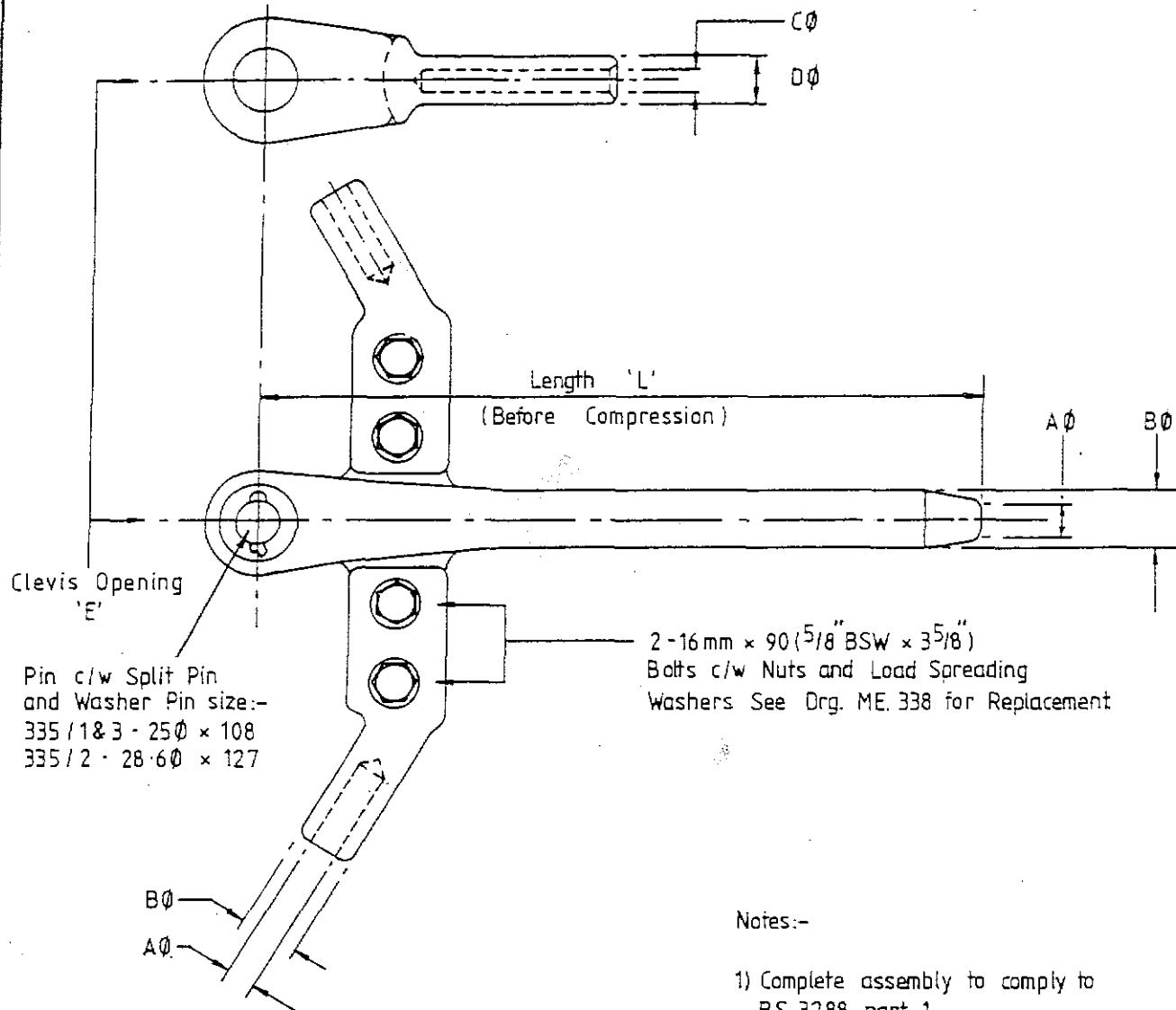
APPROVED

MEB

Midlands Electricity plc

DRG No

M. 335

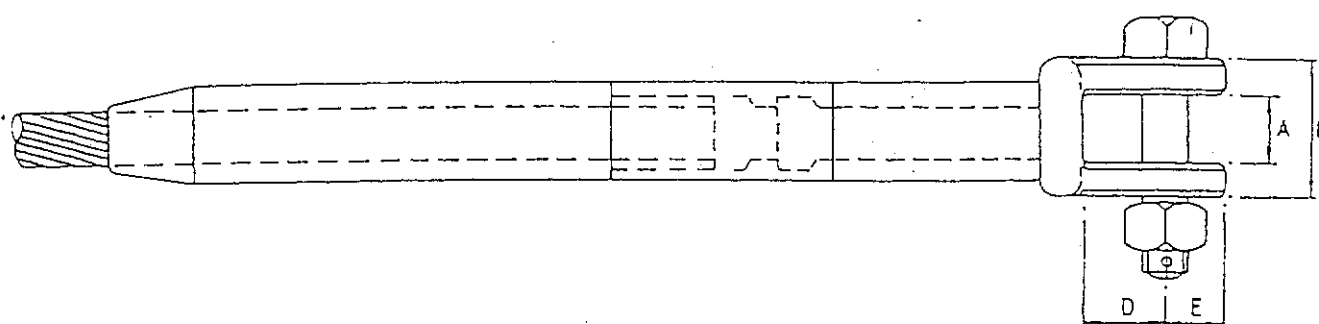
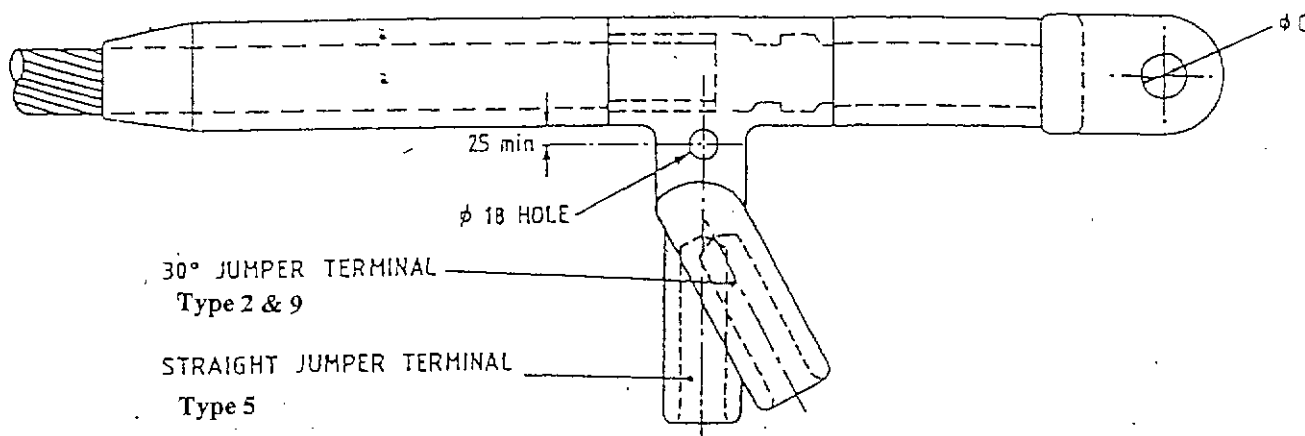


Notes:-

- 1) Complete assembly to comply to BS 3288 part 1.
- 2) Separate angled and straight jumper assembly to drg. ME. 333
- 3) Bolts & Washers to drg. ME 338

M. REF.	CEGB COMMODITY	CONDUCTOR SIZE		MANUFACTURER REF.		DIMENSIONS (mm.)						DIE REF
		SIZE	STRAND	BOWTHORPE	UNIDARE	AØ	BØ	CØ	DØ	L	E	
335/1	Body 4301163 Term 4301727	175mm ²	30/2-79 7/2-79	DDA2W1953	WYQ-220	21.4	38.1	9.10	18	279	35.3	A9 S9
335/2	Body 4301171 Term 4301735	400mm ²	54/3-18 7/3-18	DDA2W2863	WYQ-241	30.1	47.6	9.9	19.2	333.4	41.3	A11 S10
335/3	—	100mm ²	26/4-72 7/1-78	DDA2W1443	—	17.1	38.1	6.35	18	279	35.3	A9 S9
335/5	—	150mm ²	30/2-59 7/2-59	DDA2W1813	—	19.8	38.1	9.10	18	279	35.3	A9 S9

SCALE	N.T.S.	CONDUCTOR DEAD-END DOUBLE JUMPER CONNECTION 100mm ² , 400mm ² ACSR CONDUCTORS.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	16.11.82		B	
DRAWN	R.A.C.		Bore Dies corrected	DRG. No.
TRACED	HPS		R.A.C. 19-11-86	M. 335
CHECKED				
APPROVED				
C. D. W Ltd. 3297		ESI STANDARD		ESI DRG. No.



FLAG CAN BE INLINE OR TRANSVERSE
CAST OR MANUFACTURED

Size mm	A min	B max	C Dia	D min	E max	BOLT TO BS 3288
100						
150	33	59	27	35	33	M24 x 100
175						
400	41	70	29	48	59	M27 x 110

M. REF	CONDUCTOR		MANUFACTURER			DIMENSIONS				LENGTH L	DIE REF
	SIZE	STRAND				A φ	B φ	C φ	D φ		
336/2	175mm ²	30/2-79Al 7/2-79St									St - S9 Al - A9
336/5	70mm ²	12/2-79Al 7/2-79St									St - S9 Al - A9
336/9	400mm ²	54/3-18Al 7/3-18St									St - S10 Al - A11

TITLE

CONDUCTOR DEAD-END
(INTEGRAL JUMPER)
70mm² - 175mm² - 400mm² A.C.S.R.

SCALE

DATE

DRAWN

TRACED

CHECKED

APPROVED

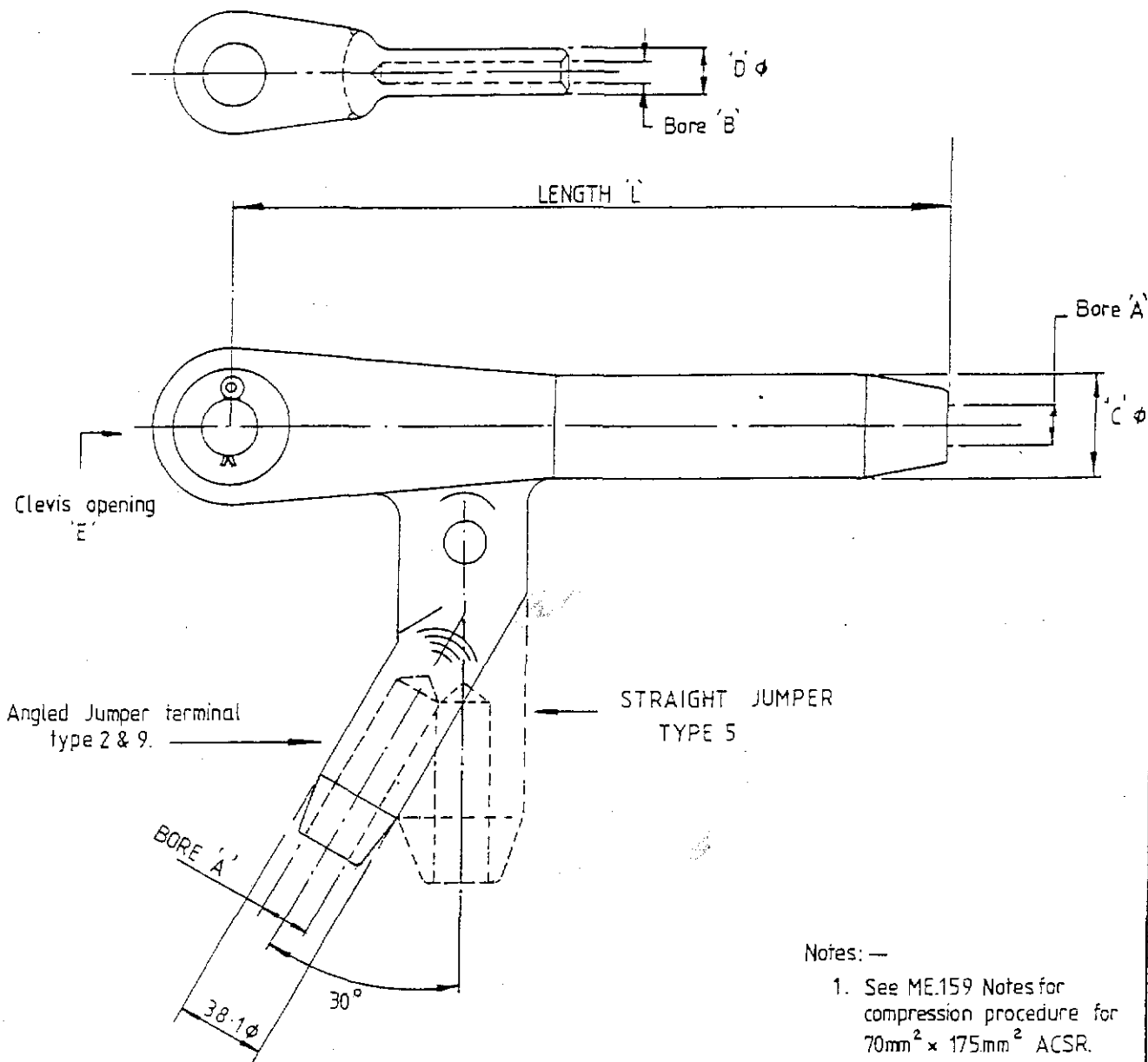
MEB

Midlands Electricity plc

DRG No

M. 336

ADDED BY
DATE

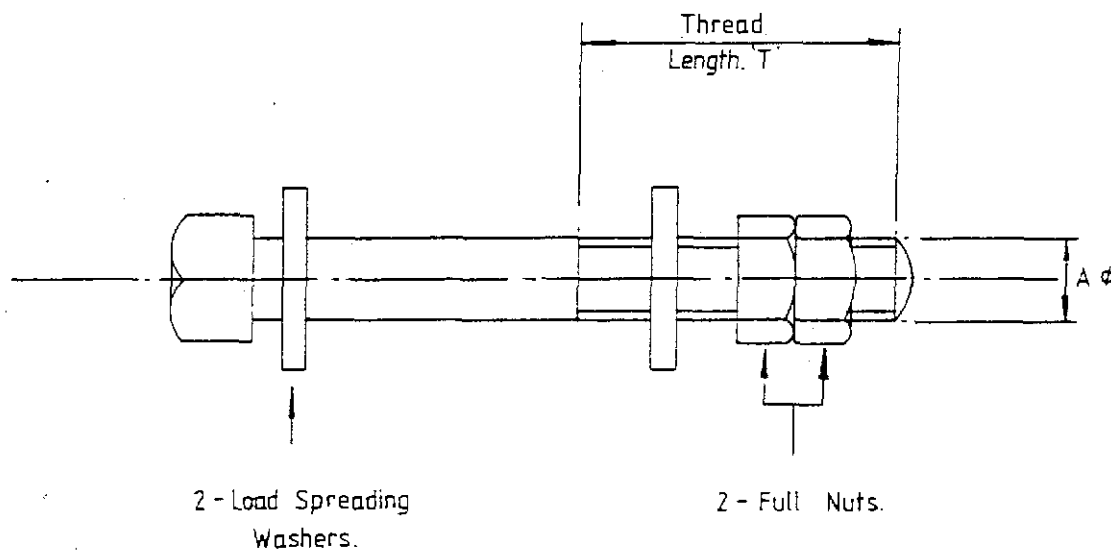


Notes: —

1. See ME.159 Notes for compression procedure for $70\text{mm}^2 \times 175\text{mm}^2$ ACSR.
2. Clevis pins to drg N°M 509

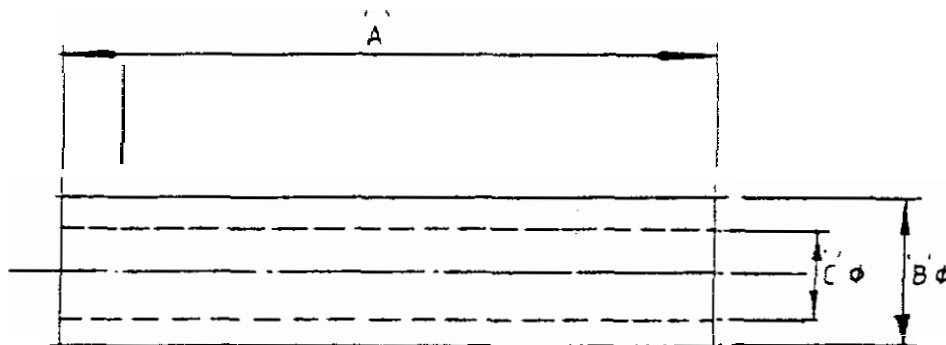
M. REF	C.E.G.B. COMMODITY CODE	CONDUCTOR		MANUFACTURER			DIMENSIONS					LENGTH L	DIE REF
		SIZE	STRAND	BICC Burnd	Bowthorpe	Unidare	A ϕ	B ϕ	C ϕ	D ϕ	E		
336/2	—	175mm^2	30/2-79Al 7/2-79St	DT 78-104R YDWT175 R1T30-CK8	DIA1953	Z JAY220	21.4	9.1	38.1	18.0	35.3	279.0 + 1.5mm - 0.00	St - S9 Al - A9
336/5	—	70mm^2	12/2-79Al 7/2-79St	DT 78-100R YDWT070 R1T-CK8	DIS 1395		17.1	9.1	38.1	18.0	35.3	279.0 + 1.5mm - 0.00	St - S9 Al - A9
336/9	—	400mm^2	S4/3-18Al 7/3-18St		DIA 2862.	Z JAY241	30.1	9.9	47.6	19.2	41.3	362.0	St - S11 Al - A11

SCALE	NTS	CONDUCTOR DEAD-END (INTEGRAL JUMPER) $70\text{mm}^2 - 175\text{mm}^2 - 400\text{mm}^2$ A.C.S.R.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	16.11.82		B	
DRAWN	R.A.C.		Bore Dia's corrected RAC 19.11.86	DRG. No. M. 336
TRACED				
CHECKED				
APPROVED				
E.S.I. STANDARD.			E.S.I. DRG. N°	



M. REF	C.E.G.B. COMMODITY CODE	A ϕ	Length L	Length T	Washer			Tightening Torque.
					O.D.	I.D.	Thk	
338/1	4301792	16mm $\frac{5}{8}$ " BSW	90 ($3\frac{5}{8}$ ")	38.1	44.5	17.5	9.5	
338/2	-	12mm $\frac{1}{2}$ " BSW	65 ($2\frac{1}{2}$ ")	31.8	31.8	14.3	3.2	

SCALE	N.T.S.	REPLACEMENT BOLTS FOR DISCONNECTABLE DEAD-ENDS & NON TENSION JOINTS.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	16.11.82			
DRAWN	R.A.C.			DRG. No.
TRACED				M.338.
CHECKED				ESI. DRG. No.
APPROVED				
ESI STANDARD				



NOTE

- 1 Joints are to comply with BS. 3286 part 1
- 2 Bicon Old and Equivalent New Number Shown
- 3 * Die Ref. U568 to be used with 339/4 Type YDS26D only.

TYPE	CONDUCTOR SIZE	BICC BURNDY REF		DIMENSIONS			DIE REF	
		NEW REF	OLD REF	A	B	C	G10 (Y35)	MH100(MD6)
M. 339/1	.022 ins ² (16 mm ²) 3/-103 " .025 ins ² (22 mm ²) 7/-073 "	SF 1 C 1	YDS 1C1 (YDS3CD)	158	10.7	6.5	U203 "	W 203
M. 339/2	.025 ins ² (19 mm ²) 3/-112 " .03 ins ² (23 mm ²) 3/-124 " .035 ins ² (26 mm ²) 3/-131 " .04 ins ² (30 mm ²) 7/-093 " .04 ins ² (30 mm ²) 3/-141 "	SF 1 C 2	YDS 1C2 (YDS 2CD)	165	13.6	8.1	U204 "	W 204
M. 339/3	.05 ins ² (38 mm ²) 3/-158 " .05 ins ² (38 mm ²) 7/-103 "	SF 1 C 3	YDS 1C3 (YDS1C4)	203	15.8	9.1	U373 "	-
M. 339/4	.075 ins ² (55 mm ²) 3/-193 " .075 ins ² (55 mm ²) 7/-127 " .075 ins ² (65 mm ²) 7/-136 "	SF 1 C 4	YDS 1C4 (YDS26D)	225	18.0	11.3	U166 U568	-
M. 339/7	.1 ins ² (75 mm ²) 7/-146 "	SF 71C	YDS26 D9	244	21.7	12.2	U552	-
M. 339/8	.15 ins ² (115 mm ²) 7/-109 "	SF 81C	YDS29 D	298.5	24.38	14.83	U331	-
SEE DRG N° ME 709 FOR 175 mm ² CAD CU. MID SPAN JOINT.								

SCALE	—
DATE	10-1-83
DRAWN	R.A.C.
TRACED	
CHECKED	
APPROVED	✓

CONDUCTOR COMPRESSION JOINT
MID SPAN
CADMIUM COPPER CONDUCTOR

ISSUE	MIDLANDS
A	ELECTRICITY BOARD
Tool Ref.	DRG. No
Revised.	M 339
4.84. J.W.	

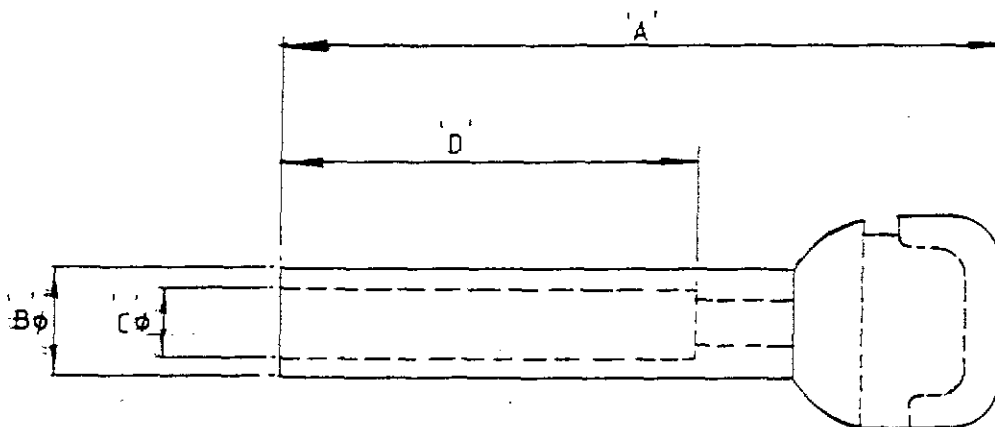
23446

EST. STANDARD

E.S.I. Drg N°

OZALID (AKI) LTD

MF



16mm socket to
B.S. 3288 part.

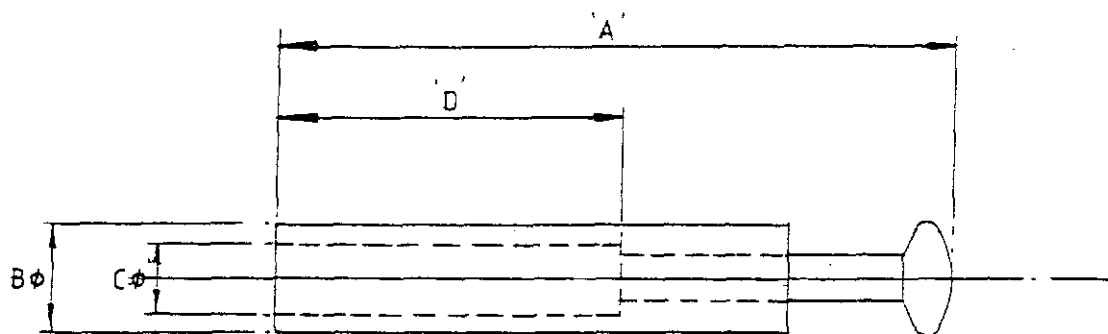
Note: 1. The joints are to comply with B.S. 3288 Pt 1.
2. Bicon Old and Equivalent New Number Shown.

TYPE	CONDUCTOR SIZE	BICC BURNDY REF		DIMENSIONS				DIE REF	
		NEW REF	OLD REF	A	B	C	D	G10 (Y35)	MH100(MD6)
M 340/1	.022 in ² (16mm ²) 3/-103"	D50-1C1	YDW1C1-BG1	186	10.7	6.5	82.6	U 203	W 203
	.025 in ² (22mm ²) 7/-073"								
M 340/2	.025 in ² (19mm ²) 3/-112"								
	.03 in ² (23mm ²) 3/-124"								
	.035 in ² (26mm ²) 3/-131"	D50-1C2	YDW1C2-BG1	186.7	13.6	8.1	82.6	U 204	W 204
	.04 in ² (30mm ²) 7/-093"								
	.04 in ² (30mm ²) 3/-141"								
M 340/3	.05 in ² (38mm ²) 3/-158"	D50-1C3	YDW1C3-BG1	198.7	15.8	9.1	104.8	U373	-
	.05 in ² (38mm ²) 7/-103"								
M 340/4	.075 in ² (55mm ²) 3/-193	D50-1C4	YDW1C4-BG1	201.9	18	11.3	111.9	U166	-
	.075 in ² (55mm ²) 7/-127								
	.075 in ² (65mm ²) 7/-136								

SCALE	—	CONDUCTOR TERMINATION COMPRESSION SOCKET DEAD END CADMIUM COPPER CONDUCTORS.	ISSUE	MIDLANDS
DATE	10.1.83		B	ELECTRICITY BOARD
DRAWN	R.A.C.		Tool Ref.	DRG. No.
TRACED			Revised	M 340.
CHECKED			4.84. JW.	
APPROVED	#			
2044G		E.S.I. STANDARD. 43-92	ESI Drg N°	OZALIO RMI LTD

MF

Mi

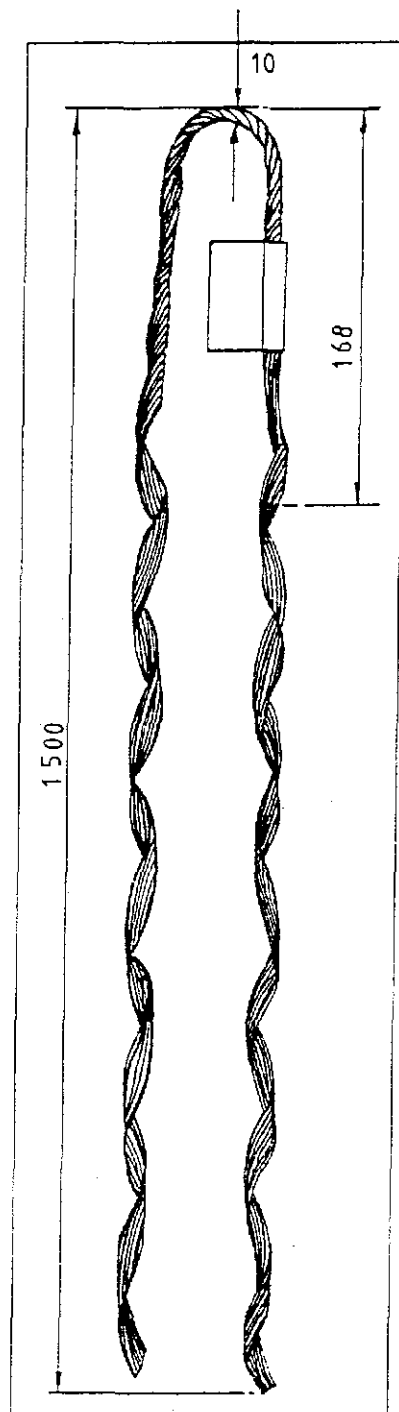


NOTES: 1- The joints are to comply with BS3288 pt 1.
2. Bicon Old and Equivalent New Number Shown.

TYPE	CONDUCTOR SIZE	BICC BURNDY REF		DIMENSIONS				DIE REF	
		NEW REF	OLD REF	A	B	C	D	G 10 (Y35)	MH100(MD6)
M 341/1	.022 in ² (16 mm ²) 31-103" .025 in ² (22 mm ²) 71-073"	D 60-1C1	YDW1C1-BG	149.9	10.7	6.5	82.6	U 203	W 203
M 341/2	.025 in ² (19 mm ²) 31-112" .03 in ² (23 mm ²) 31-124" .035 in ² (26 mm ²) 31-131" .04 in ² (30 mm ²) 71-093" .04 in ² (30 mm ²) 31-141"	D 60-1C2	YDW1C2-BG	149.9	13.6	8.1	82.6	U 204	W 204
M 341/3	.05 in ² (38 mm ²) 31-158" .05 in ² (38 mm ²) 71-103"	D 60-1C3	YDW1C3-BC	162.6	15.8	9.1	100	U 373	-
M 341/4	.075 in ² (55 mm ²) 31-193 .075 in ² (55 mm ²) 71-127 .075 in ² (65 mm ²) 71-136	D 60-1C4	YDW1C4-BG	178	18	11.3	110.7	U166	-

SCALE	—	CONDUCTOR TERMINATION COMPRESSION BALL DEAD END CADMIUM COPPER CONDUCTORS.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	10-1-83		A	
DRAWN	R.A.C.		Tool Ref.	
TRACED			Revised.	
CHECKED			4.84. J.W.	
APPROVED				
2044G		ESI STANDARD		ESI Drg No°

MF



PLP CATALOGUE NUMBER
CDE 5715026 - L

DETAILS OF HELICAL FITTING

Wire Material	Galvanised Steel
Wire Dia.	3.66 m.m.
Number of Wires	7
Colour Code	Black

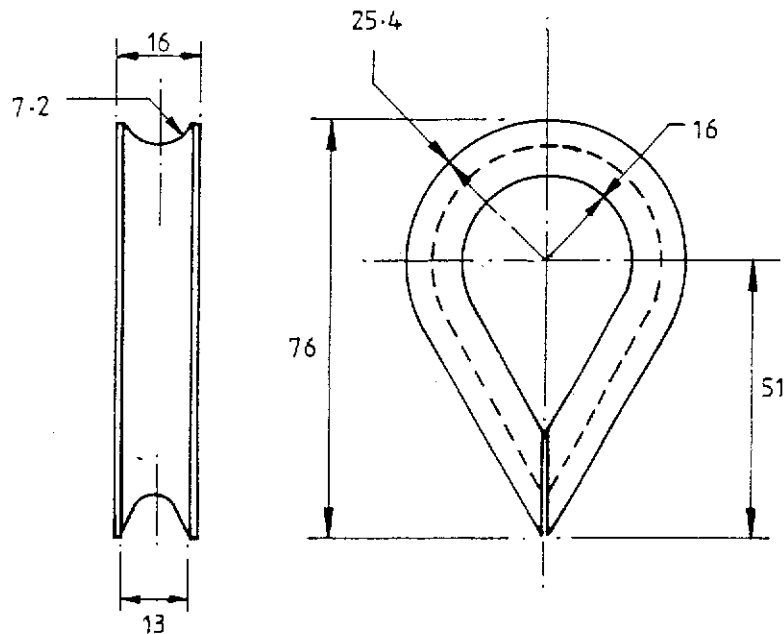
NOTE

1. Only to be used on 'Hardex' having an overall diameter within the range 19.00 to 22.00 m.m. and Steel Wire Armouring which has a left hand direction of lay.
2. For use in conjunction with Thimble M115/1.

SCALE	
DATE	7.3.86
DRAWN	
TRACED	S.J.H.
CHECKED	
APPROVED	

HELICAL TENSION TERMINATION FOR 'HARDEX'
COMBINED PILOT AND EARTH CABLE

ISSUE	MIDLANDS ELECTRICITY BOARD
DRG. No.	M. 446.



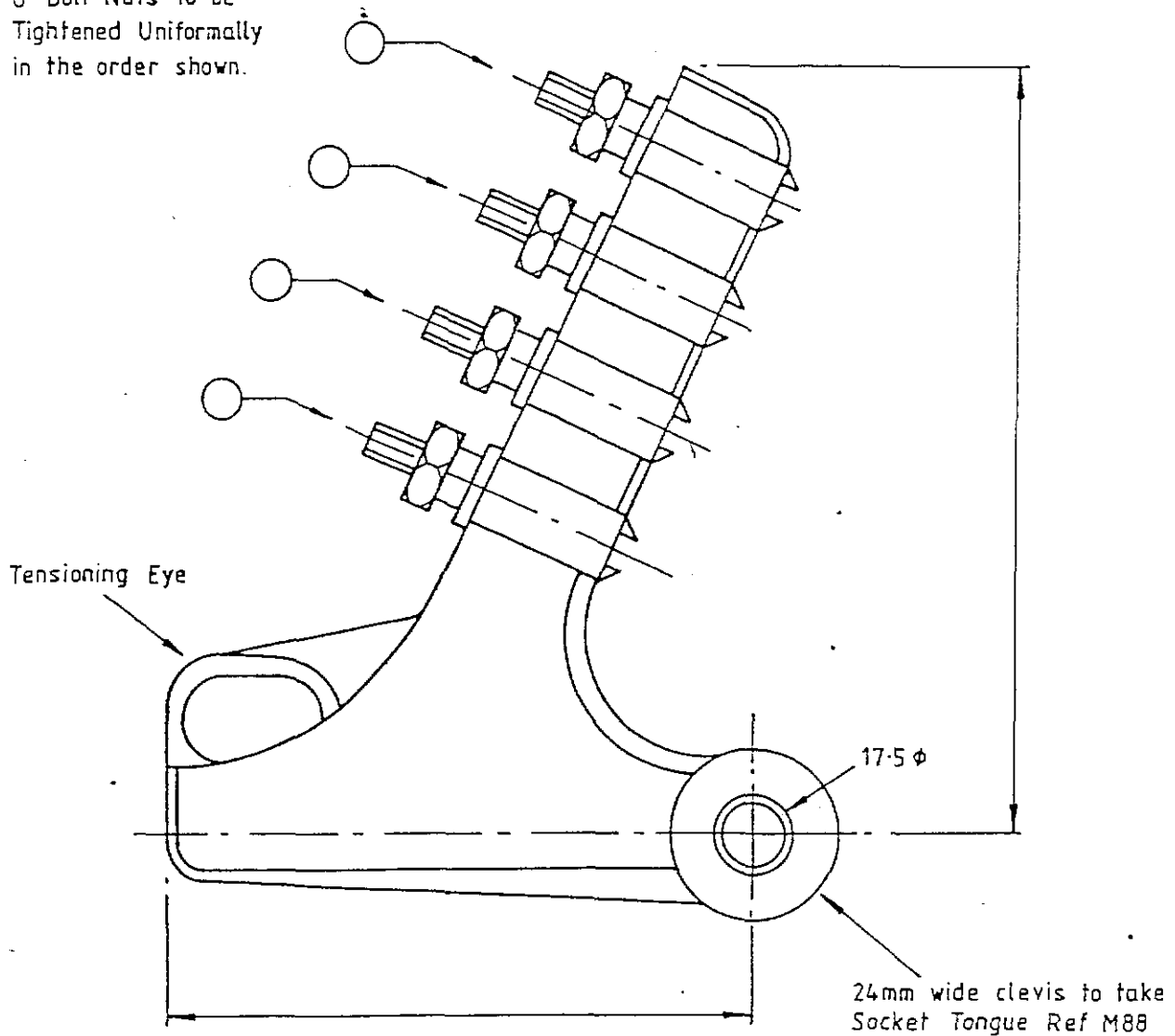
NOTES

- 1) Thimble to BS 464
- 2) Galvanized to BS 729
- 3) For Stay Terminating Thimbles see Drg. M115

SCALE		CONDUCTOR TERMINATING THIMBLE	ISSUE	MEB
DATE	21.11.88			
DRAWN				DRG. No. M461
TRACED	S.G.T.			
CHECKED				
APPROVED				

TIGHTENING SEQUENCE

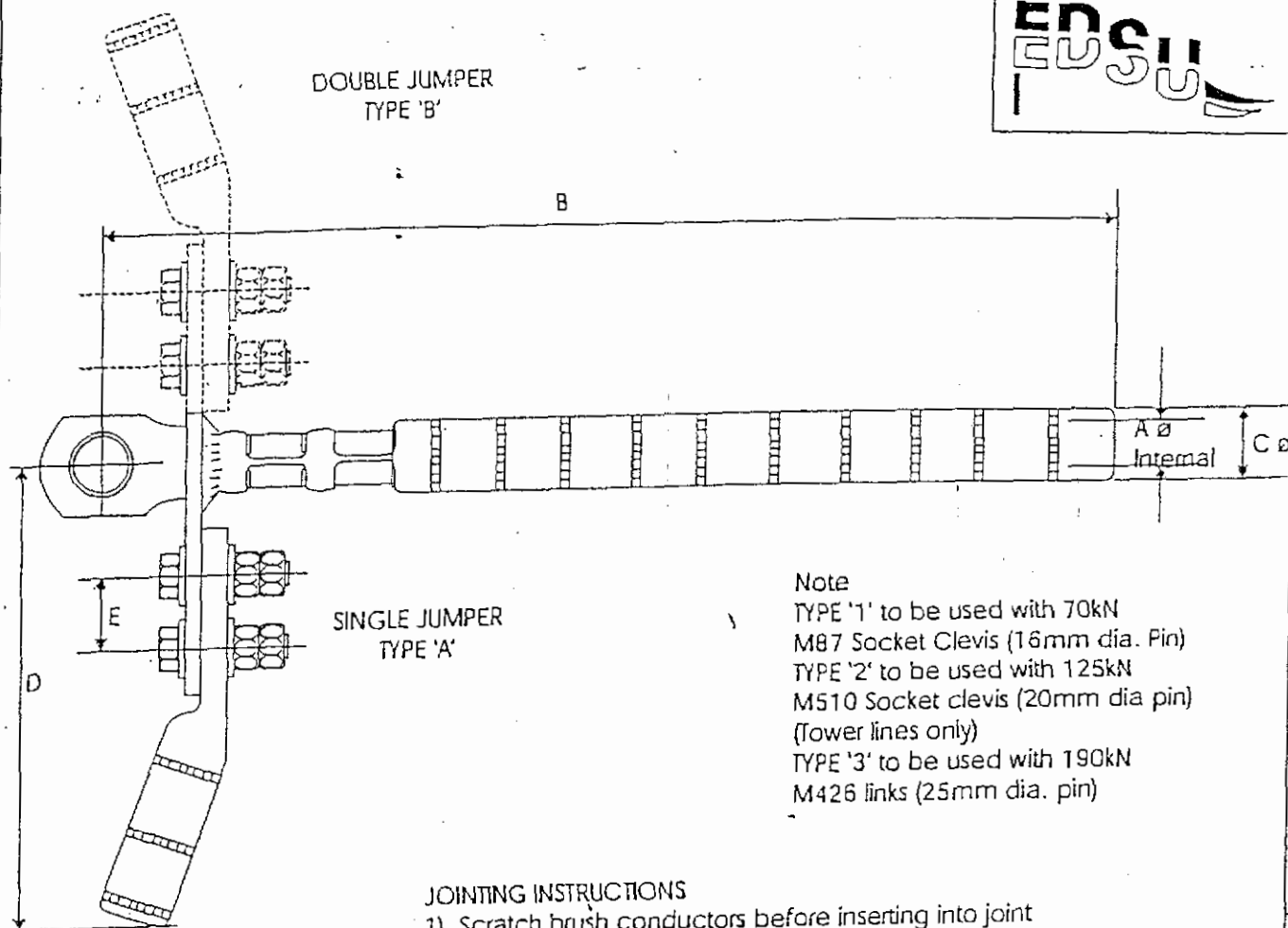
U-Bolt Nuts to be Tightened Uniformly in the order shown.



Notes:-

1. Recommended Installation Torque
2. U-Bolts to be Tightened Uniformly in the order shown Above.
3. Supplied with 52 x 16 φ Clevis Pin, Washer and Split Pin.

ADD'D BY DATE	TITLE BOLTED TERMINATING CLAMP FOR 200mm² AAAC CONDUCTORS	SCALE	—	 Midlands Electricity plc DRG No M 564
		DATE	5 - 10 - 90	
		DRAWN	A.J.B.	
		TRACED	G.S.	
		CHECKED		
		APPROVED		

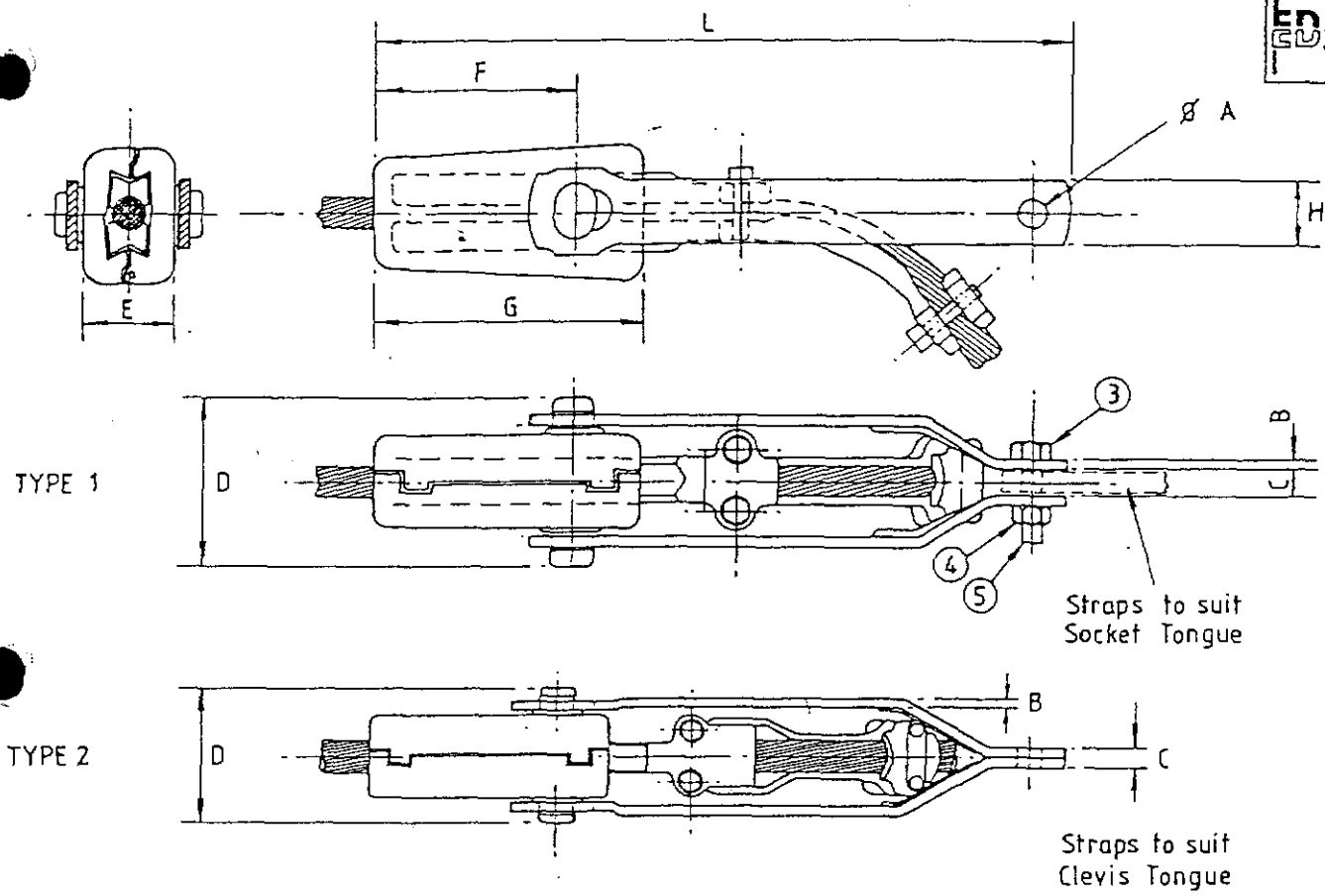


JOINTING INSTRUCTIONS

- 1) Scratch brush conductors before inserting into joint
- 2) Insert conductor fully into joint
- 3) Compress from inner position of joint and lug
- 4) Scratch brush and grease faces of jumper lug and flag
- 5) Fully tighten jumper lug bolts.

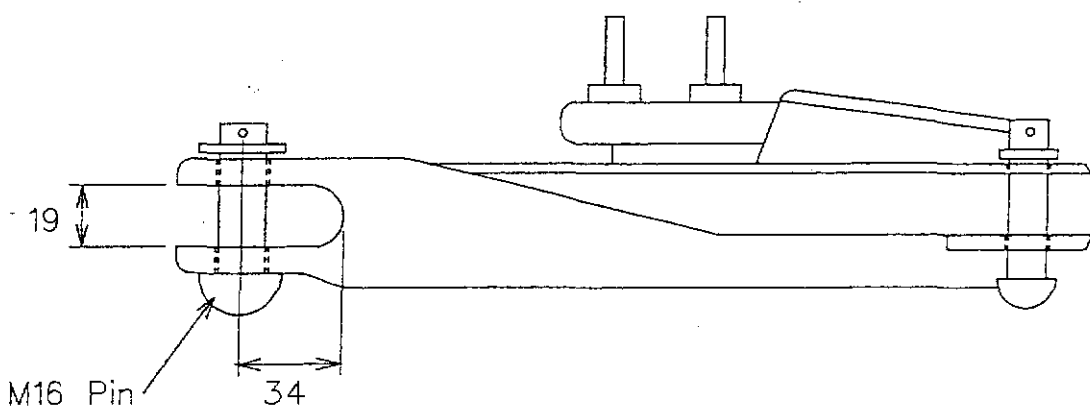
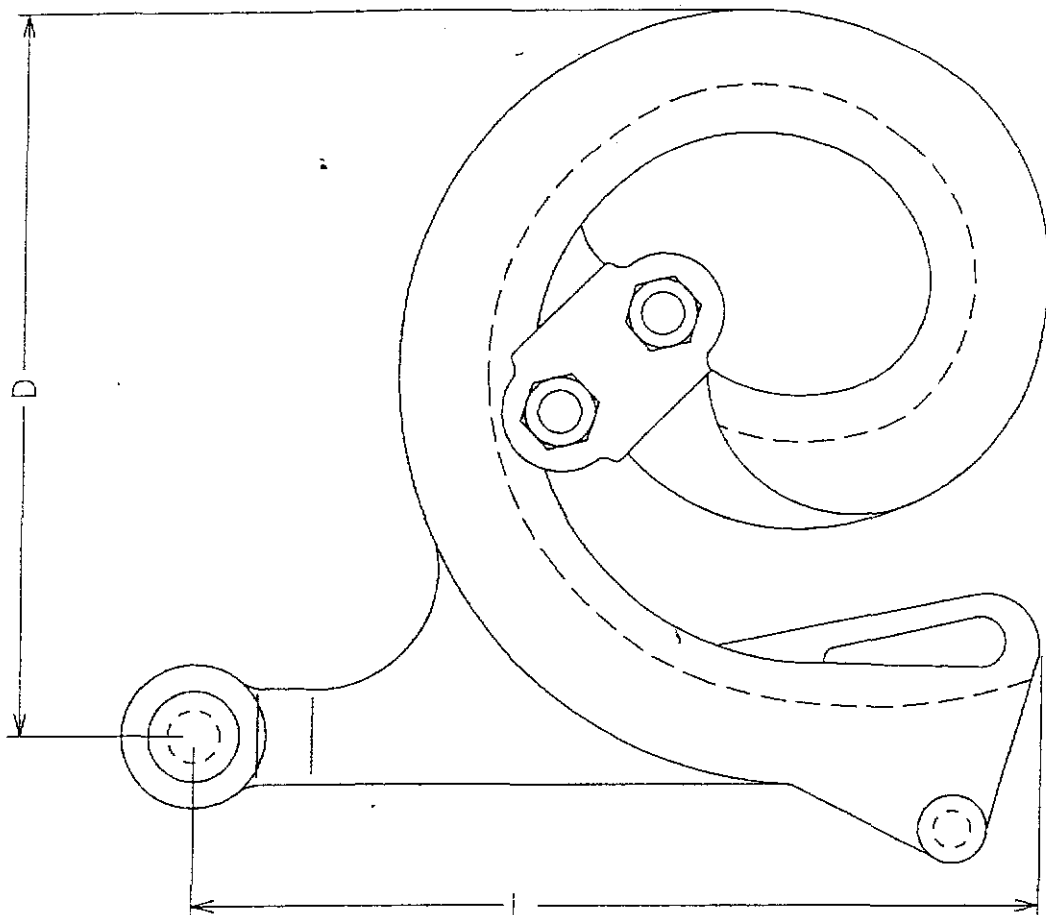
MIN	CONDUCTOR TYPE & SIZE	TONGUE JUMPER TYPE	DIMENSIONS					BOLT SIZE	DIE REF
			A	B	C	D	E		
M566/0	50mm ² AAAC	1/B						M10 X35	U659
M566/1	50mm ² AAAC	1/A						M10 X 35	U659
M566/2	100mm ² AAAC	1/A						M16 X 65	U654
M566/3	200mm ² AAAC	1/A						M16 X 65	A 9
M566/4	200mm ² AAAC	2/A						M16 X 65	A 9
M566/5	300mm ² AAAC	2/A						M16 X 65	A 10
M566/6	500mm ² AAAC	3/A						M16 X 70	A 12
M566/7	200mm ² AAAC	1/B						M16 X 65	A 9
M566/8	200mm ² AAAC	2/B						M16 X 65	A 9
M566/9	300mm ² AAAC	2/B						M16 X 65	A 10

ADDED BY DATE	TITLE CRIMPET DEAD END UNIT TYPE FOR AAAC REMOVABLE JUMPER	SCALE		 Midlands Electricity plc ORG No M566
		DATE	11/11/99	
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		



Ref	Conductor Size & Type	Type	Dimensions (mm)									Clamp Bolt Torque	Weight Kg
			A	B	C	D	E	F	G	H	L		
568/1	200 AAAC 37/2-87	1	18		18							Nm	
568/2	200 AAAC 37/2-87	1	21.5		27							Nm	
568/3	300 AAAC 37/3-53	1	21.5		27							Nm	
568/4	500 AAAC 61/3-50	1	27		32							Nm	

Wedge Clamp to comply with TPS 1/132
 Hot Dip Galvanized to BS729 (60 micron absolute min.)
 Clamp fitting instructions see drg. M1 4001 or M1 4002
 Notes 568/1 use 70KN Socket Tongue M88
 568/2 & 568/3 use 125KN Socket Tongue M251/1
 568/4 use 190KN Socket Tongue M435



MIN	L	D	Conductor	Bolt Torque	Notes
3457010	273	238	115mm (.15) Cadmium Copper 135mm (.175) Cadmium Copper 125mm (.2) H.D. Copper 160mm (.25) H.D. Copper	42Nm	Phase cond. SBR
3457011	230	170	85mm (.19) (.093) Cadmium Copper	52Nm	Earthwire SBR

C.A.D. PRODUCED

SCALE	
DATE	20.11.98
DRAWN	L.V.L.B.
TRACED	
CHECKED	N.R.W.
APPROVED	R.C.F.

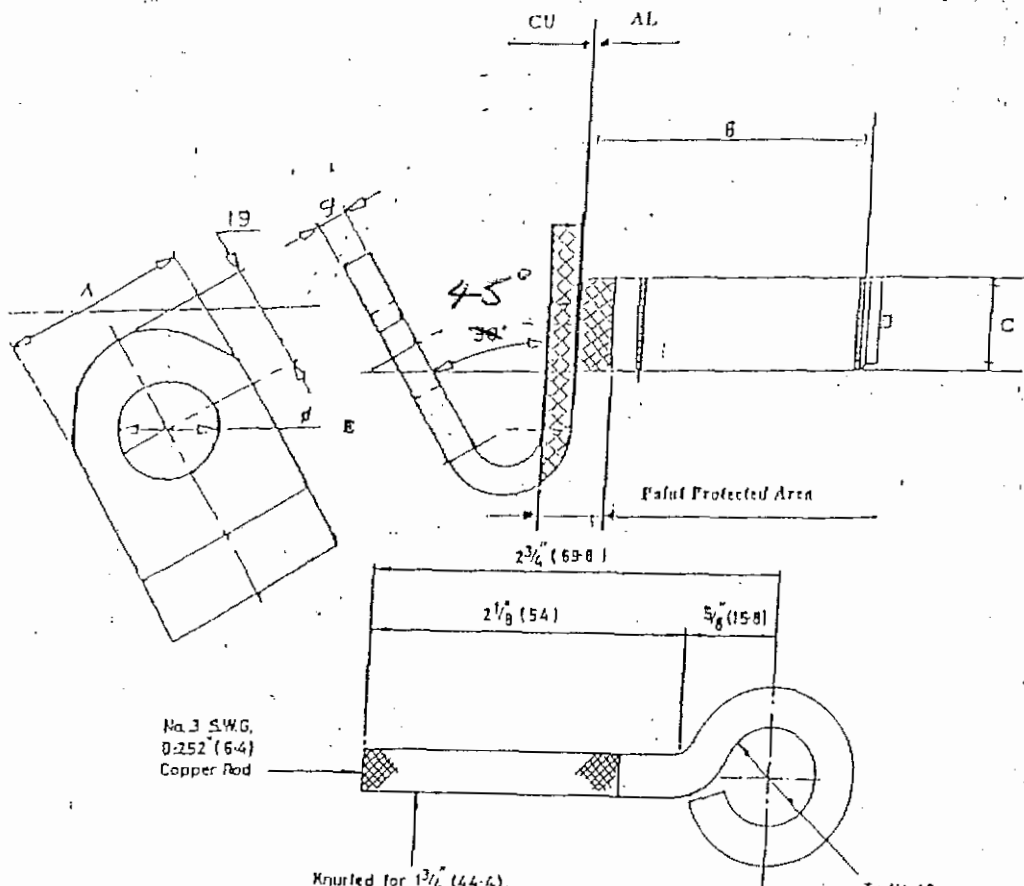
CADMIUM COPPER
110kN SNAIL CLAMP

ISSUE	
DRG. No.	M701

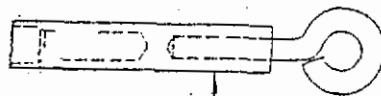
MEB
Islands Electricity plc

SECTION 7

MIN NUMBER	M	DESCRIPTION
3451950	195	SINGLE HOLE BI-METAL LUG
3452190	219	T JOINT WITH JUMPER LUGS
3452360	236	SINGLE HOLE COPPER LUG
3452370	237	SINGLE HOLE COPPER LUG
3452380	238	TWO HOLE COPPER LUG
3452730	273	CRIMPET FOR COPPER CONDUCTORS
345277	277	LINE TAP LUGS FOR FLEXIBLE 50mm AL
3452850	285	SINGLE HOLE H D ALUMINIUM LUG
3452950	295	LINE TAP LUGS FOR FLEXIBLE 35mm COPPER
3453020	302	CRIMPET COVERS
3453100	310	LUGS FOR AL ALLOY & ACSR
3453230	323	PARALLEL GROVE CLAMP
3453270	327	BI-METAL COMPRESSION TERMINAL PIN TYPE 175/400 ACSR
3453280	328	BI-METAL COMPRESSION TERMINAL STRAIGHT PALM 175/400 ACSR
3453290	329	BI-METAL COMPRESSION TERMINAL 30° PALM 175/400 ACSR
3453980	398	BI-METAL TERMINAL LUG
3453990	399	BI-METAL SPACER KIT



To fit 16 mm Stalk on
HV Bushing of Transformer
to B.E.S. T.I. or E.S.I. Std.
J5-1



Aluminium Sleeve
Compressed on to Copper Stem
by Manufacturer. The copper
stem to be coated with penetrox
'A' grease before compression

Type	Min	Conductor Size & Type	Manufacturers Ref	Dimensions					Y35 / MD6 Die Ref
195/1	3451951			A	B	C	D	E	
195/2	3451952			10.5		19		M12	
195/3	3451953							M16	W247
195/4	3451954							M16	W247
195/5	3451955							M16	

TITLE

Single Hole Bi-Metal Lug
Angled & Counter Board

SCALE

DATE

DRAWN

TRACED

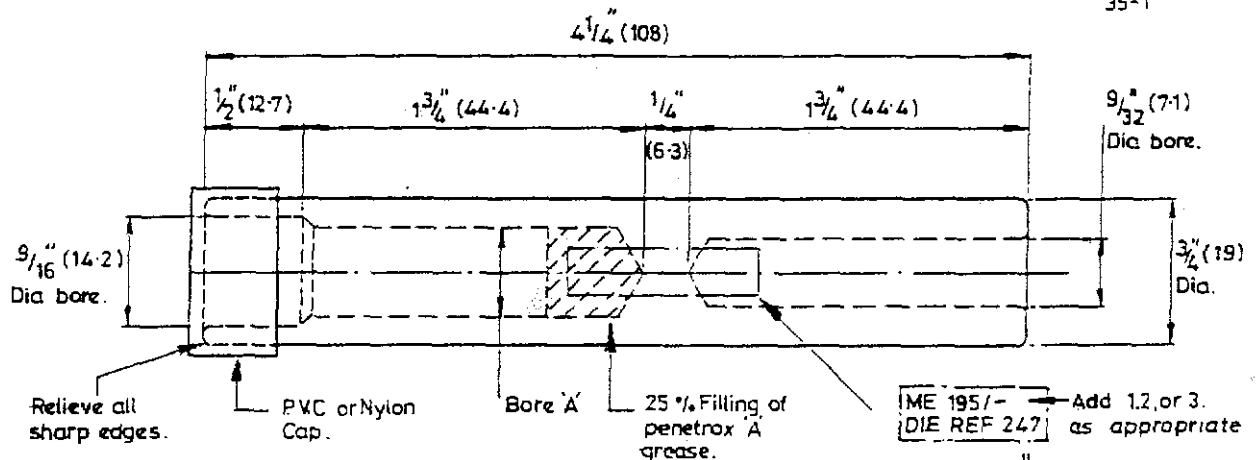
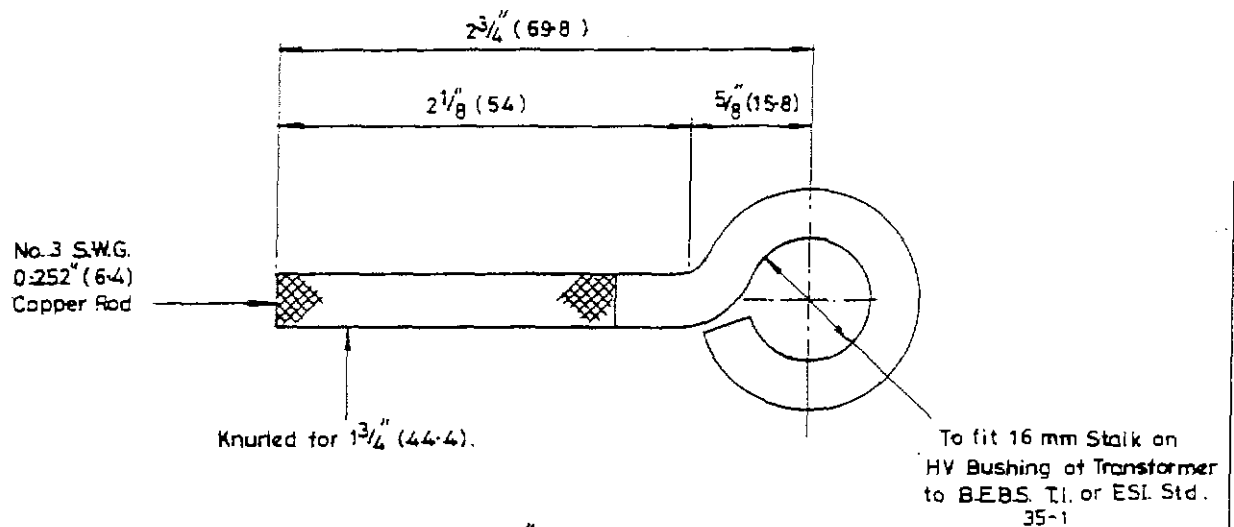
CHECKED

APPROVED

GPU
POWER UK

DWG No

M195



Aluminium Sleeve
Compressed on to Copper Stem
by Manufacturer. The copper
stem to be coated with penetrox
'A' grease before compression

METHOD OF FIXING

TYPE	CONDUCTOR	BORE 'A'	COMPRESSOR DIE DETAILS
195/2	0.050" (50) Al Alloy	13/32 (10.3)	BOWTHORPE 15TA DIE. H.T.C 3.
195/3	0.050" (50) Aluflex	7/16 (11.1)	BURNDY Y35 DIE. U247.

MATERIAL

BODY-- Fully Annealed 99.5 %
Purity Aluminium.

STEM No. 3 S.W.G. (6.4) HD HC Copper
Rod Standard Commercial
Finish.

Supplied c/w 2 Flat Brass Washers
to Drg ME 315/1

PREPARATION

Conductor and to be cleaned
and Greased immediately prior
to compressing.

MAKING OF JOINT

Remove the PVC. sheath for 1 3/4 (44.4)
and insert conductor so that insulation
enters to a depth of 1/2 (12.7). Compress
over the insulation when making the joint.

PROTECTION

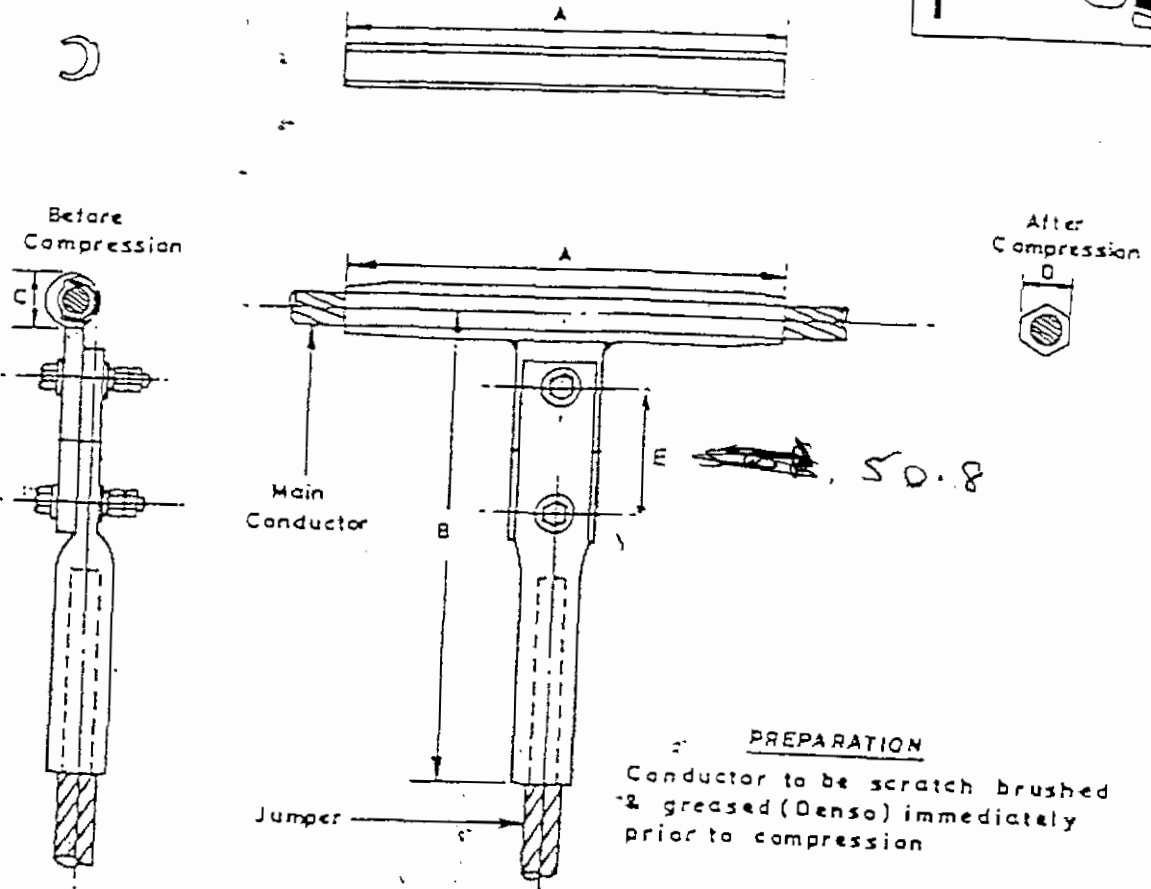
When installed the complete connector and
1/2 (13) of conductor to be coated with
Densol grease and Densol taped.

NOTE Protection for bare 0.05 in² Al. Alloy
to be the same as for Aluflex.

SCALE	1/1
DATE	13.3.70.
DRAWN	T.R.P.
TRACED	
CHECKED	
APPROVED	

NON-TENSION BI-METALIC COMPRESSION JOINT (RING END)

ISSUE	MIDLANDS ELECTRICITY BOARD
K	
Type withdrawn Die Ref. revision	DRG. No. ME. 195.



M.E TYPE	CONDUCTORS				DIMENSIONS						DIE REF
					Before Compression			After Comp	E	BOLT SIZE	
	MAIN	JUMPER			JOINT	JUMPER	A				
M.E 219/1	100 ACSR 100 AAAC	SIZE TO BE SPECIFIED WHEN ORDERING									
M.E 219/2	150 ACSR										
M.E 219/3	175 ACSR 200 AAAC										M12 A - 8
M.E 219/4	425 AAAC 400 ACSR 400 AAC										M16 A - 9
M.E 219/8	300 AAAC										M16 A - 11
M.E 219/9	500 AAAC										M16 A - 10

TITLE

COMPRESSION 'T' JOINTS
WITH JUMPER LUGS

SCALE

DATE

DRAWN

TRACED

CHECKED

APPROVED

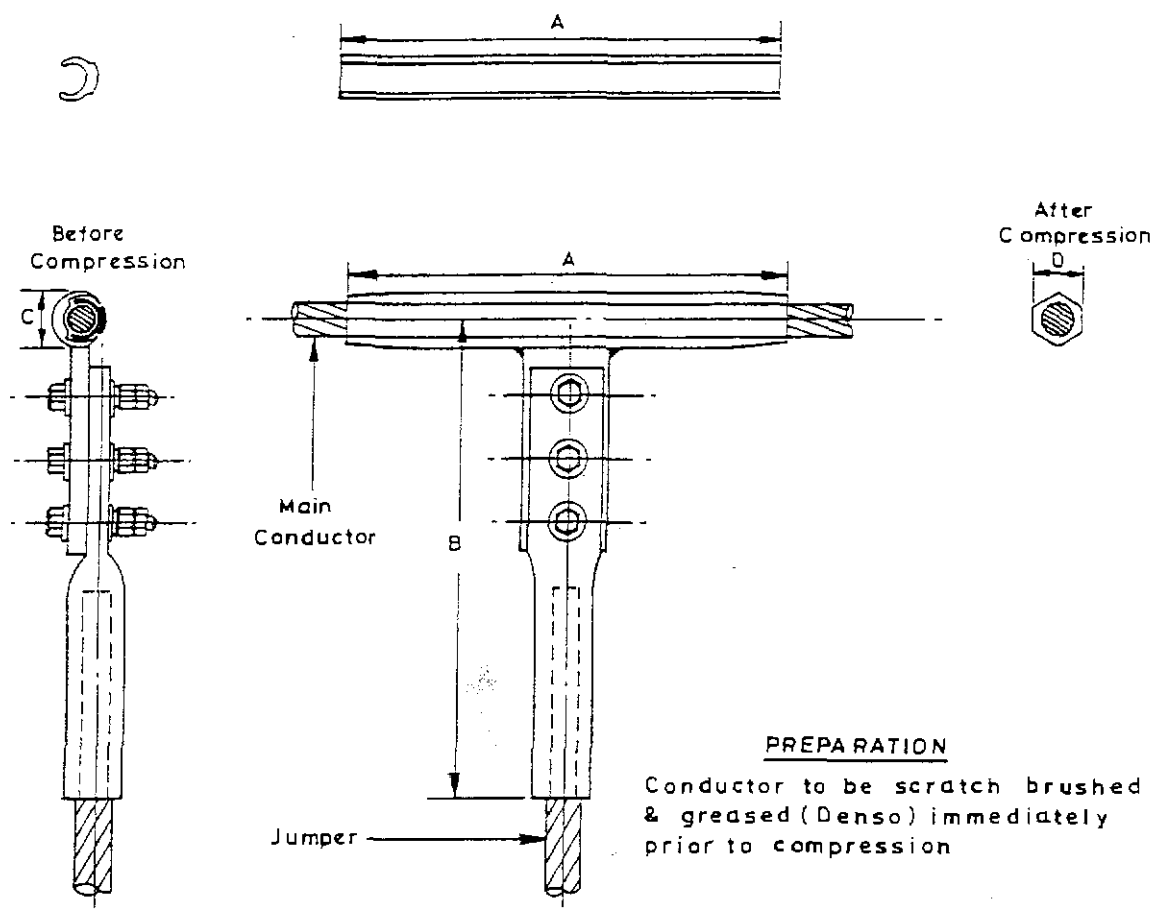
11/11/99

ORG No

MEB

Midlands Electricity plc

M.E. 219

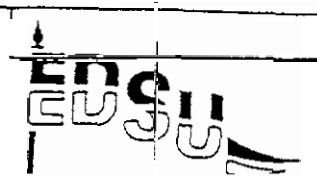
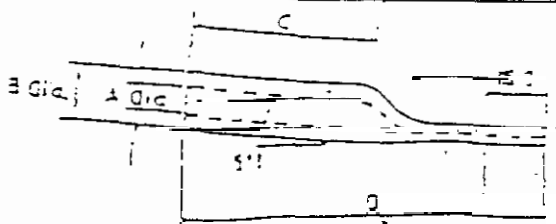


M.E. TYPE	CONDUCTORS		ALCAN CAT. Nos.		DIMENSIONS				COMPRESSOR DIE No.
					Before Compression			After Comp.	
	MAIN	JUMPER	JOINT*	JUMPER†	A	B	C	D	
M.E. 219/1	•1SCA (100mm ²)	SIZE TO BE SPECIFIED WHEN ORDERING	TJ-7-1	SAT-7	17" (432)	11 ⁵ / ₈ " (295)	1-187" (30-15)	1-001" (25-43)	4A-7
M.E. 219/2	•15SCA (150mm ²)		TJ-8-1	SAT-8	20" (508)	12 ⁵ / ₁₆ " (313)	1-312" (33-32)	1-110" (28-19)	4A-8
M.E. 219/3	•175SCA (175mm ²)		TJ-9-1	SAT-9	21 ¹ / ₂ " (546)	14 ¹ / ₈ " (359)	1-500" (38-1)	1-273" (32-33)	4A-9
M.E. 219/4	•4 Al (400mm ²)		TJ-10	SAT-10	24" (610)	15 ⁵ / ₈ " (397)	1-687" (42-85)	1-426" (36-22)	4A-10
M.E. 219/5	•4SCA (400mm ²)		TJ-11	SAT-11	25 ¹ / ₂ " (648)	16 ³ / ₁₆ " (411)	1-875" (47-63)	1-583" (40-21)	4A-11

NOTE Catalogue No. * covers Tee connector body only. Jumper terminals must be order separately under Cat. No. †

SCALE		COMPRESSION 'T' JOINT FOR S.C.A. CONDUCTORS.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	28-1-63		C	
DRAWN	A.J.B.		Redrawn & metricated.	DRG. No.
TRACED	P.G.			M.E. 219
CHECKED				
APPROVED	W.D.H.		A.R.B. 8/76	

GAF 9738



REQUIREMENTS

- 1 / Surface area required for electrical contact
- 2 / Mechanical strength of side wall
- M236/4 & 5 may require to be cast in order to meet requirements.

- NOTES
- A. The lugs must be manufactured from Electrolytic Tough Pitch H.C. Copper conforming to BS 1977 and BS 2875 C101
 - B. Each fitting to have stamped on it the conductor size & stranding
 - C. Tolerances on the Tubes as drawn
 - D. To comply with BS 3233

TYPE No	CONDUCTOR		A	B	C	D	E Min	F Dia
	STRANDING	AREA SQ						
236/0	7 / 4.9	125mm ²						M20
236/1	7 / 4.9	125mm ²						M16
236/2	7 / 4.9	125mm ²						M12
236/4	7 / 107	16mm ²						M16
236/5	7 / 107	16mm ²						M12
236/7	3 / 3.73 7 / 2.5	32&35mm ²						M16
236/8	3 / 3.73 7 / 2.5	32&35mm ²						M12
236/9	19 / 2.1	70mm ²						M10

TYPE	COMPRESSION DIE DETAILS	
	MH 100 (MD6)	G10 (Y35)
236/0	Not to be used	U169
236/1	Not to be used	U169
236/2	Not to be used	U169
236/4	W162	U162
236/5	W162	U162
236/7	W163	U163
236/8	W163	U163
236/9	Not to be used	U166

TITLE

COMPRESSION LUG
FOR COPPER CONDUCTOR
SINGLE HOLE FIXING

SCALE _____

DATE 11/11/99

DRAWN _____

TRACED _____

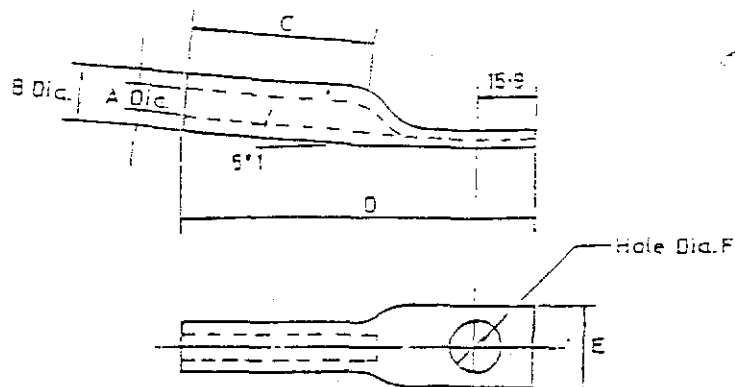
CHECKED _____

APPROVED _____

MEB
Midlands Electricity plc

M:236

ADD BY
DATE



TYPE No	CONDUCTOR		A	B	C	D	E Min	F Dia	BICCON CATALOGUE NUMBER
	STRANDING	AREA SQ IN (mm ²)							
23710	7/136"	.10 (70mm ²)	11.9	19.05	51	102	26	14.3	L7C12 (YCA 258)
23711	3/104"	.025 (16mm ²)	5.94	14.30	31.8	76.2	20.3	15.9	L2C16 (YCA5C3-62)
23712	3/147"	.05 (32mm ²)	8.71	14.30	44.5	88.9	20.3	15.9	L4C16 (YCA2C5-62)
	7/104"	.058 (35mm ²)							
23713	7/136"	.10 (70mm ²)	11.89	19.05	50.8	101.6	25.9	15.9	L7C16 (YCA 258-1)
23714	3/104"	.025 (16mm ²)	5.94	14.30	31.8	76.2	20.3	9.5	L2C10 (YCA5C3-37)
23715	3/147"	.05 (32mm ²)	8.71	14.30	44.5	88.9	20.3	9.5	L4C10 (YCA 2C5-37)
	7/104"	.058 (35mm ²)							
23716	7/166"	.15 (100mm ²)	13.87	22.22	34.9	100	30.2	15.9	L8C16 (YCA 275-1)
23717	7/193"	.2 (125mm ²)	14.86	23.83	61.9	109.5	33.3	15.9	L9C16 (YCA 291-1)
23718	7/116"	.075 (50mm ²)	9.52	14.47	37.7	76.2	23.0	14.3	L6C12 (YCA 244)
23719	7/136"	.10 (70mm ²)	11.9	19.05	51	102	26	21	L7C20 (YCA 258-3)

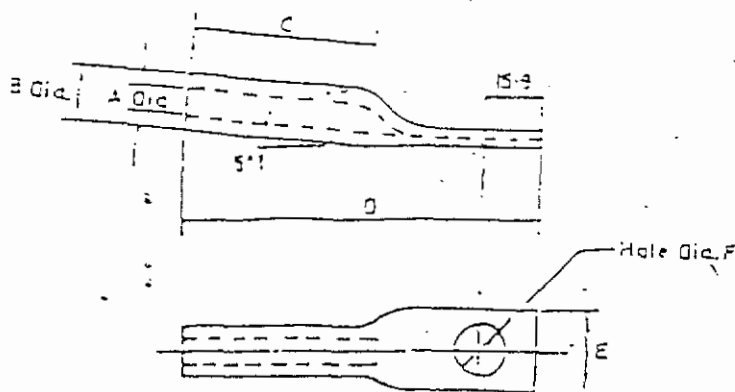
- NOTES
- The lugs must be manufactured from Electrolytic Tough Pitch H.C. Copper conforming to B.S.1977 and B.S.2871.C101.
 - Each fitting to have stamped on it the conductor size & stranding
 - Tolerances on the Tubes as drawn are $\pm .004"$ (.10) on the O.D. and $\pm .003"$ (.076) on the I.D. Machine tolerances to be $\pm .002"$ (.051)
 - To comply with B.S.3288
 - Biccon Old and Equivalent Number Shown

TYPE	COMPRESSION DIE DETAILS	
	MH100 (MD6)	G10 (Y35)
23710		U-166
23711	W-162	U-162
23712	W-163	U-163
23713		U-166
23714	W-162	U-162
23715	W-163	U-163
23716		U-168
23717		U-169
23718	W-164	U-164
23719		U-166

SCALE	N.T.S.
DATE	6.10.63
DRAWN	R.A.C.
TRACED	H.P.S.
CHECKED	
APPROVED	J.D.H.

COMPRESSION LUG
FOR COPPER CONDUCTOR
SINGLE HOLE FIXING

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
	A	
Biccon Ref. Rev.		DRG. No
4.84	1W	M 237

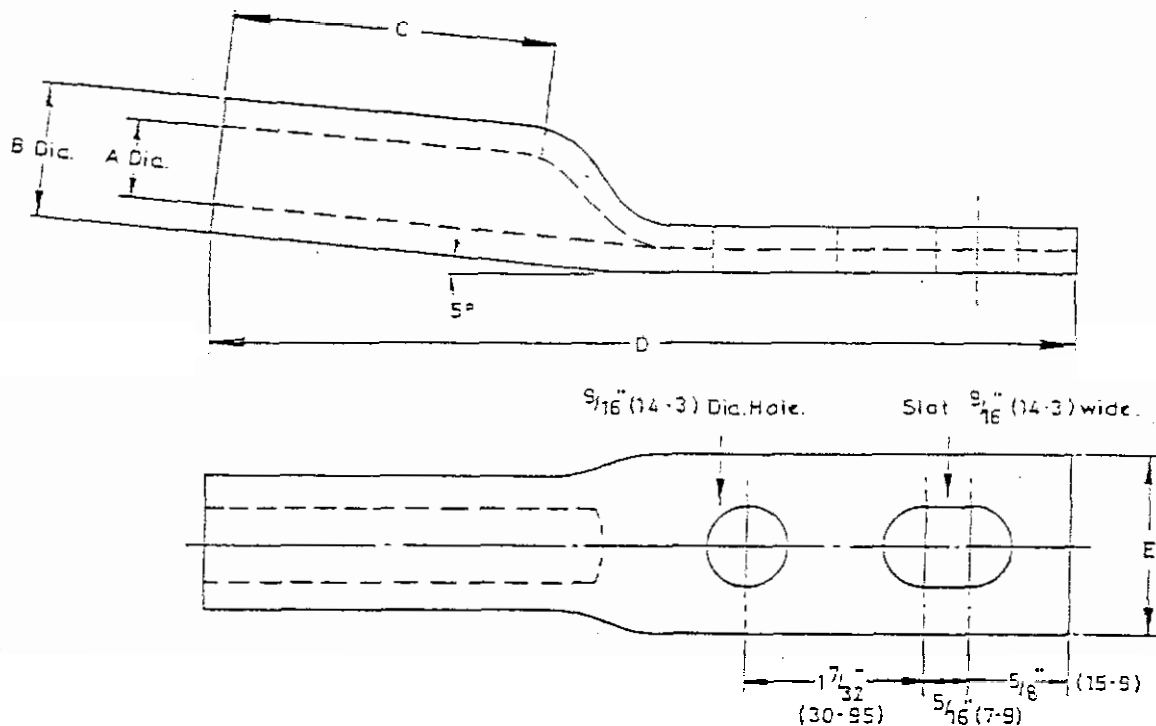


- NOTES
- The lugs must be manufactured from Electrolytic Tough Pitch H.C. Copper conforming to B.S.1377 and B.S.2371 C101
 - Each fitting to have stamped on it the conductor size & stranding
 - Tolerances on the Tubes as drawn
 - To comply with B.S.3233

TYPE No	CONDUCTOR		A	B	C	D	E Min	F Dia	
	STRANDING	AREA SQ in (mm ²)							
237/0	7/.136in 7/3.45mm	.10 (70mm ²)						14.3	
237/3	7/.136in 7/3.45mm	.10 (70mm ²)						15.9	
237/6	7/.166in 7/4.22mm	.15 (100mm ²)						15.9	
237/8	7/.116in 7/2.95mm	.075 (50mm ²)						14.3	
237/9	7/.136in 7/3.45mm	.10 (70mm ²)						21	

TYPE	COMPRESSION DIE DETAILS	
	MH 100 (MD6)	G10 (Y35)
237/0	Not to be used	U - 166
237/3	Not to be used	U - 166
237/6	Not to be used	U - 168
237/8	W - 164	U - 164
237/9	Not to be used	U - 166

DATE 11/11/99 TIME 10:00	TITLE COMPRESSION LUG FOR COPPER CONDUCTOR SINGLE HOLE FIXING	SCALE		MEB Midlands Electricity plc ORG No M.237
		DATE	11/11/99	
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		



TYPE No.	CONDUCTOR		A	B	C	D	E Min.	BICCON CATALOGUE NUMBER
	STRANDING	AREA SQ. IN.						
38/1	7/136"	10 (70 mm ²)	.468" (11.89)	.750" (19.05)	2" (50.8)	5 1/2" (139.7)	1.02" (25.9)	L7C12D (YCA 258-2B)
38/2	7/166"	15 (100 mm ²)	.546" (13.87)	.938" (23.82)	2 1/2" (63.5)	6" (152.4)	1.25" (31.8)	L8C12D (YCA 275-2B)
38/3	7/193"	20 (125 mm ²)	.609" (15.47)	.938" (23.82)	3" (76.2)	6 1/2" (165.1)	1.37" (34.8)	L9C12D (YCA 291-2B)
238/4	7/116"	7.5 (50 mm ²)	.552" (14.01)	.447" (11.37)	3 1/2" (89.1)	13 1/8" (336.5)	2.5" (63.5)	L6C12D (YCA 244-2B)

- NOTES**
- The lugs must be manufactured from Electrolytic Tough Pitch H.C. Copper conforming to B.S. 1977 and B.S. 2871: C101.
 - Each fitting to have stamped on it the conductor size & stranding.
 - Tolerances on the Tubes as drawn are $\pm .004"$ (-10) on the O.D. and $\pm .003"$ (-0.076) on the I.D. Machine tolerances to be $\pm .002"$ (-0.051).
 - The Joint after compression must meet the requirements of BS 3288 Pt 1.
 - Biccon Old and Equivalent New Numbers Shown

TYPE	COMPRESSOR DIE DETAILS	
	G10 (Y35)	MH100 (MD6)
238 / 1	U-166	-
238 / 2	U-168	-
238 / 3	U-169	-
238 / 4	U-164	W 164

SCALE	N.T.S.
DATE	19.5.65
DRAWN	W.D.H.
TRACED	P.G.
CHECKED	
APPROVED	GAF

COMPRESSION LUG
FOR COPPER CONDUCTOR
TWO HOLE FIXING.

ISSUE	MIDLANDS ELECTRICITY BOARD
C	DRG. No.
Biccon Ref. Revised.	M.E. 238.
4.84. J.W.	

MF MF

MDO 15428 E		ES1 STANDARD 63-92		ES1 DRG N°	
APPROVED BY DATE	CRIMPET RECOMMENDATIONS FOR COPPER CONDUCTORS				
	SCALE	DATE	DRAWN	TRACED	CHECKED
	16-5-96				
TITLE		MIDLANDS ELECTRICITY PLC			
		M273			

MIN	No.	Compression Dia/Number of Crimps	Fig.	Type	Manufacturer	Reference	Length	Height	Width	Gap
273/0		U-E	2							
273/1		W-BG	2							
273/2		W-C	2							
273/3		W-C	2							
273/4		U-E	2							
273/5		U-F	2							
273/6		U-F	2							
273/7		U-F	2							
273/8		U-F	2							
273/9		U-F	2							



CONDUCTOR	Type & Size (mm²)	Earth Rod	SEE DRAWING NO. M D 72	14	18	25	35	50	70	100	125	18	25	35	50	70	120
H	14	18	25	35	50	70	100	125	18	25	35	50	70	100	125	18	25
D	14	18	25	35	50	70	100	125	18	25	35	50	70	100	125	18	25
A	14	18	25	35	50	70	100	125	18	25	35	50	70	100	125	18	25
N	14	18	25	35	50	70	100	125	18	25	35	50	70	100	125	18	25
E	14	18	25	35	50	70	100	125	18	25	35	50	70	100	125	18	25
L	14	18	25	35	50	70	100	125	18	25	35	50	70	100	125	18	25
D	14	18	25	35	50	70	100	125	18	25	35	50	70	100	125	18	25



CONDUCTOR		ALL ALUMINIUM ALLOY				A C S R						H. D. AL			Hybrid Core	
Type & Size (mm)		25	50	100	200	25	40	50	100	150	175	25	50	100	25	35
Bar	J/O swg	SEE DRAWING No. MD 72														
A	25	274/0	274/1	274/8	274/8	274/0	274/1	274/1	274/8	274/9	274/8					
A	50	274/1	274/2	274/4	274/8	274/1	274/2	274/2	274/4	274/9	274/8					
A	100	274/8	274/4	274/7	174/1	274/8	274/4	274/4	274/7	174/1	174/1					
C	200	274/8	274/8	174/1	174/1	274/8	274/8	274/8	174/1	174/1	174/1					
A	25	274/0	274/1	274/8	274/8	274/0	274/1	274/1	274/8	274/9	274/8					
A	40	274/1	274/2	274/4	274/8	274/1	274/2	274/2	274/4	274/9	274/8					
C	50	274/1	274/2	274/4	274/8	274/1	274/2	274/2	274/4	274/9	274/8					
S	100	274/8	274/4	274/7	174/1	274/8	274/4	274/4	274/7	174/1	174/1					
R	150	274/9	274/9	174/1	174/1	274/9	274/9	274/9	174/1	174/1	174/1					
	175	274/8	274/8	174/1	174/1	274/8	274/8	274/8	174/1	174/1	174/1					
A	25								274/0	274/1	274/5	274/0	274/0			
A	50								274/1	274/2	274/4	274/1	274/1			
C	100								274/5	274/4	274/7	274/8	274/8			

FIG 3
CARLOCK
CRIMPET

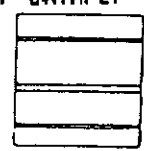
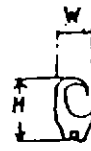
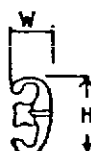
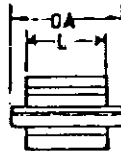


FIG 4 UNIVERSAL
CRIMPET

FIG 5
H-CRIMPET

MIN No.	Compression Die/Number of Crimps		G10/Y35		Fig. Type	Manufacturers Reference	Dimension (mm)			
	MH100/MD8	No	No	No			O/A	Length	Height	Width
274/0	W - 0	4		U - 0	1	3				
274/1	W - 0	4		U - 0	1	3				
274/2	W - D3	4		U - D3	1	3				
274/3	W - D3	4		U - D3	1	3				
274/4				U - D3	2	3				
274/5				U - D3	2	4				
274/6				U - N	2	4				
274/7				U - D3	2	4				
274/8				U - D3	1	3				
274/9				U - N	2	4				
174/1				U - N	3	5				

TITLE

CRIMPET RECOMMENDATIONS
FOR ALUMINIUM CONDUCTORS

SCALE

DATE

DRAWN

TRACED

CHECKED

APPROVED

16-5-94

C. E.

MEB

Midlands Electricity plc

DRG No

M274 / 174

AMDO 15628 E

ESI STANDARD 43-92

ESI DRG No

CRIMPET RECOMMENDATIONS FOR HARD DRAWN AND ANNEALED COPPER CONDUCTORS.

CONDUCTOR		HARD DRAWN COPPER (mm ²)								ANNEALED COPPER (mm ²)						
TYPE & SIZE (mm ²)	3/0 swg	14	16	22	25	50	70	100	125	16	25	35	70	95	120	
HARD	3/0 swg	SEE	DRG	N°	MD	72										
14					273/2	273/2		273/0	-							
16	DRG			273/1	273/3	273/3	273/0	273/5	273/5	273/1	273/2	273/2	273/0			
22				273/2	273/3	-	-	-	273/0	273/5	273/6	273/2	273/3	273/3	273/0	273/8
25	NO	-	-	-	-	-	-	273/5	273/6	273/2	273/3	273/0	273/5			
50		-	-	273/0	273/0	273/0	273/4					273/0	273/4		273/8	
70	MD 72	273/0	-	-	-	-	273/4	273/6	273/7	273/0	273/0	273/4	273/6		273/8	
100		-	-	273/5	273/5	273/5	273/6	273/7	-							
125		-	-	273/5	273/6	273/6	273/7	-	273/9						273/9	
ANNEALED	16	-	-	273/1	273/2	273/2		273/0								
25	-	-	-	273/2	273/3	273/3		273/0								
35	-	-	-	273/2	273/3	273/0	273/0	273/4								
70	-	-	-	270/0	273/0	273/5	273/4	273/6								
95	-	-	-													
120	-	-	-		273/8		273/8	273/8	273/9							

CRIMPET RECOMMENDATIONS FOR AL ALLOY ACSR CONDUCTORS

CONDUCTOR	3/0	25AA	50AA	100	150	175
TYPE & SIZE (mm ²)	SOLID	25ACSR	50ACSR	ACSR	ACSR	ACSR
3/0 SOLID	SEE	DRG	N°	MD 72	-	274/6
25 AA or ACSR	DRG	274/0	274/1	274/8	274/9	274/6
50 AA 40x50 ACSR	N°	274/1	274/2	274/4	-	-
100 ACSR	N° 72	274/8	274/4	274/7	-	-
150	-	274/9	274/9	-	-	-
175	-	274/6	274/6	-	-	-

CRIMPET RECOMMENDATIONS FOR HARD DRAWN ALUMINIUM

CONDUCTOR SIZE mm ²	25	50	100
25	274/0	274/1	274/5
50	274/1	274/2	274/4
100	274/5	274/4	274/7

YC COPPER
CRIMPET

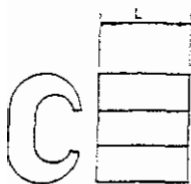


FIG 1

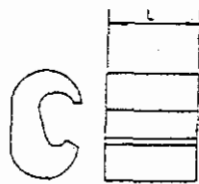


FIG 2

VP CABELOCK
CRIMPET

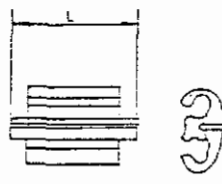


FIG 3



YPC UNIVERSAL
CRIMPET

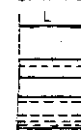


FIG 4

ME REF.	BICCON REF	FIG.	O/A LENGTH	DIE REF	N° OF	CRIMPS
273/0	TC 8C (YC 26 C2)	2	27	U-E	-	3
1	TC 5C (YC 4 C4)	1	19	U-BG	2	1
2	TC 6C (YC 2 C4)	2	22	U-C	2	1
3	TC 7C (YC 2 C2)	1	27	U-C	2	1
4	TC 9C (YC 26 C26)	1	27	U-E	-	3
5	TC 10C (YC 26 C2)	2	30	U-F	-	3
6	TC 11C (YC 26 C26)	2	30	U-F	-	3
7	TC 12C (YC 26 C28)	1	30	U-F	-	3
8	TC 13C (YC 26 C29)	1	30	U-708	-	4
9	TC 14C (YC 29 C291)	1	52	U-708	-	6
274/0	TC 1A (YP 2 U3)	3	38	U-O	4	1
1	TC 2A (YP 26 AU2)	3	41	U-O	4	1
2	TC 3A (YP 25 U25)	3	41	U-O3	4	1
3	TC 4A (YP 27 AU4)	3	41	U-O3	4	1
4	TC 8A (YP 26 U26)	3	89	U-O3	9	2
5	TC U1A (YPC 26 U4)	4	41	U-O3	4	1
6	TC U6A (YPC 38R 26U)	4	54	U-N	-	2
7	TC U2A (YPC 28U 28)	4		U-O3	-	2
8	TC 7A (YP 26 U2)	3		U-O3	4	1
9	TC U3A (YPC 39R 26U)	4	54	U-N	-	2

NOTE

CABELOCK OR UNIVERSAL CRIMPETS
SHOULD NOT BE USED FOR
BI-METALLIC CONNECTIONS

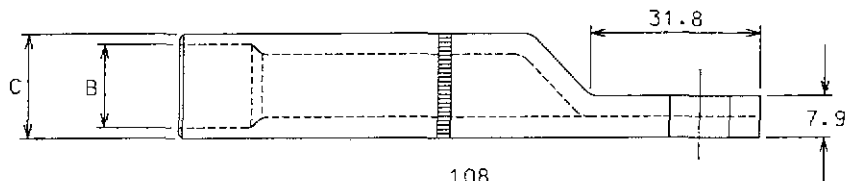
SCALE	1
DATE	11.5.8.79
DRAWN	RAC
TRACED	SGT
CHECKED	A
APPROVED	

COPPER AND ALUMINIUM
CRIMPET RECOMMENDATIONS.

ISSUE	ISSUE
G	F MF
Biccon Ref. revised.	ME REF N° 2
4.86.	BICC N° Modified
	R.A.C. 17-6-81

MIDLANDS
ELECTRICITY BOARD
ME 273/4

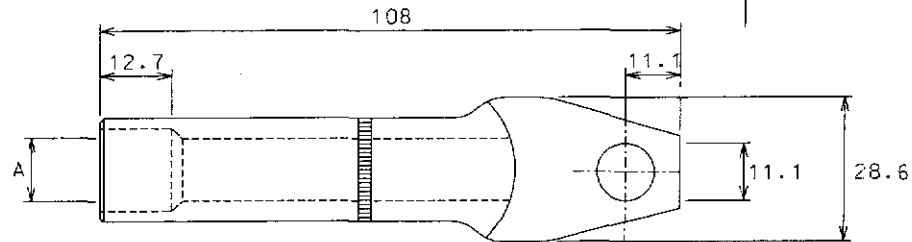
M277/2



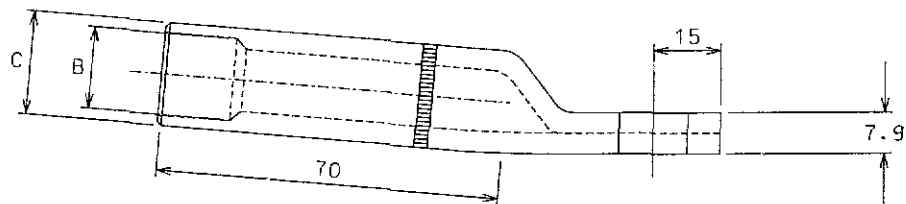
L/L TAPPING LUG

MATERIAL

Aluminium Tube
or Bar.
99.5% min. purity.



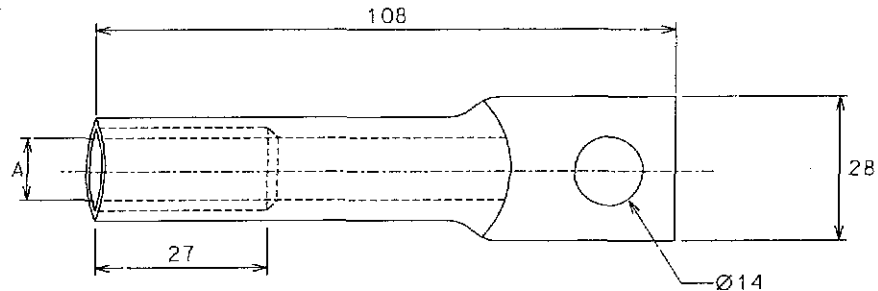
M277/3



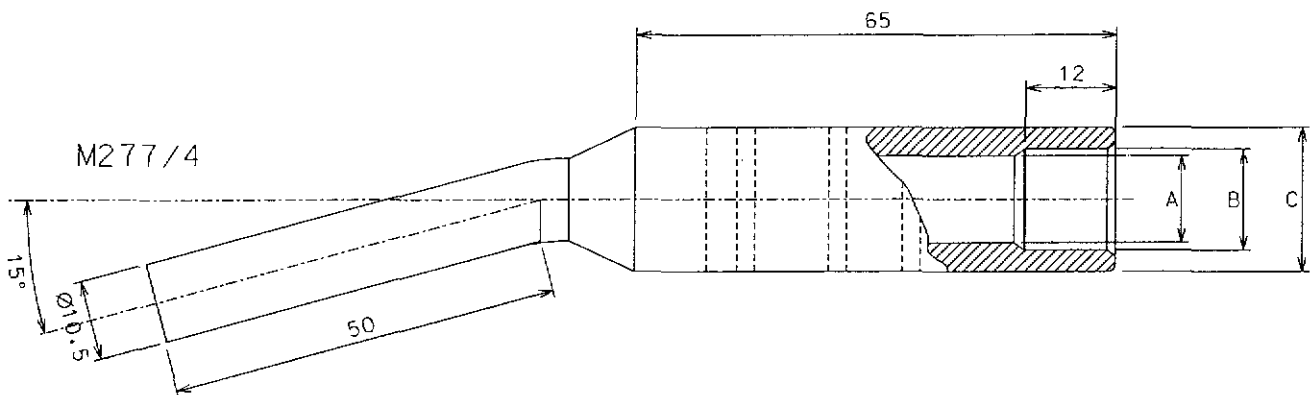
EARTHING EQUIPMENT LUG

MATERIAL

Aluminium Tube
or Bar.
99.5% min. purity.



M277/4



JOINT	Conductor	Die Details G10 (Y35) Only	A	B	C
M 277/2	50mm ² Aluflex	U - 659	11.1	14.3	20.3
M 277/3	50mm ² Aluflex	U - 659	11.1	14.3	20.3
M 277/4	50mm ² Aluflex	U - 659	11.8	13.5	20

SCALE	—
DATE	06.06.03
DRAWN	ADD
TRACED	—
CHECKED	KD
APPROVED	11.06.03

LINE TAP COMPRESSION LUG FOR
FLEXIBLE 50mm² AL. CONDUCTORS

ISSUE DATE

Produced by
Primary CAD

Aquila

ORG. No.

M277

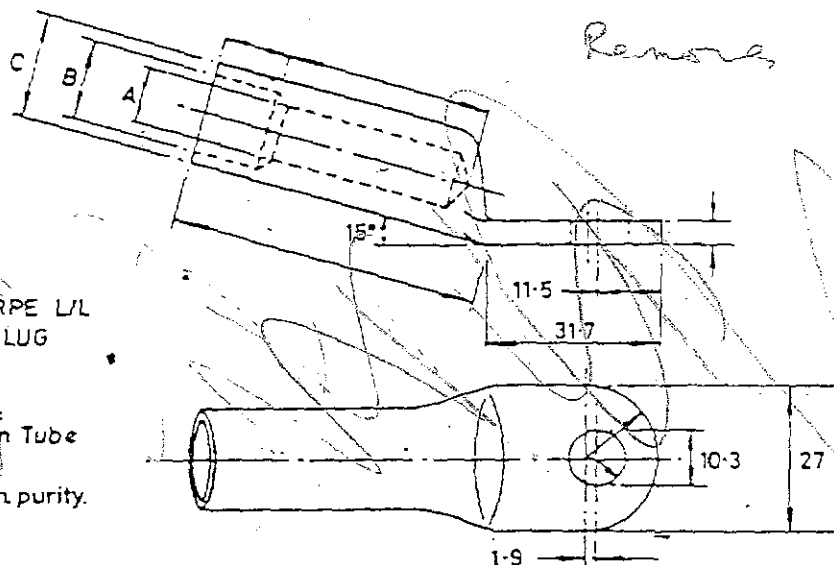
DRG. No.

M277

A4

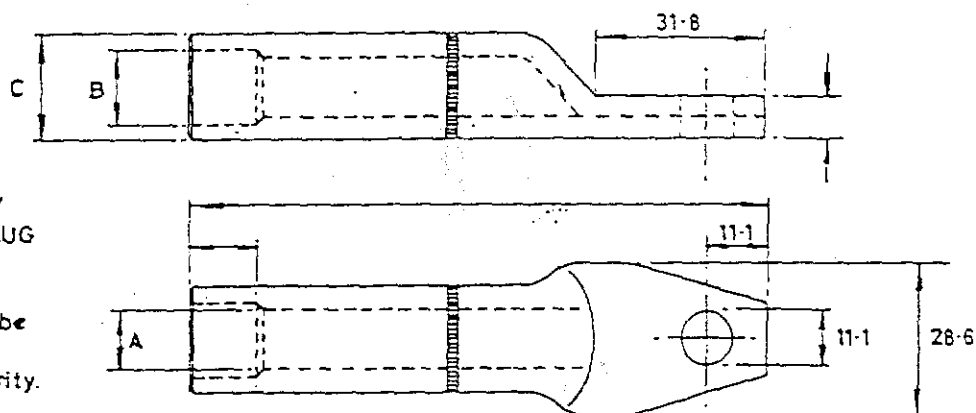
ME 277/1
BOWTHORPE L/L
TAPPING LUG

MATERIAL
Aluminium Tube
or Bar.
99.5% min. purity.



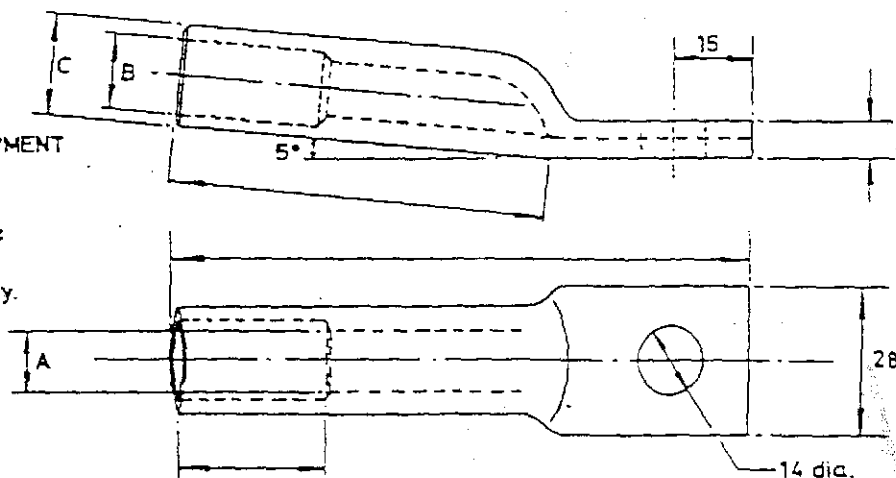
ME 277/2
BICC BURNDY
L/L TAPPING LUG

MATERIAL
Aluminium Tube
or Bar.
99.5% min. purity.



ME 277/3
BICC BURNDY
EARTHING EQUIPMENT
LUG.

MATERIAL
Aluminium Tube
or Bar.
99.5% min. purity.



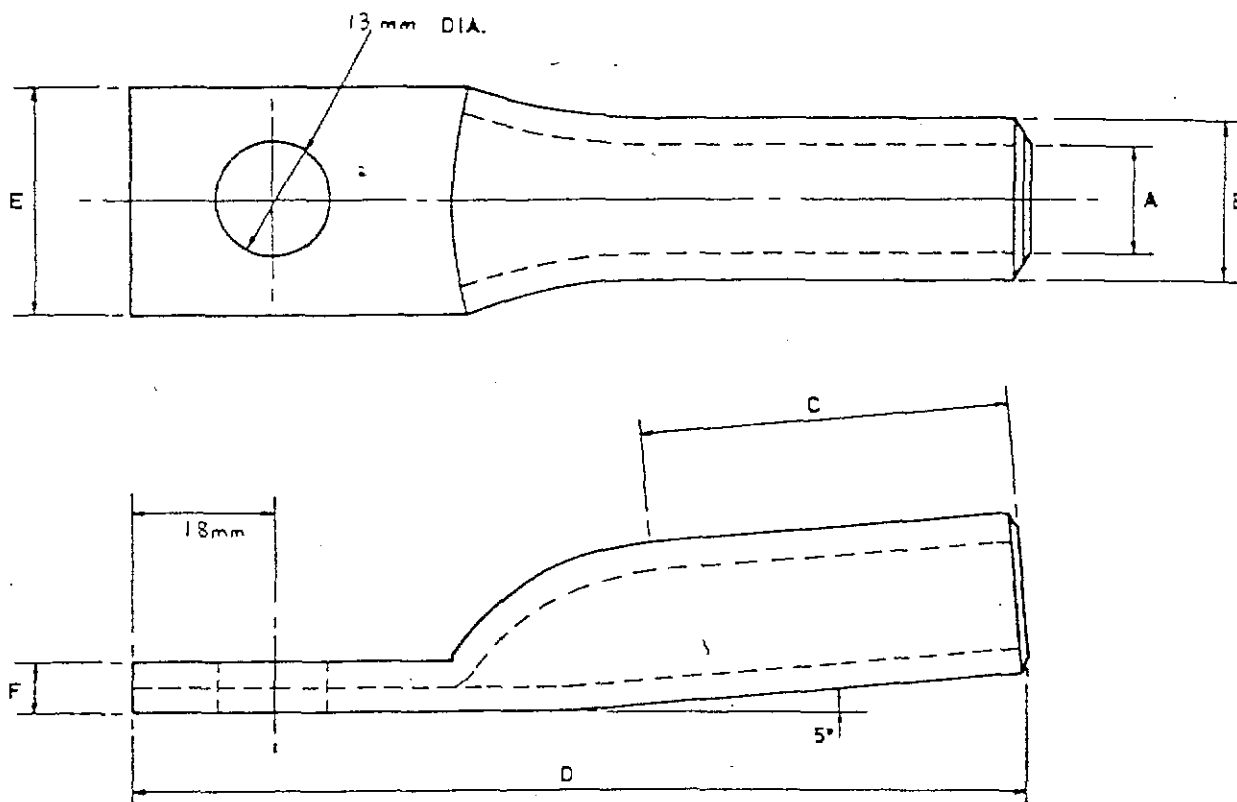
JOINT	CONDUCTOR	COMPRESSOR DIE DETAILS			DIMENSIONS mm			MANUFACTURER & REF
		USE G10 (Y35) TOOL ONLY			A	B	C	
277/1	50mm ² Aluflex	U 247						
277/2		U 659						
277/3		U 659						

Note: For connections to Earthing Equipment use ME 277/3

SCALE	1:1
DATE	25.4.78
DRAWN	RAC
TRACED	PG
CHECKED	Id

LINE TAP COMPRESSION LUG FOR
FLEXIBLE 50mm² AL. CONDUCTORS

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
G	F (M)	
DIE REV	RE-DRAWN & ME 277/3 ADDED	DRG No
SL	JW	ME 277



TYPE	CONDUCTOR			DIMENSIONS IN (mm.)						DIE N°	
	SIZE	STRANDING		A	B	C	D	E	F	G10(Y35)	MHTCOM06)
M.E. 285/1	05in. ²	7/122"								U-243	W-243
	50mm. ²	7/3-10mm									
M.E. 285/2	1in. ²	7/173"								U-249	W-249
	100mm. ²	7/4-39mm									
⊗ M.E. 285/3	95mm. ²	ABC								U-658	W-658

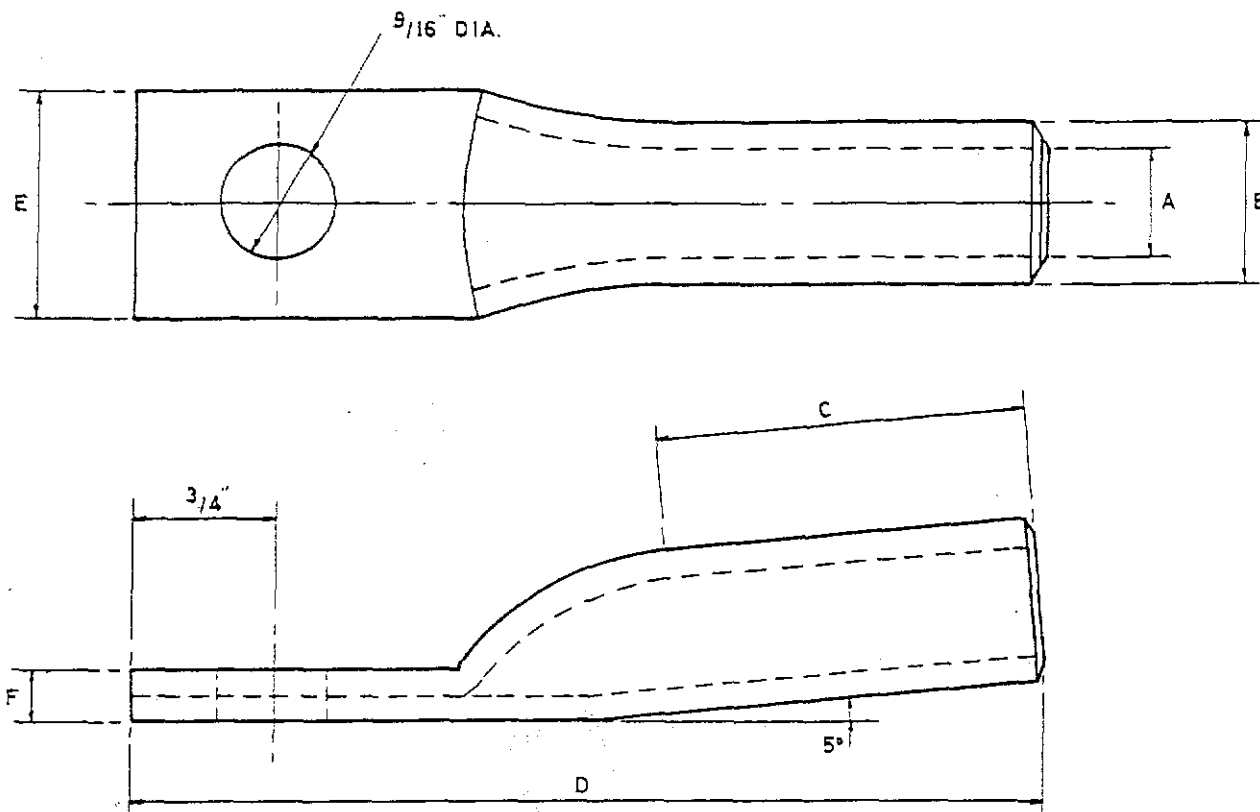
⊗ Barrel recessed

to allow for conductor insulation

SCALE	FULL SIZE
DATE	29.6.73
DRAWN	R. J. K.
TRACED	G. A. H.
CHECKED	
APPROVED	<i>[Signature]</i>

COMPRESSION LUG
FOR H. D. ALUMINIUM CONDUCTORS

ISSUE	MIDLANDS ELECTRICITY BOARD
C	
95mm. ² ABC added.	DRG. No. ME 285
11.12.89 AIR	



TYPE	CONDUCTOR		BICCON. CAT. N°	DIMENSIONS IN (m.m.)						DIE N°	
	SIZE	STRANDING		A	B	C	D	E	F	G10 (Y35)	MHTD00(MD6)
M.E.285/1	.05in. ²	7/1.122"	L9A12	25/64"	5/8	23/8	5"	7/8	3/16"	U-243	W-243
	50mm. ²	7/3.10mm	(YCA 249 A)	(9.9)	(15.9)	(60.3)	(127)	(22.2)	(4.8)		
M.E.285/2	.1in. ²	7/1.173"	L25A12	9/16"	27/32	2"	43/4"	13/16"	1/4"	U-249	W-249
	100mm. ²	7/4.39mm	(YCA 279 A)	(14.3)	(21.4)	(50.8)	(120.7)	(30.2)	(6.35)		

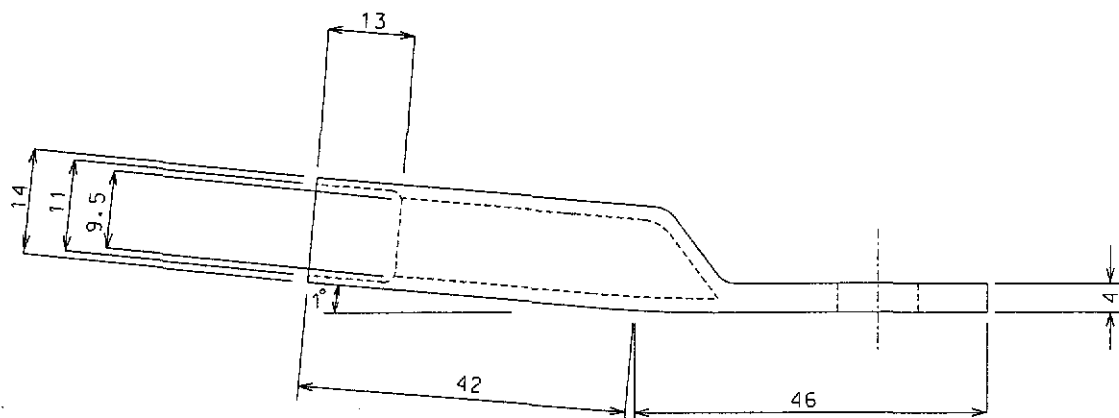
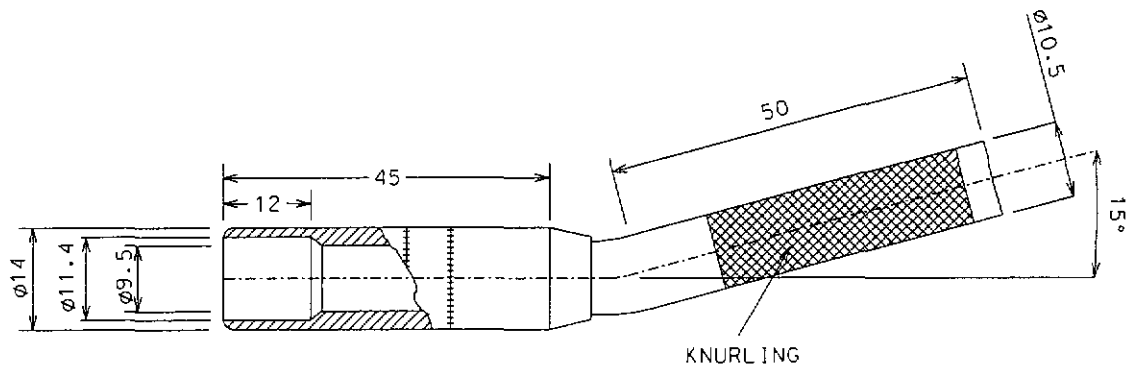
NOTE: BICCON OLD AND EQUIVALENT NEW NUMBER SHOWN

SCALE	FULL SIZE
DATE	29.6.73
DRAWN	R. J. K.
TRACED	G. A. H.
CHECKED	
APPROVED	

COMPRESSION LUG
FOR H. D. ALUMINIUM CONDUCTORS

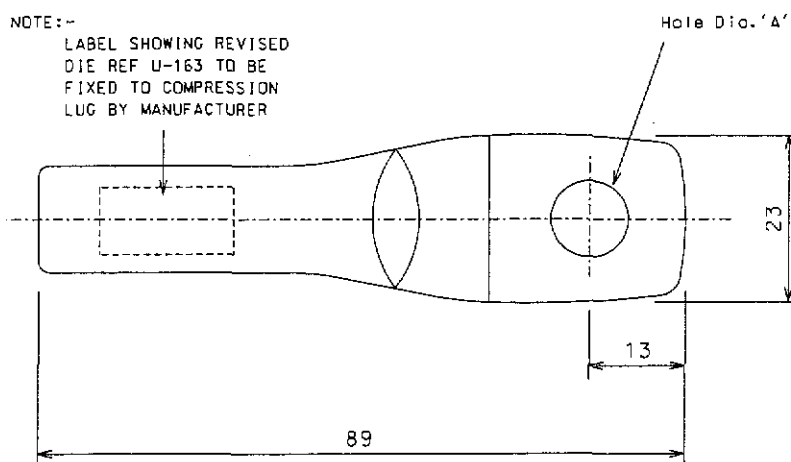
ISSUE	B
BICCON REF REV	
4.84. 1W.	

MIDLANDS
ELECTRICITY BOARD
DRG. No.
ME 285



NOTE:-

LABEL SHOWING REVISED
DIE REF U-163 TO BE
FIXED TO COMPRESSION
LUG BY MANUFACTURER



JOINT	Conductor	Die Details G10 (Y35) Only	Hole Dia 'A'	USE
M 295/1	35mm ² Cu flex	U - 163	11 mm	L/L Tap
M 295/2	35mm ² Cu flex	U - 163	13mm	L/L Tap
M 295/3	35mm ² Cu flex	U - 163	17mm	H.V Trans F Connection
M 295/4	35mm ² Cu flex	U - 164	10.5mm Pin	L/L Tap

SCALE	—
DATE	06.06.03
DRAWN	MJB
TRACED	—
CHECKED	11/06/03
APPROVED	KD

COMPRESSION LUG FOR
CU. LIVE LINE TAPPING CABLE.

ISSUE	DATE
A	08.07.04
MINOR MOD	
KD/ADD	

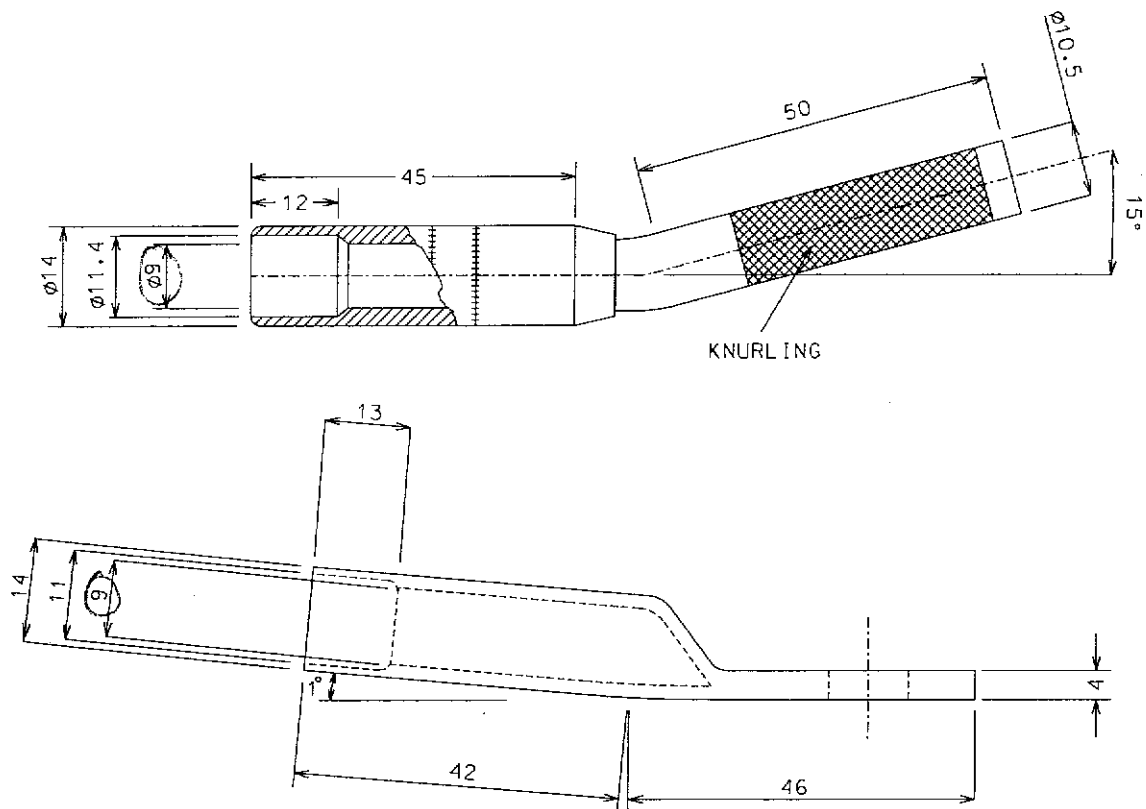
Produced by Primary CAD
Central Networks A company of
DRG. No. M295

DRG. No.

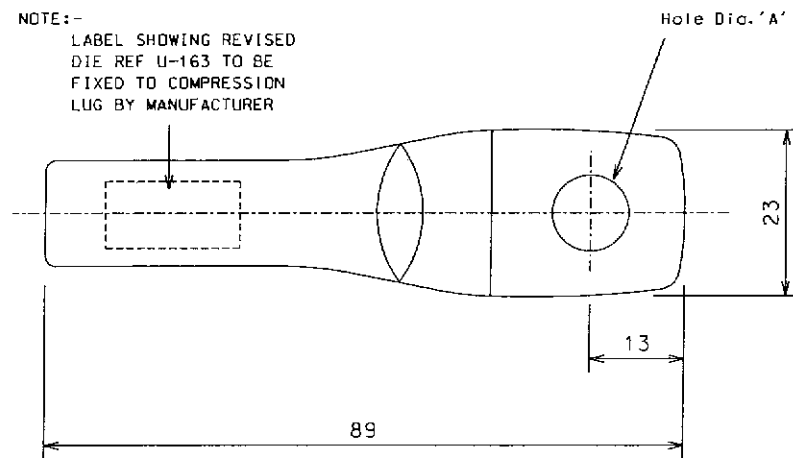
M295

A4

Supers 6060



NOTE:-
LABEL SHOWING REVISED
DIE REF U-163 TO BE
FIXED TO COMPRESSION
LUG BY MANUFACTURER



JOINT	Conductor	Die Details G10 (Y35) Only	Hole Dia 'A'	USE
M 295/1	35mm ² Cu flex	U - 163	11 mm	L/L Tap
M 295/2	35mm ² Cu flex	U - 163	13mm	L/L Tap
M 295/3	35mm ² Cu flex	U - 163	17mm	H.V Trans F Connection
M 295/4	35mm ² Cu flex	U - 164	10.5mm Pin	L/L Tap

SCALE	—
DATE	06-06-03
DRAWN	MJB
TRACED	—
CHECKED	11/06/03
APPROVED	KD

COMPRESSION LUG FOR
CU. LIVE LINE TAPPING CABLE.

ISSUE DATE

—

—

—

—



Aquila

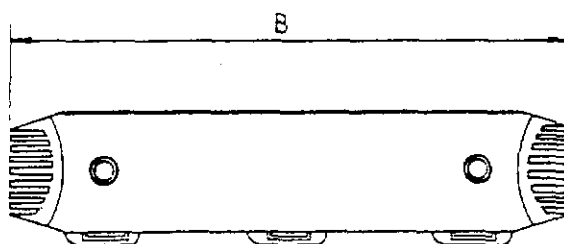
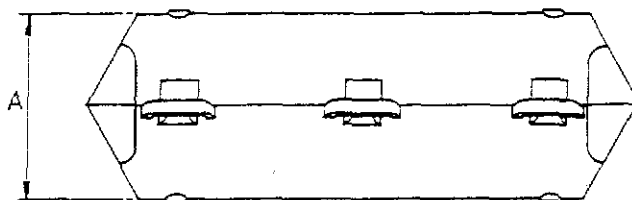
DRG. No.

M295

DRG. No.

M295

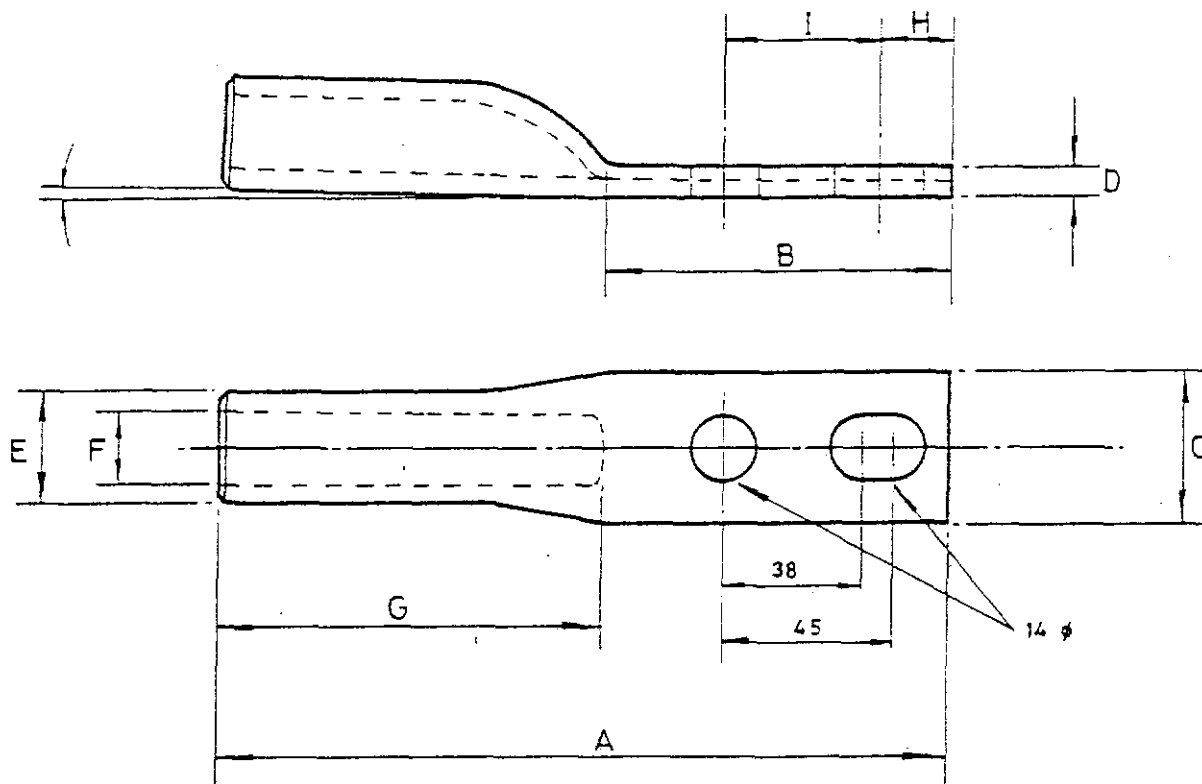
A4



MEB REF.	BICCON CAT No.	A	B	C
M. 302/1	C1 (CCO)	37	109	26
M. 302/2	C2 (CCD)	46	167	30

Crimpet cover filled with Plastic Compound before closing
to prevent ingress of moisture into cable sheath.

SCALE	N.T.S.	CONDUCTOR COMPRESSION JOINT CRIMPET COVERS	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	5.8.81		A	DRG. No.
DRAWN	R.A.C.		References revised.	M. 302
TRACED	A.E.		R.A.C. 10-85	
CHECKED				
APPROVED	<i>[Signature]</i>	E.S.I. STANDARD 43-92		E.S.I. Drg. No. 439231



TYPE	BOLT HOLES	CONDUCTOR TYPE & SIZE	MAN. REF. NUMBER	A	B	C	D	E	F	G	H	I	Y 35 DIE REF.
ME 310/1	SINGLE BOLT	25mm ² AL. AL.	L12 RL12 YCA3R1L	121	40	25	6	17	8	80	19		U-698
ME 310/2	DOUBLE BOLT	25mm ² A.C.S.R.	L12RL12D YCA3R1L-2B	160	80	25	6	17	8	80	19	42	
ME 310/3	SINGLE BOLT	50mm ² AL. AL.	L20RL12 YCA248RL	120	36	38	8	20	11	80	19		U-659
ME 310/4	DOUBLE BOLT	50mm ² A.C.S.R.	L20RL12D YCA248RL-2B	160	80	28	8	20	11	80	19	42	
ME 310/5	SINGLE BOLT	100mm ² A.C.S.R.	L34RL12 YCA279RL-BG	128	36	34	10	26	15	90	19		U-654
ME 310/6	DOUBLE BOLT		L34RL12D YCA279-2B-BG	170	78	36	9	26	15	90	19	42	
ME 310/7	DOUBLE BOLT	150mm ² A.C.S.R.	L44R12D YCA319R-2B	186	80	40	8	29	20	105	19	42	U-316
ME 310/8	DOUBLE BOLT	175mm ² A.C.S.R.	L51R12D YCA335R-2B-BG	193	80	46	12	33	21	100	19	42	U-317 U-756

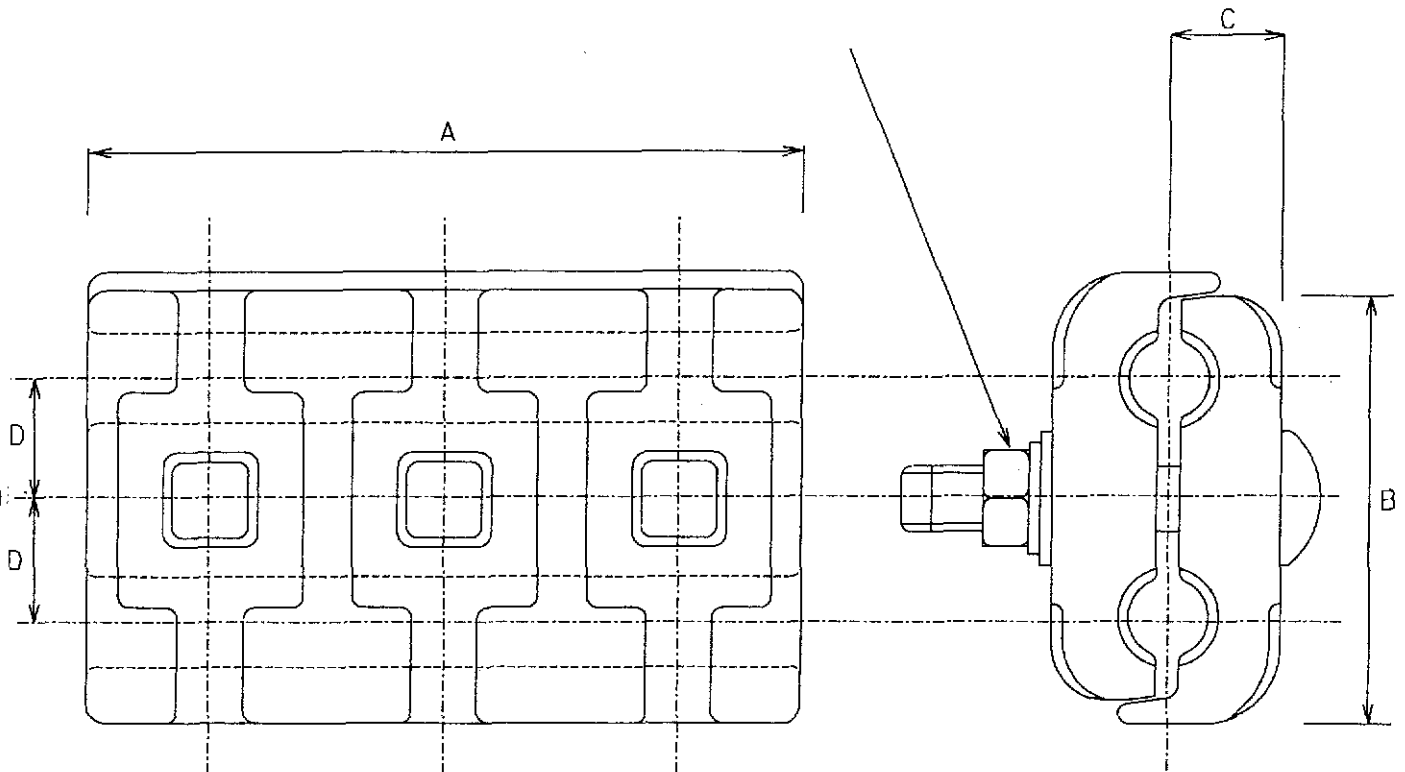
NOTES 1. To comply with BS.3288 Pt2
 2. BICCON NEW & EQUIVALENT OLD NUMBER SHOWN.
 3. 310/8 Type L51R12D (YCA33 R-2B-BG) replaces
 Previous Joint L46R12D (YCA337R-2B)
 Although L51R12D is marked suitable
 for 200mm² ACSR it is also recommended by BICCON
 for 175 mm² ACSR (30-7 Strand).

SCALE	N.T.S.
DATE	5.9.77
DRAWN	R.A.C.
TRACED	
CHECKED	<i>[Signature]</i>
APPROVED	<i>[Signature]</i>

COMPRESSION LUGS FOR
 AL ALLOY & ACSR CONDUCTORS

ISSUE	MIDLANDS ELECTRICITY BOARD
A	DRG. No
Type B Ref. Changed.	M310

Carriage Bolt With Full
Nut, Washer & Spring Washer



Material: Aluminium Clamp. Steel Bolts

Marking: OLF Code Clamp Size Makers
Identity & Date (Month/Year)

NOTE

Each Clamp Has 3 Carriage Bolts
One Of Which Is Extended For
Earth Bond Connection.

			Dimensions. mm				
MIN No	Conductor	Drg. No	A	B	C	D	E
3453230	70mm2 HORES/HORSE ACSR	A3-93AA00	130	70	16	19	M12
3453230	100MM2 DOG/DOG ACSR	A3-93AA00	130	70	16	19	M12
3453230	100MM2 OAK/DAK AAAC	A3-93AA00	130	70	16	19	M13
3453231	150MM2 WOLF/WOLF ACSR	A3-93AA04	140	78	16	22	M16
3453232	175MM2 LYNX/LYNX ACSR	A3-93AA02	140	78	16	22	M16
3453233	400MM2 ZEBRA/ZEBRA ACSR	A3-93AA01	200	102	28	29	M20
3453234	200MM2 POPLAR/POPLAR AAAC	A3-93AA03	140	78	16	22	M16

SCALE	-
DATE	05.02.03
DRAWN	M.J.B
TRACED	-
CHECKED	KD
APPROVED	13.02.03

PARALLEL GROOVE CLAMP

ISSUE DATE

Produced by
Primary CAD

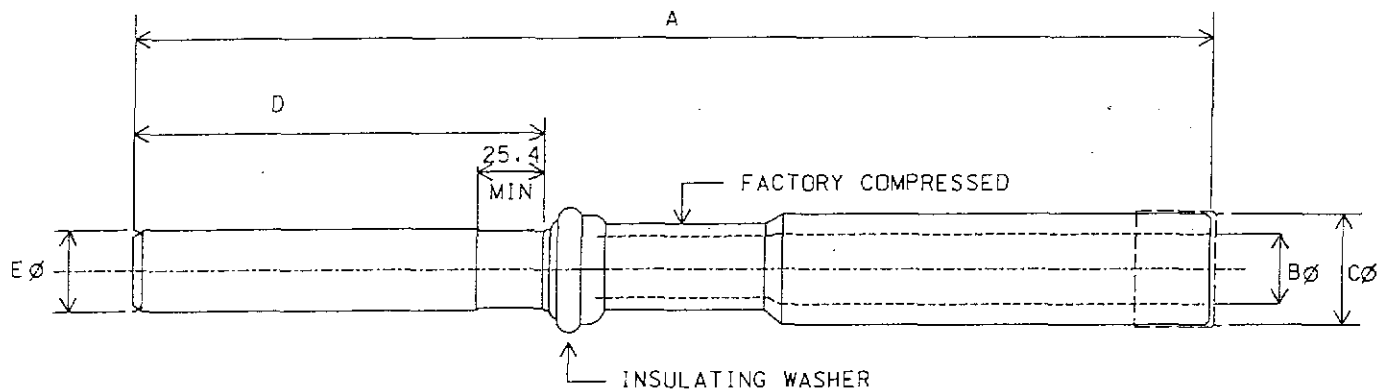


Aquila

Drg. No.

M 323

Drg. No. M 323



MATERIAL:-

PIN - TELLURIUM
 INSULATING WASHER - NEAPRENE
 SLEEVE - ALUMINIUM 99.5% PURITY

M.REF	CONDUCTOR		A	B	C	D	E	MDN REF	DIE REF
	SIZE	STRAND							
327/0	175mm ² ACSR 200mm ² AAAC	30/2.79AL-7/2.79ST 37/2.87							A9
327/2	400mm ² ACSR 425mm ² AAAC	54/2.18AL-7/3.18ST 37/4.14	395	30.5	48	153	35	85ET05	A11
327/4	300mm ² AAAC	37/3.53	390	28.8	48	165	35	85ET17	A10

SCALE	
DATE	11/11/99
DRAWN	
TRACED	
CHECKED	
APPROVED	

BI-METAL COMPRESSION
 TERMINAL PIN TYPE

ISSUE	DATE
A	14.12.03
MINOR MODS	
PA/KD	



Aquila

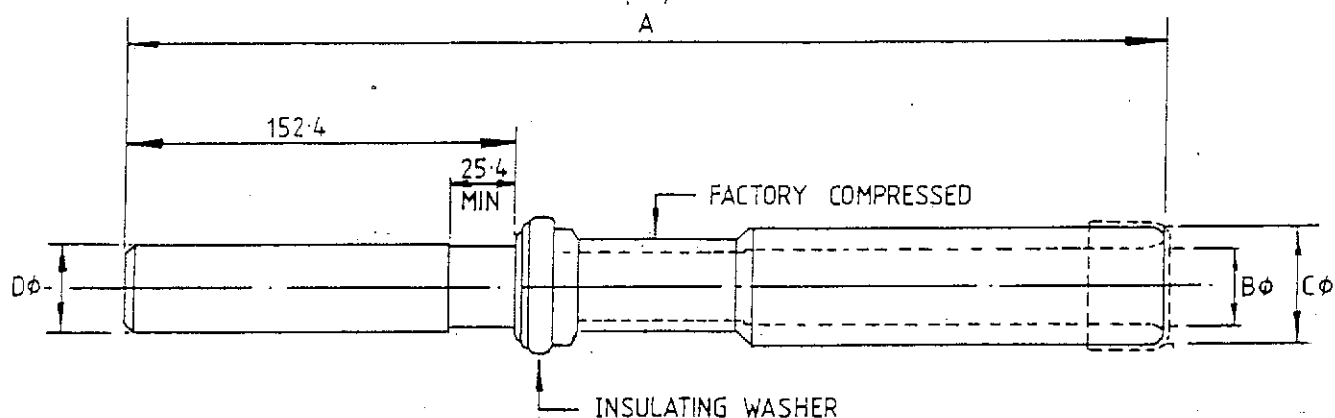
DRG. No.

M327

DRG. No.

M327

A4

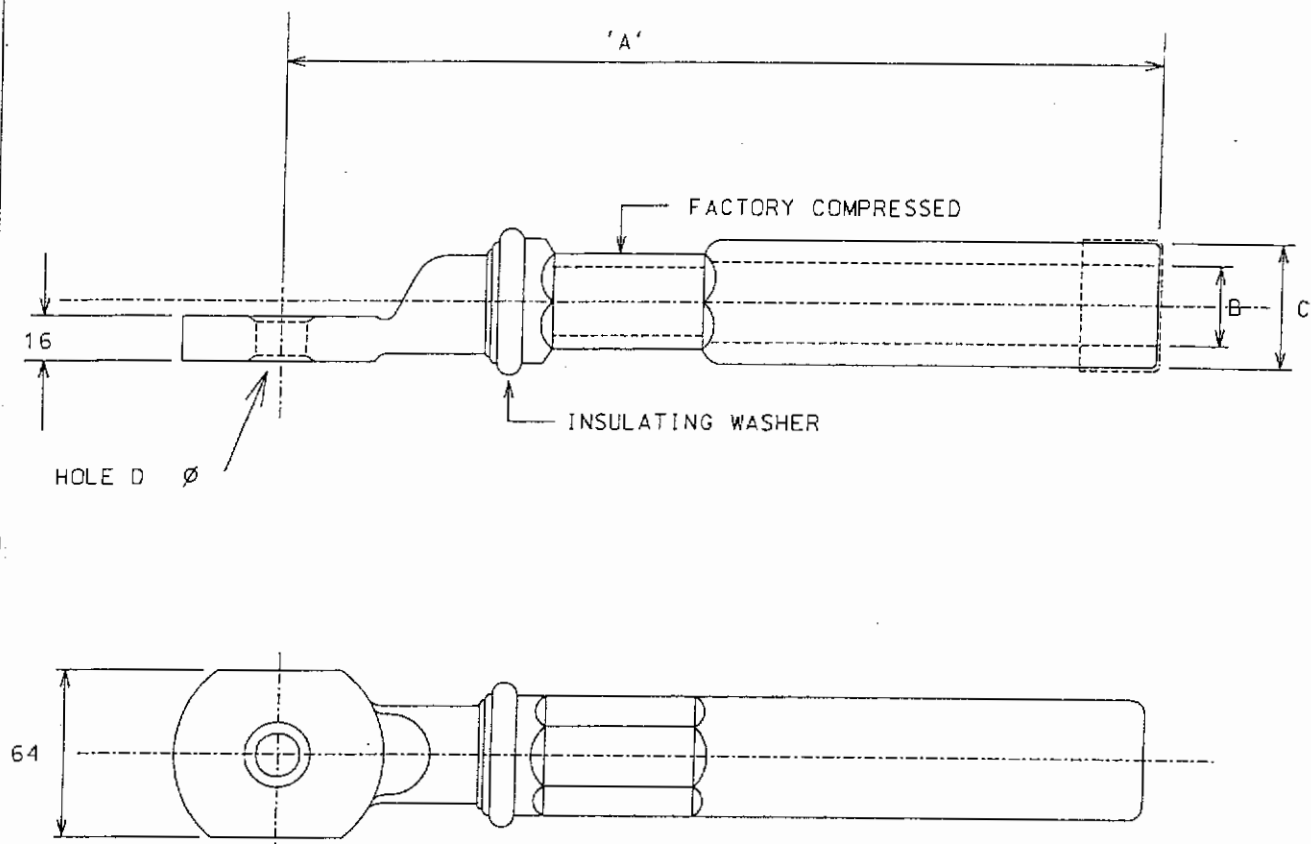


MATERIAL:—

PIN - TELLURIUM
 INSULATING WASHER - NEAPRENE.
 SLEEVE - ALUMINIUM 99.5% PURITY.

M REF	C.E.G.B. COMMODITY CODE	CONDUCTOR		MANF. REFERENCE		LENGTH A	B φ	C φ	D φ	DIE REF
		SIZE	STRAND.	UNIDARE	BOWTHORPE					
327/0	4301901	175mm ² ACSR	37/2-79	511101	PIT1953	381	21.8	38.1	25.4	A9
327/2	4301906	400mm ² ACSR	61/3-18	511102	PIT2862	429	30.2	47.5	34.9	A11.

SCALE	NTS	BI-METAL COMPRESSION TERMINAL PIN TYPE 175mm ² & 400mm ² ACSR CONDUCTORS: E.S.I. STANDARD	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE			B	
DRAWN	R.A.C.		D φ Added.	DRG. No.
TRACED			R.A.C. 86.	M. 327.
CHECKED				
APPROVED				ESI Drg N ^o



MATERIAL:-

PALM - BRONZE BS.1400.LG 2-C

INSULATING WASHER - NEOPRENE

SLEEVE - ALUMINIUM 99.5% PURITY

M. REF	CONDUCTOR				LENGTH A	B Dia	C Dia	DIE REF.
	SIZE	STRAND	MDN REF	D				
328/0	175mm ² ACSR 200mm ² AAAC	30/2.79AL-7/2.79ST 37/2.87	85ER 06 - 5	22	297	21.8	38	A9
328/2	400mm ² ACSR	54/3.18AL 7/3.18ST	85ET09	22	315	30.5	48	A11
328/4	300mm ² AAAC	37/3.53	85ET13	22	320	28.8	48	A10

FILE	-
DATE	17.02.03
DRAWN	PA
TRACED	-
CHECKED	KD
APPROVED	11.03.03

BI-METAL COMPRESSION
TERMINAL STRAIGHT PALM

ISSUE DATE

Produced by
Primary CAD

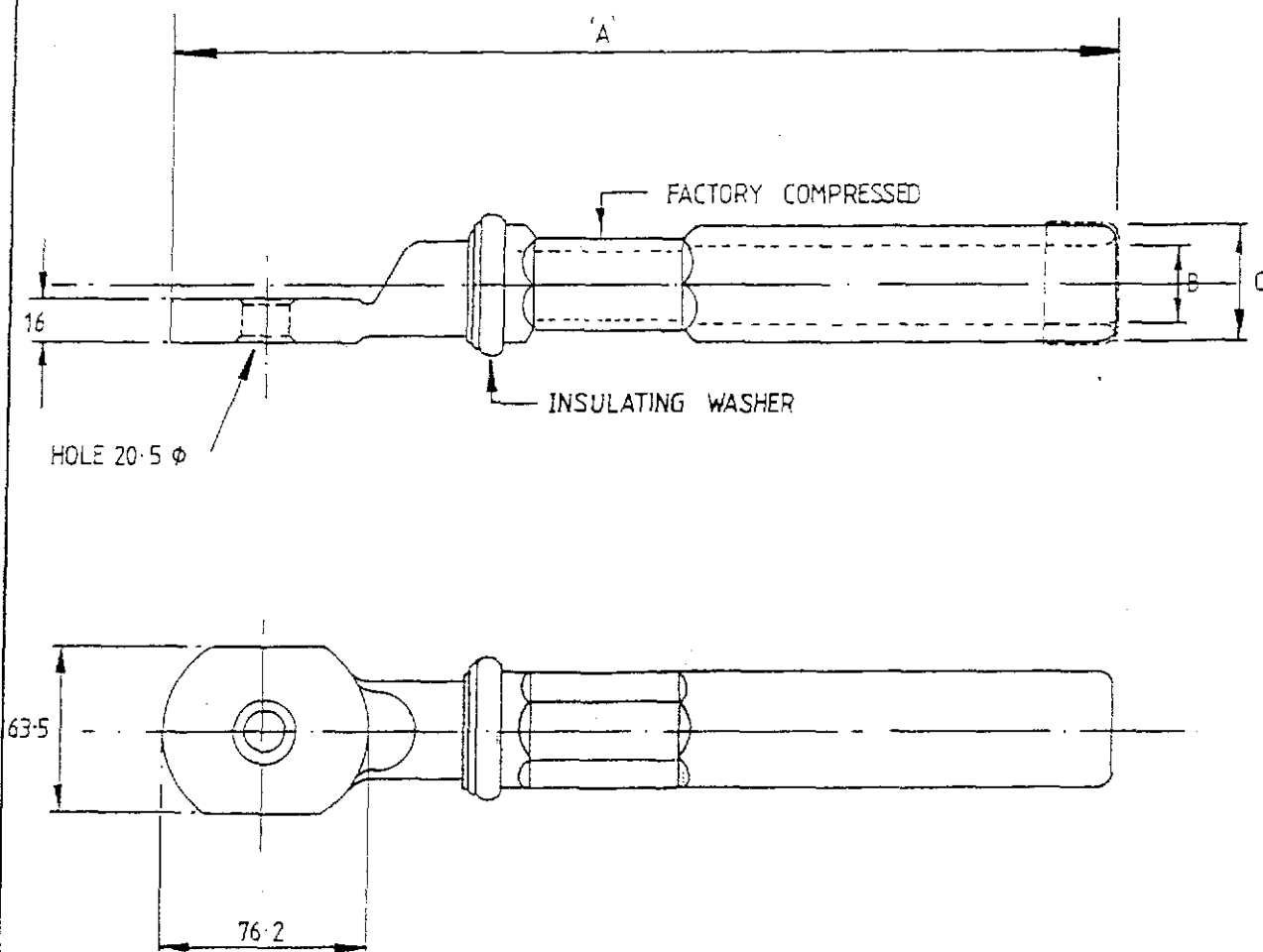


Aquila

DRG. No.

M328

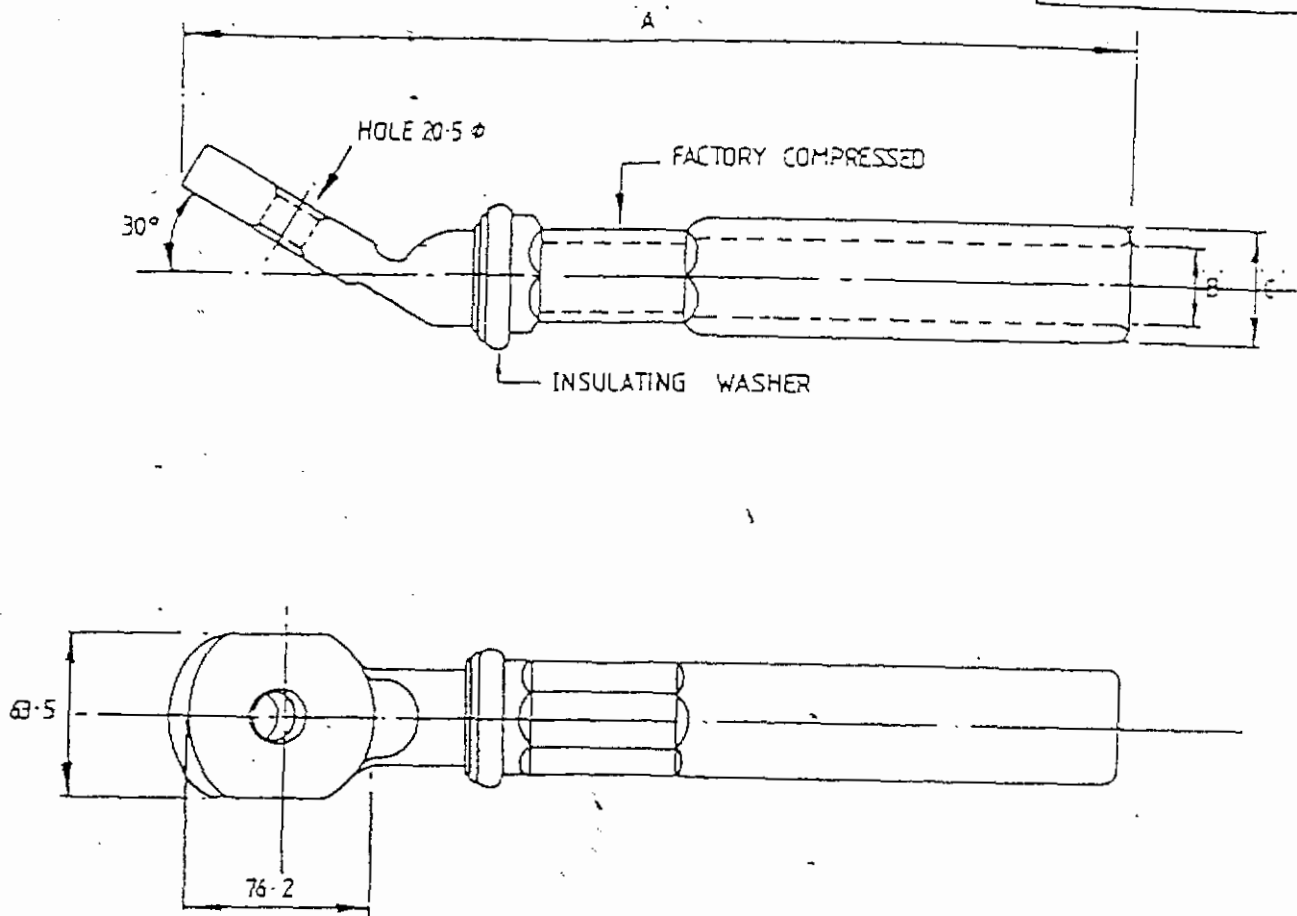
DRG. No. M328



MATERIAL: —
 PALM - BRONZE BS.1400: LG 2 - C.
 INSULATING WASHER - NEOPRENE.
 SLEEVE - ALUMINIUM 99.5% PURITY.

M. REF	C.E.G.B COMMODITY	CONDUCTOR		MANF. REFERENCE		LENGTH		DIE REF
		SIZE	STRAND	UNIDARE	BOWTHORPE	A	B ϕ	
328/0	4301911	175mm ² ACSR	37/2-79	511201	PTS1953	340	21.8	A9
328/2	4301921	400mm ² ACSR	61/3-18	511202	PTS2862	387	30.2	A11

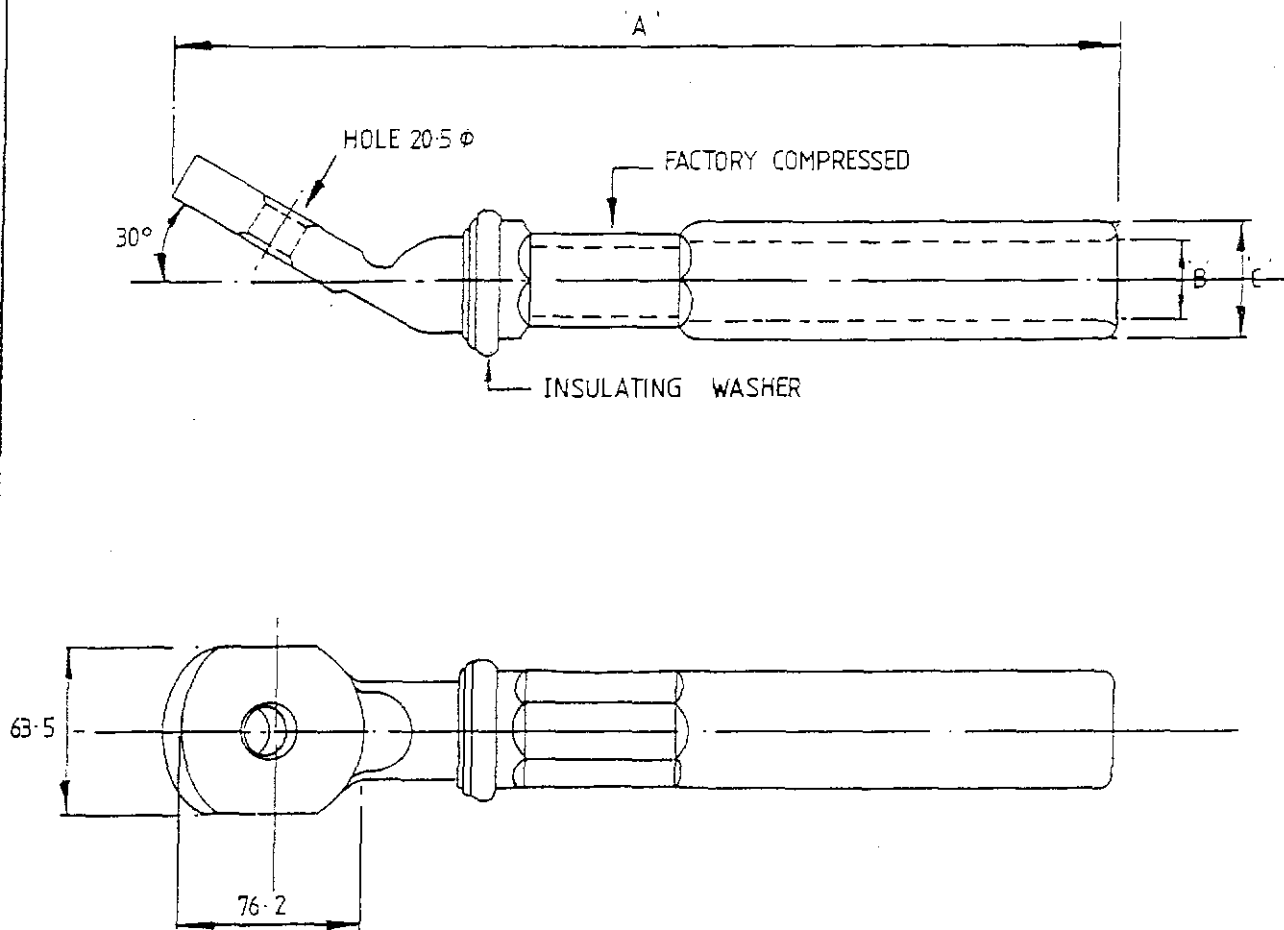
SCALE	N.T.S.	BI METAL COMPRESSION TERMINAL STRAIGHT PALM 175mm ² and 400mm ² ACSR CONDUCTORS	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	26-7-82		A	
DRAWN	R.A.C.		Bowthorpe	DRG. No. M. 328
TRACED			Ref. Added.	
CHECKED			CAR. 4/84	
APPROVED	<i>[Signature]</i>	ESI STANDARD		ESI Drg N°



MATERIALS: —
 PALM - BRONZE BS 1400: LG 2 - C
 INSULATING WASHER - NEAPRENE
 SLEEVE - ALUMINIUM 99.5% PURITY.

M.REF.	CONDUCTOR				LENGTH A	B Dia	C Dia	DIE REF.
	SIZE	STRAND						
329/0	175mm ² ACSR 200mm ² AAAC	30/2.79AL-7/2.79ST 37/2.87						A9
329/2	400mm ² ACSR	54/3.18 AL 7/3.18 ST						A11
329/3	400mm ² AAC	37/3.78						A11
329/4	300mm ² AAAC	37/3.53						A10

ADD'D BY DATE	TITLE	SCALE	 Midlands Electricity plc M.329	
	BI-METAL COMPRESSION TERMINAL 30° PALM	DATE		11/11/99
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		



MATERIALS: —

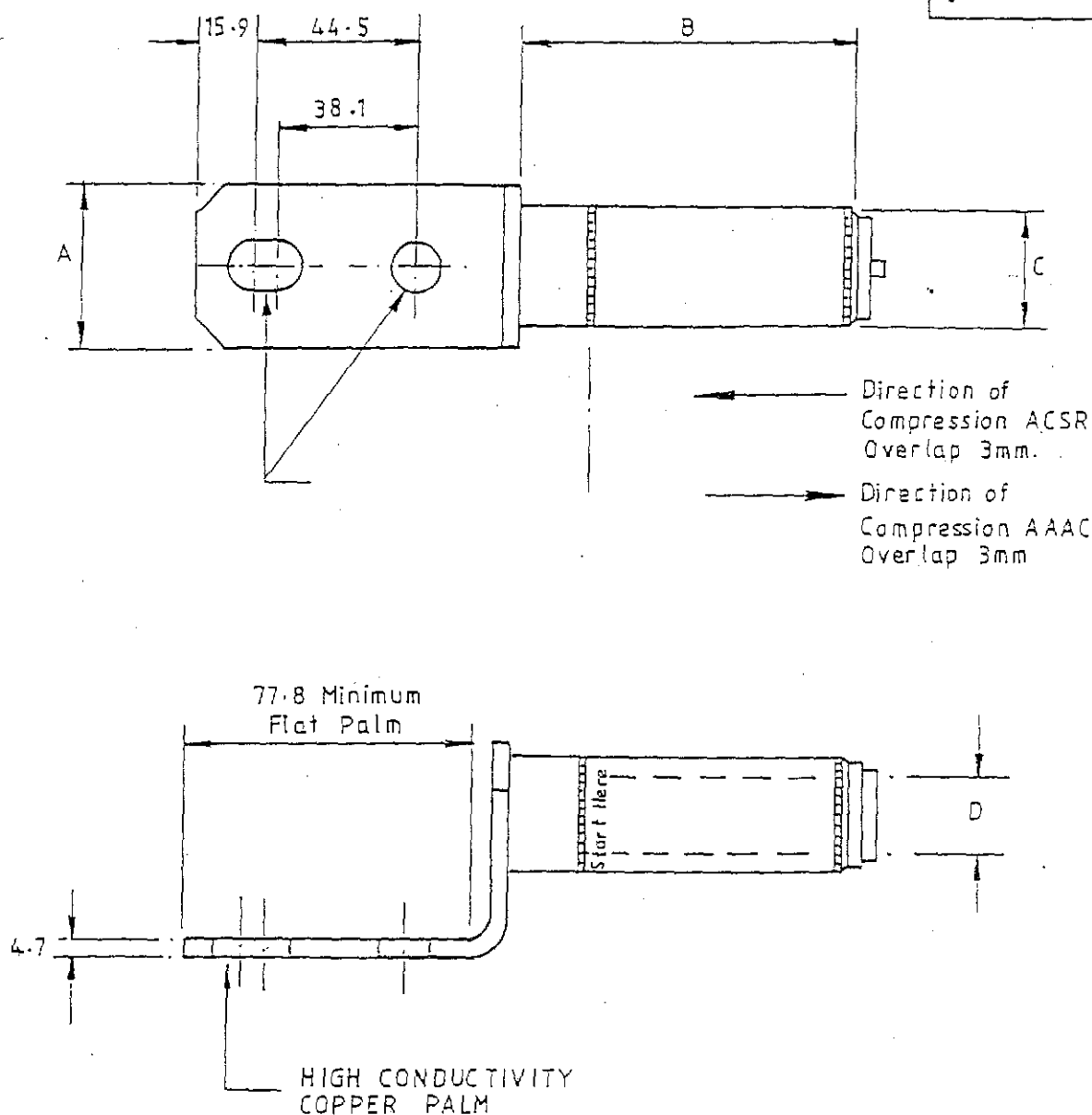
PALM - BRONZE BS 1400 LG 2 - C

INSULATING WASHER - NEAPRENE

SLEEVE - ALUMINIUM 99.5% PURITY.

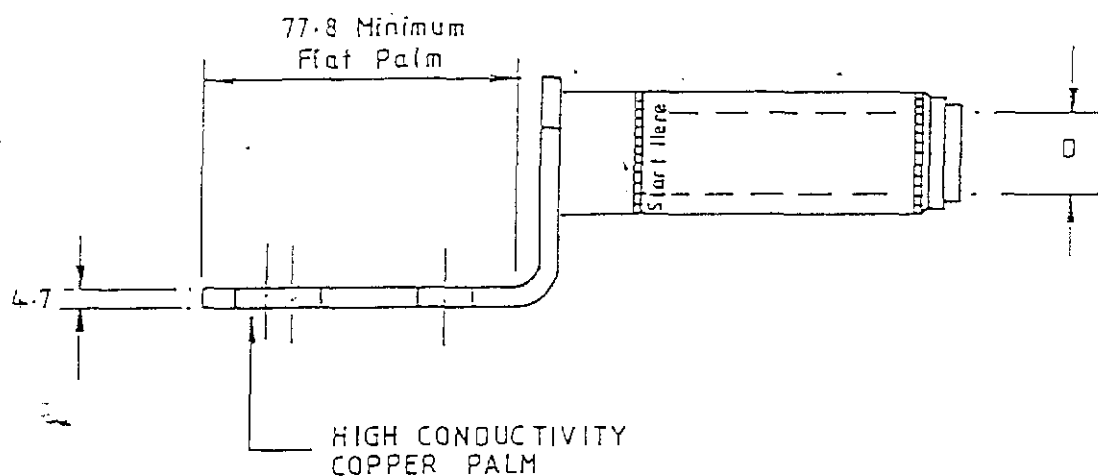
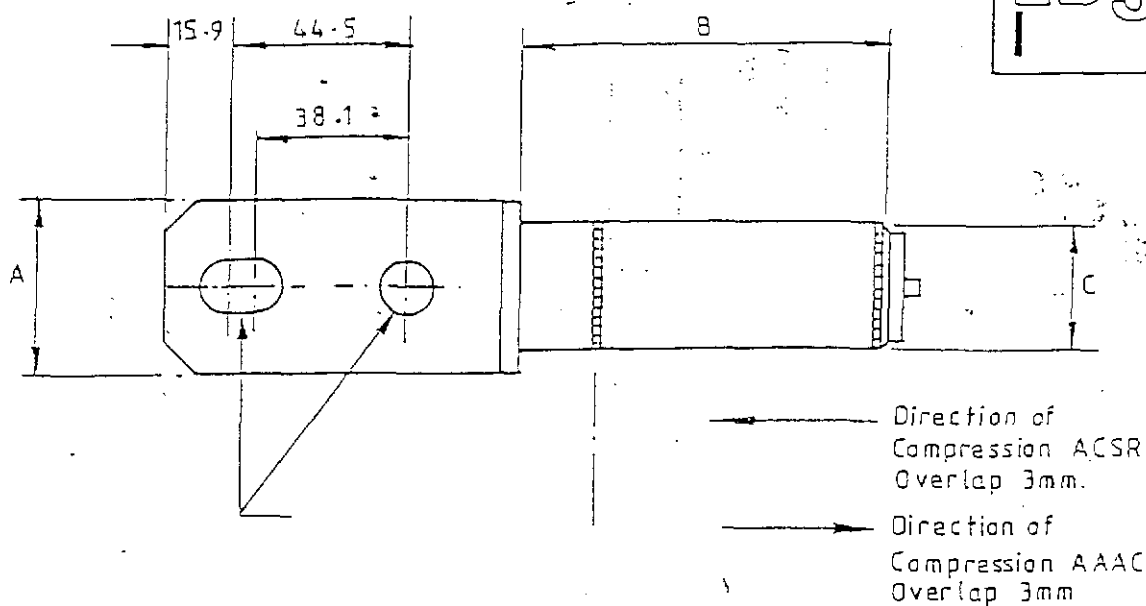
M. REF	CE.G.B COMMODIFY	CONDUCTOR		MANF. REFERENCE		LENGTH A	B ϕ	C ϕ	DIE REF.
		SIZE	STRAND	UNIDARE	BOWTHORPE				
329/0	4301916	175mm ² ACSR	37/2-79	511301	PTA1953	340	21.8	31.8	A9
329/2	4201926	400mm ² ACSR	61/3-18	511302	PTA2862	387	30.2	47.5	A11

SCALE	N.T.S.	BI-METAL COMPRESSION TERMINAL 30° PALM 175mm² & 400mm² ACSR CONDUCTORS.	ISSUE	MIDLAND ELECTRICITY BOARD
DATE	26-7-82		A	
DRAWN	R.A.C.		Bowthorpe	DRG. No. M.329
TRACED			Ref. Added	
CHECKED			CAR. 4/84	
APPROVED	#	ESI STANDARD.	ESI Drg N°	



Type	Min	Conductor Size and Type	Manufacturers Ref	Dimensions (mm)				Y 35 Die Ref
				A	B	C	D	
398/1	3453981	50 AAAC 50 ACSR	YCA050RL12	31.8	79	20.3	11.1	U 659
398/2	3453982	100 AAAC 100 ACSR	LB 34RL12D	31.8	83	25.9	15.5	U 654
398/3	3453983	200 AAAC 175 ACSR	LB 51R12D	44.5	100	33.3	21.0	U 756

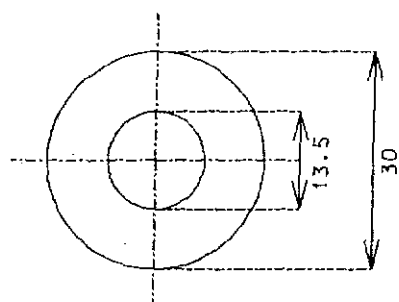
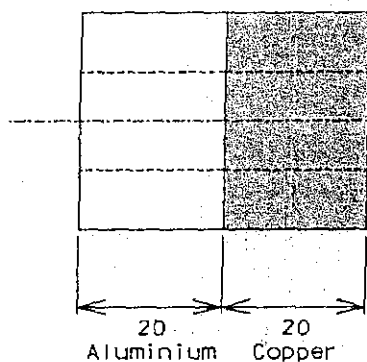
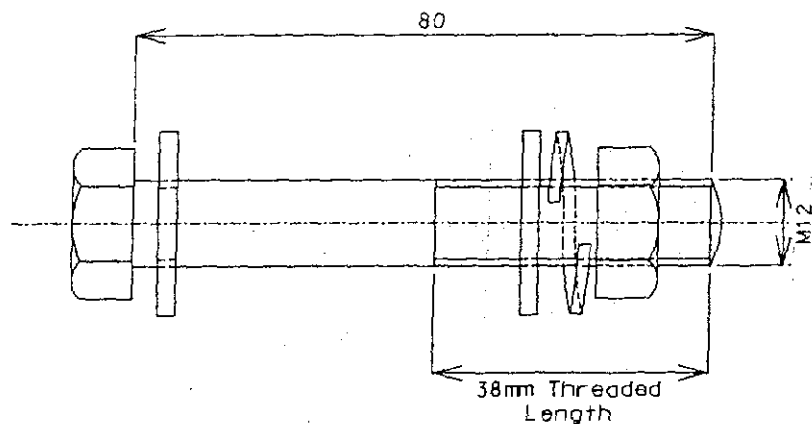
A ADDED BY AJB/ELW DATE 30.7.96 398/1 DETAILS ADDED.	TITLE BI- METALLIC TERMINAL LUG	SCALE		MEB Midlands Electricity plc M 398 
		DATE	1.6.93	
		DRAWN	A J B	
		TRACED	SGT	
		CHECKED	RLT	
		APPROVED		



HOLES TO BE M12

TYPE	MIN	CONDUCTOR SIZE & TYPE	MANUFACTURERS REF	DIMENSIONS				Y35 DIE REF
				A	B	C	D	
398/1	3453981	50 AAAC 50 ACSR						U659
398/2	3453982	100 AAAC 100 ACSR						U654
398/3	3453983	200 AAAC 175 ACSR						U756

ADD'D BY DATE	TITLE BI - METALLIC TERMINAL LUG	SCALE		 Midlands Electricity plc
		DATE	11/11/99	
		DRAWN		
		TRACED		DRG No M.398
		CHECKED		
		APPROVED		



MATERIAL

Friction Welded Spacer
Aluminium 99.5% purity
Copper 99.9% purity
Etched Finish

Hardware
Stainless Steel (18/8)
Tightening Torque - 47KN

Kit Composition
3 Friction Welded Spacers
3 M12x80 Bolt (38mm Treaded)
6 M12 Full Nuts
6 M12 Washers (Form A)
3 M12 Spring Washers (Type B)

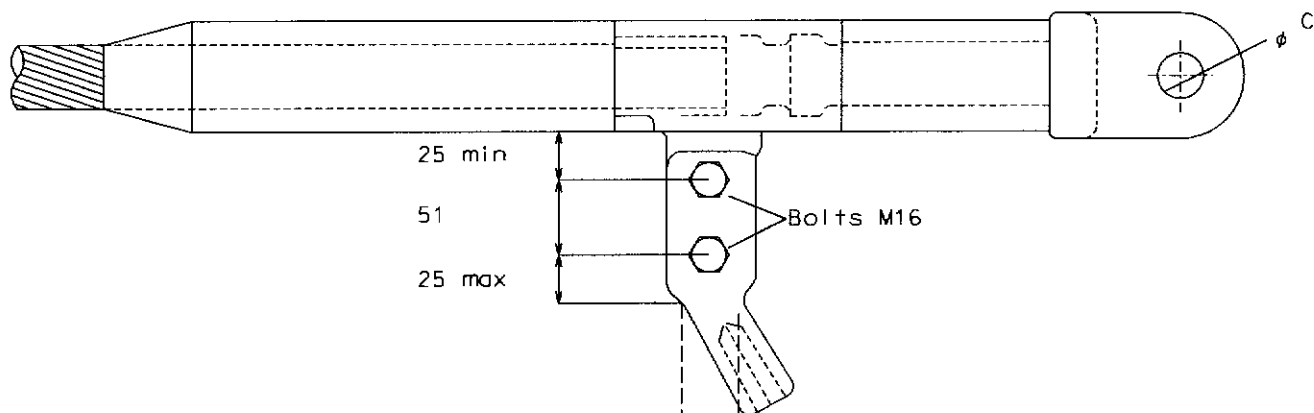
SCALE	1-1
DATE	17/8/95
DRAWN	RCF
TRACED	SGT
CHECKED	
APPROVED	RCF

BI - METAL SPACER KIT FOR CABLE TERMINATIONS

ISSUE	DATE
A	8/11/00
RE-DRAWN NEW BORDER	

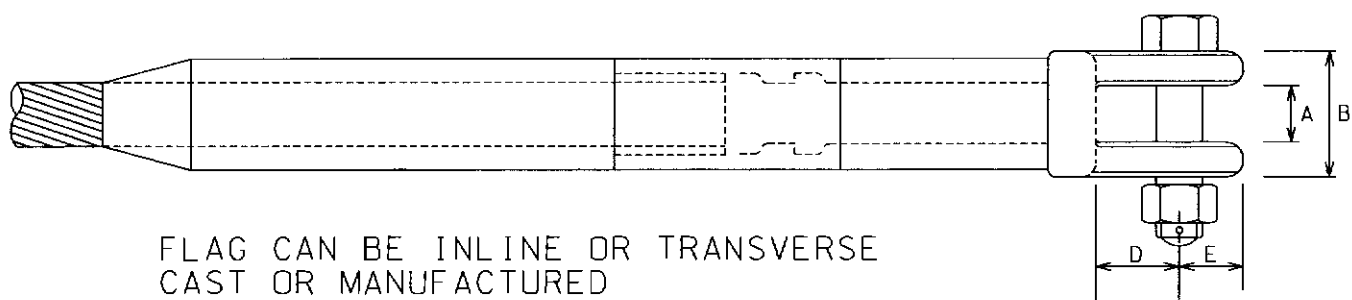
Produced by Output Services	
ORG. No.	M399

ORG. No. M399



Straight Jumper Terminal
Type 5 only

Separate angled and straight jumper
assembly to drg M333



FLAG CAN BE INLINE OR TRANSVERSE
CAST OR MANUFACTURED

A min	B max	C Dia	D min	E max	BOLT TO BS 3288
33	59	27	35	33	M24x100

ME. REF	CONDUCTOR SIZE		MANUFACTURER REF	BORE DIA		DIE REF	
	SIZE	STRAND		ALUM "A"	STEEL "B"	AL	ST
159/1	150mm ²	30/2.59Al 7/2.59St	CLNR- 9.204	20.24	8.73	A9	S9
159/2	175mm ²	30/2.79Al 7/2.79St	CLNR- 9.21	21.83	9.13	A9	S9
159/3	100mm ²	26/2.27Al 7/1.78St	CLNR- 9.180	18.50	7.14	A9	S9
159/4	100mm ²	6/4.72Al 7/1.57St	CLNR- 9.185	17.86	6.74	A9	S9
159/5	70mm ²	12/2.79Al 7/2.79St	CLNSR- 9.178CB	17.86	9.13	A9	S9

Produced by
Primary CAD

SCALE	-
DATE	12-12-03
DRAWN	MJB
TRACED	-
CHECKED	23-12-03
APPROVED	KD

CONDUCTOR DEAD-END (BOLTED
JUMPER) 70mm² - 175mm²
ACSR CONDUCTORS

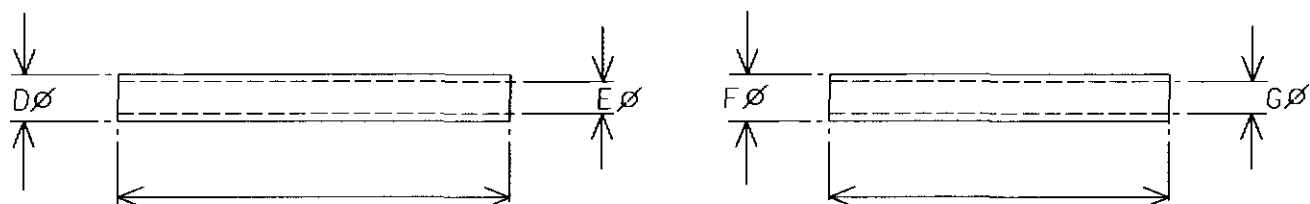
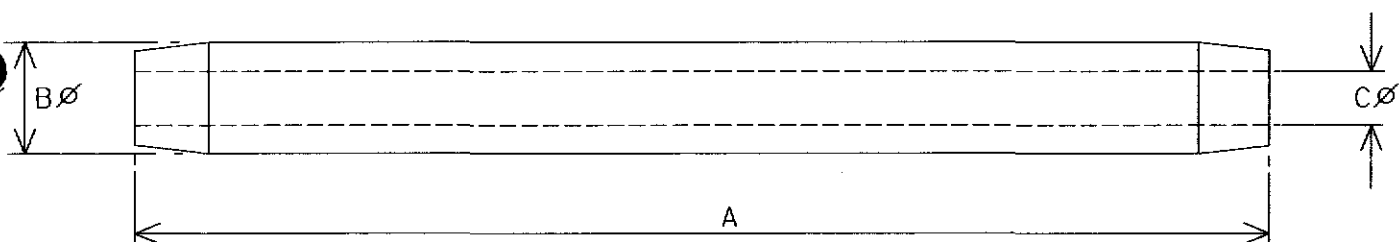
ISSUE	DATE



Aquila

DRG. No.

M159



MEB .REF.	C.E.G.B. COMMODITY CODE	CONDUCTOR SIZE (mm)	JOINT SLEEVE			STEEL SLEEVE		ALUMINIUM LINER	
			LENGTH A (mm)	O/D B (mm)	I/D C (mm)	LENGTH D (mm)	O/D E (mm)	I/D F (mm)	LENGTH G (mm)
161/0	4303117	175mm ²	572	38	21.8	18	9.1	-	-
161/1	4303109	70mm ²	420	38	22 / 17	18	9.1	21	17
161/2	-	100mm ² (13 STAND)	420	38	22 / 17	18	6.74	21	17.5
161/3	-	150mm ²	465	38	24.6	18	9.3 / 8.73	-	-
161/4	-	100mm ² (33 STAND)	420	38	22 / 17	18	6.5 / 7.14	24	17.3
161/5	4303123	400mm ²	760	48	30.5	19.2	9.9	-	-

MEB REF.	MANUFACTURER REF No	DIE REF.
	ESKAY'S	
161/0	LJ-9.21/S9	A9-S9
161/1	LJ-9.178/S9	A9-S9
161/2	LJ-9.185/S9	A9-S9
161/3	LJ-9.204/S9	A9-S9
161/4	LJ-9.180/S9	A9-S9
161/5	LJ-11.300	A11-S10

- NOTES**
1. To comply with BS 3288. Part. 1.
 2. For jointing instruction See M161. Notes.

E.S.I. STANDARD 43.92

Produced by Primary CAD

SCALE	-
DATE	28.7.82
DRAWN	R.A.C.
TRACED	S.J.H.
CHECKED	
APPROVED	H

CONDUCTOR COMPRESSION JOINT
TENSION
70mm² TO 400mm² ACSR CONDUCTOR

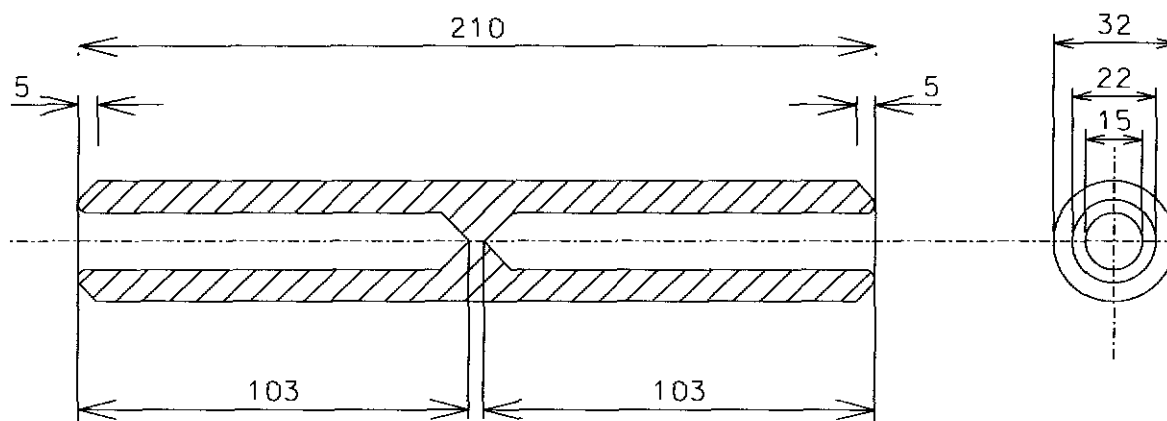
ISSUE	DATE	ISSUE	DATE
D	11.12.03	C	12.2.03
MINOR MODS		MOD OHLF ADDED	
M.J.B		PA	



Aquila

DRG. No.

M161



MEB REF	Compression Tool	Die Ref	N° of Crimps
M.169	G10 (Y35)	U 317	10

NOTE:-

To be pre-filled with PENETROX grease and capped to comply with BS 3288 Part 1.

JOINTING INSTRUCTIONS

1. Cut ends of conductor square.
2. Apply tape.
3. Wire brush conductor ends.
4. Insert conductor and compress overall commencing at centre of joint & work outwards.
5. Apply double wrap of Denso tape after compression

ESKAY'S REF

ME.169

Produced by
Primary CAD

SCALE	1-2
DATE	16.11.82
DRAWN	R. A. C
TRACED	
CHECKED	
APPROVED	

**NON-TENSION COMPRESSION
CONNECTOR 100mm² ACSR**

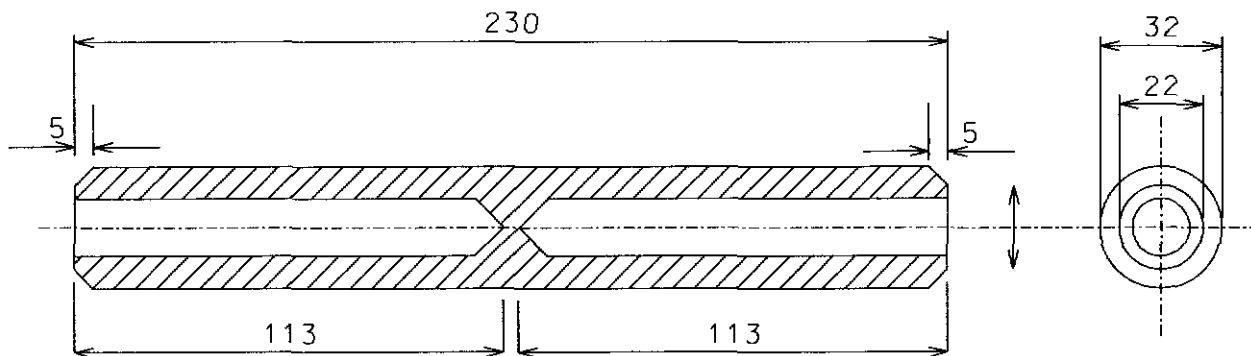
ISSUE	DATE
B	11.12.03
REDRAWN ON CAD	
MINOR MODS	
M. J. B	



Aquila

DRG. No.

M169



MEB REF	CONDUCTOR	DIMENSION 'A'	COMP TOOL	Die Ref	N° of Crimps
M.170/1	150mm ² ACSR	19	G10 (Y35)	U 317	12
M.170/2	175mm ² ACSR	21	G10 (Y35)	U 317	12

NOTE:-

To be pre-filled with PENETROX grease and capped to comply with BS 3288 Part 1.

JOINTING INSTRUCTIONS

1. Cut ends of conductor square.
2. Apply tape.
3. Wire brush conductor ends.
4. Insert conductor and compress overall commencing at centre of joint & work outwards.
5. Apply double wrap of Denso tape after compression

ESKAY'S REF

ME.170/1

ME.170/2

SCALE	N.T.S
DATE	11.11.83
DRAWN	R.A.C
TRACED	
CHECKED	
APPROVED	

NON TENSION COMPRESSION 150mm²
& 175mm² ACSR.

ISSUE	DATE
A	11.12.03
REDRAWN ON CAD	
MINOR MODS	
M.J.B	

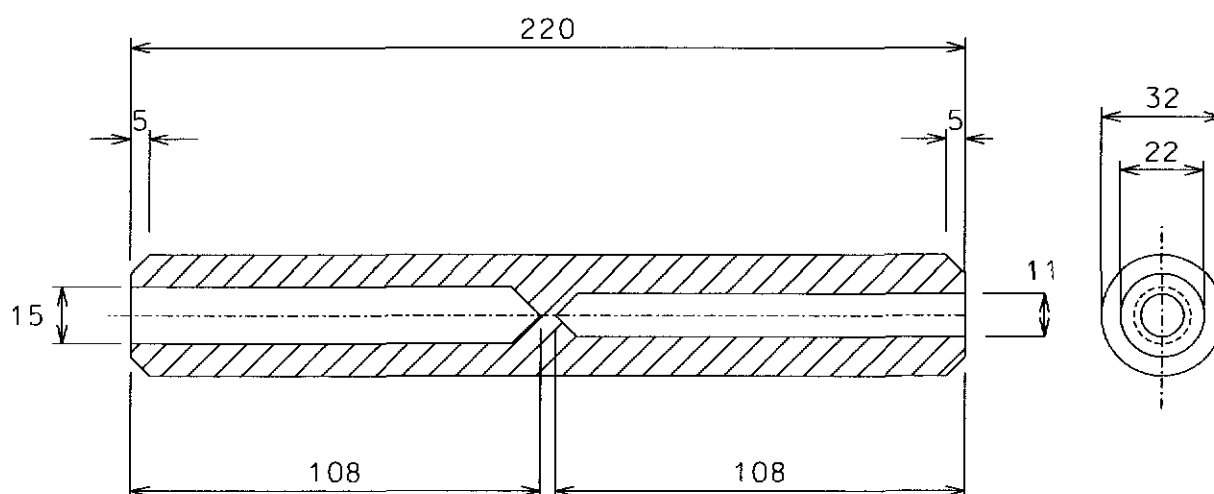
Produced by
Primary CAD



Aquila

DRG. No.

M170



MEB REF	Compression Tool	Die Ref	N° of Crimps
M.171	G10 (Y35)	U 317	10

NOTE:-

To be pre-filled with PENETROX grease and capped to comply with BS 3288 Part 1.

JOINTING INSTRUCTIONS

1. Cut ends of conductor square.
2. Apply tape.
3. Wire brush conductor ends.
4. Insert conductor and compress overall, commencing at centre of joint & work outwards.
5. Apply double wrap of Denso tape after compression.

ESKAY'S REF

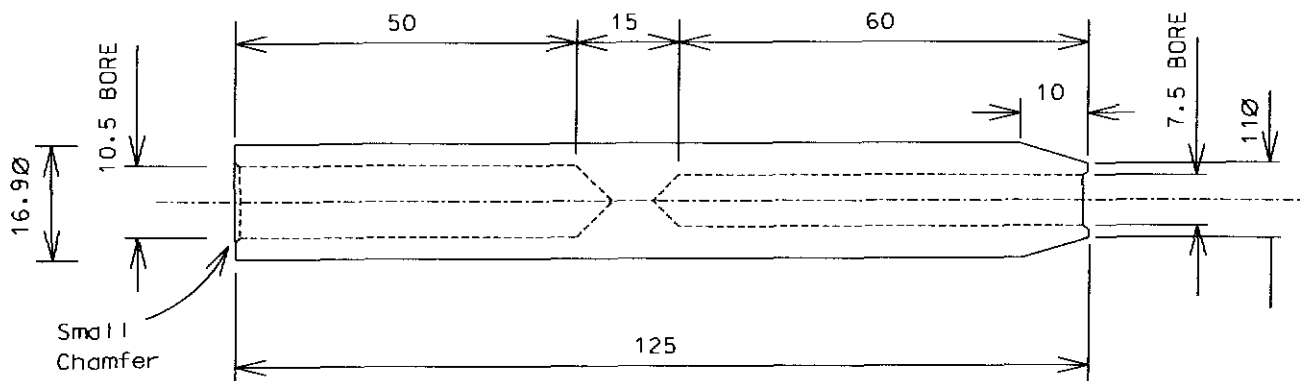
ME.171

SCALE	1-2
DATE	15.11.82
DRAWN	R.A.C
TRACED	
CHECKED	
APPROVED	

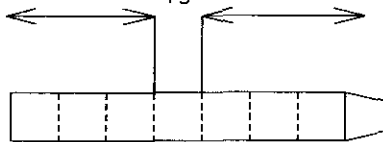
NON TENSION COPPER TO ALUMINIUM
COMPRESSION CONNECTOR 100 mm²
A.C.S.R. TO 70mm² COPPER.

ISSUE	DATE
B	11.12.03
REDRAWN ON CAD	
MINOR MODS	
M.J.B.	

Produced by Primary CAD
 Aquila
DRG. No. M171



3 compressions 15 3 compressions



Do not
Compress

Use U698 die
G10 (Y35) tool

Material :- aluminium alloy
fill with grit-loaded grease to
maximise mechanical performance

Mechanical Performance :- To meet the requirements of BS3288 (Pt1)
for tension joints.

ESKAY'S REF

ME.173

Produced by
Primary CAD

SCALE	-
DATE	26.02.91
DRAWN	B.B.
TRACED	G.S.
CHECKED	
APPROVED	

O/H LINE RECONDUCTORING TENSION JOINT FOR
CONNECTING NEW 50mm² AAAC (HAZEL) TO
OLD 25mm² AAAC (ALMOND) OR
16mm² HD COPPER (3/2.65)

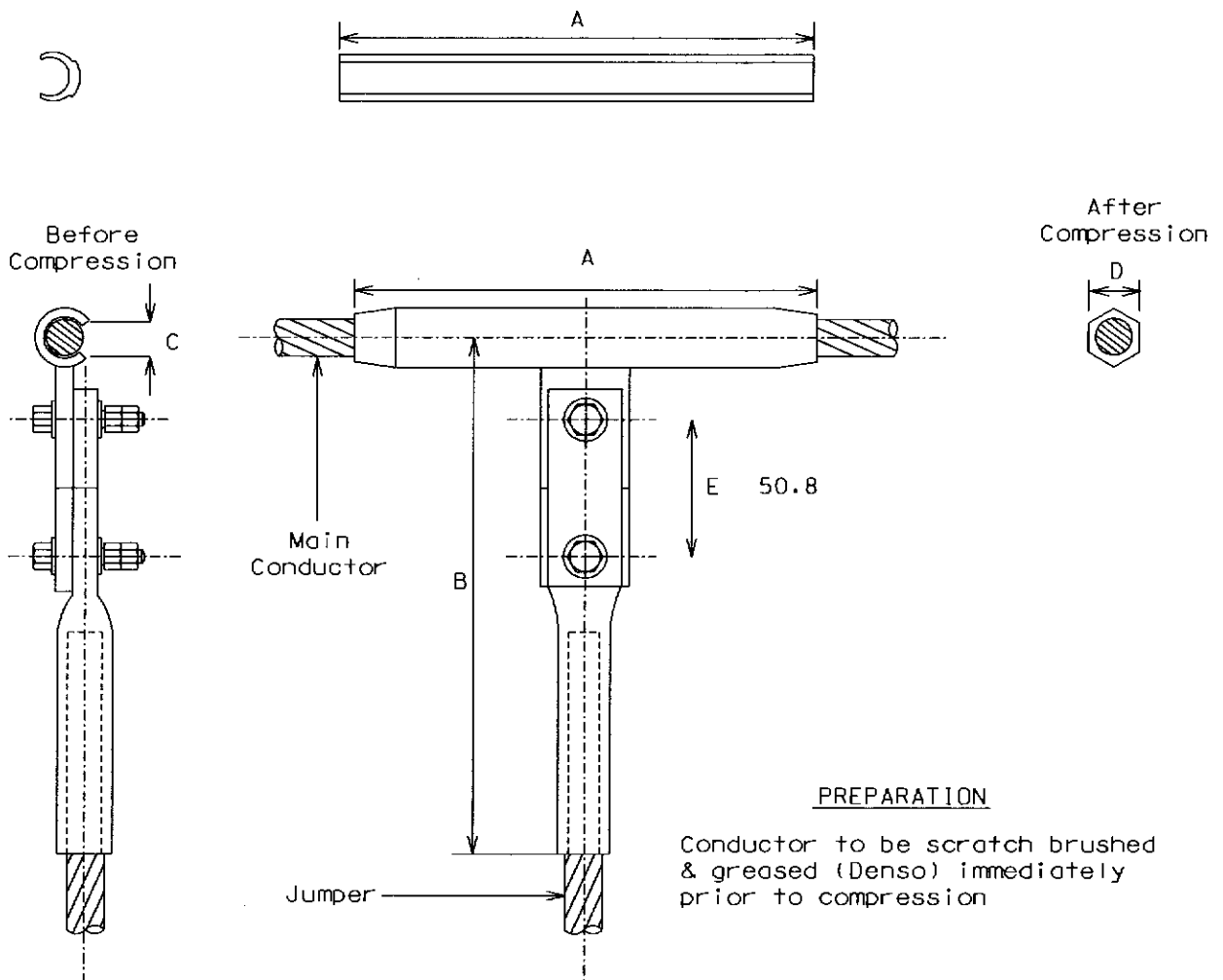
ISSUE	DATE
A	11.12.03
REDRAWN ON CAD	
MINOR MODS	
M.J.B	



Aquila

DRG. No.

M173



PREPARATION

Conductor to be scratch brushed & greased (Denso) immediately prior to compression

M TYPE	CONDUCTORS		ESKAY'S REF		DIMENSIONS						DIE REF
	MAIN	JUMPER	JOINT	JUMPER	BEFORE COMPRESSION			AFTER COMP	E	BOLT SIZE	
					A	B	C	D			
M.219/3	175 ACSR 200 AAAC	SIZE TO BE SPECIFIED WHEN ORDERING	LCTC- 5/9.21	SRJ- 5/9.21	300	242	21	32.33	51	M16	A-9
M.219/4	425 AAAC 400 ACSR 400 AAC		LCTC- 5/11.21	SRJ- 5/11.21	300	244	30	40.21	51	M16	A-11
M.219/8	300 AAAC		LCTC- 5/10.262	SRJ- 5/10.262	300	254	26	36.22	51	M16	A-10
M.219/9	500 AAAC		LCTC- 5/12.332	SRJ- 5/12.332	300	258	33	36.22	51	M16	A-12

SCALE	-
DATE	16.12.03
DRAWN	MJB
TRACED	-
CHECKED	23.12.03
APPROVED	KD

COMPRESSION 'T' JOINTS WITH JUMPER LUGS

ISSUE DATE

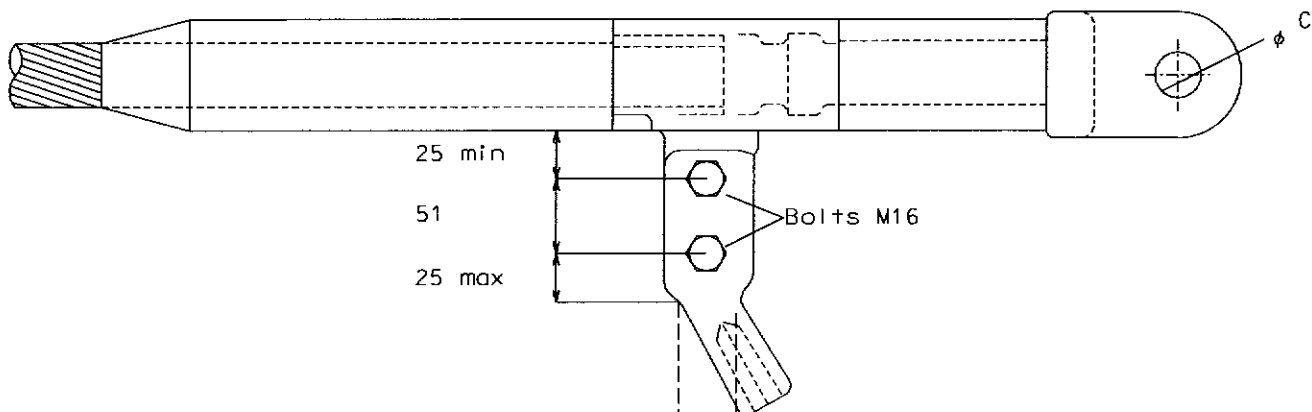
Produced by
Primory CAD



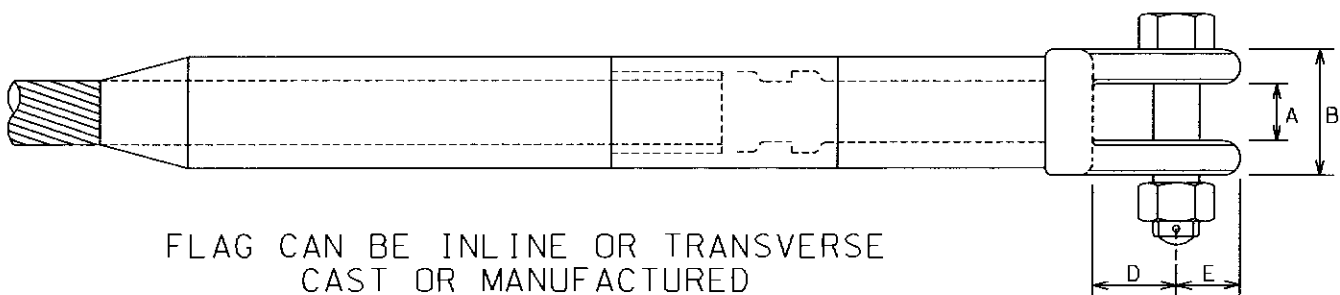
Aquila

DRG. No.

M219



Straight Jumper Terminal
Type 2



FLAG CAN BE INLINE OR TRANSVERSE
CAST OR MANUFACTURED

A min	B max	C Dia	D min	E max	BOLT TO BS 3288
43	70	29	48	59	M27x100

MEB .REF.	C.E.G.B. COMODITY CODE	CONDUCTOR		MANUFACTURER REF	DIE REF	
		SIZE	STRAND		ALUMINIUM	STEEL
253/2	BODY 4301123 TERM 4301783	400mm ² ACSR	54/3.18 AI 7/3.18 ST	CLNSR- 11.21	A11	S10

SCALE	-
DATE	12-12-03
DRAWN	MJB
TRACED	-
CHECKED	23-12-03
APPROVED	KD

CONDUCTOR DEAD-END 400mm²
ACSR CONDUCTORS

ISSUE DATE

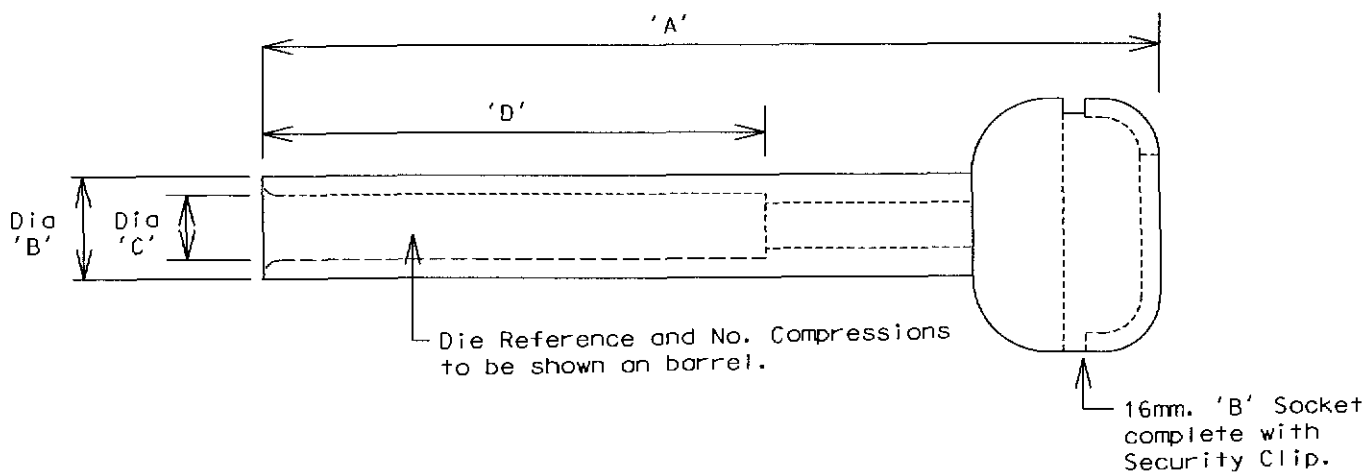
Produced by
Primary CAD



Aquila

ORG. No.

M253



- NOTES:-
1. Joint must meet the requirements of B.S.3288 Pt1.
 2. The joint must be compressed in accordance with the manufacturers instructions impressed on the sleeve.
 3. Ensure that the joint is filled with 'Penatrax' (AL Joints) or the inside face is treated with Berilium (Cu Joints) before Installation.

M. REF. No		CONDUCTOR SIZE		LENGTH A (mm)	DIA B (mm)	DIA C (mm)	DIA D (mm)	ESKAY'S REF	DIE REF	
		SIZE	STRANDING						MD0/MD6	G10/Y35
287/1	H.D. COPPER	16mm ²	3/2.65mm	165	10.0	5.95	65	CUSDN-0.059	W162	U162
		.025in ²	3/.104in							
287/2		32mm ²	3/3.75mm	180	13.0	8.50	80	CUSDN-1.083	W163	U163
		.05in ²	3/.147in							
287/3	H.D. COPPER	70mm ²	7/3.55mm	207	18.0	11.20	107	CUSDN-4.112	-	U165
		.10in ²	7/.136in							
287/4		50mm ²	7/3.00mm	197	15.9	9.50	97	CUSDN-3.095	-	U373
		.075in ²	7/.116in							
287/5	AL ALLOY	25mm ²	7/2.34mm	205	17.0	8.5	78	SDN-698085	W698	U698
		.025in ²	7/.092in							
287/6		50mm ²	7/3.30mm	229	22.0	11.5	102	SDN-659115	W659	U659
		.05in ²	7/.130in							
287/5	ACSR (S.C.A.)	25mm ²	6/2.35mm AL -1/2.36mm ST	205	17.0	8.5	78	SDN-698085	W698	U698
		.025in ²	6/0.93" AL -1/0.93" ST							
287/6		50mm ²	6/3.35mm AL -1/3.35mm ST	229	22.0	11.5	102	SDN-659115	W659	U659
		.05in ²	6/.132" AL -1/.132" ST							
287/9	ACSR (S.C.A.)	100mm ²	7/1.57mm ST -6/4.72mm AL	289	25.4	15.9	156	SDN-654159	-	U654
		.10in ²	7/0.62" ST -6/.185" AL							

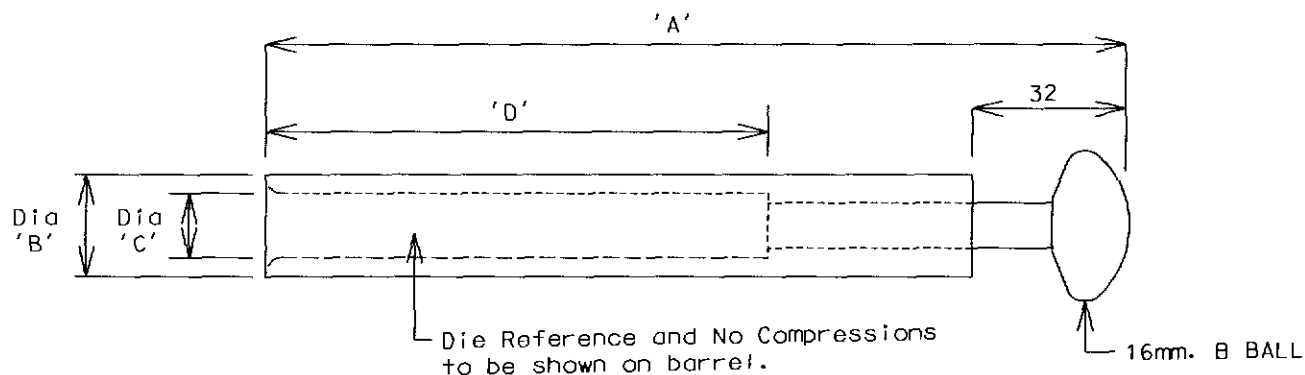
Dimensions are in millimetres unless otherwise stated.

SCALE	N.T.S.
DATE	27.11.73
DRAWN	A.R.B
TRACED	D.F.B
CHECKED	-
APPROVED	-

CONDUCTOR TERMINATION COMPRESSION SOCKET DEAD ENDCOPPER, AL ALLOY & ACSR CONDUCTORS

ISSUE	DATE
B	16.12.03
MINOR MODS	
REDRAWN ON CAD	
M.J.B	

Produced by Primary CAD	
Aquila	
DRG. No.	M287



NOTES:-

1. Joint must meet the requirements of B.S.3288 Pt1.
2. The joint must be compressed in accordance with the manufacturers instructions impressed on the sleeve.
3. Ensure that the joint is filled with 'Penatrax' (AL Joints) or the inside face is treated with Berilium (Cu Joints) before Installation.

M.REF.No	CONDUCTOR SIZE		LENGTH A (mm)	DIA B (mm)	DIA C (mm)	DIA D (mm)	ESKAY'S REF	DIE REF	
	SIZE	STRANDING						MH100/MD6	G10/Y35
288/5	25mm ²	7/2.34mm	187	17.0	8.5	78	698085	W698	U698
	.025in ²	7/.092in							
288/6	50mm ²	7/3.30mm	211	22.0	11.5	102	659115	W659	U659
	.5in ²	7/.130in							
288/9	100mm ²	7/4.65mm	271	25.4	15.9	156	654159	NOT TO BE USED	U654
288/5	25mm ²	6/2.36mm AL + 1/2.36mm ST	187	17.0	8.5	78	698085	W698	U698
	.025in ²	6/.093in AL + 1/.093in ST							
288/6	50mm ²	6/3.35mm AL + 1/3.35mm ST	211	22.0	11.5	102	659115	W659	U659
	.5in ²	6/.132in AL + 1/.132in ST							
288/9	100mm ²	7/1.57mm ST + 6/4.72mm AL	271	25.4	15.9	156	654159	NOT TO BE USED	U654
	.10in ²	7/.062in ST + 6/.186in AL							

SCALE	-
DRAWN	11.11.99
TRACED	-
CHECKED	-
APPROVED	-

CONDUCTOR TERMINATION
COMPRESSION BALL DEAD END
ACSR & AAAC

ISSUE	DATE
A	11.12.03
REDRAWN ON CAD	
MINOR MODS	
M.J.B	

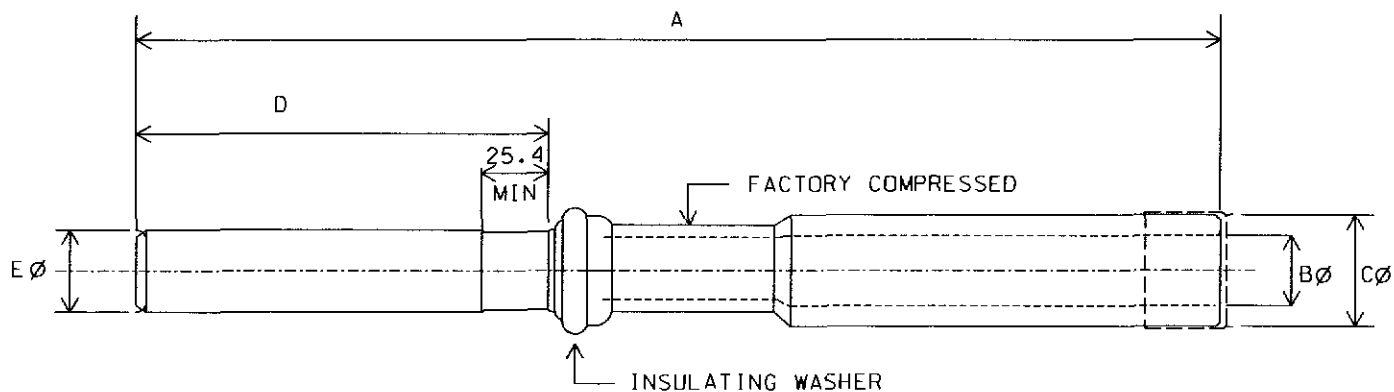


Aquila

DRG. No.

M288

Produced by
Primory CAD



MATERIAL:-

PIN - TELLURIUM
 INSULATING WASHER - NEAPRENE
 SLEEVE - ALUMINIUM 99.5% PURITY

MEB .REF.	CONDUCTOR		A	B	C	D	E	ESKAY'S LBPT-	DIE REF
	SIZE	STRAND						MDN REF	
327/0	175mm ² ACSR 200mm ² AAAC	30/2.79AL - 7/2.79ST 37/2.87	319	21.83	38.1	152	25.4	9.21	A9
327/2	400mm ² ACSR 425mm ² AAAC	54/3.18AL - 7/3.18ST 37/4.14	338	30.5	48	153	35	11.21	A11
327/4	300mm ² AAAC	37/3.53	324	28.8	43	165	35	10.262	A10

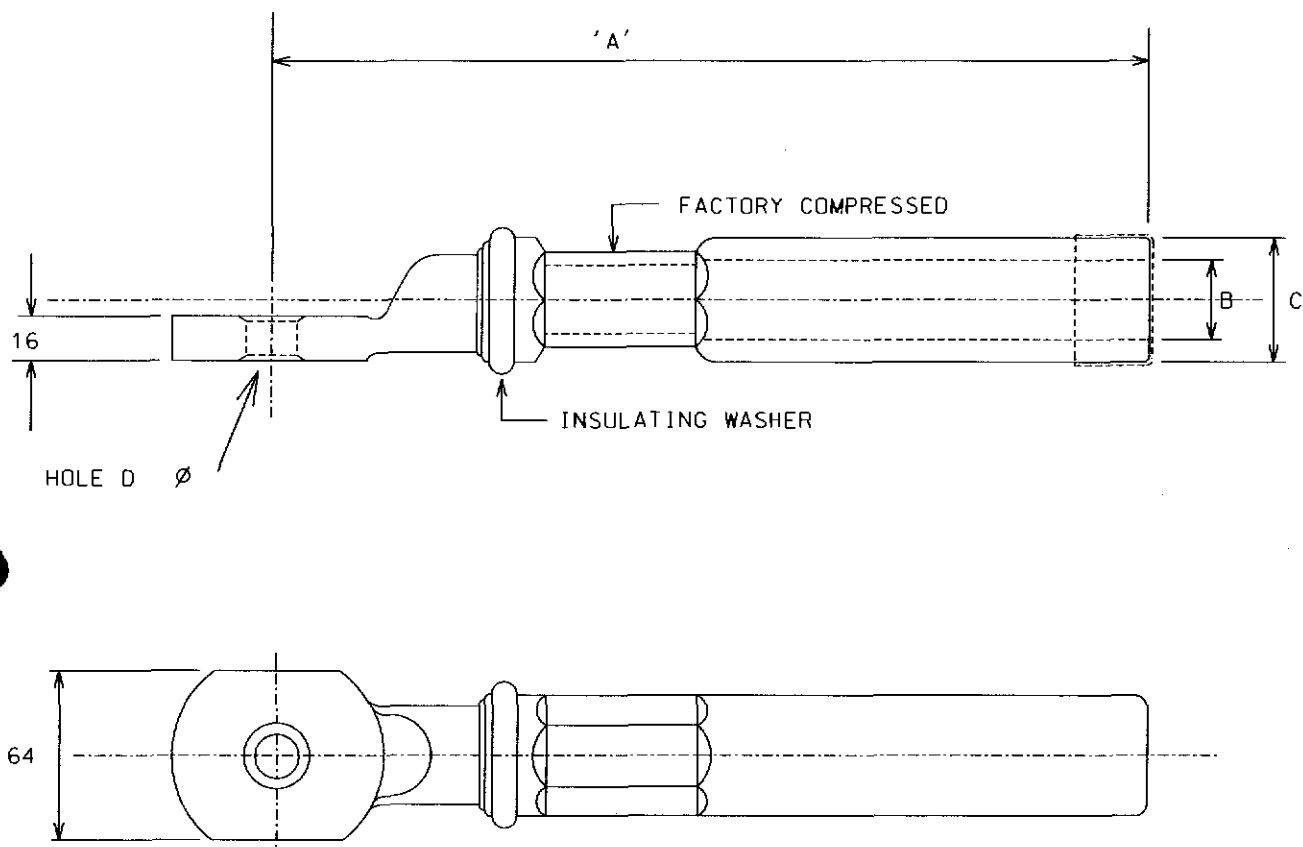
Produced by
Primary CAD

SCALE	
DATE	11/11/99
DRAWN	
TRACED	
CHECKED	
APPROVED	

**BI-METAL COMPRESSION
 TERMINAL PIN TYPE**

ISSUE	DATE
A	14.12.03
MINOR MODS	
PA/KD	

 Aquila	DRG. No.
	M327



MATERIAL:-

PALM - BRONZE BS.1400.LG 2-C

INSULATING WASHER - NEOPRENE

SLEEVE - ALUMINIUM 99.5% PURITY

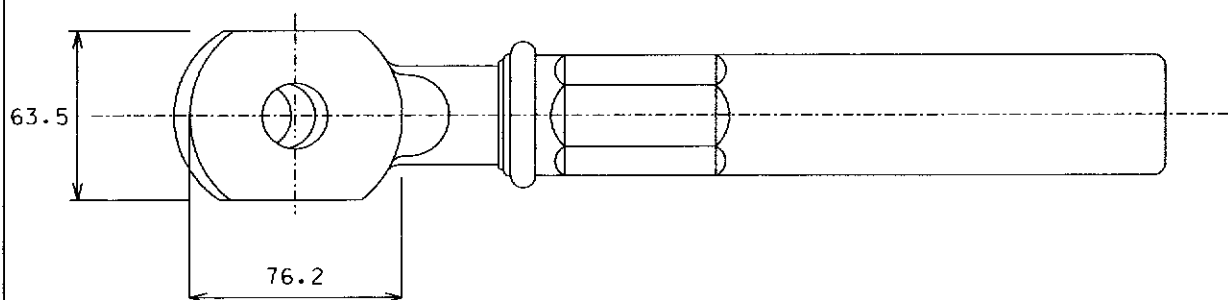
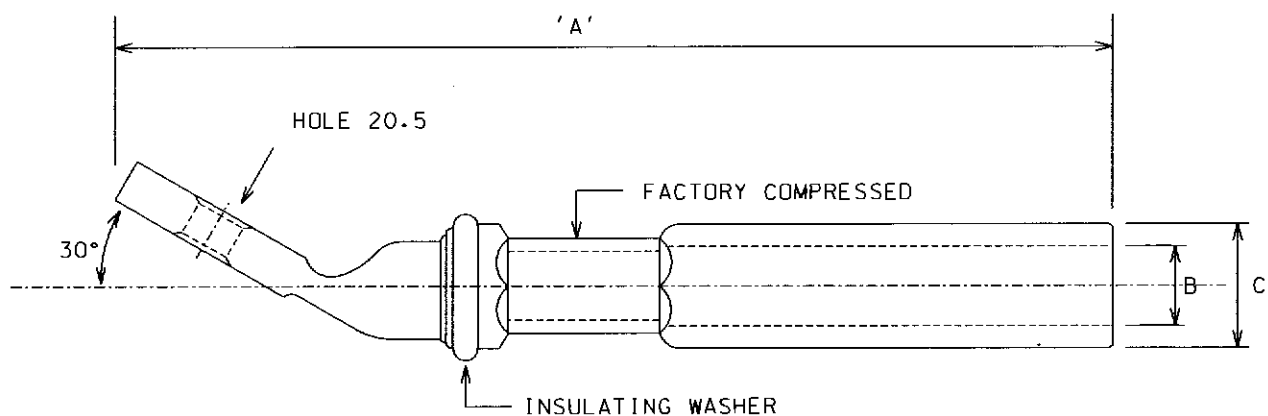
MEB REF.	CONDUCTOR		ESKAY'S		LENGTH A	B Dia	C Dia	DIE REF
	SIZE	STRAND	MDN REF	D				
328/0	175mm ² ACSR 200mm ² AAAC	30/2.79AL-7/2.79ST 37/2.87	LBT- 9.21	22	228	21.8	38	A9
328/2	400mm ² ACSR	54/3.18AL 7/3.18ST	LBT- 11.21	22	255	30.5	48	A11
328/4	300mm ² AAAC	37/3.53	LBT- 10.262	22	227	26.5	43	A10

SCALE	-
DATE	17.02.03
DRAWN	PA
TRACED	-
CHECKED	KD
APPROVED	11.03.03

**BI-METAL COMPRESSION
TERMINAL STRAIGHT PALM**

ISSUE	DATE
A	08.12.03
MINOR MODS	
M.J.B	

Produced by Primory CAD	
 Aquila	
DRG. No.	
M328	



MATERIAL:-

PALM - BRONZE BS.1400.LG 2-C

INSULATING WASHER - NEOPRENE

SLEEVE - ALUMINIUM 99.5% PURITY

MEB REF.	CONDUCTOR		REF NO	LENGTH A	B Dia	C Dia	DIE REF
	SIZE	STRAND	ESKAY'S				
329/0	175mm ² ACSR 200mm ² AAAC	30/2.79AL-7/2.79ST 37/2.87	LBT- 9.21/30°	236	21.83	38.10	A9
329/2	400mm ² ACSR	54/3.18AL 7/3.18ST	LBT- 11.21/30°	287	30.16	47.63	A11
329/3	400mm ² AAAC	37/3.78	LBT- 11.20/30°	284	28.58	47.63	A11
329/4	300mm ² AAAC	37/3.53	LBT- 10.262/30°	258	26.50	42.86	A10

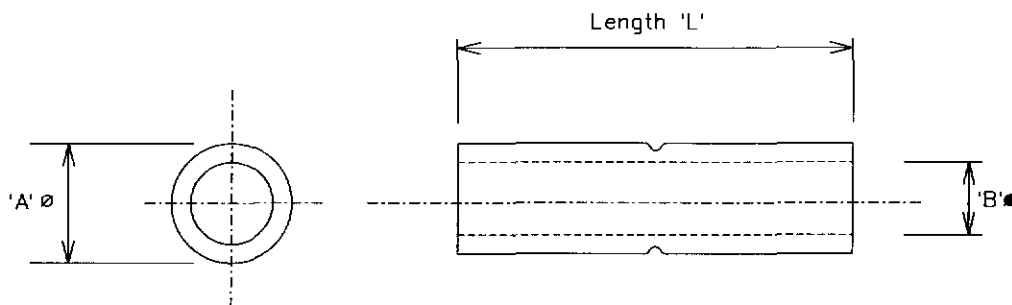
Produced by
Primary CAD

SCALE	-
DATE	11.11.99
DRAWN	-
TRACED	-
CHECKED	-
APPROVED	-

BI-METAL COMPRESSION
TERMINAL 30° PALM

ISSUE	DATE
A	12.12.03
MINOR MODS	
M. J. B.	

 **Aquila**
DRG. No.
M329



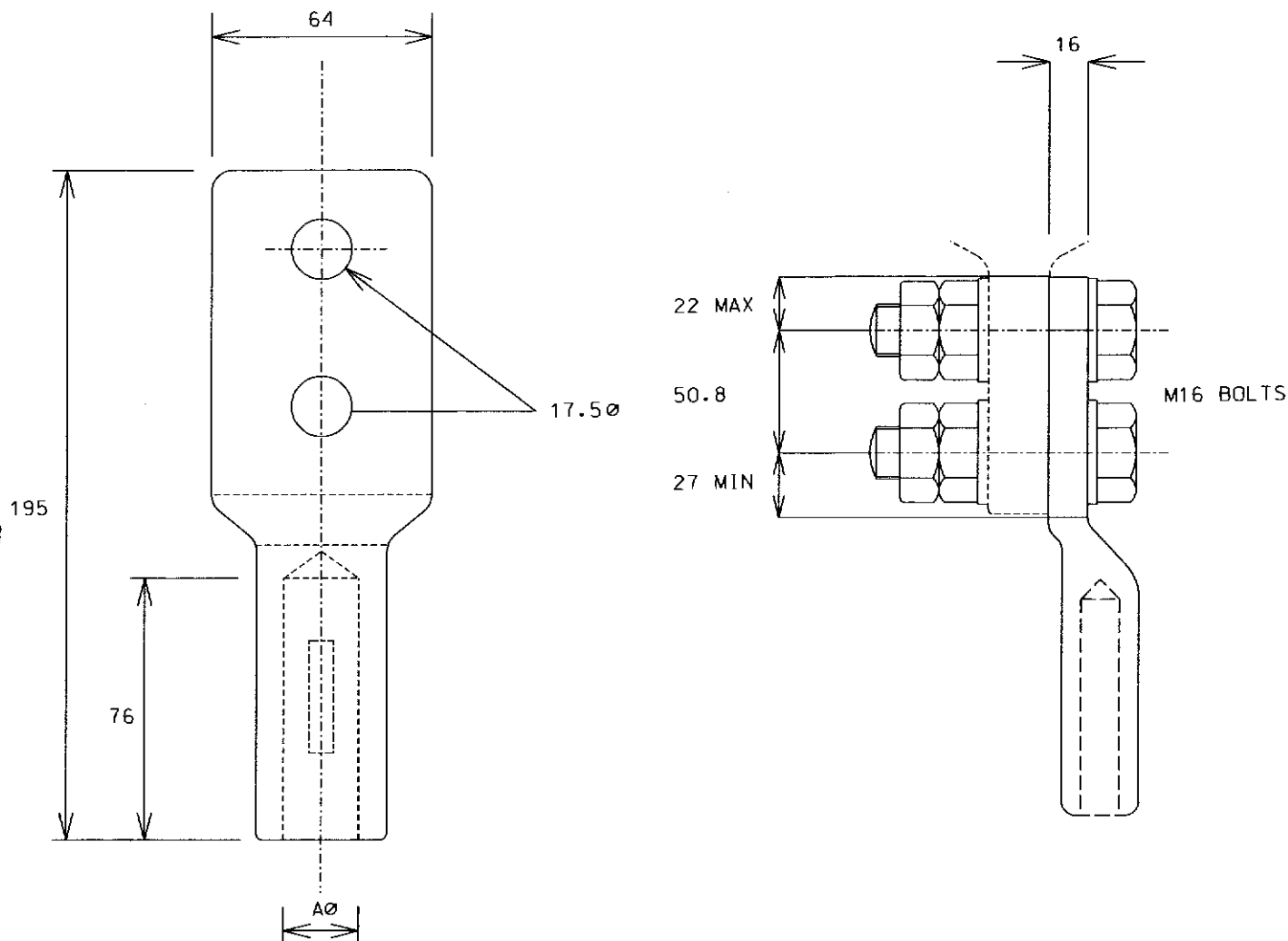
M REF.	C.E.G.B.	CONDUCTOR SIZE		ESKAY'S	DIMENSIONS			DIE REF
		SIZE	STRAND		A	B	Length L	
M330/0	4302230	175mm ² ACSR	30/2.79AL 7/2.79ST	LNTJ-9.21	38.1	21.8	178	A9
M330/4	4302235	400mm ² ACSR	54/3.18AL 7/3.18ST	LNTJ-11.21	47.6	30.2	178	A11

SCALE	—
DATE	16.11.82
DRAWN	R.A.C
TRACED	
CHECKED	
APPROVED	

CONDUCTOR COMPRESSION JOINT
NON-TENSION JOINT
175 & 400mm² ACSR

ISSUE	DATE
C	11.12.03
MINOR MODS	
M.J.B	

Produced by Primary CAD	
	Aquila
DRG. No. M330	



M.REF.	C.E.G.B. COMMODITY CODE	CONDUCTOR SIZE		MANUFACTURER REF ESKAY'S	AØ	DIE REF
		SIZE	STRAND			
333/7	43-01-775	175mm ² ACSR	30/2.79Al 7/2.79St	LSRJ-9.21	21.83	A9
333/8	-	150mm ² ACSR	30/2.59Al 7/2.59St	LSRJ-9.204	20.24	A9
333/9	-	70mm ² ACSR	12/2.79Al 7/2.79St	LSRJ-9.178	17.86	A9

NOTES:-

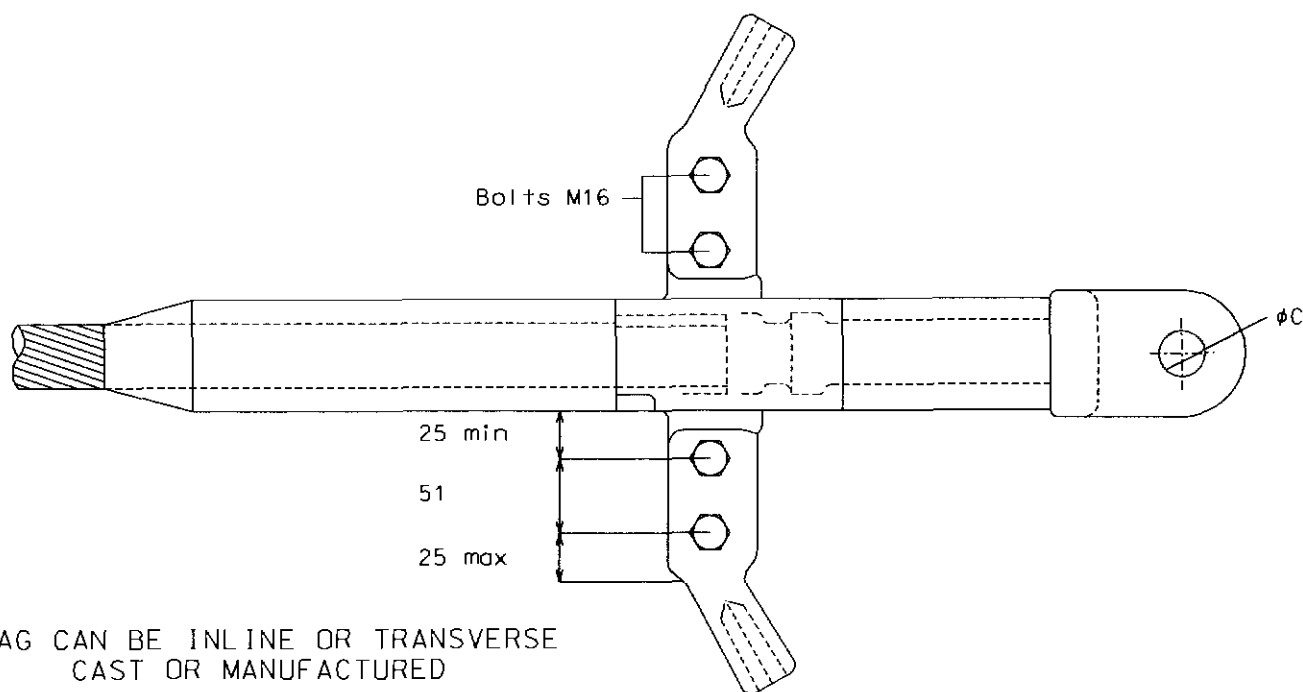
1. See M 159 notes for compression procedure.
2. Terminal supplied with Nuts, Bolts and Washers.

SCALE	-
DATE	-
DRAWN	-
TRACED	-
CHECKED	-
APPROVED	-

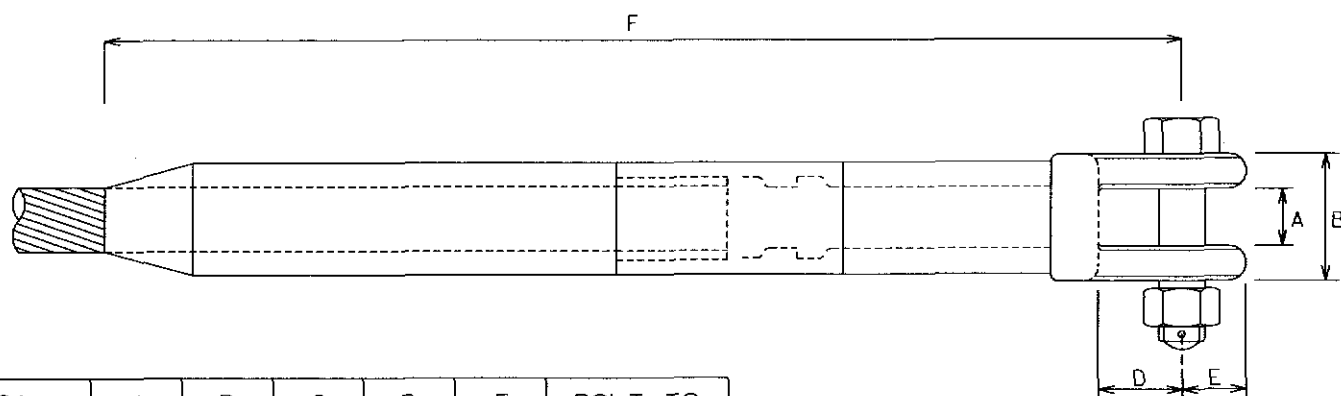
CONDUCTOR DEAD-END JUMPER
TERMINAL STRAIGHT FOR
ACSR CONDUCTORS

ISSUE	DATE
A	11.12.03
REDRAWN ON CAD MINOR MODS	
M.J.B	

Produced by Primary CAD	
 Aquila	
DRG. No.	
M333 sh+2	



FLAG CAN BE INLINE OR TRANSVERSE
CAST OR MANUFACTURED



Size mm	A min	B max	C Dia	D min	E max	BOLT TO BS 3288
100 150 175	33	59	27	35	33	M24x100
400	41	70	29	48	59	M27x110

M. REF.	CONDUCTOR SIZE		MANUFACTURER REF		DIMENSIONS (mm)					DIE REF
	SIZE	STRAND	ESKAY'S	LENGTH F	AØ	BØ	CØ	DØ	EØ	
335/1	175mm ²	30/2.79 7/2.79	TCLNR- 9.21	360	35	55	27	40	28	A9 S9
335/2	400mm ²	54/3.18 7/3.18	TCLNR- 11.21	420	41	71	29	51	29	A11 S10
335/3	100mm ²	26/4.72 7/1.78	TCLNR- 9.180	341	35	55	27	40	28	A9 S9
335/4	150mm ²	30/2.59 7/2.59	TCLNR- 9.204	358	35	55	27	40	28	A9 S9

Produced by
Primary CAD

SCALE	-
DATE	11-12-03
DRAWN	MJB
TRACED	-
CHECKED	23-12-03
APPROVED	KD

CONDUCTOR DEAD-END DOUBLE
JUMPER CONNECTION 100mm², 400mm²
ACSR CONDUCTORS

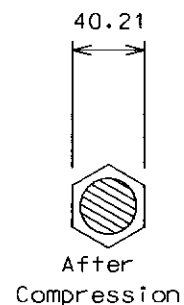
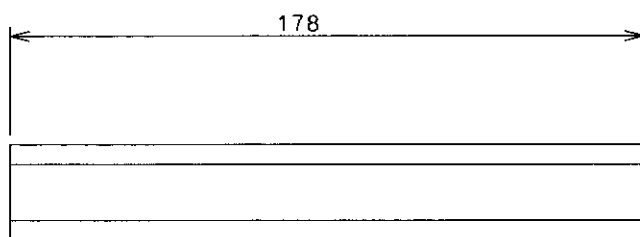
ISSUE DATE



Aquila

DRG. No.

M335



MATERIAL:- 1.875" (47.6) Dia 99.5% purity Aluminium to Specification E.1B BS. 1476.

PREPARATION:- Conductor to be Cleaned & Greased ('Denso') immediately prior to Compression.

COMPRESSION:- Hydraulic Compression Die No. A-11.

ESKAY'S REF
LANTJ-11.20

SCALE	N. T. S.
DATE	16.09.63
DRAWN	-
TRACED	L. H.
CHECKED	-
APPROVED	W. A. H.

NON-TENSION COMPRESSION JOINT
0.40° Cu EQUIV (400mm²)
ALUMINIUM CONDUCTORS

ISSUE	DATE
A	12.12.03
REDRAWN ON CAD MINOR MODS	
M. J. B.	

Produced by Primary CAD	
	Aquila
DRG. No. M243	

SECTION 8

MIN NUMBER	M	DESCRIPTION
3451200	120	NON-TENSION ALUMINIUM JOINT
3451350	135	NON-TENSION ALUMINIUM TO COPPER JOINT
3451690	169	NON-TENSION CONNECTOR 100mm ACSR
3451700	170	NON-TENSION CONNECTOR 150 & 175mm ACSR
3451710	171	NON-TENSION CONNECTOR 100mm ACSR TO 70mm COPPER
3451720	172	NON-TENSION CONNECTOR COPPER TO ALUMINIUM
3452060	206	NON-TENSION CONNECTOR COPPER
3452430	243	NON-TENSION CONNECTOR 0.4 COPPER EQUIV 400 ALUM
3452630	263	NON-TENSION CONNECTOR 240 COPPER TO 400 SCA
3453300	330	NON-TENSION CONNECTOR 175 & 400mm ACSR
3454840	484	INSULATED TWINSHIELDS

FIG I

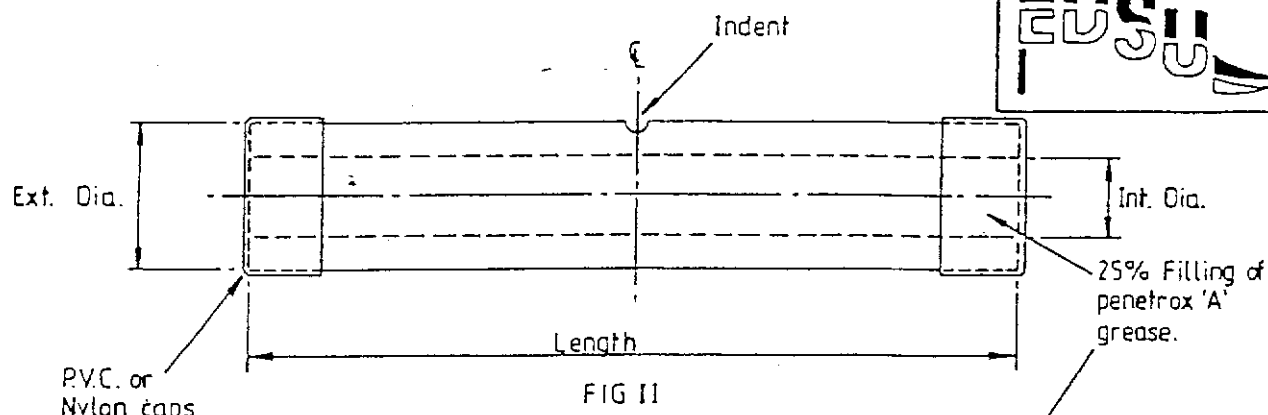
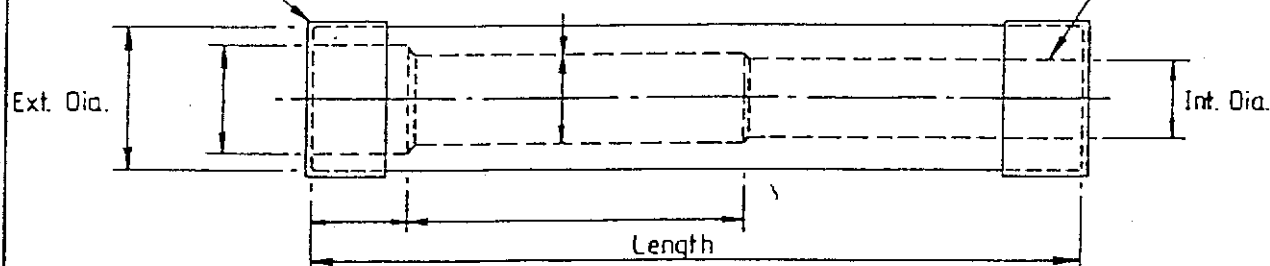


FIG II



MEB Ref	Conductor Sizes (mm ²)	Fig No.	Manufacturer Ref.		Dimensions (mm)			Die/No. of Compressions				60 Ton No.
					Length	Int. Dia.	Ext. Dia.	MD6	No.	Y35/G10	No.	
	25 AAAC											
120/1	25 AI	I						W 247		U 247		N/A
	25 ACSR											
	50 AAAC											
120/2	50 AI	I						W 247		U 247		N/A
	50 ACSR											
120/3	100 AI	I						W 247		U 247		N/A
	50 AAAC											
120/4	to Pilot Bar	I						W 247		U 247		N/A
	50 Aluflex											
120/5	to Pilot Bar	I						NOT		U 247		N/A
	150 AAAC											
	50 Aluflex											
120/6	to 25 AAAC	II						TO		U 247		N/A
	100 AAAC											
120/7	100 ACSR	I						BE		U 654		N/A
120/8	200 AAAC	I						USED		N/A		A-9
120/9	300 AAAC	I						N/A		N/A		A-10
120/0	500 AAAC	I						N/A		N/A		A-12

B. 10000 BY A.J.B/P.W.
DATE 4-5-94.
Table revised.
Refracted.

TITLE

NON-TENSION ALUMINIUM
COMPRESSION JOINT

SCALE

DATE

DRAWN

TRACED

CHECKED

APPROVED

S-6-90

A.J.B.

P.J.W.

MEB

Midlands Electricity plc

DRG No

M. 120

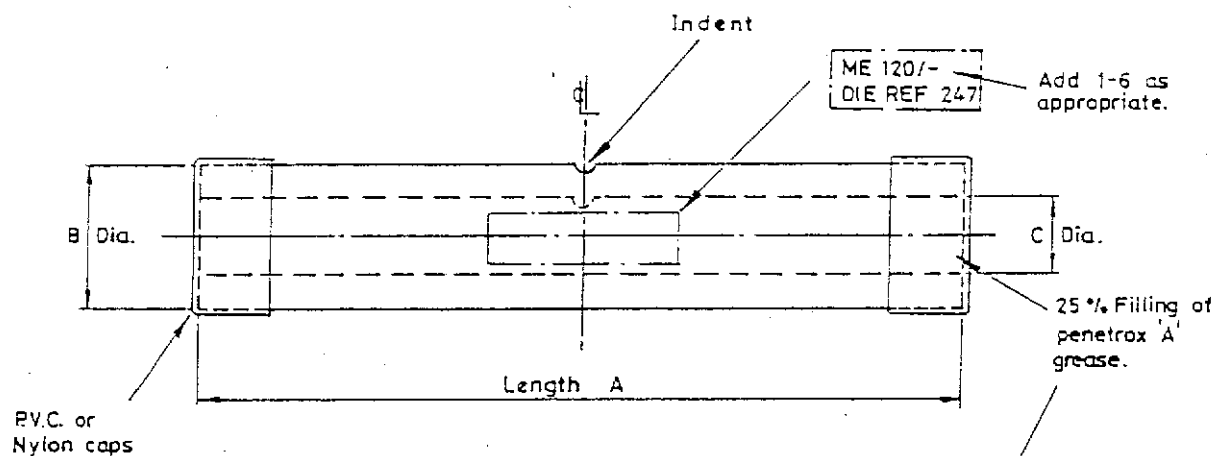


FIG I

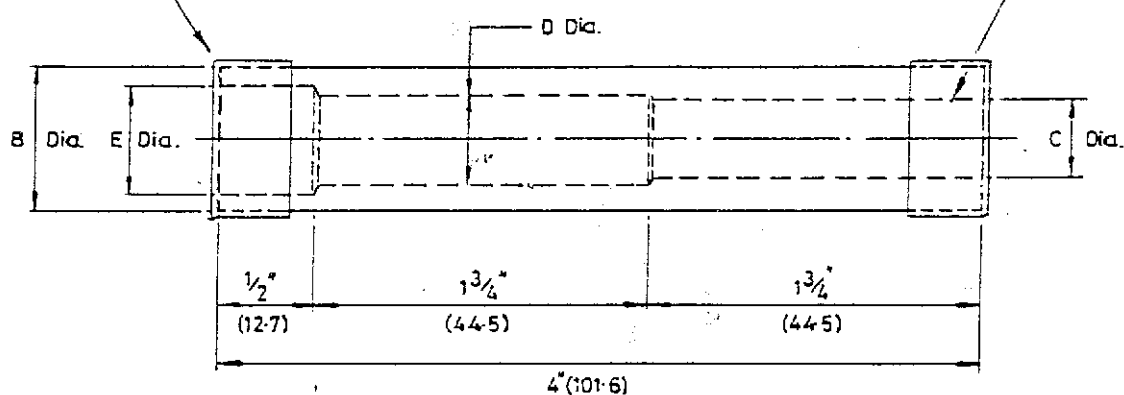


FIG II

JOINT No	CONDUCTOR SIZE in ² (mm ²)	FIG No	DIMENSIONS in (mm)					COMPRESSION DIE DETAILS	
			A	B Dia	C Dia	D Dia	E Dia	HAND TOOL	HYDRAULIC TOOLS
								BICCON MH100 (MD6)	BICCON G10 (Y35)
120/1	.025 (25) Aluminium .025 (25) Al Alloy	I	2 3/4 (69)	.75 (19.05)	.290 (7.36)	—	—	W 247	U 247
120/2	.05 (50) Aluminium .05 (50) Al Alloy		4 (101.6)		13/32 (10.32)	—	—		
120/3	.10 (100) Aluminium		5 (127)		.523 (13.5)	—	—		
120/4	Pilot Bar to .05 (50) Al Alloy.		4 (101.6)		13/32 (10.32)	—	—		
120/5	Pilot Bar .05 (50) AlA to .05 (50) Aluflex.	II	4 (101.6)		13/32 (10.3)	15/32 (11.9)	9/16 (14.3)	NOT TO BE USED	
120/6	.025 (25) Al Alloy to .05 (50) Aluflex.				.290 (7.36)	15/32 (11.9)	9/16 (14.3)		

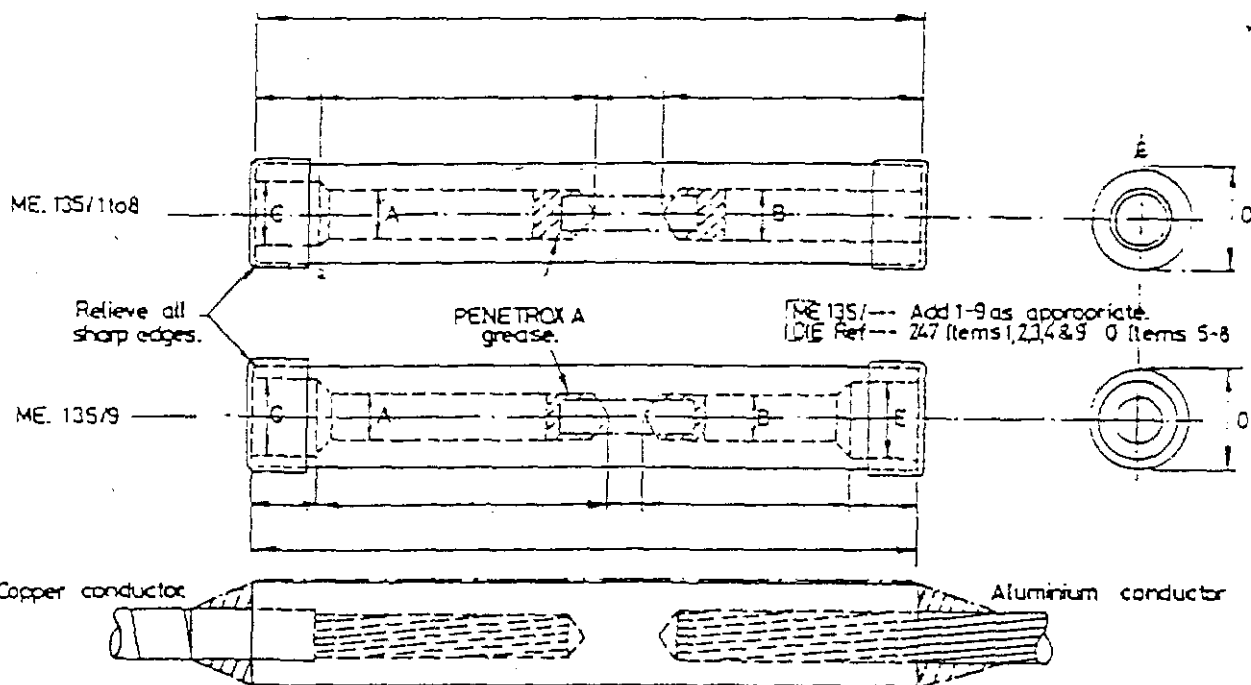
PREPARATION Apply Denso Paste and scratch brush conductor through paste.

MATERIAL Fully Annealed 99.5 % Purity Aluminium

SCALE	1/1
DATE	20.3.70.
DRAWN	W.D.H.
TRACED	T.R.P.
CHECKED	
APPROVED	

NON-TENSION ALUMINIUM COMPRESSION JOINT

ISSUE	MIDLANDS ELECTRICITY BOARD
I	
Comp'n dtls. revised	DRG. No.
J.W. 3/84.	ME.120.



TYPE N°	CONDUCTOR AND SIZE		DIAMETERS					COMPRESSION DIE DETAILS	
								HAND TOOL	HYDRAULIC
	Copper		Aluminium		A	B	C	D	E
	in	mm	in	mm					
135/1	.0225	14	.025	25					
	.025	16	.025AC	25					
135/2	.05	32	.05ALAI	50					
	.058	32	.05AI	50					
	.06	35	.05ACSP	50					
135/3	.07	36	.07	55					
	.08	38	.08	50					
135/4	.1	25	.1	100					
135/5	.2	125	.05	50					
135/6	.1	25	.1	100					
135/7	.15	100	.1	25					
135/8	.1	25	.05	50					
135/9	.05	32	.05	50 Alflex					
	.05	35cuflex							

Notes:—

Material:—

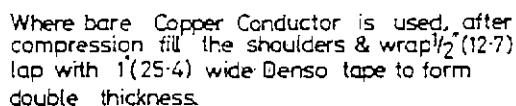
99.5% purity Aluminium
rod to specification E18
BS 1476.

1. The connector must be arranged so that moisture cannot drain from the copper conductor on to the aluminium sleeve.
2. Greasing. Before making the joint partially fill the connector and coat the conductors with Densol paste.

SCALE	FULL
DATE	20-11-79
DRAWN	RAC
TRACED	
CHECKED	
APPROVED	

NON-TENSION COMPRESSION
CONNECTOR FOR COPPER
TO ALUMINIUM JOINTS.

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
	0	
	One Ret Added. H.P.S. 2-4-84	DRG. No. ME 135.

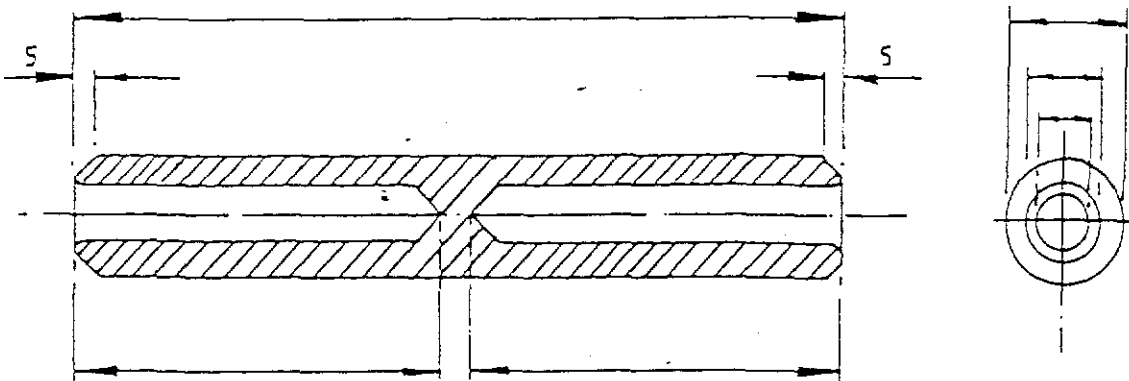


TYPE Nº	CONDUCTOR AND TYPE SIZE				DIAMETERS					COMPRESSION DIE DETAILS	
	Copper		Aluminium		A	B	C	D	E	HAND TOOL	HYDRAULIC
	in	mm	in	mm						MH100 (MD6)	G10(Y35)
135/1	.0225 .025	14 16	.025 .025AC	25 25	15/64 (5.55)	9/32 (7.14)	21/64 (8.33)	3/4 (19.05)		W247	U247
135/2	.05 .058 .06	32 32 35	.05ALAI .05AI .05ACSR	50 50 50	11/32 (8.73)	13/32 (10.3)	29/64 (11.51)	"		W247	U247
135/3	.10 .1019	136 (108)	.05	50	7/16 (11.1)	25/32 (9.92)	37/64 (14.68)	"		W247	U247
135/4	"	"	.10	100	"	17/32 (13.49)	"	"		W247	U247
135/5	.2	125	.05	50	5/8 (15.86)	25/64 (9.92)	27/32 (21.43)	1" (25.4)		0	U0
135/6	"	"	.10	100	5/8 (15.86)	17/32 (13.49)	"	"		0	U0
135/7	.15	100	.10	"	33/64 (13.1)	17/32 (13.49)	3/4 (19.05)	"		0	U0
135/8	"	"	.05	50	33/64 (13.1)	25/64 (9.92)	"	"		0	U0
135/9	.05 .05	32 35	.05	50 Allflex	11/32 (8.73)	15/32 (11.9)	29/64 (11.51)	3/4 (19.05)	9/16 (14.3)	W247	U247

Material:—

1. The connector must be arranged so that moisture cannot drain from the copper conductor on to the aluminium sleeve.
2. Greasing. Before making the joint partially fill the connector and coat the conductors with 'Oenso' paste.
3. With P.V.C. insulated conductors remove 2"(50.8) of insulation or 2(50.8) & 1 3/4"(39.45) as for 135/9. Insert conductor so that insulation enters to a depth of 1/2"(12.7) and compress over the insulation when making the joint.

SCALE	FULL	NON-TENSION COMPRESSION CONNECTOR FOR COPPER TO ALUMINIUM JOINTS.	ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	20-11-79			0	
DRAWN	RAC				
TRACED					
CHECKED					
APPROVED					
			Dis. Ref. Added. H.P.S. 2-4-84	ORG. No. ME 135.	



MEB Ref	Compression Tool	Die Ref	Nº of Crimps
M.169	G 10 (Y35)	U 317	10

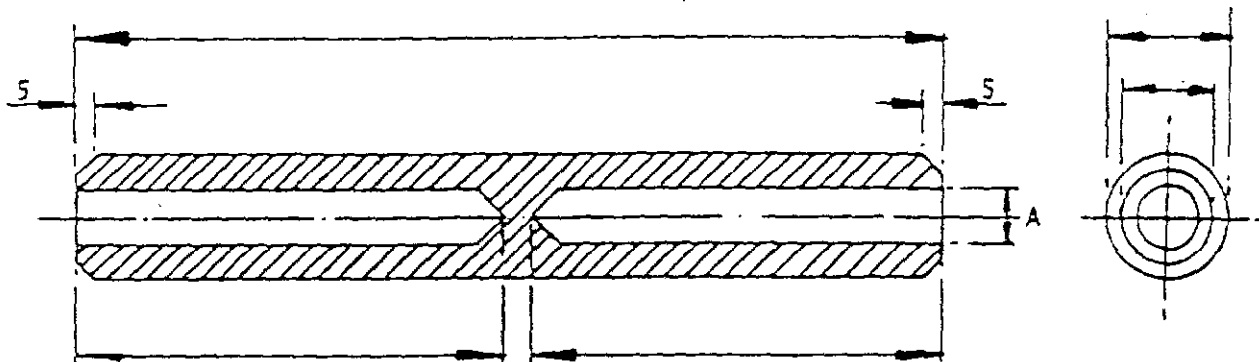
NOTE:—

To be pre-filled with PENETROX grease and capped to comply with BS 3288 Part 1.

JOINING INSTRUCTIONS

1. Cut ends of conductor square.
2. Apply tape
3. Wire brush conductor ends.
4. Insert conductor and compress overall commencing at centre of joint & work outwards.
5. Apply double wrap of Denso tape after compression.

SCALE	1-2	NON-TENSION COMPRESSION CONNECTOR 100mm ² ACSR.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	16-11-82		A	
DRAWN	RAC		Comp. Tool Revised	DRG. No.
TRACED			I.W. 3/84	M.169
CHECKED	H			
APPROVED	S			
C. D. W Ltd. 3297 /		E.S.I STANDARD 43-42	E.S.I Drg Nº	



TYPE	CONDUCTOR	DIMENSION 'A'	COMP TOOL	DIE REF	Nº OF CRIMPS
M170/1	150mm ² ACSR.		G 10 (Y35)	U317	12
M170/2	175mm ² "		G 10 (Y35)	"	11

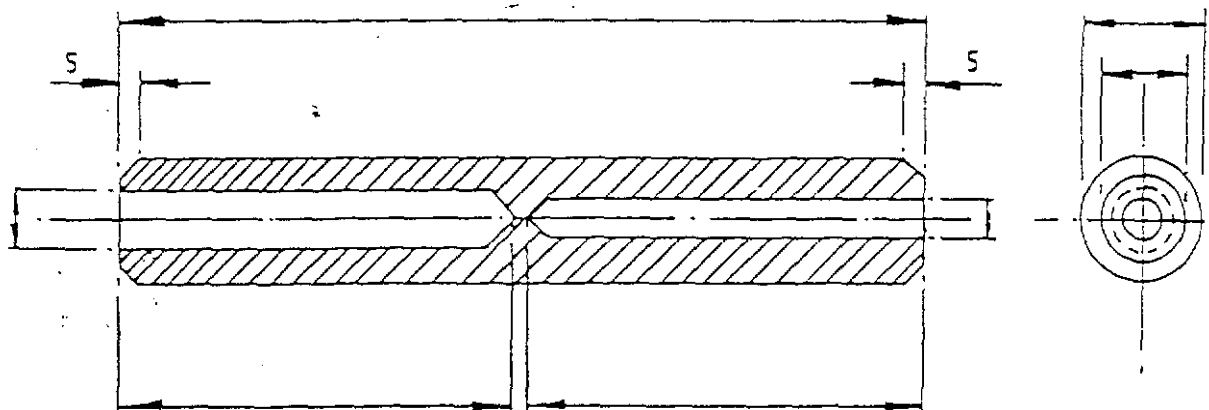
NOTE:—

To be pre-filled with PENETROX grease and capped to comply with BS 3288 Part 1.

JOINING INSTRUCTIONS

- 1 Cut ends of conductor square.
- 2 Apply tape
- 3 Wire brush conductor ends.
- 4 Insert conductor and compress overall.
- 5 Apply double wrap of 'Densa' tape after compression.

SCALE	M.C.S.	NON-TENSION COMPRESSION 150mm ² & 175mm ² ACSR.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	11.11.83		C	
DRAWN	RAC		Dimension Alteration	ORG. No.
TRACED			RAC 13-886	M.170.
CHECKED				
APPROVED		FSI STANDARD 43-92		ESI Drg N°



MEB Ref.	Compression Tool	Die Ref	Nº of Crimps
M.171	G10 (Y35)	U 317	1 Ø

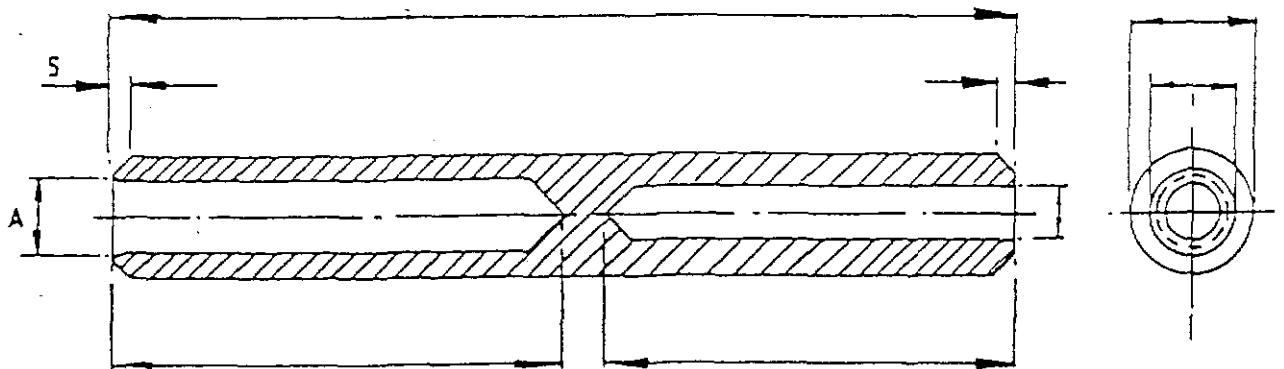
NOTE:—

To be pre-filled with PENETROX grease and capped to comply with BS.3288 Part 1.

JOINTING INSTRUCTIONS

1. Cut ends of conductor square
2. Apply tape
3. Wire brush conductor ends
4. Insert conductor and compress overall, commencing at centre of joint & work outwards.
5. Apply double wrap of Denso tape after compression.

SCALE	1:2	NON TENSION COPPER TO ALUMINIUM COMPRESSION CONNECTOR 100 mm ² ACSR. TO 70 mm ² COPPER.	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M.171
DATE	6-11-82		A	
DRAWN	RAC		Comp Tool Revised.	
TRACED			J.W. 3/84	
CHECKED				
APPROVED	J.F.			
E.S.I. STANDARD 43-92			ESI Drg No	



TYPE	CONDUCTOR		DIMENSION 'A'	COMP TOOL	DIE REF	N° OF CRIMPS
	SCA	COPPER				
M 172/1	150mm ²	125mm ²		G10 (Y35)	U317	12
M 172/2	175mm ²	.		G10 (Y35)	"	11

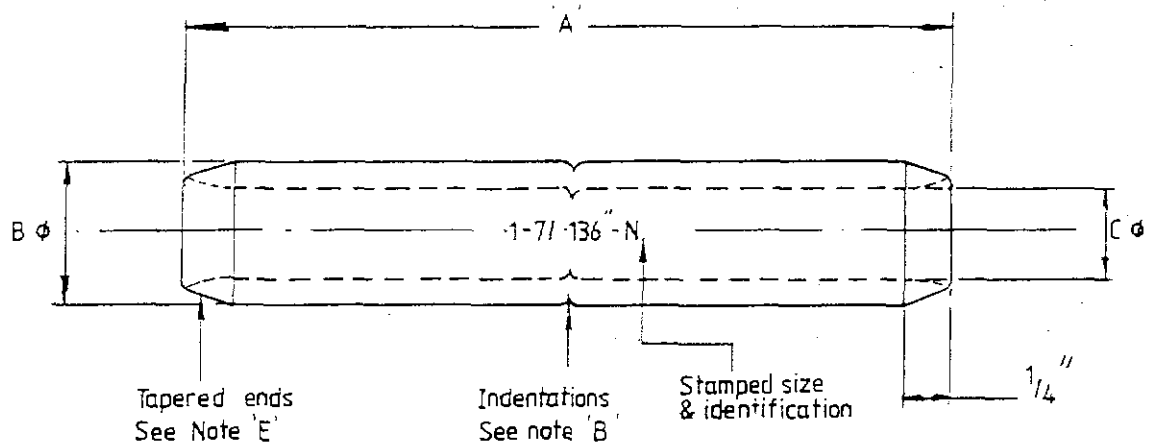
NOTE:—

To be pre-filled with PENETROX grease and capped to comply with BS 3288 Part 1.

JOINTING INSTRUCTIONS

1. Cut ends of conductor square.
2. Apply tape
3. Wire brush conductor ends.
4. Insert conductor and compress overall.
5. Apply double wrap of 'Densa' tape after compression.

SCALE	1:2	NON-TENSION COPPER TO ALUMINIUM COMPRESSION CONNECTOR.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	16-11-82		A	ORG. No.
DRAWN	RAC		Comp. Tool Revised	M.172
TRACED			J.W. 3/84	
CHECKED				
APPROVED				
C.D.W. Ltd. 3297		E.S.I. STANDARD 43 - 92	E.S.I. Org N°	



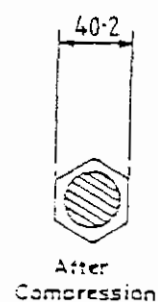
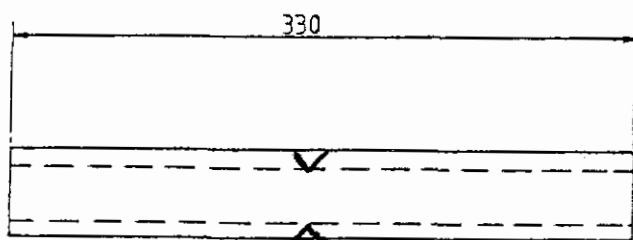
NOTES.

- A. The Joints must be manufactured from Electrolytic Tough pitch H.C. Copper conforming to B.S. 1977 and 2871 C101.
- B. Two indentations diametrically opposite must be provided to form a conductor stop. Care must be taken to see that the Bore Axis remains straight and is not deformed by this operation.
- C. Each fitting to have stamped on it the conductor size and stranding followed by the letter 'N' for Non - Tension.
- D. Tolerance on the Tubes as drawn are ± 0.04 (0.1mm) on the O/Dia and ± 0.03 (0.075 mm) on the I/Dia machine tolerance ± 0.02 (0.05mm)
- E. The Bore must be tapered at each end for a distance of $\frac{3}{16}$ " (4.8) and the edge radiused. The outside must be chamfered - off for a distance of $\frac{1}{4}$ " (6.35).

TYPE	CONDUCTOR TYPE AND SIZE m ² (mm ²)		MANUFACTURER REF.	DIMENSIONS (mm)			COMP'N. DIE DETAILS	
							HYDRAULIC	HAND TOOL
			BICCON		A	B	C	(G10 Y35) MH100 (MD6)
206/0	.0225	(14)	USE BICCON JOINT M233/0					
206/1	.025	(16)	USE BICCON JOINT M233/1					
206/2	.05	(32)	USE BICCON JOINT M233/2					
	.058	(35)						
206/3	.10	(70)	SN - 7C (YCS 258)	102	19	12	U166	W-166
206/4	.15	(100)	SN - 8C (YCS 275)	127	24	14	U168	—
206/5	.20	(125)	SN - 9C (YCS 291)	152	24	15	U169	—
206/6	.075	(50)	SN - 6C (YCS 244)	79	14.5	9.5	U164	W-164
206/7	.25	(150)	SN - 10C (YCS 303)	92	28	17.4	U170	—

NOTE: — Biccon new and equivalent old numbers shown
The joints after compression must meet the requirements of B.S.3288 Part 1.

SCALE	NTS	NON-TENSION COPPER COMPRESSION CONNECTOR.	ISSUE	MIDLANDS
DATE	7-1-81		B	ELECTRICITY BOARD
DRAWN	RAC		206/0 Added. RAC. 10-85	DRG. No. M. 206
TRACED				
CHECKED				
APPROVED				
E.S.I. STANDARD			ESI DRG. No.	



MATERIAL:- 1.875" (47.6) Dia. 99.5% purity Aluminium to Specification ESB BS. 1475.

PREPARATION:- Conductor to be Cleaned & Greased ('Densa') immediately prior to Compression.

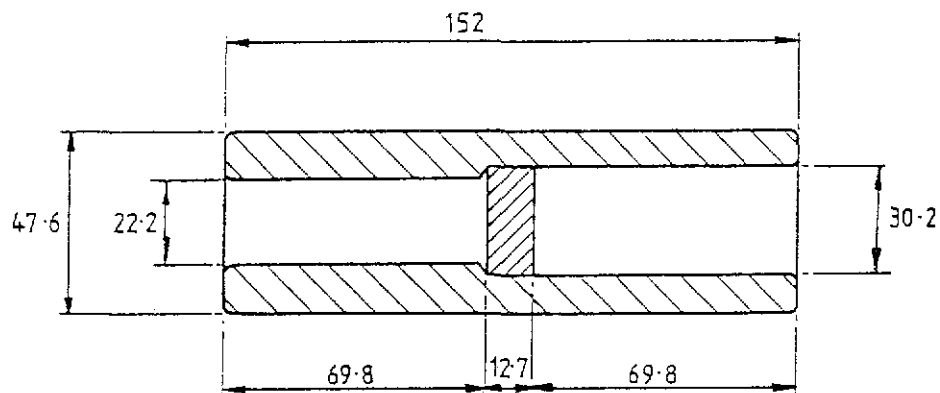
COMPRESSION:- Hydraulic Compression Die No. 4A-11.

SCALE	1 N.T.S.
DATE	11/6/9/53.
DRAWN	
TRACED	L.H.
CHECKED	
APPROVED	W.D.H.

GAF 5738

NON-TENSION COMPRESSION JOINT
0.40" Cu. EQUIV. (400mm²) ALUMINIUM
CONDUCTORS.

ISSUE	MIDLANDS ELECTRICITY BOARD
ORG. No.	M.243.



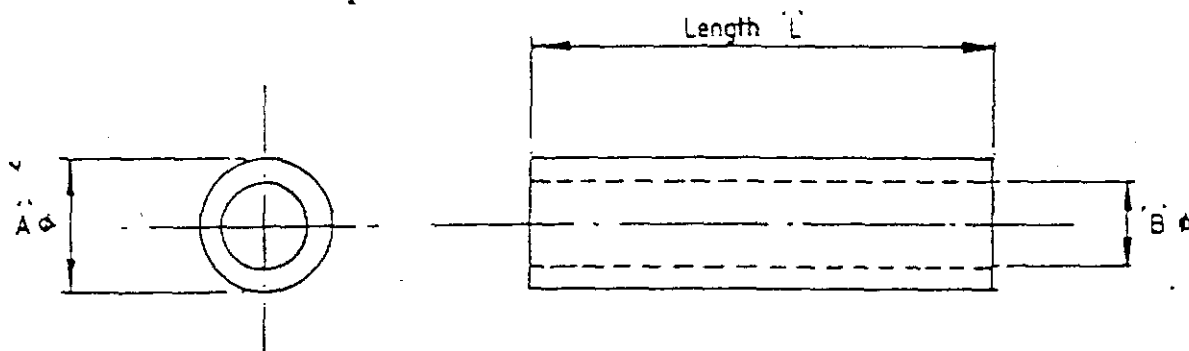
MATERIAL 1.875" (47.6) Dia. Aluminium Rod. 99.5 % min. Purity to E.I.B. B.S. 1476.

PREPARATION 1.) The connector must be arranged so that moisture cannot drain from the copper conductor on to the Aluminium sleeve.
2.) Greasing - Before making the joint, partially fill the connector and coat the conductors with 'DENSO' paste.

COMPRESSION Hydraulic Compression die No. 4A-11.

PROTECTION After compression wrap joint and 25mm of conductor on each side with DENSO tape to form double layer.

SCALE		NON TENSION ALUMINIUM TO COPPER CONNECTOR FOR 0.4in ² (400mm ²) SCA, to 0.4in ² (240mm ²) Cu	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	23.8.76.			
DRAWN	W.D.H.			
TRACED	L.C.H.			
CHECKED				
APPROVED	W.D.H.			DRG. No. M. 263.



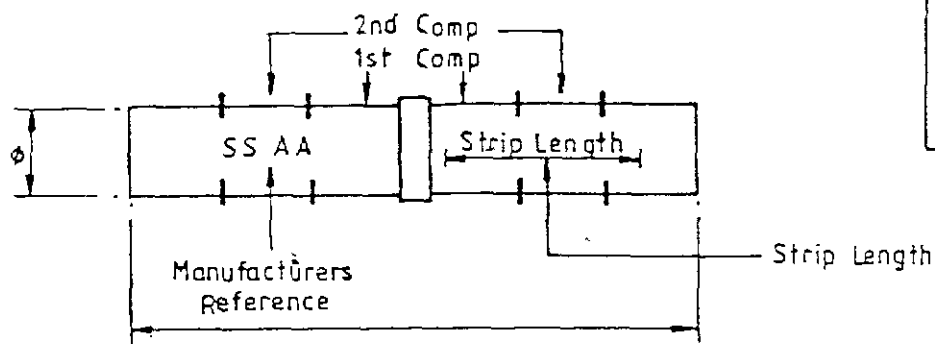
M. REF	C.E.G.B.	CONDUCTOR SIZE		DIMENSIONS			DIE REF
		SIZE	STRAND		A ϕ	B ϕ	
M. 330/0	4302230	175 mm ² ACSR	30/2.79 Al 7/2.79 St		38.1	21.8	A9
M. 330/4	4302235	400 mm ² ACSR	54/3.18 Al 7/3.18 St		47.6	30.2	A11

SCALE	N.T.S.
DATE	16.11.82
DRAWN	R.A.C.
TRACED	
CHECKED	
APPROVED	

Conductor Compression Joint. Non-Tension
Standard Length - Non-Disconnectable
175 & 400 mm² ACSR. Conductor.

E.S.I. STANDARD.

ISSUE	MIDLANDS ELECTRICITY BOARD
B	DRG. No.
Length L & Dim B' modified RAC 87	M. 330
	E.S.I. DRG. No.



MIN	MAN REF	CONDUCTOR TYPE AND SIZE	
3454841		BLACK END	BLACK END
		14mm ² (.0225in ²) Copper	14mm ² (.0225in ²) Copper
		16mm ² (.025in ²) Copper	16mm ² (.025in ²) Copper
		25mm ² (.04in ²) Copper	25mm ² (.04in ²) Copper
		25mm ² (.025in ²) Aluminium	25mm ² (.025in ²) Aluminium
3454842		SILVER END	BLACK END
		32mm ² (.05in ²) Copper	14mm ² (.0225in ²) Copper
		35mm ² (.058in ²) Copper	16mm ² (.025in ²) Copper
		35mm ² (.06in ²) Copper	25mm ² (.04in ²) Copper
		50mm ² (.05in ²) Aluminium	25mm ² (.025in ²) Aluminium
3454843		SILVER END	SILVER END
		32mm ² (.05in ²) Copper	32mm ² (.05in ²) Copper
		35mm ² (.058in ²) Copper	35mm ² (.058in ²) Copper
		35mm ² (.06in ²) Copper	35mm ² (.06in ²) Copper
		50mm ² (.05in ²) Aluminium	50mm ² (.05in ²) Aluminium

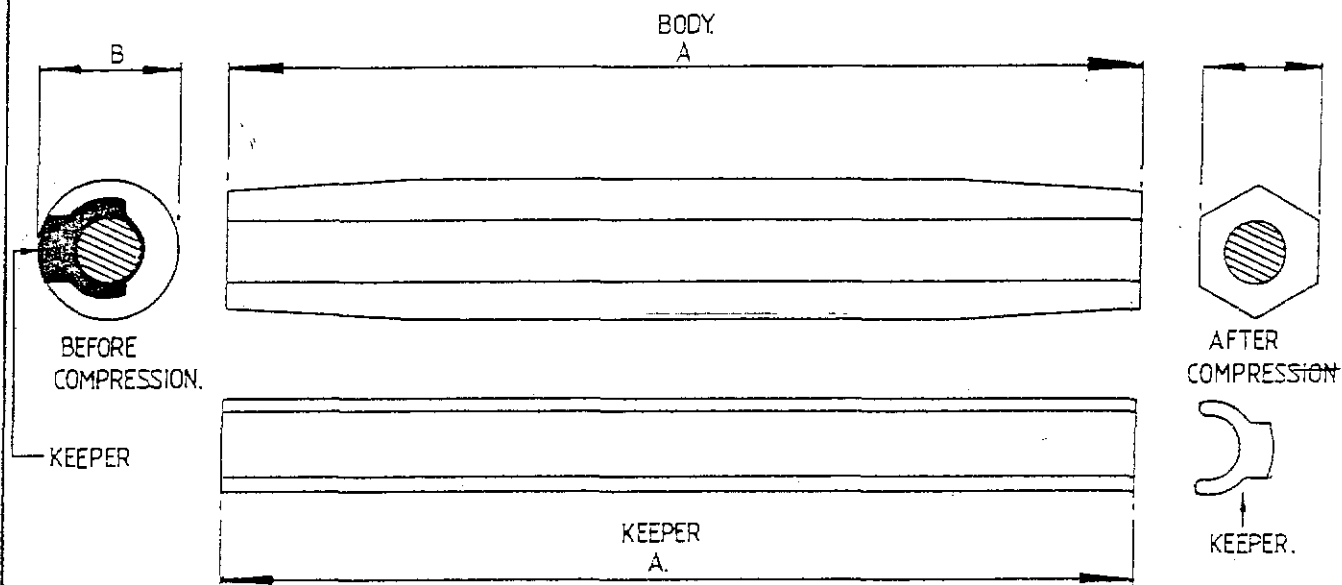
INSTALLATION NOTES

- 1) PVC or Bare Aluminium to PVC or Bare Copper combinations.
Joint to be set vertically with Copper at bottom of Joint.
Half Joint and 50mm (2inch) of Copper Conductor to be covered with double wrap of VM Tape.
- 2) PVC Copper to Bare Copper Conductor combination
Joint to be set vertically with Bare Conductor at bottom of Joint
complete Joint and 50mm (2 inch) of both Conductors to be covered with double wrap of VM Tape.
- 3) Conductor to be stripped where applicable for a distance of 37mm (1 1/2") as marked on Joint Sleeve.
- 4) Copper Conductor to be thoroughly cleaned with BICC - CCB1 scratch brush min 3471581
- 5) Aluminium Conductor to be thoroughly cleaned with BICC - CCB2 scratch brush min 3471582
- 6) End Caps to be pierced using rounded Podger not a knife.
- 7) Conductor to be fully inserted in Joint and compressed with two compression on each side using MHSORB compression tool min 3470760 fitted with ZBG Die min 3470761.

SCALE	-	INSULATED TWINSHIELDS COMPRESSION JOINT FOR COPPER & ALUMINIUM CONDUCTORS.	ISSUE	 DRG. No. M484 (MF)
DATE	13.6.89			
DRAWN	WDH			
TRACED	SGT			
CHECKED	WAB			
APPROVED				

SECTION 9

MIN NUMBER	M	DESCRIPTION
3453210	321	CONDUCTOR REPAIR SLEEVE (COMPRESSION TYPE) 70,175,400mm ² ACSR
3453220	322	CONDUCTOR REPAIR SLEEVE (PREFORMED TYPE)

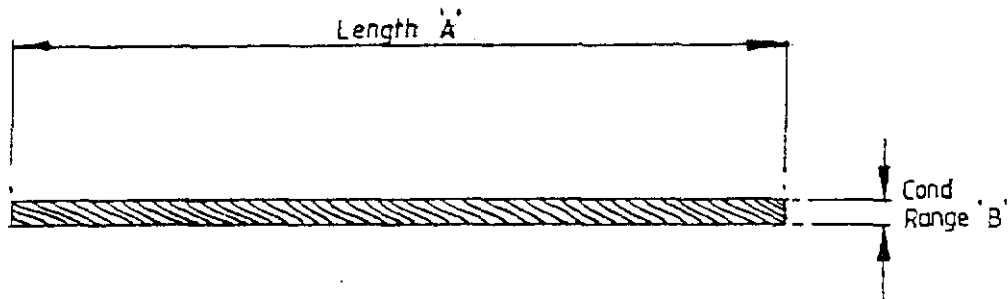


M.E.B. REF	CONDUCTOR SIZE	UNIDARE O.H.L. FITTINGS CAT N°	DIMENSIONS (mm)			DIE REF	C.E.G.B. CC N°
			A	B	C		
321/1	70mm ²	ARS - 9B	240	38.1	32.3	A-9	4302417
321/3	175mm ²	ARS - 9A	280	38.1	32.3	A-9	4302427
321/4	400mm ²	ARS - 11A	305	47.6	40.2	A-11	4302435

NOTES: —

1. To comply with B.S. 3288 Pt 1
2. For jointing Instructions see M. 321 Notes.

SCALE	N.T.S.	CONDUCTOR REPAIR SLEEVE (COMPRESSION TYPE) 70mm ² , 175mm ² + 400mm ² ACSR CONDUCTORS	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	23.7.82			
DRAWN	R. A. C.			DRG. No.
TRACED	J. W.			M. 321
CHECKED				
APPROVED		E.S.I. STANDARD		E.S.I. DRG. N°



Notes: — Maximum Permissible Damage
33% Outer Strands.

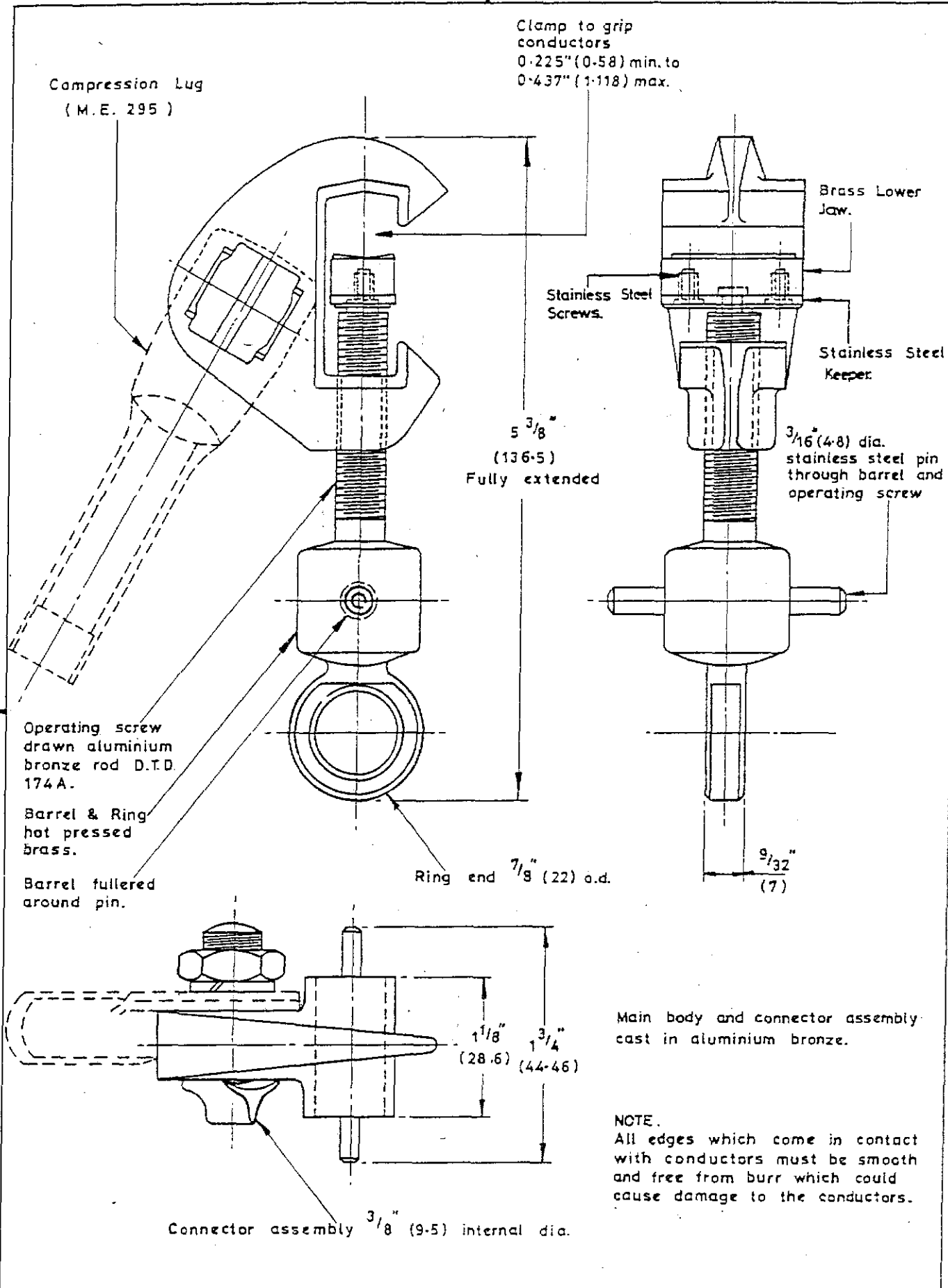
REF	COND. TYPE & SIZE	PREFORMED L/D REF	CONDUCTOR DIA RANGE (B)	LENGTH(A)
M 322/5	70 ACSR 100 ACSR	RS 5303144	(13-84 - 14-42)	890
M 322/6	150 ACSR	RS 5303185	(17-81 - 18-53)	1140
M 322/7	175 ACSR	RS 5303201	(19-33 - 20-13)	1220
M 322/8	400 HDA	RS 5303267	(25-63 - 26-69)	1520
M 322/9	400 ACSR	RS 5303289	(27-74 - 28-87)	1730
M 322/0	300 ACSR			

SCALE	N.T.S.	CONDUCTOR FITTING PREFORMED REPAIR SLEEVES FOR ACSR CONDUCTORS.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	16-2-83		A	
DRAWN	R.A.C.		Range of 322/7 Corrected. C.A.R. 29.9.83	DRG. No.
TRACED				M-322
CHECKED	#			
APPROVED	#	ESI STANDARD.		ESI DRG. N°

(MF)

SECTION 10

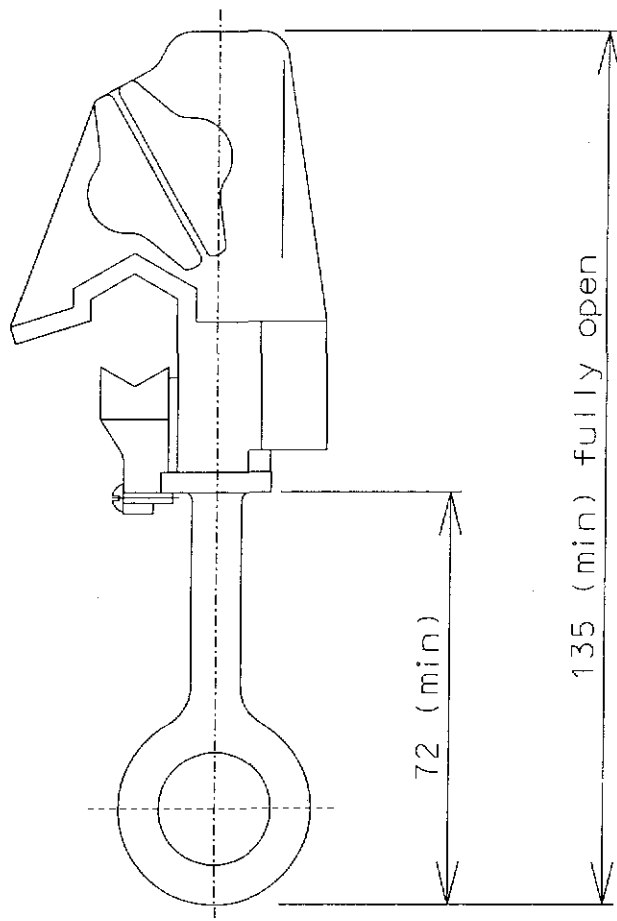
MIN NUMBER	M	DESCRIPTION
3451920	192	LIVE LINE TAP FOR COPPER CONDUCTORS UP TO 70mm ²
3451921	192/1	LIVE LINE TAP FOR COPPER
3451980	198	LIVE LINE TAP FOR AAAC & ACSR CONDUCTORS UP TO 100mm ²
3451981	198/1	LIVE LINE TAP FOR ALLOY - ACSR
3452790	279	LIVE LINE TAPPING BAIL



SCALE	1:1
DATE	20.3.75
DRAWN	A.R.B.
TRACED	M.R.T.
CHECKED	
APPROVED	

LIVE LINE TAP TYPE H1/B/2
FOR COPPER CONDUCTORS
UP TO 70mm² (0.1 ins²)

ISSUE	MIDLANDS ELECTRICITY BOARD
H.M.F.	
Description of parts corrected R.A.C. 6-8-79	DRG. No. ME 192



Current Rating 400 Amps
Weight 310 grams

Use Spigot Lug M295/4 (Supplied separately)

NOTE:
Tested to BS 3288 and ESI 43-92 1983
Electrical and Mechanical

All dimensions are in millimetres unless otherwise stated

SCALE	NTS
DATE	29/05/03
DRAWN	ADD
TRACED	
CHECKED	05/06/03
APPROVED	KD

LIVE LINE TAP FOR COPPER CONDUCTORS

ISSUE	DATE



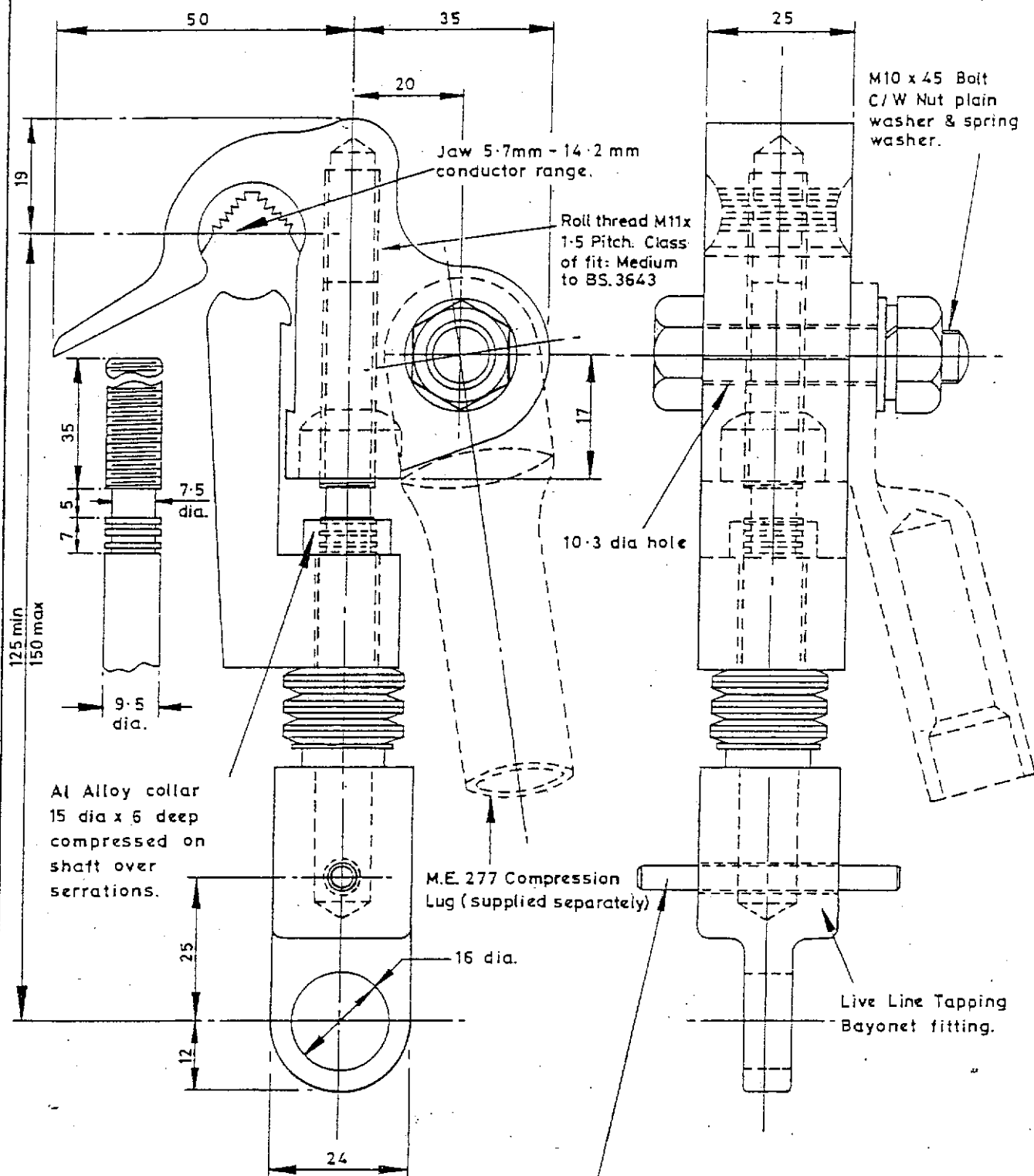
Aquila

DRG. No.

M192/1

DRG. No.

M192/1



MATERIAL

Body - Extruded Al Alloy
 Bayonet fitting - Cast Al Alloy
 Shaft & Retaining Pins - Stainless Steel
 Bolt, Nut & Washer - Stainless Steel

200A Continuous Rating

3/16" (4.7) dia. x 45 Stainless Steel pin
 through barrel. Barrel fullered around pin.

MANUFACTURERS TEST

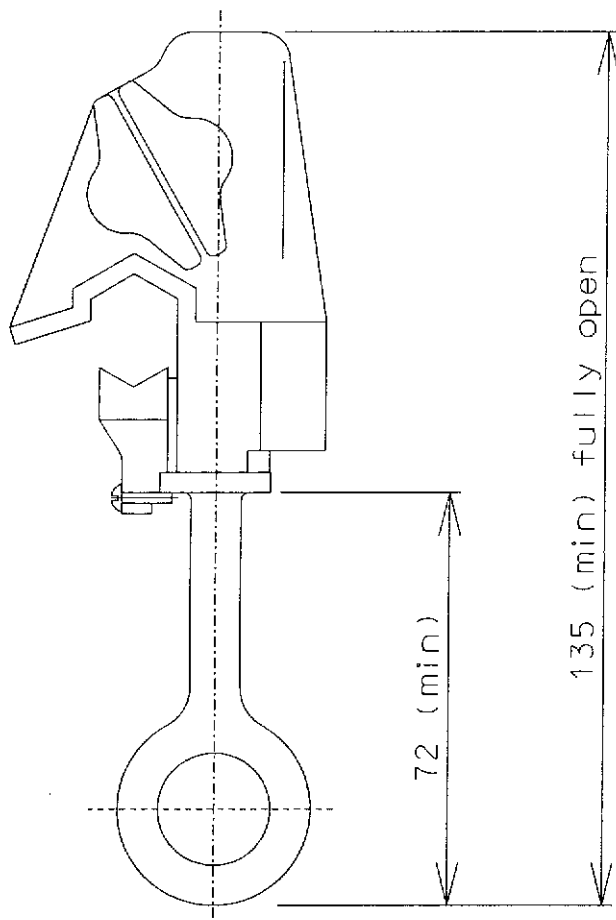
5% to be subjected to 20 N/m Torque (opening)
 and 30 N/m Torque (closing) without deformation.

SCALE	1:1
DATE	26.10.82
DRAWN	A.R.B.
TRACED	P.G.
CHECKED	
APPROVED	

LIVE LINE TAP
 FOR AL ALLOY OR A.C.S.R. CONDUCTORS
 UP TO 100 mm²

E.S.I. STANDARD

ISSUE	MIDLANDS ELECTRICITY BOARD
	DRG. No.
	M.198
	E.S.I. DRG N°



Current Rating 400 Amps
Weight 190 grams

Use Spigot Lug M277/4 (Supplied separately)

NOTE:
Tested to BS 3288 and ESI 43-92 1983
Electrical and Mechanical

All dimensions are in millimetres unless otherwise stated

SCALE	NTS
DATE	29/05/03
DRAWN	ADD
TRACED	
CHECKED	05/06/03
APPROVED	KD

LIVE LINE TAP
FOR AL ALLOY OR A.C.S.R CONDUCTORS

ISSUE	DATE

Produced by
Primary CAD



Aquila

DRG. No.

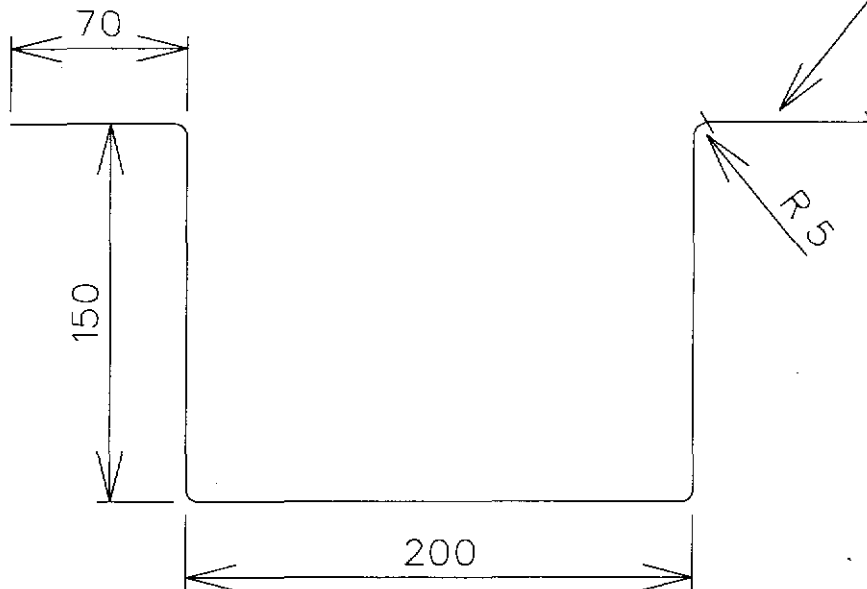
M198/1

DRG. No.

M198/1

A4

KNURLING TO BOTH ENDS



M 279/1 and M 279/2

MATERIAL -

M 279/1 690mm x 9.5mm dia
Aluminium Alloy Rod
Spec H.E.30.WP

M 279/2 690mm x 9.5mm dia
Copper Rod

TYPE	MANUFACTURER	MANUFACTURERS REF.	CONDUCTOR TYPE & SIZE mm ²	CRIMPET REF No.	DIE REF & No OF CRIMPS	
					MD.6 & MD.6-8	Y35 & Y35L
M 279/1	AQUILA	M 279/1	25mm ² Al Alloy	ME. 274/3	-	U-D3 (1)
			50mm ² " "	" "/2	-	U-D3 (1)
			100mm ² A.C.S.R	" "/9	-	U-N (2)
M 279/2	AQUILA	M 279/2	16mm ² Copper	ME. 273/4	-	U-0
			32mm ² " "	" "/"	-	U-0
			70mm ² " "	" "/"	-	U-0
			100mm ² " "	YC. 28C 26	-	U-F

AS DRG No HS.702

SCALE	1:3
DATE	16.11.82
DRAWN	R.A.C.
TRACED	
CHECKED	J.M.B.
APPROVED	

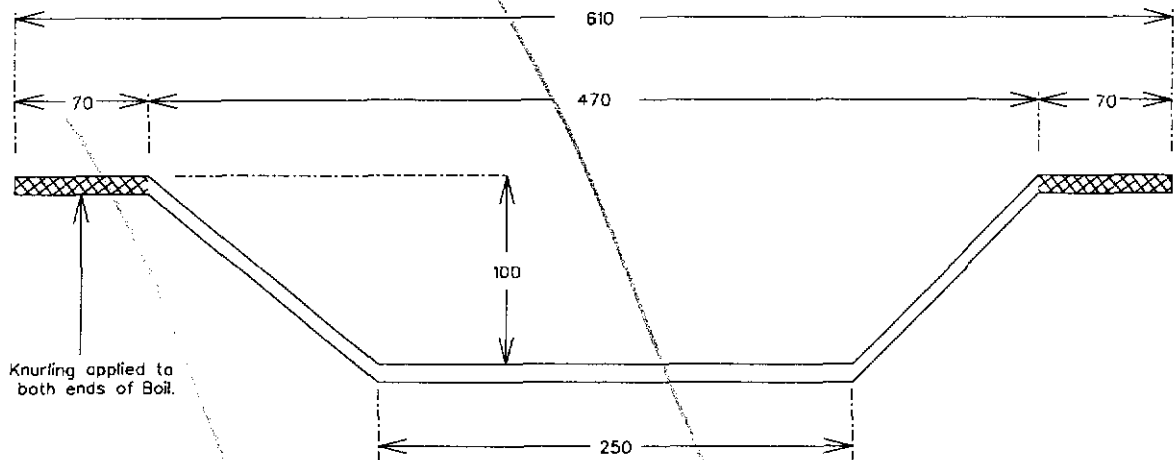
L.L. TAPPING BAIL.

ISSUE	DATE
B	09.04.03
New Diagram Re-Scaled	
KD/ADD	

Produced by Primary CAD	
 Aquila	
DRG. No.	
M279	

DRG. No.

M279



MATERIAL -

M 279/1 690mm x 9.5mm dia
Aluminium Alloy Rod
Spec H.E.30.WP

M 279/2 690mm x 9.5mm dia
Copper Rod

TYPE	MANUFACTURER	MANUFACTURERS REF.	CONDUCTOR TYPE & SIZE mm ²	CRIMPET REF No.	DIE REF & No OF CRIMPS	
					MD.6 & MD.6-8	Y35 & Y35L
M 279/1	AQUILA	M 279/1	25mm ² Al Alloy 50mm ² " " 100mm ² A.C.S.R	ME. 274/3 " "/2 " "/9	- -	U-D3 (1) U-D3 (1) U-N (2)
M 279/2	AQUILA	M 279/2	16mm ² Copper 32mm ² " 70mm ² " 100mm ² "	ME. 273/4 " "/" " "/" YC. 28C 26	-	U-0 U-0 U-0 U-F

AS DRG No HS.702

SCALE	1:5
DATE	16.11.82
DRAWN	R.A.C.
TRACED	
CHECKED	J.W.B.
APPROVED	

L.L. TAPPING BAIL.

ISSUE	DATE
A	09.01.03
REDRAWN ON CAD	
MINOR MODS.	
MJB/ADD/KD	



Aquila

DRG. No.

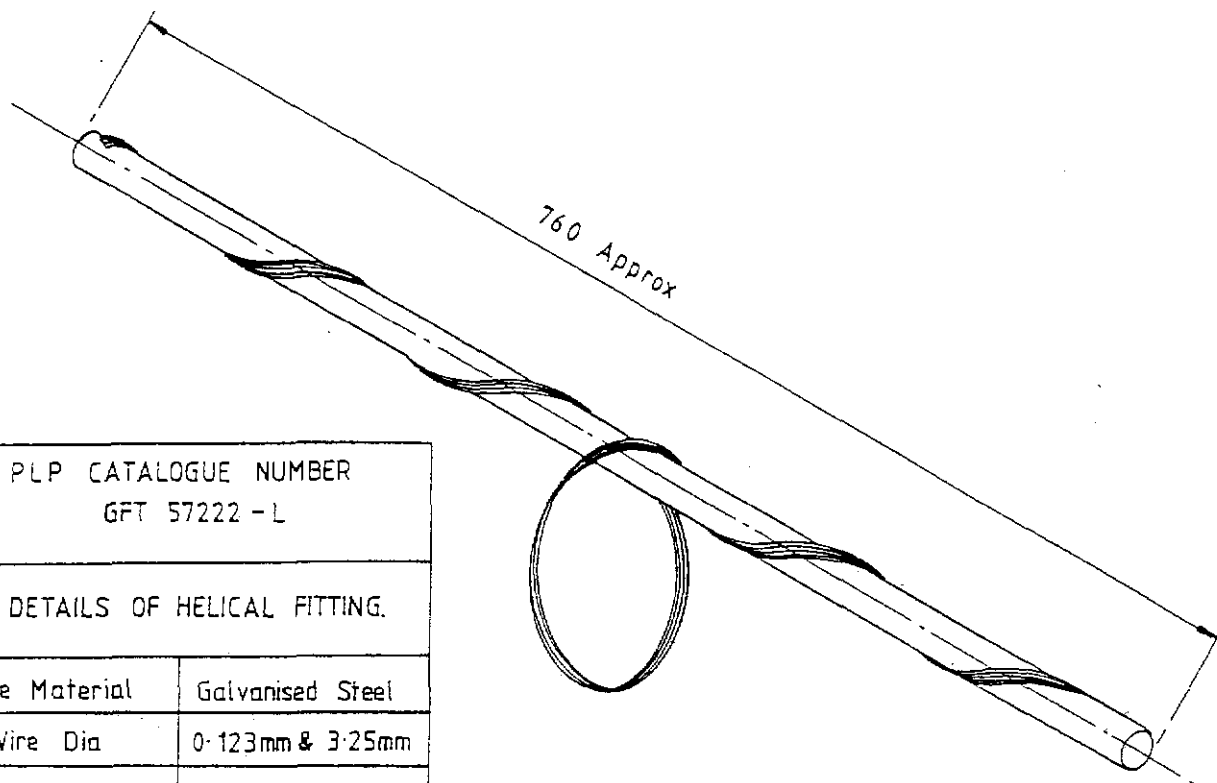
M279

DRG. No.

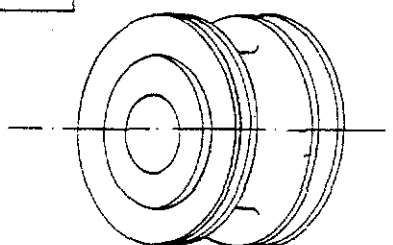
M279

SECTION 11

MIN NUMBER	M	DESCRIPTION
3454480	448	HELICAL FITTINGS - HARDEX INTERMEDIATE
3454630	463	HELICAL FITTINGS - BARE 11kv SIDE TIES
3454640	464	HELICAL FITTINGS - 11kv BARE CONDUCTOR DOUBLE TOP TIES
3454650	465	HELICAL FITTINGS - COPPER 11kv SIDE TIES
3454660	466	HELICAL FITTINGS - PVC L.V. TERMINATIONS/DEAD ENDS
3454670	467	HELICAL FITTINGS - 11kv COPPER CONDUCTOR DOUBLE TOP TIES
3454680	468	HELICAL FITTINGS - PVC SIDE TIES
3454683	468/3	HELICAL FITTINGS - PVC SIDE TIES WITH SPIRAL
3454690	469	HELICAL FITTINGS - SINGLE TOP TIE TIES FOR COPPER
3454700	470	HELICAL FITTINGS - BARE L.V. TERMINATIONS/DEAD ENDS
3454710	471	HELICAL FITTINGS - 33/66kv BARE CONDUCTOR SIDE TIES
3454720	472	HELICAL FITTINGS - BARE L.V. INTERMEDIATE/ANGLE TIES
3454730	473	HELICAL FITTINGS - SINGLE TOP TIE TIES FOR ALUMINIUM
3454740	474	HELICAL FITTINGS - SINGLE TOP TIE TIES FOR PVC
3454743	474/3	HELICAL FITTINGS - SINGLE TOP TIE TIES FOR PVC WITH SPIRAL
3454750	475	HELICAL FITTINGS - 33/66kv BARE CONDUCTOR SINGLE TOP TIES
3454760	476	HELICAL FITTINGS - 33/66kv BARE CONDUCTOR DOUBLE TOP TIES
3454770	477	HELICAL FITTINGS - PVC L.V. INTERMEDIATE/ANGLE TIES
3454780	478	HELICAL FITTINGS CONCENTRIC CABLE DEAD ENDS
3455710	571	HELICAL FITTINGS CONDUCTOR DEAD END GRIPS A.A.A.C
3455720	572	SOCKET THIMBLE H.V HELICAL DEAD ENDS 70 Kn
3455722	572/2	THIMBLE CLEVIS 125 Kn
3455723	572/3	DOUBLE EYE LINK 120kN
3455740	574	HELICAL FITTINGS CONDUCTOR DEAD END GRIPS H.D COPPER



PLP CATALOGUE NUMBER GFT 57222 - L	
DETAILS OF HELICAL FITTING.	
Wire Material	Galvanised Steel
Wire Dia	0.123mm & 3.25mm
Number of wires	4
Colour Code	Red.



NOTE

1. Only to be used on 'Hardex' having an overall diameter within the range 19.20 m.m. to 21.73 m.m. and Steel Wire Armouring which has a left hand direction of lay.
2. For use in conjunction with Spool Insulator M.447.

SCALE	-	HELICAL FITTING FOR 'HARDEX' COMBINED PILOT AND EARTH CABLE ON INTERMEDIATE POLES.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	7.3.86			
DRAWN	S.J.H.			DRG. No.
TRACED				M. 448.
CHECKED				
APPROVED				

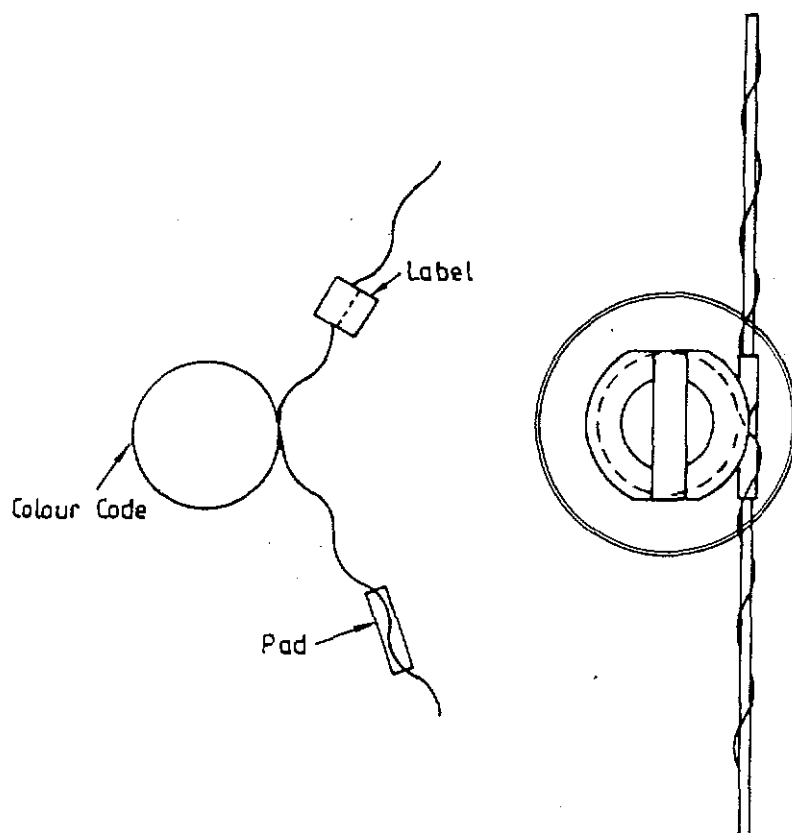
Cond. Type (Mat.)	Cond. C.S.A. mm ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia mm.	1.		2.		M.I.N. N°
					Manufacturers Ref. N°	Colour Code	Manufacturers Ref. N°	Colour Code	
AL ALLOY	25	7/2-34	7.02		GFT 5080718 RP	ORANGE			3454631
A.C.S.R.	25	7/2-36	7.08		GFT 5080720 RP	PURPLE			3454632
AL ALLOY	40	7/2-95	8.85		GFT 5080722 RP	RED			3454633
A.C.S.R.	40	7/3-00	9.00						
AL ALLOY	50	7/3-30	9.90		GFT 5080724 RP	YELLOW			3454634
A.C.S.R.	50	7/3-35	10.05						
AL ALLOY	100	7/4-65	13.95		GFT 5080730 RP	RED			3454635
A.C.S.R.	100	ALL TYPES	14.15/14.60						
A.C.S.R.	150	37/2-59	18.13		GFT 5080734 RP	BROWN			3454636
A.C.S.R.	175	37/2-79	19.53		GFT 5080736 RP	RED			3454637
AL ALLOY	200	37/2-87	20.09						

MANUFACTURERS

- ① PREFORMED LINE PRODUCTS.
- ②

NOTES

- ① INSULATOR NECK DIA 78mm.

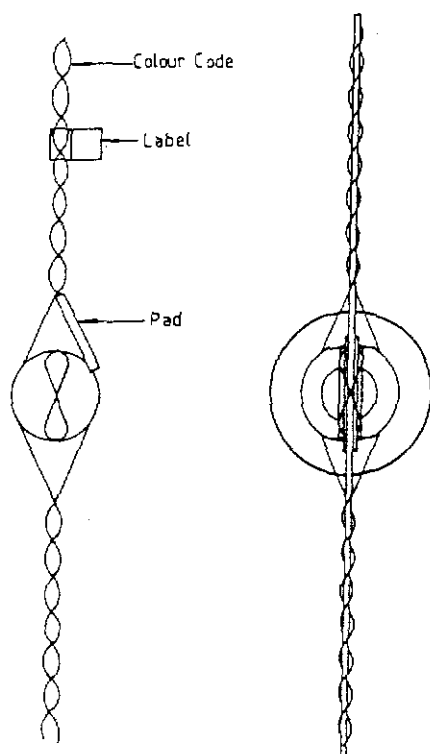


SCALE	
DATE	25.11.88
DRAWN	B.N.
TRACED	S.G.T.
CHECKED	222
APPROVED	Rg

HELICAL FITTINGS
SIDE TIES
FOR BARE ALUMINIUM CONDUCTORS
11KV. OVERHEAD LINES

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		DRG No M463

Cond. Type (Mat.)	Cond. C.S.A. mm ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia. mm.	1		2		M.I.N. No
					Manufacturers Ref. No.	Colour Code	Manufacturers Ref. No.	Colour Code	
AL ALLOY	25	7/2-34	7-02		DDT5070718 RP	ORANGE			3454641
A.C.S.R.	25	7/2-36	7-08		DDT5070720 RP	PURPLE			3454642
AL ALLOY	40	7/2-95	8-85		DDT5070722 RP	RED			3454643
A.C.S.R.	40	7/3-00	9-00						
AL ALLOY	50	7/3-30	9-90		DDT5070724 RP	YELLOW			3454644
A.C.S.R.	50	7/3-35	10-05						
AL ALLOY	100	7/4-65	13-95		DDT5070730 RP	RED			3454645
A.C.S.R.	100	ALL TYPES	14-15/14-60						
A.C.S.R.	150	37/2-59	18-13		DDT5070734 RP	BROWN			3454646
A.C.S.R.	175	37/2-79	19-53		DDT5070736 RP	RED			3454647
AL ALLOY	200	37/2-87	20-09						



MANUFACTURERS

① PREFORMED LINE PRODUCTS

NOTES:-

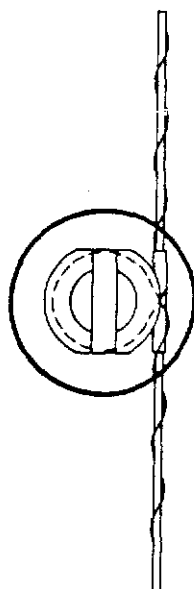
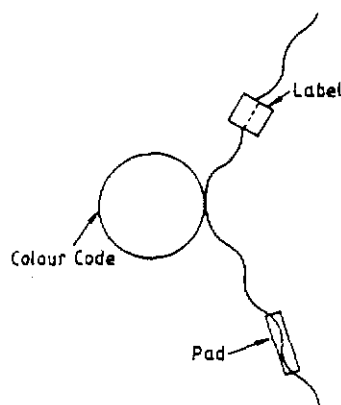
- ① INSULATOR NECK DIA 78mm
- ② DOUBLE TIE TO BE USED ONLY UNDER EXTREME DUTY CONDITIONS

SCALE	---
DATE	23-11-88
DRAWN	B.N.
TRACED	P.J.N.
CHECKED	2/12
APPROVED	R.M.V.

HELICAL FITTINGS
DOUBLE TOP TIES
FOR BARE ALUMINIUM CONDUCTORS
11KV OVERHEAD LINES

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		ORG. No. M464

Cond. Type (Mat.)	Cond. C.S.A. mm. ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia. mm.	1		2		M.I.N. No.
					Manufacturers Ref. No.	Colour Code	Manufacturers Ref. No.	Colour Code	
H.D. Cu	14	7/1-60	4.80	-	GFT5680714 RP	BLUE			3454651
H.D. Cu	16	3/2-65	5.69	-	GFT5680716 RP	BROWN			3454652
CAD. Cu	22	7/2-00	6.00	-					
CAD. Cu	30	7/2-30	6.90	-	GFT5680718 RP	ORANGE			3454653
H.D. Cu	35	7/2-50	7.50	-	GFT5680720 RP	PURPLE			3454654
CAD. Cu	38	7/2-60	7.80	-					
H.D. Cu	32	3/3-75	8.05	-	GFT5680722 RP	RED			3454655
H.D. Cu	50	7/3-00	9.00	-					
CAD. Cu	55	7/3-20	9.60	-	GFT5680724 RP	YELLOW			3454656
H.D. Cu	70	7/3-55	10.65	-	GFT5680726 RP	BLUE			3454657
CAD. Cu	75	7/3-70	11.10	-					
H.D. Cu	100	7/4-30	12.90	-	GFT5680728 RP	ORANGE			3454658
CAD. Cu	115	19/2-77	13.85	-	GFT5680730 RP	RED			3454659
H.D. Cu	125	19/2-90	14.50	-					



MANUFACTURERS:-

① PREFORMED LINE PRODUCTS

NOTES:-

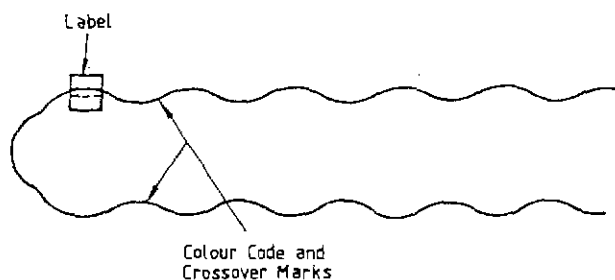
① INSULATOR NECK DIA. 78mm.

SCALE	-
DATE	28.11.88
DRAWN	B.N.
TRACED	C.A.R.
CHECKED	Q.A.
APPROVED	K.W.

HELICAL FITTINGS
SIDE TIES
FOR BARE COPPER CONDUCTORS
11kV OVERHEAD LINES

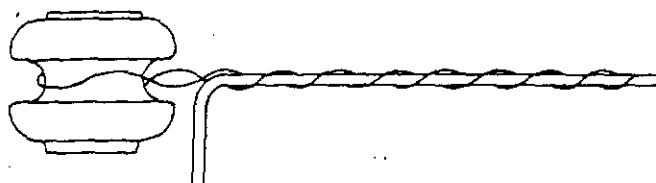
ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		DRG No M 465

Cond. Type (at.)	Cond. C.S.A. mm ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia. mm.	1		2		M.I.N. No
					Manufacturers Ref. No.	Colour Code	Manufacturers Ref. No.	Colour Code	
H.D. Cu	70	7/3-55	10-65	13-50	LVDE5715577 R	YELLOW			3454664
ALUM	50	7/3-10	9-30	11-70	LVDE5715537 R	GREEN			3454666
ALUM	100	7/4-39	13-17	16-00	LVDE5715543 R	BROWN			3454667



MANUFACTURERS

① PREFORMED LINE PRODUCTS

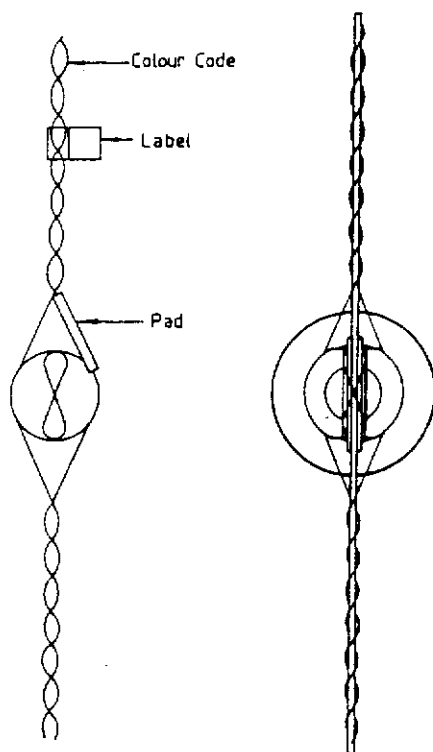


SCALE	
DATE	30-11-88
DRAWN	B.N.
TRACED	P.J.N.
CHECKED	223
APPROVED	EnV

HELICAL FITTINGS
TERMINATION DEAD END TIES
FOR P.V.C COVERED CONDUCTORS
L.V. OVERHEAD LINES

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		DRG No M466

Cond. Type (Met.)	Cond. C.S.A. mm ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia. mm.	1		2		M.I.N. N°.
					Manufacturers Ref. N°.	Colour Code	Manufacturers Ref. N°.	Colour Code	
H.D. Cu	14	7/1-60	4-80	-	DDT5670714RP	BLUE			3454671
H.D. Cu	16	3/2-65	5-69	-	DDT5670716RP	BROWN			3454672
CAD Cu	22	7/2-00	6-00	-					
CAD Cu	30	7/2-30	6-90	-	DDT5670718 RP	ORANGE			3454673
H.D. Cu	35	7/2-50	7-50	-	DDT5670720 RP	PURPLE			3454674
CAD Cu	38	7/2-60	7-80	-					
H.D. Cu	32	3/3-75	8-05	-	DDT5670722 RP	RED			3454675
H.D. Cu	50	7/3-00	9-00	-					
CAD Cu	55	7/3-20	9-60	-	DDT5670724 RP	YELLOW			3454676
H.D. Cu	70	7/3-55	10-65	-	DDT5670726 RP	BLUE			3454677
CAD Cu	75	7/3-70	11-10	-					
H.D. Cu	100	7/4-30	12-90	-	DDT5670728 RP	ORANGE			3454678
CAD Cu	115	19/2-77	13-85	-	DDT5670730 RP	RED			3454679
H.D. Cu	125	19/2-90	14-50	-					



MANUFACTURERS

① PREFORMED LINE PRODUCTS

NOTES:-

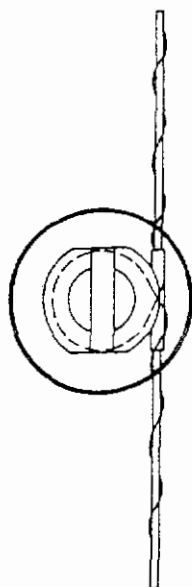
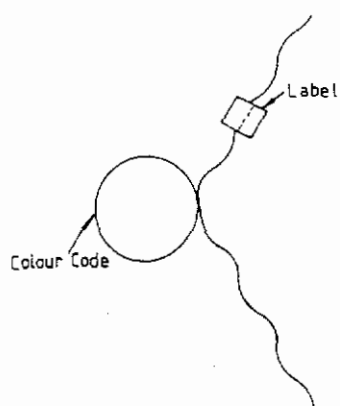
- ① INSULATOR NECK DIA 78mm
- ② DOUBLE TIE TO BE USED ONLY UNDER EXTREME DUTY CONDITIONS

SCALE	
DATE	23-11-88
DRAWN	B.N.
TRACED	P.N.
CHECKED	22
APPROVED	16

HELICAL FITTINGS
DOUBLE TOP TIES
FOR BARE COPPER CONDUCTORS
11kV OVERHEAD LINES

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		DRG. No M467

Cond. Type Mat.	Cond. C.S.A. mm ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia. mm.	1		2		M.I.N. No
					Manufacturers Ref. No.	Colour Code	Manufacturers Ref. No.	Colour Code	
H.D. Cu	16	3/2-65	5-69	9-30	GFT5780728 R	ORANGE			3454681
CAD Cu	22	7/2-00	6-00	9-60					
H.D. Cu	32	3/3-75	8-05	12-10	GFT5780732 R	PURPLE			3454683
H.D. Cu	70	7/3-55	10-65	14-70	GFT5780734 R	BROWN			3454684
AL ALLOY	25	7/2-34	7-02	10-60	GFT5780730 R	RED			3454682
A.C.S.R.	25	7/2-36	7-08	10-70					
AL ALLOY	40	7/2-95	8-85	12-80	GFT5780732 R	PURPLE			3454683
A.C.S.R.	40	7/3-00	9-00	12-95					
AL ALLOY	50	7/3-30	9-90	13-90	GFT5780732 R	PURPLE			3454683
A.C.S.R.	50	7/3-35	10-05	14-10	GFT5780734 R	BROWN			3454684
AL ALLOY	100	7/4-65	13-95	18-00	GFT5780736 R	RED			3454685
A.C.S.R.	100	ALL TYPES	14-15	18-20					
C.S.R.	150	37/2-59	18-13	22-20	GFT5780740 R	GREEN			3454686
A.C.S.R.	175	37/2-79	19-53	23-60					
AL ALLOY	200	37/2-87	20-09	24-10	GFT5780740 R	GREEN			3454686



MANUFACTURERS

① PREFORMED LINE PRODUCTS

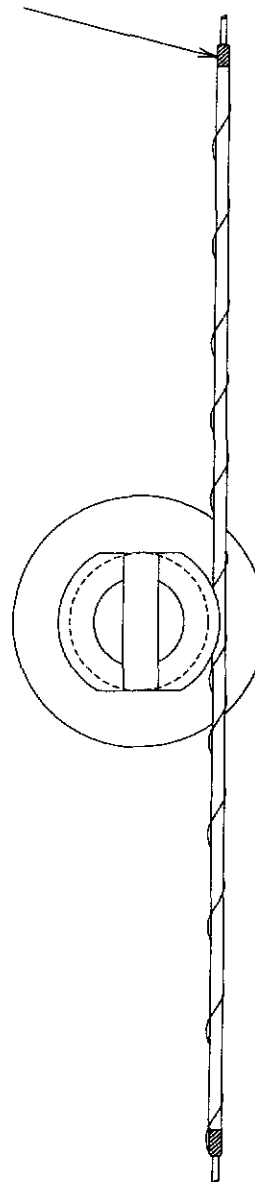
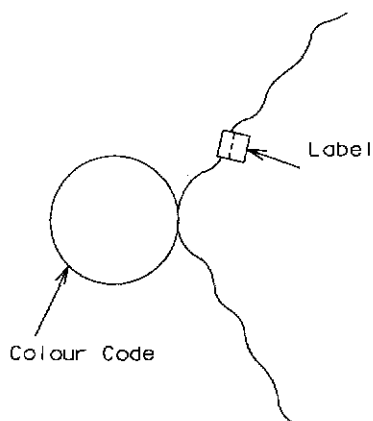
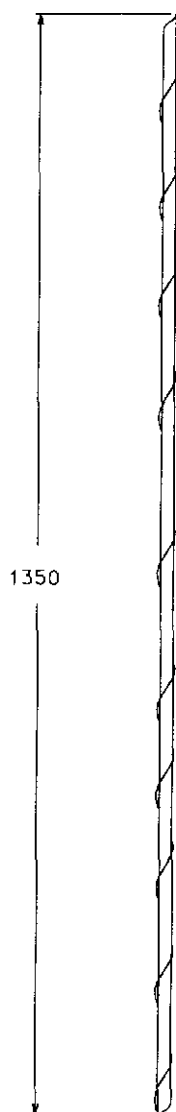
NOTES:-

① INSULATOR NECK DIA 78mm

SCALE		HELICAL FITTINGS SIDE TIES FOR P.V.C COVERED CONDUCTORS 11KV OVERHEAD LINES	ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	29-11-88				
DRAWN	B.N.				
TRACED	P.J.N.				DRG No M468
CHECKED	225				
APPROVED	125				

Cond. Type (Material)	Cond. mm ²	Cond. Stranding	Cond. O/S Dia mm	Cond. O/S Dia mm	Manufacturers Ref.N°	Colour Code	Spiral Wrap Length mm
H.D Cu	32	3/3.75	8.05	12.1	GFT5780732R	PURPLE	1350
AL ALLOY	40	7/2.95	8.85	12.8	GFT5780732R	PURPLE	1350
A.C.S.R.	40	7/3.00	9	12.95	GFT5780732R	PURPLE	1350
AL ALLOY	50	7/3.30	9.9	13.9	GFT5780732R	PURPLE	1350

P.V.C. BLACK TAPE
THE ENDS OF WRAP



Helical Side Tie and Semi-conductive Spiral wrap.
Supplied in a sealed plastic bag.to prevent contamination of the wrap.

Spiral Wrap to be kept clean and free of dirt.

SCALE	-
DATE	08.04.03.
DRAWN	ADD
TRACED	-
CHECKED	
APPROVED	

HELICAL FITTING SIDE TIE
FOR P.V.C. COVERED CONDUCTOR 11kV
WITH SEMI-CONDUCTING SPIRAL WRAP

ISSUE	DATE



Aquila

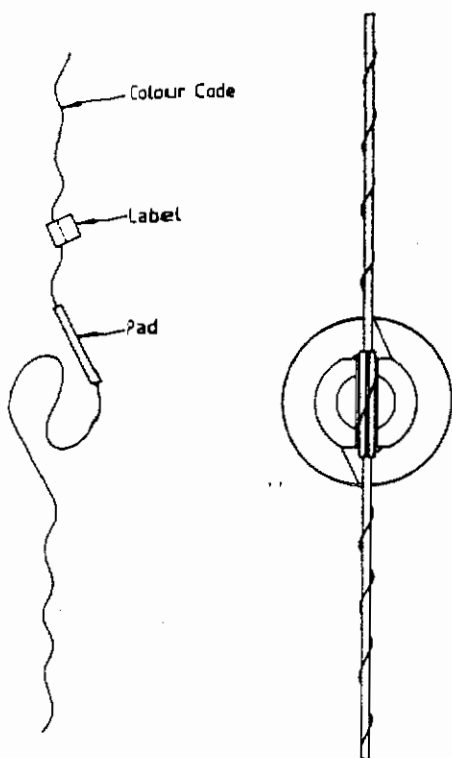
DRG. No.

M 468/3

Produced by
Primary CAD

DRG. No. M 468/3

Cond. Type (Mat.)	Cond. C.S.A. mm ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia. mm.	1		2		M.I.N. N°
					Manufacturers Ref. N°.	Colour Code	Manufacturers Ref. N°.	Colour Code	
H.D. Cu	14	7/1-60	4-80	-	DT 5670714 RP	BLUE			3454691
H.D. Cu	16	3/2-65	5-69	-	DT 5670716 RP	BROWN			3454692
CAD Cu	22	7/2-00	6-00	-					
CAD Cu	30	7/2-30	6-90	-	DT 5670718 RP	ORANGE			3454693
H.D. Cu	35	7/2-50	7-50	-	DT 5670720 RP	PURPLE			3454694
CAD Cu	38	7/2-60	7-80	-					
H.D. Cu	32	3/3-75	8-05	-	DT 5670722 RP	RED			3454695
H.D. Cu	50	7/3-00	9-00	-					
CAD Cu	55	7/3-20	9-60	-	DT 5670724 RP	YELLOW			3454696
H.D. Cu	70	7/3-55	10-65	-	DT 5670726 RP	BLUE			3454697
CAD Cu	75	7/3-70	11-10	-					
H.D. Cu	100	7/4-30	12-90	-	DT 5670728 RP	ORANGE			3454698
CAD Cu	115	19/2-77	13-85	-	DT 5670730 RP	RED			3454699
H.D. Cu	125	19/2-90	14-50	-					



MANUFACTURERS

① PREFORMED LINE PRODUCTS

NOTES:-

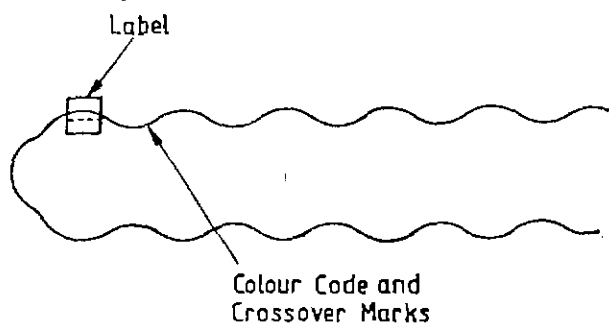
① INSULATOR DIA. 78mm

SCALE	---
DATE	22-11-88
DRAWN	B.N.
TRACED	P.J.N.
CHECKED	225
APPROVED	B.N.

HELICAL FITTINGS
SINGLE TOP TIES
FOR BARE COPPER CONDUCTORS
11KV OVERHEAD LINES

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		ORG No. M469

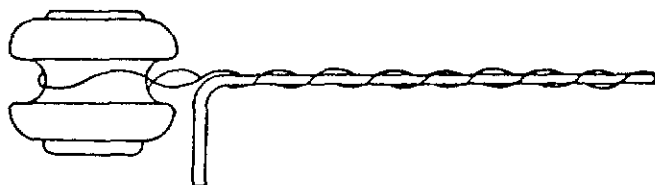
Cond. Type (Mat.)	Cond. C.S.A. mm ²	Cond. Strand	Cond. O/S Dia. mm	Cond. Cover O/S Dia. mm	1.		2.		M.I.N. N°
					Manufacturers Ref. N°	Colour Code	Manufacturers Ref. N°	Colour Code	
H.D.Cu.	32 35	3/3-75 7/2-50	8.05 7.50	-	LVDE5615409R	BLACK			3454703
H.D.Cu.	70	7/3-55	10.65	-	LVDE5615417R	WHITE			3454704
ALUM	50	7/3-10	9.30	-	LVDE5015407R	ORANGE			3454706
ALUM	100	7/4-39	13.17	-	LVDE5015413R	RED			3454707



MANUFACTURERS

- ① PREFORMED LINE PRODUCTS.
- ②

NOTES



HELICAL FITTINGS
TERMINATION / DEAD END TIES
FOR BARE CONDUCTORS
L.V. OVERHEAD LINES

ISSUE

ISSUE

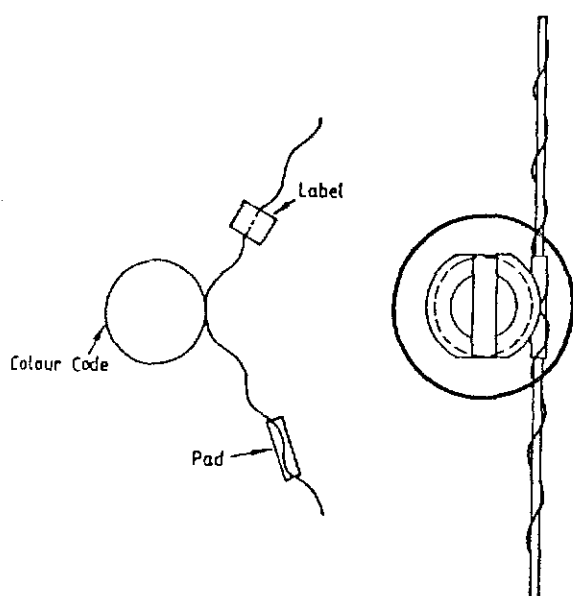
MIDLANDS
ELECTRICITY BOARD

35mm² Cond.
added.
19.12.89 A.J.B.

ORG No M470

SCALE
DATE 30.11.88
DRAWN B.N.
TRACED S.G.T.
CHECKED *qib*
APPROVED *hvj*

Cond. Type (Mat.)	Cond. C.S.A. mm. ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia. mm.	Manufacturers Ref. N°	Colour Code	Manufacturers Ref. N°	Colour Code	M.I.N. N°
CAD. Cu	38	7/2-60	7-80	-	GFT 5681420 RP	PURPLE			3454711
HD. Cu	32	3/3-75	8-05	-	GFT 5681422 RP	RED			3454712
HD. Cu	50	7/3-00	9-00	-					
CAD. Cu	55	7/3-20	9-60	-	GFT 5681424 RP	YELLOW			3454713
HD. Cu	70	7/3-55	10-65	-	GFT 5681426 RP	BLUE			3454714
CAD. Cu	75	7/3-70	11-10	-					
H.D. Cu	100	7/4-30	12-90	-	GFT 5681428 RP	ORANGE			3454715
CAD. Cu	115	19/2-77	13-85	-	GFT 5681430 RP	RED			3454716
HD. Cu	125	19/2-90	14-50	-					
A.C.S.R.	60	7/3-66	10-98	-	GFT 5081426 RP	BLUE			3454717
ALLOY	100	7/4-65	13-95	-	GFT 5081430 RP	RED			3454718
A.C.S.R.	100	ALL TYPES	14-15/14-60	-					
A.C.S.R.	150	37/2-59	18-13	-	GFT 5081434 RP	BROWN			3454719
A.C.S.R.	175	37/2-79	19-53	-	GFT 5081436 RP	RED			3454710
AL ALLOY	200	37/2-87	20-09	-					



MANUFACTURERS:-

- ① PREFORMED LINE PRODUCTS

NOTES:-

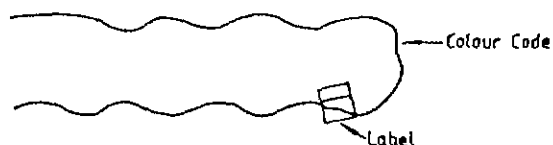
- ① INSULATOR NECK DIA. 120mm.

SCALE	---
DATE	29-11-88
DRAWN	B.N.
CHECKED	P.A.O.
APPROVED	228

HELICAL FITTINGS
SIDE TIES
FOR BARE CONDUCTORS
33 & 66 k.V. OVERHEAD LINES

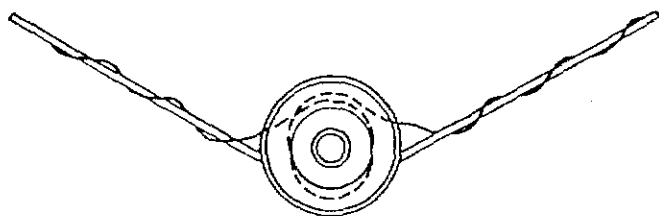
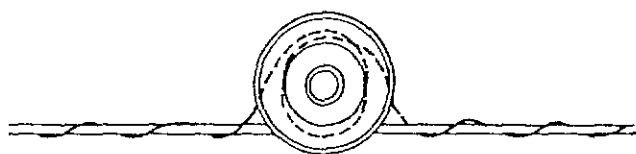
ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
	A	
	Mod. to M.I.N. Nos. 2-11-89 A.J.B.	DRG. No. M471

Cond. Type (Mat.)	Cond. C.S.A. mm. ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia. mm.	1		2		M.I.N. N°
					Manufacturers Ref. N°	Colour Code	Manufacturers Ref. N°	Colour Code	
H.D. Cu	32 35	3/3-75 7/2-50	8-05 7-50	-	LVT 5623869 RN	BLACK			3454723
H.D. Cu	70	7/3-55	10-65	-	LVT 5623877 RN	WHITE			3454724
ALUM	50	7/3-10	9-30	-	LVT 5023867 RN	ORANGE			3454726
ALUM	100	7/4-39	13-17	-	LVT 5023873 RN	RED			3454727



MANUFACTURERS :-

① PREFORMED LINE PRODUCTS

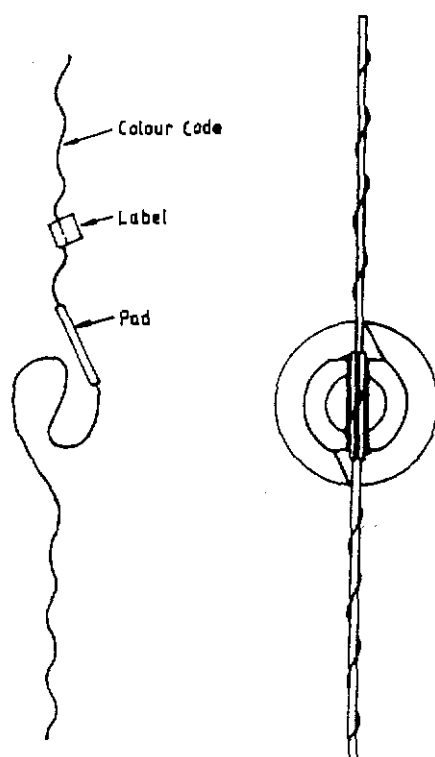


HELICAL FITTINGS
INTERMEDIATE / ANGLE TIES
FOR BARE CONDUCTORS
L.V. OVERHEAD LINES

SCALE	—
DATE	29-11-88
DRAWN	S.N.
CHECKED	P.A.O.
APPROVED	(Signature)

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
	A	
	35mm ² Cond added. 19-12-89 A.J.B.	DRG No. M472

Cond. Type (Mat.)	Cond. C.S.A. mm. ²	Cond. Strand	Cond. O/S Dia. m.m.	Cond. Cover O/S Dia. m.m.	1		2		M.I.N. No.
					Manufacturers Ref. No.	Colour Code	Manufacturers Ref. No.	Colour Code	
AL. ALLOY	25	7/2-34	7.02	-	DT5070718RP	ORANGE			3454731
A.C.S.R.	25	7/2-36	7.08	-	DT5070720RP	PURPLE			3454732
AL. ALLOY	40	7/2-95	8.85	-	DT5070722RP	RED			3454733
A.C.S.R.	40	7/3-00	9.00	-					
AL. ALLOY	50	7/3-30	9.90	-	DT5070724RP	YELLOW			3454734
A.C.S.R.	50	7/3-35	10.05	-					
AL. ALLOY	100	7/4-65	13.95	-	DT5070730RP	RED			3454735
A.C.S.R.	100	ALL TYPES	14.15/14.60	-					
A.C.S.R.	150	37/2-59	18.13	-	DT5070734RP	BROWN			3454736
A.C.S.R.	175	37/2-79	19.53	-	DT5070736RP	RED			3454737
AL. ALLOY	200	37/2-87	20.09	-					



MANUFACTURERS

① PREFORMED LINE PRODUCTS

NOTES

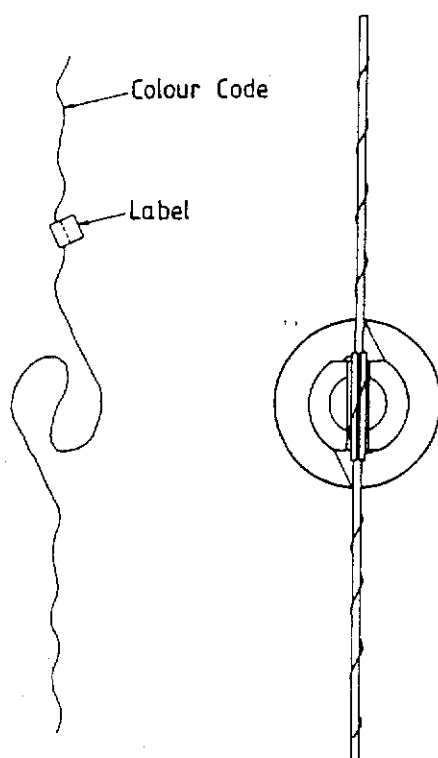
① INSULATOR NECK DIA 78 mm.

SCALE	
DATE	29.11.88
DRAWN	BN
TRACED	S.J.H.
CHECKED	BN
APPROVED	BN

HELICAL FITTINGS
SINGLE TOP TIES
FOR BARE ALUMINIUM CONDUCTORS
11kV OVERHEAD LINES

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		DRG No M473

Cond. Type (at)	Cond. C.S.A. mm. ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia. mm.	1		2		M.I.N. N°
					Manufacturers Ref. N°	Colour Code	Manufacturers Ref. N°	Colour Code	
H.D. Cu	16	3/2-65	5.69	9.30	DT5770728R	ORANGE			345474 1
CAD Cu	22	7/2-00	6.00	9.60					
H.D. Cu	32	3/3-75	8.05	12.10	DT5770732R	PURPLE			345474 3
H.D. Cu	70	7/3-55	10.65	14.70	DT5770734R	BROWN			345474 4
AL ALLOY	25	7/2-34	7.02	10.60	DT5770730R	RED			345474 2
A.C.S.R.	25	7/2-36	7.08	10.70					
AL ALLOY	40	7/2-95	8.85	12.80	DT5770732R	PURPLE			345474 3
A.C.S.R.	40	7/3-00	9.00	12.95					
AL ALLOY	50	7/3-30	9.90	13.90	DT5770732R	PURPLE			345474 3
A.C.S.R.	50	7/3-35	10.05	14.10	DT5770734R	BROWN			345474 4
AL ALLOY	100	7/4-65	13.95	18.00	DT5770736R	RED			345474 5
A.C.S.R.	100	ALL TYPES	14.15	18.20					
C.S.R.	150	37/2-59	18.13	22.20	DT5770740R	GREEN			345474 6
C.S.R.	175	37/2-79	19.53	23.60					
AL ALLOY	200	37/2-87	20.09	24.10	DT5770740R	GREEN			345474 6



MANUFACTURERS

① PREFORMED LINE PRODUCTS

NOTES

① INSULATOR NECK DIA. 78 mm.

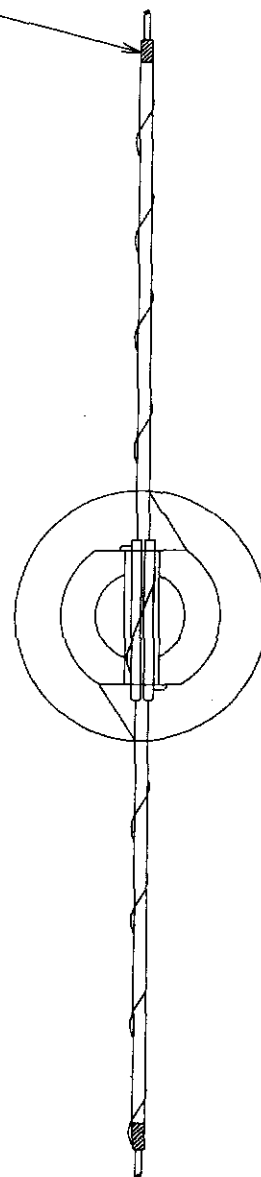
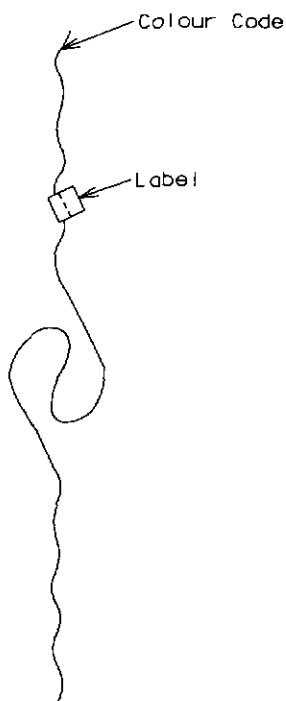
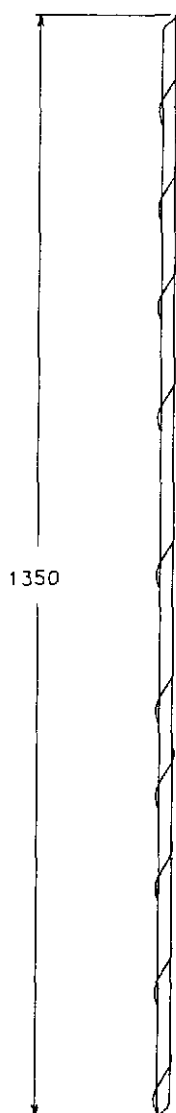
SCALE	—
DATE	25-11-88
DRAWN	B.N.
TRACED	G.H.R.
CHECKED	G.H.R.
APPROVED	G.H.R.

HELICAL FITTINGS
SINGLE TOP TIES
FOR P.V.C COVERED CONDUCTORS
11kV OVERHEAD LINES

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		DRG No
		M474

Min No	Cond. Type (Material)	Cond. mm ²	Cond. Stranding	Cond. O/S Dia mm	Cond. O/S Dia mm	Manufacturers Ref. N°	Colour Code	Spiral Wrap Length mm
3454743	H.D Cu	32	3/3.75	8.05	12.1	K480/1	PURPLE	1350
3454743	AL ALLOY	40	7/2.95	8.85	12.8	K480/1	PURPLE	1350
3454743	A.C.S.R.	40	7/3.00	9	12.95	K480/1	PURPLE	1350
3454743	AL ALLOY	50	7/3.30	9.9	13.9	K480/1	PURPLE	1350

ENDS TO BE SECURED
WITH BLACK P.V.C. TAPE



Helical Top Tie and Semi-Conductive Spiral Wrap are supplied together in a sealed plastic bag.

Prior to application the Spiral Wrap and Perform are to be clean and free from contamination

SCALE	-
DATE	25.11.88
DRAWN	B.N.
TRACED	G.H.R.
CHECKED	
APPROVED	

HELICAL FITTING SINGLE TOP TIE
FOR P.V.C. COVERED CONDUCTOR 11KV COMPLETE
WITH SEMI-CONDUCTING SPIRAL WRAP

ISSUE	DATE
A	17.05.01
Redrawn on C.A.D.	
D.B.	



Aquila

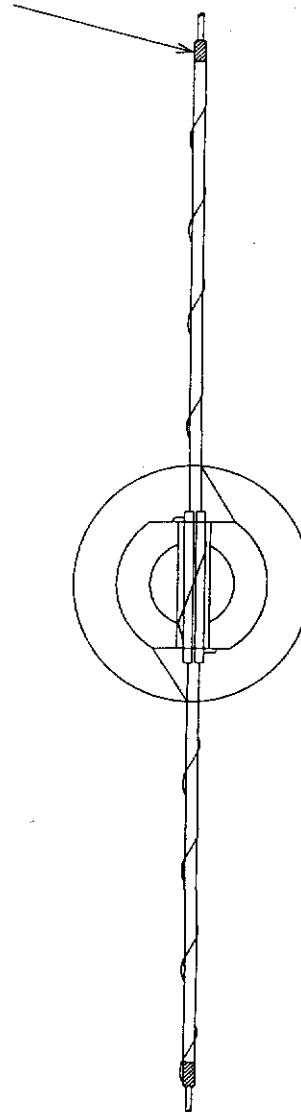
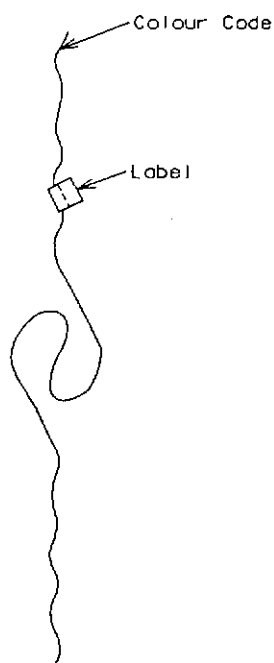
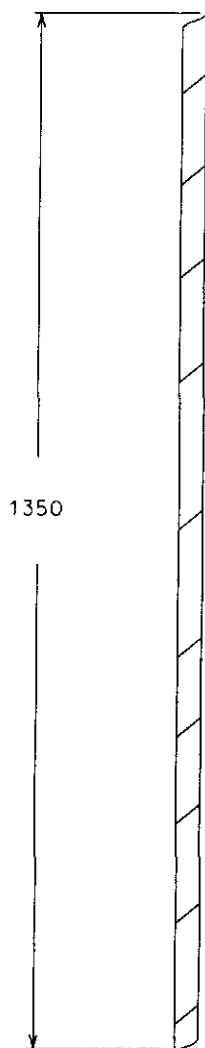
DRG. No.

M 474/3

DRG. No. M 474/3

Min No	Cond. Type (Material)	Cond. mm ²	Cond. Stranding	Cond. O/S Dia mm	Cond. Cover O/S Dia mm	Manufacturers Ref. N°	Colour Code	Spiral Wrap Length mm
3454743	H.D Cu	32	3/3.75	8.05	12.1	K480/1	PURPLE	1350
3454743	AL ALLOY	40	7/2.95	8.85	12.8	K480/1	PURPLE	1350
3454743	A.C.S.R.	40	7/3.00	9	12.95	K480/1	PURPLE	1350
3454743	AL ALLOY	50	7/3.30	9.9	13.9	K480/1	PURPLE	1350

ENDS TO BE SECURED
WITH BLACK P.V.C. TAPE



Helical Top Tie and Semi-Conductive Spiral Wrap are supplied together in a sealed plastic bag.

Prior to application the Spiral Wrap and Preform are to be clean and free from contamination

SCALE	-
DATE	05.06.01
DRAWN	D.B.
TRACED	
CHECKED	K.D.
APPROVED	07.06.01

HELICAL FITTING SINGLE TOP TIE
FOR P.V.C. COVERED CONDUCTOR 11KV COMPLETE
WITH SEMI-CONDUCTING SPIRAL WRAP

ISSUE DATE

Produced by
Output Services

GPU
POWER UK

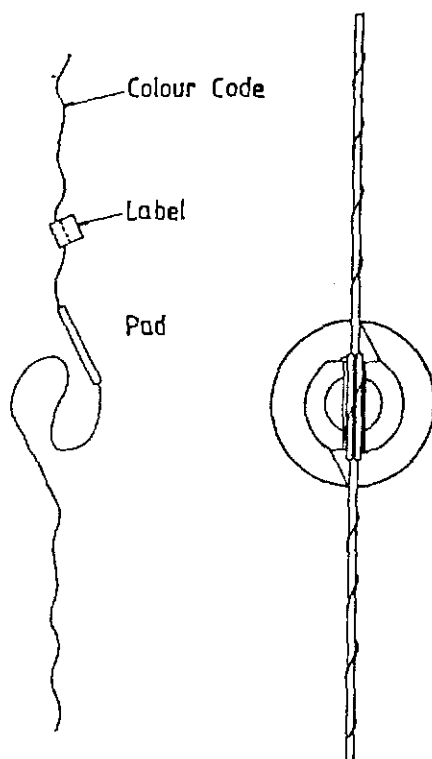
DRG. No.

M 474/3

DRG. No. M 474/3

A4

Cond. Type (Mat.)	Cond. CSA mm. ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia. mm.	1		2		MIN. N°
					Manufacturers Ref. N°	Colour Code	Manufacturers Ref. N°	Colour Code	
CAD Cu	38	7/2-60	7-80		DT5671420RP	PURPLE			3454751
H.D. Cu	32	3/3-75	8-05		DT5671422RP	RED			3454752
H.D. Cu	50	7/3-00	9-00						
CAD Cu	55	7/3-20	9-60		DT5671424RP	YELLOW			3454753
H.D. Cu	70	7/3-55	10-65		DT5671426RP	BLUE			3454754
CAD Cu	75	7/3-70	11-10						
H.D. Cu	100	7/4-30	12-90		DT5671428RP	ORANGE			3454755
CAD Cu	115	19/2-77	13-85		DT5671430RP	RED			3454756
H.D. Cu	125	19/2-90	14-50						
AC.S.R.	60	7/3-66	10-98		DT5071426RP	BLUE			3454757
AL ALLOY	100	7/4-65	13-95		DT5071430RP	RED			3454758
AC.S.R.	100	ALL TYPES	14-15/14-60						
AC.S.R.	150	37/2-59	18-13		DT5071434RP	BROWN			3454759
AC.S.R.	175	37/2-79	19-53						
AL ALLOY	200	37/2-57	20-09		DT5071436RP	RED			3454750



MANUFACTURERS

① PREFORMED LINE PRODUCTS

NOTES

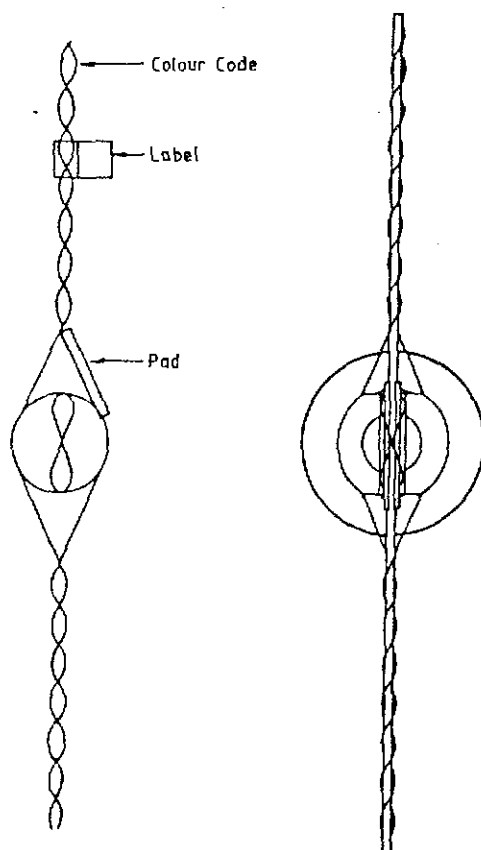
① INSULATOR NECK DIA. 120 mm.

SCALE	—
DATE	25-11-88
DRAWN	BN
CHECKED	GHR
APPROVED	QJB

HELICAL FITTINGS
SINGLE TOP TIES
FOR BARE CONDUCTORS
33 & 66 kV. OVERHEAD LINES

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
	A	
	Mod. to MIN Nos. 2-11-89 AJB	ORG. No. M475

Cond. Type (Mat.)	Cond. C.S.A. mm. ²	Cond. Strand	Cond. O/S Dia. mm.	Cond. Cover O/S Dia. mm.	1		2		M.I.N. No.
					Manufacturers Ref. No.	Colour Code	Manufacturers Ref. No.	Colour Code	
CAD. Cu.	38	7/2-60	7-80	-	DDT5671420RP	PURPLE			3454761
HD. Cu.	32	3/3-75	8-05	-	DDT5671422RP	RED			3454762
HD. Cu.	50	7/3-00	9-00	-					
CAD. Cu.	55	7/3-20	9-60	-	DDT5671424RP	YELLOW			3454763
HD. Cu.	70	7/3-55	10-65	-	DDT5671426RP	BLUE			3454764
CAD. Cu.	75	7/3-70	11-10	-					
HD. Cu.	100	7/4-30	12-90	-	DDT5671428RP	ORANGE			3454765
CAD. Cu.	115	19/2-77	13-85	-	DDT5671430RP	RED			3454766
HD. Cu.	125	19/2-90	14-50	-					
A.C.S.R.	60	7/3-66	10-98	-	DDT5071426RP	BLUE			3454767
ALLOY	100	7/4-65	13-95	-	DDT5071430RP	RED			3454768
A.C.S.R.	100	ALL TYPES	14-15/14-60	-					
A.C.S.R.	150	37/2-59	18-13	-	DDT5071434RP	BROWN			3454769
A.C.S.R.	175	37/2-79	19-53	-	DDT5071436RP	RED			3454760
AL. ALLOY	200	37/2-87	20-09	-					



MANUFACTURERS.

- ① PREFORMED LINE PRODUCTS

NOTES.

- ① INSULATOR NECK DIA. 120mm.

SCALE	-
DATE	1-12-88
DRAWN	B.N.
TRACED	C.A.R.
CHECKED	qbb
APPROVED	BN

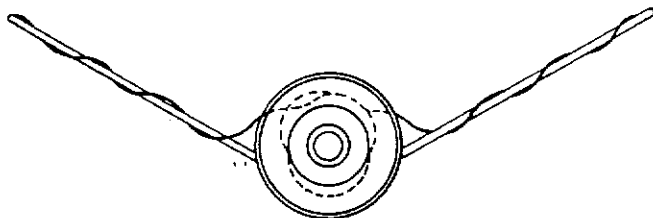
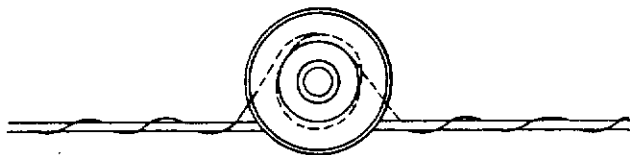
HELICAL FITTINGS
DOUBLE TOP TIES
FOR BARE CONDUCTORS
33 & 66kV OVERHEAD LINES

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
	A	
	Mod. to MIN Nos. 2.11.89 A.18	DRG. No. M476

Cond Type (Mat.)*	Cond C.S.A mm ²	Cond Strand	Cond O/S Dia. m.m.	Cond Cover O/S Dia. m.m.	1		2		M.I.N. N°
					Manufacturers Ref N°	Colour Code	Manufacturers Ref N°	Colour Code	
H D Cu	70	3/3-55	10-65	13-50	LVT5723877RN	YELLOW			345 4774
ALUM	50	7/3-10	9-30	11-70	LVT5723837RN	GREEN			345 4776
ALUM	100	7/4-39	13-17	16-00	LVT5723843RN	BROWN			345 4777

MANUFACTURERS

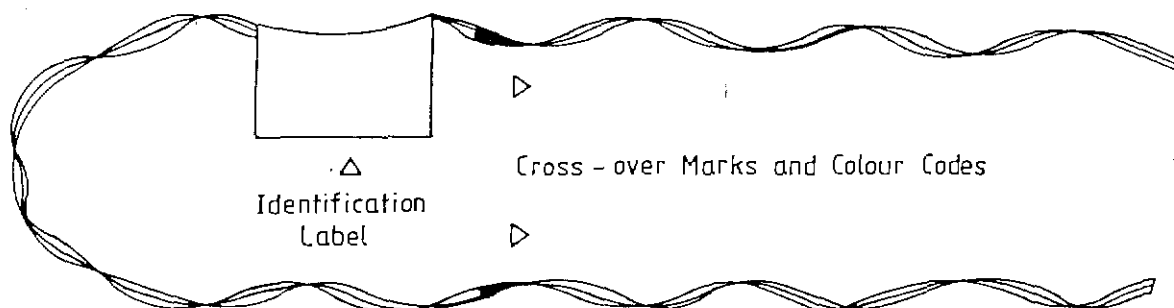
① PREFORMED LINE PRODUCTS



DATE	1:12:00
DRAWN	B.N.
TRACED	S.J.H.
CHECKED	PPB
APPROVED	PPB

HELICAL FITTINGS
INTERMEDIATE / ANGLE TIES
FOR P.V.C. COVERED CONDUCTORS
L.V. OVERHEAD LINES

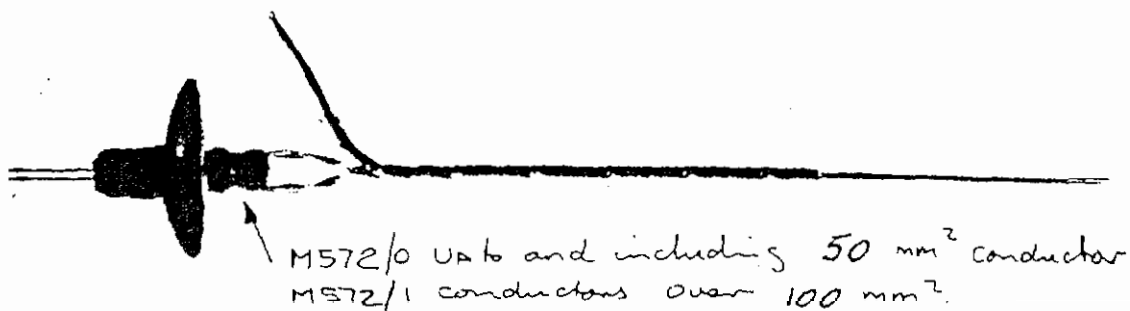
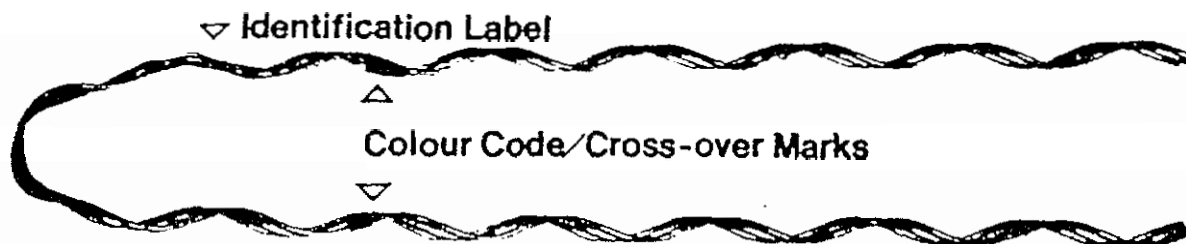
ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		DRG No M 477



MIN N°	Conductor	Conductor Dia (mm)	Max Span (metres)	Manufacturer	Manufacturers Reference	Colour Code	Fitting Length (mm)
3454781	25mm ² Hybrid Concentric Cable	13.0	20	Preformed Line Products	LTCND5709147-R	Brown	550
3454782	16mm ² Copper Concentric Cable	12.1	20	Preformed Line Products	LTCND5710121-R	Green	550

Note:- Limited Tension Helical Fittings should only be used on single spans where the Span Length is less than 20metres.

ADDED BY DATE	TITLE HELICAL FITTING CONCENTRIC CABLE DEAD ENDS	SCALE	—	 Midlands Electricity plc ORG No M478
		DATE	9-7-93	
		DRAWN	A.J.B.	
		TRACED	P.J.W.	
		CHECKED	<i>RCJ</i>	
		APPROVED		



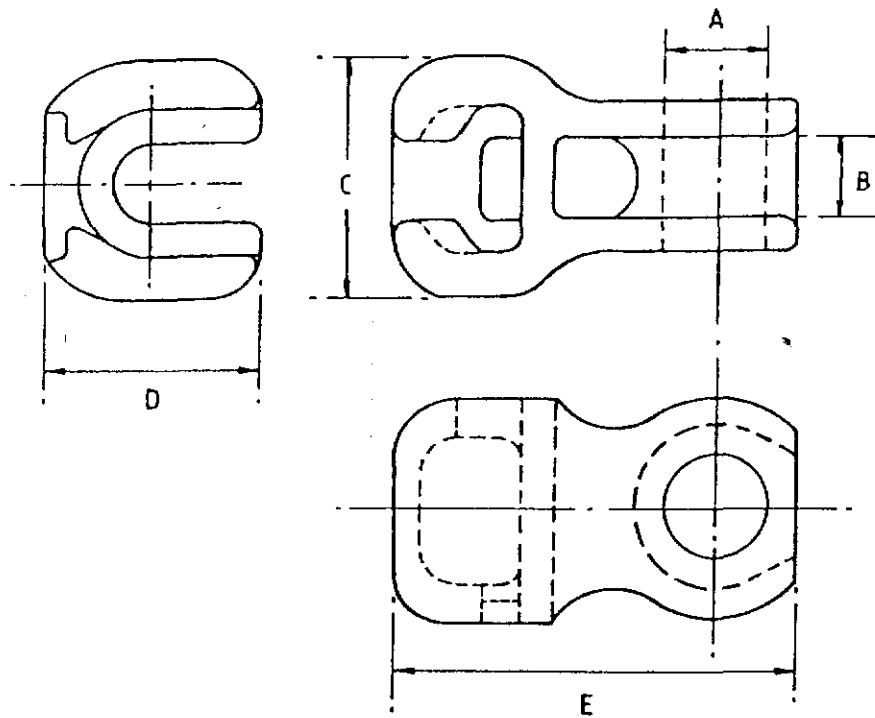
INSTALLATION : Installation Procedure covered in Method OHL24 Fig 9.

NOTE : Conductor Grip MUST NOT Be Re-used after the Original Installation

M.REF	MIN	CONDUCTOR TYPE AND SIZE	LENGTH	MAN. REF.	COLOR CODE
571/0	3455710	25 mm ² AAAC	559	DBE5011714R	YELLOW
571/1	3455711	50 mm ² AAAC	686	DBE5011717R	YELLOW
571/2	3455712	100 mm ² AAAC	889	DBE5011720R	PURPLE

CONDUCTOR DATA END GRIP
AAAC CONDUCTORS

M571



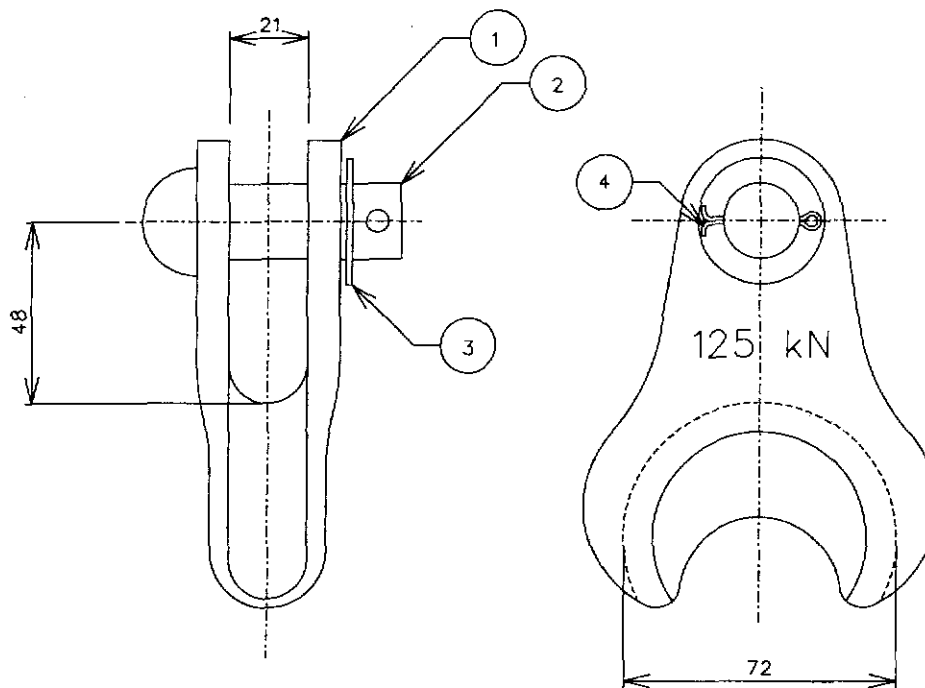
M. REF	MIN.	DIMENSIONS					MAN. REF.
		A	B	C	D	E	
572/0	3455720	24	19	56	51	94	STT6260001
572/1	3455721	30	22	56	51	115	STT6260012

COUPLING

16mm SOCKET TO B.S.3288

GALV. TO B.S.729

ADD BY DATE	TITLE SOCKET THIMBLE HV HELICAL DEAD END GRIPS MFL 70 KN	SCALE	—	 Midlands Electricity plc DRG No M572
		DATE	5.10.92	
		DRAWN	A.J.B.	
		TRACED	E.L.W.	
		CHECKED		
		APPROVED		



NOTES

1. PIN/WASHER:- HOT DIP GALVANISED TO BS729
2. THIMBLE CLEVIS CAST IN MALLEABLE IRON TO BS 6681 GRADE P 55-04

ITEM	DESCRIPTION
1	THIMBLE CLEVIS
2	Ø 20 x 65 lg ROUND HEAD PIN - Gr 8.8
3	M20 GALV. WASHER
4	SPLIT PIN

FILE	1:2
DATE	05.08.02
DRAWN	MJB
TRACED	-
CHECKED	-
APPROVED	-

Thimble Clevis 125kN

ISSUE	DATE



Aquila

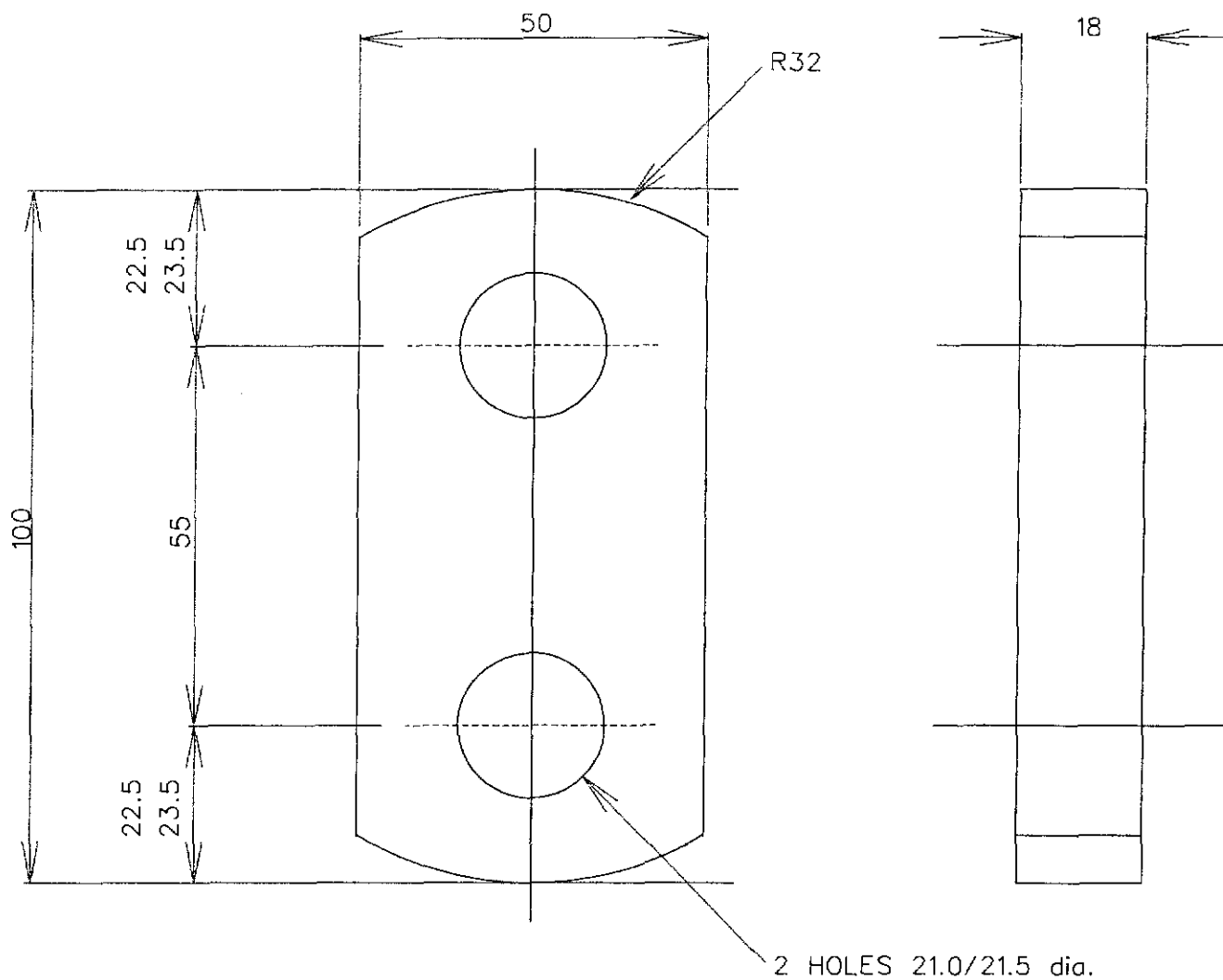
DRG. No.

M572/2

DRG. No.

M572/2

A4



NOTES

1. HOT DIP GALVANISE TO BS729

All dimensions are in millimeters unless otherwise stated.

SCALE	1:4
DATE	02/08/02
DRAWN	GC
TRACED	
CHECKED	-
APPROVED	-

DOUBLE EYE LINK
120kN RATING

ISSUE	DATE



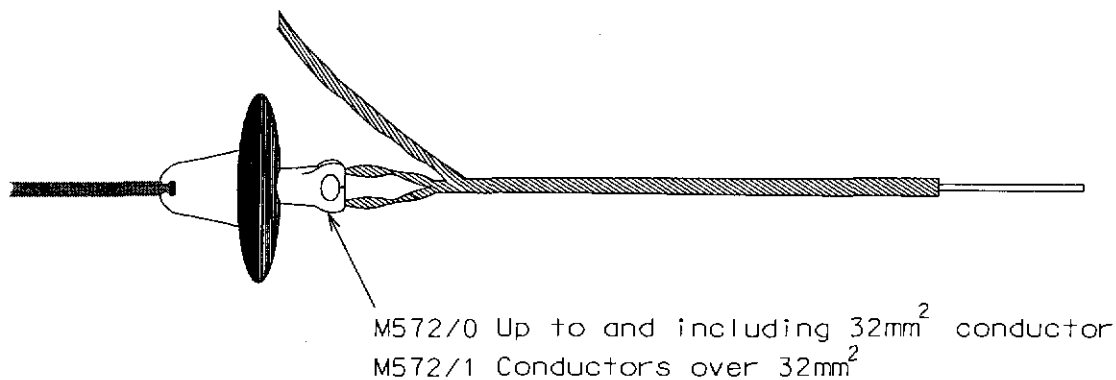
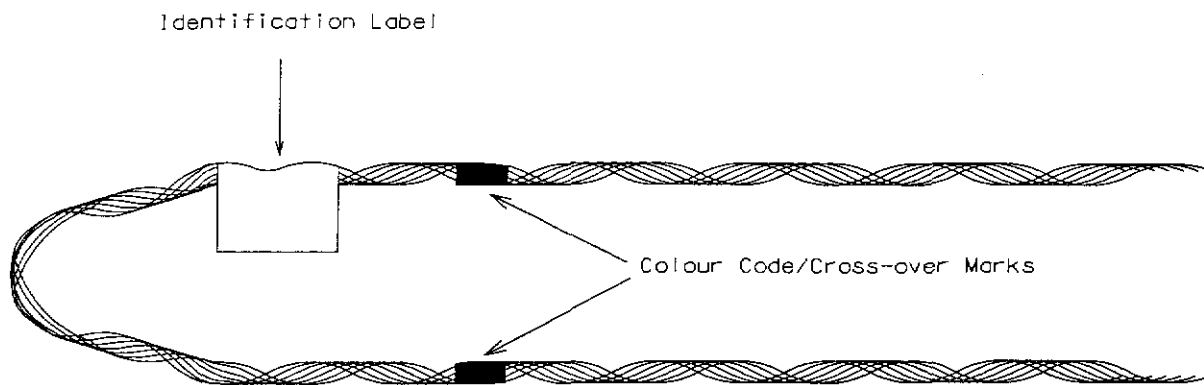
Aquila

DRG. No.

M572/3

DRG. No.

M572/3



INSTALLATION: Installation Procedure covered in method DHL 24 Fig.9.

NOTE: Conductor Grip MUST NOT be re-used after the original installation

M.REF	MIN	CONDUCTOR TYPE AND SIZE	LENGTH	M.REF.	COLOUR CODE
574/0	3455740	16mm ² H/D COPPER	457	DDE4611812R	BROWN
574/1	3455741	25mm ² H/D COPPER	559	DDE4611814R	YELLOW
574/2	3455742	32mm ² H/D COPPER	635	DDE4611815R	RED
574/4	3455744	70mm ² H/D COPPER	686	DDE4611817R	YELLOW
574/6	3455746	100mm ² H/D COPPER	838	DDE4611819R	ORANGE
574/7	3455747	125mm ² H/D COPPER	889	DDE4611821R	BLUE

SCALE	-
DATE	25.05.01
DRAWN	D.B.
TRACED	
CHECKED	K.D.
APPROVED	07.06.01

CONDUCTOR DEAD END GRIP
H/D COPPER CONDUCTORS

ISSUE DATE

Produced by Output Services

GPU
POWER UK

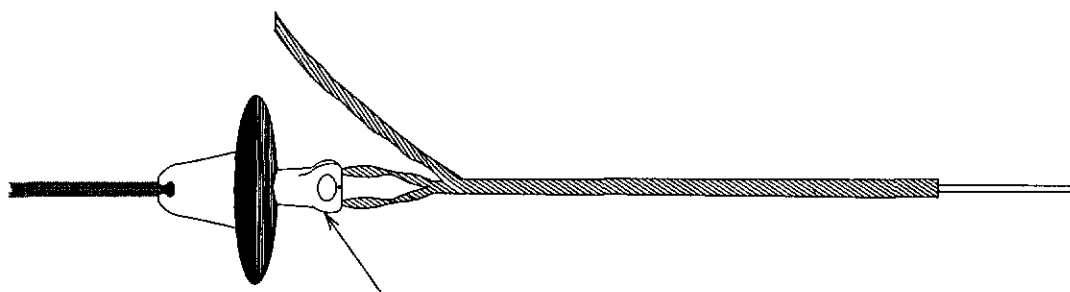
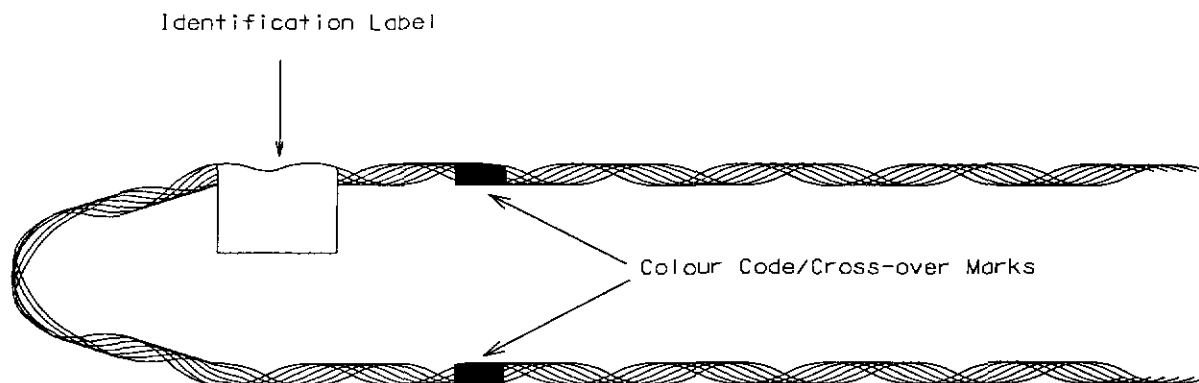
DRG. No.

M574

DRG. No.

M574

A4



M572/1 LARGE SOCKET THIMBLE

NOTE: Conductor Grip MUST NOT be re-used after the original installation

M.REF	MIN	CONDUCTOR TYPE AND SIZE	LENGTH	M.REF.	COLOUR CODE
575/0	3455750	.15 H/D CAD COPPER 19/2.77	889	DOE4611820R	PURPLE

3455745 *1 4/0 CBD COPPER

SCALE	-
DATE	22.05.01
DRAWN	D.B.
TRACED	
CHECKED	K.D.
APPROVED	07.06.01

CONDUCTOR DEAD END GRIP
.15 H/D CAD COPPER

ISSUE	DATE

Produced by
Output Services

GPU
POWER UK

DRG. No.
M575

DRG. No.

M575

A4

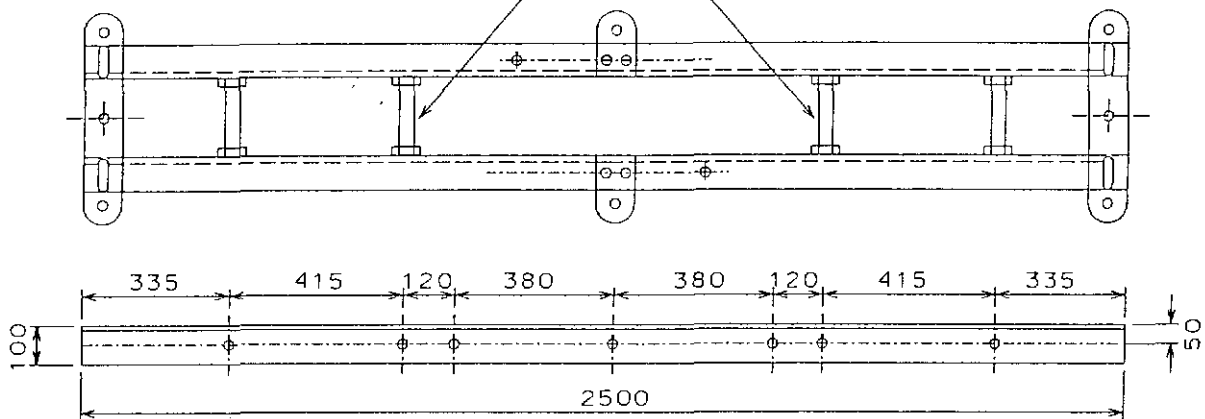
SECTION 12

MIN NUMBER	M	DESCRIPTION
3450041	4/1	SECTION ANGLE CROSSARM SINGLE/H POLES (HEAVY CONSTRUCTION)
3450050	5	ANGLE STRUT
3450480	48	ANGLE CROSSARM
3450491	49/1	ANGLE CROSSARM
3450492	49/2	ANGLE CROSSARM
3450500	50	ANGLE CROSSARM
3450510	51	INTERMEDIATE CROSSARM H POLE 11& 33 kv (HEAVY CONSTRUCTION)
3450560	* 56	LINE CROSSARM 1M SPACING
3450560	* 56	LINE CROSSARM 1M SPACING MODIFIED
3450561	* 56/1	LINE CROSSARM SINGLE PHASE
3450570	* 57	CROSSARM TERMINAL
3450570	* 57	CROSSARM TERMINAL MODIFIED
3450571	* 57/1	CROSSARM TERMINAL SINGLE PHASE
3450580	58	SECTION ANGLE CROSSARM 1M SPACING
3450581	* 58/1	SECTION ANGLE CROSSARM SINGLE PHASE
3450590	59	TIE STRAP
3450630	63	SQUARE CURVED WASHER
3450730	73	SQUARE TAPERED CURVED WASHER
3450762	76/2	AUTO RECLOSER & TRANSFORMER MOUNTING CROSSARM (H-POLE)
3450820	82	H-POLE TERMINAL CROSSARM / LINE
3450961	96/1	LINE CROSSARM 11 & 33 kv HEAVY
3451000	100	CHANNEL CROSSARM 66kv
3451010	101	BRACED ANGLE CROSSARM
3451032	103/2	BRACED TERMINAL CROSSARM WITH PROVISION FOR SURGE DIVERTERS (66kv HEAVEY CONSTRUCTION
3451033	103/3	TERMINAL CROSSARM 66 Kv
3451060	106	TERMINATING STRAP
3451070	107	11 & 33kv H-POLE THRO.(PIN) CROSSARM (HEAVY)
3451450	145	LINE CROSSARM (TERMINAL)
3451460	146	LINE TERMINATING STRAP
3451480	148	TERMINAL CROSSARM 11kv HEAVY CONSTRUCTION
3451490	149	BRACED TERMINAL CROSSARM 33kv HEAVY CONSTRUCTION
3451500	150	LINE STRAP
3451501	150/1	LANDING STRAP TERMINAL
3451510	151	LONG SECTION STRAP HEAVY CONSTRUCTION
3451511	151/1	LANDING STRAP SECTION
3451520	152	SHORT SECTION STRAP (HEAVY CONSTRUCTION)
3451530	153	BRACED ANGLE CROSSARM
3451540	154	11 & 33kv PILOT INSULATOR SPINDLE
3451560	156	SINGLE CROSSARM LINE SECTION STRAP
3451860	186	SECTION CROSSARM
3451870	187	SECTION CROSSARM
3451880	188	LINE CROSSARM 66kv DUAL (SUSP)
3451890	189	LINE CROSSARM LONG 66 Kv DUAL (SUSP)
3451910	191	LINE CROSARM 1.2M SPACING SINGLE & H-POLE
3451940	194	SECTION ANGLE CROSSARM 1.2M SPACING SINGLE & H-POLE
3451960	196	SECTION ANGLE CROSSARM 1.4M SPACING SINGLE & H-POLE
3451961	196/1	CROSSARM STRUT FOR 1.4 SPACING SECTION ANGLE & 1.2 SPACING TERMINAL CROSSARMS
3452220	222	DUAL CIRCUIT CROSSARM
3452260	226	ANGLE STRUT
3452270	227	LINE CROSSARM
3452280	228	D; SHAPED TAPERED WASHER
3452460	246	CROSSARM TIE STRAP

SECTION 12

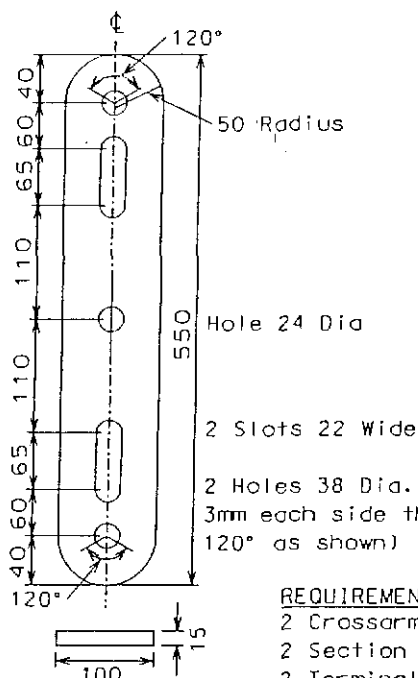
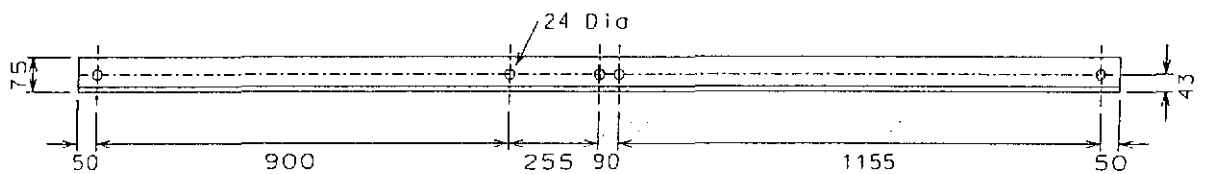
MIN NUMBER	M	DESCRIPTION
3452571	257/1	BRACED ANGLE CROSSARM 66kV UP TO 200mm ² AAAC
3452580	258	SUSPENSION TYPE CHANNEL CROSSARM 66kV
3452590	259	LONG SECTION STRAP 125kN
3452591	259/1	LONG SECTION STRAP (MIN FAILING LOAD 70kN)
3452600	260	SHORT SECTION STRAP 125kN
3452601	260/1	SHORT SECTION STRAP (MIN FAILING LOAD 70kN)
3452610	261	LONG TERMINAL STRAP 125kN
3452620	262	SHORT TERMINAL STRAP 125kN
3452680	268	SECTION STRAP (MIN FAILING LOAD 20kN)
3453150	315	BRASS WASHER
3453170	317	SQUARE FLAT WASHER
3453180	318	CROSSARM STRAP FOR USE WITH M107 400mm H.D.A
3453190	319	ROUND FLAT WASHER
3453620	362	POLE BOLTS 16mm DIA
3453630	363	POLE BOLTS 20mm DIA
3453640	364	ASSEMBLY BOLTS 12mm DIA
3453650	365	ASSEMBLY BOLTS 16mm DIA
3453660	366	ASSEMBLY BOLTS 20mm DIA
3454000	400	TERMINAL CROSSARM 1.2M SPACING SINGLE POLES 100mm ² MAX
3456050	605	TOWER FITTING STEP BOLTS

Only required for 200AAAC



Holes 22 Dia
unless otherwise stated

100x75x12 B.S.U.A.



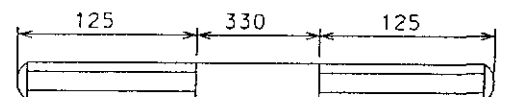
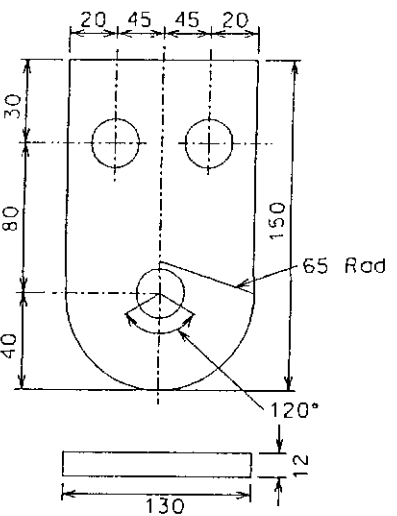
STRAP SECTION
(ESI No 439525)

2 Holes 22 Dia

Hole 38 Dia

(Edges chamfered
3mm each side
through 120°
as shown)

STRAP TERMINAL
(ESI No. 439528)



M20 Tie Bolt threaded 125 each end
C/W 4 Nuts

REQUIREMENTS

- 2 Crossarm Members
- 2 Section Straps
- 2 Terminal Straps
- 4 Tie Bolts
- 12 M20x60 bolts C/W
Nut and Plain Washer
- 4 Crossarm Struts M5 or M196/1

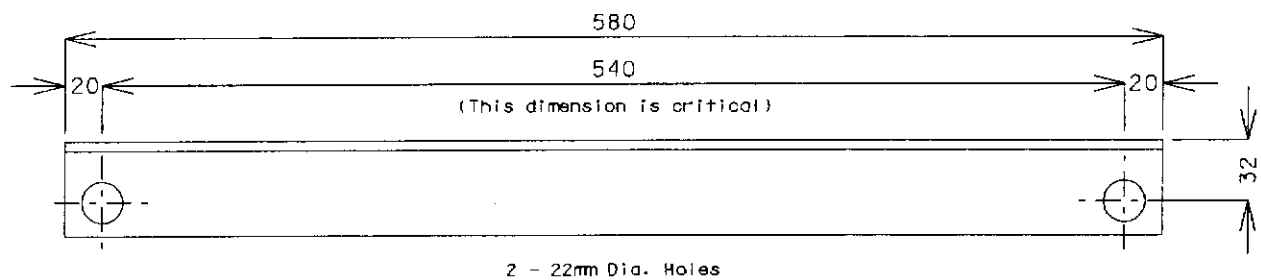
C.A.D. PRODUCED

SCALE	-
DATE	6.10.99
DRAWN	A.J.E.
TRACED	
CHECKED	K.P.D.
APPROVED	

HEAVY CONSTRUCTION
SECTION ANGLE CROSSARM
1.2m SPACING
SINGLE & H POLES
100/200AC

ISSUE	Midlands Electricity plc
DRG. No.	M4/1

DRG. No. M4/1



MATERIAL

50 x 50 x 6 M.S. Angle to B.S. 4.

FINISH

Galv. to B.S. 729
after cutting and drilling.

NOTE:

For use on Heavy Construction.

All dimensions are in millimeters unless otherwise stated.

Produced by
Output Services

SCALE	1-4
DATE	4/9/00
DRAWN	ADD
TRACED	
CHECKED	13/10/00
APPROVED	KD

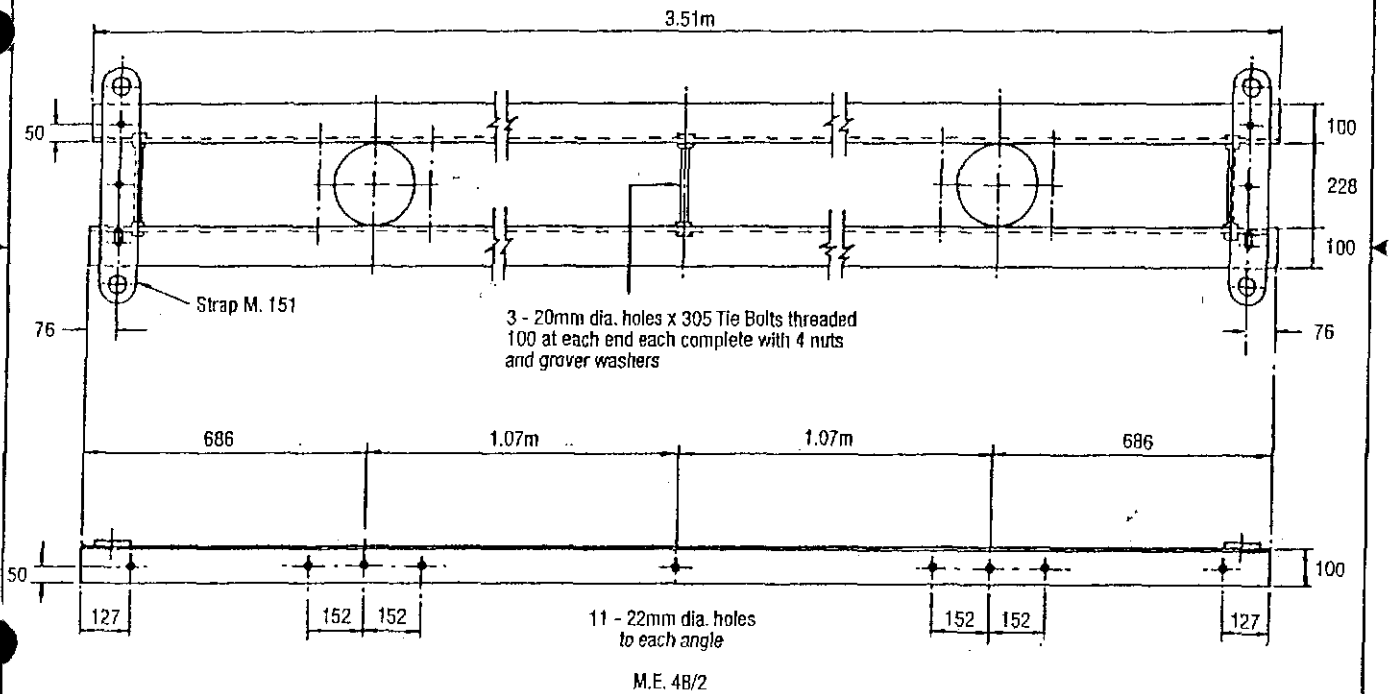
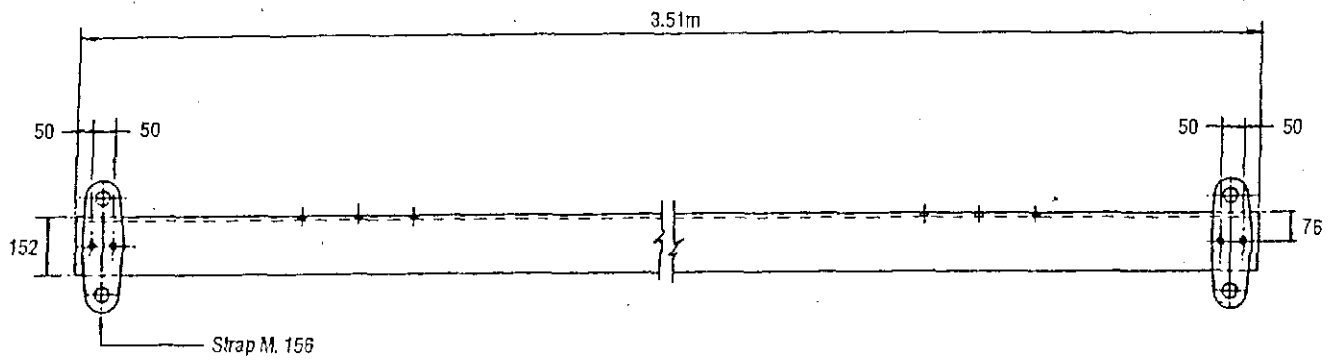
TIE STRUP
11KV HEAVY CONSTRUCTION/
33KV CONSTRUCTION

ISSUE	DATE

GPU POWER UK
DRG. No. M5

DRG. No.

M5



MATERIAL

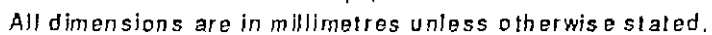
- 48/1 152 x 100 x 10 M.S. Angle to B.S. 4
For conductors up to and including 100m² copper or copper equivalent
- 48/2 100 x 100 x 10 M.S. Angle to B.S. 4
For conductors up to and including 200m² copper or copper equivalent

FINISH

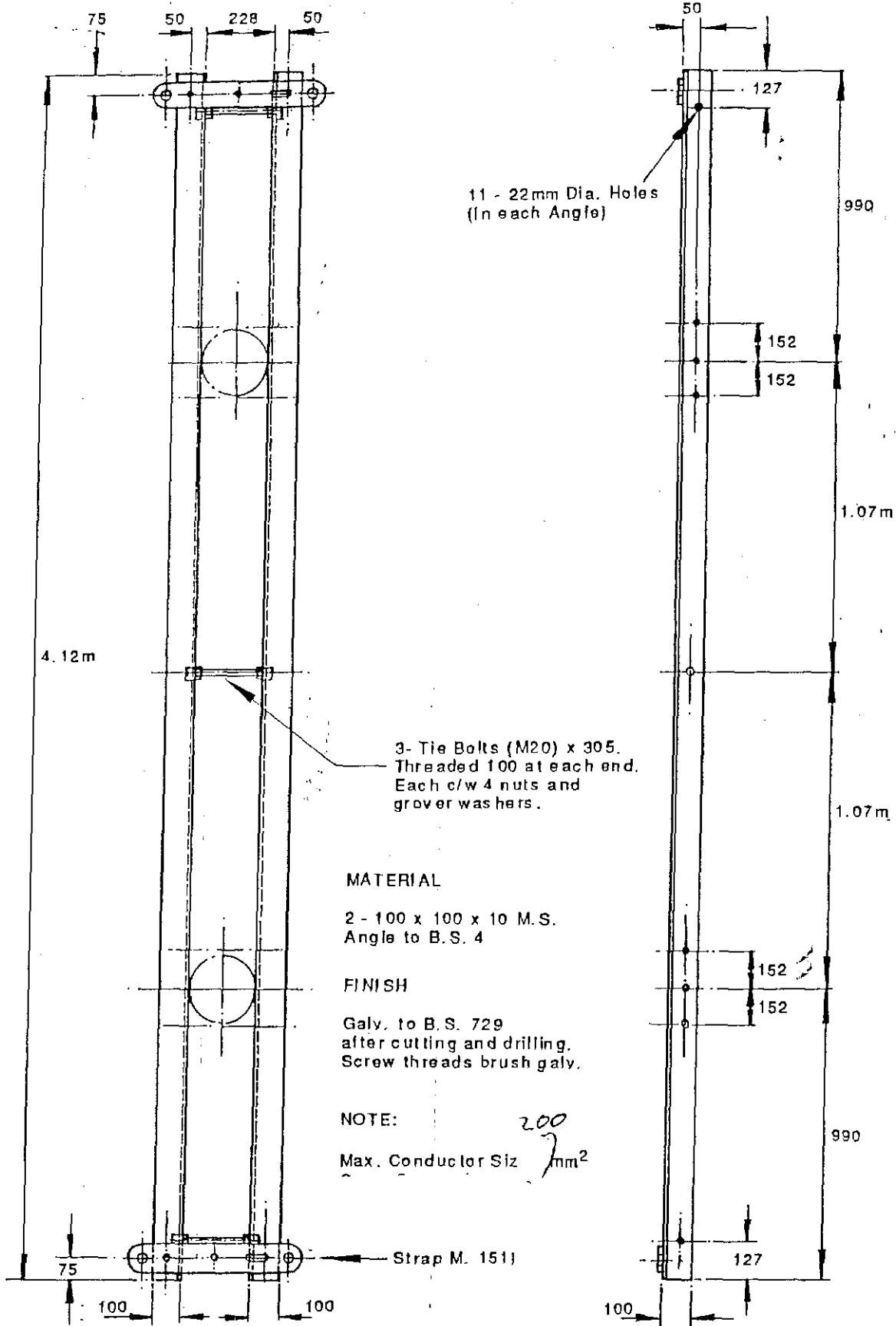
Galv. to B.S. 729 after cutting and drilling.
Screw threads brush Galv.

All dimensions are in millimetres unless otherwise stated

DRAWN BY: DATE:	TITLE ANGLE CROSSARM	SCALE	1 : 20	 Midlands Electricity plc
		DATE	20/2/73	
		DRAWN		
		TRACED		
		CHECKED		DRG. No. M. 48
		APPROVED		

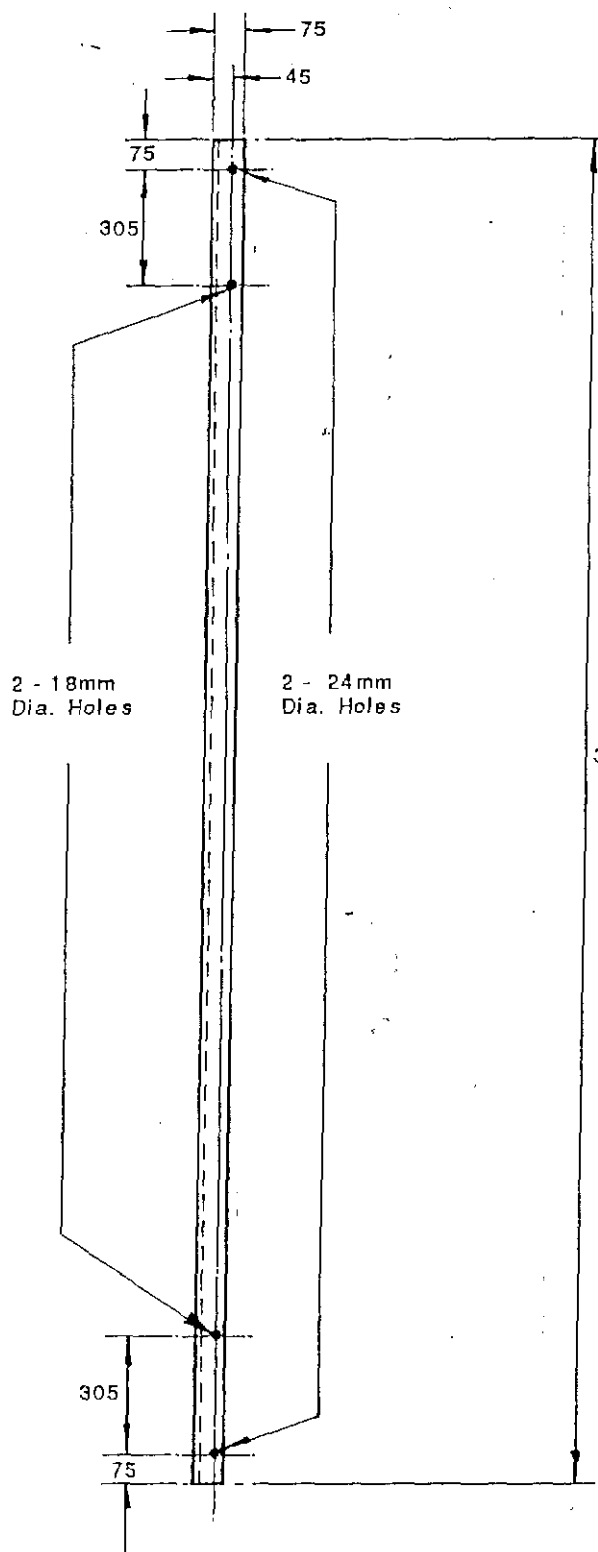


AMDO 15628 E



All dimensions are in millimetres unless otherwise stated.

ADDED BY DATE	TITLE	SCALE	1/20	 Midlands Electricity plc
		DATE	25/8/93	
		DRAWN		
		TRACED		ORG No M. 49/2
		CHECKED		
		APPROVED		



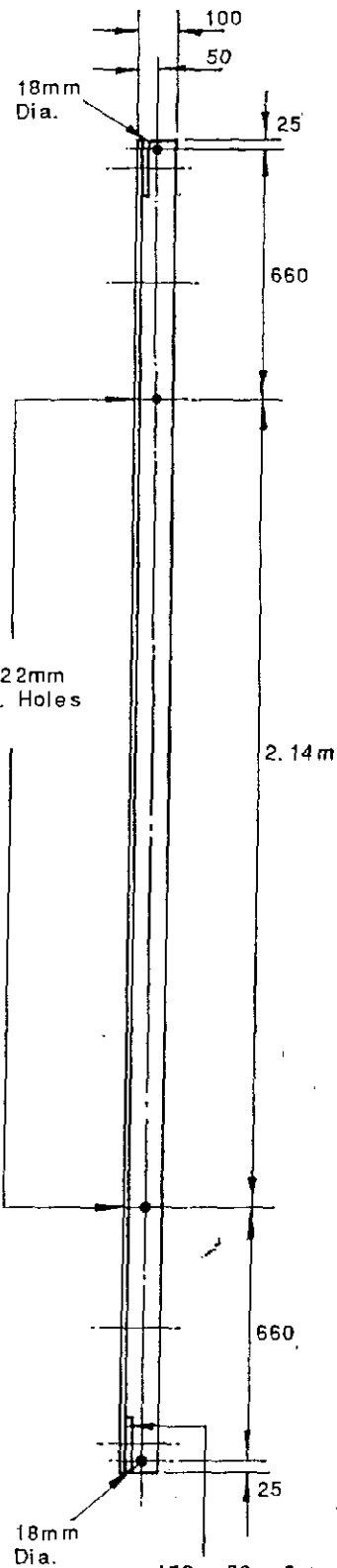
2 - 18mm
Dia. Holes

2 - 24mm
Dia. Holes

3.51m

MATERIAL
100 x 75 x 10 M.S.
Angle to B.S. 4
2 - 152 x 50 x 6 M.S. Flat

FINISH
Galv. to B.S. 729
after cutting and drilling.



2 - 22mm
Dia. Holes

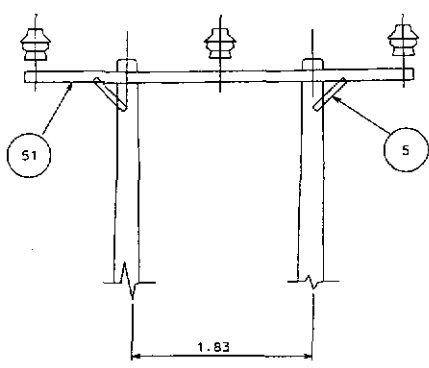
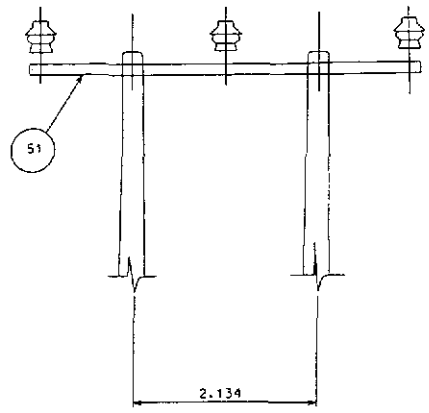
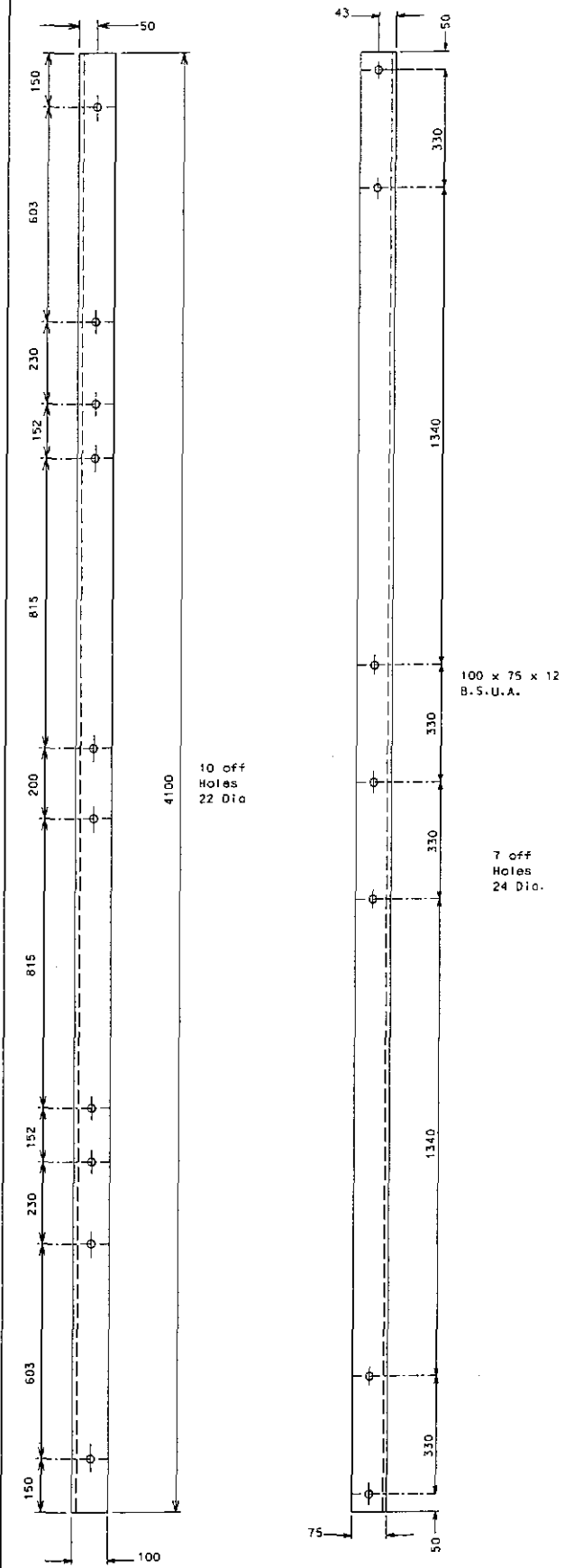
2.14m

18mm
Dia.

152 x 50 x 6
reinforcing plate
welded in position
at each end.

All dimensions are in millimetres unless otherwise stated.

ADDED BY DATE	TITLE	SCALE	1/20	 Midlands Electricity plc
		DATE	25/8/93	
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		
ANGLE CROSSARM		ORG No		M. 50



Support Straps M5 to be fitted on H poles with 1.83 (6') centres.

Material
Mild Steel to BS4360 grade 43A
100 x 75 x 12 B.S.U.A. to BS 4848

FINISH
Hot Dip Galvanised to BS729
after fabrication and drilling.

ALL DIMENSIONS IN mm

SCALE	NTS
DATE	06.09.00
DRAWN	DB
TRACED	
CHECKED	27.09.00
APPROVED	KD

HEAVY CONSTRUCTION
11KV AND 33KV.
H POLE INTERMEDIATE CROSSARM

ISSUE	DATE

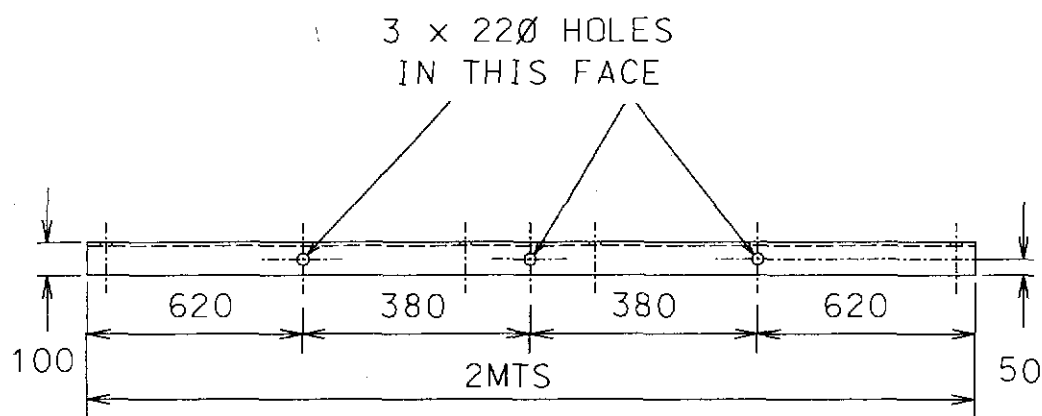
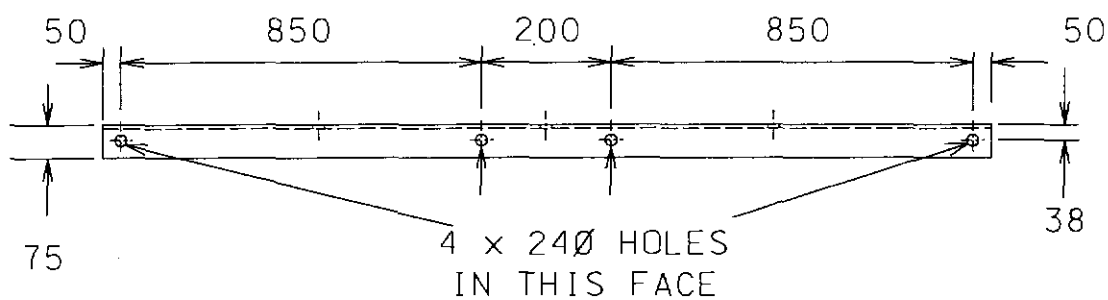
Produced by
Output Services

GPU
POWER UK

DRG. No.
M 51

DRG. No. M 51

A4



Material : 100 x 75 x 10 BSUA

Finish : Galv. to B.S. 729 after cutting & drilling

SCALE	—
DATE	17.09.01
DRAWN	D.B.
TRACED	—
CHECKED	08.10.01
APPROVED	K.D.

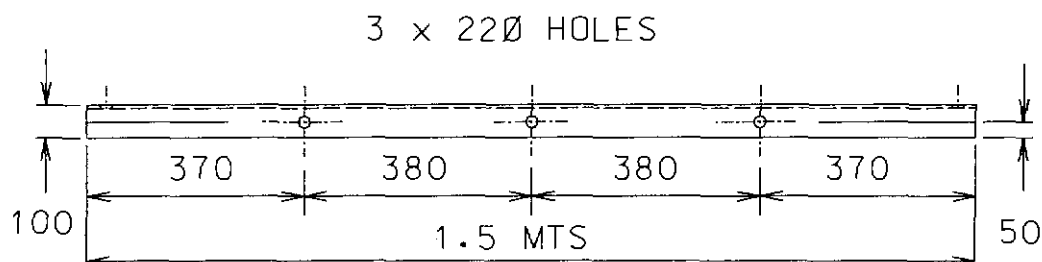
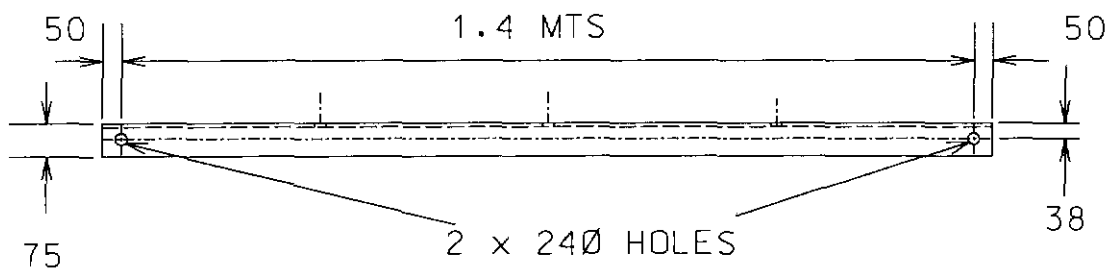
LINE CROSSARM
MODIFIED M 56 CROSSARM

ISSUE	DATE

Produced by Output Services	
	
ORG. No. M56	

DRG. No.

M56



Material : 100 x 75 x 10 BSUA

Finish : Galv. to B.S. 729 after cutting & drilling

SCALE	—
DATE	17.09.01
DRAWN	D.B.
TRACED	—
CHECKED	08.10.01
APPROVED	K.D.

SINGLE PHASE
LINE CROSSARM
MODIFIED M56 CROSSARM

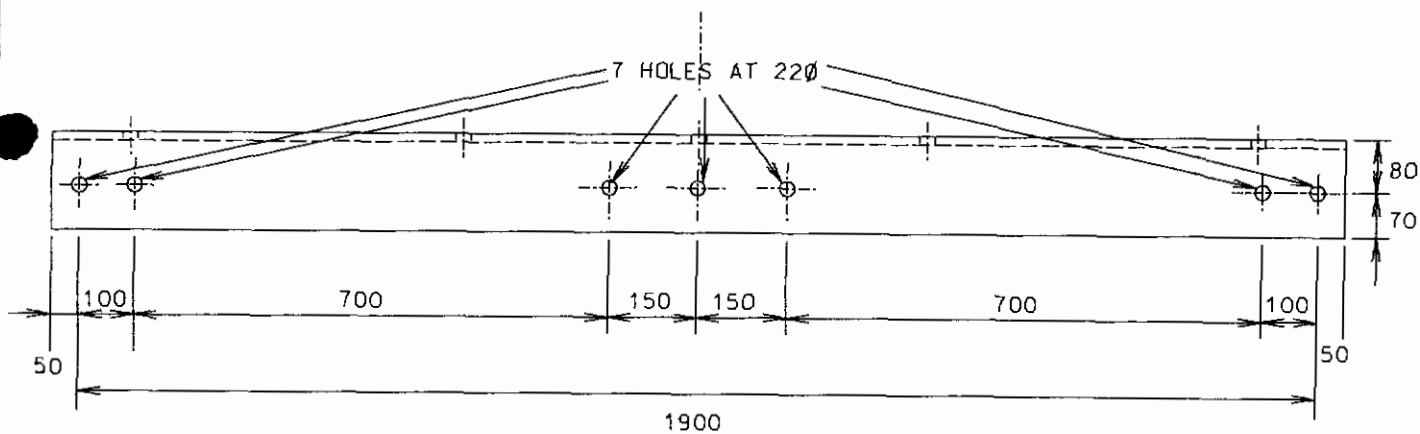
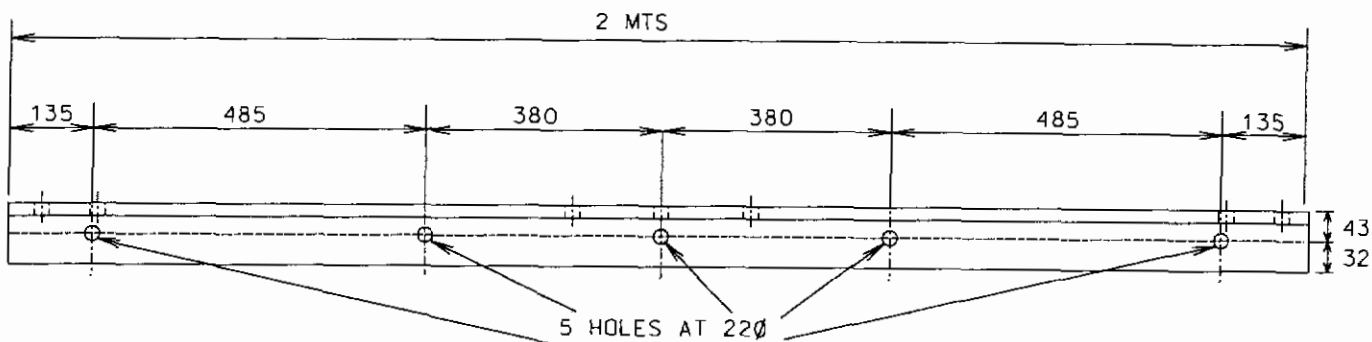
ISSUE	DATE

Produced by Output Services	
	
DRG. No.	M56/1

DRG. No.

M56/1

A4



MATERIAL
150 x 75 x 10 B S U A

MODIFIED
CROSSARM-TERMINAL
ESI DR No 439503
ESI STANDARD 43-95

SCALE	~
DATE	12.09.01
DRAWN	D.B.
TRACED	~
CHECKED	08.10.01
APPROVED	K.D.

ISSUE DATE

ISSUE DATE

ISSUE DATE

ISSUE DATE

ISSUE DATE

ISSUE DATE

ISSUE DATE

ISSUE DATE

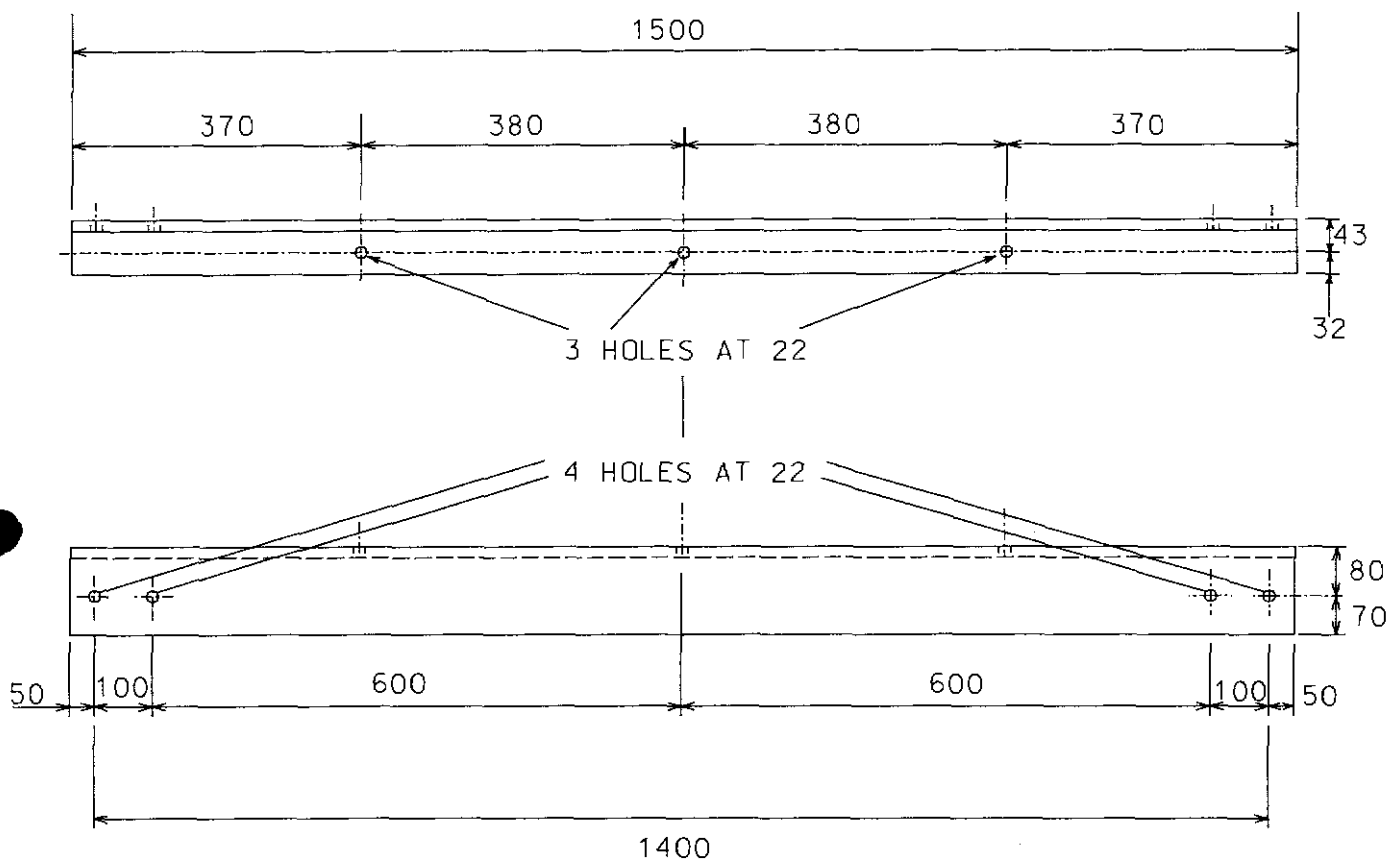
ISSUE DATE

Produced by
Output Services

GPU
POWER UK

DRG. No.
M57

DRG. No. M57



SCALE	-
DATE	11.09.01
DRAWN	D.B.
TRACED	-
CHECKED	08.10.01
APPROVED	K.D.

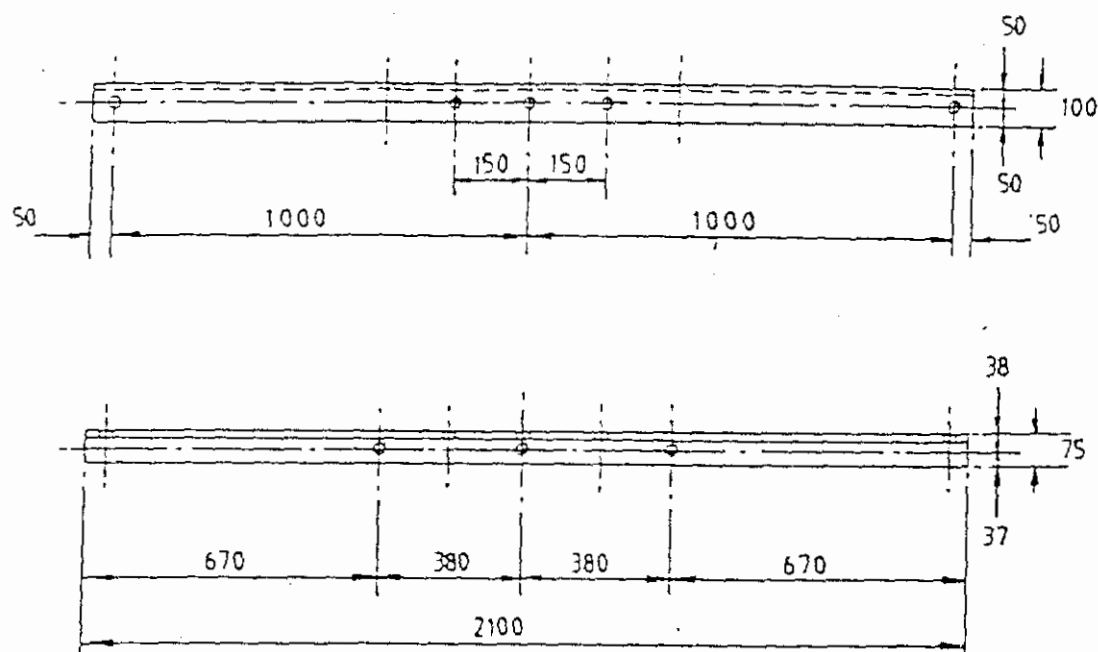
SINGLE PHASE M57/1
ESI DR.No 439503 TERMINAL

ISSUE	DATE

Produced by Output Services	
DRG. No. M57/1	
ESI Org No 439503	

DRG. No.

M57/1



All holes 22 ϕ

MATERIAL
100 x 75 x 10 BSUA

FINISH
To be galv. to B.S. 729
after cutting and drilling

TITLE

SECTION ANGLE CROSSARM
1M SPACING
ESi DR No 439502

SCALE

DATE

DRAWN

TRACED

CHECKED

APPROVED

13-2-92

A.J.B.

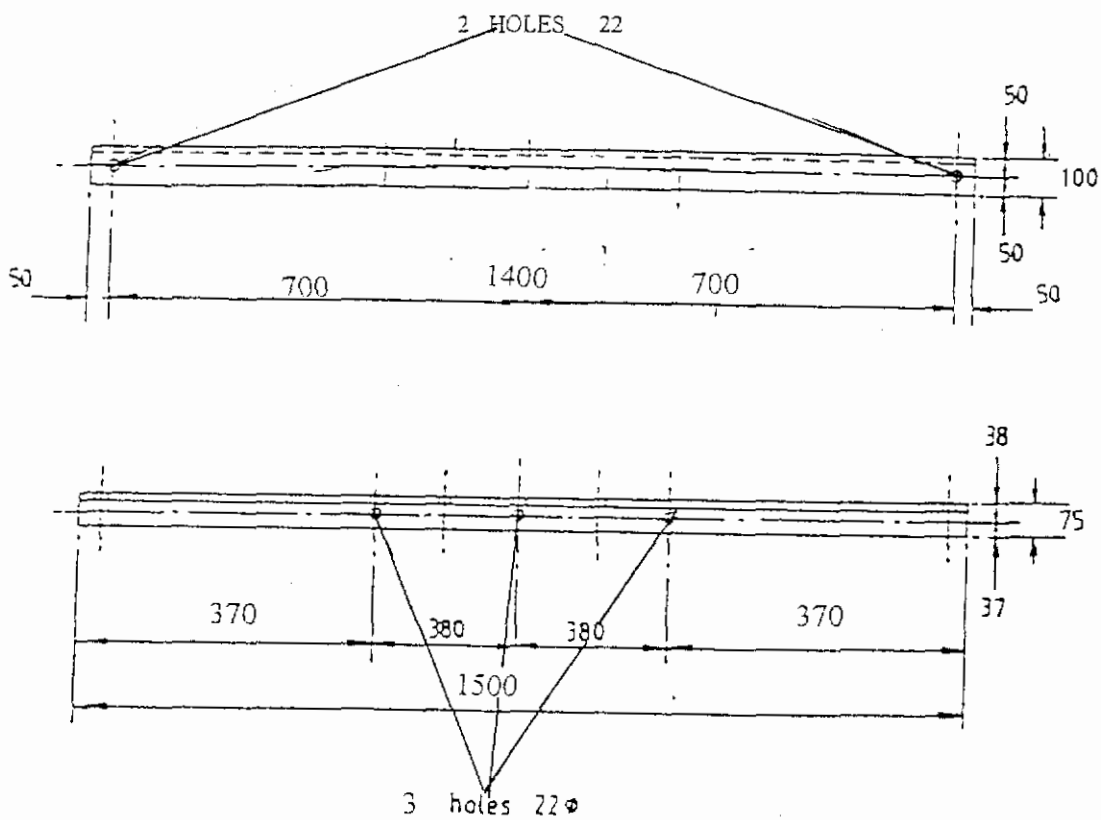
P.J.W.

DWG No

MEB

Midlands Electricity plc

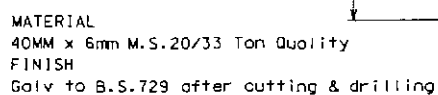
M 58

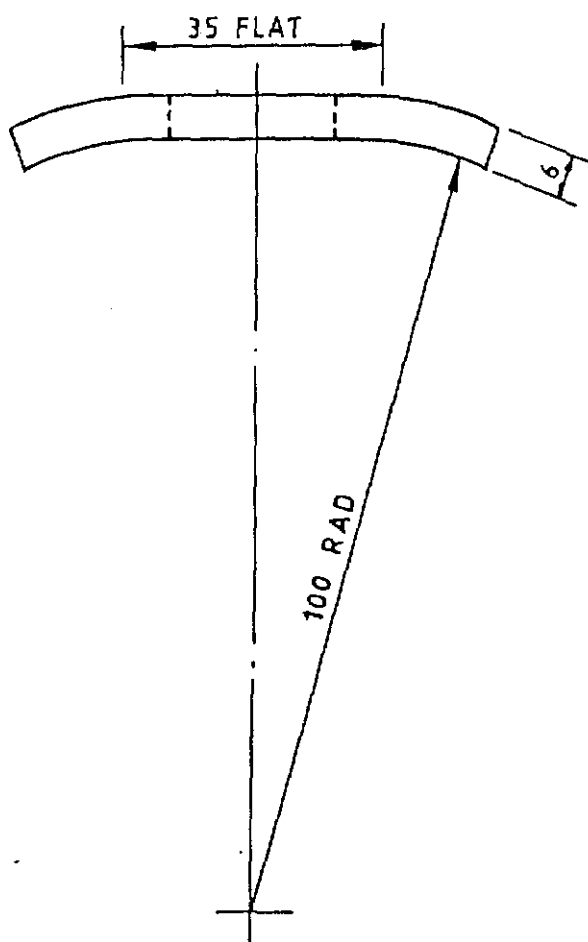
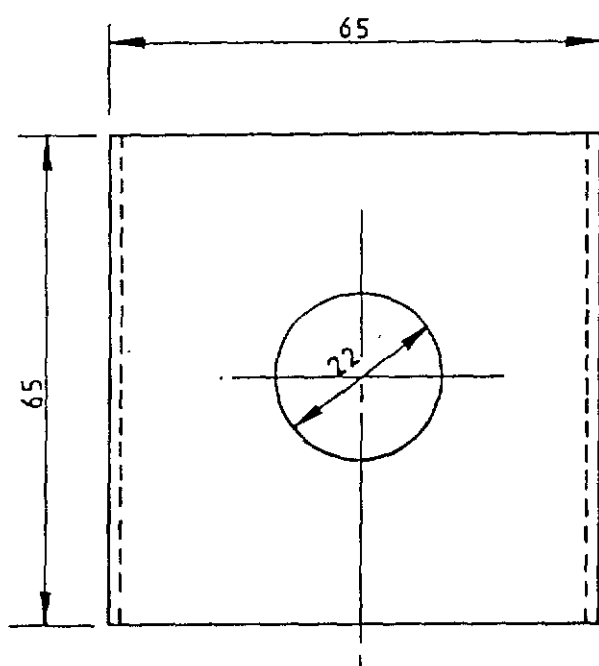


MATERIAL
100 x 75 x 10 BSUA

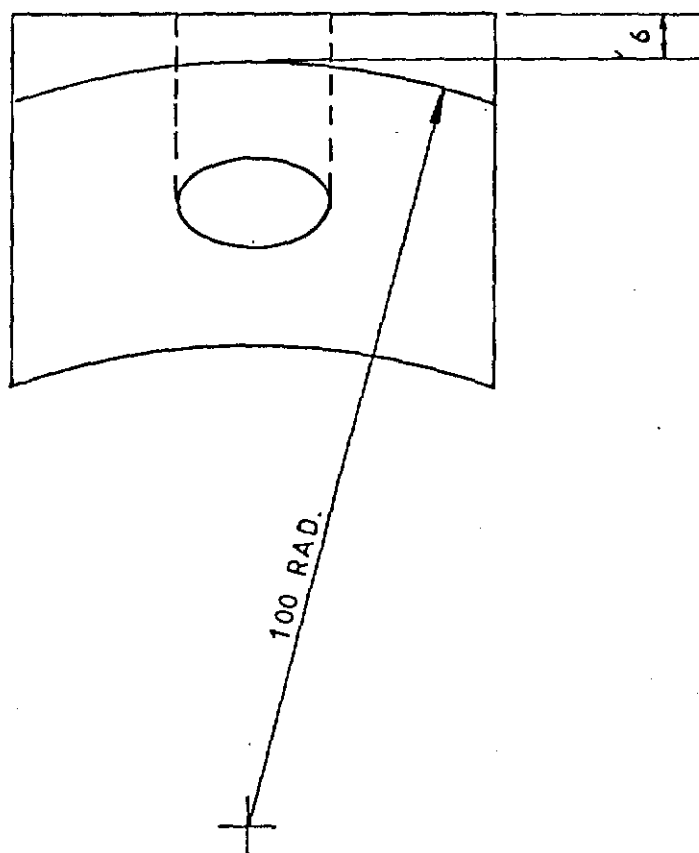
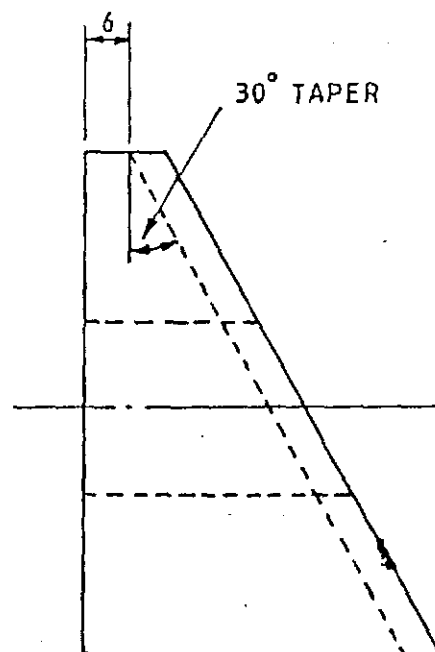
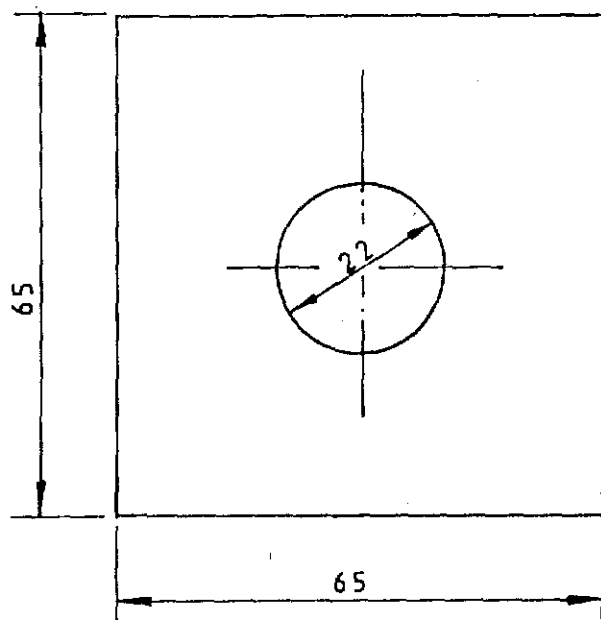
FINISH
To be galv. to B.S. 729
after cutting and drilling

E 10000 1 B/R JW 13-2-92 DIMENSIONED REVISED RETRACED DRG.	TITLE	SINGLE PHASE ONLY SECTION ANGLE CROSSARM	SCALE	---	MEB Midlands Electricity plc M 58/1
			DATE	13-2-92	
			DRAWN	A.J.B.	
			TRACED	P.J.W.	
			CHECKED		
		ESi DR No 439502	APPROVED		





SCALE	N.T.S.	SQUARE CURVED WASHER	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE				
DRAWN				DRG. No.
TRACED				M. 63
CHECKED				
APPROVED	<i>[Signature]</i>			
C. D. W Ltd. 3287		ESI. STANDARD 43 96		ESI. DRG. No 439605

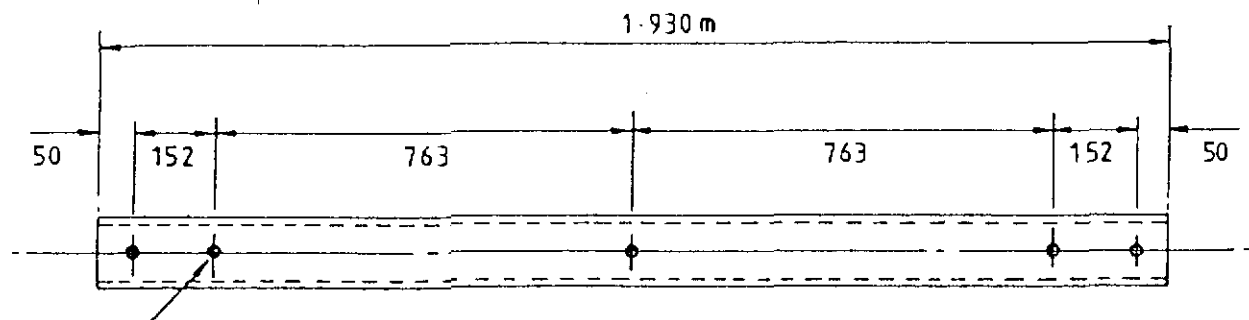


SCALE	N.T.S.
DATE	
DRAWN	
TRACED	
CHECKED	
APPROVED	

SQUARE TAPERED CURVED
WASHER

ISSUE	MIDLANDS ELECTRICITY BOARD
	DRG. No.
	M.73

EST DGN No. 439607



5 - 22 Dia. Holes

MATERIAL

2 off per set
 127 x 64 x 14.9 Kg/m
 M.S. Channel to B.S.4 Part 1
 28/33 ton quality

FINISH

Galv. to B.S. 729 after
 cutting & drilling

NOTES

1. For Mounting Transformers
 use 4 - Gusset Plates (M75).
 Transformer to be held with
 2 - Fixing straps (M2)
 see Section 20
2. Use ME183 for BTH Type
 Recloser and as Hoisting Channel
 for all Reclosers see Section 1.

ADDED BY
 DATE

TITLE

HIGH SPEED AUTO RECLOSER &
 TRANSFORMER MOUNTING CROSSARM
 ('H' POLE)

SCALE

DATE

DRAWN

TRACED

CHECKED

APPROVED

6.1.93

A.J.B.

E.L.W.

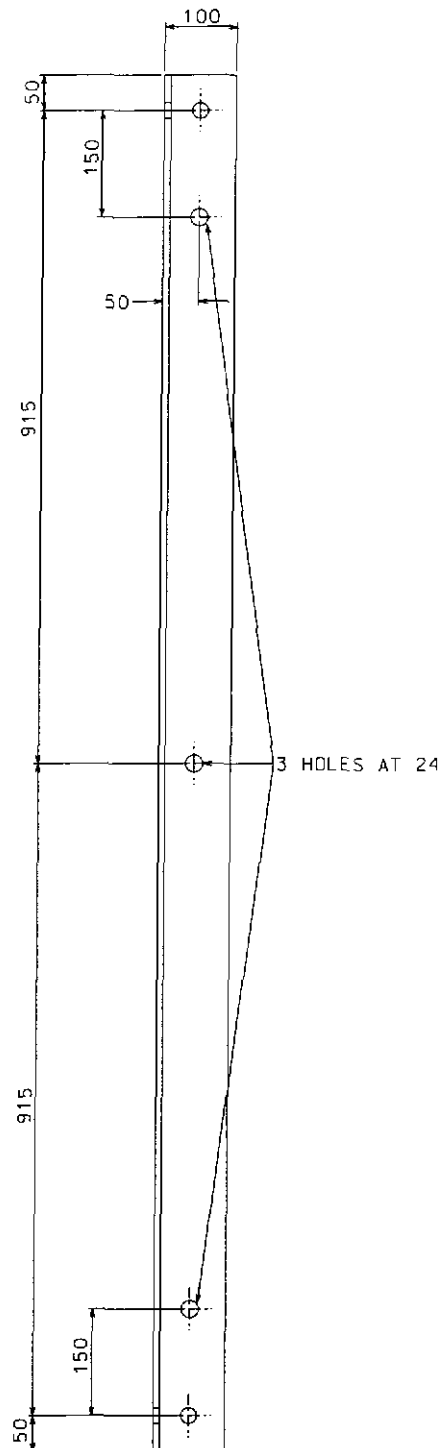
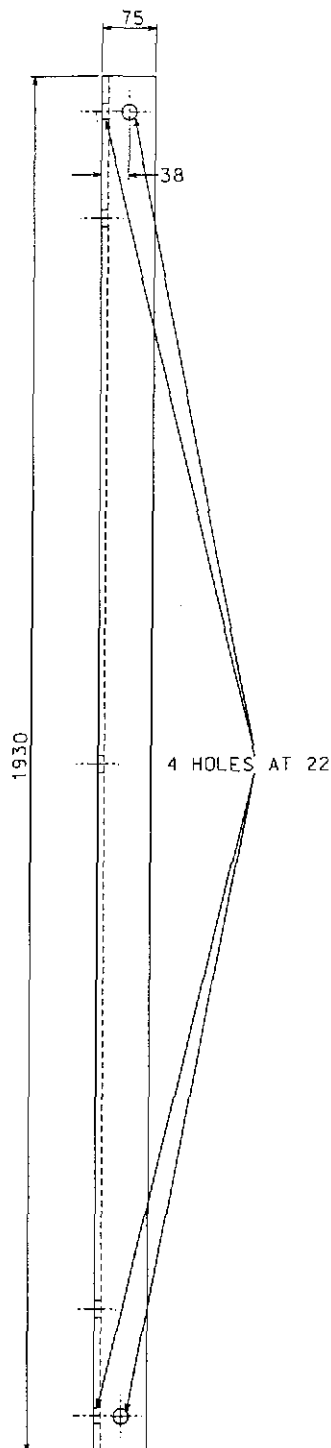
REC

MEB

Midlands Electricity plc

DWG No

M. 76 / 2



MATERIAL
100x75x9 8.4lb/ft M.S. Angle
to B.S.4 28/33 Ton Quality

FINISH
Galv to B.S.729 after
cutting & drilling

Unless otherwise stated all dimensions are in mm

Produced by
Output Services

SCALE	1:10
DATE	5.9.00
DRAWN	A.J.E.
TRACED	
CHECKED	27/9/00
APPROVED	KD

H POLE TERMINAL CROSSARM
H POLE LINE CROSSARM

ISSUE	DATE

GPU
POWER UK

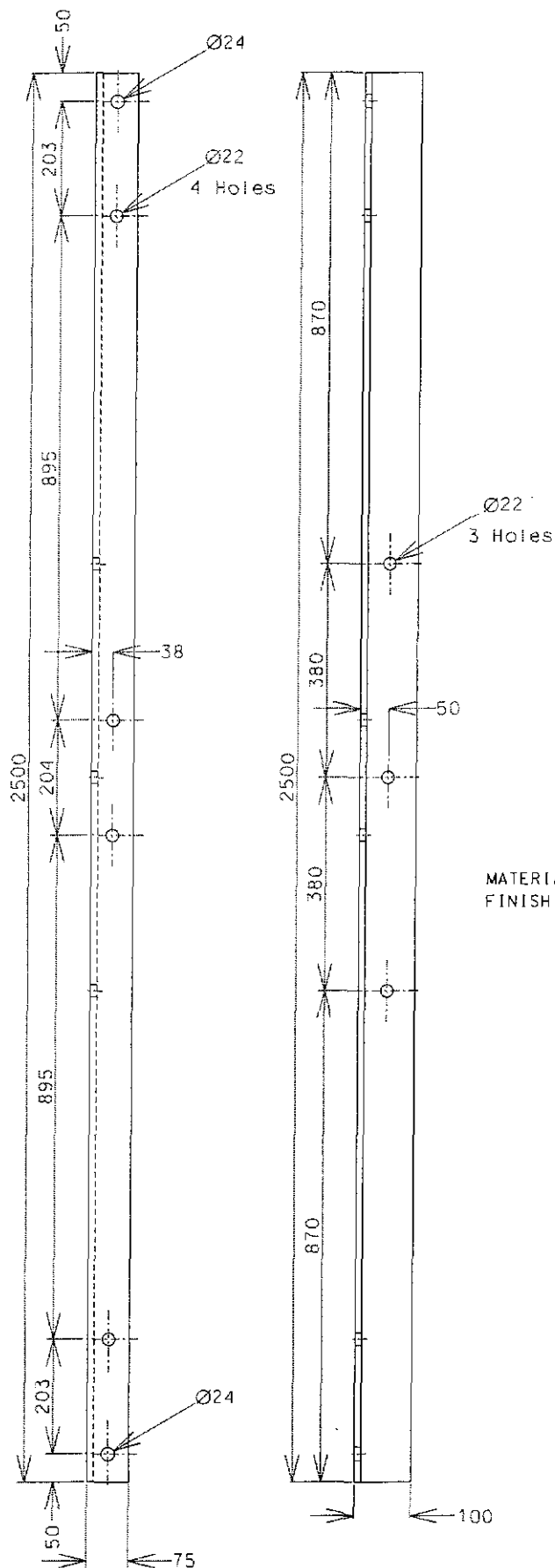
DRG. No.

M82

DRG. No.

M82

A4



MATERIAL 100 x 75 x 12 BSUA
FINISH Galvanised to BS729
After Cutting & Drilling

SCALE	NTS
DATE	7/9/00
DRAWN	A00
TRACED	
CHECKED	27/9/00
APPROVED	KD

LINE CROSSARM
(11 & 33KV HEAVY)

ISSUE	DATE

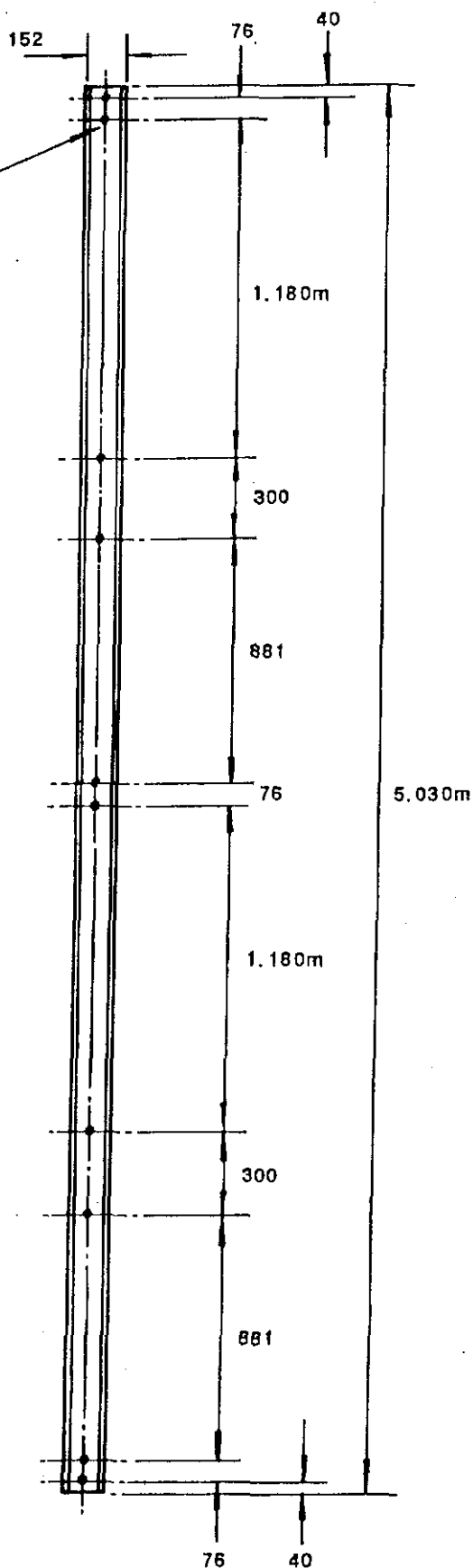
Produced by Output Services	
	
DRG. No. M96/1	

DRG. No.

M96/1

A4

10 - 22mm Dia. Holes



MATERIAL

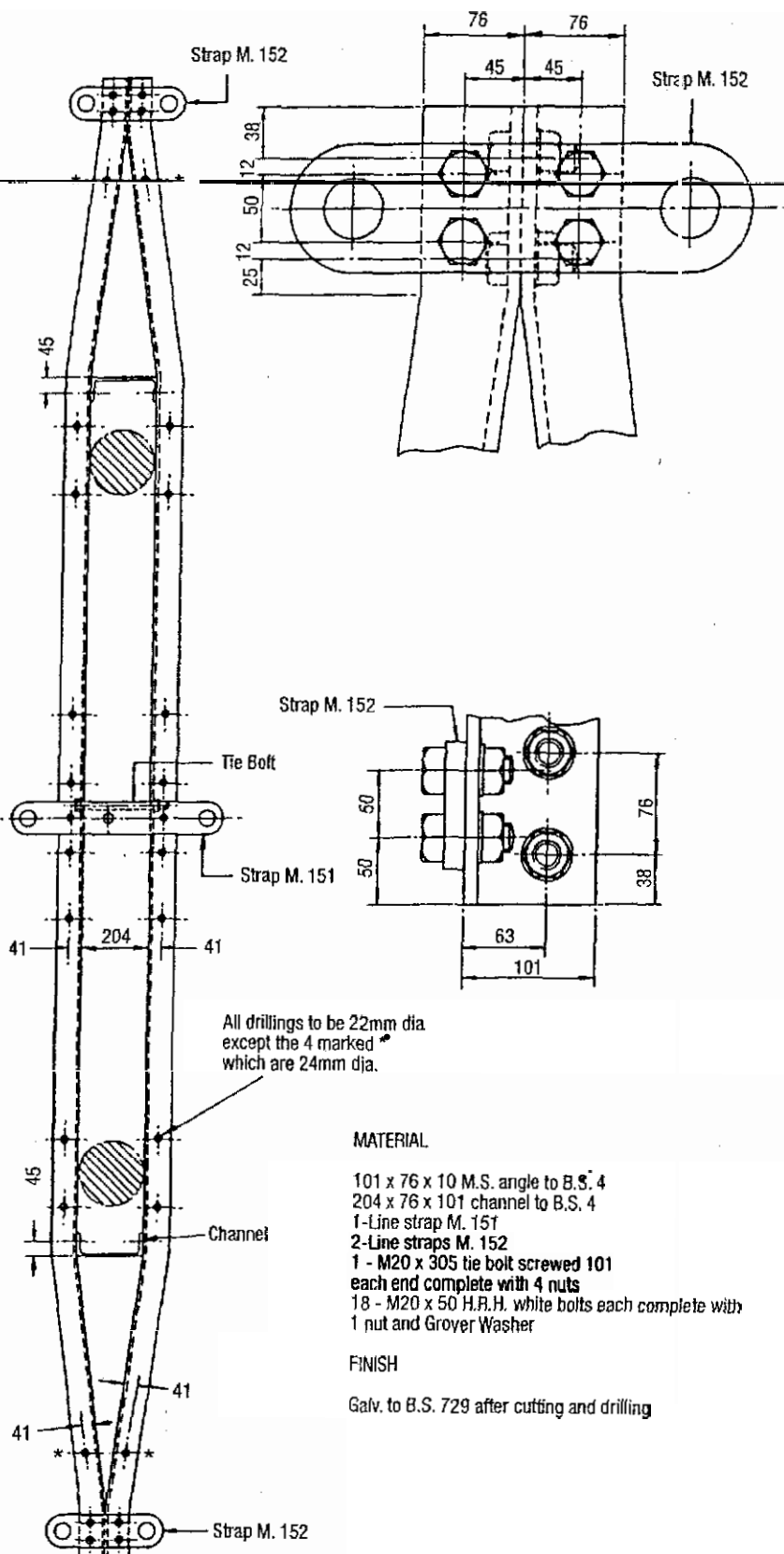
152 x 75 M.S.
Channel to B.S. 4

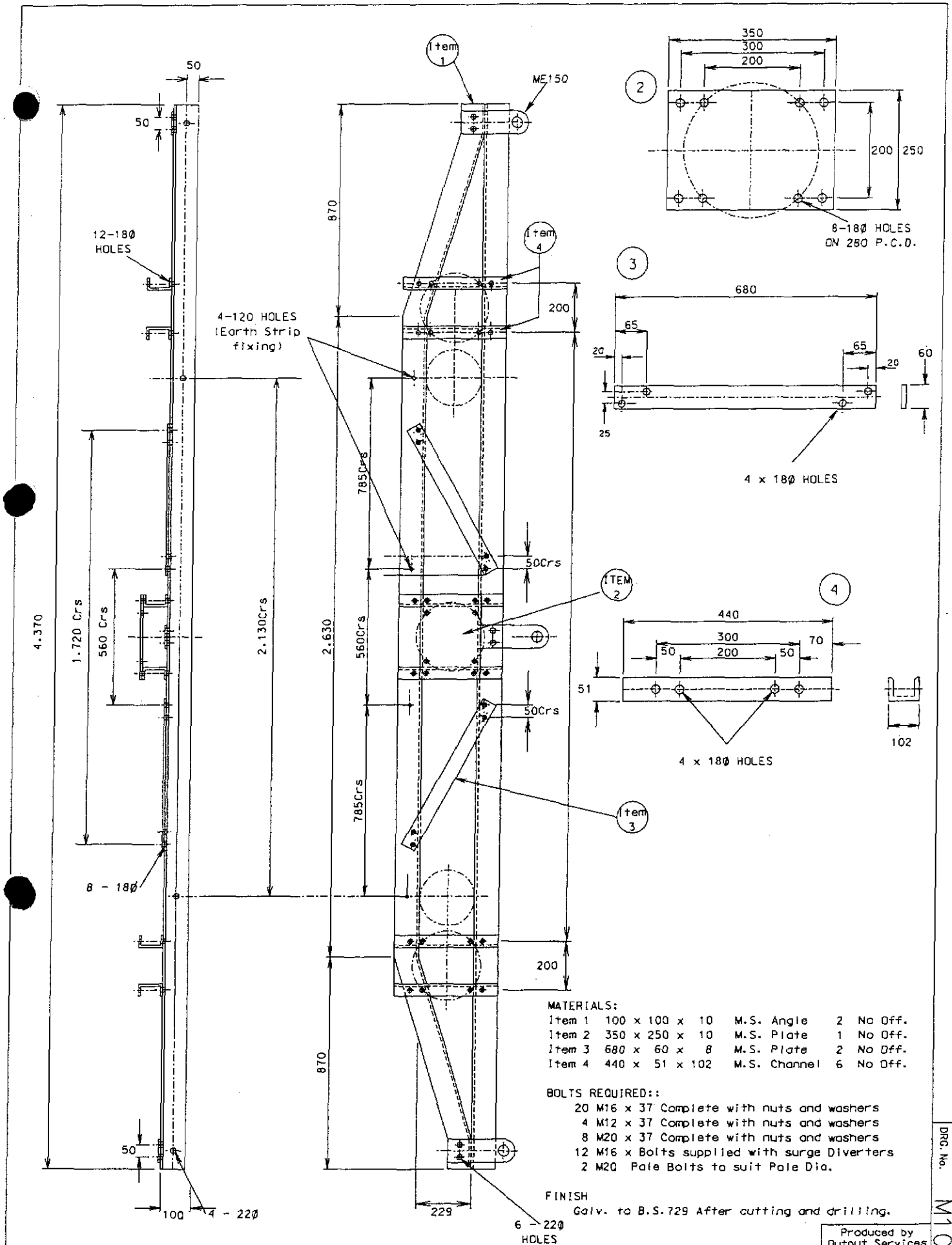
FINISH

Galv. to B.S. 729
after cutting and drilling.

All dimensions are in millimetres unless otherwise stated.

DRAWN BY DATE	TITLE CHANNEL CROSSARM (66kV SUSP)	SCALE	1/25	 Midlands Electricity plc DRG No M. 100
		DATE	12/8/93	
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		





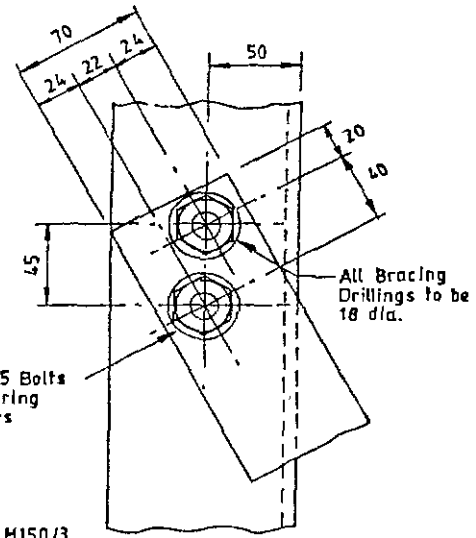
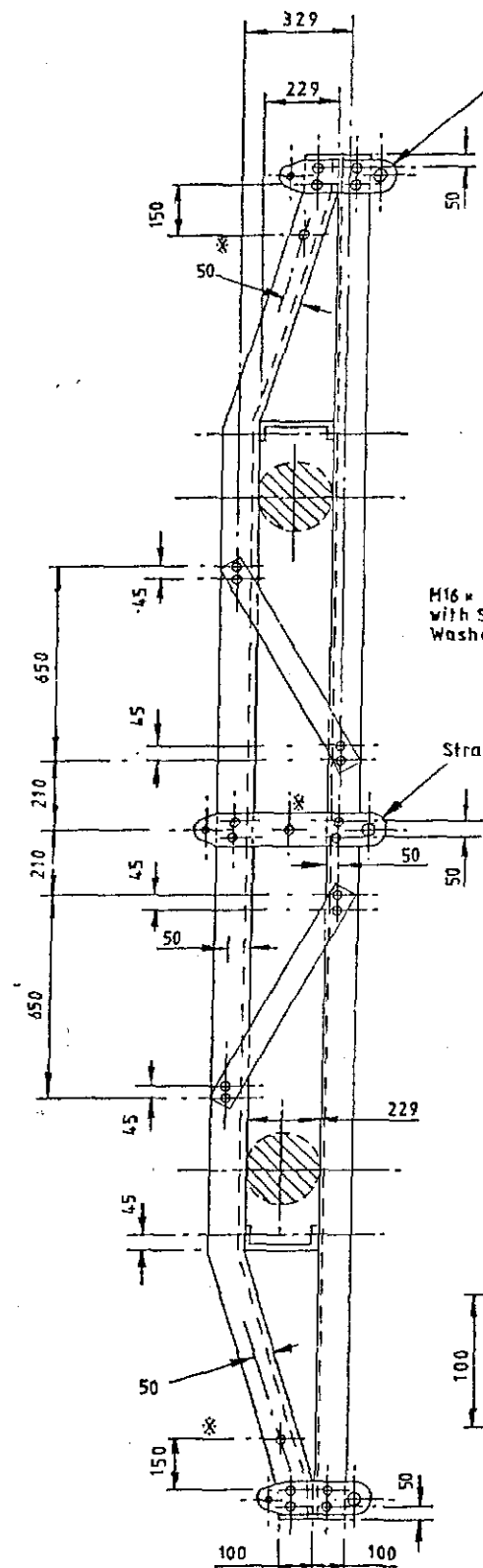
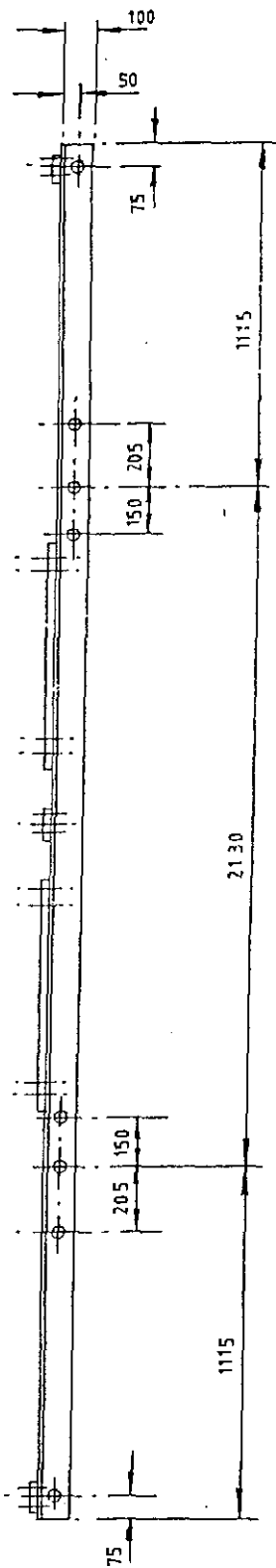
SCALE	1:20
DATE	04.10.01
DRAWN	D.B.
TRACED	-
CHECKED	-
APPROVED	-

BRACED TERMINAL CROSSARM WITH
PROVISION FOR SURGE DIVERTERS
(66KV HEAVY CONSTRUCTION)

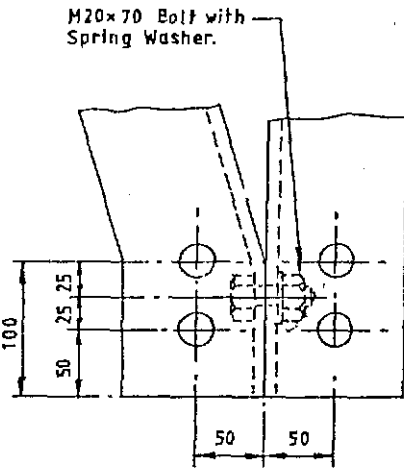
ISSUE	DATE

Produced by Output Services
GPU POWER UK
DRG. No. M103/2

DRG. No. M103/2



All holes to be 22 dia. except bracing drillings which should be 18 dia. & pilot insulator holes marked * to be 24 dia.



MATERIAL

100 x 100 x 10 BSA Angle to B.S. 4.
229 x 76 Channel to B.S. 4.
70 x 8 x 770 MS Flat

6 - M20 x 70 Bolts C/W Nut & Washer
8 - M16 x 45 Bolts C/W Nut & Washer
12 - M20 x 45 Bolts C/W Nut & Washer
Assembly Bolts to ESI 43 - 96

2 - Straps M 150 / 2
1 - Strap M 150 / 3

FINISH

Galvanised to BS. 729 after cutting & drilling.

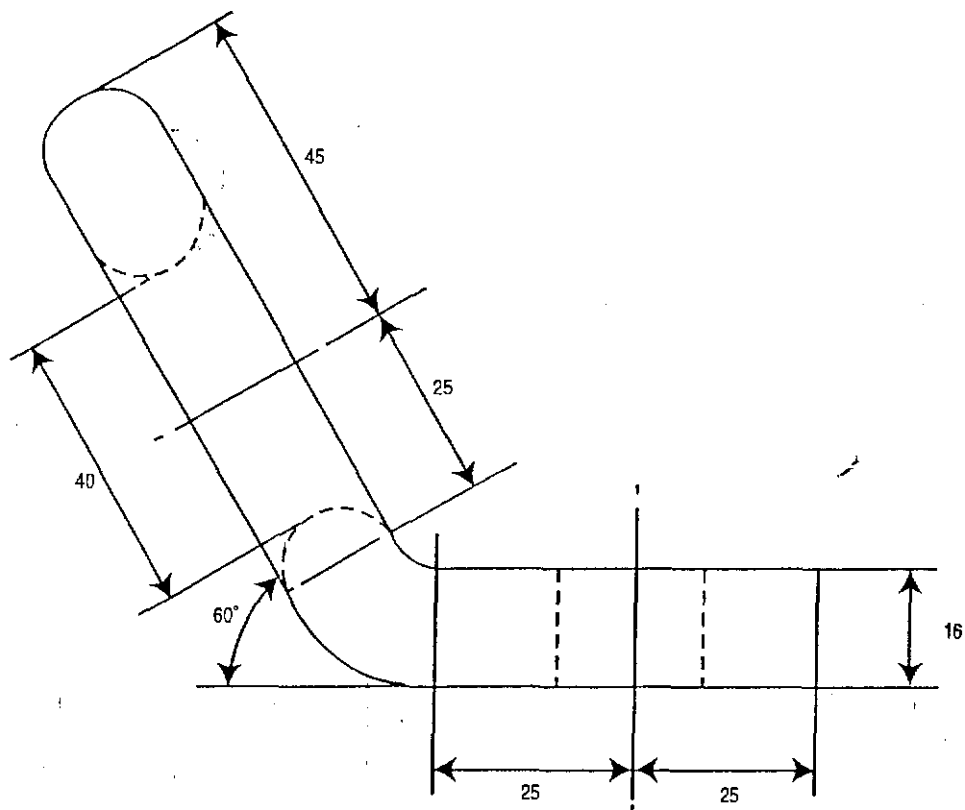
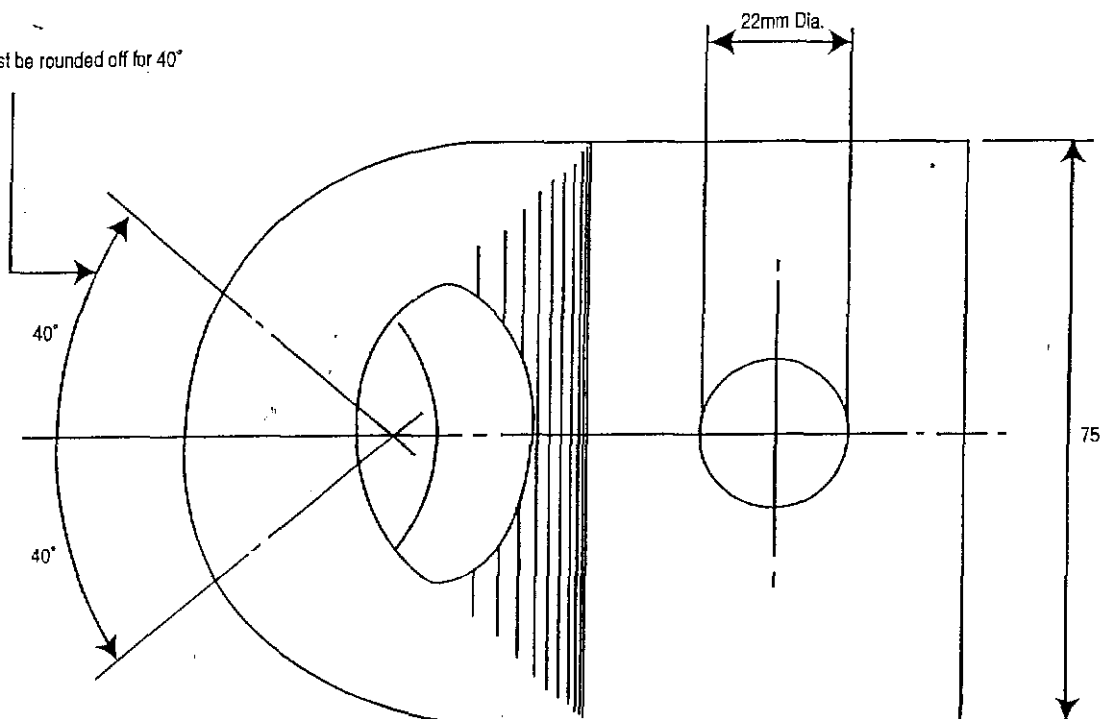
**TERMINAL CROSSARM
(66 kV)**

SCALE	---
DATE	20-2-95
DRAWN	A.J.B.
TRACED	C.E.
CHECKED	W.D.H.
APPROVED	

MEB Midlands Electricity plc
ONE No M. 103/3

NOTE:

Edges must be rounded off for 40°



MATERIAL

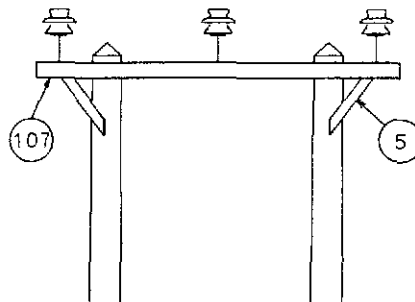
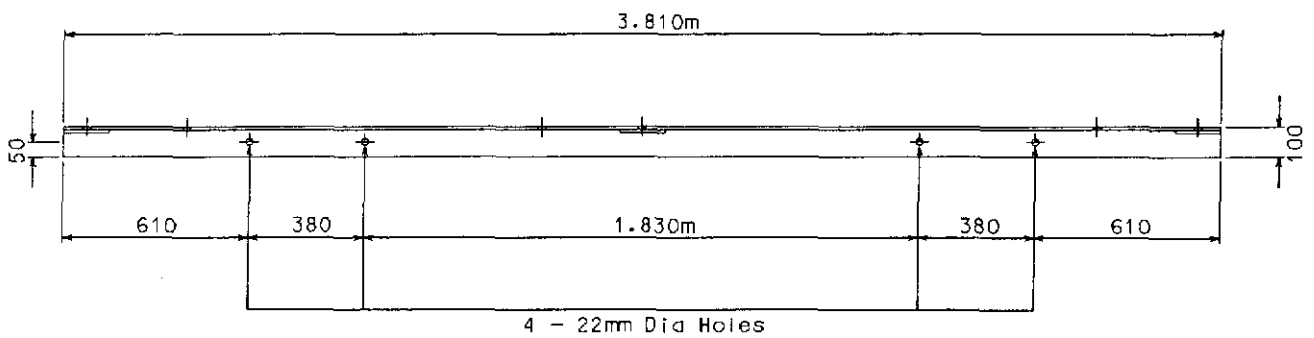
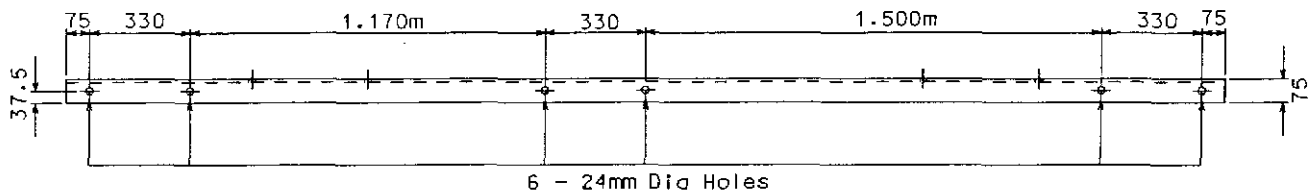
75 x 16 M.S.

FINISH

Galv. to B.S. 729 after cutting and drilling.

All dimensions are in millimetres unless otherwise stated

ADDED BY: DATE:	TITLE TERMINATING STRAP	SCALE	1/1	 Midlands Electricity plc ORG. No. M. 106
		DATE	12/8/93	
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		



MATERIAL
100x75x10 MS ANGLE TO BS4-
150x50x10 MS FLAT
GALV TO BS729
AFTER CUTTING AND DRILLING

NOTE
MS ANGLE STRUTS TO BE FITTED

All dimensions are in millimetres unless otherwise stated

SCALE	-
DATE	7.9.00
DRAWN	A.J.E.
TRACED	
CHECKED	13.10.00
APPROVED	KD

HEAVY CONSTRUCTION 11 & 33kV
"H" POLE THRO, (PIN) CROSS ARM

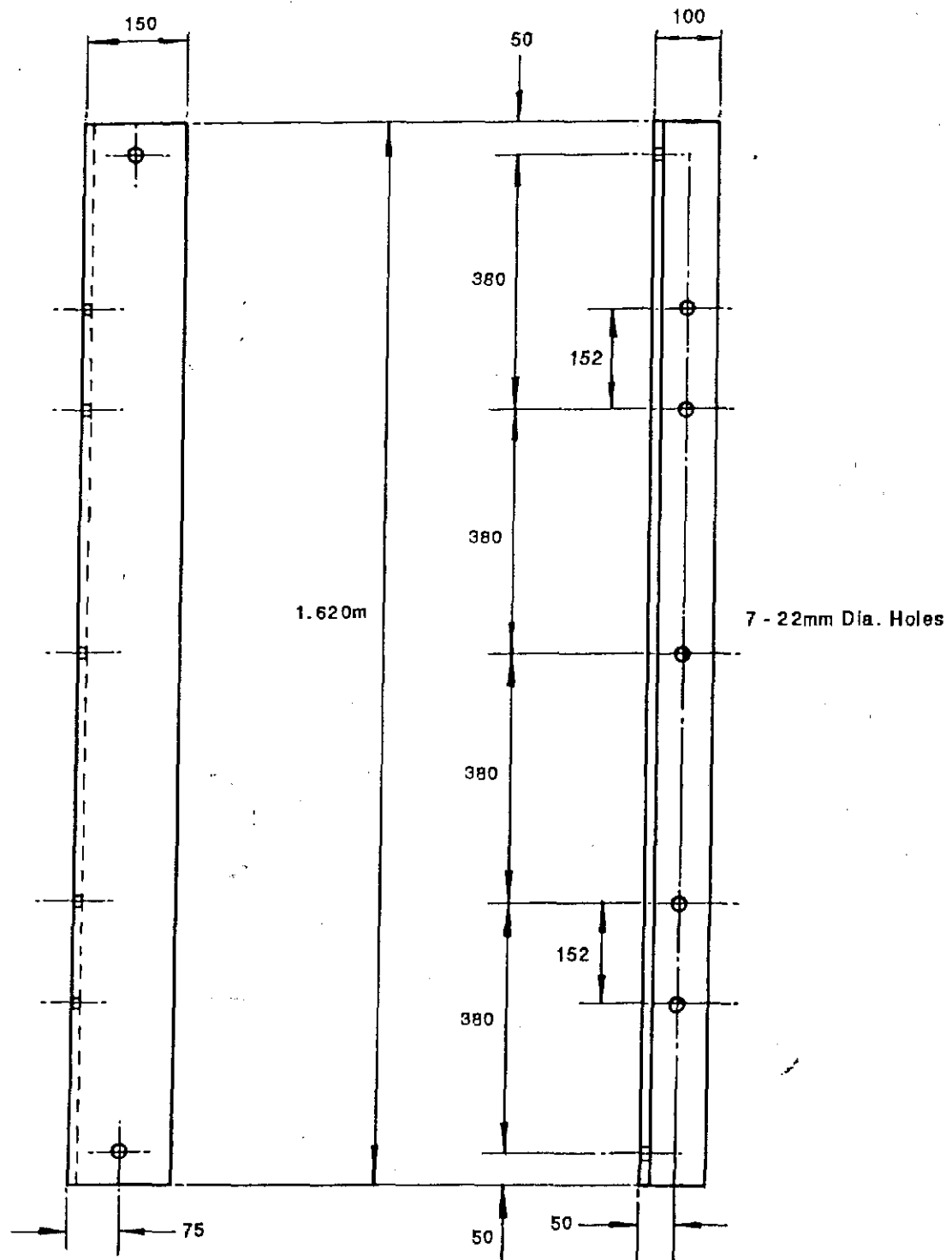
ISSUE	DATE

Produced by Output Services
GPU POWER UK
DRG. No. M107

DRG. No.

M107

A4



MATERIAL

150 x 100 x 12 M.S.
Angle to B.S.4

FINISH

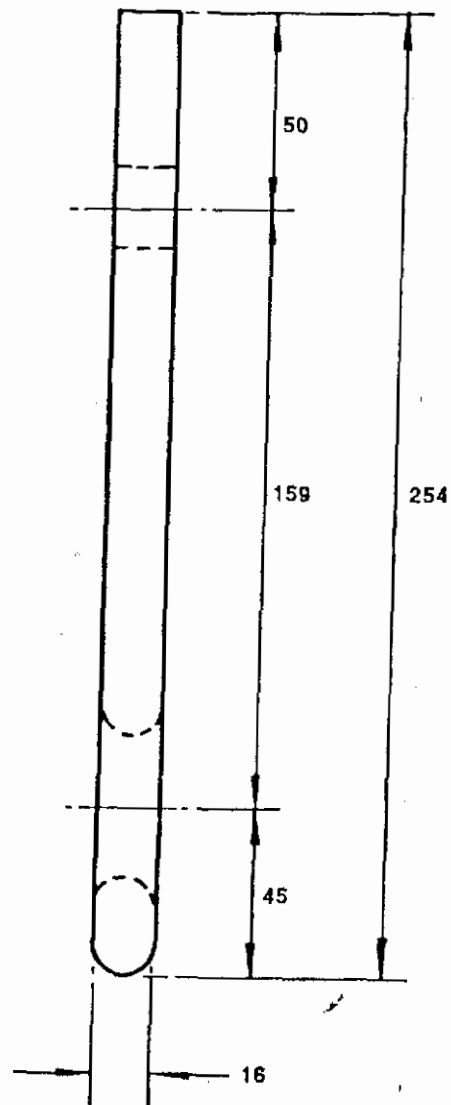
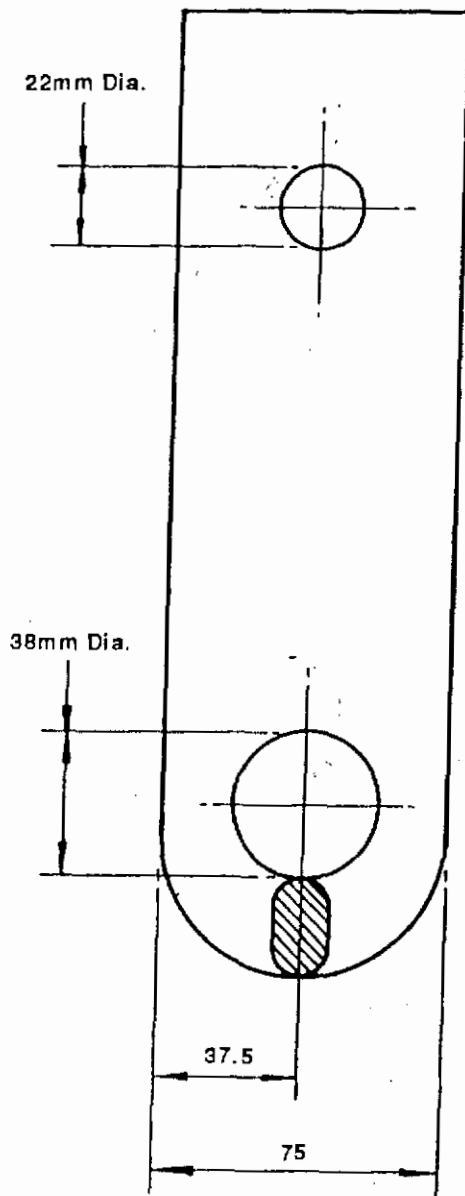
Galv. to B.S. 729
after cutting and drilling.

MAX. CONDUCTOR SIZE

Use limited to conductors up to and
including 100mm² or 100mm²
Cu. equiv. S.C.A.

All dimensions are in millimetres unless otherwise stated.

DRAWN BY DATE	TITLE	SCALE	1/10	 Midlands Electricity plc
		DATE	12/8/93	
		DRAWN		
		TRACED		ORG No M. 145
		CHECKED		
		APPROVED		

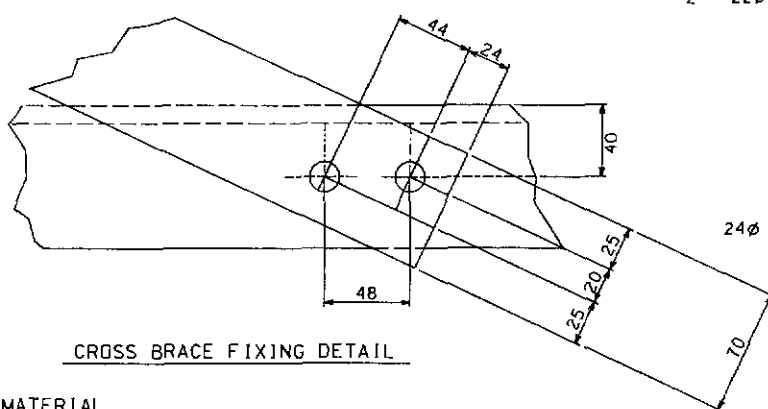
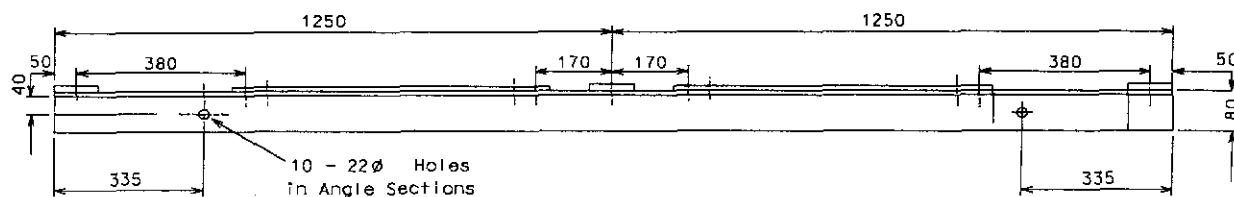
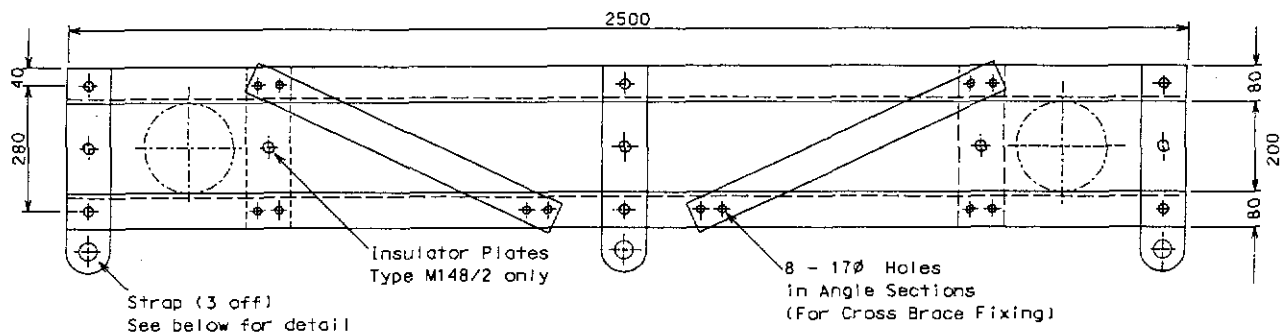


MATERIAL
16 x 75 M.S.

FINISH
Galv. to B.S. 729
after cutting and drilling.

All dimensions are in millimetres unless otherwise stated.

APPROVED BY DATE	TITLE LINE TERMINATING STRAP	SCALE	1/2	 Midlands Electricity plc
		DATE	12/8/93	
		DRAWN		
		TRACED		DWG No
		CHECKED		M. 146
		APPROVED		



MATERIAL

80 x 80 x 10 x 11.9 kg. M.S. Angle to B.S.4

70 x 80 M.S. Flat

100 x 15 M.S. Flat

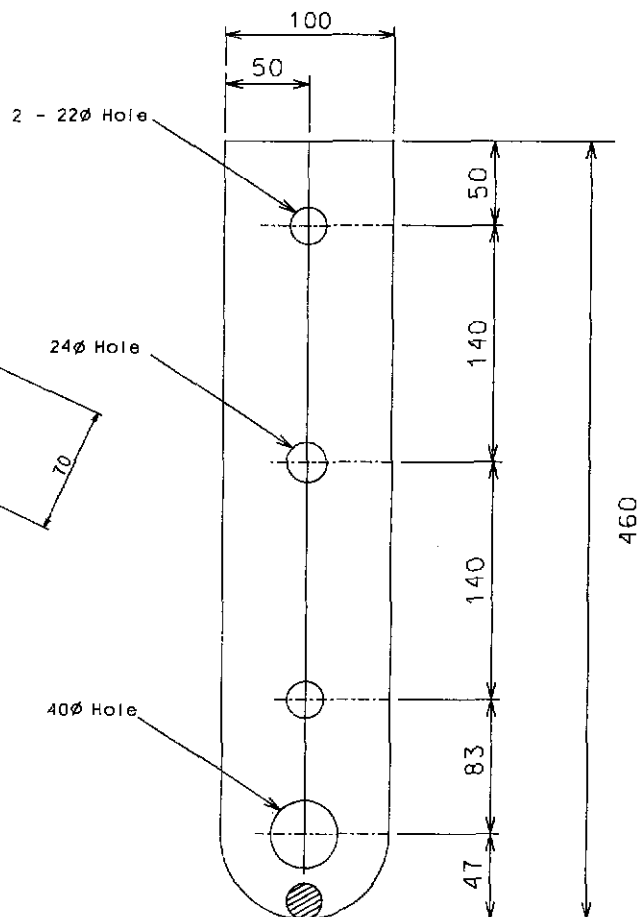
28/33 ton quality

6 - M20 x 50 HRH Bolts C/W Nut and grover washer

8 - M16 x 45 HRH Bolts C/W Nut and grover washer

FINISH

Galvanised to B S 729 after cutting and drilling.



Type M148/2 only - Insulator plate

MATERIAL

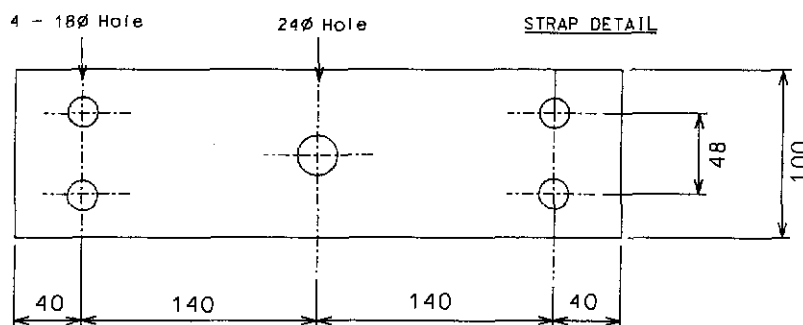
100 x 12 M S Plate

2 off C/W 4-M16 x 45 Bolts

& 4-8 mm Packing Washers

FINISH

Galv. to B S 729 after cutting and drilling.



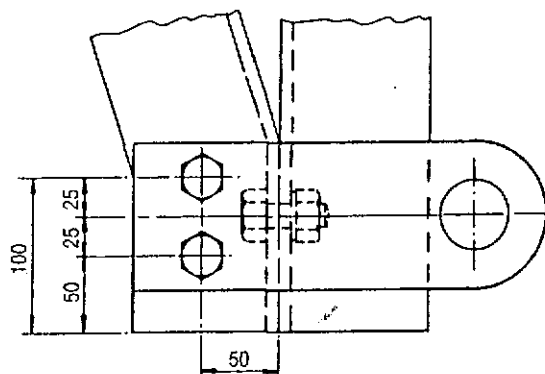
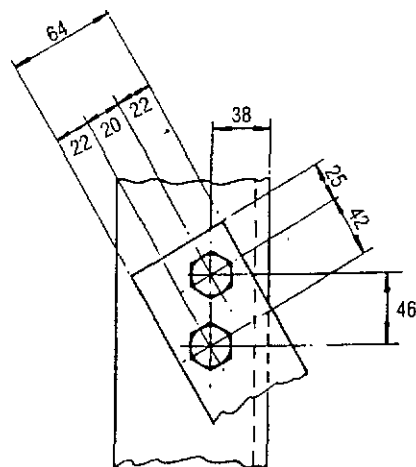
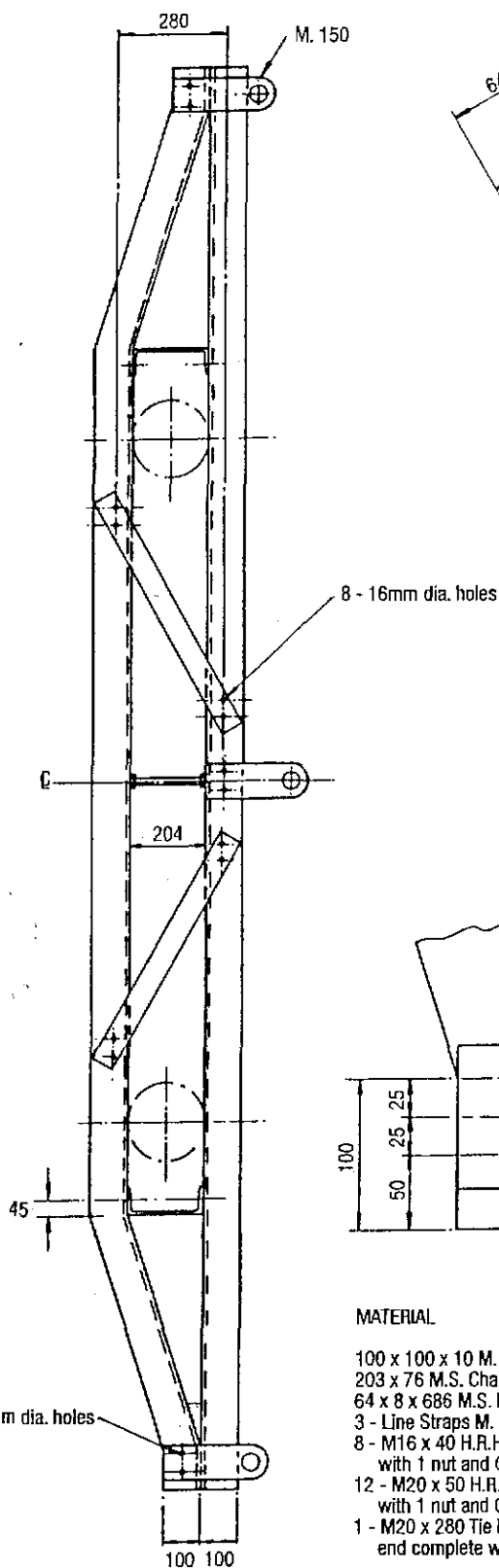
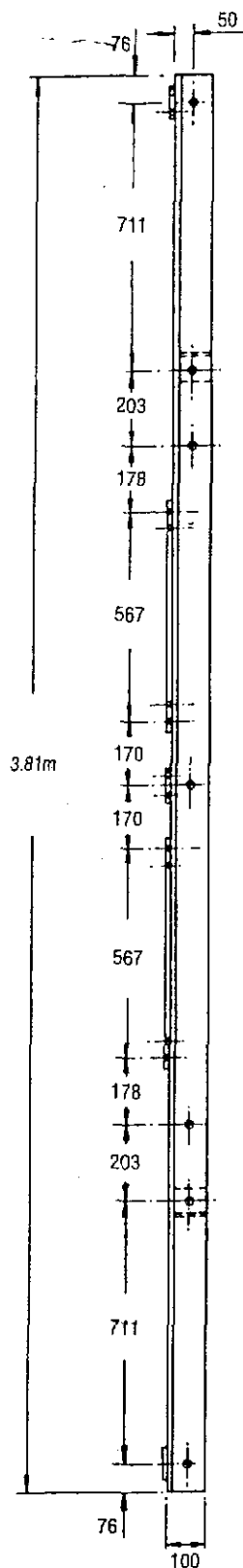
SCALE	NTS
DRAWN	14.09.00
TRACED	D.B.
CHECKED	27.09.00
APPROVED	KD

TERMINAL CROSSARM 11KV HEAVY CONSTRUCTION

ISSUE	DATE
A	01.07.02
MINOR MODS	
M.J.B. / K.D.	

Produced by Output Services
Aquila
DRG. No. M148

DRG. No. M148



MATERIAL

100 x 100 x 10 M.S. Angle
 203 x 76 M.S. Channel
 64 x 8 x 686 M.S. Flat
 3 - Line Straps M. 150
 8 - M16 x 40 H.R.H. White bolts each complete with 1 nut and Grover Washer
 12 - M20 x 50 H.R.H. White bolts each complete with 1 nut and Grover Washer
 1 - M20 x 280 Tie Bolt screwed 100 at each end complete with 4 nuts.

FINISH

Galv. to B.S. 729 after cutting and drilling

All dimensions are in millimetres unless otherwise stated

TITLE

BRACED TERMINAL CROSSARM
(33 k.V. HEAVY CONSTRUCTION)

SCALE 1/20-1/5

DATE 27/2/73

DRAWN

TRACED

CHECKED

APPROVED

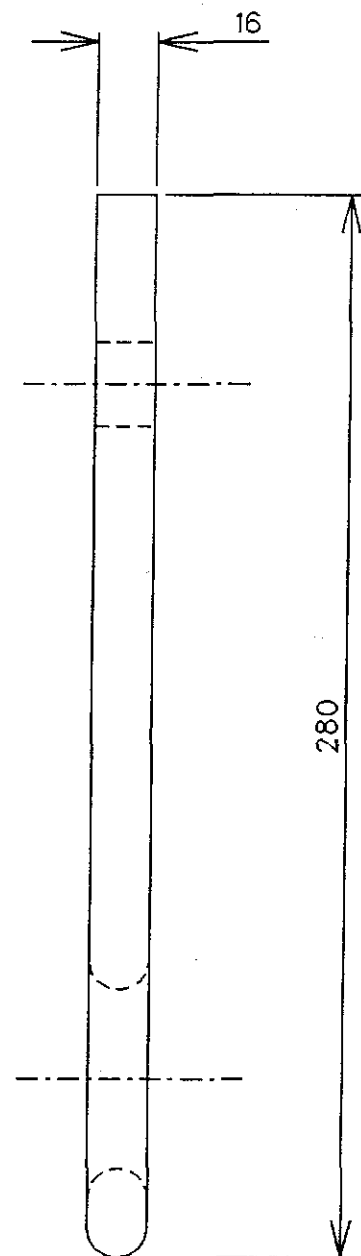
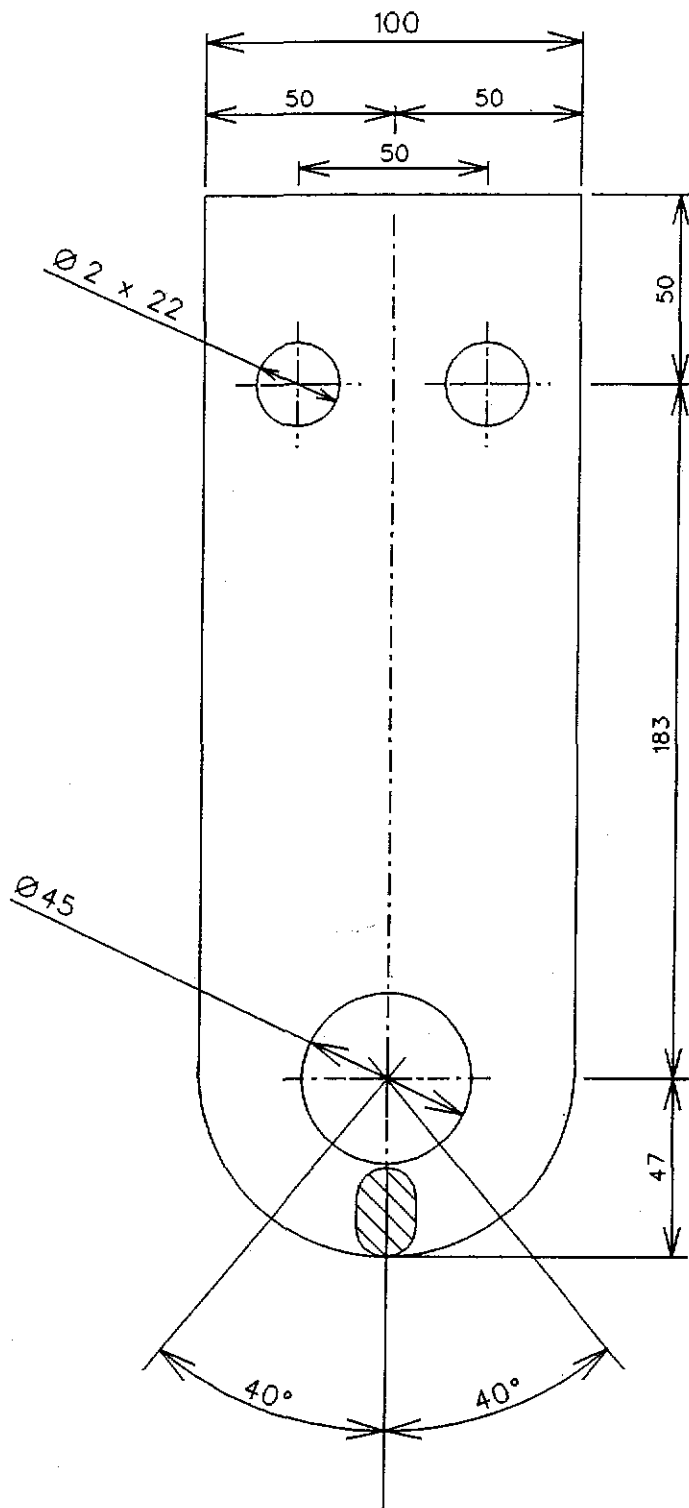
MEB

Midlands Electricity plc

ORG. No.

M. 149

ADDED BY
DATE



MATERIAL

280 x 100 x 16 M.S.

FINISH

Galv. to B.S. 729
after cutting and drilling.

Edges must be rounded
off for a distance of 40°
as shown in section.

All dimensions are in millimetres unless otherwise stated.

SCALE	1-2
DATE	12/8/93
DRAWN	-
TRACED	-
CHECKED	-
APPROVED	-

LINE STRAP

ISSUE	DATE
A	28/6/02
Redrawn on CAD Minor Mods.	
GSC/KD	



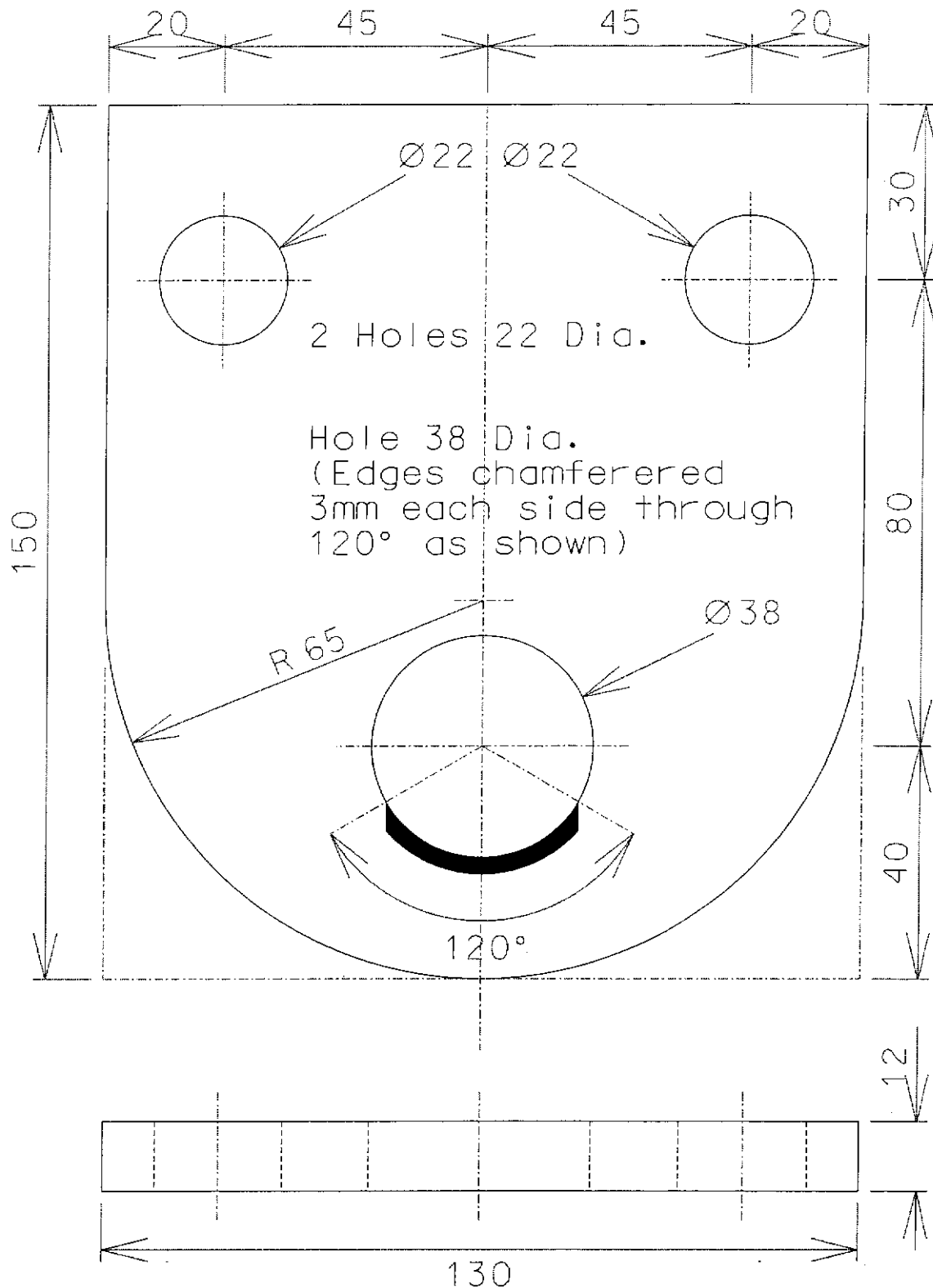
Aquila

DRG. No.

M150

DRG. No.

M150



Supplied with crossarm kits.

ESI 439528

SCALE	1-1
DATE	7/9/00
DRAWN	ADD
TRACED	
CHECKED	7/9/00
APPROVED	ADD

LANDING STRAP
TERMINAL

ISSUE	DATE

Produced by
Output Services

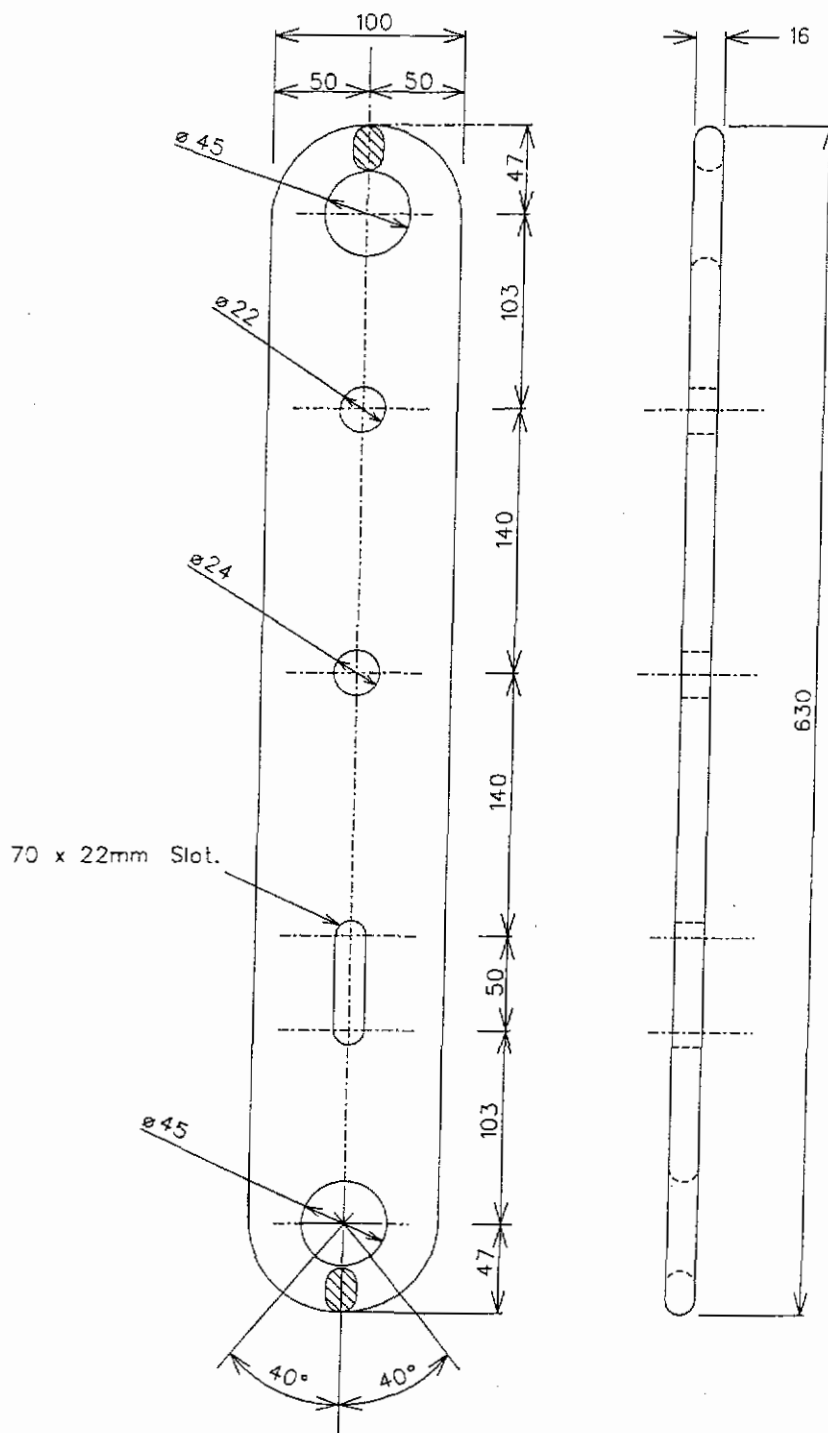
GPU
POWER UK

DRG. No.
M150/1

DRG. No.

M150/1

A4



MATERIAL

630 x 100 x 16 Flat M.S.

FINISH

Galv. to B.S. 729
after cutting and drilling.

Edges must be rounded
off for a distance of 40°
as shown in section.

All dimensions are in millimetres unless otherwise stated.

SCALE	1-4
DATE	12/8/93
DRAWN	-
TRACED	-
CHECKED	-
APPROVED	-

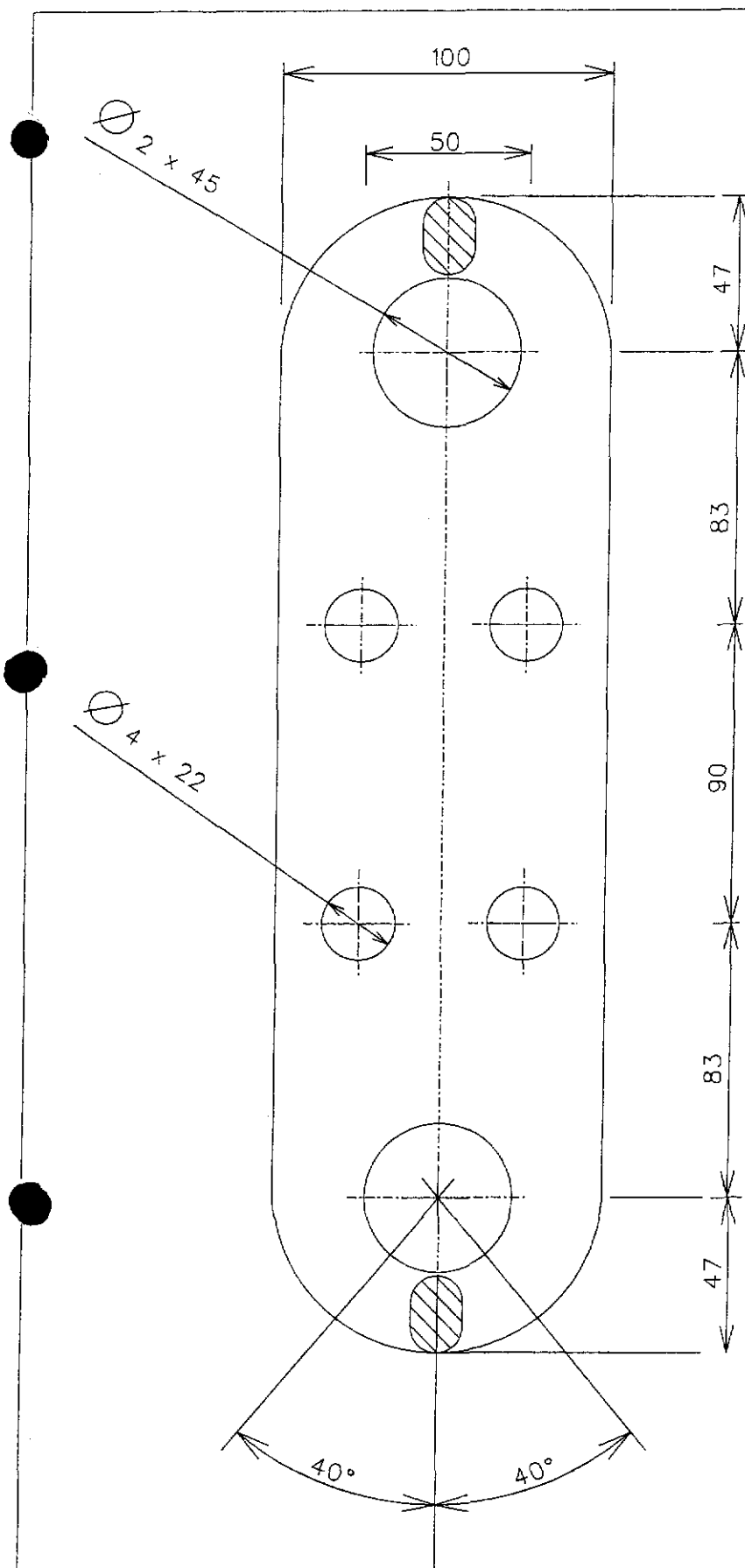
LONG SECTION STRAP
(HEAVY CONSTRUCTION)

ISSUE	DATE
A	1/7/02
Redrawn on CAD Minor Mods.	
GSC/KD	

Produced by Output Services	
 Aquila	
DRG. No.	
M151	

DRG. No.

M151



FINISH

Galv. to B.S. 729
after cutting and drilling.

MATERIAL

Edges must be rounded
off for a distance of 40°
as shown in section.

All dimensions are in millimetres unless otherwise stated.

SCALE	1-2
DRAWN	12/8/93
TRACED	-
CHECKED	-
APPROVED	-

SHORT SECTION STRAP (HEAVY CONSTRUCTION)

ISSUE	DATE
A	28/6/02
Redrawn on CAD Minor Mods.	
GSC/KD	



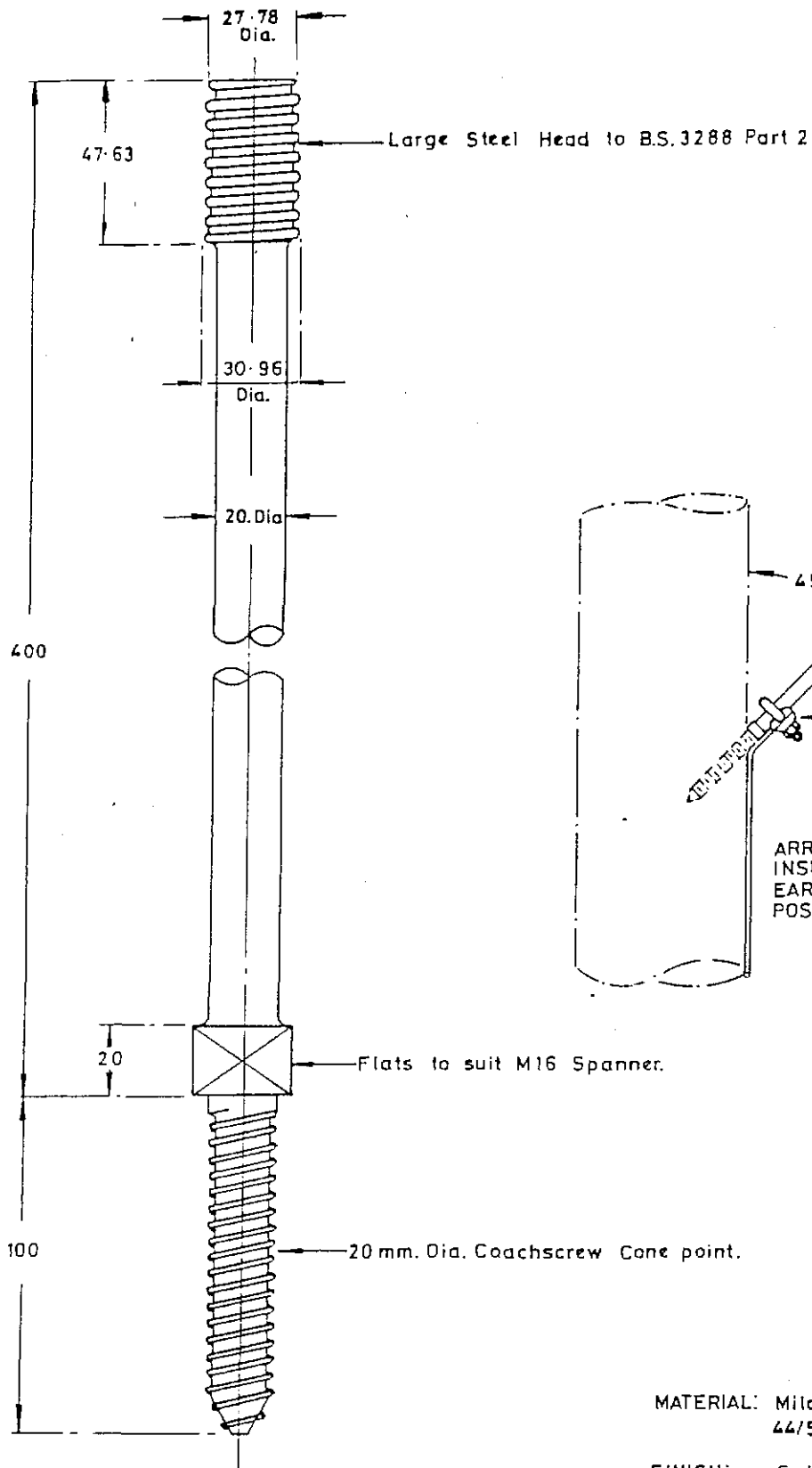
Aquila

DRG. No.

M152

DRG. No.

M152



MATERIAL: Mild Steel to B.S. 4360
44/52 kg/mm.² quality.

FINISH: Galv'd. to B.S. 729

NOTE: TO B.S. 3288 PART II

SCALE	1/2
DATE	23.7.71
DRAWN	R.N.
TRACED	A.C.
CHECKED	
APPROVED	

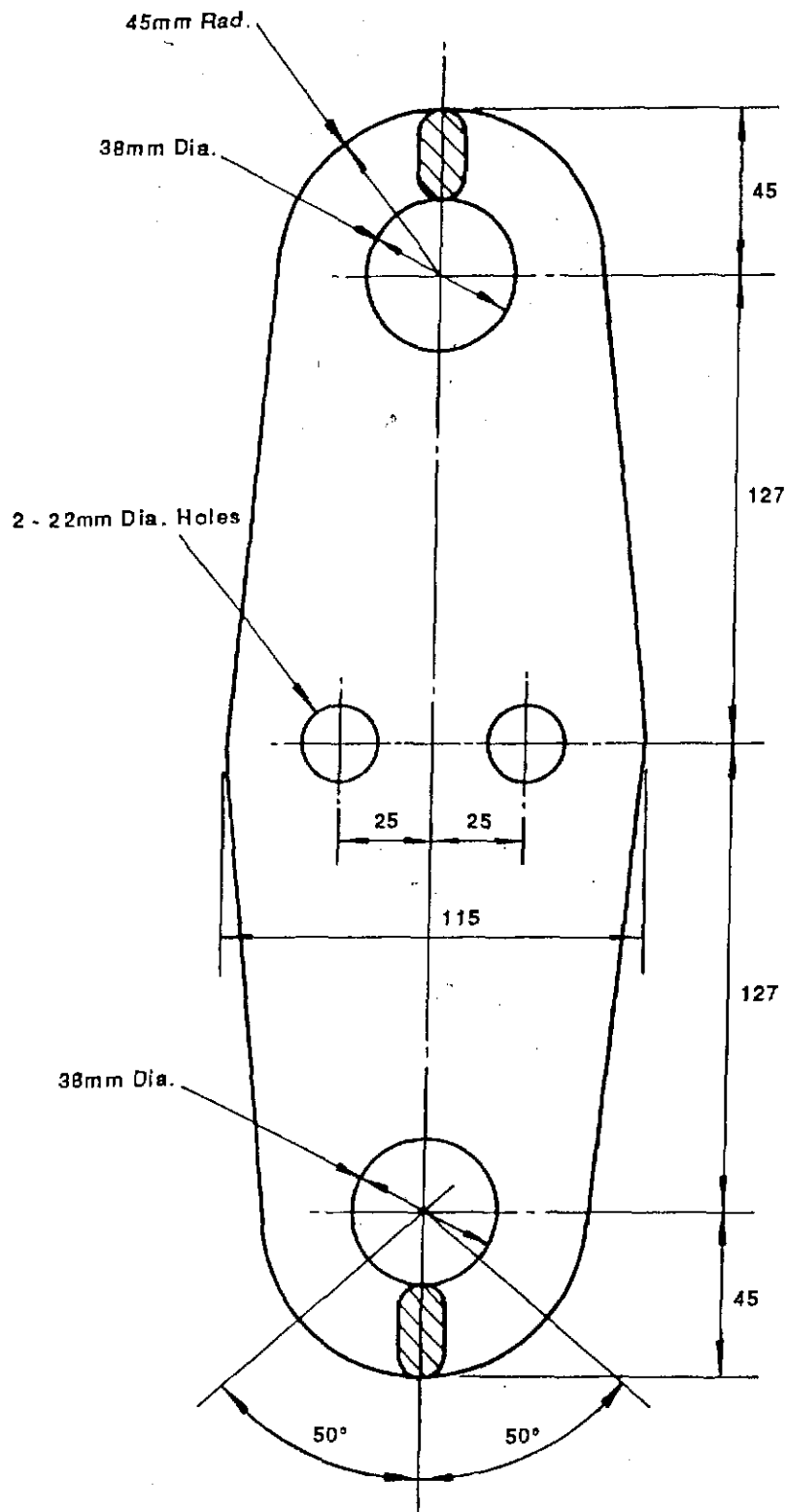
11 kV. & 33 kV PILOT INSULATOR SPINDLE.

ISSUE
G
Bulldog grip rel. changed 21/1/74

MIDLANDS
ELECTRICITY BOARD

DRG. No.

M 154



FINISH

Galv. to B.S. 729
after cutting and drilling.

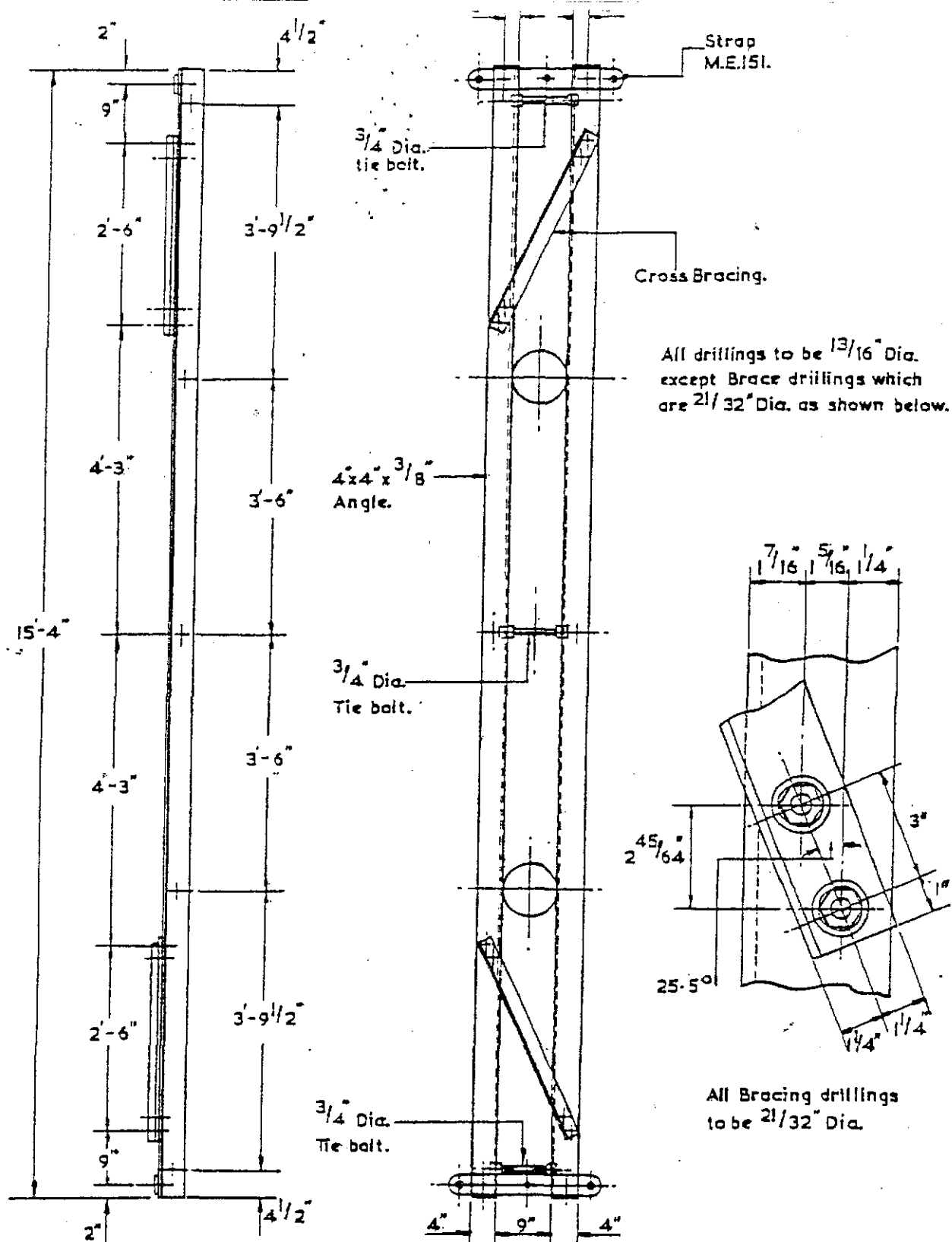
MATERIAL

344 x 115 x 16 M.S. strap

Edges must be rounded
off for a distance of 50°
as shown in section

All dimensions are in millimetres unless otherwise stated.

DRAWN BY DATE	TITLE SINGLE CROSSARM LINE SECTION STRAP	SCALE	1/2	MEB Midlands Electricity plc ORG No M. 156
		DATE	12/8/93	
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		



MATERIAL.

4x4 x 3/8" (9.7 lb) M.S. Angle to B.S. 4.
 2 1/2 x 1 1/2 x 5/16" (3.9 lb) M.S. Angle to B.S. 4.
 28/33 ton quality.
 2-LINE STRAPS TO M.E.151.
 3-3/4" W x 12" Tie Bolts screwed 4" each end c/w 4 Nuts.

4 - 3/4" W x 2" H.R.H. Bolts ea. c/w 1 Nut & Grover Washer.
 8 - 5/8" W x 1/2" H.R.H. Bolts ea. c/w 1 Nut & Grover Washer.

FINISH.

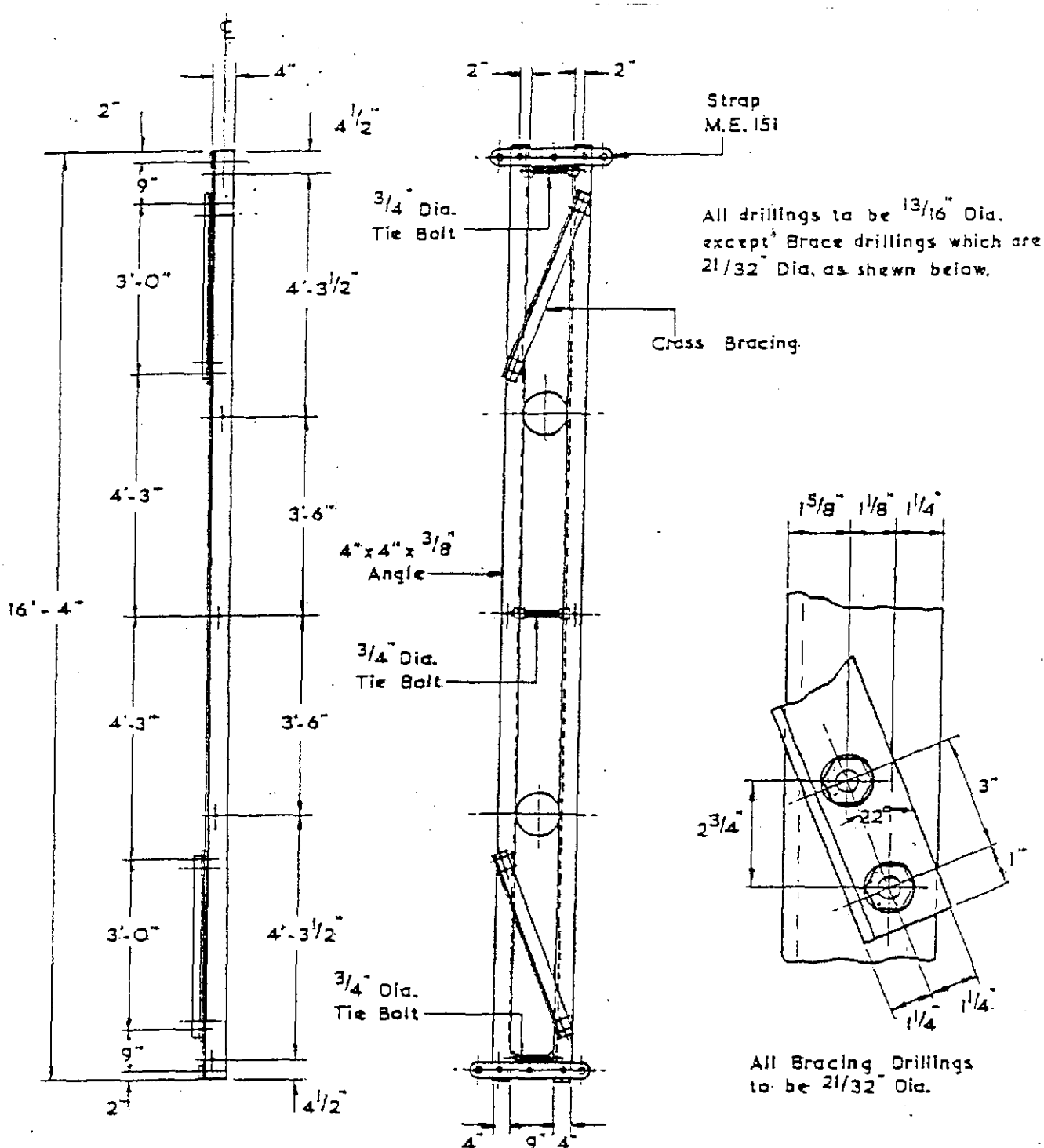
Galv. to B.S. 729 after cutting & drilling.

SECTION CROSSARM.

SCALE	1/24	ORIENTATION	ISSUE	ISSUE	ISSUE
DATE	23-11-62				
CRAWN	A.T.				
TRACED	YED 3-1-63				
CHECKED	24/6				
APPROVED.	YED				
			Redrawn 23-11-62		

DRAWING No.

M.E.186



MATERIAL

4" x 4" x 3/8" (9-71b) M.S. Angle to B.S. 4.

2 1/2" x 2" x 5/16" (4-41b) M.S. Angle to B.S. 4.

2B/33 ton quality

2 Line Straps to M.E. 151.

3 - 3/4" W x 12" Tie Bolt screwed 4" each end c/w 4 Nuts

4 - 3/4" W x 2" H.R.H. Bolts ea. c/w 1 Nut & Grover Washer

8 - 5/8" W x 1 1/2" H.R.H. Bolts ea. c/w 1 Nut & Grover Washer.

FINISH

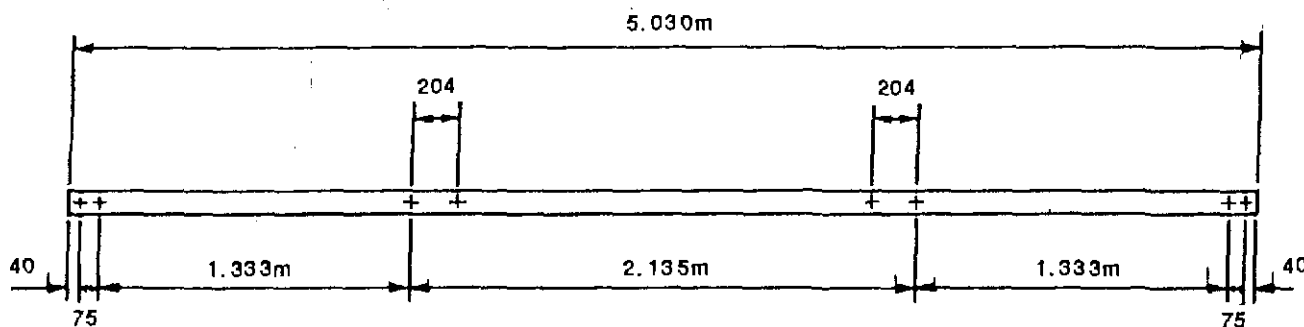
Galv. to B.S. 729 after cutting & drilling.

SECTION CROSS ARM.

SCALE	1/32	ORIENTATION	ISSUE	ISSUE	ISSUE
DATE	23.11.62		A.M.F.		
DRAWN	A.T.				
TRACED	Y.E.D.				
CHECKED	Y.E.D.		Redrawn		
APPROVED	Y.E.D.				

DRAWING No.

M.E. 187



8 - 22mm Dia. Holes

FINISH

Galv. to B.S. 729
after cutting and drilling.

MATERIAL

127 x 65 M.S. Channel to B.S.4

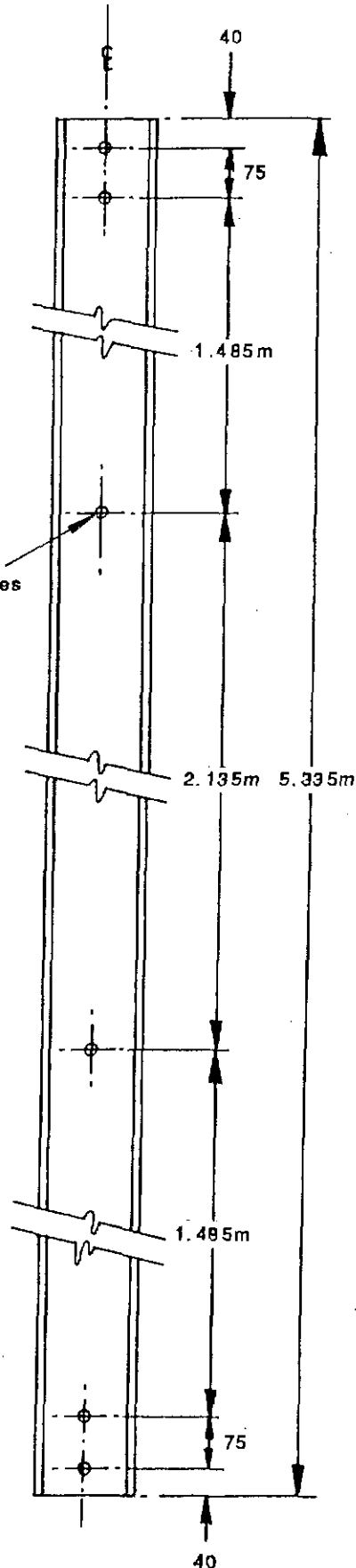
For conductors up to and including
100mm² copper or S.C.A.

152 x 75 M.S. Channel to B.S.4.

For conductors 150mm² and 200mm²
copper or 175mm² S.C.A.

All dimensions are in millimetres unless otherwise stated.

ADD'D BY DATE	TITLE LINE CROSSARM FOR 66kV DUAL CIRCUIT (SUSP)	SCALE	1/30	 Midlands Electricity plc
		DATE	12/8/93	
		DRAWN		
		TRACED		
		CHECKED		
		APPROVED		DRG No M. 188



For conductors up to and including
100mm² copper or S.C.A.
152 x 75 M.S. Channel to B.S.4.
For conductors 150mm² and 200mm²
copper or 175mm² S.C.A.

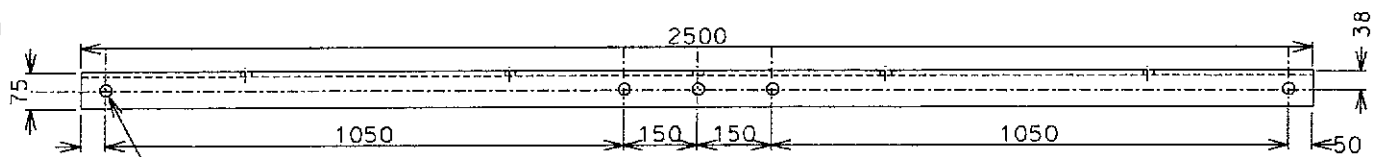
DATE _____
ACCEPTED BY _____

SCALE	1/10
DATE	12/8/93
DRAWN	
TRACED	
CHECKED	
APPROVED	

MEB
Midlands Electricity plc

DANG NG

M. 189



LINE CROSSARM
1.2M SPACING
SINGLE AND 'H' POLES

SCALE	NTS
DATE	11/9/00
DRAWN	ADD
TRACED	
CHECKED	21/9/00
APPROVED	KD

ISSUE	DATE
A	14.12.01

TEXT REMOVED.
TEXT ADDED.

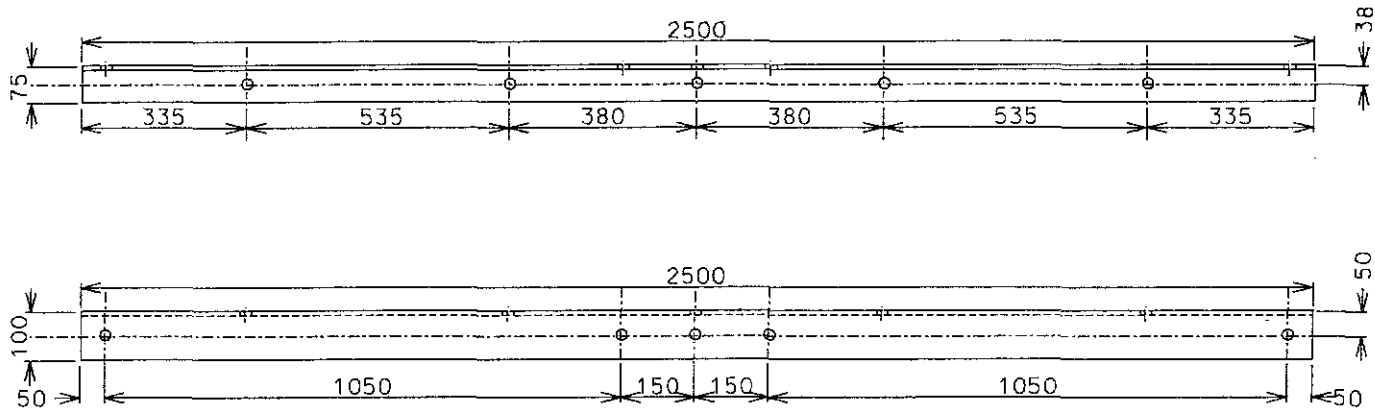
D.B.

Produced by Output Services	
DRG. No. M191	

DRG. No.

M191

A4



All Holes 22 Diameter

SECTION ANGLE CROSSARM
1.2M SPACING
SINGLE AND 'H' POLES

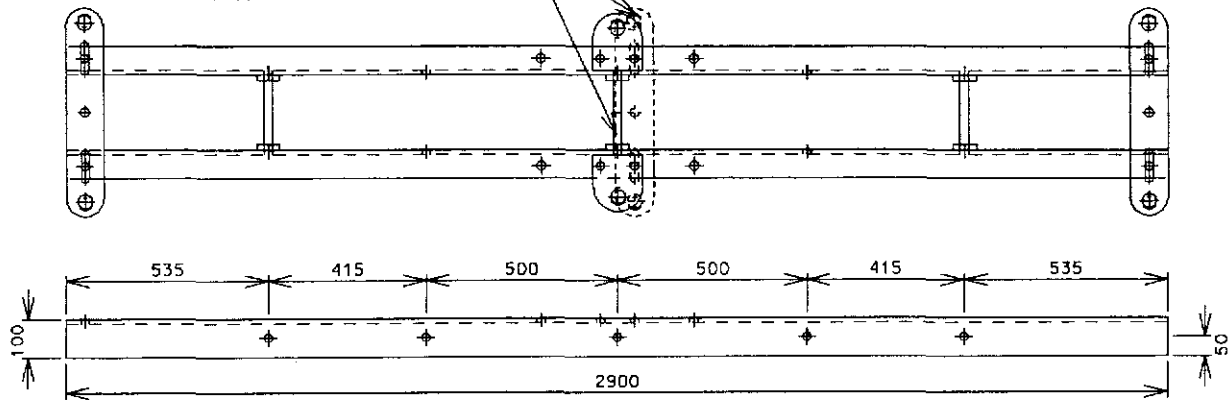
SCALE	NTS
DATE	12/9/00
DRAWN	ADD
TRACED	
CHECKED	21/9/00
APPROVED	KD

ISSUE	DATE
A	18.12.01
TEXT ADDED.	
D.B.	

Produced by Output Services	
DRG. No.	M194

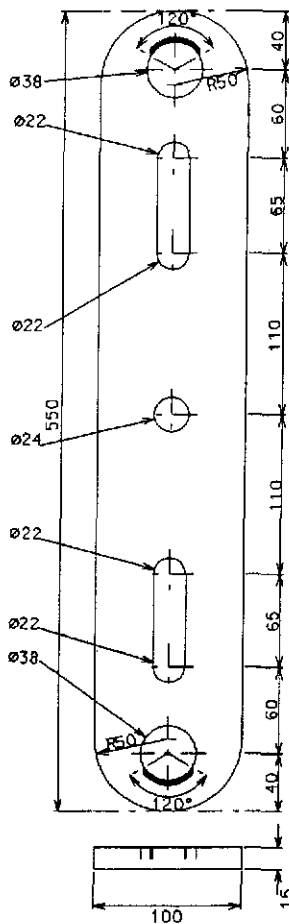
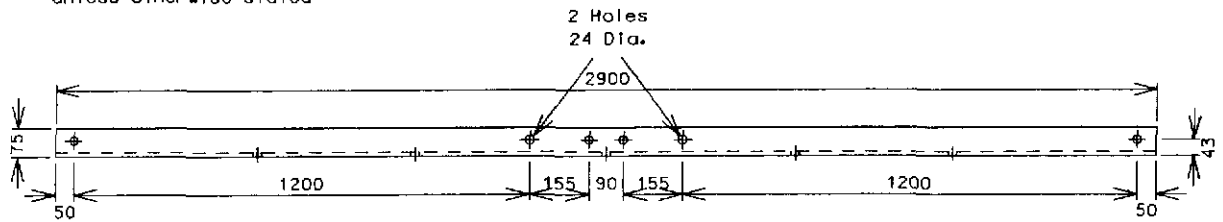
DRG. No. M194

Section Strap & Tie Bolt
to be fitted when used for H-Poles



Holes 22 Dia.
unless otherwise stated

100 x 75 x 12 B.S.U.A.



50 Radius

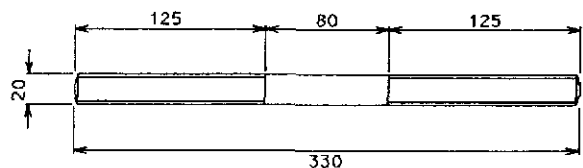
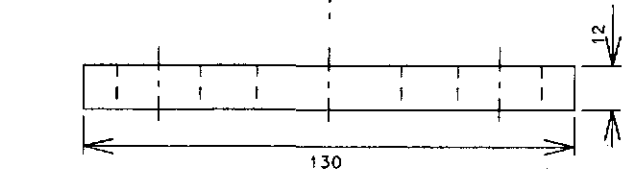
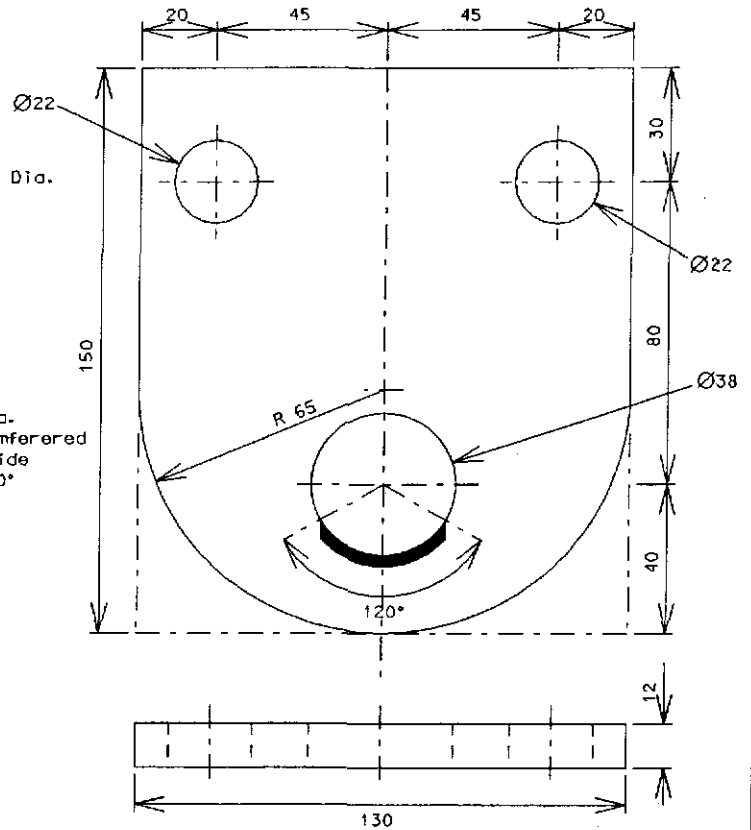
2 Holes 22 Dia.

Hole 24 Dia.

2 Slots 22 Wide

2 Holes 38 Dia.
(Edges Chamfered 3mm
each side through 120°
as shown)

REQUIREMENTS:
2 Crossarm Members
3 Section Straps
2 Terminal Straps
2 Tie Bolts
12 M20 x 60 Bolts C/W
Nut and Plain Washer
4 Crossarm Struts to
Drg. M195/1



M20 Tie Bolt threaded 125 each end C/W 4 Nuts.

DATE	NTS
DATE	15/9/00
DRAWN	ADD
TRACED	
CHECKED	13/10/00
APPROVED	KD

SECTION ANGLE CROSSARM
1.4M SPACING
SINGLE - H - POLES

ISSUE DATE

ISSUE DATE

DRG. No.

M196

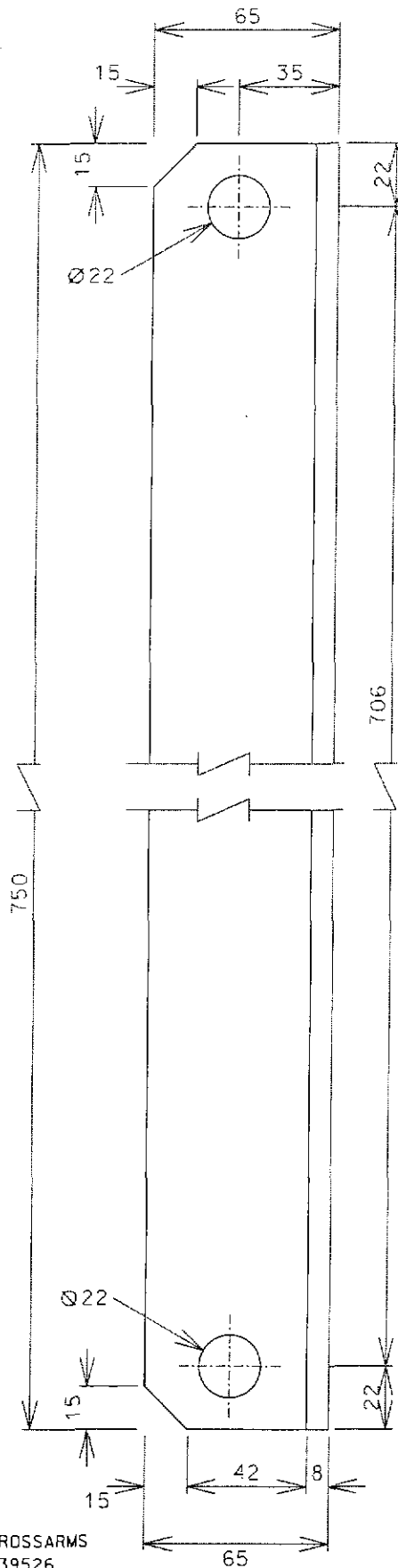
Produced by
Output Services

GPU
POWER UK

DRG. No.

M196

44



65 x 50 x 8 B.S.U.A.

2 Holes 22 Dia.

NOTE:
FOR USE WITH M196 & M400 CROSSARMS
(4 REQUIRED) AS ESI STD. 439526

SCALE	NTS
DATE	13/9/00
DRAWN	ADD
TRACED	
CHECKED	21/9/00
APPROVED	KD

CROSSARM STRUT FOR
1.4M SPACING SECTION ANGLE CROSSARM
AND 1.2M SPACING TERMINAL CROSSARM

ISSUE	DATE

Produced by
Output Services

GPU
POWER UK

ORG. No.

M196/1

ORG. No.

M196/1

A4

MATERIAL
M.S. Angle to B.S. 4.
28/33 ton quality

FINISH
Galv. to B.S. 729 after cutting
& drilling. All joints to be
pickletight.

Drg. No.	Conductor Size	Angle Section	Type	'X'
M.E.222/1	1" S.C.A.	5" x 3" x 3/8"	A	3'-9"
			B	4'-3"
M.E.222/2	1.75" S.C.A.	5" x 3" x 1/2"	A	3'-9"
			B	4'-3"

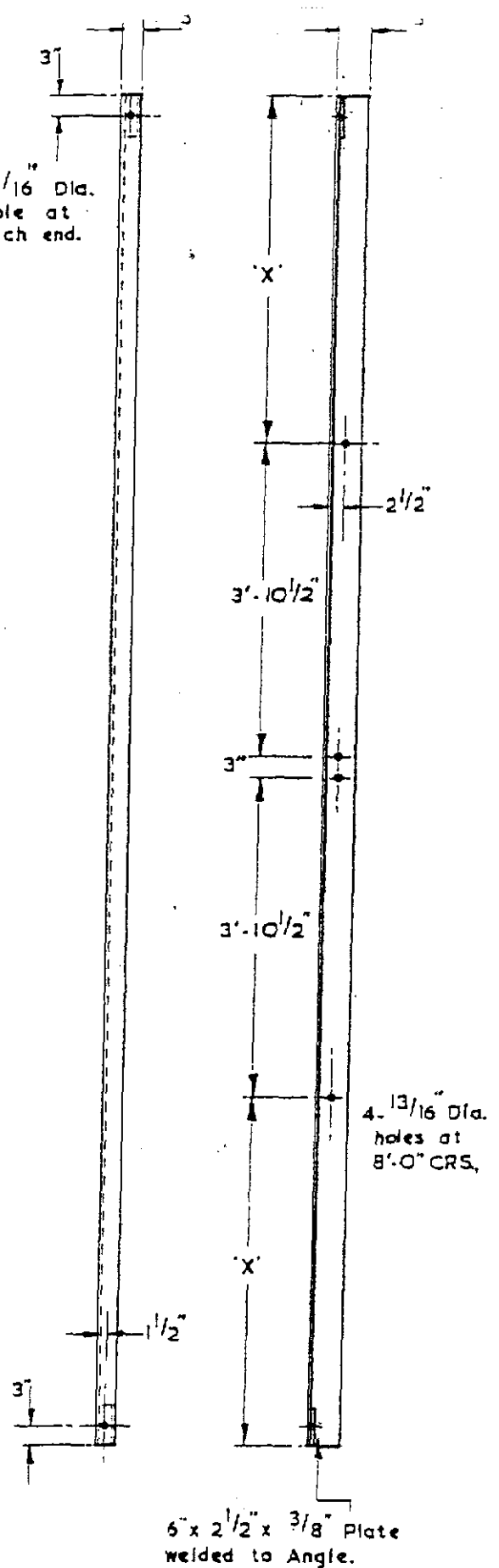
DUAL CIRCUIT CROSSARM

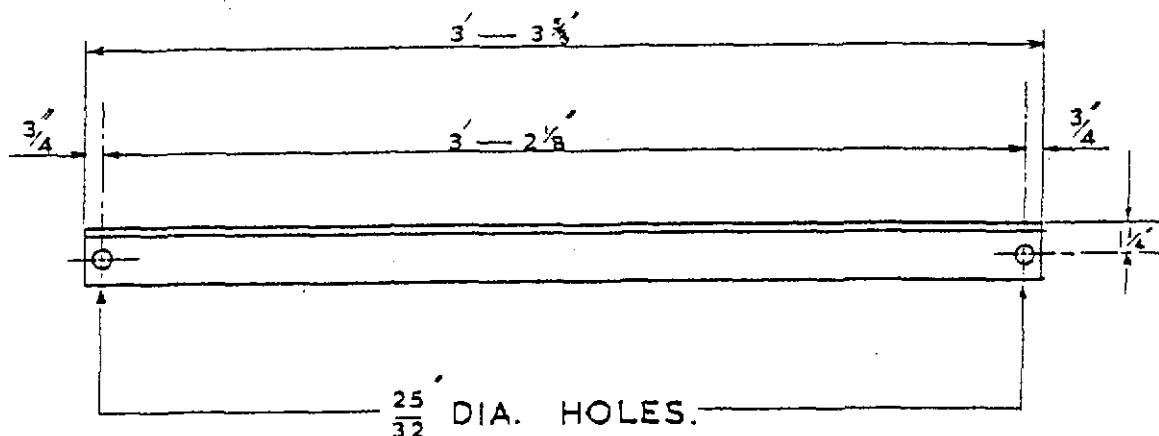
SCALE	1/24	ORIENTATION	ISSUE	ISSUE	ISSUE
DATE	27.11.62		CMF		
DRAWN	A.T.				
TRACED	Y.E.D.				
CHECKED	W.S.		Redrawn		
APPROVED	Y.S.		27.11.62		

DRAWING No.

M.E. 222.

15/16" Dia.
hole at
each end.





MATERIAL

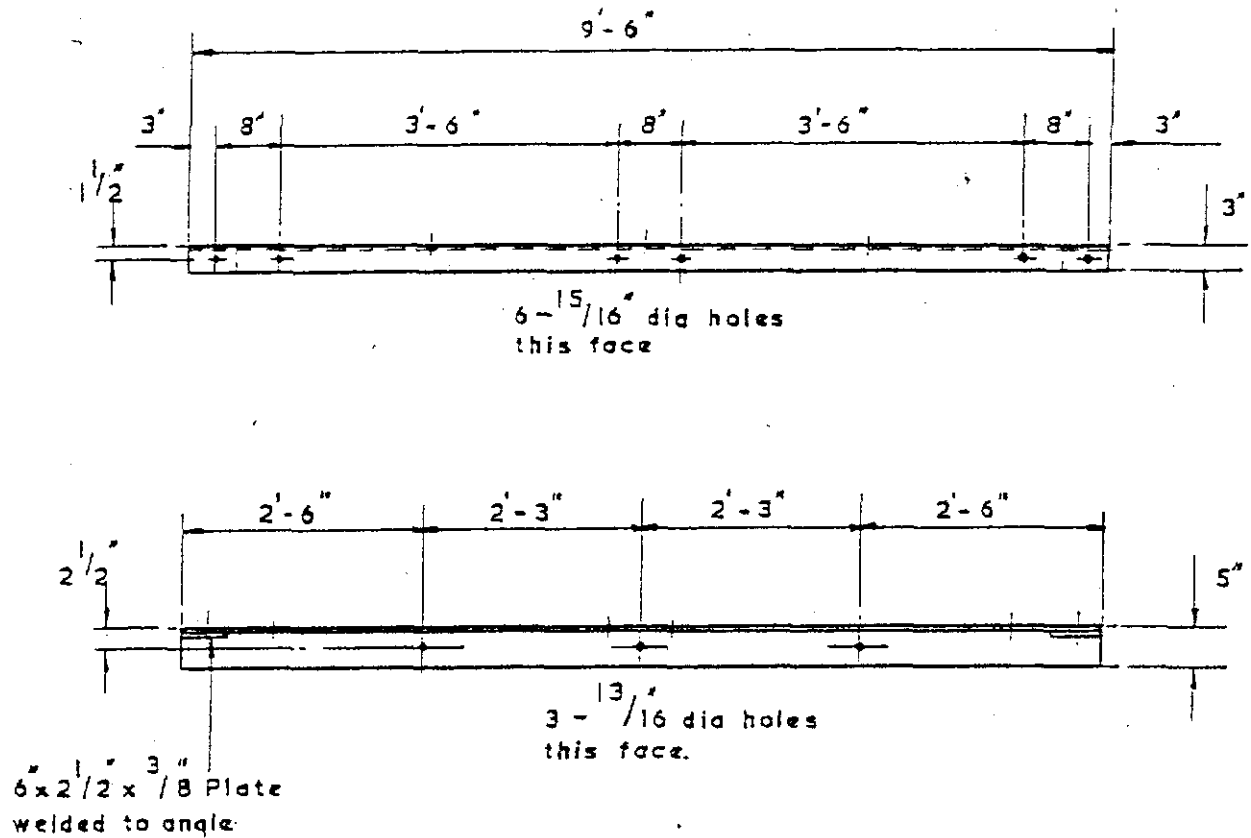
2 1/4" x 2 1/4" x 5/16". M.S. ANGLE 28/33 TON QUALITY.
GALV² TO B.S. 729. AFTER CUTTING & DRILLING
FOR USE WITH CROSSARM M.E. 227.

NOTE

HEAVY CONSTRUCTION 4' — 6"
CONDUCTOR SPACING.

ANGLE STRUT

SCALE	1/8	ORIENTATION	ISSUE	ISSUE	ISSUE	DRAWING No. M.E. 226.
DATE	19.5.52					
DRAWN	MG					
TRACED						
CHECKED	<i>W.G.C.</i>					
APPROVED	<i>W.G.C.</i>					



NOTE

For use with .175" S.C.A. 66 kv line on
Lapp Insulators 4'-6" Conductor spacing

MATERIAL.

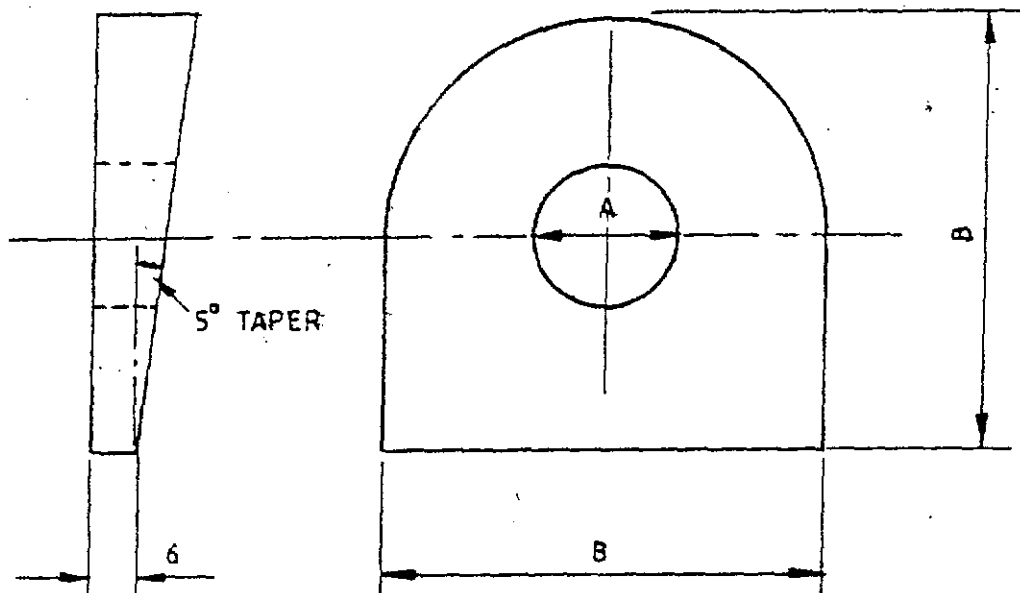
5" x 3" x 3/8" x 9'-6" M.S. Angle to B.S.4.
28/33 ton quality.

FINISH

Galv to BS 729 after cutting, drilling &
welding. All joints to be pickle treat.

LINE CROSSARM

SCALE	1/24	ORIENTATION	ISSUE	ISSUE	ISSUE	DRAWING No. M.E.227.
DATE	28.11.62.					
DRAWN	A.T.					
TRACED	Y.E.D.					
CHECKED	<i>[Signature]</i>					
APPROVED	<i>[Signature]</i>					



FOR USE WITH METRIC BOLT	HOLE DIA. A	OVERALL SIZE B
M12	14	32
M16	18	38
M20	22	45

SCALE	NTS.
DATE	13-8-84
DRAWN	R.A.C.
TRACED	C.A.R.
CHECKED	<i>[Signature]</i>
APPROVED	<i>[Signature]</i>

D SHAPED TAPERED
WASHER

ISSUE

MIDLANDS
ELECTRICITY BOARD

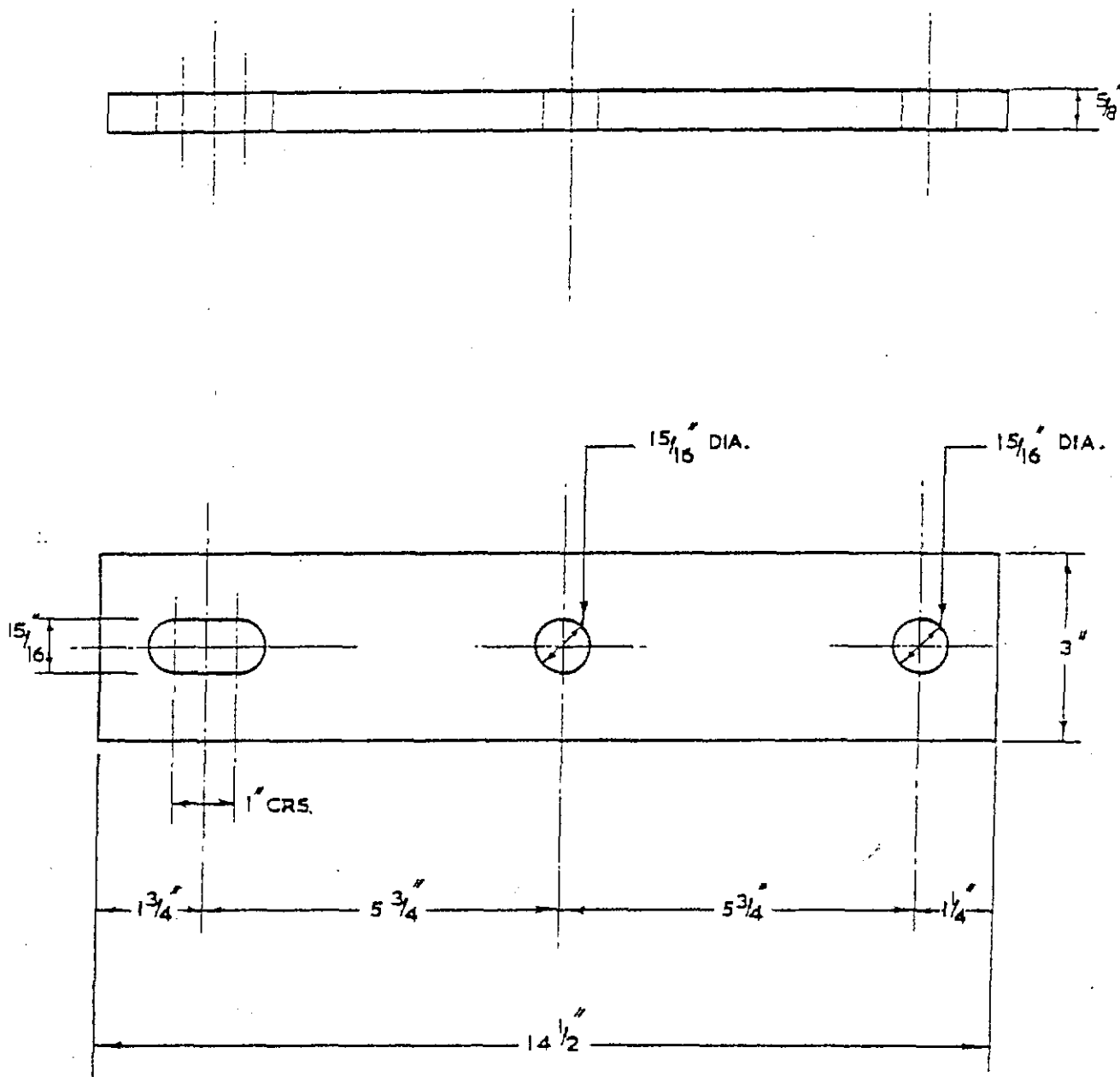
ORG. No.

M.228

ES. D. W. 43.3287

ESI. STANDARD 43 96

ESI. ORG. No 439606



MATL. :- $14\frac{1}{2} \times 3 \times \frac{5}{8}$ MS Flat.

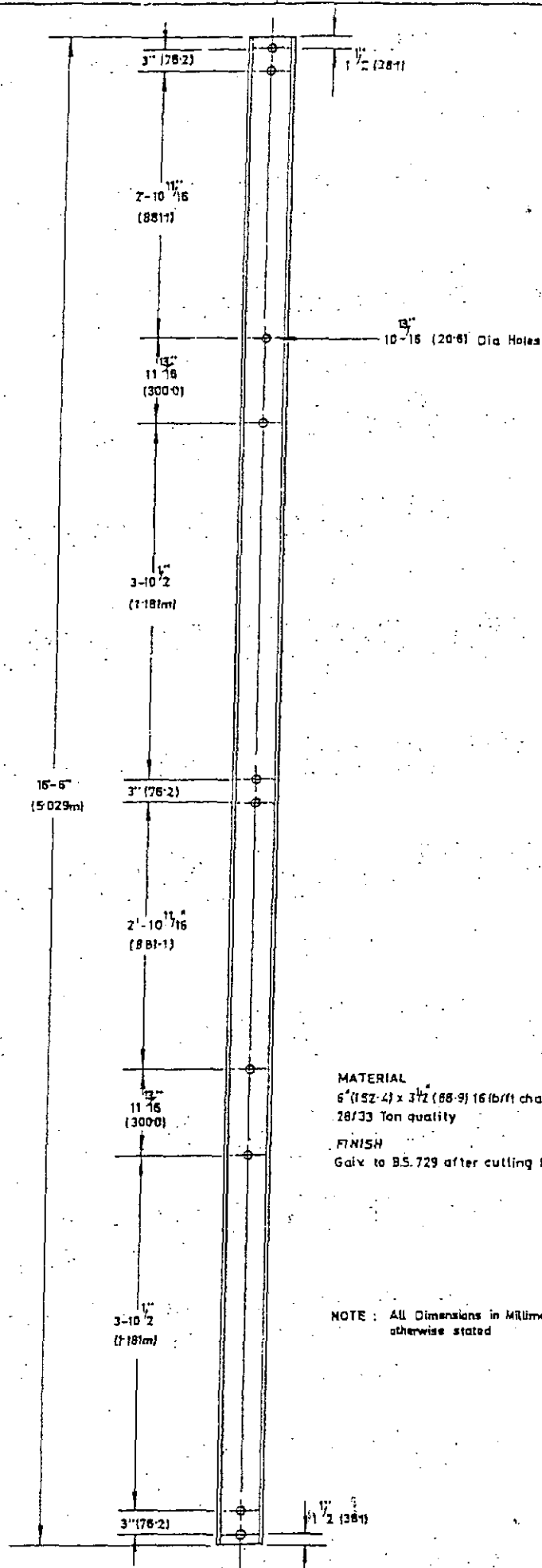
FINISH :- Galv to BS. 729
after cutting & drilling

NOTE :- Supplied with 2- $\frac{7}{8} \times 2$ Bolts c/w Nuts & Washers.

SCALE	3 / 8
DATE	29.4.64
DRAWN	N.E.O.
TRACED	
CHECKED	
APPROVED	

CROSSARM TIE STRAP.

ISSUE	
	MIDLANDS ELECTRICITY BOARD
DR'S No.	
	M.E. 246.



MATERIAL
 6" (152.4) x 3 1/2" (88.9) 16 lb/ft channel to B.S.4
 28/33 Ton quality

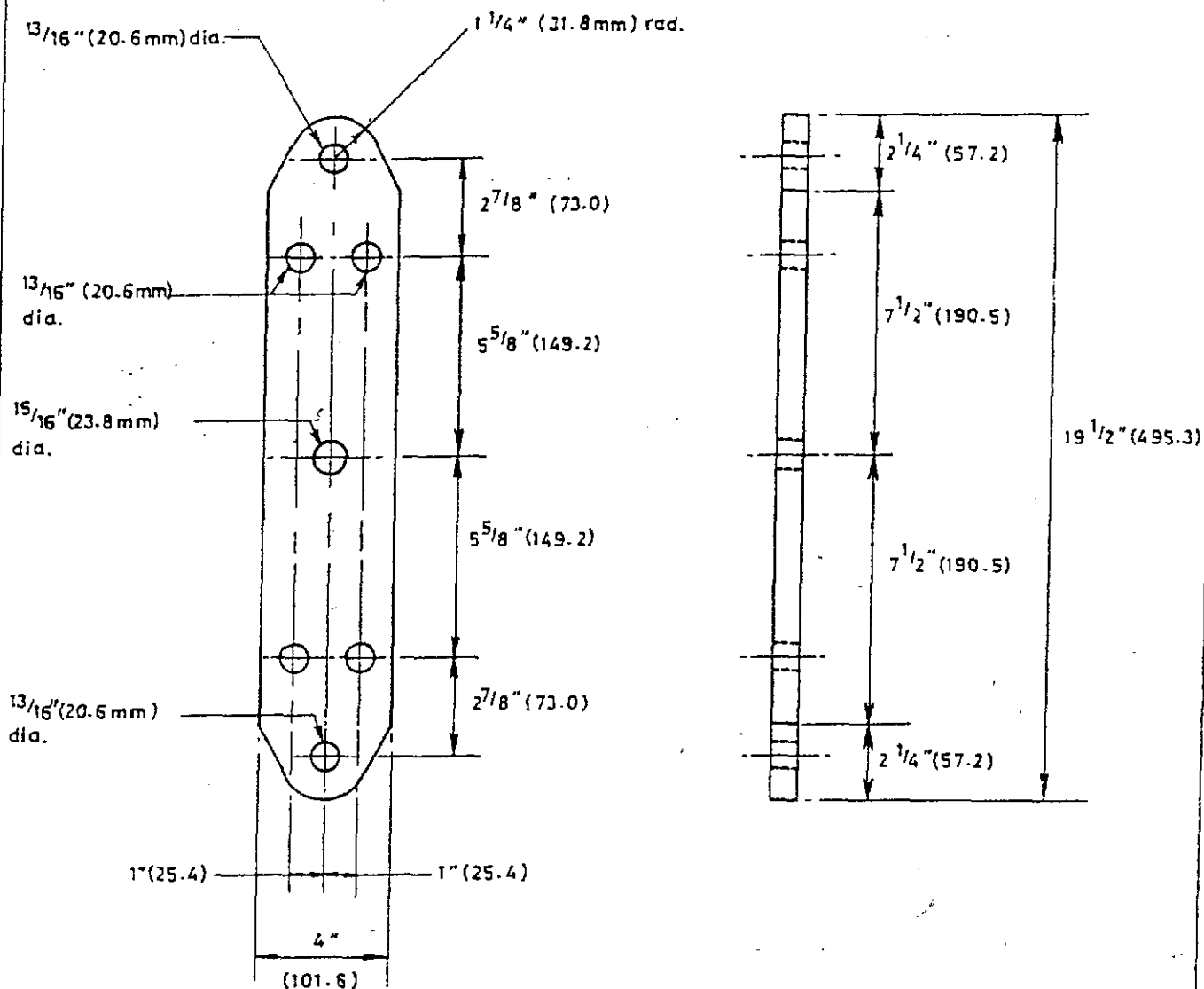
FINISH
 Galv to B.S. 729 after cutting & drilling.



SCALE	1/20
DATE	20-1-77
DRAWN	W.B.
CHECKED	J.H.F.
APPROVED	J.F.

SUSPENSION TYPE CHANNEL CROSSARM
 (66 kV 0.4 in² (400 m²) Cu EQUIVALENT SCA)

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
	A	
	Redrawn to Metric 9.8.78 A.B.B.	DRG. No. ME 258



MATERIAL

4" (101.6) x 3/4" (19) M.S. to B.S. 4360 Grade 435.

FINISH

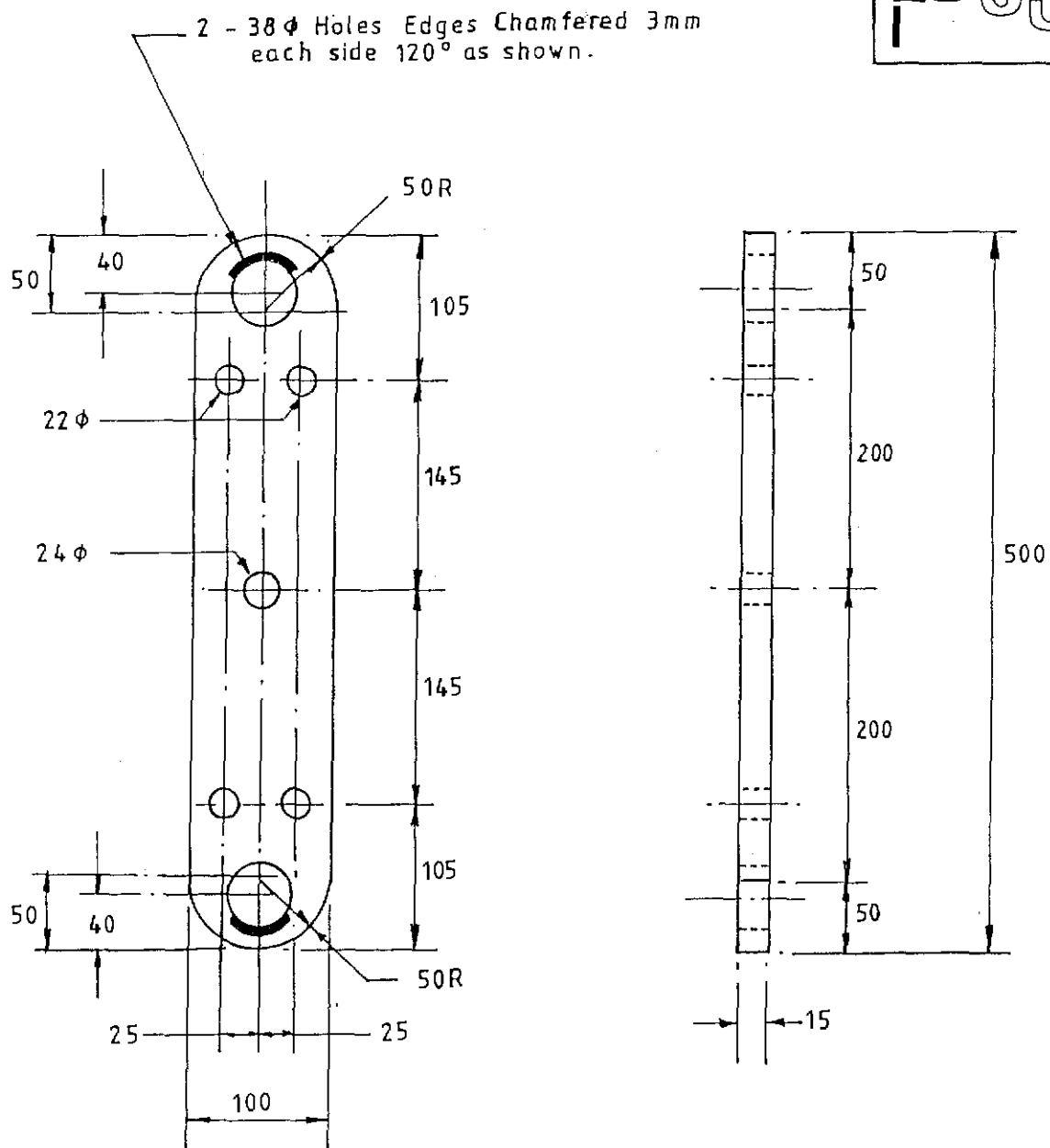
Galvanised to B.S. 729 after cutting and drilling.

SCALE	1/5
DATE	10.10.72.
DRAWN	W.B.
TRACED	A.W.
CHECKED	
APPROVED	

LONG SECTION STRAP
MINIMUM FAILING LOAD 125kN 12,700Kgf

ISSUE
1
Re-drawn to metric.

MIDLANDS ELECTRICITY BOARD
DRG. No. M.E. 259



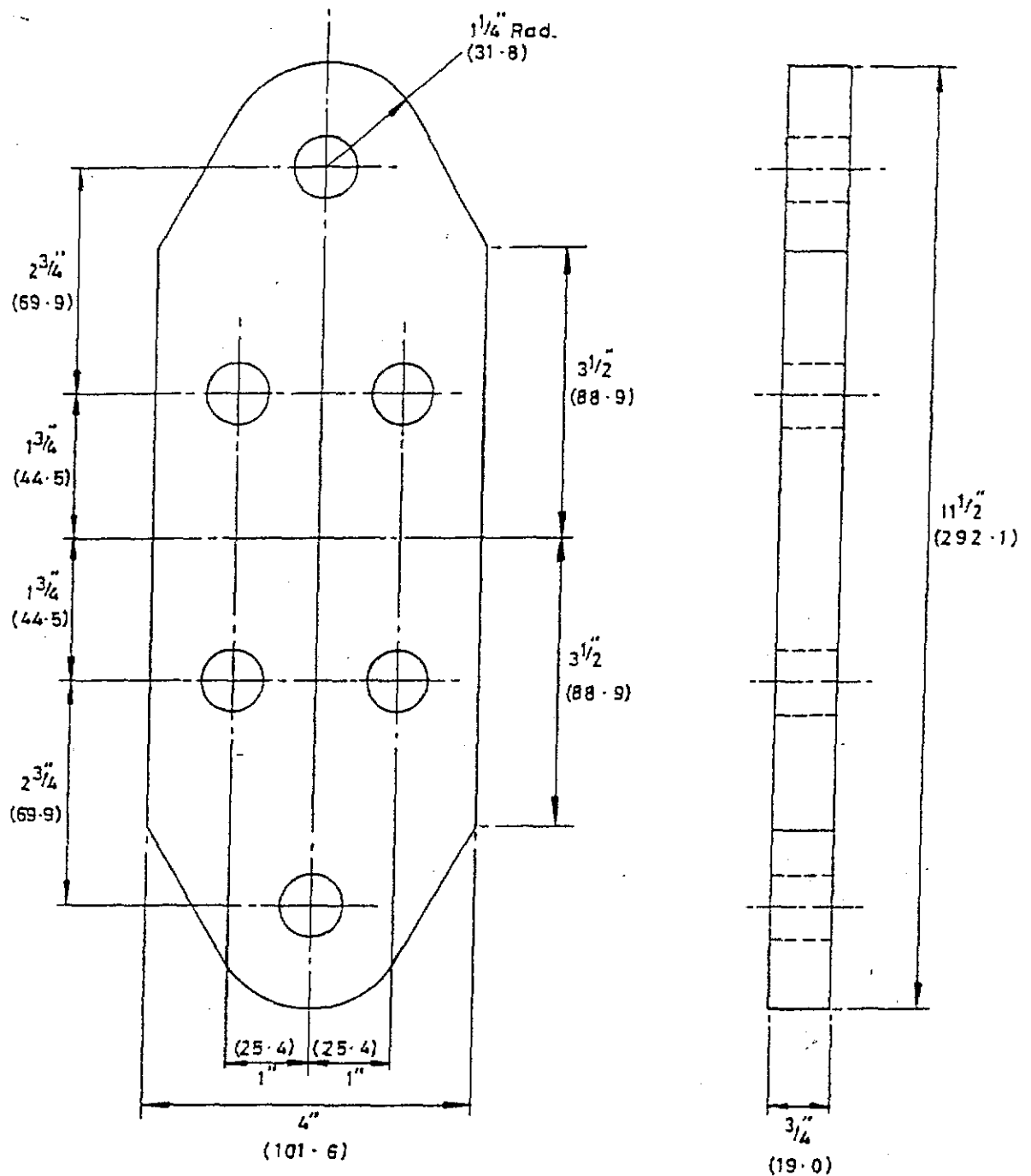
MATERIAL

10 x 15 Flat M.S. to B S 4360 Grade 435

FINISH

Galvanised to B.S. 729 after cutting and drilling.

ADDED BY DATE	TITLE LONG SECTION STRAP MINIMUM FAILING LOAD 70 kN (7135 kgf)	SCALE		 Midlands Electricity plc ORG No M259/1
		DATE	1.11.93	
		DRAWN	A.J.B.	
		TRACED	S.G.T.	
		CHECKED		
		APPROVED		



MATERIAL

4" (101.6) x 3/4" (19) M.S. to B.S. 4360 Grade 435.

FINISH

Galvanised to B.S. 729 after cutting and drilling.

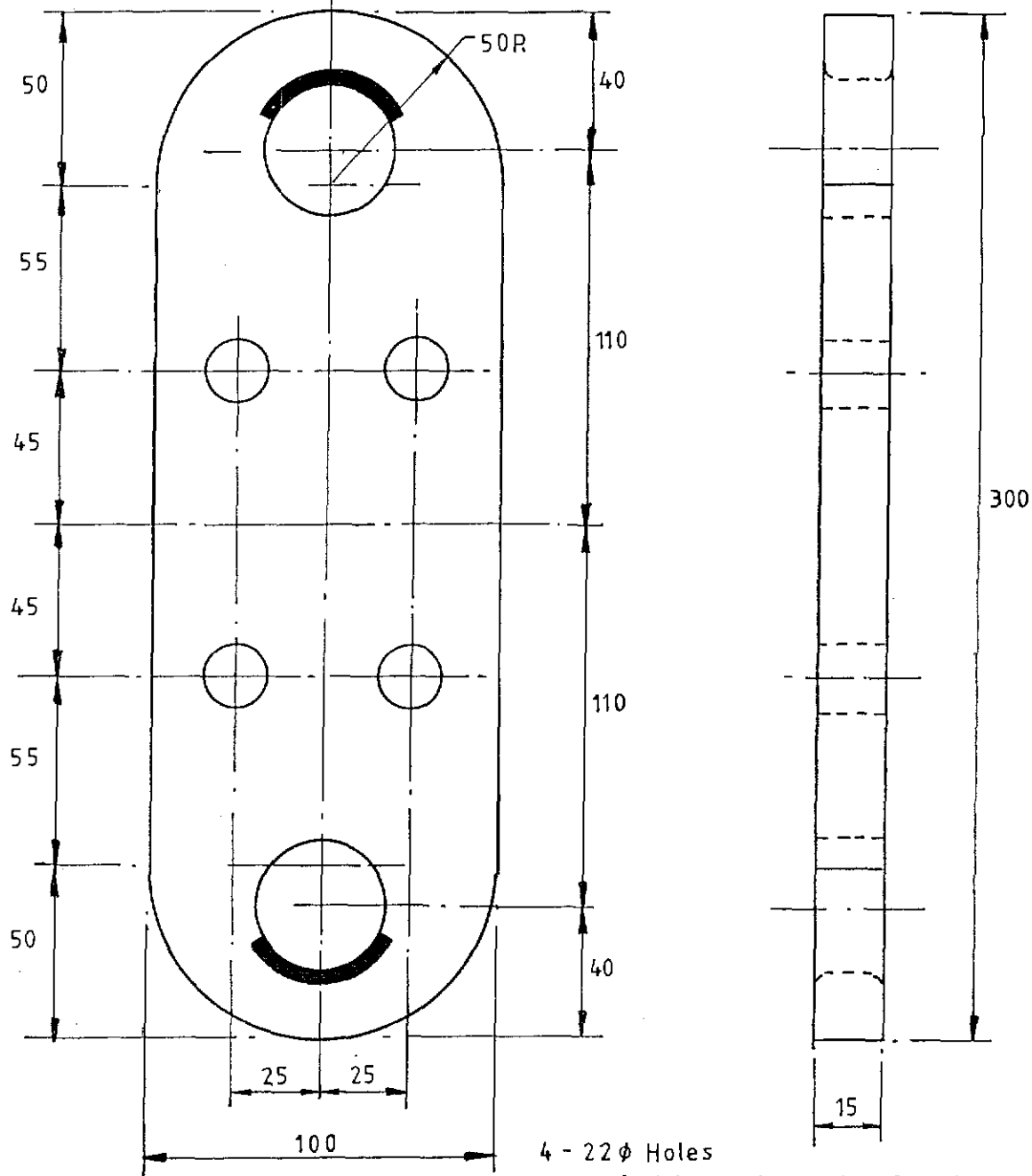
SCALE	1 / 2
DATE	17.10.72
DRAWN	
TRACED	C. E.
CHECKED	
APPROVED	H

OZALID 16339/

SHORT SECTION STRAP
MINIMUM FAILING LOAD 125kN 12,700Kgf

ISSUE
Re drawn metric equiv. added

MIDLANDS ELECTRICITY BOARD
DRG. No.
ME. 260



4 - 22 ϕ Holes

2 38 ϕ Holes

Edges Chamfered 3mm each side 120° as shown.

MATERIAL

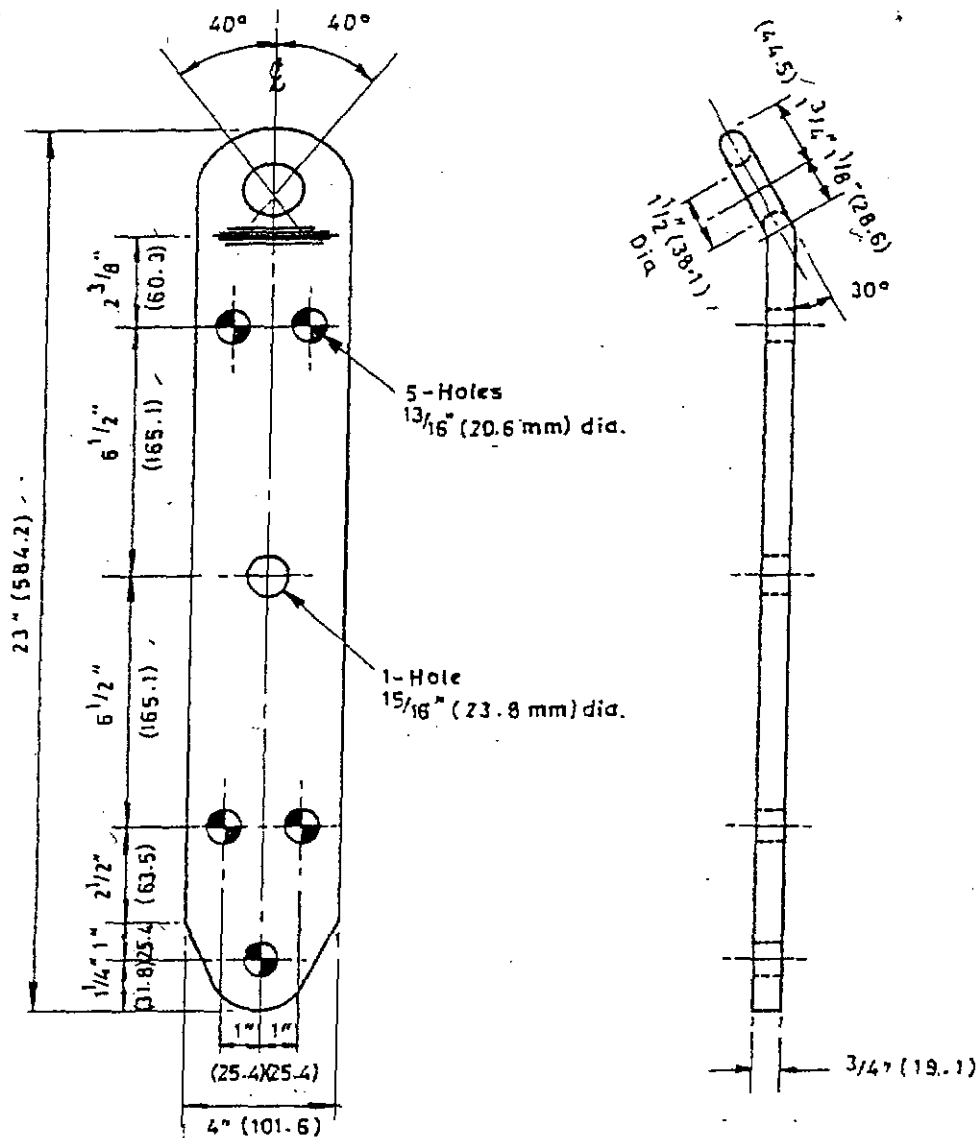
100 x 15 Flat M.S. to B.S. 4360 Grade 435

FINISH

Galvanised to B.S. 729 after cutting and drilling

ADD'D BY DATE	TITLE SHORT SECTION STRAP MINIMUM FAILING LOAD 70kN (7135kgf)	SCALE		 Midlands Electricity plc DRG No M 260/1
		DATE	1 11 93	
		DRAWN	A.J. B	
		TRACED	S.G. T.	
		CHECKED		
		APPROVED		

NOTE:- Edges to be rounded off
40° each side of ϕ



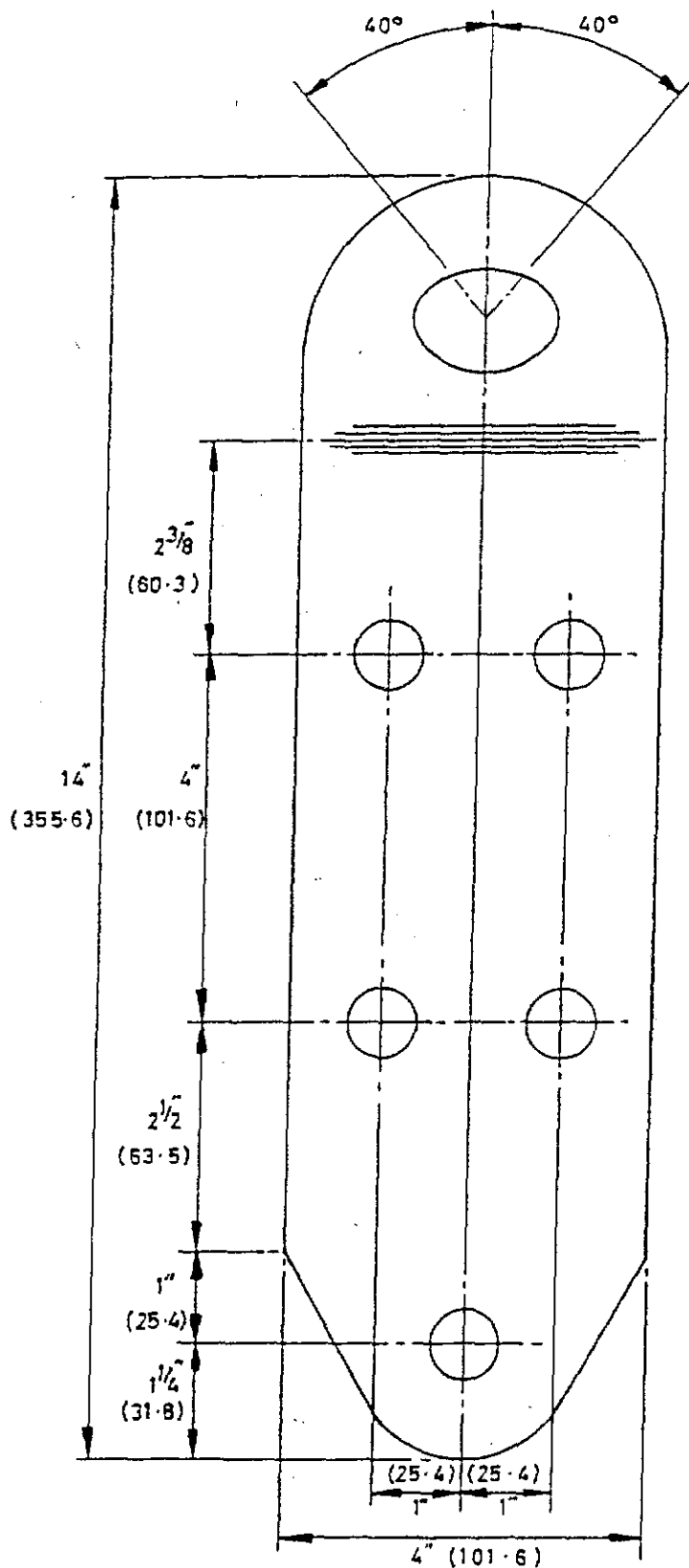
MATERIAL 4" (101.6) x 3/4" (19)
28/33 Ton/in² (44/55 Kg/mm²) Quality

FINISH Galvanised to B.S. 729
after cutting & drilling

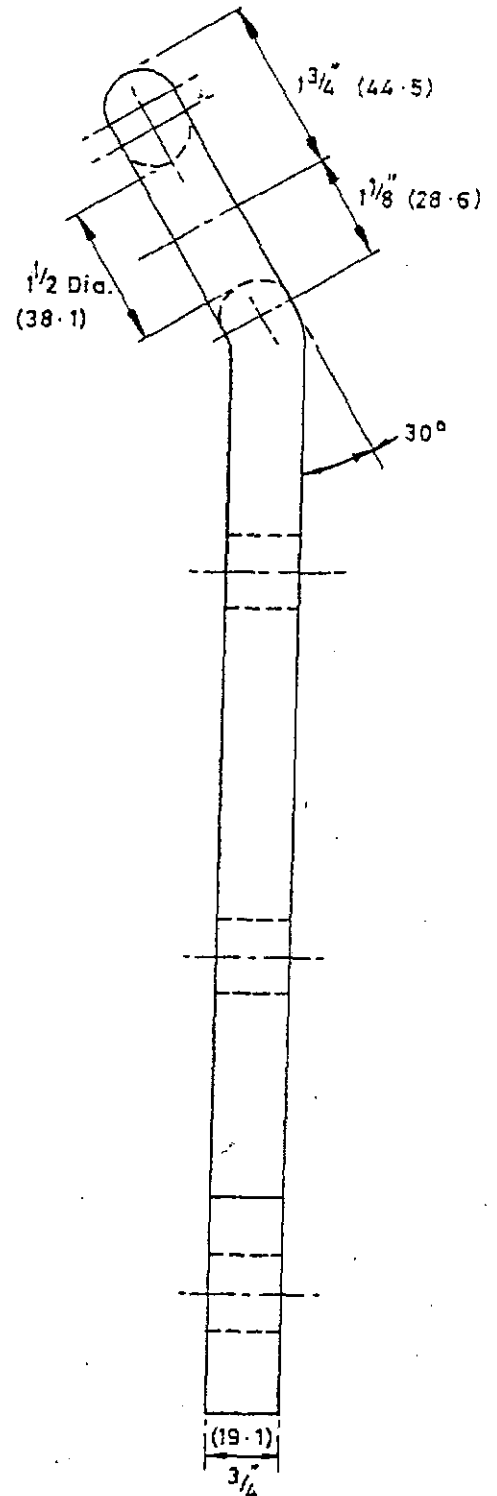
SCALE	1/5
DATE	10.10.72
DRAWN	F.A.H.
TRACED	A.W.
CHECKED	
APPROVED	

LONG TERMINAL STRAP
MINIMUM FAILING LOAD 125kN 12,700Kgf.

ISSUED	MIDLANDS ELECTRICITY BOARD
Redrawn & converted to metric.	DRG. No. M.E. 261



NOTE: Edges to be rounded off
40° each side of C/L.



5-13/16 (20.6) Dia. Holes

MATERIAL

4" (101.6) x 3/4" (19) M.S. to B.S. 4360 Grade 435

FINISH

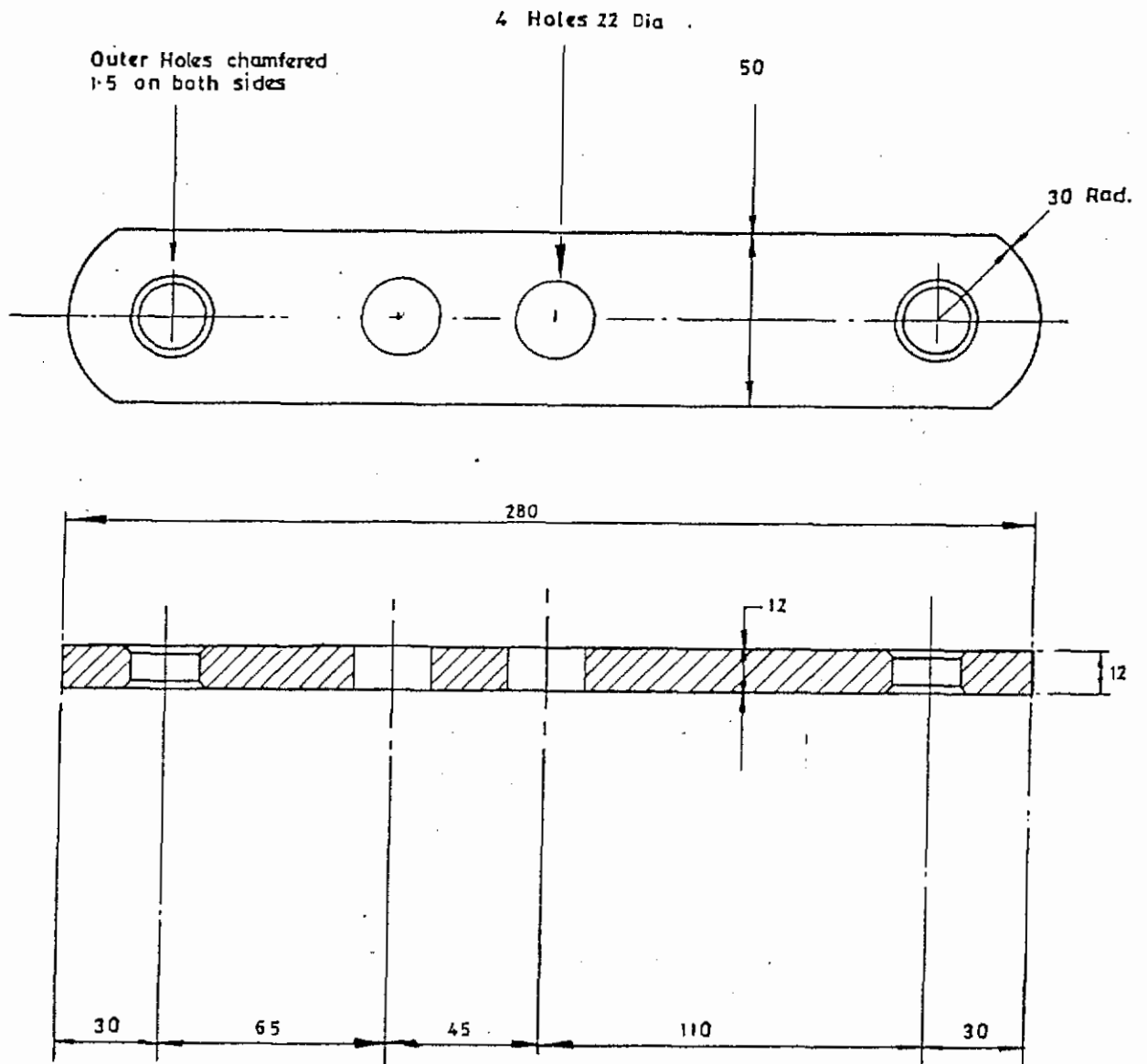
Galvanised to B.S. 729 after cutting and drilling

SCALE	1 / 2
DATE	18-10-72
DRAWN	
TRACED	C. E.
CHECKED	
APPROVED	

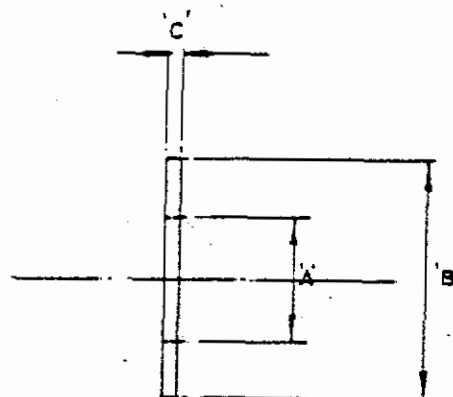
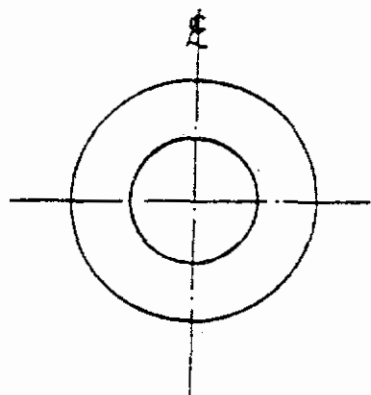
SHORT TERMINAL STRAP
MINIMUM FAILING LOAD 125kN 12,700 Kgf

ISSUE
Redrawn metric equiv. added

MIDLANDS
ELECTRICITY BOARD
DRG. No.
M.E. 262.



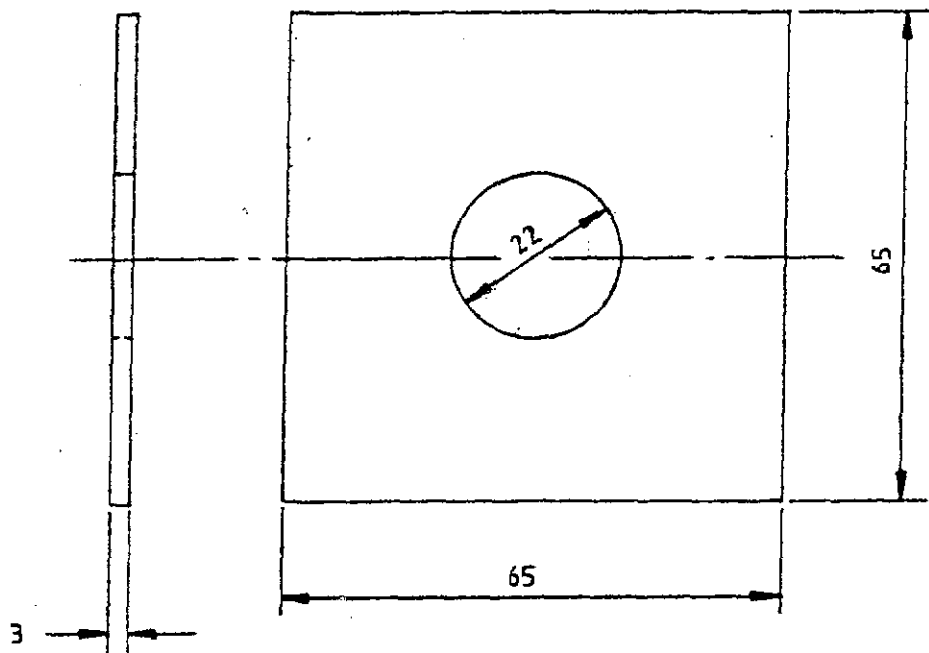
SCALE	1:2	SECTION STRAP MINIMUM FAILING LOAD 20 kN (1835 Kgf.) E.S.I. STANDARD 43-95	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	26-10-82			
DRAWN	R.N.			
TRACED	M.R.J.			DRG. No. M.268.
CHECKED				
APPROVED	<i>H</i>			E.S.I. DRG. No. 439507



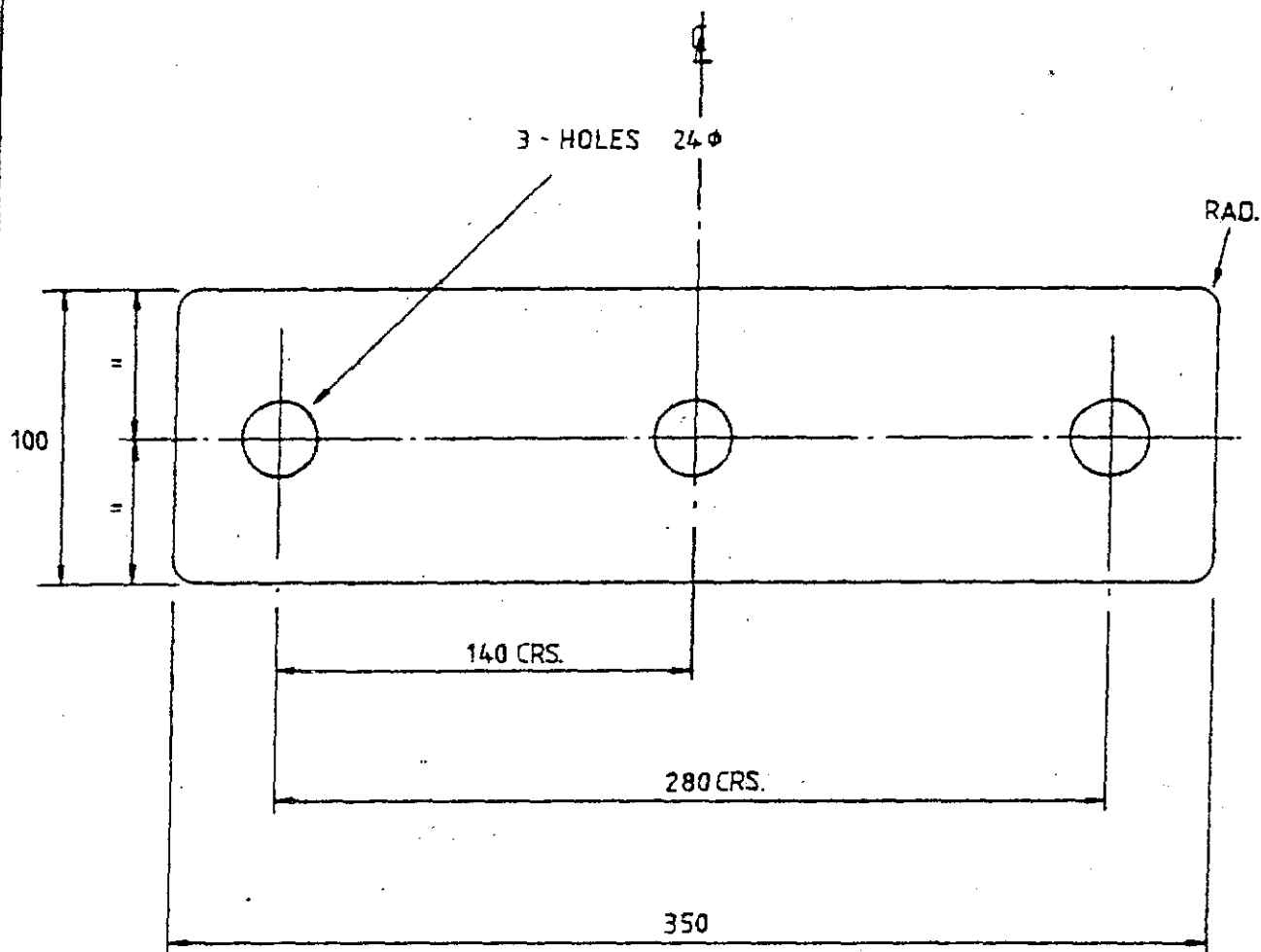
TYPE	A Dia(mm)	B Dia(mm)	C (mm)	USE
M. 315/1.	16.54 min	31.5 min	1.83	ME 195/123 ME 295/3
M. 315/2.	22.89 min.	41.02 min	"	ME 237

TO BS. 3410: 1961.
Table 3.

SCALE	FULL	BRASS WASHER.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	26-10-82			
DRAWN	R.A.C.			DRG. No.
TRACED				M: 315.
CHECKED	<i>[Signature]</i>			
APPROVED	<i>[Signature]</i>	E.S.I. STANDARD		E.S.I. DRG. No.



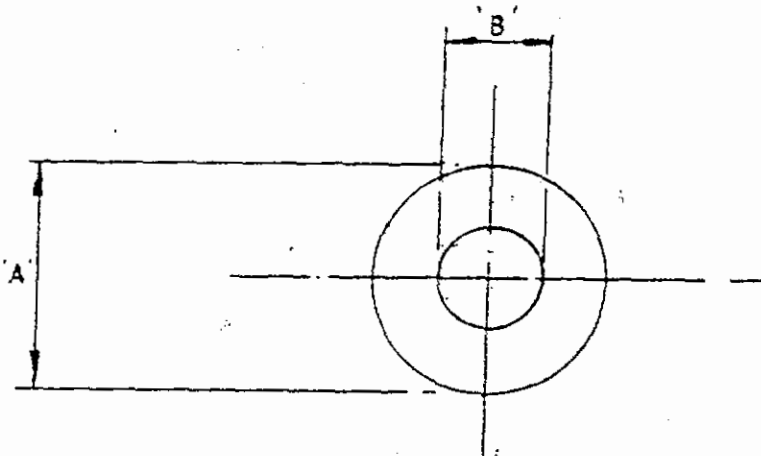
SCALE	NTS.	SQUARE FLAT WASHER	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	13-8-84			ORG. No.
DRAWN	R.A.C.			M.317
TRACED	C.A.R.			
CHECKED	<i>[Signature]</i>			
APPROVED	<i>[Signature]</i>			
D. W. Lm. 3297 / 1		ESI STANDARD 43 96		ESI ORG. N° 439604



MATERIAL :- 100 x 12 STEEL PLATE
TO B.S. 4360 1972
GRADE 50^B

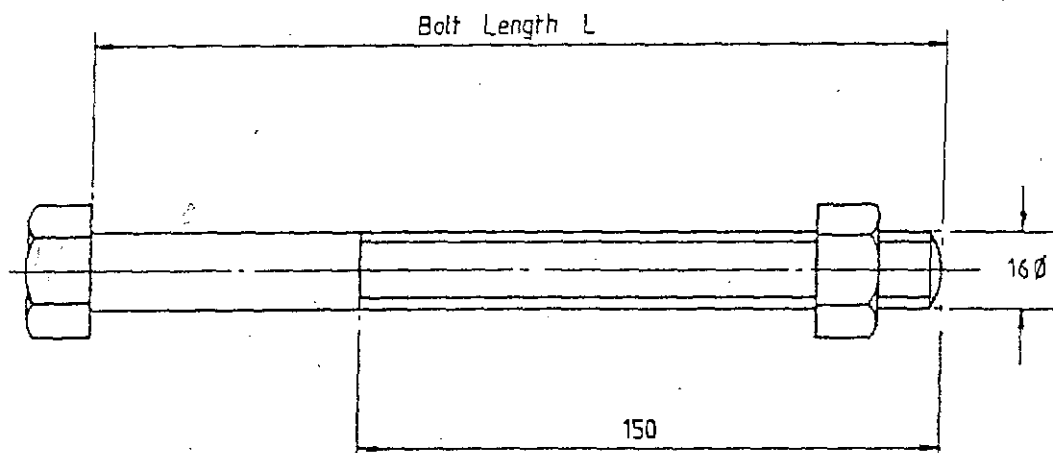
FINISH :- GALV'D TO B.S. 729
AFTER CUTTING AND
DRILLING.

SCALE	NTS.	CROSSARM STRAP FOR USE WITH ME 107 CROSSARM AND 400mm ² H.D.A. CONDUCTOR.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	2-8-84			
DRAWN	R.A.C.			DRG. No. M. 318.
TRACED	C.A.R.			
CHECKED				
APPROVED				



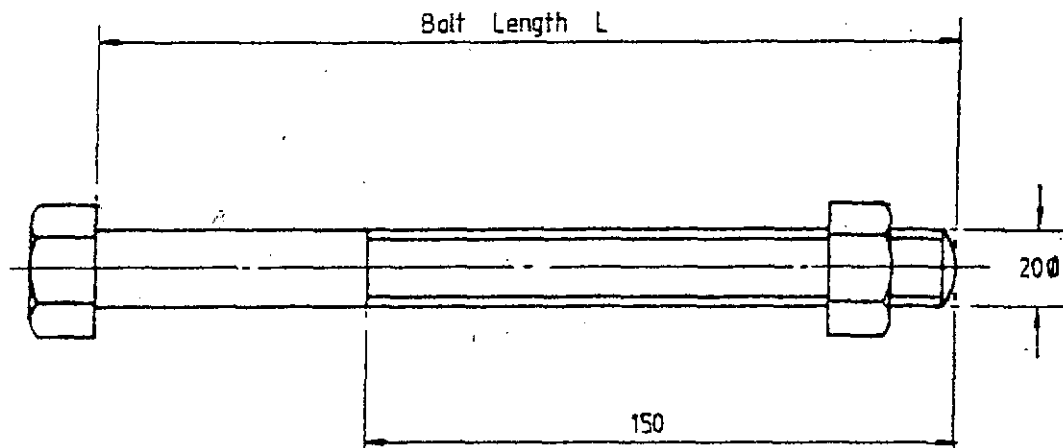
FOR USE WITH METRIC BOLTS.	DIA A	DIA B
M 12	24	14
M 16	30	18
M 20	37	22

SCALE	N.T.S.	ROUND FLAT WASHER.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	18-8-84			
DRAWN	RAC.			
TRACED				
CHECKED				
APPROVED	<i>[Signature]</i>		DRG. No.	M. 319
C. O. W. Ltd. 3287		ESI. STANDARD 43-96	ESI. DRG. N°	



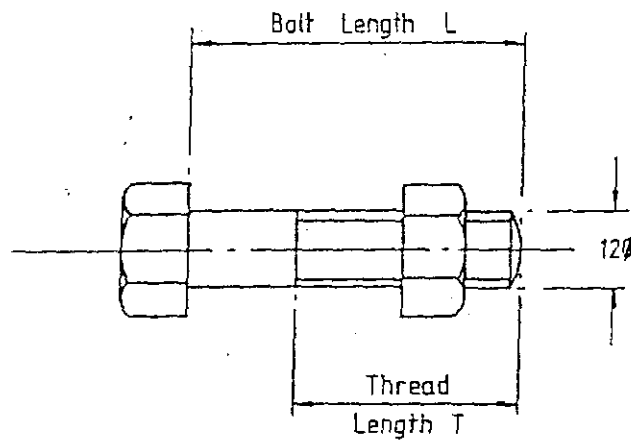
TYPE	BOLT LENGTH L (mm)
362 / 1	200 (8")
362 / 2	220 (9")
362 / 3	260 (10")
362 / 4	300 (12")
362 / 5	375 (15")
362 / 6	450 (18")
362 / 7	530 (21")
362 / 8	610 (24")

SCALE	-	POLE BOLTS 16mm DIAMETER	ISSUE	MIDLANDS ELECTRICITY BOARD ORG. No. M 362
DATE	25.11.83			
DRAWN				
TRACED	J.N.			
CHECKED	D.T.			
APPROVED	J.N.			
C. D. W Ltd. 3297		E.S.I. STANDARD 43-96		



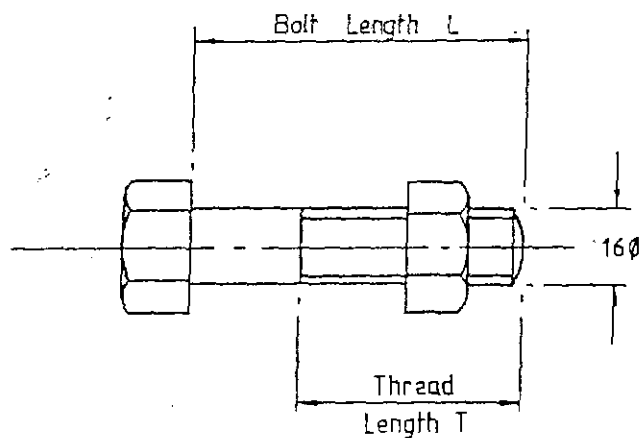
TYPE	BOLT LENGTH L (mm)
363 / 1	200 (8")
363 / 2	220 (9")
363 / 3	260 (10")
363 / 4	300 (12")
363 / 5	375 (15")
363 / 6	450 (18")
363 / 7	530 (21")
363 / 8	610 (24")
363 / 9	760 (30")

SCALE	-	POLE BOLTS 20mm. DIAMETER E.S.I. STANDARD 43-96	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M 363 (MF)
DATE	25. 11. 83.		A	
DRAWN			363/4 Length	
TRACED	J.W.		Corrected	
CHECKED	D.T.		CAR. 8-8-84.	
APPROVED	<i>[Signature]</i>			
D. W. L. 3297				



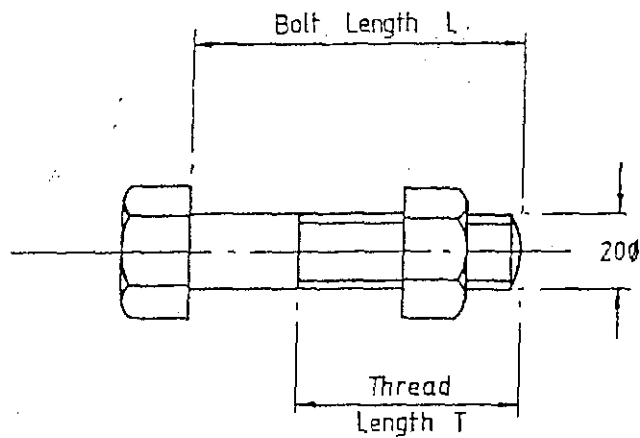
TYPE	BOLT LENGTH L (mm)	THREAD LENGTH T (mm)
364/0	30 (1 1/4")	30
364/1	45 (1 3/4")	30
364/2	60 (2 1/2")	30
364/3	75 (3")	30
364/5	100 (4")	30
364/7	140 (5 1/2")	36
364/8	260 (10")	50
364/9	300 (12")	50

SCALE	—	ASSEMBLY BOLTS 12mm DIAMETER	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	29.11.83			
DRAWN				DRG. No.
TRACED	H.P.S.			M364
CHECKED	D.T.			
APPROVED	<i>[Signature]</i>			
C. D. W Ltd. 3297		ESI STANDARD 43-96		



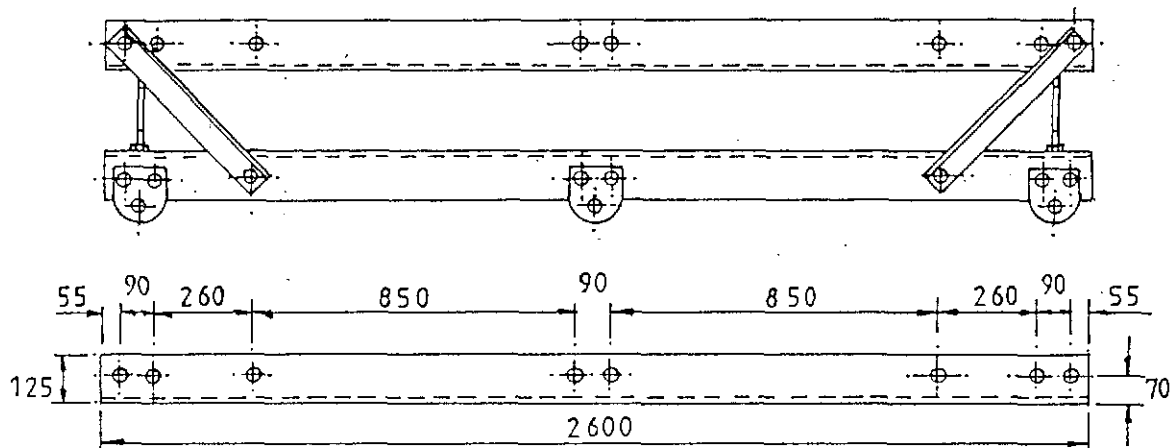
TYPE	BOLT LENGTH L (mm)	THREAD LENGTH T (mm)
365/1	50 (2")	38
365/2	70 (3")	38
365/3	90 (3 1/2")	38
365/4	110 (4 3/4")	38
365/5	130 (5")	44
365/8	180 (7")	44

SCALE	—	ASSEMBLY BOLTS 16mm. DIAMETER	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	29-11-83			
DRAWN				ORG. No. M365
TRACED	HPS			
CHECKED	S.F.			
APPROVED	[Signature]			
C. D. W Ltd. 3297		E.S.I. STANDARD 43-96		



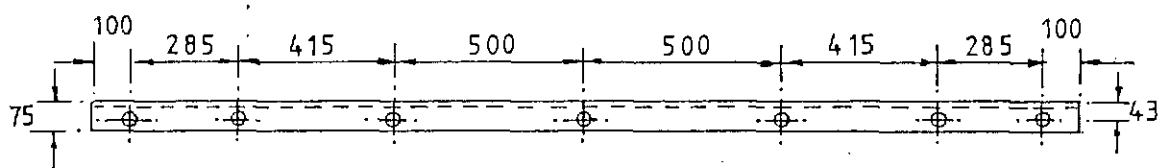
TYPE	BOLT LENGTH L (mm)	THREAD LENGTH T (mm)
366/1	45 (1 ³ / ₄ ")	46
366/2	60 (2 ¹ / ₂ ")	46
366/3	80 (3")	46
366/4	100 (4")	46
366/5	130 (5")	52
366/7	180 (7")	52

SCALE	—	ASSEMBLY BOLTS 20mm DIAMETER	ISSUE	MIDLANDS ELECTRICITY BOARD ORG. No. M366
DATE	29.11.83			
DRAWN				
TRACED	H.P.S.			
CHECKED	D.T.			
APPROVED	<i>[Signature]</i>			
E. D. W Ltd. 3291		E.S.I. STANDARD 43-96		

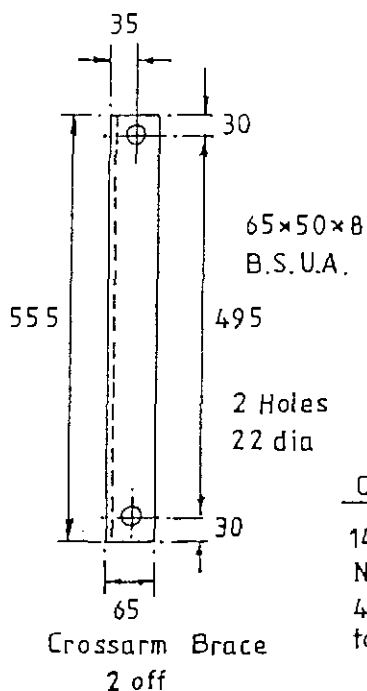


Holes 22 dia.

125 x 75 x 12 BSUA

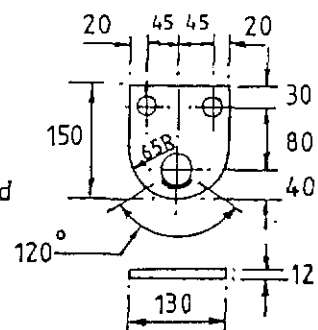


Crossarm Member
2 off



Crossarm Brace
2 off

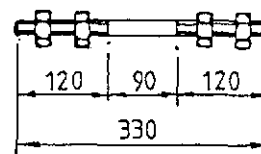
2 Holes
22 dia.
Hole 38 dia.
(Edges chamfered
3mm each side
through 120°
as shown)



Terminal Strap
3 off

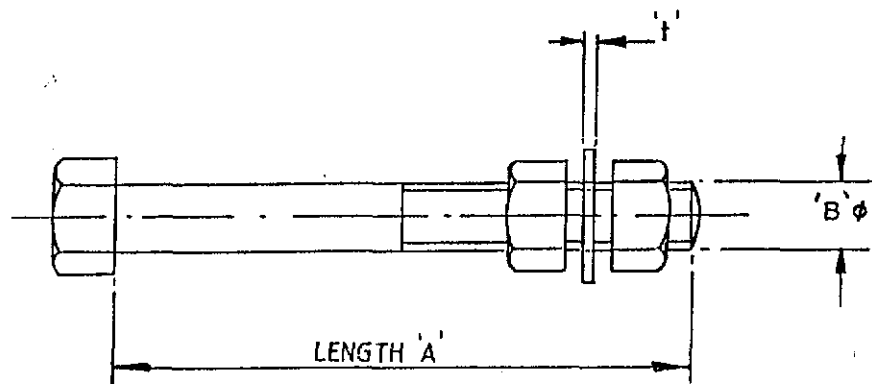
OTHER REQUIREMENTS

14 - M20 x 60 Bolts c/w
Nut and Plain Washer
4 - Crossarm Struts
to Drg. M196/1



M20 Tie Bolt
2 off

ADDED BY DATE	TITLE	TERMINAL CROSSARM 1-2M SPACING SINGLE POLES CONDUCTORS UP TO 100mm ²	SCALE	—	 Midlands Electricity plc DRG No M400
			DATE	16.11.93	
			DRAWN	A J B	
			TRACED	SGT	
			CHECKED		
			APPROVED		



ME REF	C.E.G.B CC N°	LENGTH 'A'	DIA 'B'	THICKNESS 't'
605 / 1	4384080	5 1/2" (140mm)	5/8" BSW (M16)	1/8" (3.2)
605 / 2	0208348	6" (150mm)	3/4" BSW (M20)	1/8" (3.2)

MATERIAL:---
MILD STEEL
GALVANISED
TO BS 729.

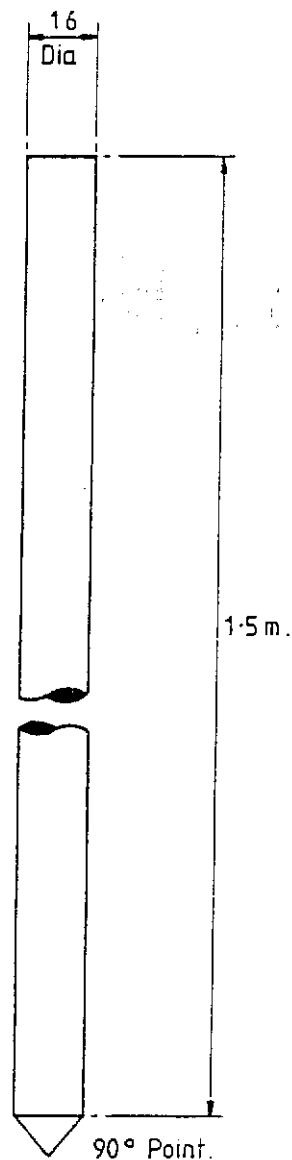
SCALE	N.T.S.	TOWER FITTINGS STEP BOLTS.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	3-8-82.			
DRAWN	R.A.C.			DRG. No.
TRACED				ME 605
CHECKED				
APPROVED				

7044G

7828/B

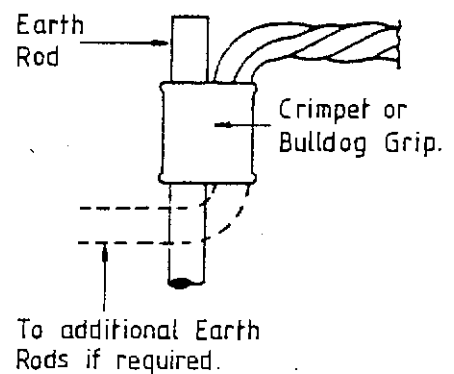
SECTION 13

MIN NUMBER	M	DESCRIPTION
3450930	93	NON EXTENSIBLE GALVANISED STEEL EARTH ROD
3451240	124	WIRE ROPE GRIP
3451930	193	EARTHING CLAMP
3452250	225	STEEL TOWER EARTH CONDUCTOR TO STEEL BONDS



NOTE.

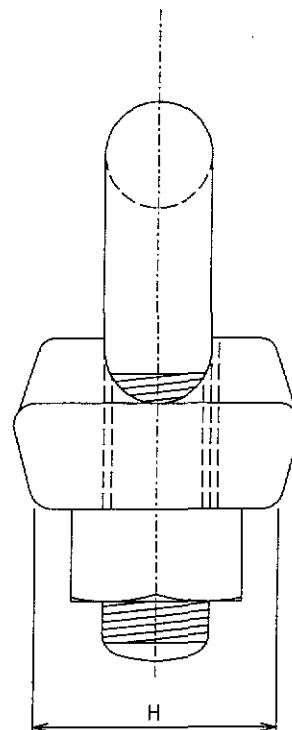
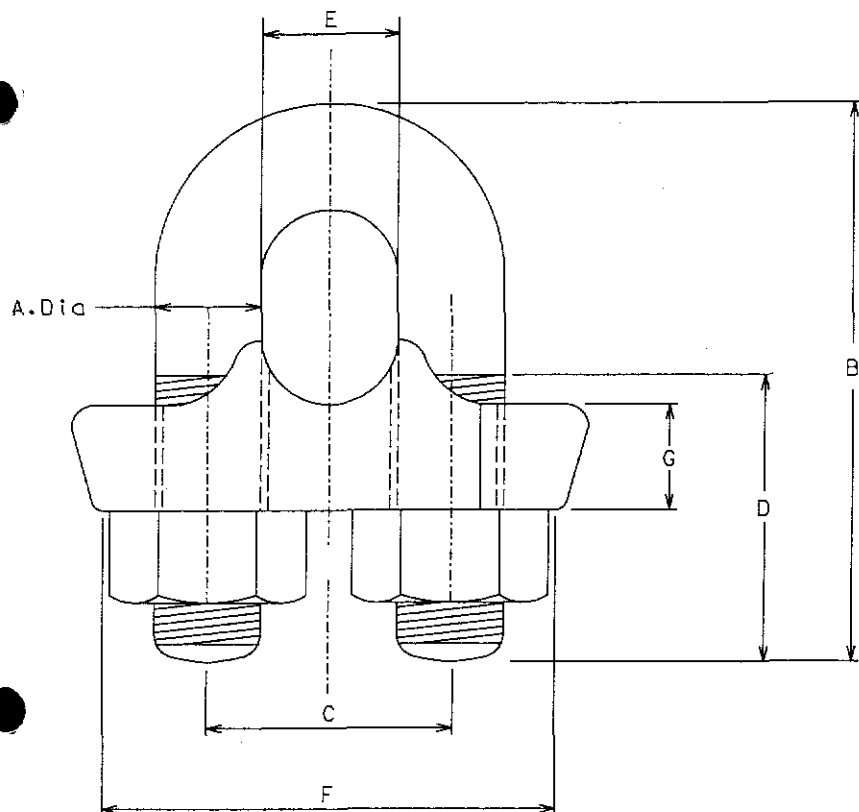
Earth rod connections to be protected with Denso Tape.



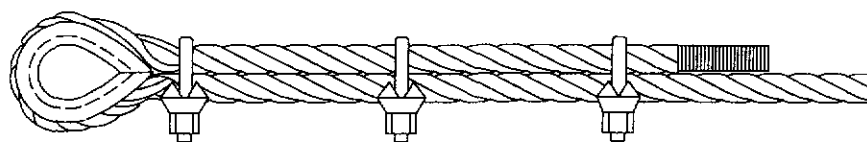
EARTH ROD CONNECTIONS.

CONDUCTOR	CRIMPET REF.	DIE REF. Y35 CRIMPS	Nº OF CRIMPS	BULLDOG GRIP
16mm ² Cu	ME 273 /8	D3	2	M124/4
32mm ² Cu	ME 273 /8	D3	2	
35mm ² Cu	ME 273 /8	D3	2	
70mm ² Cu	ME 273 /8	D3	2	

SCALE		NON EXTENSIBLE GALVANISED STEEL EARTH ROD.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	2-6-81		A	
DRAWN	R. A. C.		Crimpet Ref	DRG. No
TRACED	A. E.		Changed &	M93.
CHECKED			35mm ² Cu 87	
APPROVED	#	E.S.I. STD 43-94		E.S.I. Drg No. 439402

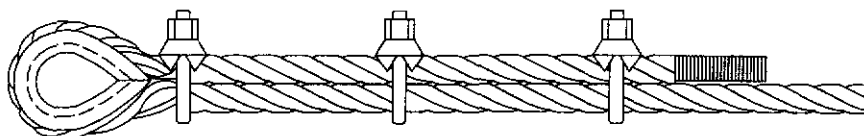


METHOD OF FITTING



RIGHT

'U' Bearing on Rope Tail



WRONG

'U' Bearing on Working Rope

MATERIAL

28/33 ton quality M.S.
Manufactured to B.S.462 &
Galv. to BS 729.

TYPE	ROPE		DIMENSIONS (mm)							
	CIRC	DIA	A	B	C	D	E	F	G	H
124/4	50	16	14	74	32	38	18	60	14	32
124/6	30	9	10	48	20.5	25	10.5	40	10	23

SCALE	1:1
DRAWN	26.10.82
TRACED	R.A.C.
CHECKED	J.W.
APPROVED	-

WIRE ROPE GRIP

ISSUE	DATE
A	21.12.01
TYPES REMOVED.	
REDRAWN ON C.A.D. D.B.	

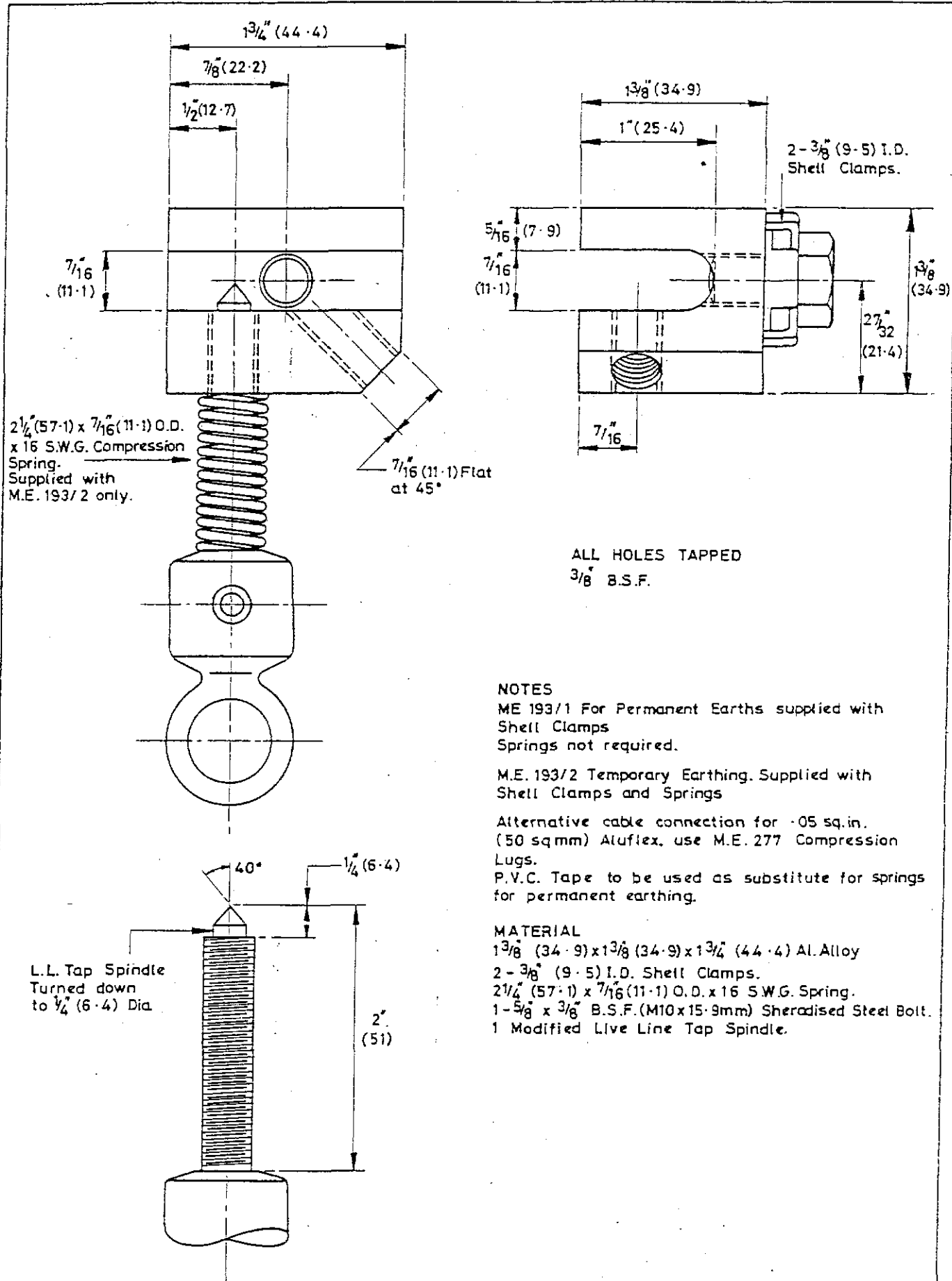
Produced by
Output Services

GPU
POWER UK

DRG. No.
M124

DRG. No.

M124

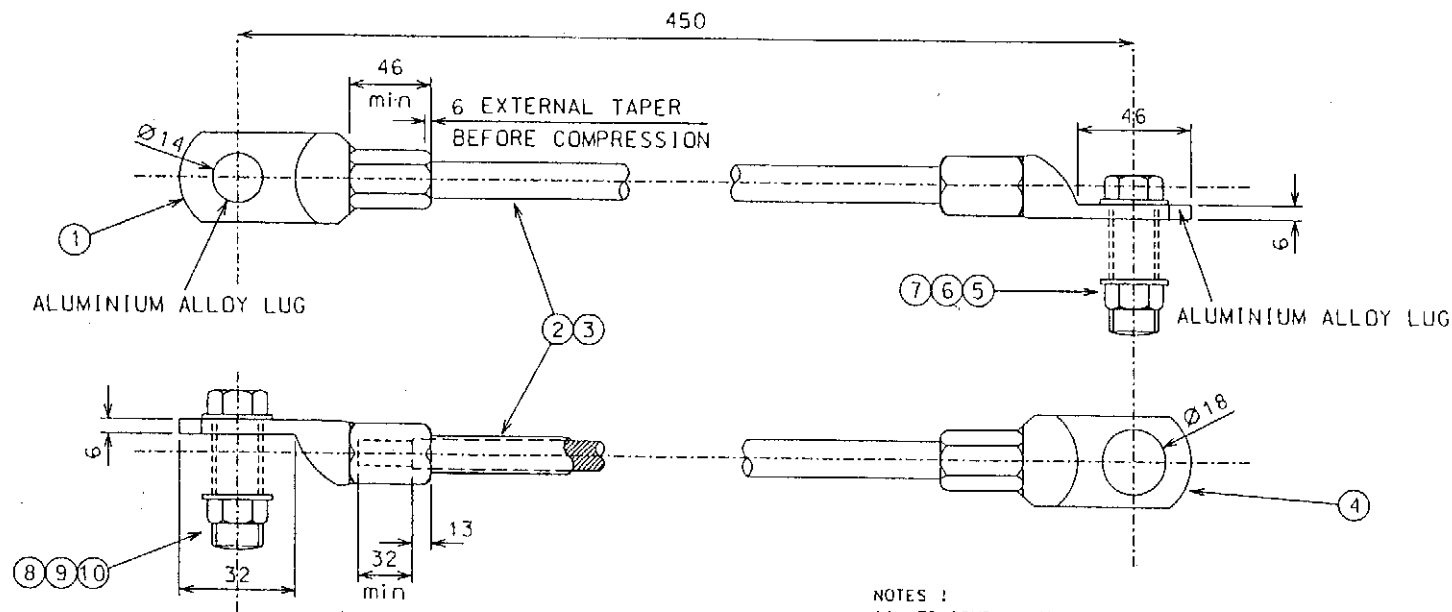


SCALE	1 / 1
DATE	4.6.70
DRAWN	W.D.H.
TRACED	C.E.
CHECKED	
APPROVED	

EARTHING CLAMP

ISSUE	MIDLANDS ELECTRICITY BOARD
C.E.	DRG. No.
Material changed 6.6.73 C.E.	M.E. 193.

ITEM	QTY	DESCRIPTION	MATERIALS
1	1	LUG	ALUMINIUM ALLOY TO BS2898 EIE-M
2	1	CONDUCTOR	STRANDED ALUMINIUM ALLOY, 120mm ² NOMINAL ALUMINIUM AREA.
3	1	SHEATH	NEOPRENE OR BLACK P.V.C.
4	1	LUG	ALUMINIUM ALLOY TO BS2898 EIE-M
5	1	M16 x 45 SET SCREW	MILD STEEL TO BS4190 - 4.6. GALVANISED TO BS1461 & NGTS 3.4.6
6	1	M16 HEX NUT	MILD STEEL TO BS4190 - 4 & GALVANISED TO BS1461 & NGTS 3.4.6
7	2	M16 FLAT WASHERS	MILD STEEL TO BS4320 - A GALVANISED TO BS1461 & NGTS 3.4.6
8	1	M12 x 30 SET SCREW	MILD STEEL TO BS4190 - 4.6 GALVANISED TO BS1461 & NGTS 3.4.6
9	1	M12 HEX NUT	MILD STEEL TO BS4190 - 4.6 GALVANISED TO BS1461 & NGTS 3.4.6
10	2	M12 FLAT WASHERS	MILD STEEL TO BS4190 - 4.6 GALVANISED TO BS1461 & NGTS 3.4.6



NOTES :

- 1). TO COMPLY WITH NGTS 3.4.6
- 2). MAX RESISTANCE VALUE - 120 MICROHMS
- 3). TYPE REF. MUST BE STAMPED ONTO LUGS.
- 4). END OF CONDUCTOR TO BE DIPPED IN APPROVED CONDUCTOR GREASE AFTER TRIMMING SHEATH BUT BEFORE COMPRESSION OF LUGS.
- 3). BOLT & NUTS TO BE LIGHTLY GREASED.

SCALE	-
DATE	6.3.01
DRAWN	G.B.
TRACED	
CHECKED	
APPROVED	

STEEL TOWER EARTH CONDUCTOR TO STEELWORK BOND
450MM/14/18MM DIAMETER HOLES/TYPE 2

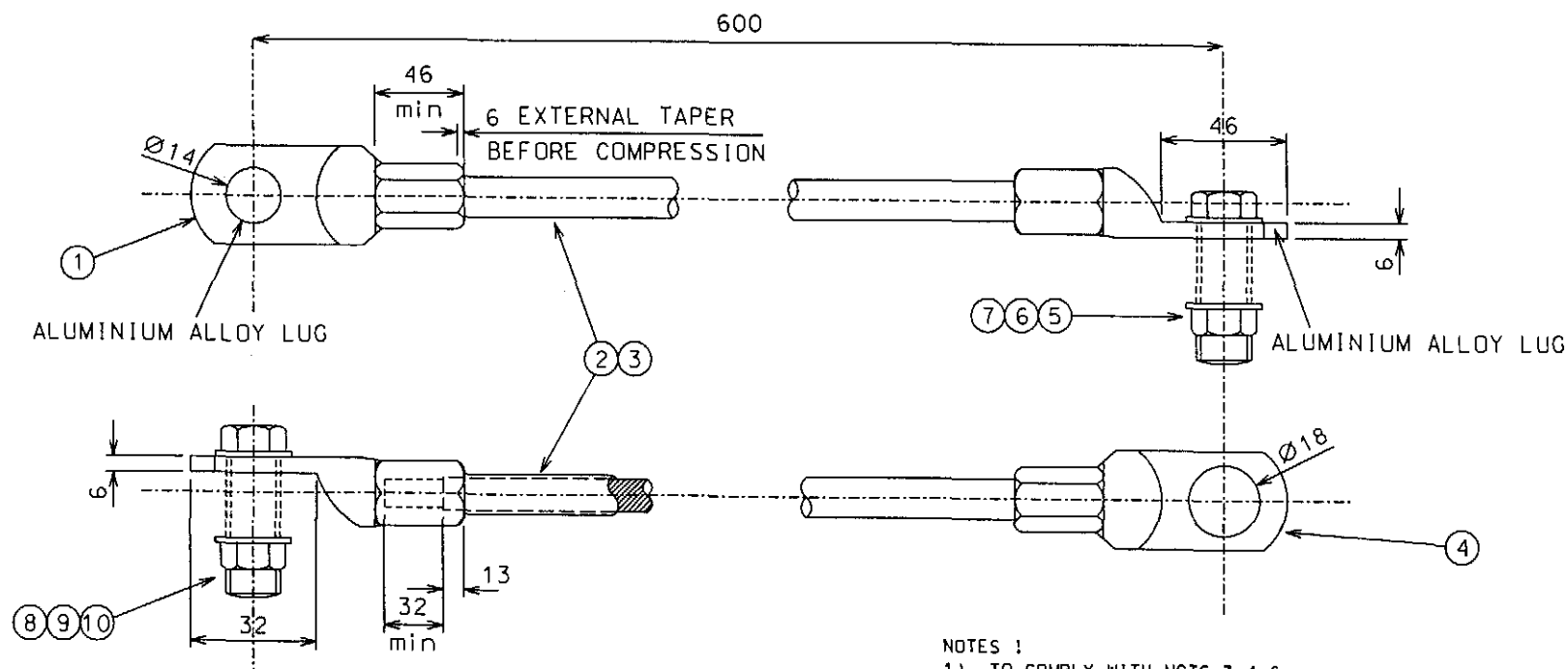
ISSUE	DATE	ISSUE	DATE	ISSUE	DATE

Produced by Output Services
GPU POWER UK
ORG. No. M225

ORG. No.

M225

A4



- 1). TO COMPLY WITH NGTS 3.4.6
- 2). MAX RESISTANCE VALUE - 160 MICROHMS
- 3). TYPE REF. MUST BE STAMPED ONTO LUGS.
- 4). END OF CONDUCTOR TO BE DIPPED IN APPROVED CONDUCTOR GREASE AFTER TRIMMING SHEATH BUT BEFORE COMPRESSION OF LUGS.
- 3). BOLT & NUTS TO BE LIGHTLY GREASED.

SCALE	-
DATE	06.03.01
DRAWN	D.B.
TRACED	
CHECKED	
APPROVED	

STEEL TOWER EARTH CONDUCTOR TO
STEELWORK BOND
600MM/14/18MM DIAMETER HOLES/TYPE 3

ISSUE	DATE	ISSUE	DATE	ISSUE	DATE

Produced by
Output Services

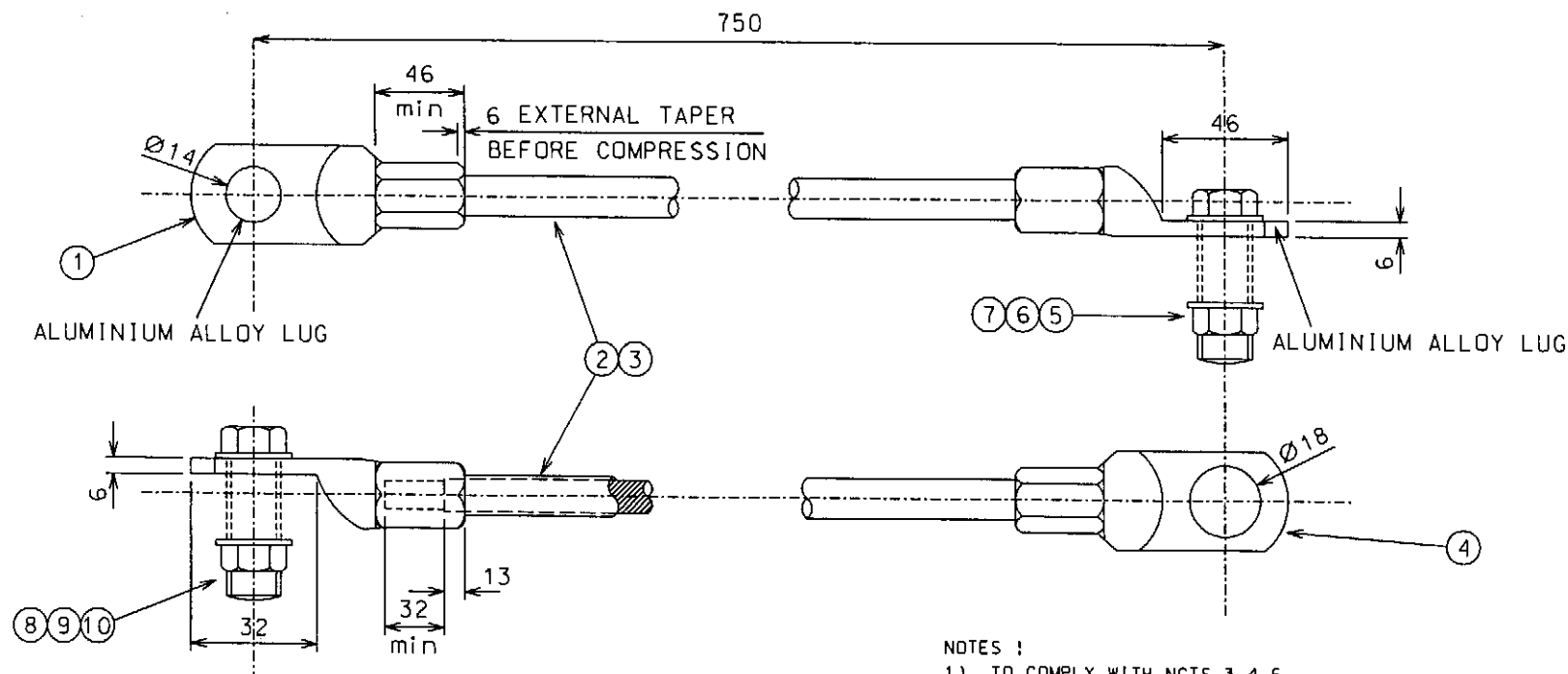


GPU
POWER UP

ORG. No.

M225-1

ITEM	QTY	DESCRIPTION	MATERIALS
1	1	LUG	ALUMINIUM ALLOY TO BS2898 EIE-M
2	1	CONDUCTOR	STRANDED ALUMINIUM ALLOY, 120mm ² NOMINAL ALUMINIUM AREA.
3	1	SHEATH	NEOPRENE OR BLACK P.V.C.
4	1	LUG	ALUMINIUM ALLOY TO BS2898 EIE-M
5	1	M16 x 45 SET SCREW	MILD STEEL TO BS4190 - 4.6. GALVANISED TO BS1461 & NGTS 3.4.6
6	1	M16 HEX NUT	MILD STEEL TO BS4190 - 4 & GALVANISED TO BS1461 & NGTS 3.4.6
7	2	M16 FLAT WASHERS	MILD STEEL TO BS4320 - A GALVANISED TO BS1461 & NGTS 3.4.6
8	1	M12 x 30 SET SCREW	MILD STEEL TO BS4190 - 4.6 GALVANISED TO BS1461 & NGTS 3.4.6
9	1	M12 HEX NUT	MILD STEEL TO BS4190 - 4.6 GALVANISED TO BS1461 & NGTS 3.4.6
10	2	M12 FLAT WASHERS	MILD STEEL TO BS4190 - 4.6 GALVANISED TO BS1461 & NGTS 3.4.6



NOTES :

- 1). TO COMPLY WITH NGTS 3.4.6
- 2). MAX RESISTANCE VALUE - 200 MICROHMS
- 3). TYPE REF. MUST BE STAMPED ONTO LUGS.
- 4). END OF CONDUCTOR TO BE DIPPED IN APPROVED CONDUCTOR GREASE AFTER TRIMMING SHEATH BUT BEFORE COMPRESSION OF LUGS.
- 5). BOLT & NUTS TO BE LIGHTLY GREASED.

SCALE	-
DATE	06.03.01
DRAWN	D.B.
TRACED	
CHECKED	
APPROVED	

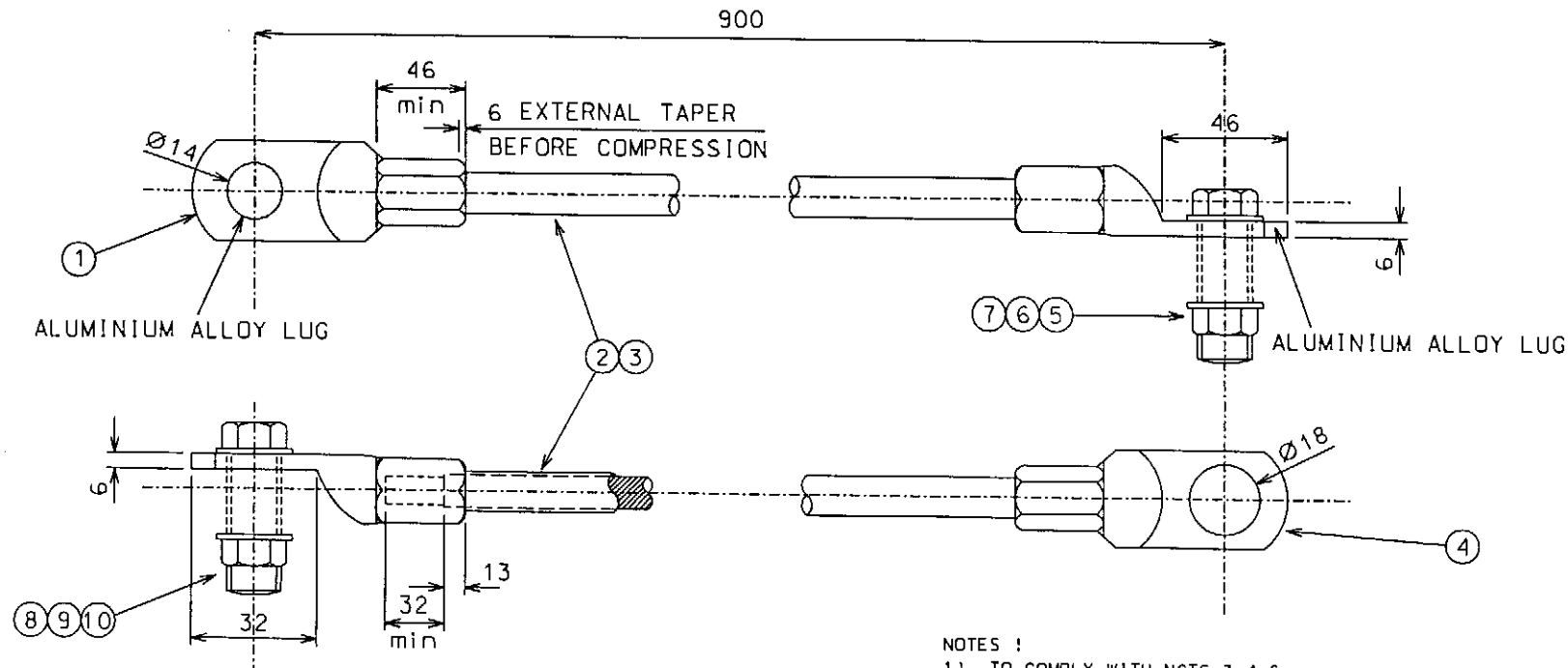
STEEL TOWER EARTH CONDUCTOR TO
STEELWORK BOND
750MM/14/18MM DIAMETER HOLES TYPE 4

ISSUE	DATE	ISSUE	DATE	ISSUE	DATE

Produced by Output Services	
DRG. No.	M225-2

DRG. No. M225-2

ITEM	QTY	DESCRIPTION	MATERIALS
1	1	LUG	ALUMINIUM ALLOY TO BS2898 EIE-M
2	1	CONDUCTOR	STRANDED ALUMINIUM ALLOY, 120mm ² NOMINAL ALUMINIUM AREA.
3	1	SHEATH	NEOPRENE OR BLACK P.V.C.
4	1	LUG	ALUMINIUM ALLOY TO BS2898 EIE-M
5	1	M16 x 45 SET SCREW	MILD STEEL TO BS4190 - 4.6. GALVANISED TO BS1461 & NGTS 3.4.6
6	1	M16 HEX NUT	MILD STEEL TO BS4190 - 4 & GALVANISED TO BS1461 & NGTS 3.4.6
7	2	M16 FLAT WASHERS	MILD STEEL TO BS4320 - A GALVANISED TO BS1461 & NGTS 3.4.6
8	1	M12 x 30 SET SCREW	MILD STEEL TO BS4190 - 4.6 GALVANISED TO BS1461 & NGTS 3.4.6
9	1	M12 HEX NUT	MILD STEEL TO BS4190 - 4.6 GALVANISED TO BS1461 & NGTS 3.4.6
10	2	M12 FLAT WASHERS	MILD STEEL TO BS4190 - 4.6 GALVANISED TO BS1461 & NGTS 3.4.6



NOTES :

- 1). TO COMPLY WITH NGTS 3.4.6
- 2). MAX RESISTANCE VALUE - 240 MICROHMS
- 3). TYPE REF. MUST BE STAMPED ONTO LUGS.
- 4). END OF CONDUCTOR TO BE DIPPED IN APPROVED CONDUCTOR GREASE AFTER TRIMMING SHEATH BUT BEFORE COMPRESSION OF LUGS.
- 3). BOLT & NUTS TO BE LIGHTLY GREASED.

SCALE	-
DATE	06-03-01
DRAWN	D.B.
TRACED	
CHECKED	
APPROVED	

STEEL TOWER EARTH CONDUCTOR TO
STEELWORK BOND
900MM/14/18MM DIAMETER HOLES TYPE 5

ISSUE	DATE	ISSUE	DATE	ISSUE	DATE

Produced by
Output Services

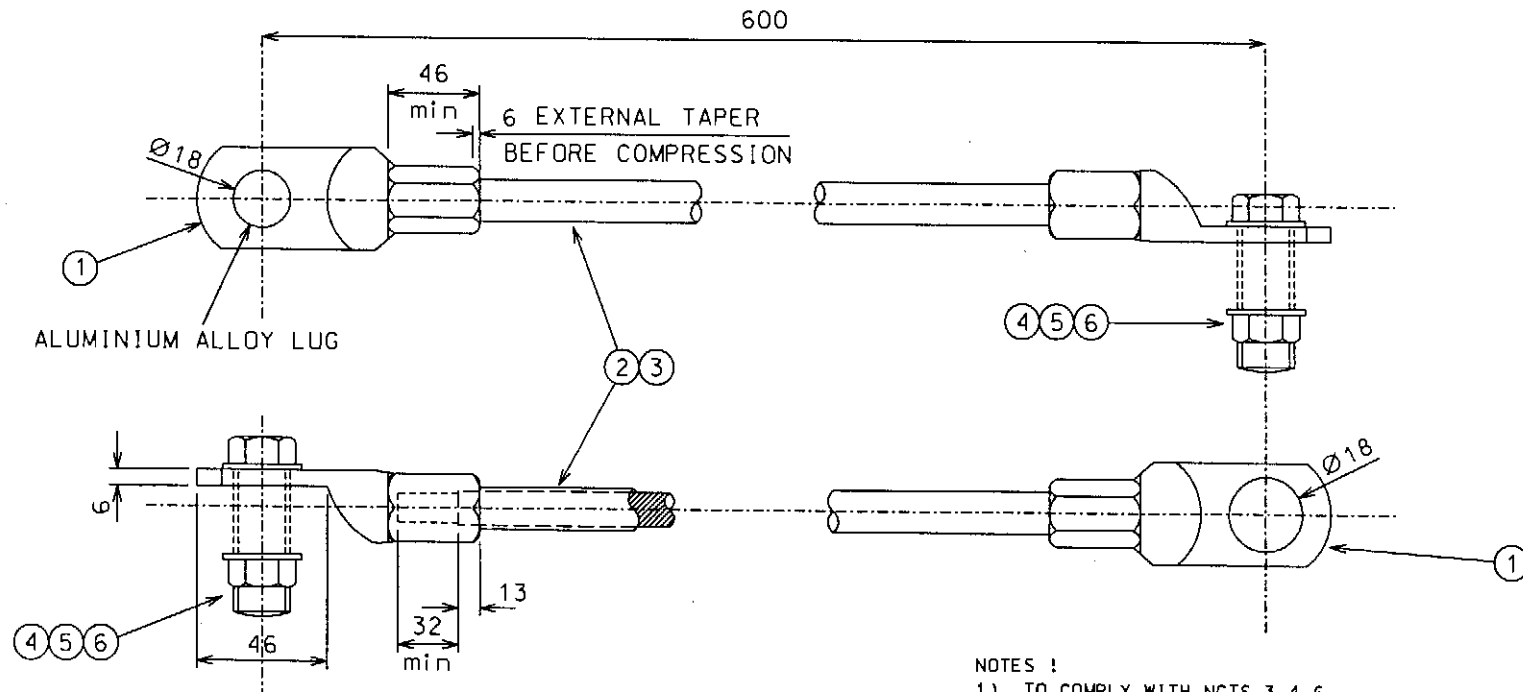
GPU
POWER UK

DRG. No.

M225-3

DRG. No. M225-3

ITEM	QTY	DESCRIPTION	MATERIALS
1	2	LUG	ALUMINIUM ALLOY TO BS2898 E1E-M
2	1	CONDUCTOR	STRANDED ALUMINIUM ALLOY, 120mm ² NOMINAL ALUMINIUM AREA.
3	1	SHEATH	NEOPRENE OR BLACK P.V.C.
4	2	M16 x 45 SET SCREW	MILD STEEL TO BS4190 - 4.6. GALVANISED TO BS1461 & NGTS 3.4.6
5	2	M16 HEX NUT	MILD STEEL TO BS4190 - 4 & GALVANISED TO BS1461 & NGTS 3.4.6
6	4	M16 FLAT WASHERS	MILD STEEL TO BS4320 - A GALVANISED TO BS1461 & NGTS 3.4.6



NOTES !

- 1). TO COMPLY WITH NGTS 3.4.6
- 2). MAX RESISTANCE VALUE - 160 MICROHMS
- 3). TYPE REF. MUST BE STAMPED ONTO LUGS.
- 4). END OF CONDUCTOR TO BE DIPPED IN APPROVED CONDUCTOR GREASE AFTER TRIMMING SHEATH BUT BEFORE COMPRESSION OF LUGS.
- 3). BOLT & NUTS TO BE LIGHTLY GREASED.

SCALE	-
DATE	06.03.01
DRAWN	D.B.
TRACED	
CHECKED	
APPROVED	

STEEL TOWER EARTH CONDUCTOR TO
STEELWORK BOND
600MM/18MM DIAMETER HOLES TYPE 8

ISSUE	DATE	ISSUE	DATE	ISSUE	DATE

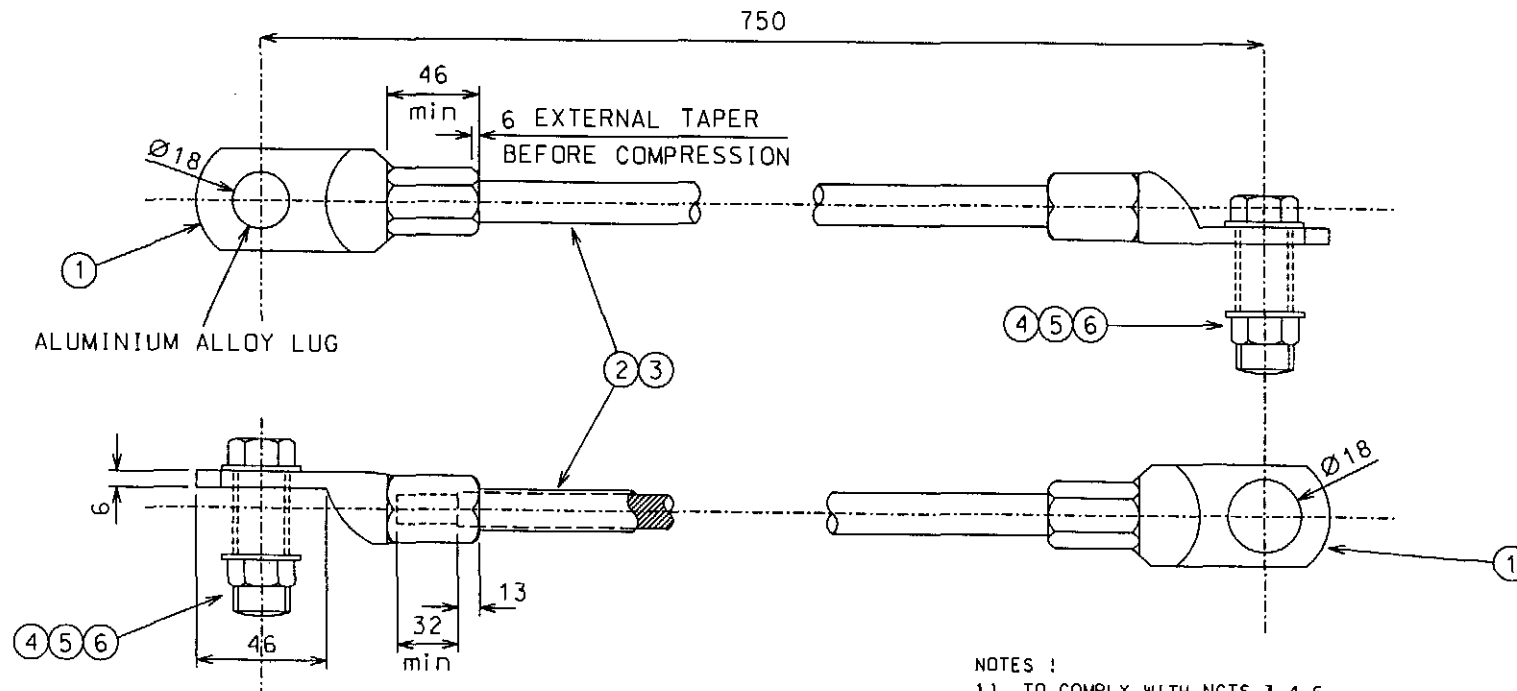
Produced by Output Services	
DRG. No.	M225-5

DRG. No.

M225-5

A4

ITEM	QTY	DESCRIPTION	MATERIALS
1	2	LUG	ALUMINIUM ALLOY TO BS2898 EIE-M
2	1	CONDUCTOR	STRANDED ALUMINIUM ALLOY, 120mm ² NOMINAL ALUMINIUM AREA.
3	1	SHEATH	NEOPRENE OR BLACK P.V.C.
4	2	M16 x 45 SET SCREW	MILD STEEL TO BS4190 - 4.6. GALVANISED TO BS1461 & NGTS 3.4.6
5	2	M16 HEX NUT	MILD STEEL TO BS4190 - 4 & GALVANISED TO BS1461 & NGTS 3.4.6
6	4	M16 FLAT WASHERS	MILD STEEL TO BS4320 - A GALVANISED TO BS1461 & NGTS 3.4.6



NOTES :

- 1). TO COMPLY WITH NGTS 3.4.6
- 2). MAX RESISTANCE VALUE - 200 MICROHMS
- 3). TYPE REF. MUST BE STAMPED ONTO LUGS.
- 4). END OF CONDUCTOR TO BE DIPPED IN APPROVED CONDUCTOR GREASE AFTER TRIMMING SHEATH BUT BEFORE COMPRESSION OF LUGS.
- 5). BOLT & NUTS TO BE LIGHTLY GREASED.

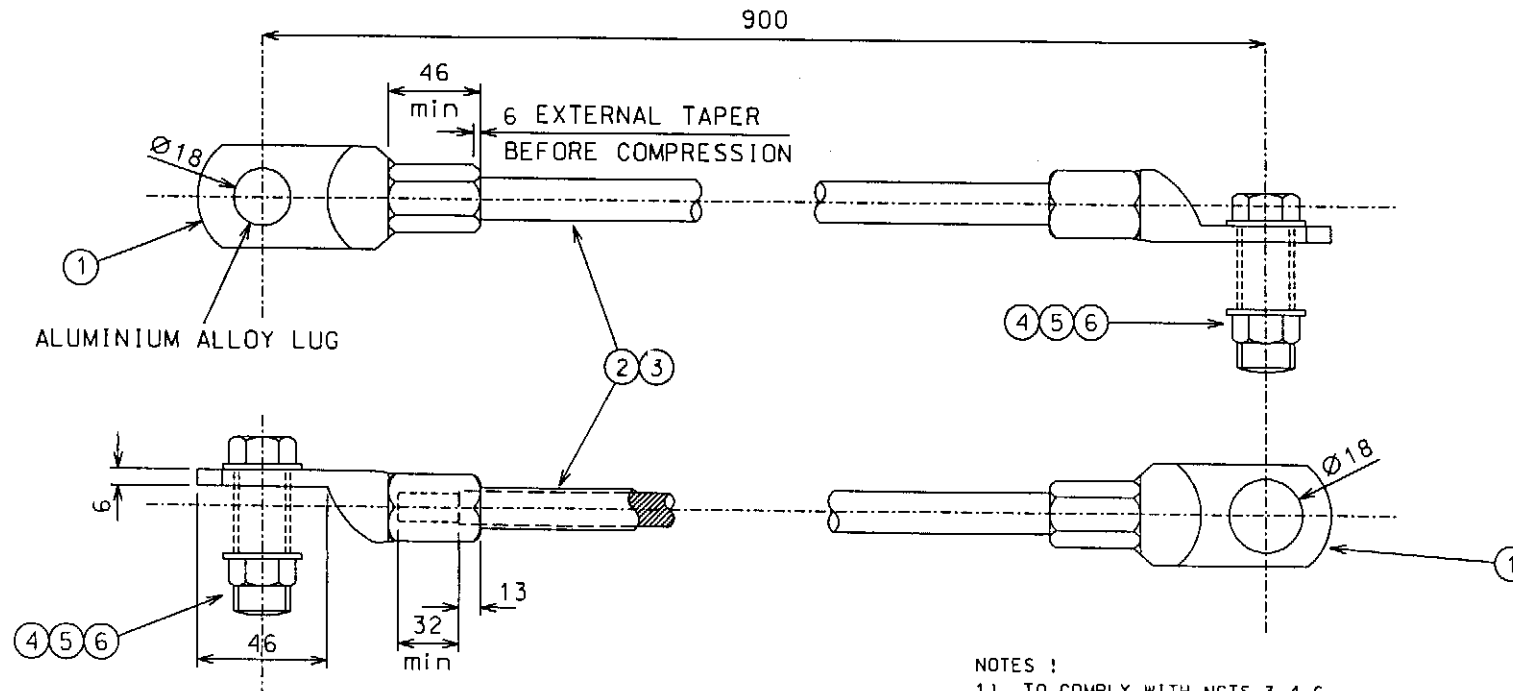
SCALE	—
DATE	06.03.01
DRAWN	D.B.
TRACED	
CHECKED	
APPROVED	

STEEL TOWER EARTH CONDUCTOR TO
STEELWORK BOND
750MM/18MM DIAMETER HOLES TYPE 9

ISSUE	DATE	ISSUE	DATE	ISSUE	DATE

Produced by Output Services	
DRG. No.	M225-6

ITEM	QTY	DESCRIPTION	MATERIALS
1	2	LUG	ALUMINIUM ALLOY TO BS2898 ETE-M
2	1	CONDUCTOR	STRANDED ALUMINIUM ALLOY. 120mm ² NOMINAL ALUMINIUM AREA.
3	1	SHEATH	NEOPRENE OR BLACK P.V.C.
4	2	M16 x 45 SET SCREW	MILD STEEL TO BS4190 - 4.6. GALVANISED TO BS1461 & NGTS 3.4.6
5	2	M16 HEX NUT	MILD STEEL TO BS4190 - 4 & GALVANISED TO BS1461 & NGTS 3.4.6
6	4	M16 FLAT WASHERS	MILD STEEL TO BS4320 - A GALVANISED TO BS1461 & NGTS 3.4.6



NOTES :

- 1). TO COMPLY WITH NGTS 3.4.6
- 2). MAX RESISTANCE VALUE - 240 MICROHMS
- 3). TYPE REF. MUST BE STAMPED ONTO LUGS.
- 4). END OF CONDUCTOR TO BE DIPPED IN APPROVED CONDUCTOR GREASE AFTER TRIMMING SHEATH BUT BEFORE COMPRESSION OF LUGS.
- 3). BOLT & NUTS TO BE LIGHTLY GREASED.

SCALE	—
DATE	06.03.01
DRAWN	D.B.
TRACED	
CHECKED	
APPROVED	

STEEL TOWER EARTH CONDUCTOR TO
STEELWORK BOND
900MM/18MM DIAMETER HOLES TYPE 10

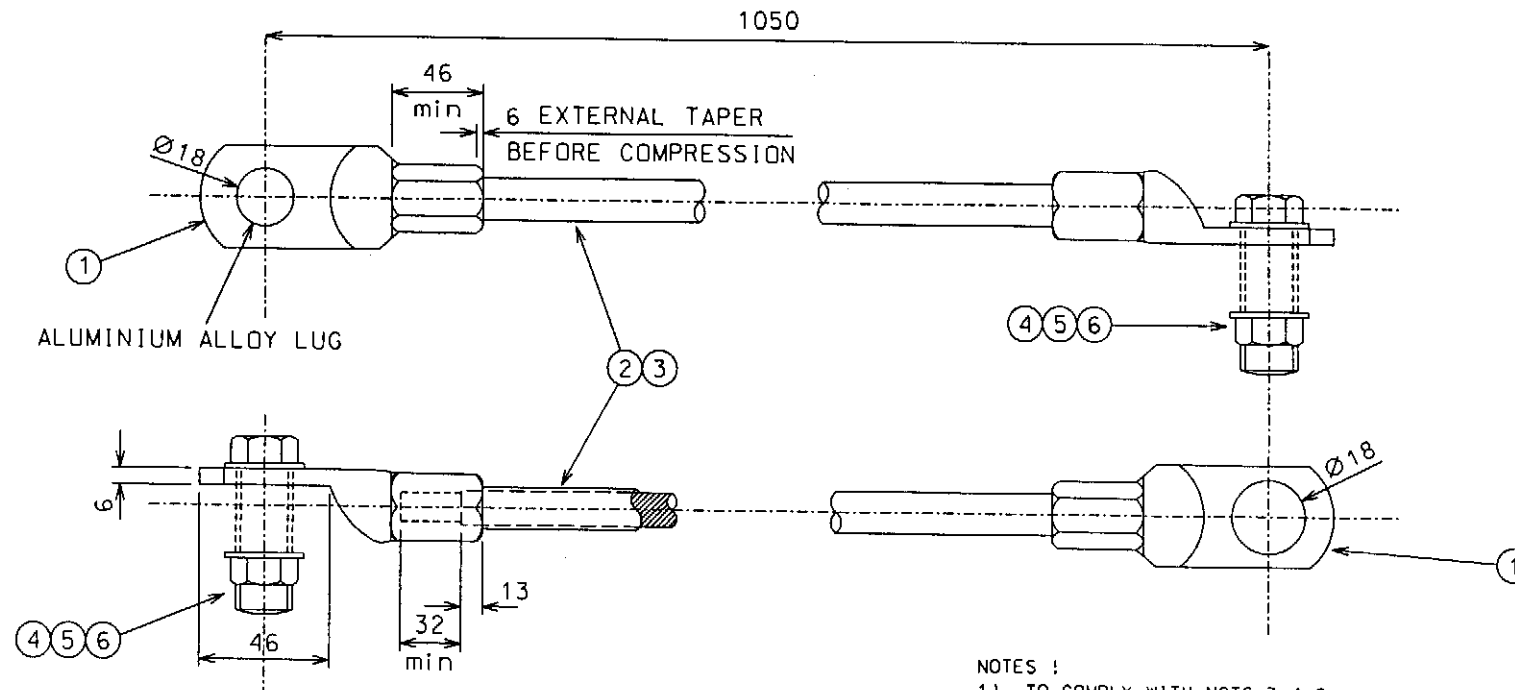
ISSUE	DATE	ISSUE	DATE	ISSUE	DATE

Produced by Output Services	
DRG. No.	M225-7

DRG. No.

M225-7

ITEM	QTY	DESCRIPTION	MATERIALS
1	2	LUG	ALUMINIUM ALLOY TO BS2898 EIE-M
2	1	CONDUCTOR	STRANDED ALUMINIUM ALLOY, 120mm ² NOMINAL ALUMINIUM AREA.
3	1	SHEATH	NEOPRENE OR BLACK P.V.C.
4	2	M16 x 45 SET SCREW	MILD STEEL TO BS4190 - 4.6. GALVANISED TO BS1461 & NGTS 3.4.6
5	2	M16 HEX NUT	MILD STEEL TO BS4190 - 4 & GALVANISED TO BS1461 & NGTS 3.4.6
6	4	M16 FLAT WASHERS	MILD STEEL TO BS4320 - A GALVANISED TO BS1461 & NGTS 3.4.6



NOTES !

- 1). TO COMPLY WITH NGTS 3.4.6
- 2). MAX RESISTANCE VALUE - 280 MICROHMS
- 3). TYPE REF. MUST BE STAMPED ONTO LUGS.
- 4). END OF CONDUCTOR TO BE DIPPED IN APPROVED CONDUCTOR GREASE AFTER TRIMMING SHEATH BUT BEFORE COMPRESSION OF LUGS.
- 5). BOLT & NUTS TO BE LIGHTLY GREASED.

SCALE	-
DATE	06.03.01
DRAWN	D.B.
TRACED	
CHECKED	
APPROVED	

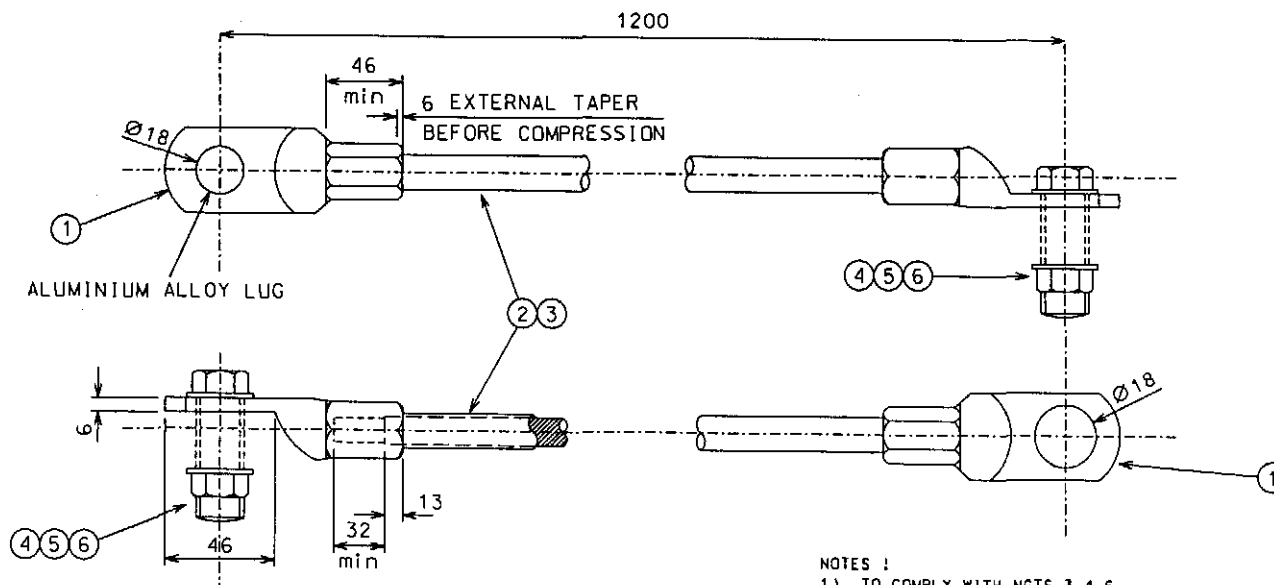
STEEL TOWER EARTH CONDUCTOR TO
STEELWORK BOND
1050MM/18MM DIAMETER HOLES TYPE 11

ISSUE	DATE	ISSUE	DATE	ISSUE	DATE

Produced by Output Services	
DRG. No.	M225-8

DRG. No. M225-8

ITEM	QTY	DESCRIPTION	MATERIALS
1	2	LUG	ALUMINIUM ALLOY TO BS2898 EIE-M
2	1	CONDUCTOR	STRANDED ALUMINIUM ALLOY, 120mm ² NOMINAL ALUMINIUM AREA.
3	1	SHEATH	NEOPRENE OR BLACK P.V.C.
4	2	M16 x 45 SET SCREW	MILD STEEL TO BS4190 - 4.6. GALVANISED TO BS1461 & NGTS 3.4.6
5	2	M16 HEX NUT	MILD STEEL TO BS4190 - 4 & GALVANISED TO BS1461 & NGTS 3.4.6
6	4	M16 FLAT WASHERS	MILD STEEL TO BS4320 - A GALVANISED TO BS1461 & NGTS 3.4.6



NOTES :

- 1). TO COMPLY WITH NGTS 3.4.6
- 2). MAX RESISTANCE VALUE - 320 MICROHMS
- 3). TYPE REF. MUST BE STAMPED ONTO LUGS.
- 4). END OF CONDUCTOR TO BE DIPPED IN APPROVED CONDUCTOR GREASE AFTER TRIMMING SHEATH BUT BEFORE COMPRESSION OF LUGS.
- 5). BOLT & NUTS TO BE LIGHTLY GREASED.

SCALE	---
DATE	06.03.01
DRAWN	D.B.
TRACED	
CHECKED	
APPROVED	

STEEL TOWER EARTH CONDUCTOR TO
STEELWORK BOND
1200MM/18MM DIAMETER HOLES TYPE 12

ISSUE	DATE	ISSUE	DATE	ISSUE	DATE

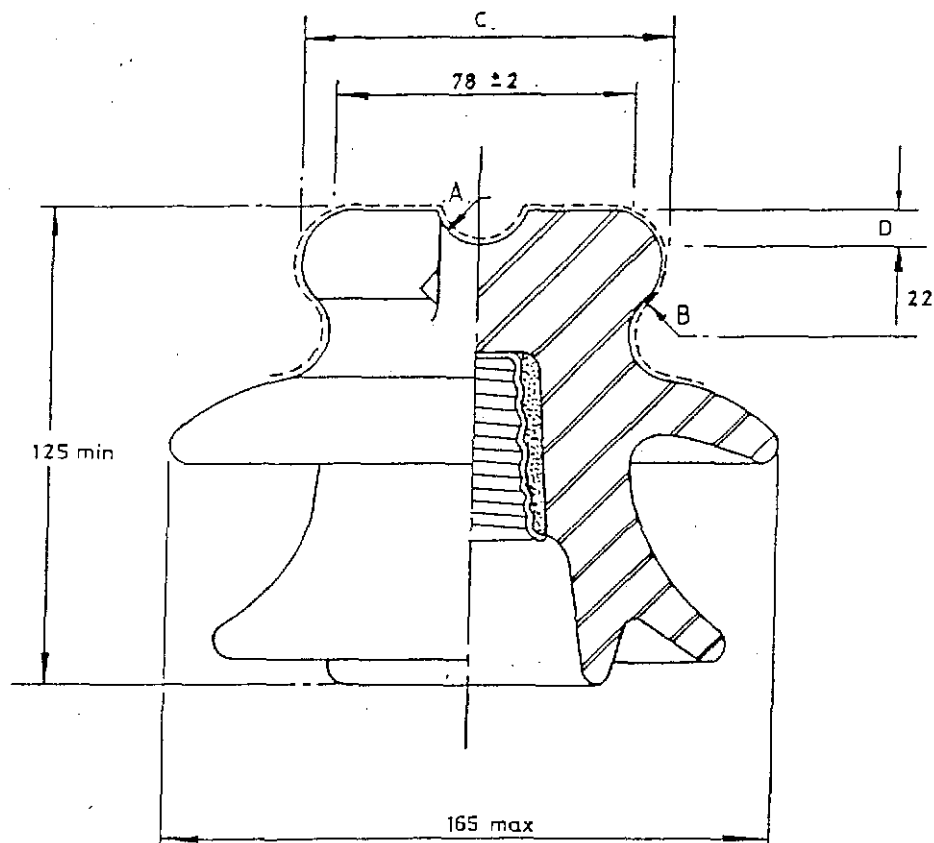
Produced by Output Services
GPU POWER UK
DRG. No. M225-9

DRG. No. M225-9

A4

SECTION 14

MIN NUMBER	M	DESCRIPTION
3450640	64	11kV PIN INSULATOR
3450645	64/5	11kV PIN INSULATOR LONG CREEPAGE
3450850	85	DISC INSULATOR M.F.L 70kN
3451320	132	33kV PIN INSULATOR
3451651	165/1	33kV LINE POST LAPP INSULATOR
3451652	165/2	33kV LINE POST POLYMER INSULATOR
3451680	168	66kV LINE POST LAPP INSULATOR
3452541	254/1	125kN DISC INSULATOR 146 CENTRES
3453010	301	80kN HIGH IMPACT DISC
3453091	309/1	125kN DISC INSULATOR PORCELAIN
3453092	309/2	125kN DISC INSULATOR GLASS
3453130	313	11kV PIN INSULATOR HIGH IMPACT
3454171	417/1	11kV COMPOSITE INSULATOR PIG-TAIL HOOK 20kN LIGHT CONSTRUCTION
3454172	417/2	11kV COMPOSITE INSULATOR SOCKET & BALL 70kN
3454180	418	DISC INSULATOR M.F.L 80kN 132kV SUSPENSION
3454190	419	DISC INSULATOR M.F.L 120kN 132kV SUSPENSION
3454200	420	DISC INSULATOR M.F.L 190kN 132kV SUSPENSION - 132kV TOWER LINES TWIN 400 mm ² A.C.S.R CONDUCTORS
3454240	424	L.V SUPPORT INSULATOR
3454241	424/1	L.V BRACKET INSULATOR
3455400	540	INSULATOR SECURITY CLIP



1. Dimensions and characteristics to comply with B.S.137 for a nominal system voltage (3 phase) of 12 k.V. British Practice

2. Tolerances except where specified on drawings are to comply with B.S. 137.

3. Thimble to be fitted suitable for large steel head Fig 1 B.S. 3288 Part 2 and should have a minimum tin content of 4.8% by weight.

4. Insulator to be indelibly marked in accordance with B.S.137.

Semi Conducting glazed head.

Porcelain to be glazed Brown.

7. The firing date (month and year) to be indelibly marked on insulator also makers identification mark.

Ref.	Rad. A	Rad. B	C	D
064/1	25	14	102	16

COUPLING

WEIGHT /UNIT 2.3 kg

MINIMUM FAILING LOAD (CANT) 1089 kgf

MAXIMUM WORKING LOAD (CANT) 545 kgf

CREPAGE LENGTH - SURFACE 225 mm

" " PROTECTED 100 mm

ELECTROSTATIC CAPACITANCE

FLASHOVER VOLTAGE

D.F.O. NORMAL 80 kV

" 1 MIN 70 kV

W.F.O. NORMAL 53 kV

" 1 MIN 50 kV

I.F.O. 50% 1/50 μ s. 105 kV

" " " " 135 kV

IMPULSE WITHSTAND 50 Hz 95 kV

PUNCTURE VOLTAGE 50 Hz 130 kV

B.S. REFERENCE B.S.137

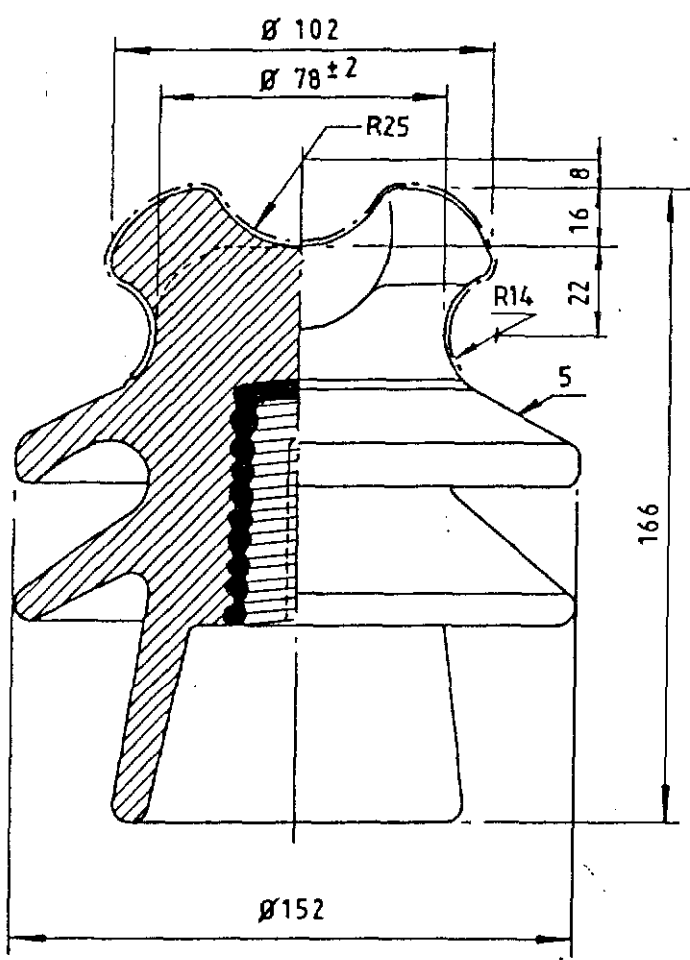
CHARACTERISTIC TABLE FOR INSULATORS

MINIMUM REQUIREMENTS.

SCALE	-	11 kV PIN INSULATOR	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	21.11.75		C	
DRAWN	A.R.B.		Note 3 & I.F.O. mods.	
TESTED	L.H.		9-8-93AJ.B.	
CHECKED	PSK			
APPROVED		E.S.I. STANDARD 43/93		DRG. No. M.64. ESJ. DRG N: 439301

MEMO

(MF)



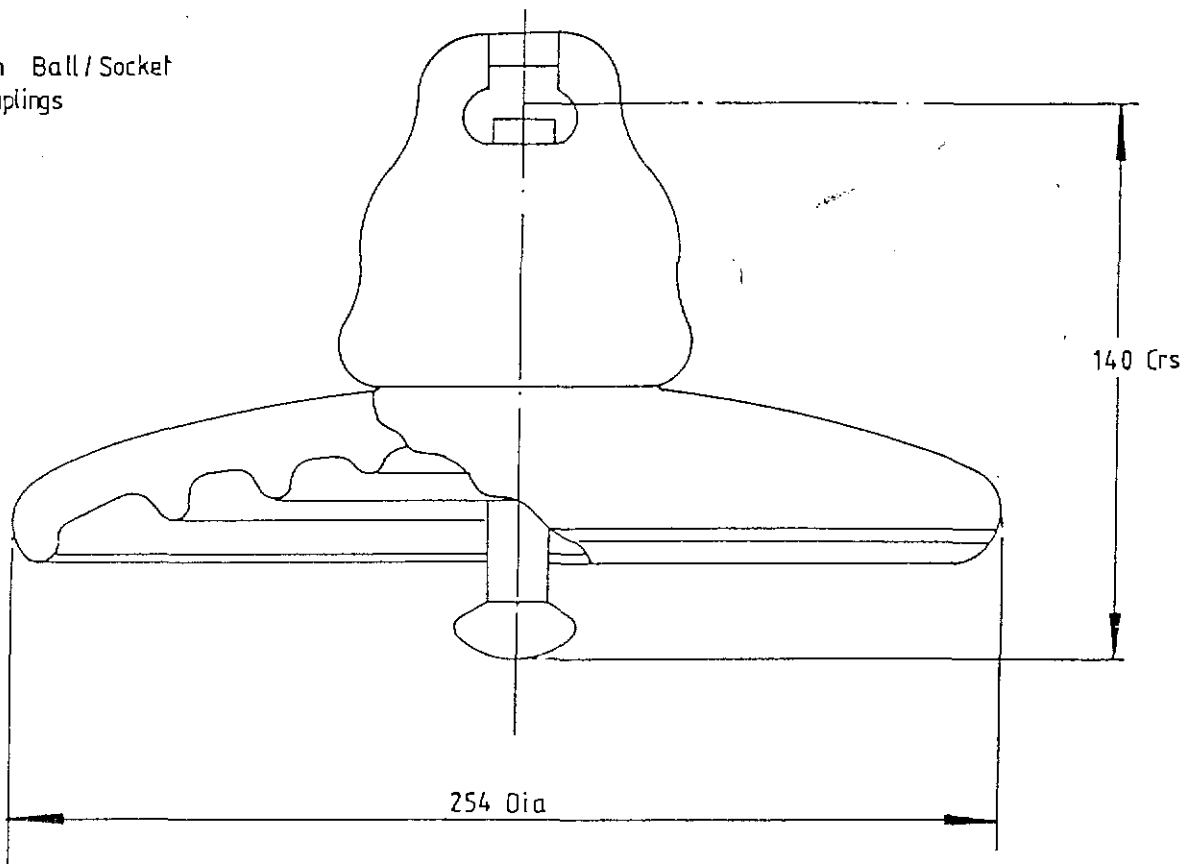
- 1. Dimensions and characteristics to comply with B.S. 137 for a nominal system voltage (3 phase) of 12kV. British Practice.
- 2. Tolerances except where specified on drawings are to comply with B.S.137.
- 3. Lead thimble to be fitted suitable for large steel head Fig.1 B.S. 3288 Part 2 and should have a minimum tin content of 4.8% by weight.
- 4. Insulator to be indelibly marked in accordance with B.S.137.
- 5. Semi Conducting glazed head.
- 6. Porcelain to be glazed Brown.
- 7. The firing date (month and year) to be indelibly marked on insulator also makers identification mark.

COUPLING	-
WEIGHT / UNIT	2.3kg.
MINIMUM FAILING LOAD (CANT)	1089kgf.
MAXIMUM WORKING LOAD (CANT)	545kgf.
CREPAGE LENGTH - SURFACE	300 mm
" " - PROTECTED	160 mm
ELECTROSTATIC CAPACITANCE	
FLASHOVER VOLTAGE	
D.F.O. NORMAL	80kV.
" 1 MIN.	75kV.
W.F.O. NORMAL	55kV.
" 1 MIN.	50kV.
I.F.O. 50% 1/50 μ s +	120 kV.
	150 kV.
IMPULSE WITHSTAND 50Hz	110 kV.
PUNCTURE VOLTAGE 50Hz	130kV.
B.S. REFERENCE	BS137

CHARACTERISTIC TABLE FOR INSULATORS
MINIMUM REQUIRMENTS

ADDED BY DATE	TITLE	11kV PIN INSULATOR (LONG CREEPAGE)	SCALE	—	M.E.B. Midlands Electricity plc M. 64/5
			DATE	6-10-92	
			DRAWN		
			TRACED	C. E.	
			CHECKED		
			APPROVED		

16 mm Ball / Socket
couplings



REF.	TYPE	WEIGHT / UNIT
85 / 1	PORCELAIN	6.5 Kg.
85 / 2	GLASS	3.5 Kg.

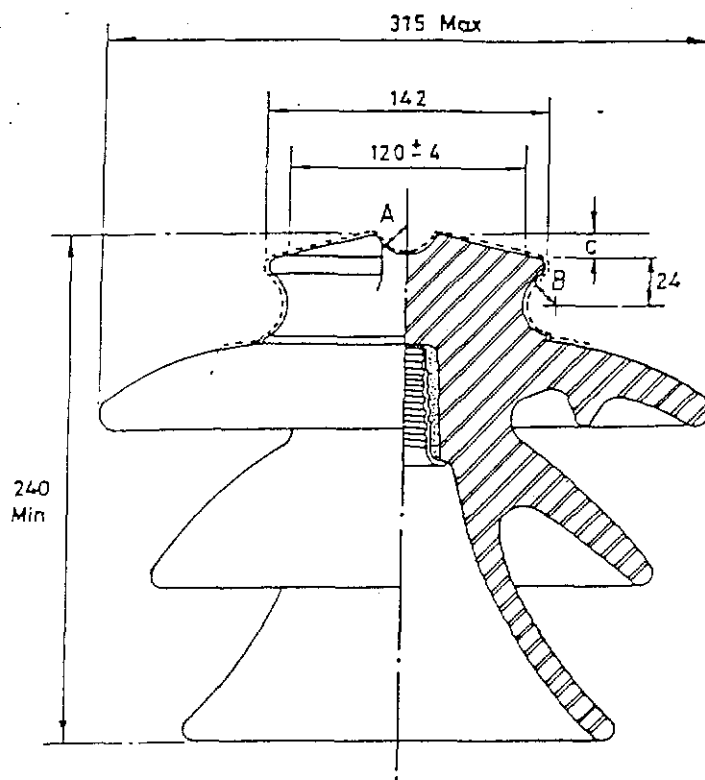
APPLICATION SUSPENSION - Wood pole / Tower lines
up to 66kV.
TENSION - Wood pole / Tower lines
up to 66kV. with
conductor M.W.L. up to
22.3kN (5000 lbf)

WEIGHT / UNIT	kg.
MINIMUM FAILING LOAD	70 kN
MAXIMUM WORKING LOAD	-
CREEPAGE LENGTH - SURFACE	298mm
" " - PROTECTED	180mm
ELECTROSTATIC CAPACITANCE	-
FLASHOVER VOLTAGE	-
D.F.O. NORMAL	74kV.
" 1 MIN.	70kV.
W.F.O. NORMAL	45kV.
" 1 MIN.	40kV.
I.F.O. 50% 1/50 μ s +	105kV.
" " " "	110kV.
IMPULSE WITHSTAND 50Hz	110kV.
PUNCTURE VOLTAGE 50Hz	125kV.
B.S. REFERENCE	B.S. 137

1. Dimensions and characteristics to comply with B.S. 137.
2. Tolerances except where specified on drawing are to comply with B.S. 137.
3. Insulator to be indelibly marked in accordance with B.S. 137.
4. Porcelain to be glazed brown.
5. See M300 for High Impact type.

CHARACTERISTIC TABLE FOR INSULATORS
MINIMUM REQUIREMENTS

SCALE	N.T.S.	DISC INSULATOR M.E.L 70kN.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	13-2-84		B	
DRAWN	R.A.C.		Tables added.	DRG. No. M. 85
TRACED				
CHECKED			7.10.92 A.J.B.	
APPROVED				
			E.S.I. STANDARD 43 - 93	E.S.I. DRG N°439306



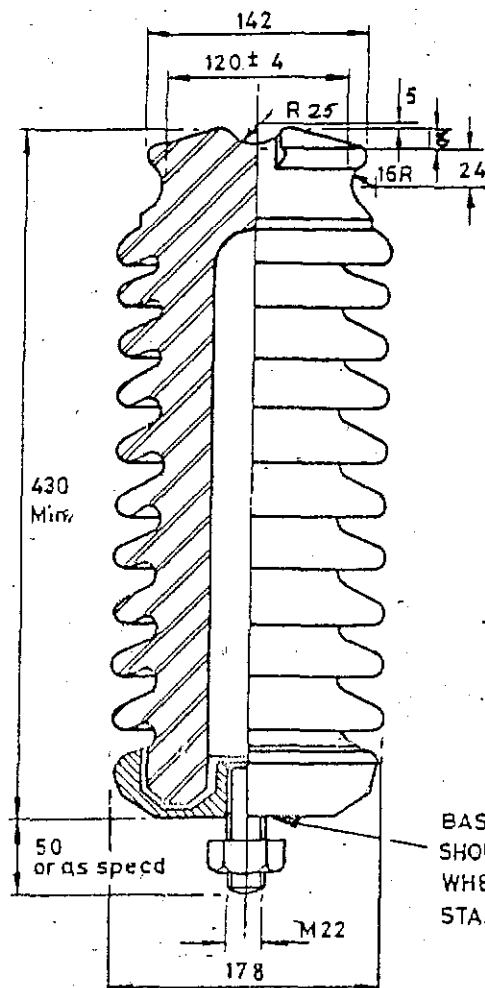
1. Dimensions and characteristics to comply with B.S.137 for a nominal system voltage (3 phase) of 36 kV British practice.
2. Tolerances except where specified on drawing are to comply with B.S.137
3. Thimble to be fitted suitable for large steel head Fig.1 B.S.3288 Part 2 and should have a minimum tin content of 4.8 % by weight.
4. Insulator to be indelibly marked in accordance with B.S.137
5. Semi Conducting glazed head.
6. Porcelain to be glazed Brown
7. The firing date (month and year) to be indelibly marked on insulator also makers identification mark.

Ref.	Rad. A	Rad. B	C
132/1	25	16	16

COUPLING	
WEIGHT /UNIT	kg
MINIMUM FAILING LOAD (CANT)	1089 kgf
MAXIMUM WORKING LOAD (CANT)	545 kgf
CREPAGE LENGTH - SURFACE	737 mm
" " PROTECTED	268 mm
ELECTROSTATIC CAPACITANCE	
FLASHOVER VOLTAGE	
O.F.O. NORMAL	140 kV
" 1 MIN	130 kV
W.F.O. NORMAL	105 kV
" 1 MIN	90 kV
I.F.D. 50% 1/50 μ s +	230 kV
" " " " "	280 kV
IMPULSE WITHSTAND 50Hz	200 kV
PUNCTURE VOLTAGE 50 Hz	230 kV
B.S. REFERENCE	B.S.137

CHARACTERISTIC TABLE FOR INSULATORS
MINIMUM REQUIREMENTS.

SCALE	-	33 kV PIN INSULATOR	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	25.10.82		C	
DRAWN	A.R.B.		Note 3 & D.F.O. mods	DRG. No.
CHECKED	L.F.M.		9-8-93 AJB	M.132.
APPROVED				
E.S.I. STANDARD		43/93	E.S.I. DRG. 439302	



1. Dimensions and characteristics to comply with B.S. 137, for a nominal system voltage (3 phase) of 36 kV British Practice
2. Tolerances except where specified on drawing are to comply with B.S. 137
3. Insulator to be indelibly marked in accordance with B.S. 137
4. Porcelain to be glazed brown.
5. This drawing illustrates a hollow core insulator, but a solid core insulator with the same performance is an acceptable alternative
6. In localities not subject to salt or industrial deposits, the pin insulator to Org. N° ME 132 to be used

COUPLING	
WEIGHT/UNIT	16.5 kg
MINIMUM FAILING LOAD (CANT)	10 kN
MAXIMUM WORKING LOAD (CANT)	5 kN
CREEPAGE LENGTH - SURFACE	900mm min
" " PROTECTED	368mm "
ELECTROSTATIC CAPACITANCE	-
FLASHOVER VOLTAGE	
D.F.O. NORMAL	145 kV
" 1 MIN	130 kV
W.F.O. NORMAL	92 kV
" 1 MIN	90 kV
I.F.O. 50% 1/50 μ s +	220 kV
" " " " -	310 kV
IMPULSE WITHSTAND 50Hz	200 kV
PUNCTURE VOLTAGE 50Hz	-
B.S. REFERENCE	B.S.137

CHARACTERISTIC TABLE FOR INSULATORS
MINIMUM REQUIREMENTS.

SCALE		33kV LINE POST INSULATOR	ISSUE	MIDLANDS
DATE	25-10-82			ELECTRICITY BOARD
DRAWN	ARB.			
CHECKED	L.F.M.			DRG. No.
APPROVED				M.165
		E.S.I. STANDARD 43-93	E.S.I. DRG N° 439307	

PRODUCT CODE: 56141-17

Semi-Conducting DMC
Pin Head to ESI 43-93

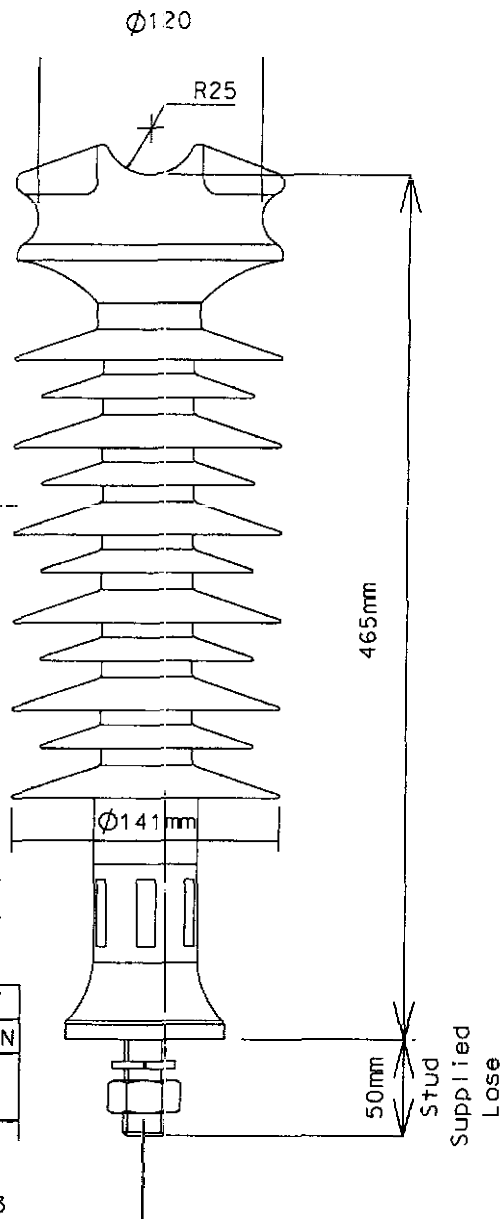
INSULATOR IDENTITY:

1. PRODUCT CODE.
2. MANUFACTURER'S IDENTITY.
3. YEAR/WEEK NUMBER BATCH IDENTITY.
4. CUSTOMER IDENTITY.

ELECTRICAL DATA				
TEST	56141-17	56141-XX	SPECIFICATION	
CREEPAGE (mm)	1100	1100	IEC 815	
AUT STRING (mm)	347	347		
MPULSE FLASHOVER (DRY) (kVp)	250	250		
MPULSE WITHSTAND (DRY) (kVp)	235	235		
AC 1min. FLASHOVER (DRY) (kV)	150	150	IEC383 :1983	
AC 1min. WITHSTAND (DRY) (kV)	137	137	BS137:1982	
AC 1min. FLASHOVER (WET) (kV)	119	119		
AC 1min. WITHSTAND (WET) (kV)	100	100		

MECHANICAL DATA				
TEST	56141-17	56141-XX	SPECIFICATION	
CANTILEVER DEFLECTION AT 10kN. LOAD (mm)	60	60	W T HENLEY	
MINIMUM TENSILE FAILING LOAD (kN)	<50	<50		
MINIMUM TENSILE TORQUE (NM)	<150	<150		

ROUTINE PRODUCTION TESTS			
COMPONENT	TEST	FREQUENCY	SPECIFICATION
MOULDED	WATER IMMERSION/ SUBSTRATE ELECTRICAL	MIN. 1%	BB. QA/P.1-A: 1988
FINAL ASSEMBLY	MECHANICAL	1%	50% SML for 1min. 120% SML for 1min. PULL TO FAIL



SCALE	-
DATE	06.02.03
DRAWN	M.J.B
TRACED	-
CHECKED	KD
APPROVED	13.02.03

36 kV. 10kN. CANTILEVER
POLYMER POST INSULATOR
WITH EPDM CLADDING

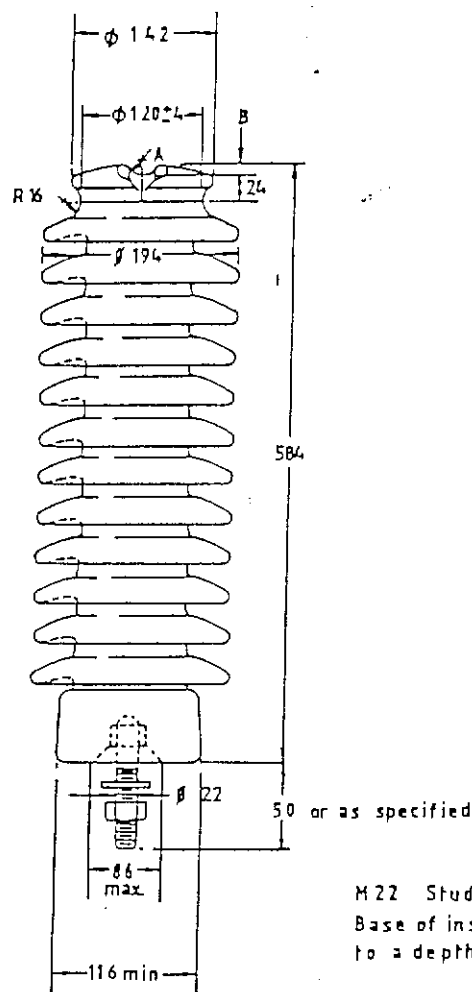
ISSUE	DATE

Produced by
Primary CAD

 Aquila

DRG. No.
M165/2

DRG. No. M165/2



1. Dimensions and characteristics to comply with B.S.137 for a nominal system voltage (3 phase) of 72 kV. British Practice.
2. Tolerances except where specified on drawing are to comply with BS.137.
3. Insulator to be indelibly marked in accordance with B.S.137.
4. Porcelain to be glazed brown.

COUPLING
 WEIGHT / UNIT 27 kg.
 MINIMUM FAILING LOAD (CANT) 10 kN
 MAXIMUM WORKING LOAD (CANT) 5 kN
 CREEPAGE LENGTH - SURFACE 1.360 m
 LENGTH - PROTECTED 500 mm
 FLASHOVER VOLTAGE
 D.F.O. NORMAL 190 kV
 D.F.O. 1 MIN.
 W.F.O. NORMAL 135 kV
 W.F.O. 1 MIN.
 I.F.O. 50% 1/50 μ s + 300 kV
 " " " " - 420 kV
 IMPULSE WITHSTAND 50 Hz 280 kV
 B.S. REFERENCE BS.137

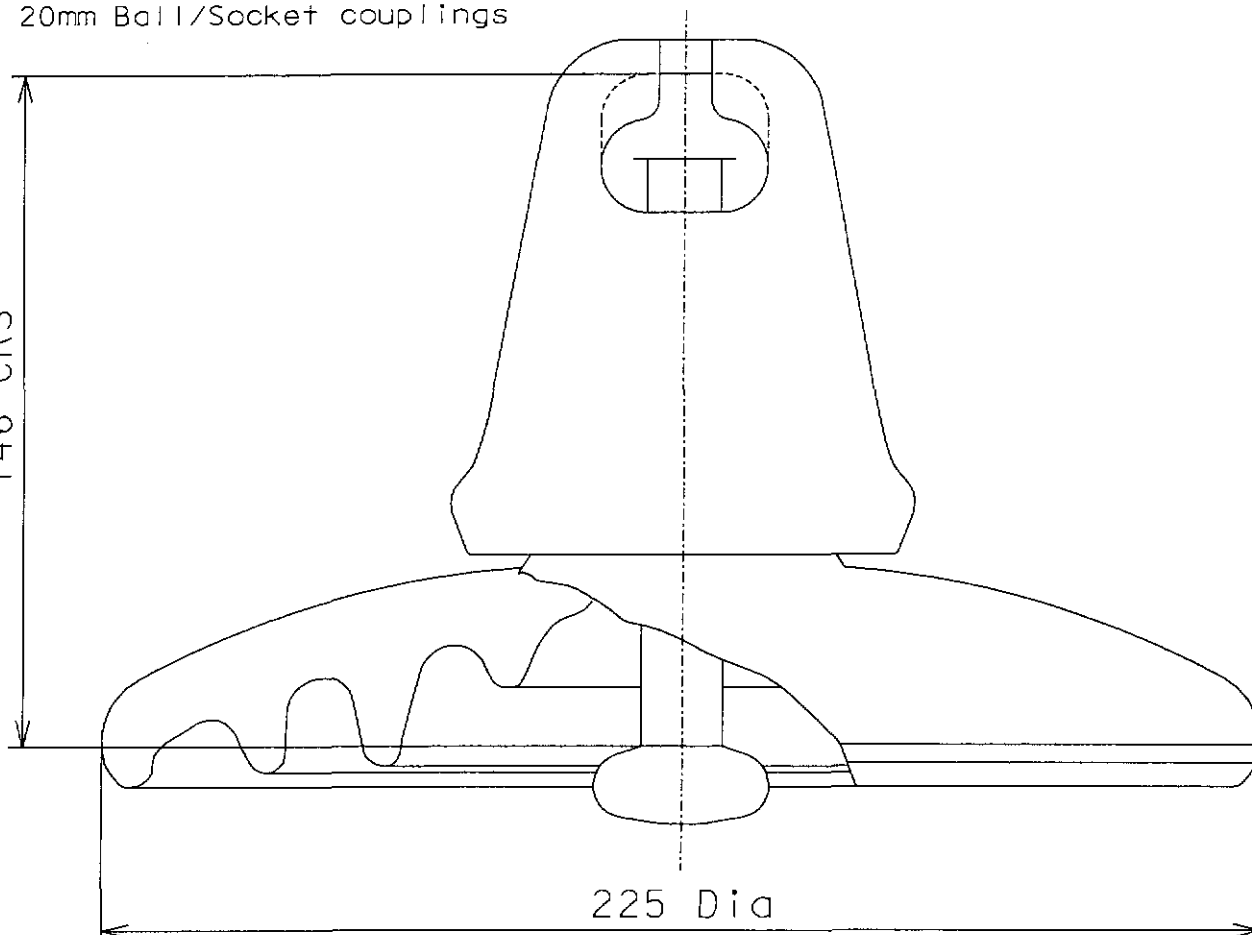
Ref.	Rad. A	Rad. B
168/1	25	16

CHARACTERISTIC TABLE FOR INSULATORS MINIMUM REQUIREMENTS

SCALE		66kV LINE POST LAPP INSULATOR	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	28.8.84.		A.	
DRAWN			Dim's added.	
TRACED	IW		7-10-92 A.J.B.	
CHECKED				
APPROVED		E.S.I. STANDARD		DRG. No. M. 168

20mm Ball/Socket couplings

146 CRS



225 Dia

MIN	TYPE	MANUFACTURER	MAN REF
3452541	PORCELAIN	MOSDORFER	269069

APPLICATION

Tension Only - Wood pole lines up to 66kV with
conductor M.W.L over 22.3kN (5000lbf)
up to 36kN (8000lbf) [Replacements only]

SCALE	1:1.6"
DATE	20-01-03
DRAWN	ADD
TRACED	
CHECKED	-
APPROVED	-

DISC INSULATOR MFL 125 Kn
146mm CENTRES

ISSUE DATE

Produced by
Output Services



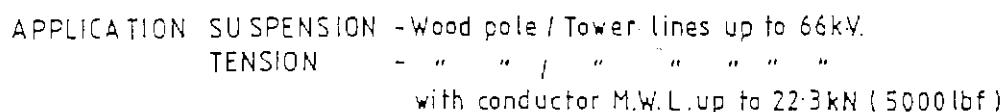
Aquila

DRG. No.

M 254

DRG. No.

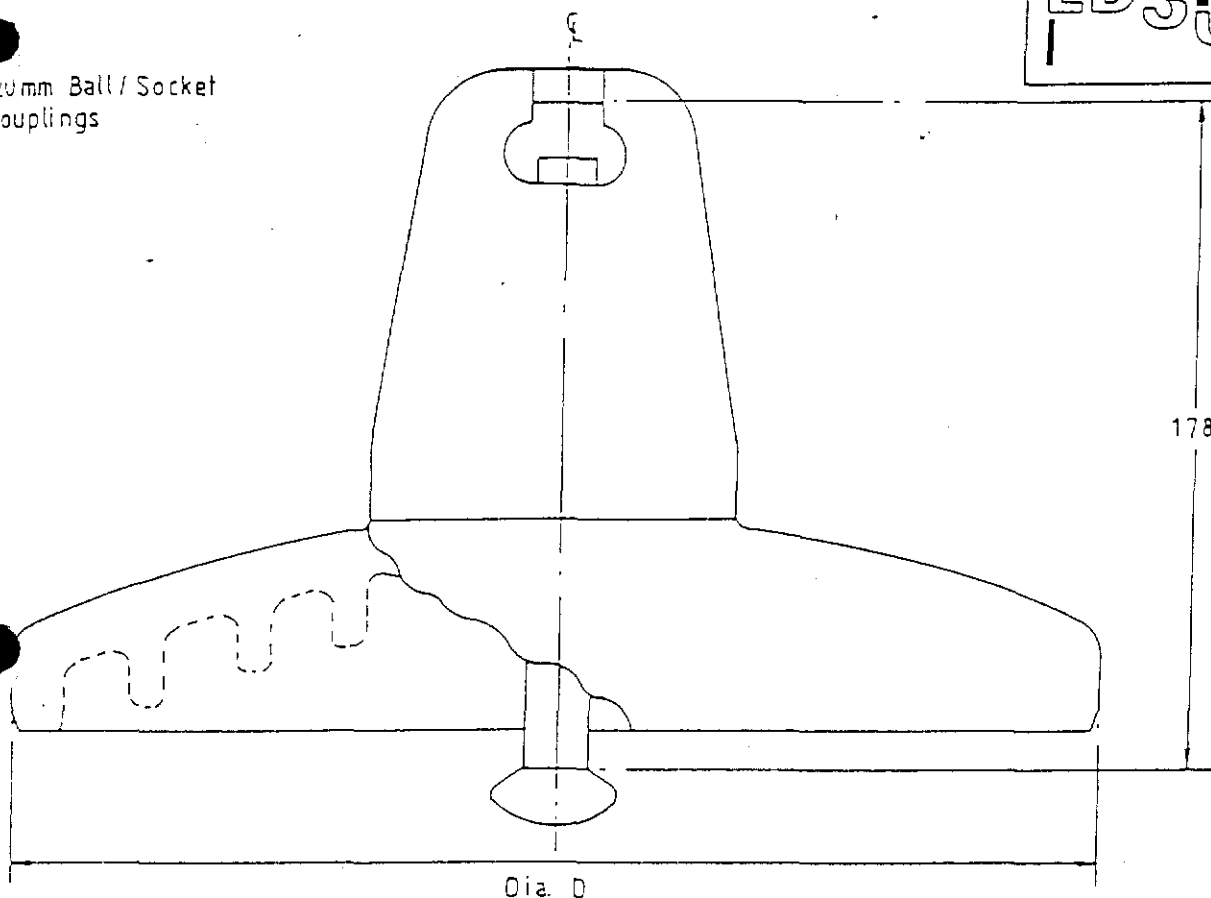
M 254



1. Dimensions and characteristics to comply with B.S.137.
2. Tolerance except where specified *on drawing* are to comply with B.S.137.
3. Insulator to be indelibly marked in accordance with B.S.137.
4. Porcelain to be glazed grey.

ADDITIONAL DAIL - 93 Mod's to table.	(25/7)	TITLE	SCALE	—	 Midlands Electricity plc	
		DISC INSULATOR MFL 80kN (HIGH IMPACT PORCELAIN)	DATE	13.10.92		DRG No M 300
			DRAWN	A.J.B.		
			TRACED	E.L.W.		
			CHECKED	<i>[Signature]</i>		
			APPROVED	<i>[Signature]</i>		
ESI STANDARD 43-93	ESI DRG No					

20mm Ball / Socket
couplings



REF.	TYPE	DIA. D	WEIGHT / UNIT
M309/1	PORCELAIN	292 mm	8.5 Kg.
M309/2	GLASS	280 mm	4.5 Kg.

APPLICATION

TENSION ONLY (a) Tower lines up to 66 kV with
conductor M.W.L. over
22.3 kN (5000 lbf.) up to
36 kN (8000 lbf.)

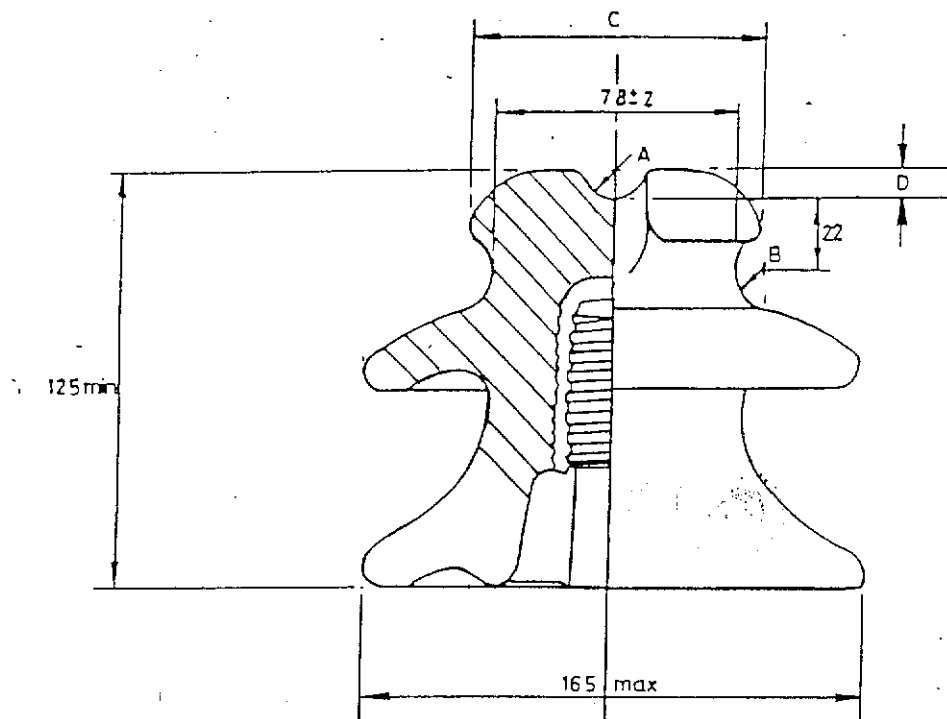
(b) 132 kV. Tower lines with conductors
up to 175 mm² A.C.S.R. conductor

MINIMUM FAILING LOAD	125 kN
MAXIMUM WORKING LOAD	-
CREEPAGE LENGTH - SURFACE	356 mm
" " - PROTECTED	232 mm
ELECTROSTATIC CAPACITANCE	-
FLASHOVER VOLTAGE	-
D.F.O. NORMAL	80 kV.
" 1 MIN.	70 kV.
W.F.O. NORMAL	50 kV.
" 1 MIN.	40 kV.
I.F.O. 50% 1/50 μ s +	120 kV.
" " " "	120 kV.
IMPULSE WITHSTAND 50 Hz	110 kV.
PUNCTURE VOLTAGE 50 Hz	130 kV.
BS REFERENCE	BS.137

1. Dimensions and characteristics to
comply with BS.137.
2. Tolerances except when specified on
drawing are to comply with BS.137
3. Insulator to be indelibly marked in
accordance with BS.137.
4. Porcelain to be glazed brown.

CHARACTERISTIC TABLE FOR INSULATORS MINIMUM REQUIREMENTS

ADD'D BY DATE 10-10-92 Table 1	TITLE	SCALE		MEB Midlands Electricity plc M 309 ESI DRG. N°
	DISC INSULATOR M.F.L. 125kN	DATE	14-10-92	
	TENSION-TOWER LINES UP TO 132kV.	DRAWN	A.J.B.	
	CONDUCTORS UP TO 175mm ² ACSR	TRACED	E.L.W.	
	ESI STANDARD 43-93	CHECKED	257	
		APPROVED		



1. Dimensions and characteristics to comply with B.S.137 for a nominal system voltage (3 phase) of 12 k.V. British Practice

2. Tolerances except where specified on drawings are to comply with B.S. 137.

3. Thimble to be fitted suitable for large steel head Fig 1 B.S. 3288 Part 2 and should have a minimum tin content of 4.8% by weight.

4. Insulator to be indelibly marked in accordance with B.S.137.

5. Semi Conducting glazed head. Porcelain to be glazed Grey

7. The firing date (month and year) to be indelibly marked on insulator also makers identification mark.

Ref.	Rad. A	Rad. B	C	D
313/1	25	14	102	16

COUPLING

WEIGHT /UNIT	2.5 kg
MINIMUM FAILING LOAD (CANT)	1089 kgf
MAXIMUM WORKING LOAD (CANT)	545 kgf
CREAPAGE LENGTH - SURFACE	225 mm
" " PROTECTED	100 mm

ELECTROSTATIC CAPACITANCE

FLASHOVER VOLTAGE

D.F.O. NORMAL	80 kV
" 1 MIN	70 kV
W.F.O. NORMAL	50 kV
" 1 MIN	50 kV
I.F.O. 50% 1/50 μ s.	105 kV
" " " " "	135 kV

IMPULSE WITHSTAND 50 Hz 95 kV

PUNCTURE VOLTAGE 50 Hz 130 kV

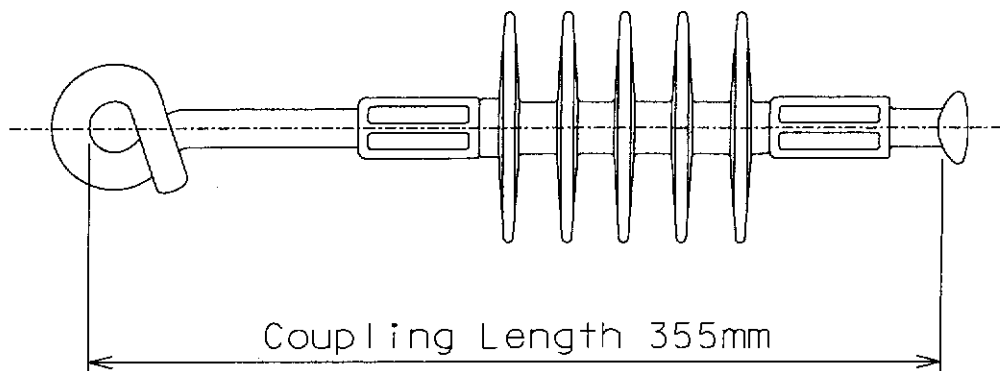
B.S. REFERENCE B.S.137

MECHANICAL IMPACT STRENGTH 21N m

CHARACTERISTIC TABLE FOR INSULATORS

MINIMUM REQUIREMENTS.

SCALE	-	11 kV PIN INSULATOR (HIGH IMPACT)	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	26.10.82		C	
DRAWN	R.A.C.		Note 3, WFO & I.F.O. mods	DRG. No.
CHECKED			9-8-9 AJB	MF M313
APPROVED				ESJ. DRG N: 439301
E.S.I. STANDARD 43/93				



Manufacturer	Man Ref
W T HENLEY	56145/09

11KV 20kn Ball/Pig-Tail Hook Composite Insulator

DRG. No. M417/1

SCALE	
DATE	31.8.00
DRAWN	A.J.E.
TRACED	
CHECKED	13.10.00
APPROVED	ADD

11kv COMPOSITE INSULATOR
PIG-TAIL HOOK
(LIGHT CONSTRUCTION)

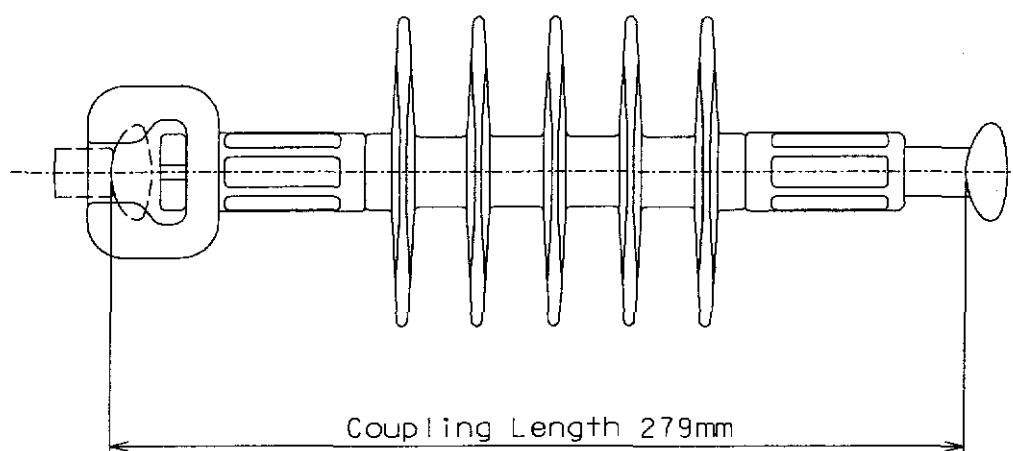
ISSUE DATE

DRG. No.

M417/1



Produced by
Output Services



Manufacturer	Man Ref
W T HENLEY	56145/06

11kv 70kn 16mm B type Composite Insulator

SCALE	
DATE	31.8.00
DRAWN	A.J.E.
TRACED	
CHECKED	13.10.00
APPROVED	KD

11kv COMPOSITE INSULATOR
SOCKET AND BALL

ISSUE	DATE
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Produced by
Output Services



DRG. No.

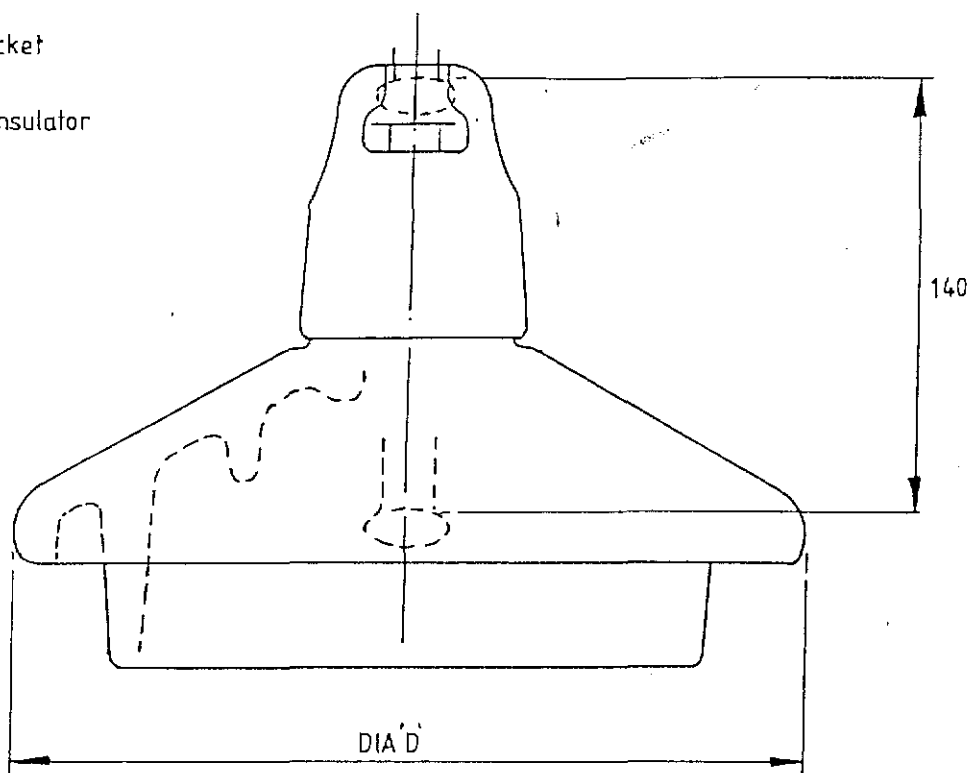
M417/2

DRG. No.

M417/2

16 mm Ball / Socket
Couplings.

M. 418/2 Glass Insulator
shown.



REF.	TYPE	DIA. D	WEIGHT / UNIT
418 / 1	PORCELAIN	280 mm	7.0 Kg.
418 / 2	GLASS	255 mm	4.5 Kg

APPLICATION

SUSPENSION ONLY - 132 kV. Tower lines with
single 175 mm² and 400 mm²
conductor, suspension,
pilot and low duty
positions.

MINIMUM FAILING LOAD	80 kN.
MAXIMUM WORKING LOAD	-
CREEPAGE LENGTH - SURFACE	390 mm
" " - PROTECTED	230 mm
ELECTROSTATIC CAPACITANCE	-
FLASHOVER VOLTAGE	-
D.F.O. NORMAL	85 kV.
" 1 MIN.	75 kV.
W.F.O. NORMAL	50 kV.
" 1 MIN.	45 kV.
I.F.O. 50% 1/50 μs +	140 kV.
" " " " -	125 kV.
IMPULSE WITHSTAND 50 Hz	115 kV.
PUNCTURE VOLTAGE 50 Hz	125 kV.
B.S. REFERENCE	B.S. 137

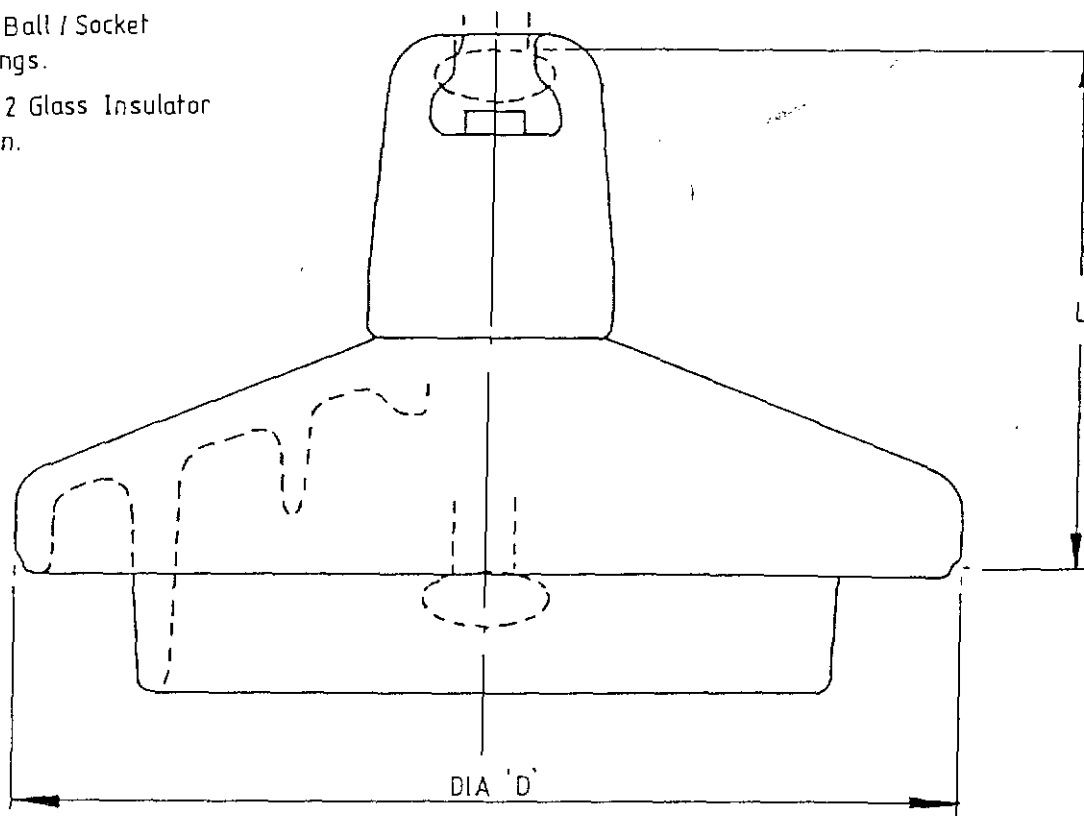
1. Dimensions and characteristics to
comply with B.S. 137.
2. Tolerances except where specified
on drawing are to comply with
B.S. 137.
3. Insulator to be indelibly marked in
accordance with B.S. 137.
4. Porcelain to be glazed brown.

CHARACTERISTIC TABLE FOR INSULATORS MINIMUM REQUIREMENTS

SCALE	N.T.S.	DISC INSULATOR M.F.L. 80 kN 132 kV SUSPENSION.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	10-2-84		A	
DRAWN	R.A.C.		Tables added.	DRG. No.
TRACED			15-10-92 A.J.B.	M. 418. (MF)
CHECKED				
APPROVED		E.S.I. STANDARD 43-93		E.S.I. DRG. N°

20mm Ball / Socket
Couplings.

M.419/2 Glass Insulator
shown.



REF.	TYPE	DIA D	L	WEIGHT/UNIT
419/1	PORCELAIN	318 mm	171 mm	11 Kg.
419/2	GLASS	320 mm	170 mm	7 Kg.

APPLICATION:-
SUSPENSION ONLY - 132kV Tower lines
with twin 175mm²
A.C.S.R. Conductors.

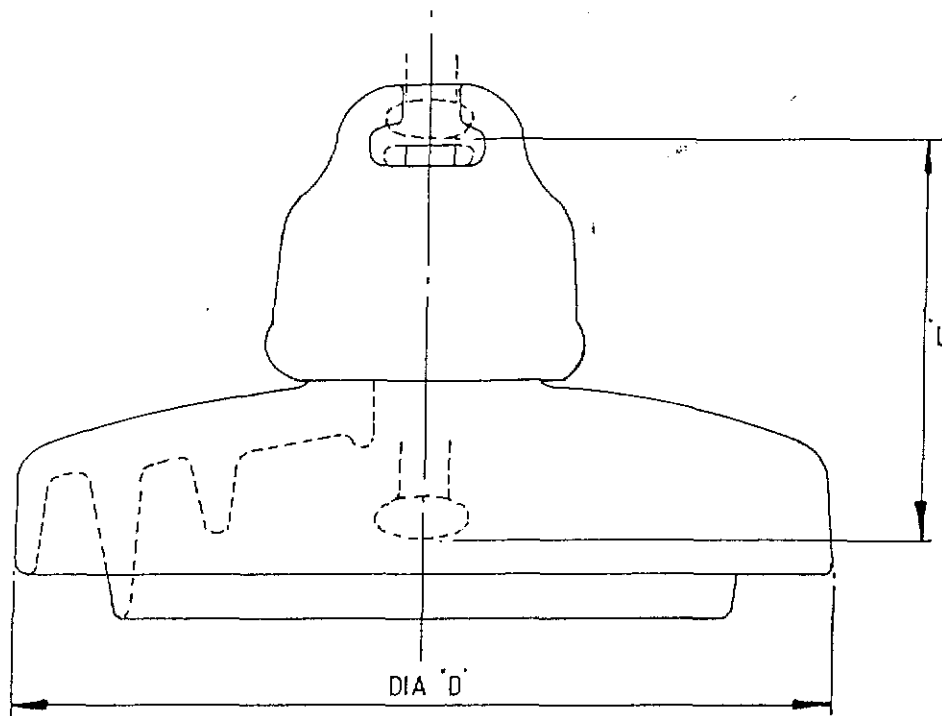
MINIMUM FAILING LOAD	125kV.
MAXIMUM WORKING LOAD	-
CREEPAGE LENGTH - SURFACE	489mm
" " - PROTECTED	286mm
ELECTROSTATIC CAPACITANCE	-
FLASHOVER VOLTAGE	-
D.F.O. NORMAL	100kV.
" 1MIN.	85kV.
W.F.O. NORMAL	50kV.
" 1MIN.	45kV.
I.F.O. 50% 1/50 μ s +	150kV.
" " " " -	140kV.
IMPULSE WITHSTAND 50Hz	130kV.
PUNCTURE VOLTAGE 50Hz	140kV.
B.S. REFERENCE	B.S.137

1. Dimensions and characteristics to comply with BS.137.
2. Tolerances except where specified on drawing are to comply with B.S.137.
3. Insulator to be indelibly marked in accordance with B.S.137.
4. Porcelain to be glazed brown.

CHARACTERISTIC TABLE FOR INSULATORS
MINIMUM REQUIREMENTS

SCALE	N.T.S.	DISC INSULATOR MEL.125kN 132kV SUSPENSION.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	10-2-84		A	
DRAWN	R.A.C		Tables added. 20-10-92 A.J.B.	DRG. No. M.419
TRACED				
CHECKED				
APPROVED				
E.S.I. STANDARD 43 - 93			E.S.I. DRG N°	

24mm Ball & Socket
coupling.



REF	TYPE	DIA. D	L	WEIGHT/UNIT
M420/1	PORCELAIN	380mm	187mm	17.7 Kg.
M420/2	GLASS	355mm	170mm	—

MINIMUM FAILING LOAD	190 kN
MAXIMUM WORKING LOAD	—
CREEPAGE LENGTH - SURFACE	584 mm
- PROTECTED	381 mm
ELECTROSTATIC CAPACITANCE	—
FLASHOVER VOLTAGE	—
D.F.O. NORMAL	100 kV.
" 1 MIN.	90 kV.
W.F.O. NORMAL	50 kV.
" 1 MIN.	45 kV.
I.F.O. 50% 1/50 μ s +	150 kV.
" " " " "	155 kV.
IMPULSE WITHSTAND 50 Hz	130 kV.
PUNCTURE VOLTAGE 50 Hz	140 kV.
B.S. REFERENCE	B.S. 137

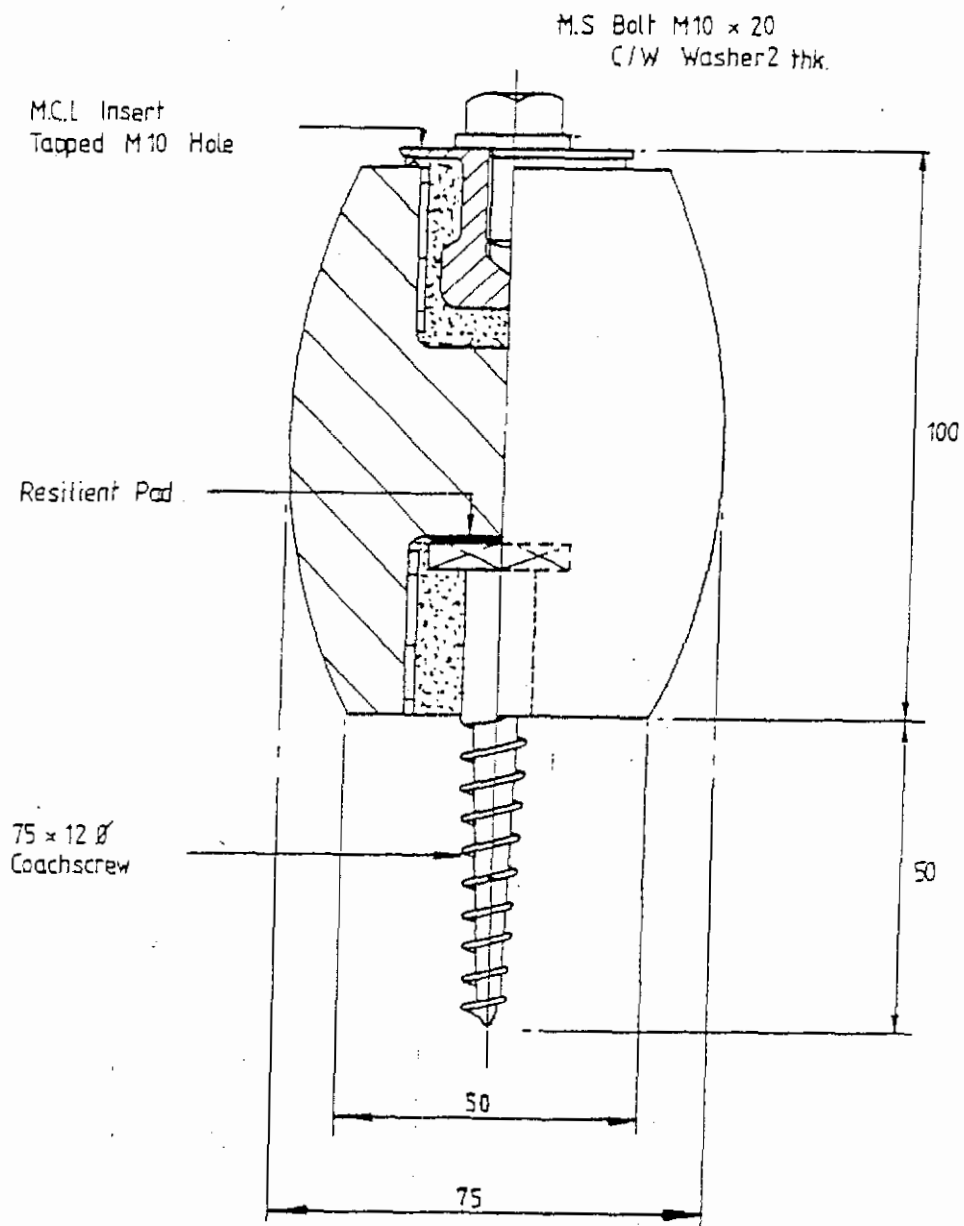
APPLICATION :- SUSPENSION ONLY -

- 400 kV. Tower lines operating at 132 kV. with twin 400 mm² A.C.S.R. conductors.
- Replacements for 132 kV. tower and 275 kV. tower lines operating at 132 kV. constructed using reclaimed 275 kV. Insulators.

- Dimensions and characteristics to comply with B.S. 137.
- Tolerances except where specified on drawing are to comply with B.S. 137.
- Insulator to be indelibly marked in accordance with B.S. 137.
- Porcelain to be glazed brown.

CHARACTERISTIC TABLE FOR INSULATORS MINIMUM REQUIREMENTS

SCALE	N.T.S.	DISC INSULATOR M.F.L. 190 kN SUSPENSION - 132 kV TOWER LINES TWIN 400 mm ² A.C.S.R. CONDUCTORS.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	15-2-84		A	
DRAWN	R.A.C.		Tables added.	
TRACED			20-10-92 A.J.B.	
CHECKED				
APPROVED		ES.I. STANDARD 43-93		DRG. No. M. 420 (MF)
				E.S.I. DRG N°



SCALE	N.T.S.
DATE	20.9.84
DRAWN	R.A.L.
TRACED	J.W.
CHECKED	J.C.
APPROVED	J.C.

L.V. SUPPORT INSULATOR

E.S.I. STANDARD 43-93

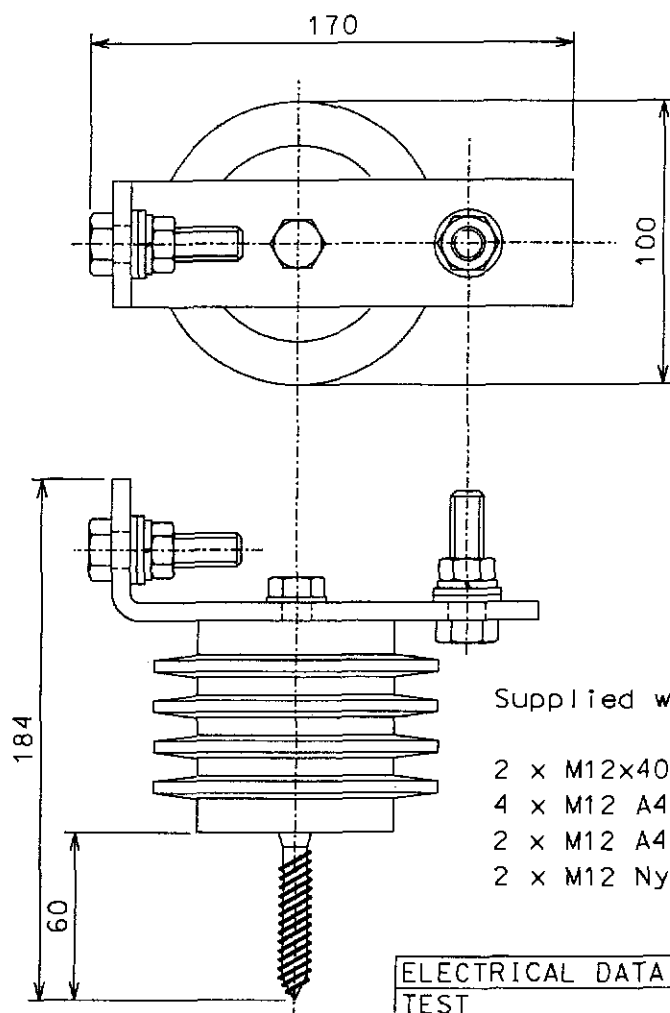
ISSUE

MIDLANDS
ELECTRICITY BOARD

DRG. No.

M 424

E.S.I. DRG. No.



INSULATOR MARKING:

1. Product Code
2. Manufacturer's Identity.
3. Year/Week number batch identity.

Supplied with:

- 2 x M12x40 A4 S/S Hex head Bolts
- 4 x M12 A4 S/S Plain Washers
- 2 x M12 A4 S/S Spring Washers
- 2 x M12 Nyloc Nuts

ELECTRICAL DATA	
TEST	VALUE
CREEPAGE (mm)	236 (30mm/kV)
TAUT STRING (mm)	141
IMPULSE FLASHOVER (DRY)	+ve 77.7kVp / -ve 146.1kVp
IMPULSE WITHSTAND (DRY)	+ve 74.2kVp / -ve 138.5kVp
AC 1 MIN. FLASHOVER (DRY)	48.8kV
AC 1 MIN. WITHSTAND (DRY)	44.0kV
AC 1 MIN. FLASHOVER (WET)	27.8kV
AC 1 MIN. WITHSTAND (WET)	23.1kV

MECHANICAL DATA (INSULATOR)		
TEST	VALUE	SPECIFICATION
GLASS TRANS. TEMP T_g (OSC)	105-115 C	
FLEXURAL STRENGTH	120-130 N/mm ²	ISO 178
TENSILE STRENGTH	80-95 N/mm ²	ISO R527
IMPACT STRENGTH	9-11 kJ/m ²	ISO 179

PRODUCT CODE: 56141-22

SCALE	-
DATE	05.02.03
DRAWN	M. J. B
TRACED	-
CHECKED	K. D
APPROVED	13.02.03

LV BRACKET INSULATOR

ISSUE DATE

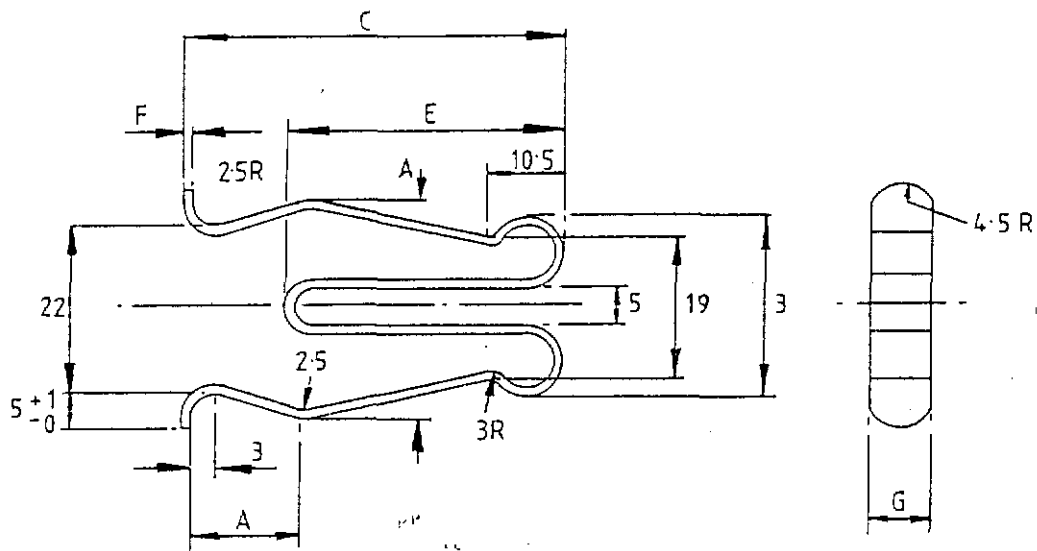


Aquila

DRG. No.

M 424/1

DRG. No. M 424/1



M. REF	CEGB CC.	SOCKET REF.	A	B	C	D	E	F	G
540/1	4312950	16B	28	24	50±1.5	15.5	36	1.5	7.9 ^{+0.2} ₋₀
540/2	4312952	20	30	25	62±1.5	42	42	2	7.0 ^{+0.2} ₋₀

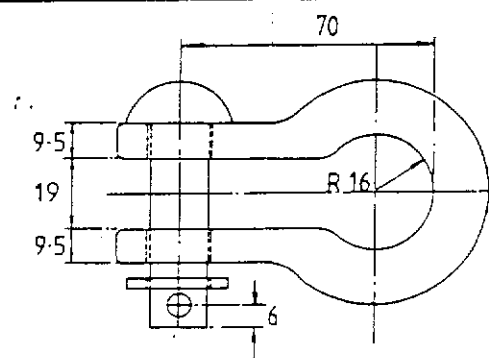
SCALE	N.T.S.	INSULATOR FITTING SECURITY CLIPS.	ISSUE	MIDLANDS ELECTRICITY BOARD ORG. No. M.540 E.S.I. ORG. N°
DATE	9-8-82			
DRAWN	R.A.C.			
TRACED				
CHECKED				
APPROVED	<i>[Signature]</i>	E.S.I. STANDARD		

SECTION 15

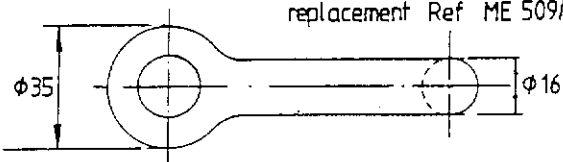
MIN NUMBER	M	DESCRIPTION
3450010	1	INSULATOR FITTING SHACKLES M.F.L 70kN
3450250	25	EYEBOLT
3450620	62	11kV INSULATOR PIN 10kN
3450670	67	BALL ENDED HOOK M.F.L
3450710	71	PILOT INSULATOR 11 & 33kV
3450840	84	33kV INSULATOR PIN 10kN
3450860	86	BALL ENDED HOOK M.F.L 70kgf
3450870	87	SOCKET CLEVIS M.F.L 70kN
3450880	88	SOCKET TONGUE M.F.L 70kN
3450890	89	ARCING HORNS
3451040	104	SUSPENSION 'U' LINK
3451050	105	GUSSET PLATE 66KV SUSP
3451080	108	M.S. EYE NUT
3451330	133	BALL ENDED EYE LINK M.F.L 70kN
3451540	154	11 & 33kV PILOT INSULATOR SPINDLE
3451570	157	INSULATOR BRACKET
3451840	184	SOCKET CLEVIS M.F.L 70kN
3451850	185	BALL CLEVIS M.F.L 70kN
3452350	235	CLAMP ADAPTORS M.F.L 70kN
3452490	249	BALL ENDED EYE LINK M.F.L 125kN
3452500	250	D' SHACKLE M.F.L 125kN
3452510	251	SOCKET TONGUE M.F.L 125 kN
3452550	255	ARCING HORNS FOR USE WITH 12700KG EQUIPMENT
3453060	306	CRANKED LINKS M.F.L 125kN
3453070	307	LANDING PLATE M.F.L 125kN
3453160	316	VANDAL SHIELD
3453430	343	EARTHWIRE FITTINGS CLEVIS CONNECTORS 70 & 175mm ² TERMINATIONS
3453970	397	INSULATOR SUPPORT PLATE
3454011	401/1	EXTENSION LINK M.F.L 70kN
3454270	427	U' BOLTS
3454280	428	EARTHWIRE SUSPENSION LINK
3454310	431	BALL ENDED EYE LINK M.F.L 190kN
3454320	432	LANDING PLATE M.F.L 190 kN
3454330	433	SHACKLE M.F.L 190kN
3454340	434	TOWER SHACKLE 400mm ² ACSR CONDUCTORS M.F.L 190kN
3454350	435	SOCKET TONGUE M.F.L 190kN
3454360	436	CRANKED LINKS M.F.L 190kN
3454431	443/1	CRANKED LINKS M.F.L 70kN
3454441	444/1	LANDING PLATE M.F.L 70kN
3455090	509	INSULATOR FITTING / CONDUCTOR TERMINATION CLEVIS PINS
3455100	510	SOCKET CLEVIS M.F.L 125kN
3455130	513	SOCKET CLEVIS M.F.L 70kN
3455500	550	INSULATOR ARCING HORN 132kV SUSPENSION LIVE END TYPE 4SC
3455510	551	INSULATOR ARCING HORN 132kV LOW DUTY EARTH END TYPES 4LA & 7LA
3455520	552	INSULATOR ARCING HORN 132kV SUSPENSION EARTH END TYPE 4SA FOR 39" GAP
3455530	553	INSULATOR ARCING HORN 132kV SUSPENSION EARTH END TYPE 4SB FOR 44" GAP
3455540	554	INSULATOR ARCING HORN 132kV SUSPENSION EARTH END TYPE 7SA FOR 39" GAP
3455550	555	INSULATOR ARCING HORN 132kV SUSPENSION EARTH END TYPE 7SB FOR 44" GAP
3455560	556	INSULATOR ARCING HORN 132kV SUSPENSION LIVE END TYPE 4TC
3455570	557	INSULATOR ARCING HORN 132kV SUSPENSION LIVE END TYPE 7TC

SECTION 15

MIN NUMBER	M	DESCRIPTION
3455580	558	INSULATOR ARCING HORN 132kV TENSION EARTH END TYPE 4TB FOR 44" GAP
3455590	559	INSULATOR ARCING HORN 132kV TENSION EARTH END TYPE 4TA FOR 39" GAP
3455600	560	INSULATOR ARCING HORN 132kV TENSION EARTH END TYPE 7TA FOR 39" GAP
3455610	561	INSULATOR ARCING HORN 132kV TENSION EARTH END TYPE 7TB FOR 44" GAP
3455620	562	INSULATOR ARCING HORN 132kV SUSPENSION LIVE END TYPE 3/2/132 SC
3455690	569	INSULATOR BRACKET

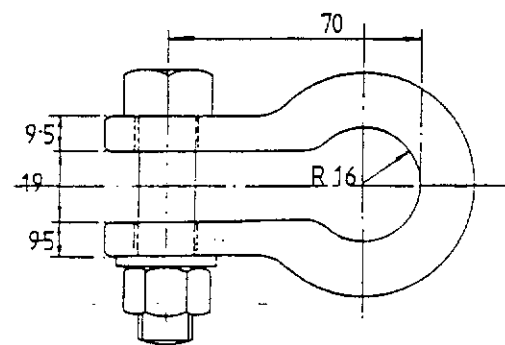


16 Dia x 55 pin with washer & 5 ϕ split pin replacement Ref ME 509/5

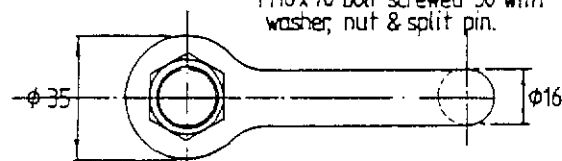


TYPE M 1/0

BS 3288 Pt Ref 15/29

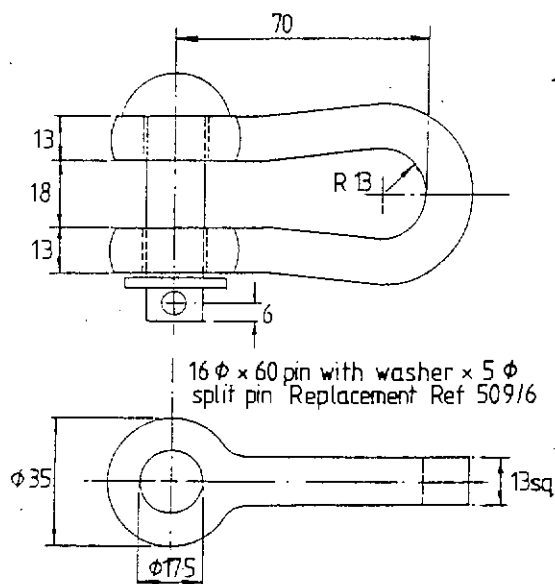


M16 x 70 bolt screwed 30 with washer, nut & split pin.

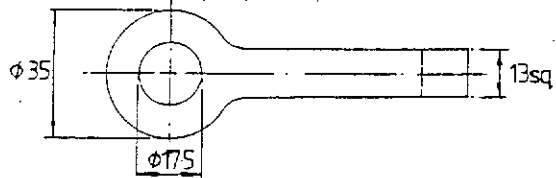


TYPE M 1/1

BS.3288 Pt Ref 15/29A

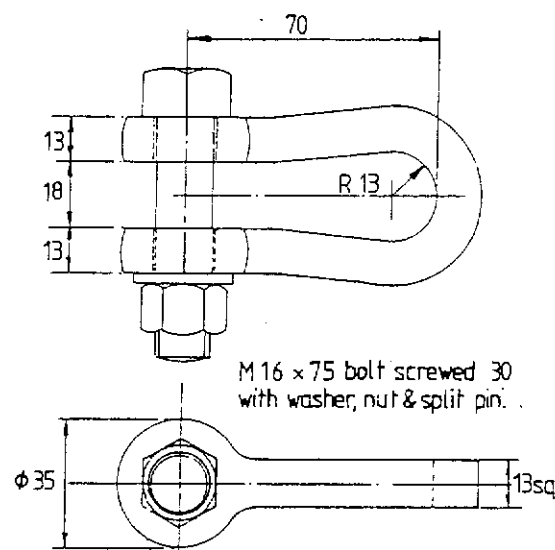


16 ϕ x 60 pin with washer x 5 ϕ split pin Replacement Ref 509/6

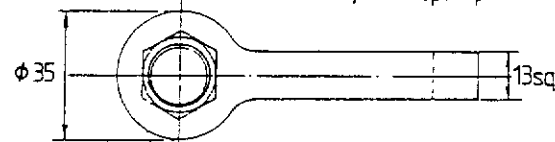


TYPE M 1/2

B.S. 3288 Pt2 Ref 15/33



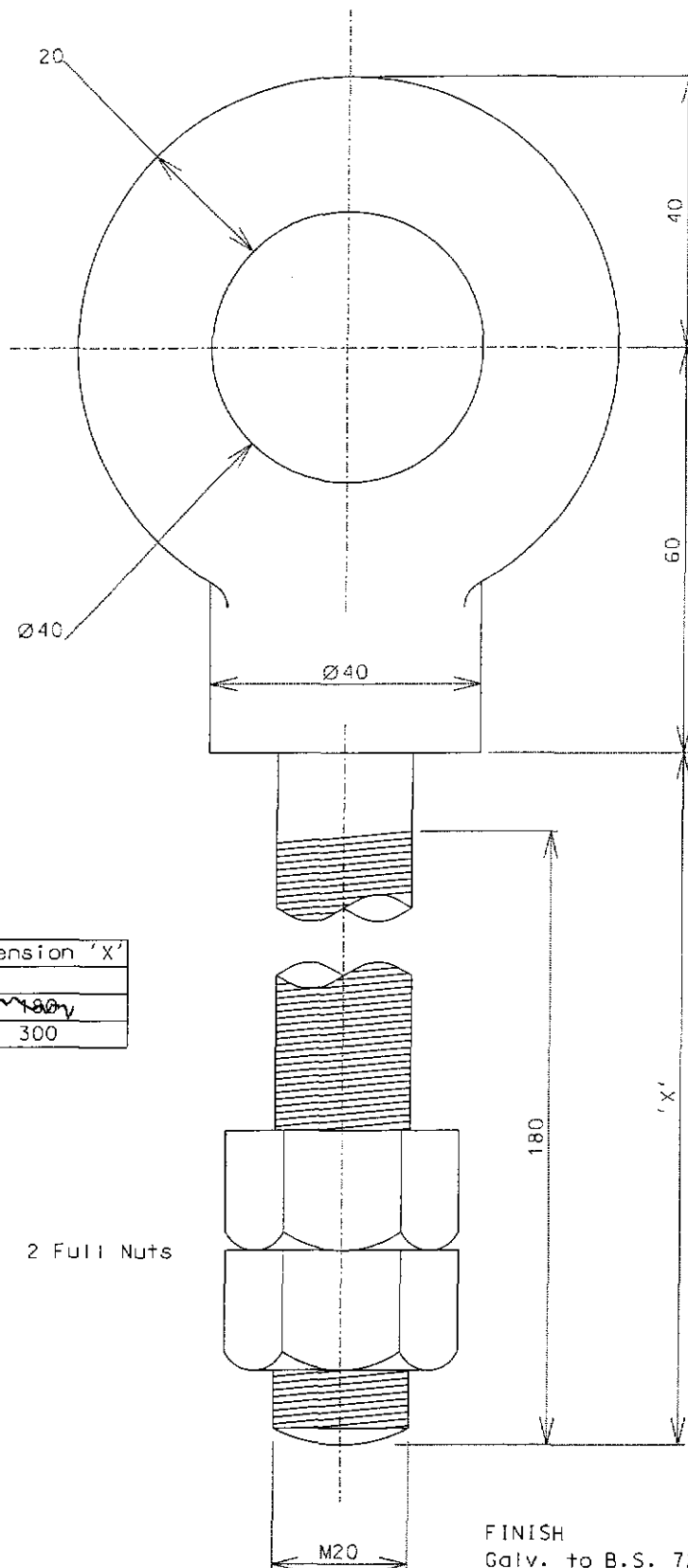
M16 x 75 bolt screwed 30 with washer, nut & split pin.



TYPE M 1/3

BS. 3288 Pt 2 15/33A.

SCALE	N.T.S.	INSULATOR FITTING SHACKLES M.E.L. 70 kN.	ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	—		A	B M.E.	
DRAWN	F.A.C.		Types 2,3&4 added R.A.C.	Type No's M was ME 12/85	
TRACED	—				
CHECKED	—				
APPROVED	—				
E.S.I. STANDARD					DRG. No. M 1.
					E.S.I. DRG. No.



Type	Dimension 'X'
3450251	180
3450252	300

2 Full Nuts

MATERIAL
Mild Steel
28/33 Ton Quality

FINISH
Galv. to B.S. 729 with all
Threading Brush Galv.

SCALE	1-1
DATE	6/9/00
DRAWN	ADD
TRACED	
CHECKED	27/9/00
APPROVED	KD

EYEBOLT

ISSUE	DATE
-------	------

DRG. No.

GPU
POWER UK

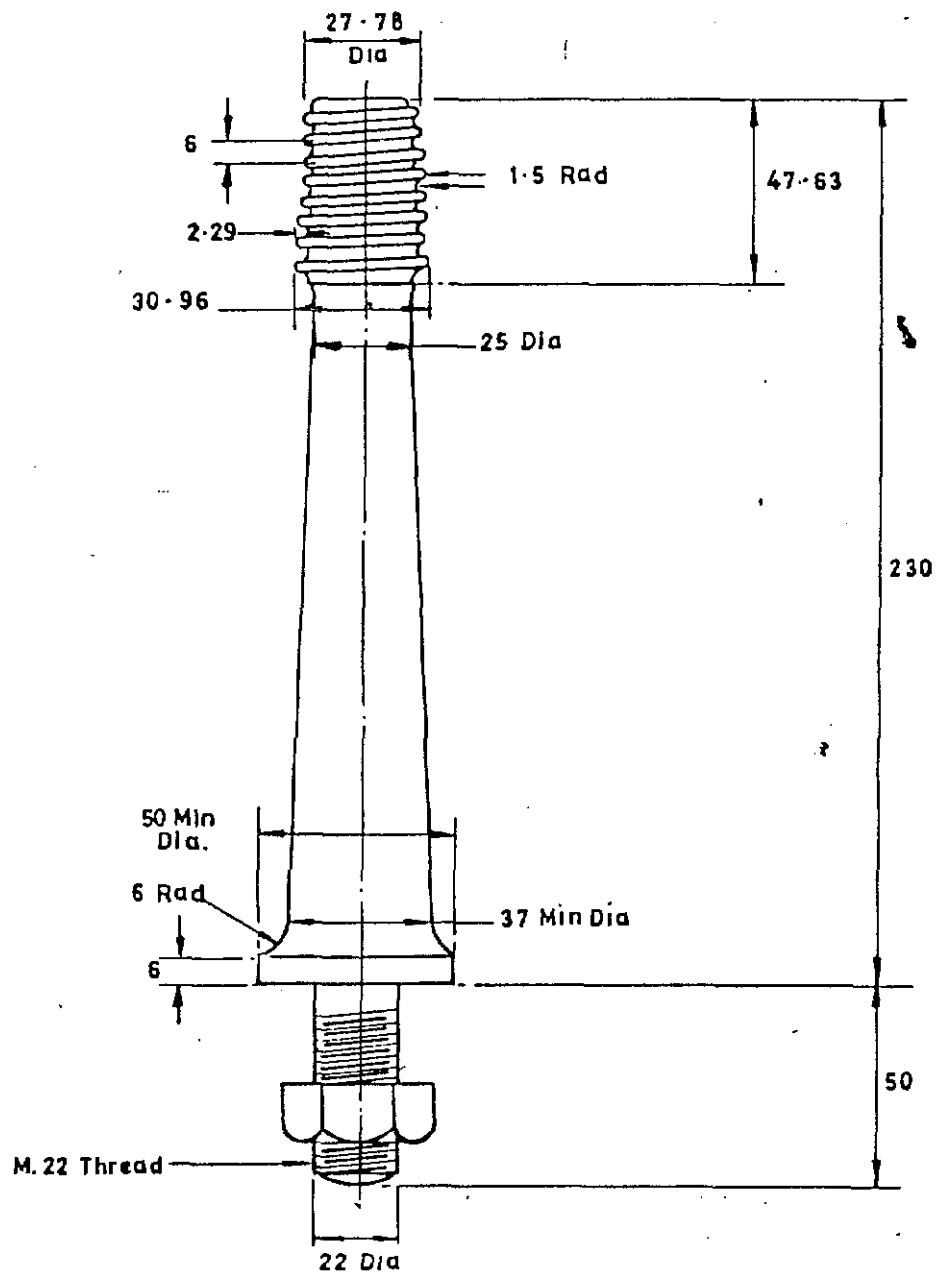
M25

Produced by
Output Services

DRG. No.

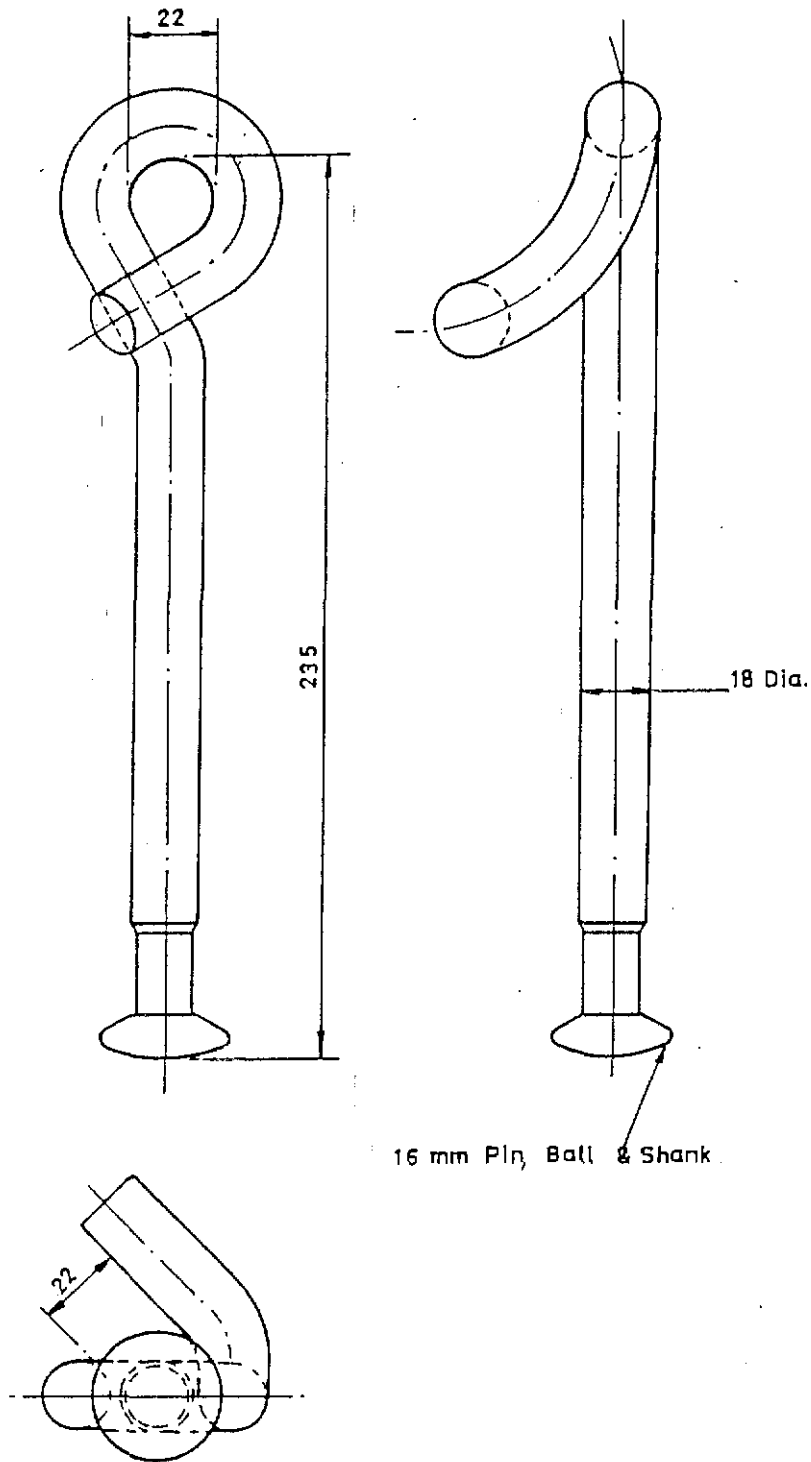
M25

A4



NOTE: TO B.S.3286 PT.1 & 2
FIG. 4 REF. 29

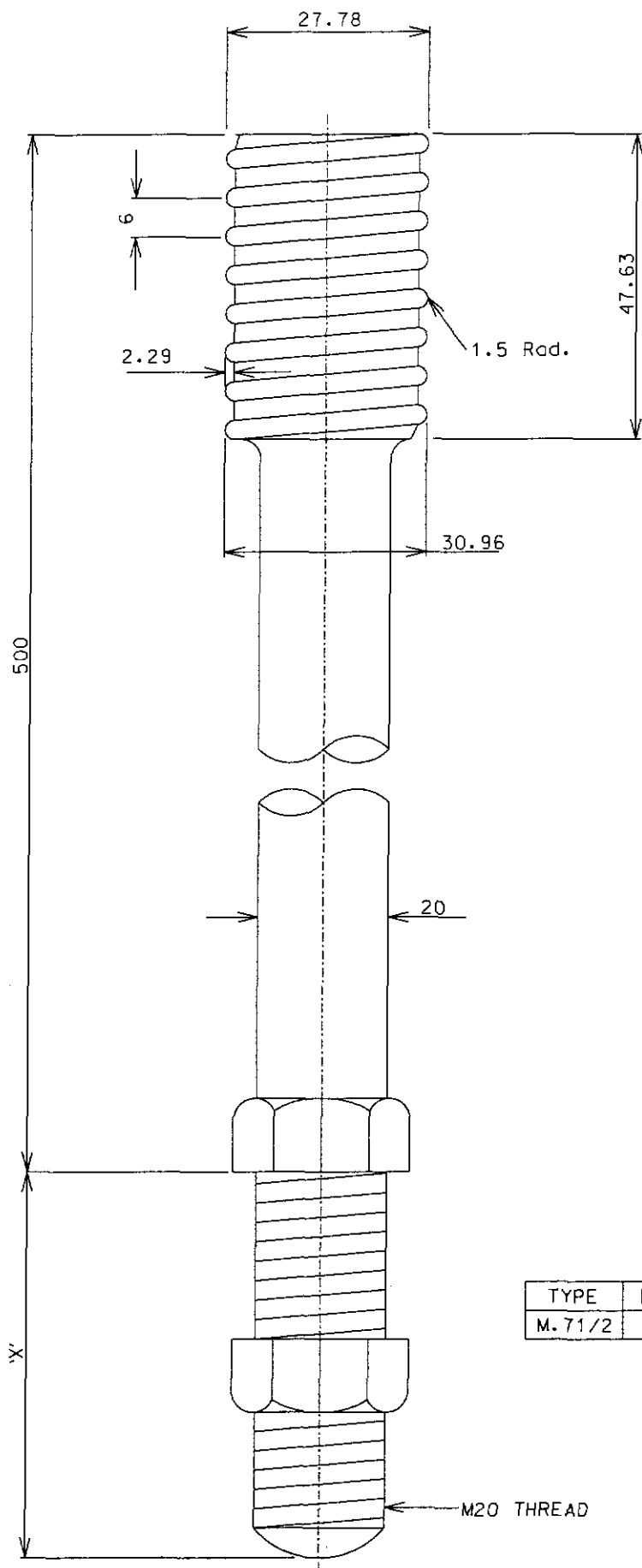
SCALE	1:2	11 kV INSULATOR PIN 10 kN (1020 Kgf)	ISSUE-	MIDLANDS ELECTRICITY BOARD
DATE				
DRAWN				DRG. No.
TRACED				M.62
CHECKED				
APPROVED				FSI DRG' N°



NOTE: To be 3288 PT.1 & 2
Fig 12 Ref 4/88

SCALE	1/2	BALL ENDED HOOK MINIMUM FAILING LOAD 1835 Kgf.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	26.10.82			
DRAWN	F.A.H.			DRG. No.
TRACED	M.R.J.			M.67
CHECKED	W			
APPROVED		E.S.I. STANDARD		E.S.I. DRG N°

M.F.



TYPE	DIMENSION 'X'
M. 71/2	150

NOTE To BS 3288 PT 1 & 2 Fig.5 Ref.50

SCALE	1:10
DATE	26.10.82
DRAWN	W.B.
TRACED	S.J.H.
CHECKED	-
APPROVED	-

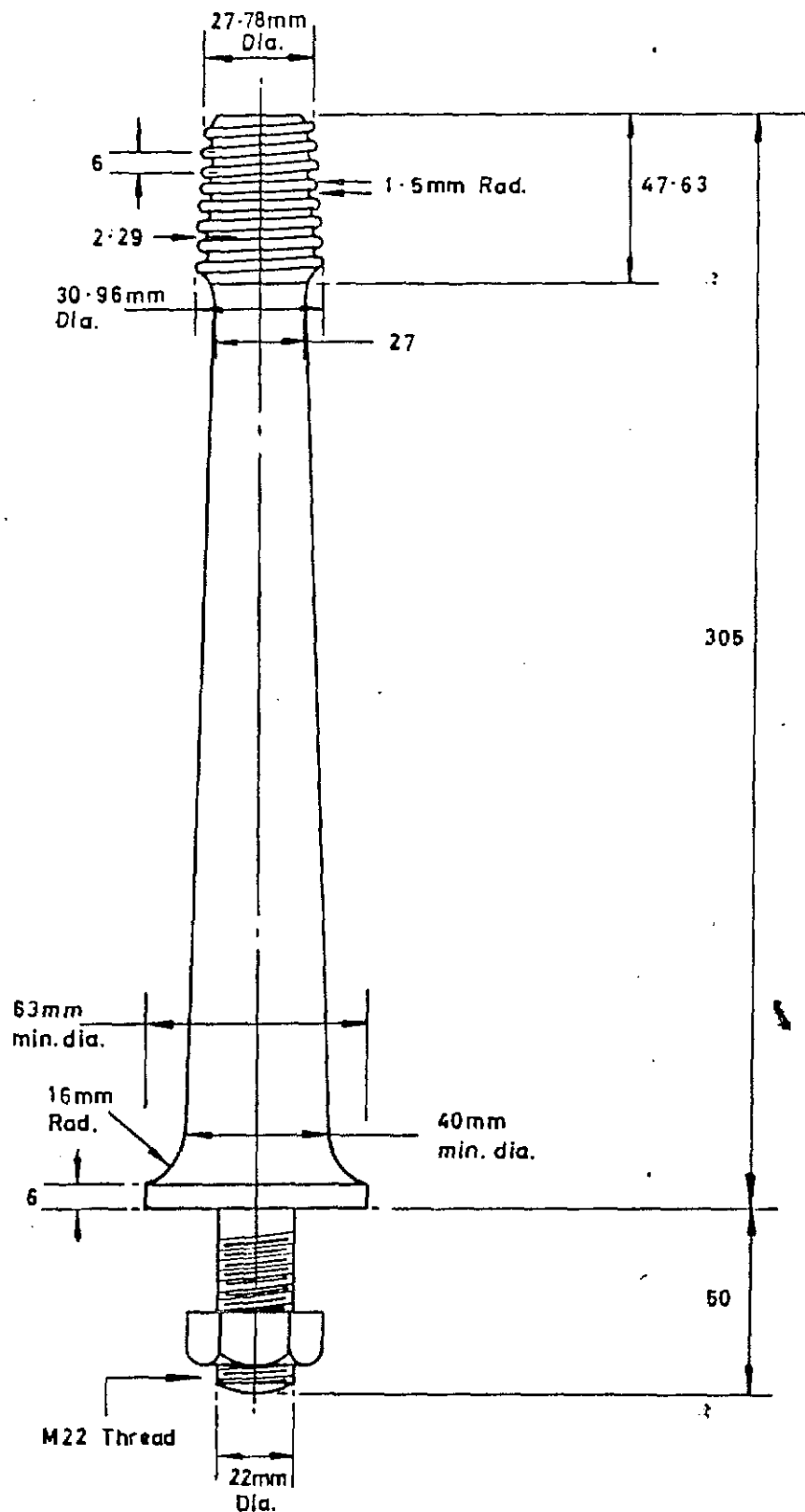
PILOT INSULATOR PINS (11 AND 33KV)

ISSUE	DATE
B	13.12.01
REDRAWN ON	C.A.D.
D.B.	

Produced by Output Services
GPU POWER UK
DRG. No. M71

DRG. No.

M71

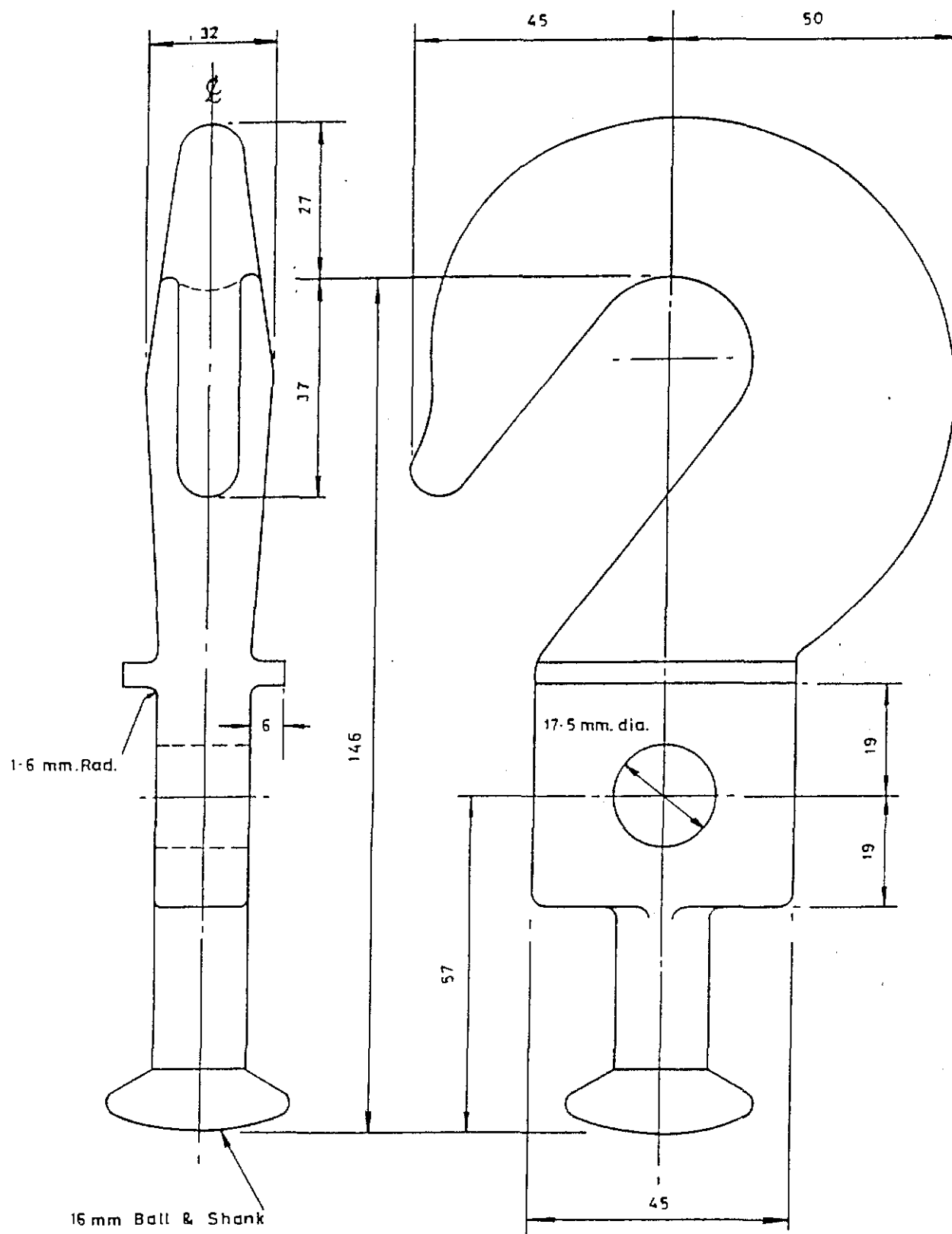


NOTE:
To B.S. 3288 P.T. 1 & 2 Fig.4. Ref. 31

SCALE	1 / 2
DATE	
DRAWN	
TRACED	
CHECKED	
APPROVED	

33kV INSULATOR PIN
10kN (1020Kgf)

ISSUE	MIDLANDS ELECTRICITY BOARD
DRG. No.	M. 84.



NOTE: To B.S. 3288. P.T. 1 & 2, Fig.13 Ref. 15/32.

SCALE	1/1
DATE	26-10-82
DRAWN	F.A.H.
TRACED	A.W.
CHECKED	
APPROVED	

BALL ENDED HOOK
MINIMUM FAILING LOAD 7140 kgf.

E.S.I. STANDARD

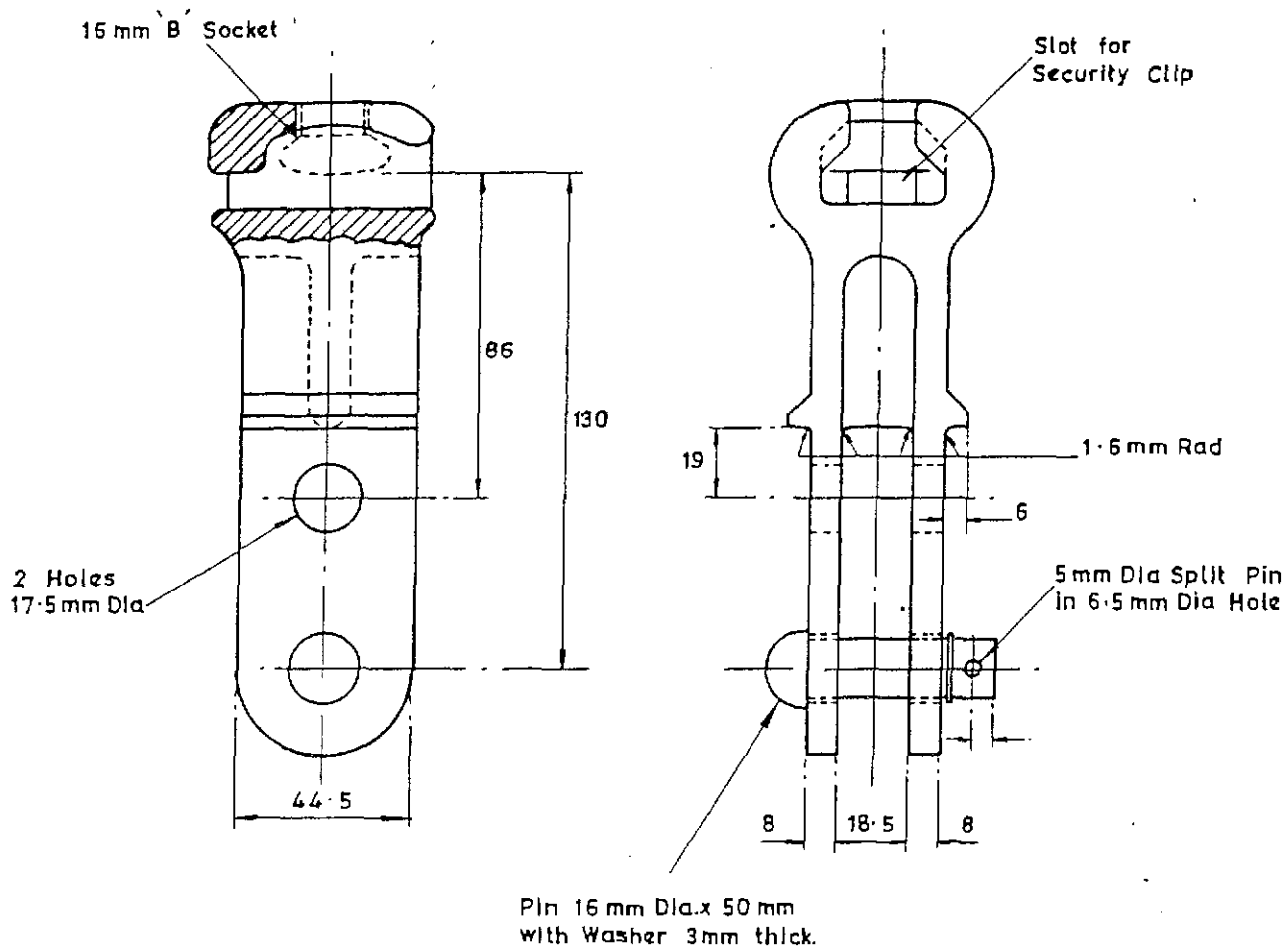
ISSUE

MIDLANDS
ELECTRICITY BOARD

DRG. No.

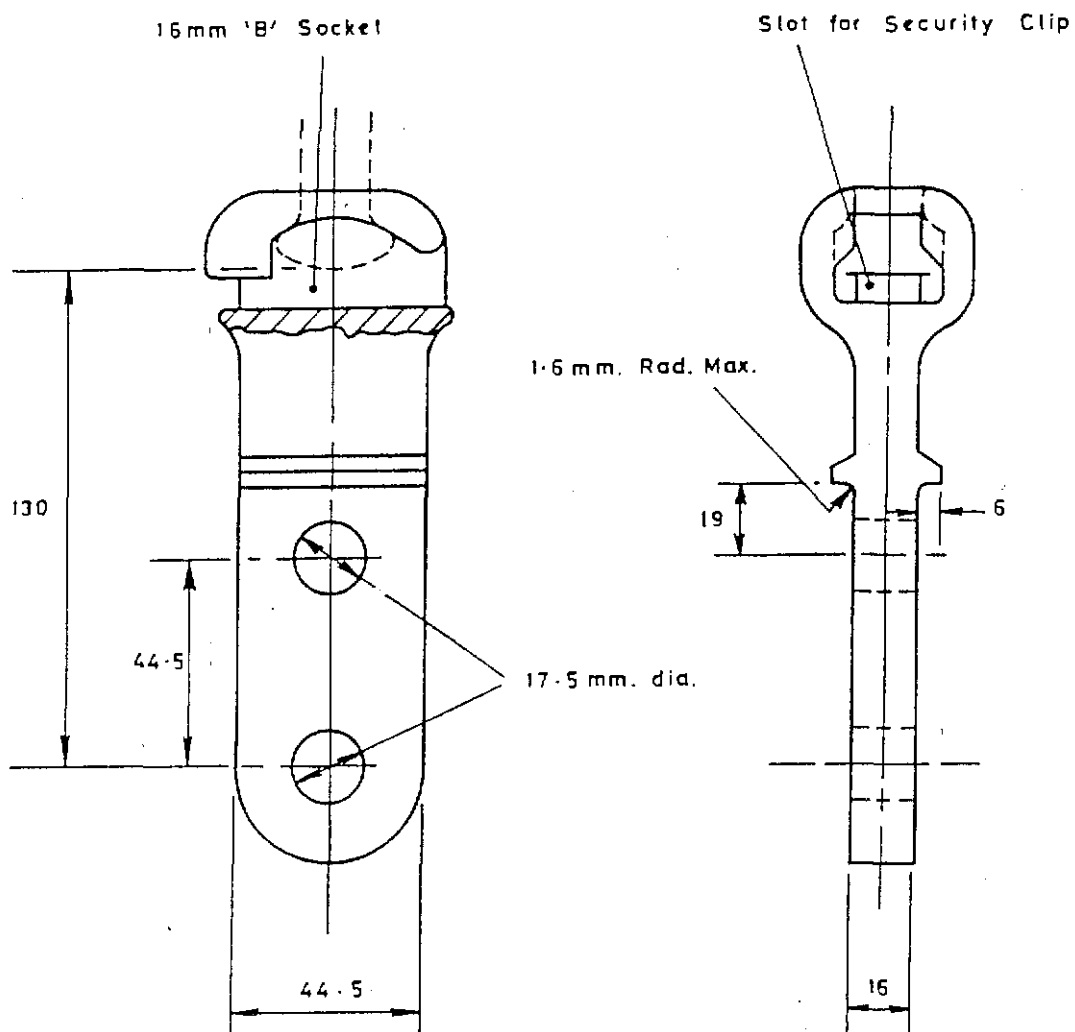
M.86.

E.S.I. DRG. No.



NOTE: To B.S 3288 P.T. 1 & 2
Fig 30 Ref 15/31

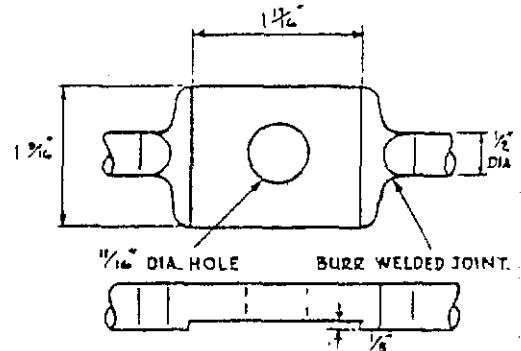
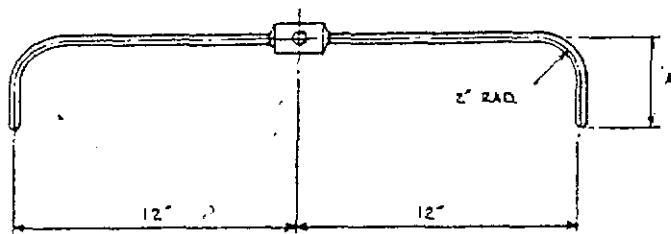
SCALE	1/2	SOCKET CLEVIS MINIMUM FAILING LOAD 70kN(7140Kgf)	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	26.10.82			DRG. No.
DRAWN	F.A.H.			M87
TRACED	M.R.J.			
CHECKED				
APPROVED	<i>[Signature]</i>	ESI. STANDARD		ESI. DRG N°



NOTE. Security Clip included.
To B.S. 3288 P.T. 1 & 2.
Fig. 36. Ref. 15 / 35.

SCALE	1 / 2	SOCKET TONGUE MINIMUM FAILING LOAD 70kN (7140 Kgf) E.S.I. STANDARD	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M.88. E.S.I. DRG. No.
DATE	26-10-82			
DRAWN	WB.			
TRACED	A.W.			
CHECKED				
APPROVED	<i>[Signature]</i>			

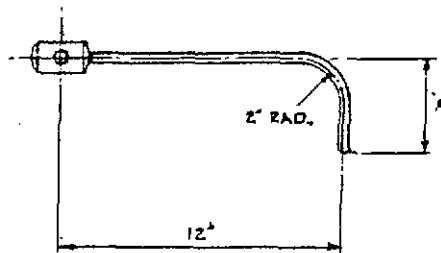
FOR SUSPENSION INSULATOR STRINGS
(33 & 66 K.V.)



DRAWING N°	A	BOLT SIZE *	BOLT SIZE *	LOCATION
M.E.89/1A	1 3/4"	5/8" x 2 1/4"	5/8" x 3"	SUSP. (LIVE END)
M.E.89/1B	6 3/8"	5/8" x 2 1/4"	5/8" x 3"	" (DEAD END) #

FORGED STEEL
FIXING BOSSES.

FOR STRAIN INSULATOR STRINGS.
(33 & 66 K.V.)



DRAWING N°	A	BOLT SIZE *	BOLT SIZE *	LOCATION.
M.E.89/2A	1 3/4"	5/8" x 2 1/4"	5/8" x 3"	STRAIN (LIVE END)
M.E.89/2B	6 3/4"	5/8" x 2 1/4"	5/8" x 3"	" (DEAD END) #

* USE 2 1/4" BOLT FOR ATTACHMENT TO HOOK, EYELINK OR TONGUE FITTING. USE 3" BOLT FOR ATTACHMENT TO CLEVIS FITTING.

OMIT ON 33 K.V.

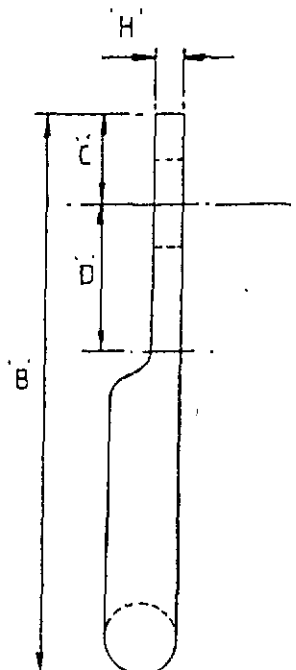
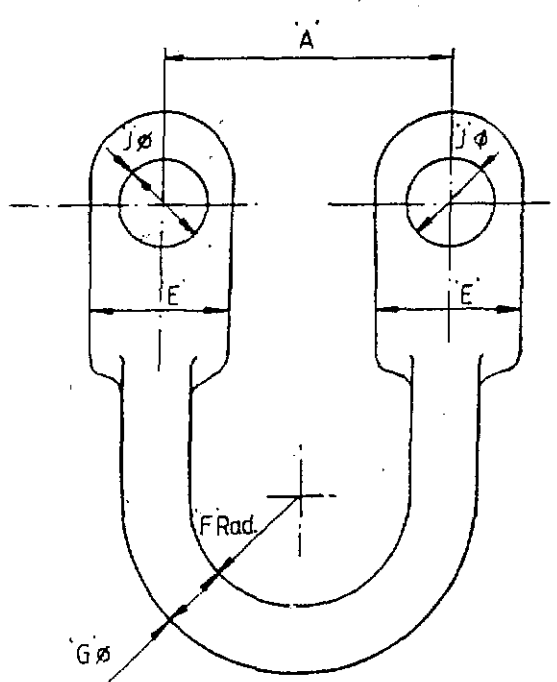
NOTE:- TO COMPLY WITH B.S. 3288 PART I

ARCING HORNS:

SCALE	1/8 INCH TO 1 FOOT	ORIENTATION	ISSUE	ISSUE	ISSUE
DATE	1 ST MARCH 1951		MF	MF	
DRAWN					
TRACED					
CHECKED					
APPROVED					
			TABLES REVISED.		
			NOTE ADDED.		

DRAWING N°

M.E. 89



ITEM	A	B	C	D	E	F	G	H	J	BOLTS	LOCATION
104/1	76.2	146	25.4	38	38	286	19	7.5	21	2-M20 x 45	66 k.V. Wood Pole Suspension
104/2	54	135	15	73	32	17	16	8	18	2-M16 x 45	66 k.V. Blaw Knox Suspension Tower

MATERIAL : Forged Steel 28/33 Ton Quality.
Bolts c/w Nut and Spring Washer.

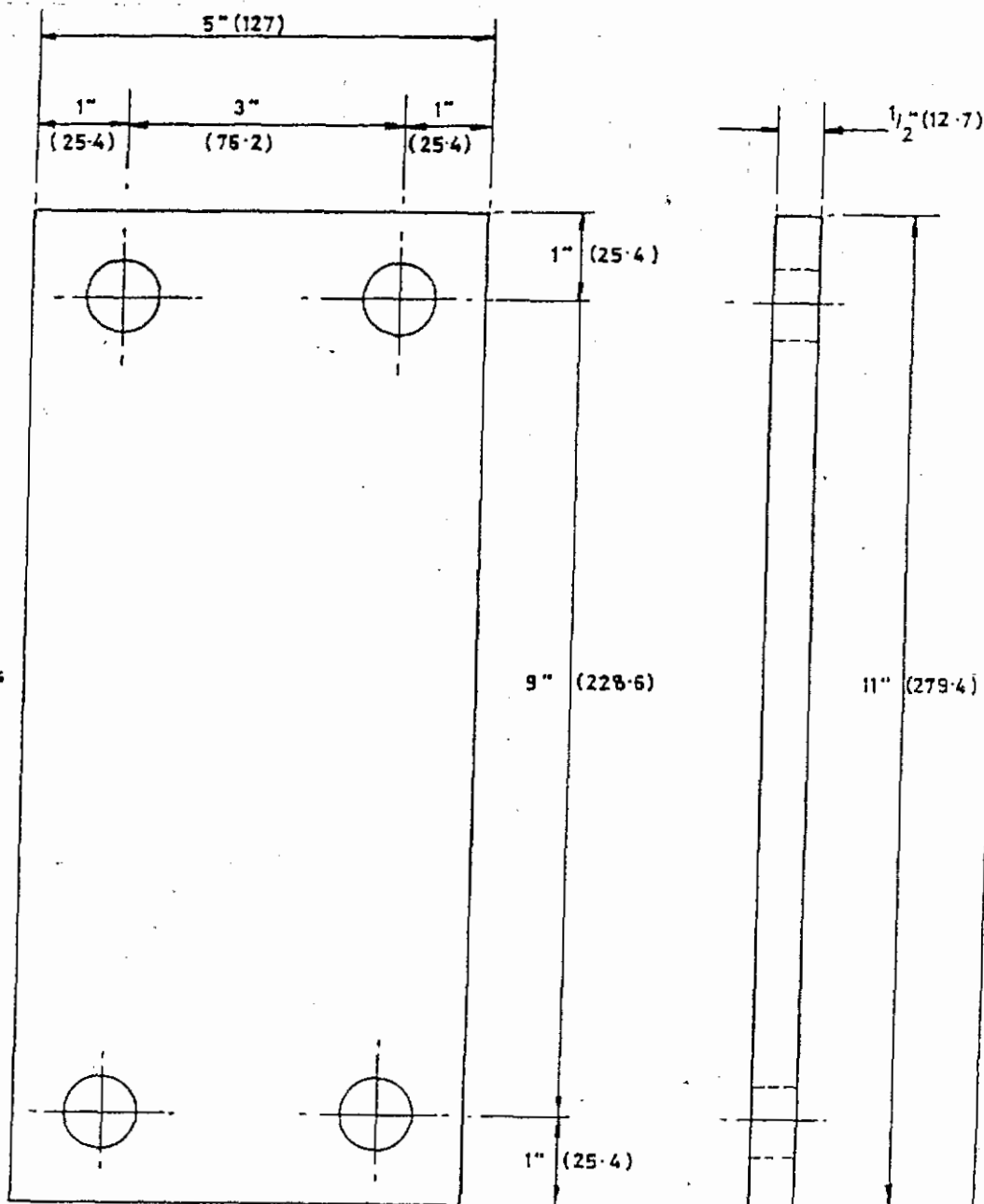
FINISH : Galv. to B.S.729

SCALE	1:20
DATE	22.1.86
DRAWN	S.J.H.
TRACED	
CHECKED	
APPROVED	<i>W.H.</i>

SUSPENSION 'U' LINK

ISSUE	MIDLANDS ELECTRICITY BOARD
	ORG. No. M.104.

4 - $\frac{13}{16}$ " (20.6) Dia Holes



MATERIAL 9" x 5" x $\frac{1}{2}$ " (228.6 x 127 x 12.7)
M.S. Plate 28/30 Ton Quality

FINISH Galv. to B.S. 729 after Cutting
& Drilling.

SCALE	1/2
DATE	12.11.62.
DRAWN	A.T.
TRACED	J.A.L.
CHECKED	
APPROVED	<i>[Signature]</i>

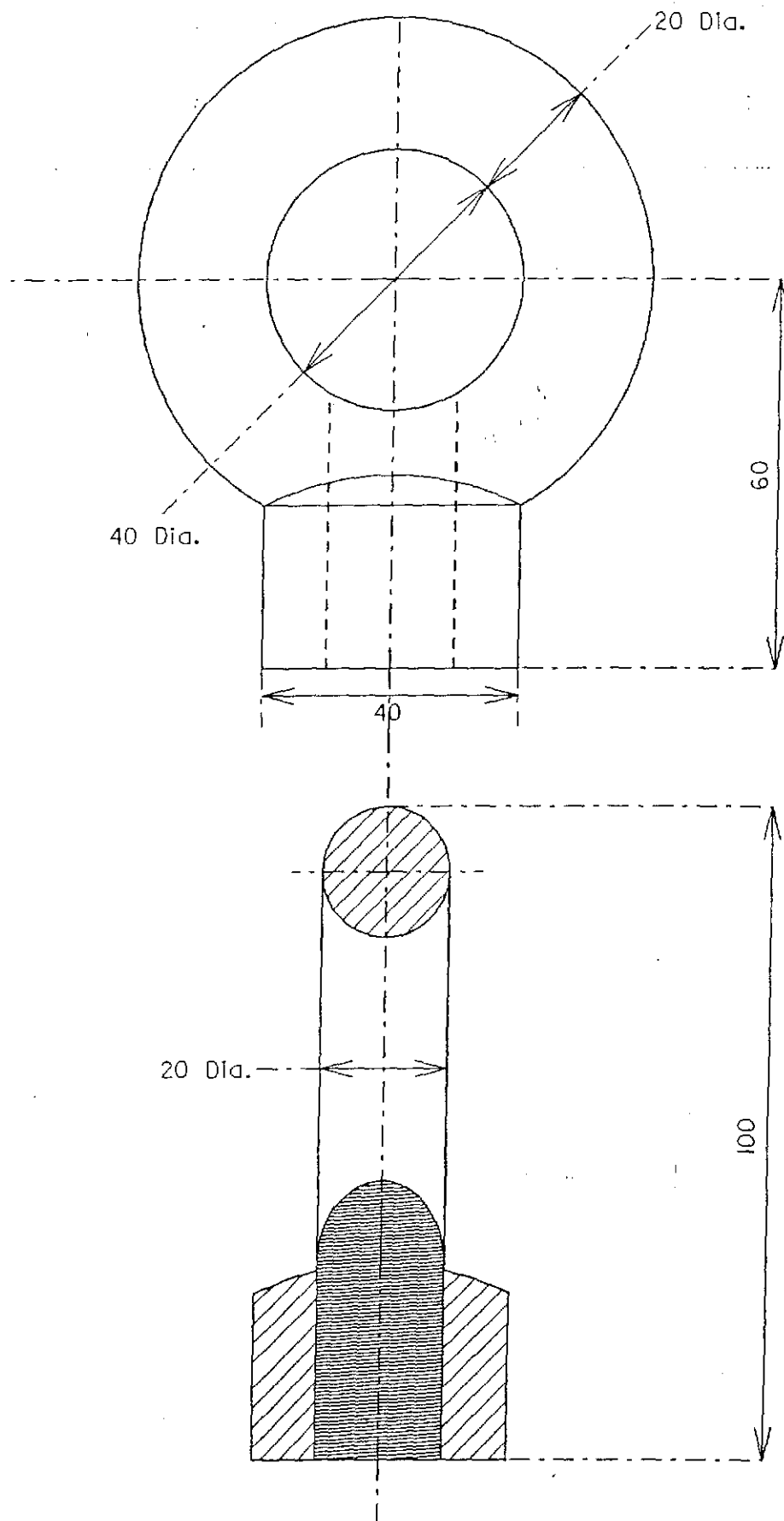
GUSSET PLATE
(66 KV Susp)

ISSUE	B.M.
Re-drawn	26.3.70.

MIDLANDS
ELECTRICITY BOARD

DRG. No.
M.E. 105.

(MF)



MATERIAL
M.S. 28/33 ton quality
FINISH
Galv to BS 729
Screw threads to be wellalied

Drilled & tapped M20

SCALE	-
DATE	9.5.00
DRAWN	A00
TRACED	
CHECKED	
APPROVED	

M.S. EYE NUT

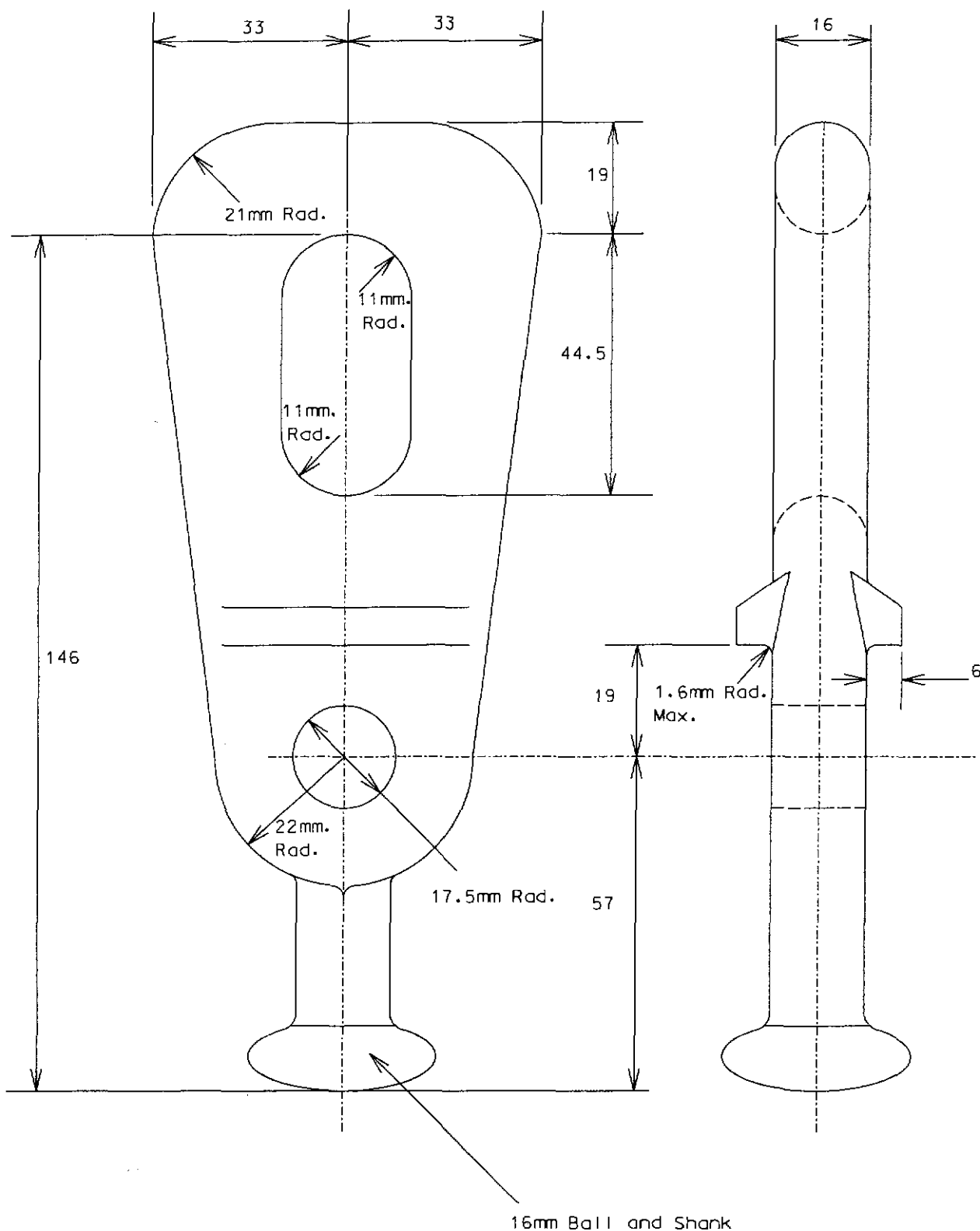
ISSUE	DATE

Produced by
Output Services

GPU
POWER UK

DRG. No. M108

DRG. No. M108



NOTE.

To B.S. 3288 P.T. 1&2 Fig.19 Ref.15/30

SCALE	1:10
DATE	26.10.82
DRAWN	W.B.
TRACED	C.E.
CHECKED	-
APPROVED	-

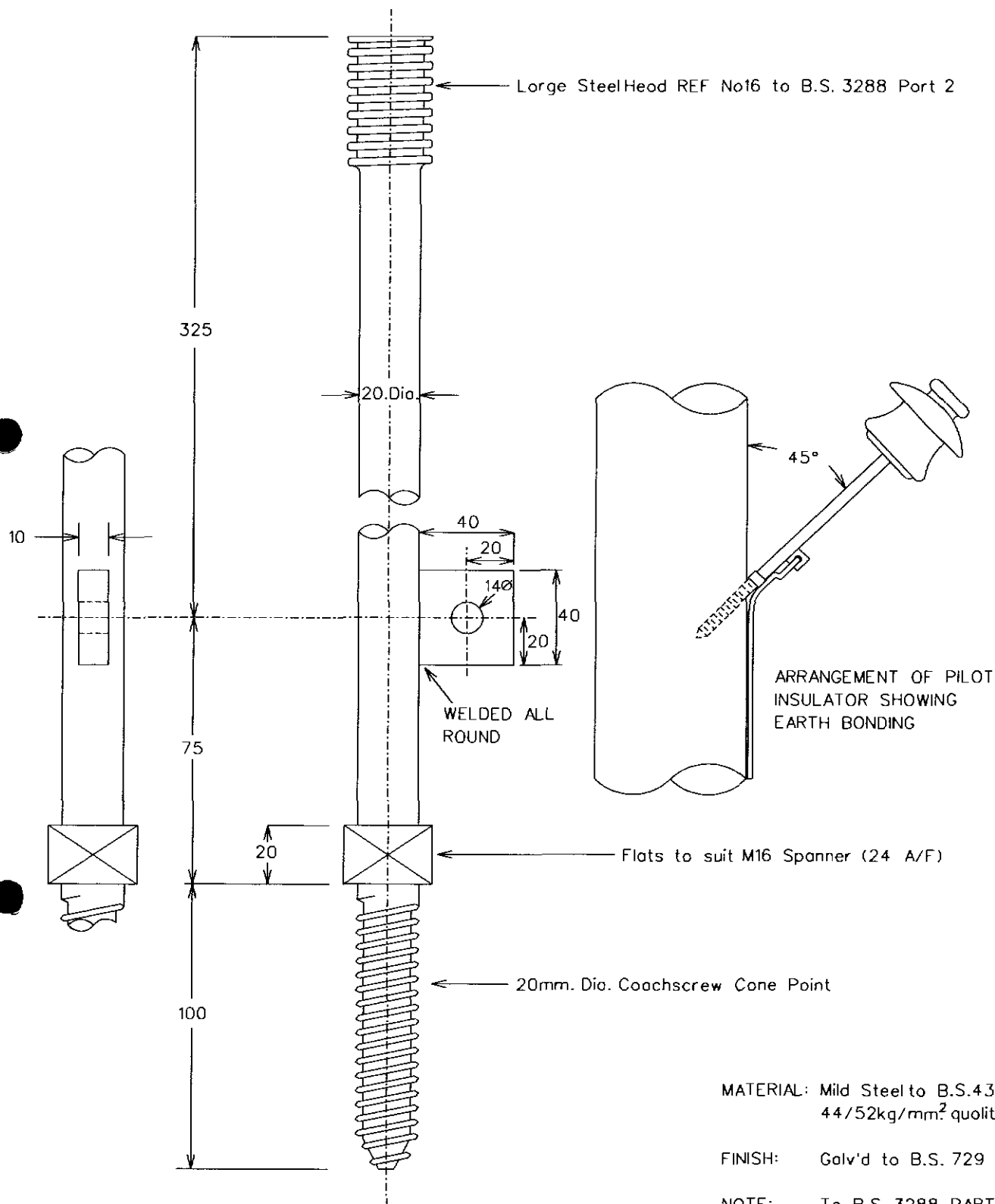
BALL ENDED EYE LINK
MINIMUM FAILING LOAD 70 KN

ISSUE	DATE

Produced by Output Services	
DRG. No.	M133

DRG. No.

M133



MATERIAL: Mild Steel to B.S.4360
44/52kg/mm² quality.

FINISH: Galv'd to B.S. 729

NOTE: To B.S. 3288 PART 2
Spanner Flats Can Be
Located Adjacent To
Screw or Head

All dimensions are in millimetres unless otherwise stated.

SCALE	1-2
DRAWN	04/03/04 M.J.B
TRACED	-
CHECKED	05/03/04
APPROVED	KD

11 kV. & 33 kV PILOT INSULATOR SPINDLE TIMBER FIXING

ISSUE	DATE

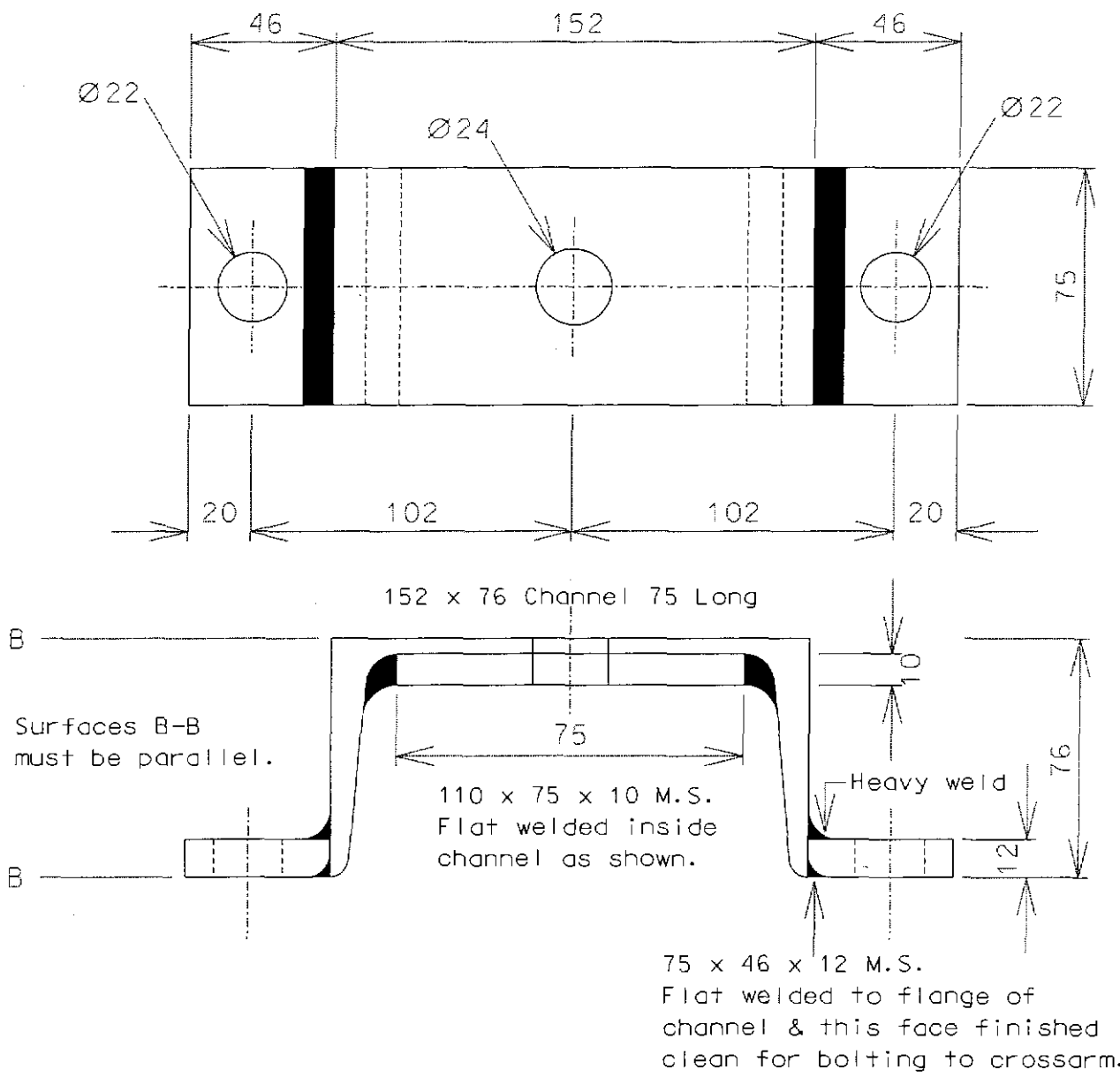


Aquila

DRG. No.

M154

Produced by
Primary CAD



MATERIAL
 152 x 76 x Channel to B.S.4
 75 x 46 x 12 Plate
 110 x 75 x 10 Plate

FINISH
 Welded joints to be pickletight.
 Galv. to B.S. 729 after cutting,
 drilling & welding.

For mounting centre insulator, 33 kV Intermediate Pole.

SCALE	1 / 2
DATE	12/9/00
DRAWN	ADD
TRACED	
CHECKED	27/9/00
APPROVED	KD

INSULATOR BRACKET

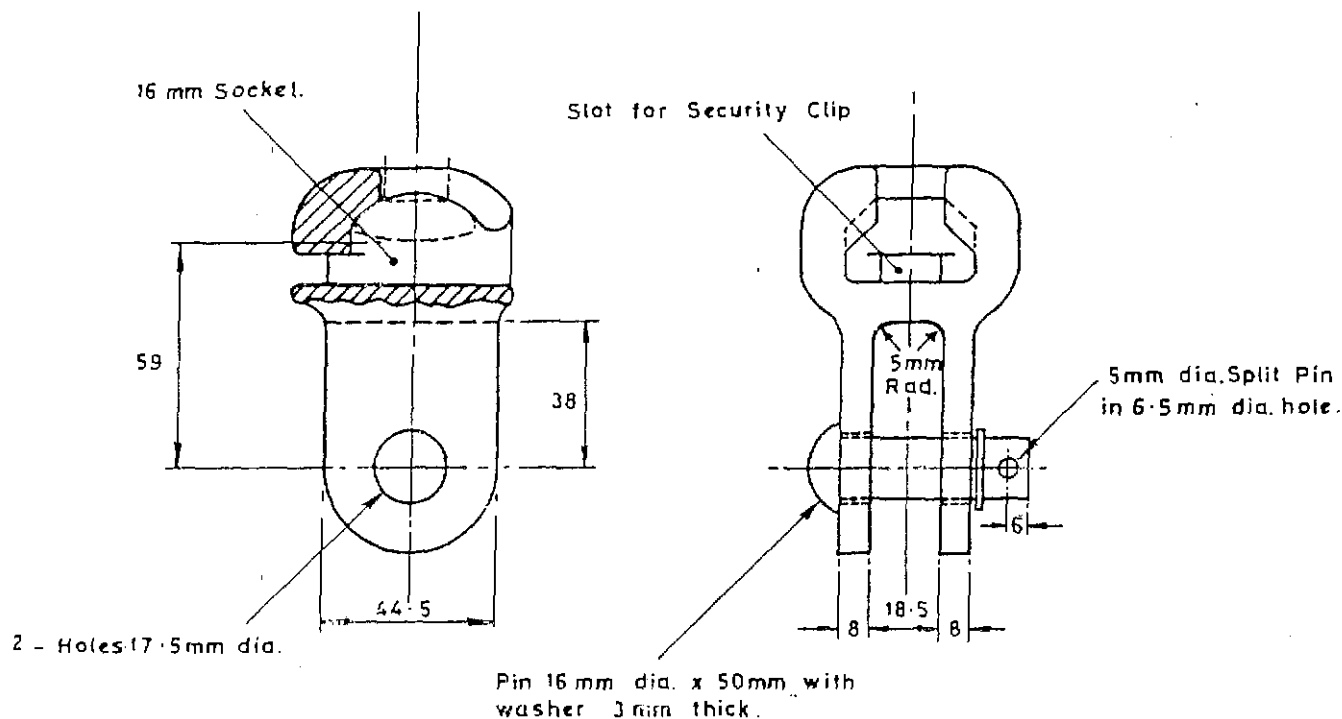
ISSUE	DATE

Produced by
Output Services

GPU
POWER UK

DRG. No.
M157

DRG. No. M157



NOTE. To B.S. 3288 P.T.1 and 2. Fig. 31. Ref. 15/84.

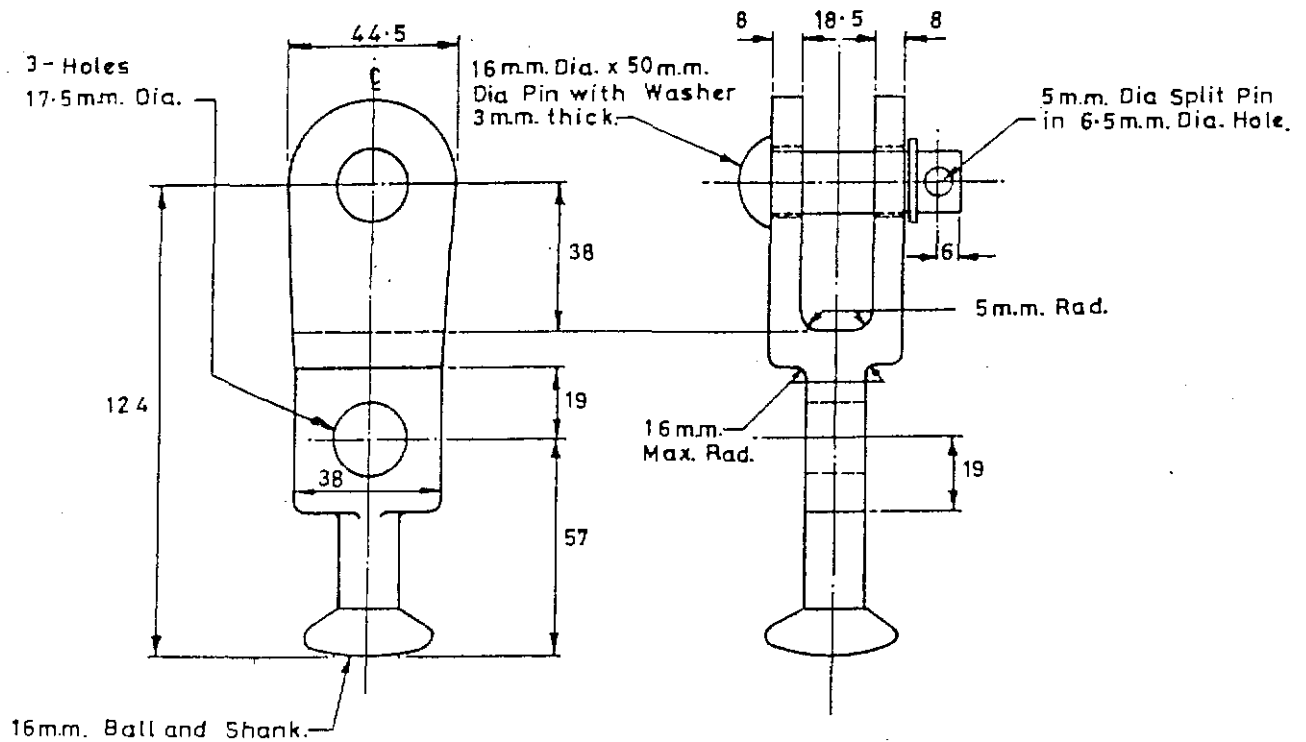
SCALE	1/2
DATE	22.9.72
DRAWN	F.A.H.
TRACED	A.W.
CHECKED	
APPROVED	<i>[Signature]</i>

OZALID 163397

SOCKET CLEVIS

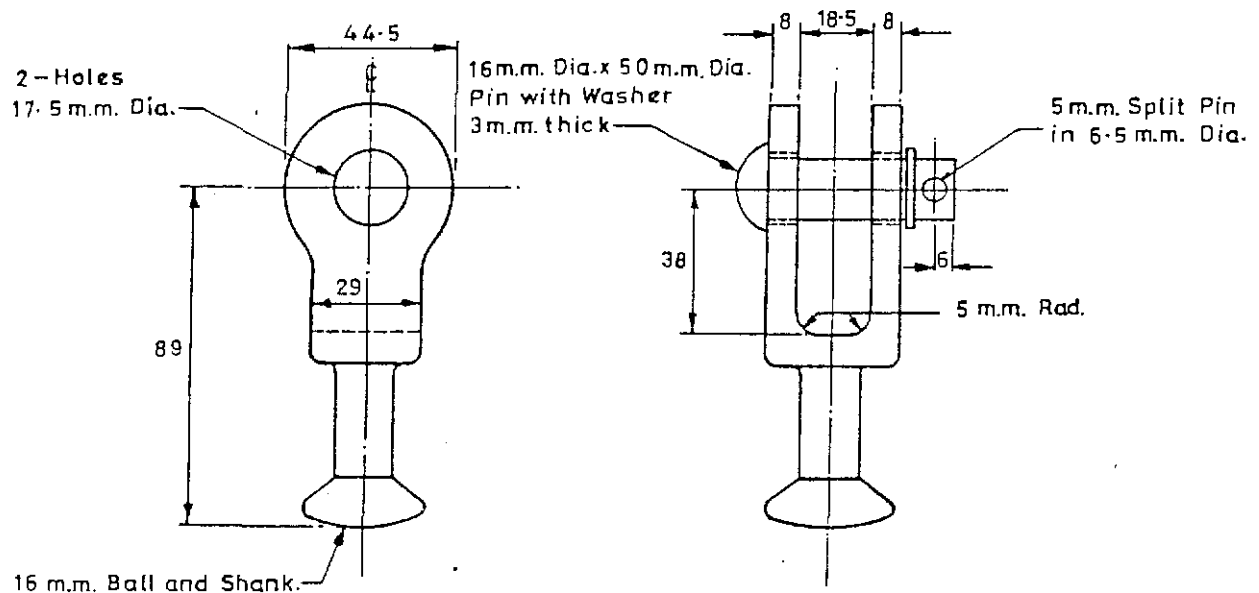
MINIMUM FAILING LOAD 70 kN.(7140 Kgf)

ISSUE -	MIDLANDS
D	ELECTRICITY BOARD.
Redrawn & converted to metric.	DRG. No.
	M .184.



ME 185/1

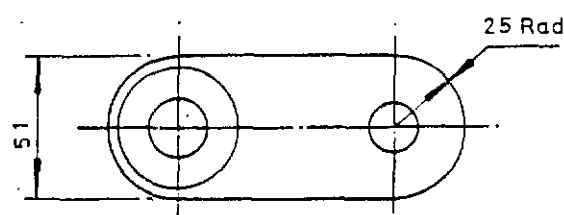
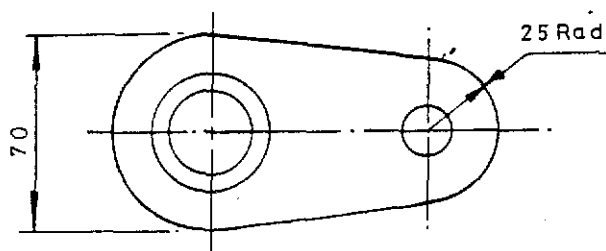
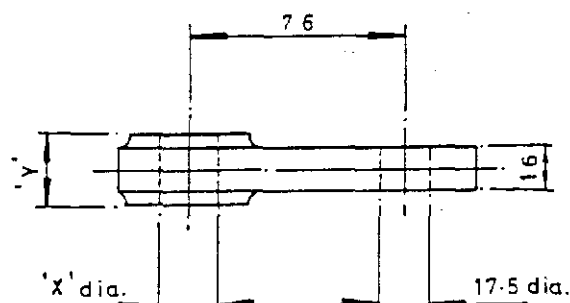
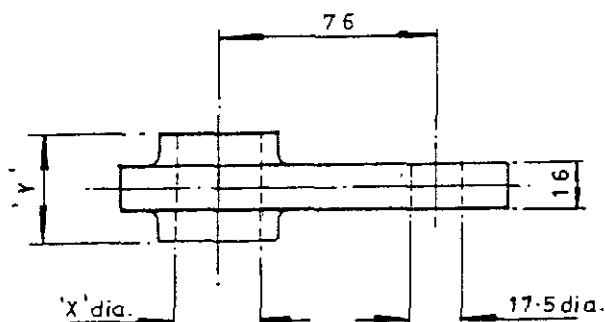
NOTE. To B.S. 3288 PT1 & 2. Fig. 23 Ref. 15/25



ME 185/2

NOTE To B.S. 3288 PT1 & 2 Fig. 24 Ref. 15/83

SCALE	1/2	BALL CLEVIS MINIMUM FAILING LOAD 70 kN (7140 kgf.)	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M 185.
DATE	20.9.72		E	
DRAWN	E.A.H.		ME N°s	
TRACED	S.J.H.		Added	
CHECKED	S.J.H.		R.A.C. 2-278	
APPROVED				



M. N°	B.S. Ref.	'X'	'Y'	M.F.L.
235/1	15/27 B	26	32	7140 kgf
235/2	15/27 C	29	38	7140 kgf

To B.S. 3288 Parts 1 & 2. Fig. 44.

M. N°	B.S. Ref.	'X'	'Y'	M.F.L.
235 / 3	15/27 A	21	25	7140 kgf

To B.S. 3288 Part 2

- M. 235/1 For use with new pattern 100 mm² to 175 mm² ACSR conductors. dead end ME 159.
M. 235/2 For use with 400 mm² A.C.S.R. dead end.
M. 235/3 For use with old pattern 150 mm² ACSR dead end.

NOTE

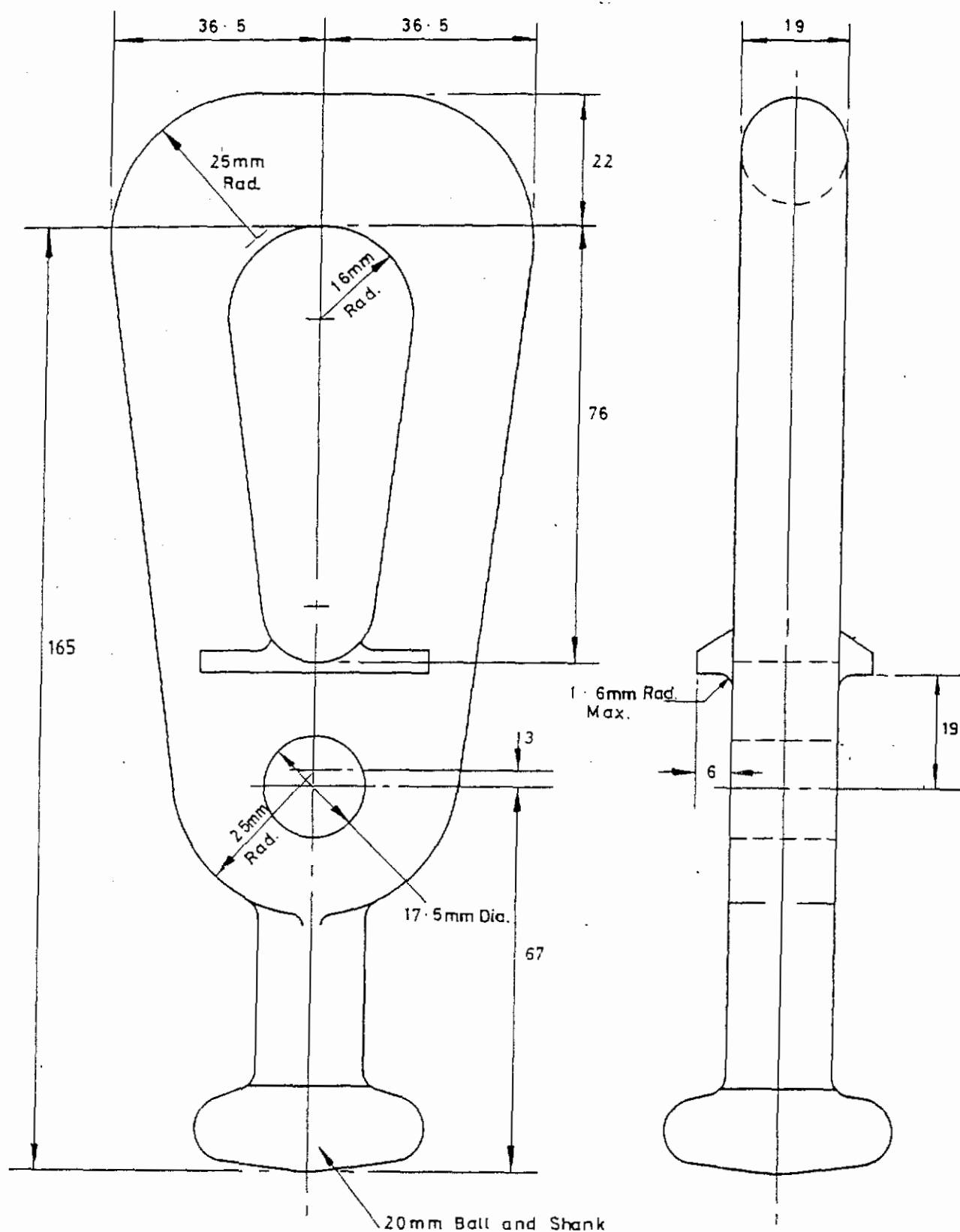
To be used with M.E.184 or M87

SCALE	-
DATE	26.10.82
DRAWN	A.R.B.
TRACED	S.J.H.
CHECKED	<i>[Signature]</i>
APPROVED	<i>[Signature]</i>

CLAMP ADAPTORS
MINIMUM FAILING LOAD
AS TABLED.

E.S.I. STANDARD

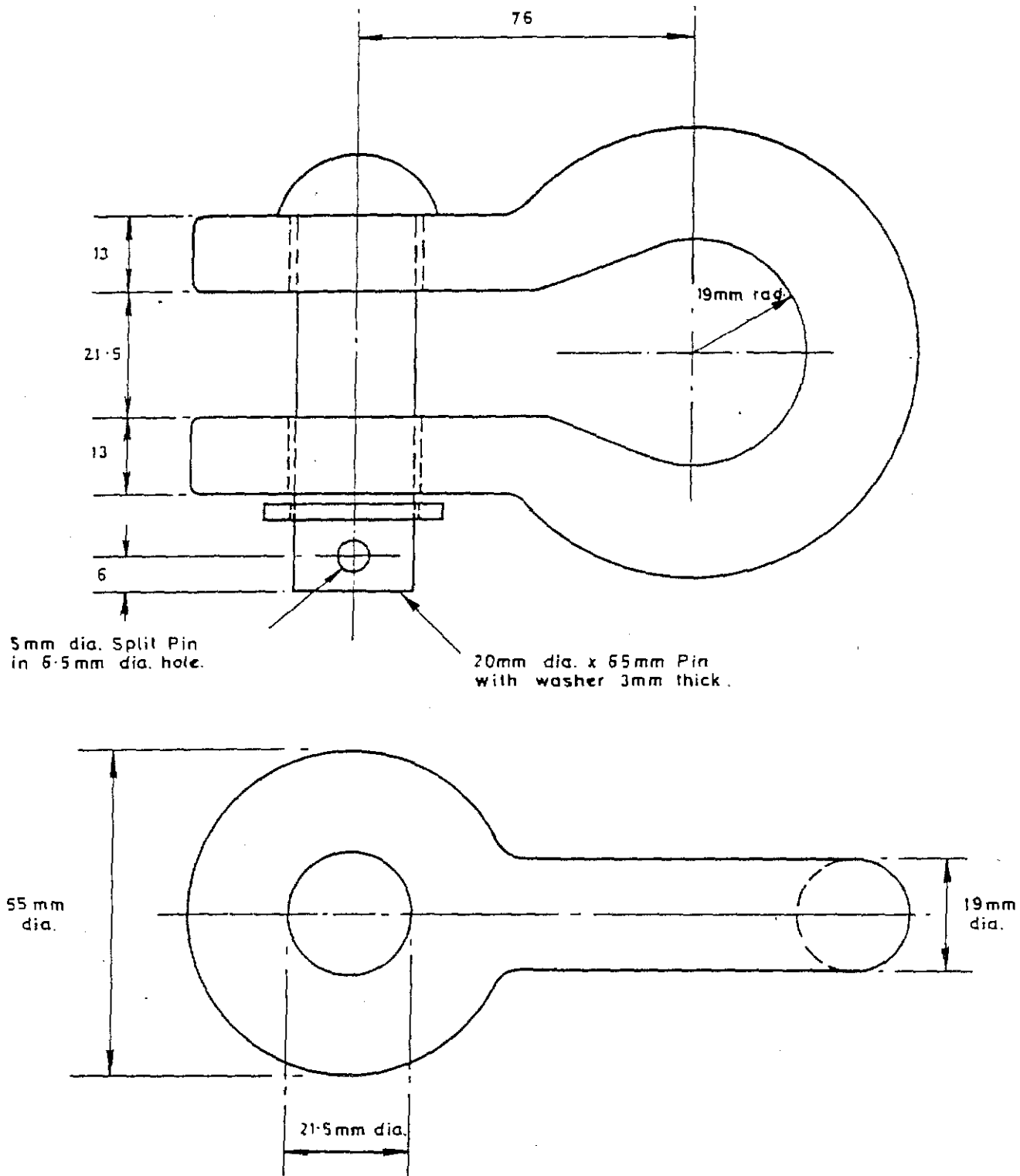
ISSUE	MIDLANDS ELECTRICITY BOARD
A.	DRG. No.
235/2	M.235.
MFL rev.	
29 10 85	E.S.I. DRG.N°



NOTE:-

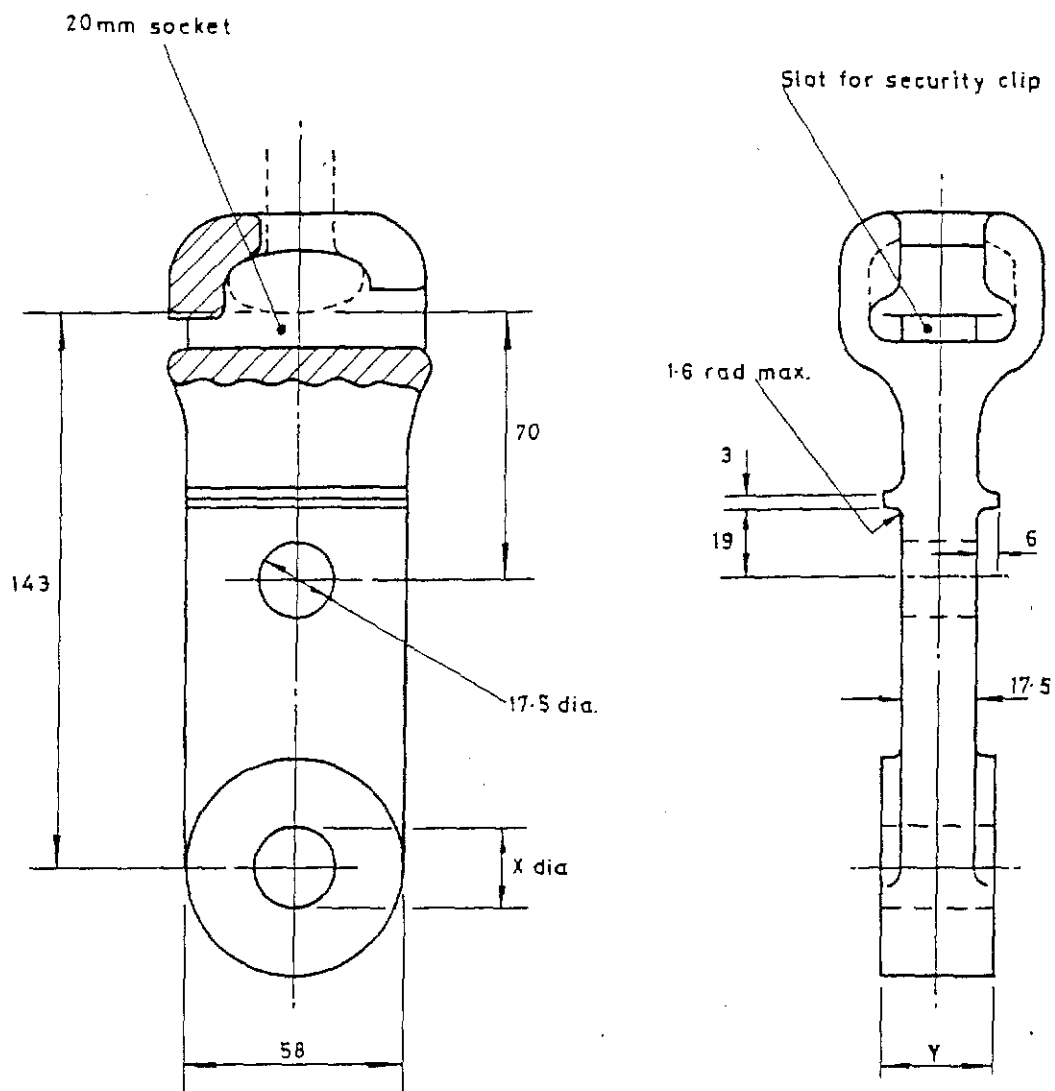
To B.S. 3288 PT 1&2 Fig.16. Ref.28/30

SCALE	1/1	BALL ENDED EYE LINK MINIMUM FAILING LOAD 12,700 Kgf. E.S.I. STANDARD	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M.249. E.S.I. DRG. No.
DATE	26.10.82			
DRAWN	W.B.			
TRACED	C.E.			
CHECKED				
APPROVED				



NOTE: To B.S. 3288 P.T. 1 and 2, Fig. 12 Ref. 28 / 29.

SCALE	1/1	'D' SHACKLE MINIMUM FAILING LOAD 12.700 Kgf.	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M.250 E.S.I. DRG. N°
DATE	26.10.82			
DRAWN	W.B.			
TRACED	A.W.			
CHECKED				
APPROVED		E.S.I. STANDARD		



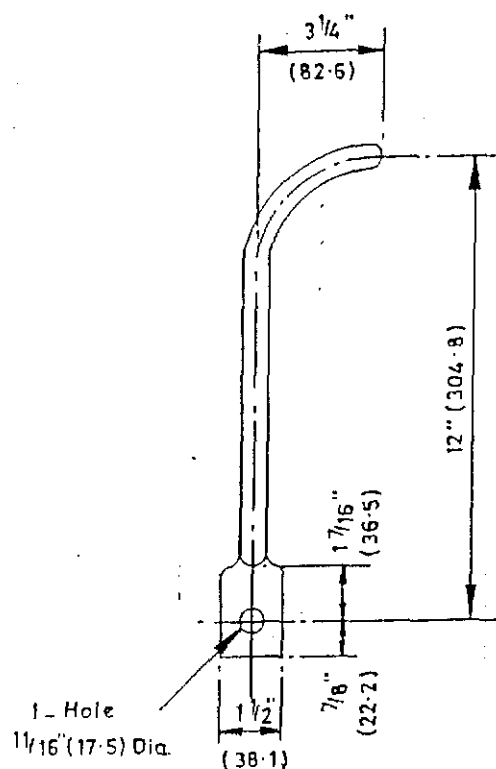
NOTE : SECURITY CLIP INCLUDED

M N°	BS REF N°	X	Y
M.251/1.	28/36A	21.5	25
M.251/2	28/36B	26	32

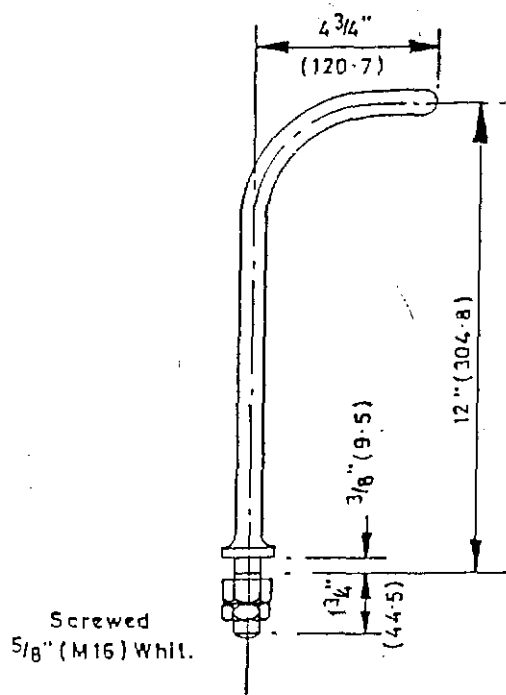
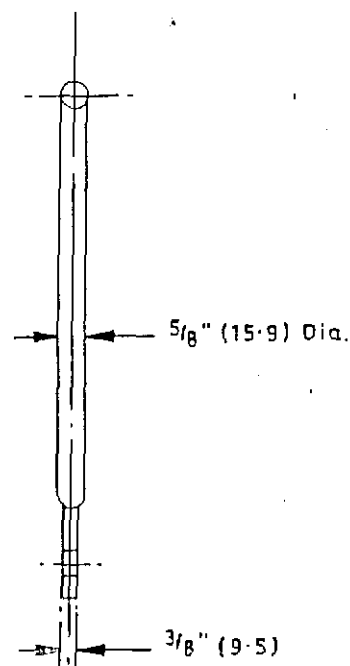
TO BS3288 FIG.41 REF.28/36

ALL DIMENSIONS IN MILLIMETRES UNLESS STATED

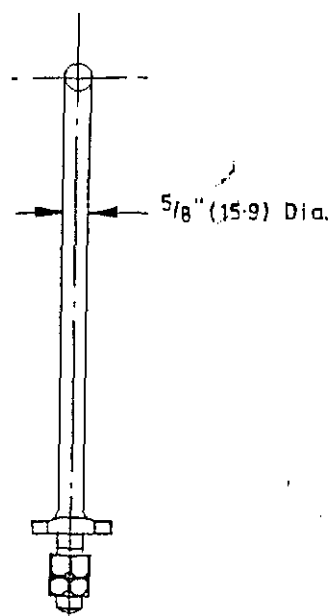
SCALE	1/2	SOCKET TONGUE MINIMUM FAILING LOAD 12700 Kgf.	ISSUE	MIDLANDS
DATE	25-10-82			ELECTRICITY BOARD
DRAWN	WB			DRG N°
TRACED	L.F.M.			M.251.
CHECKED	J.M.B.			
APPROVED		E.S.I. STANDARD.		E.S.I. DRG.N°



LIVE END



DEAD END



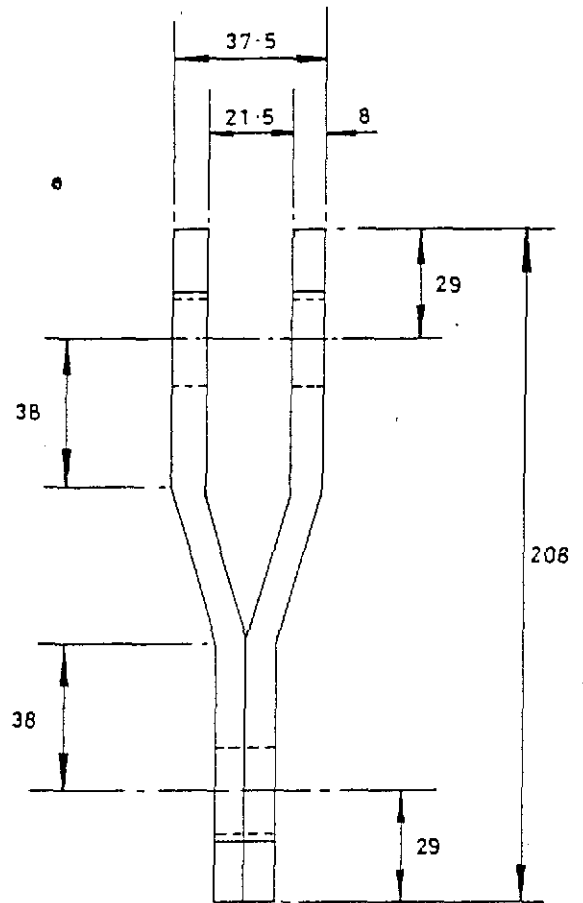
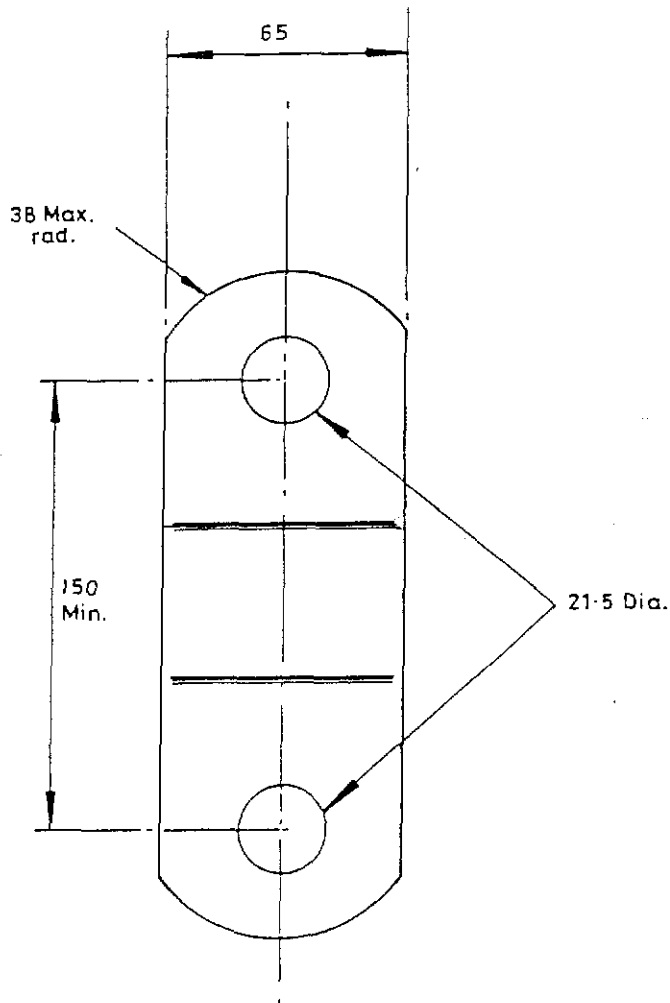
Unless otherwise indicated all dimensions are in mm.

SCALE	1/5
DATE	29-7-76
DRAWN	F.A.H.
TRACED	P.G.
CHECKED	
APPROVED	

ARCING HORNS
(FOR USE WITH 12700kg MFL EQUIP.)

ISSUE	MIDLANDS
B	ELECTRICITY BOARD
Redrawn Converted to metric.	DRG No.
29-7-76	ME 255

GAF 9736



NOTE :- FOR ASSEMBLY DRG. SEE ME 90/2.
HEAVY CONSTRUCTION STEEL TOWERS.
 $150\text{mm}^2 \times 175\text{mm}^2$ A.C.S.R.
(20mm BALL & SOCKET FITTINGS)

MINIMUM FAILING LOAD 125 kN.

TO BS. 3286 REF 28/87

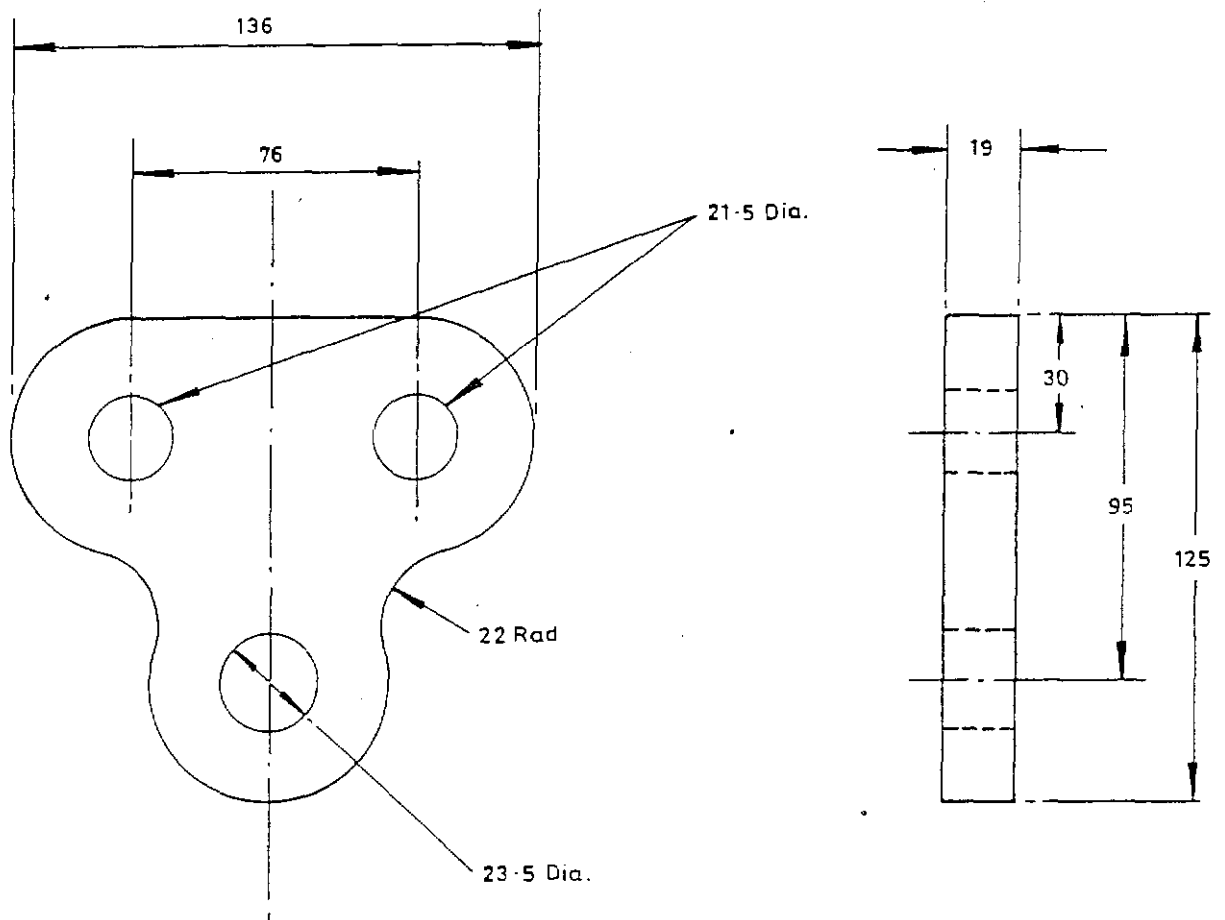
MAXIMUM WORKING LOAD 3625 Kgf (8,000 lbf)

MATERIAL:— M.S. Strip (28-33 Ton quality)

FINISH:— Galv to BS 729 of

* COMPLETE WITH 20% CLEVIS PIN SPLIT PIN & WASHER

SCALE	N.T.S.	CRANKED LINKS MFL 12700 Kgf.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	21/2/78		A	DRG. No.
DRAWN	R.A.C.		* NOTE ADDED R.A.C. 10/85	M. 306
TRACED	P.G.			
CHECKED	9/11/78			
APPROVED				
2044G		ESI STANDARD		ESI ORG N°



NOTE :- FOR ASSEMBLY DRG. SEE ME 90/2
 HEAVY CONSTRUCTION STEEL TOWERS
 150mm² x 175mm² A.C.S.R.
 (20mm BALL & SOCKET FITTINGS)

MINIMUM FAILING LOAD 125 kN.

TO BS 3288 REF 28/21

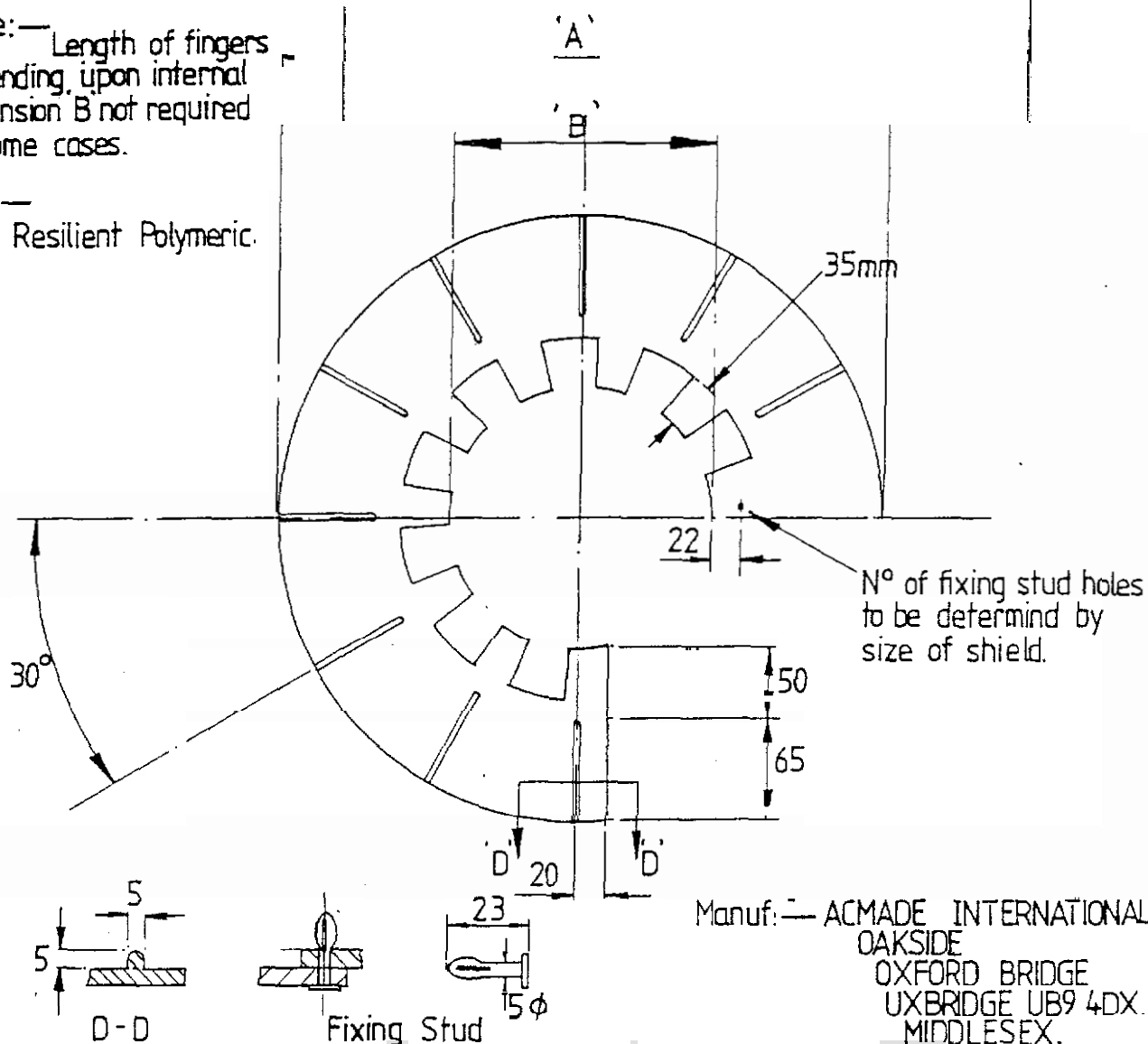
MAXIMUM WORKING LOAD 3629 Kgf (8000 lbf)

SCALE	N.I.S.	LANDING PLATE MFL 12,700 kgf.	ISSUE	MIDLANDS
DATE	21/2/78			ELECTRICITY BOARD
DRAWN	R.A.C.			DRG. No.
TRACED	P.G.			M 307
CHECKED	C.M.P.			
APPROVED	<i>[Signature]</i>			
ESI STANDARD			ESI DRG N°	

Note:— Length of fingers depending upon internal dimension B not required in some cases.

Matl:—

Resilient Polymeric.



Shield	Equipment	Equipment Drawing No	Location on Equipment	Porcelain core Dia mm.	Dimension A mm	Dimension B mm
5	Pirelli 66 kV outdoor S/E.	W1-532012	Nº 2&3 Sheds.	145 - 180	500	200
5A	MP Surge Diverter	LA 4043	All Sheds.	158 - 165	430	200
5B	Pirelli 66kV outdoor S/E.	W1-532012	Top Shed.	166	500	270
5C	33kV Line Post Insulator	ME. 165	Alternate Sheds	120	410	200
5D	BICC 33kV outdoor S/E.	D90061-99	All Sheds.	142 - 246	346	200
6	Pirrelli 66kV outdoor S/E.	W1-532012	Nº 4&5 Sheds	185 - 220	560	200
6A	" "	"	Bottom Shed	214	560	290
7	132kV outdoor S/E			165 - 200	730	225
8	" "			210 - 245	780	285

SCALE	N.T.S.
DATE	5.8.80
DRAWN	24.C
TRACED	
CHECKED	
APPROVED	

VANDAL SHIELD.

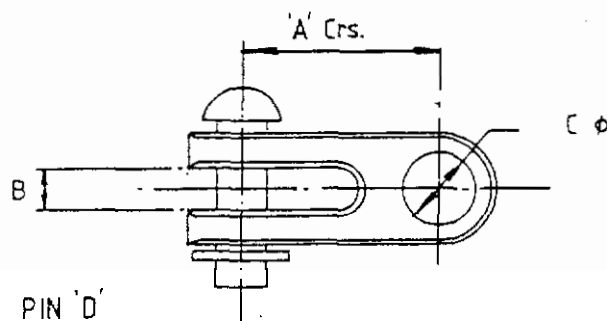
ISSUE

MIDLANDS

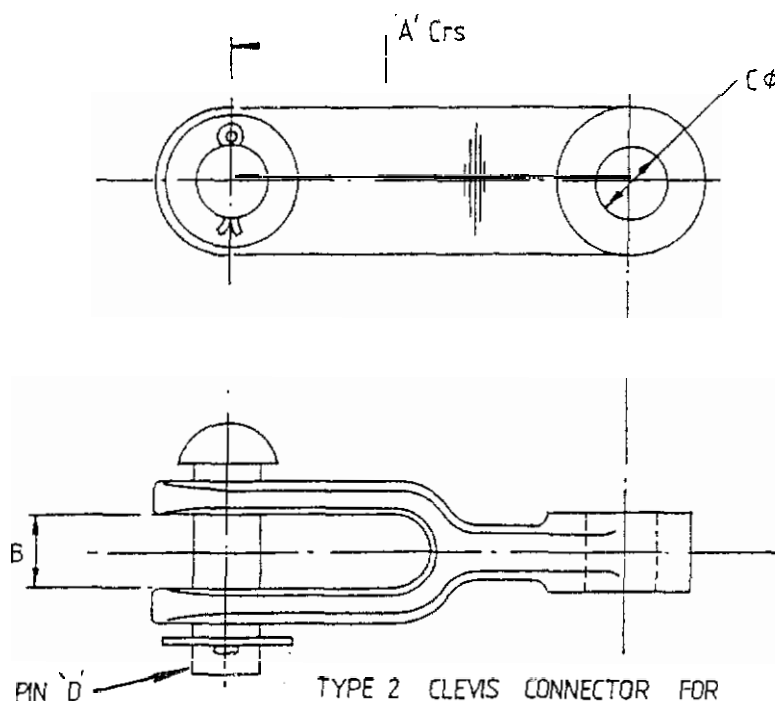
ELECTRICITY BOARD

DRG. No.

M. 316



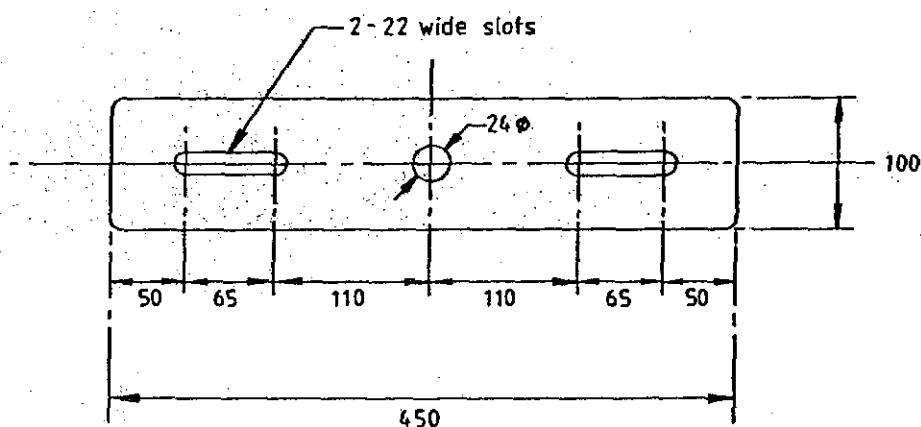
TYPE 1 CLEVIS CONNECTOR FOR
PRE - 1960 TOWERS &
L4M TOWERS.



TYPE 2 CLEVIS CONNECTOR FOR
POST 1960 TOWERS
EXCLUDING L4M TOWERS

TYPE	MANUFACTURER REF		DIMENSIONS (mm)				PIND L x φ
	BICCON	UNIDARE	A	B	C		
M 343 / 1	Style B	V-9	76	16	27		65 x 20
M 343 / 2	" 'C'	VA-9	152	28.6	27		80 x 25

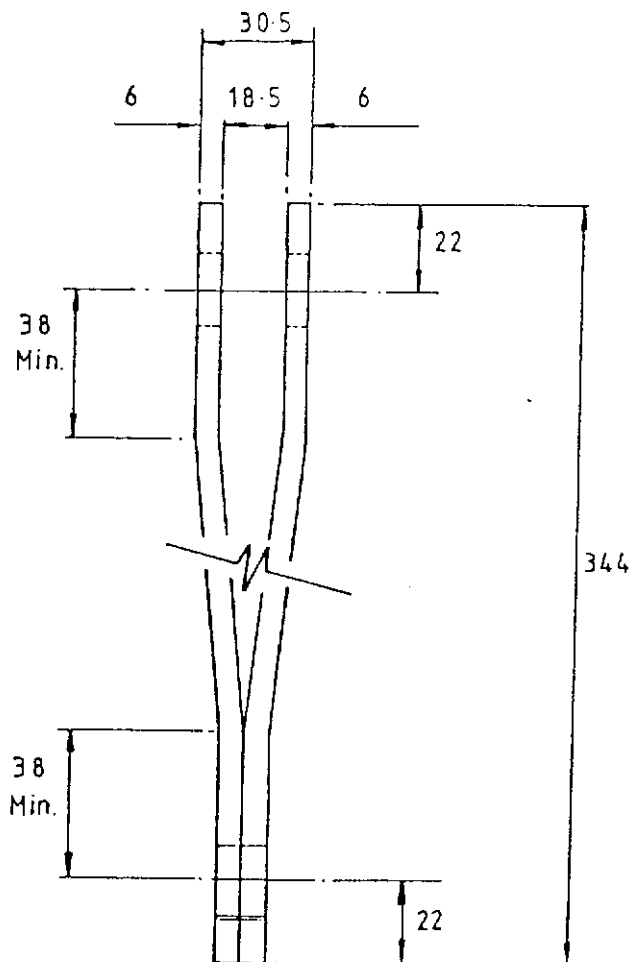
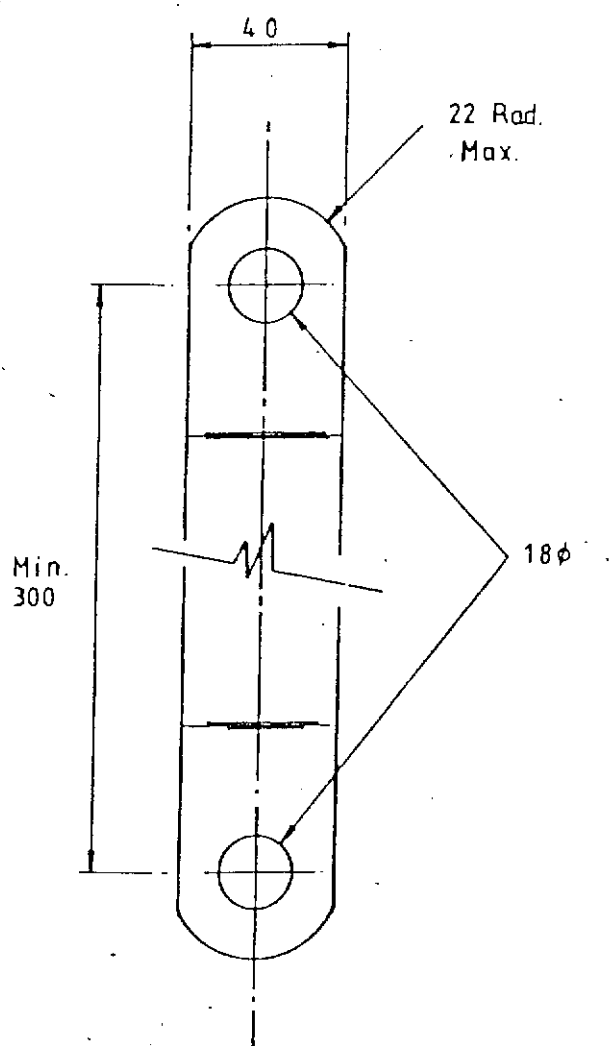
SCALE	NTS.	EARTHWIRE FITTINGS CLEVIS CONNECTORS FOR 70mm ² & 175mm ² EARTHWIRE TERMINATIONS	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M.343.
DATE	15-7-82			
DRAWN	R.A.C.			
TRACED				
CHECKED				
APPROVED	CCN	ESI. STANDARD		ESI Drg N°.



MATERIAL - 450 x 100 x 15 MS Flat
Galv. to BS729 After
Cutting & Drilling.

APPROVED BY: DATE:	TITLE INSULATOR SUPPORT PLATE	SCALE	---	MEB Midlands Electricity plc Dwg No M397
		DATE	1-6-93	
		DRAWN	A.J.B.	
		TRACED	P.J.W.	
		CHECKED		
		APPROVED		

AMDO 15628 E



MINIMUM FAILING LOAD 70 kN

TO BS. 3288 REF. 15b7

MAXIMUM WORKING LOAD 2268 kgf (5000 lbf.)

Supplied with 16φ×50 Pin, washer and split pin.

SCALE	1:20
DATE	20-2-95
DRAWN	A.J.B.
TRACED	
CHECKED	
APPROVED	WDH

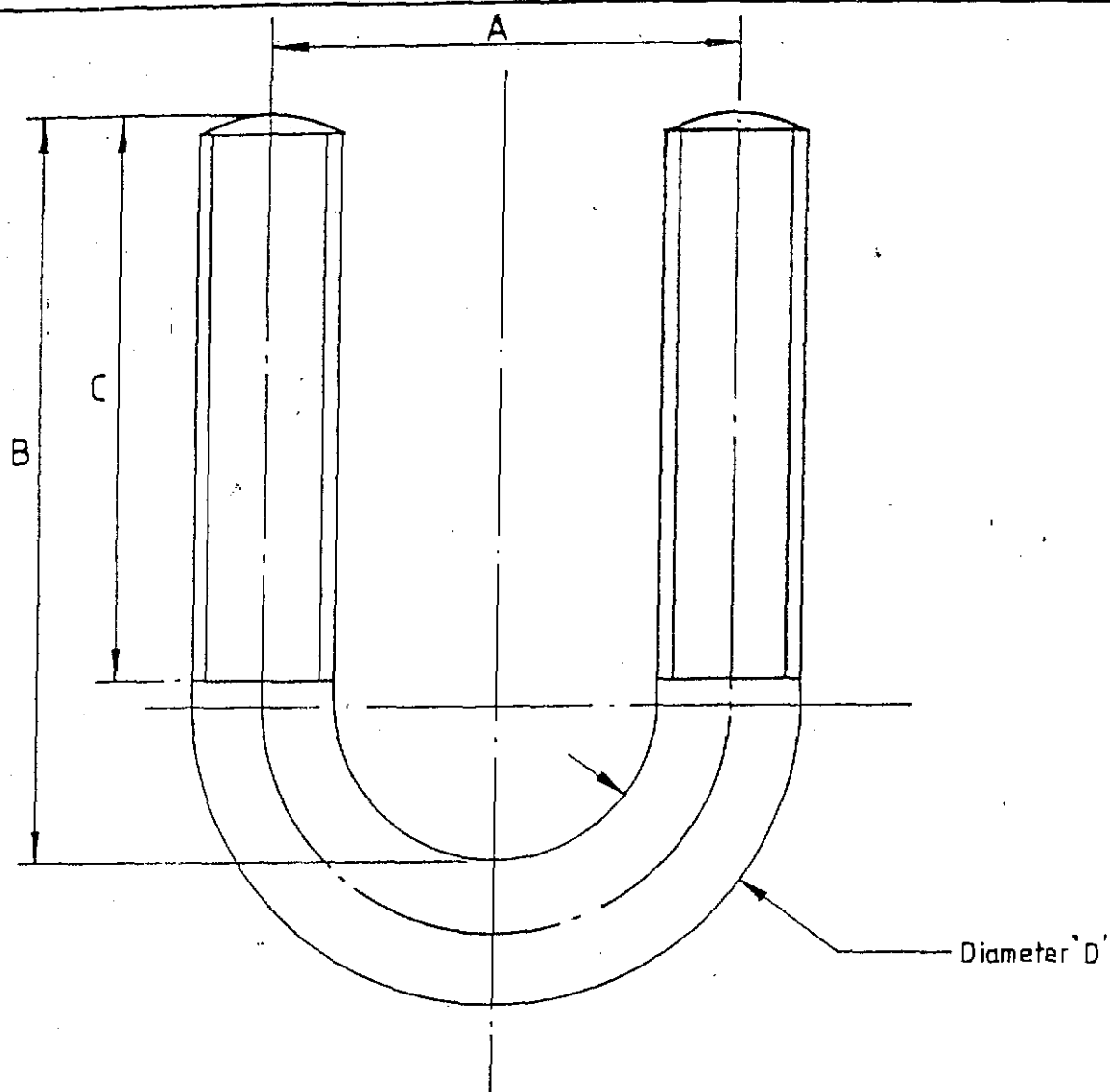
EXTENSION LINK
MFL 70kN

ISSUE

MIDLANDS
ELECTRICITY PLC

DRG. No.

M 401/1

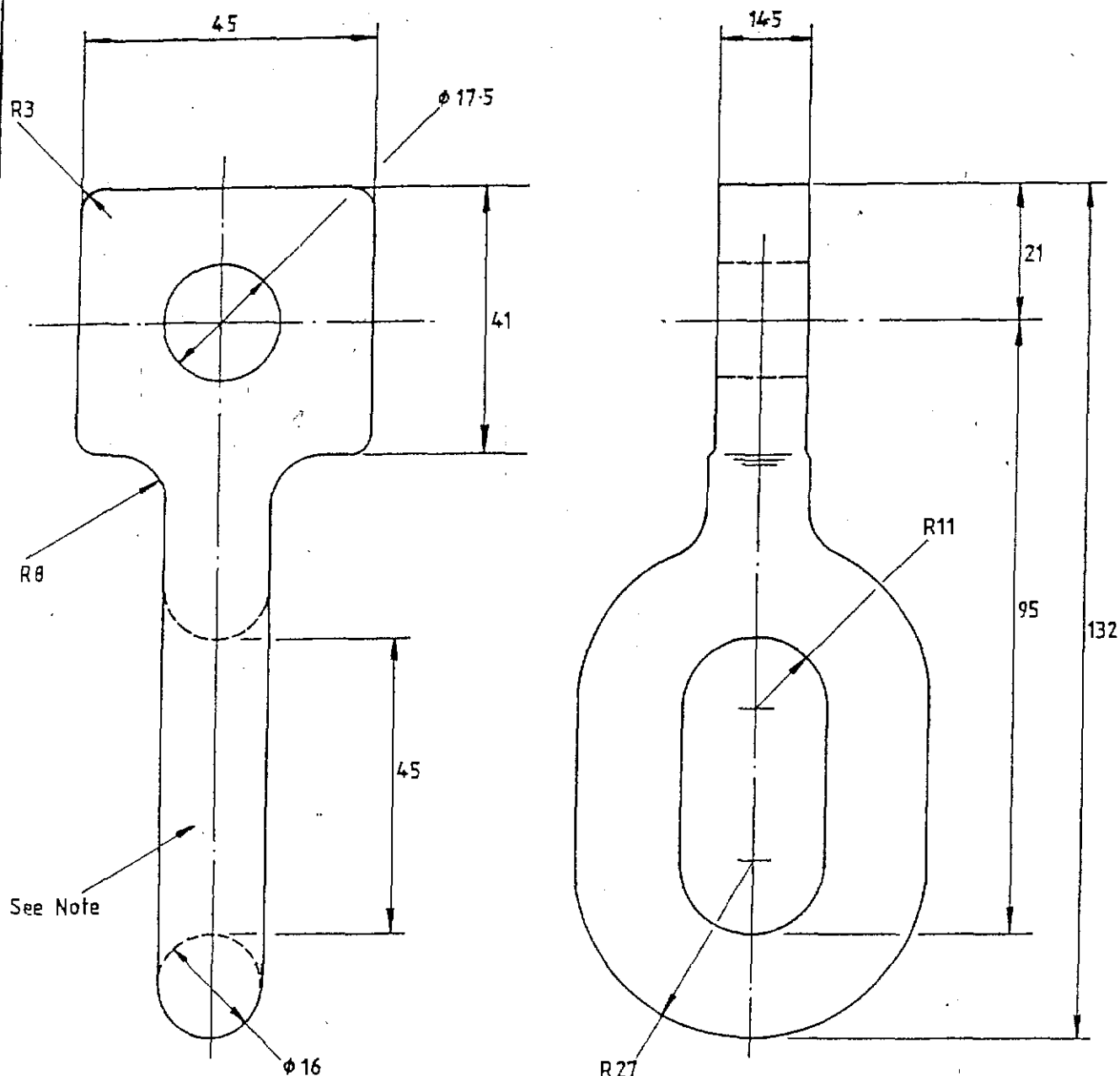


TYPE	DIMENSIONS (mm)				CEGB	APPLICATION
	A	B	C	D	CC N°	
M427/1	63.5	100	75	20	4384061	E/W SUSP. PL 16 D ² TOWERS
M427/2	51	124	75	16	—	E/W SUSP. BLAWKNOX 66kV TW
M427/3	50	200	50	16		
M427/4	57	100	50	16		
M427/5	83	165	75	16		

SUPPLIED WITH 2 PLAIN WASHERS,
4 FULL NUTS AND 2 HALF NUTS,
OR 6 FULL NUTS.

MATERIAL MILD STEEL GALV. TO B.S.729

SCALE		U BOLTS	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	21.5.85		A	
DRAWN	WDH		Types	
TRACED	S.G.T.		3-5	DRG. No. M427. —
CHECKED			Added	
APPROVED	W.D.W.			



Note Mark in position shown in 4mm high characters (i) D690
(ii) Makers identify; to be legible after galvanizing.

TYPE	C.E.G.B. CC N°	APPLICATION
M 428 /1	4384280	E/W SUSPENSION

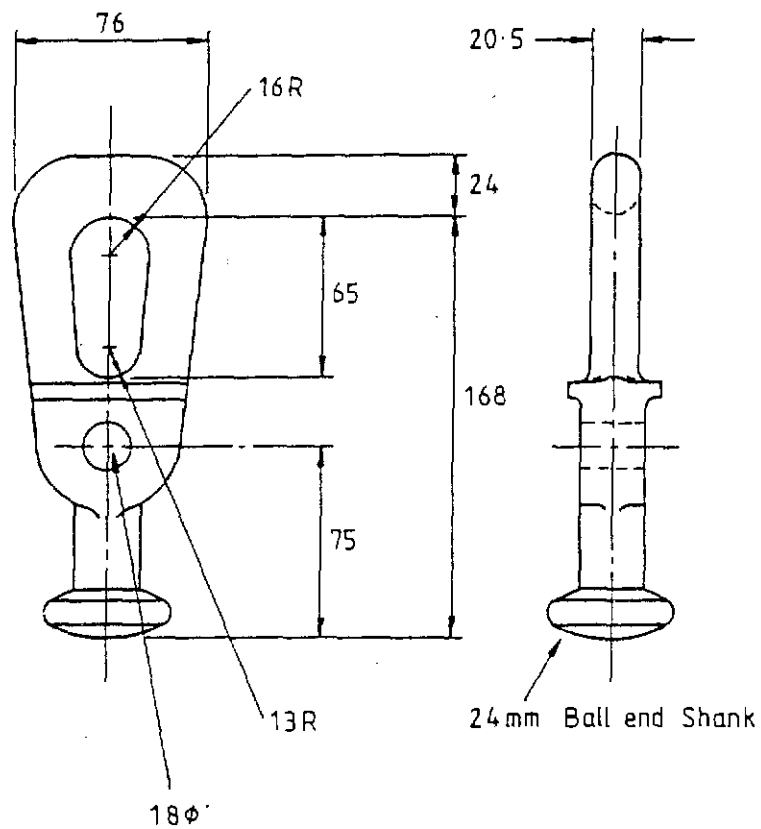
MATERIAL :- Forged Steel 070 M20
B.S. 970 Fully Normalised

FINISH :- Galvanized to B.S. 729

SCALE	FULL
DATE	24-6-85
DRAWN	R.A.C.
TRACED	G.H.R.
CHECKED	
APPROVED	

EARTHWIRE SUSPENSION LINK

ISSUE	MIDLANDS ELECTRICITY BOARD
	DRG. No.
	M 428



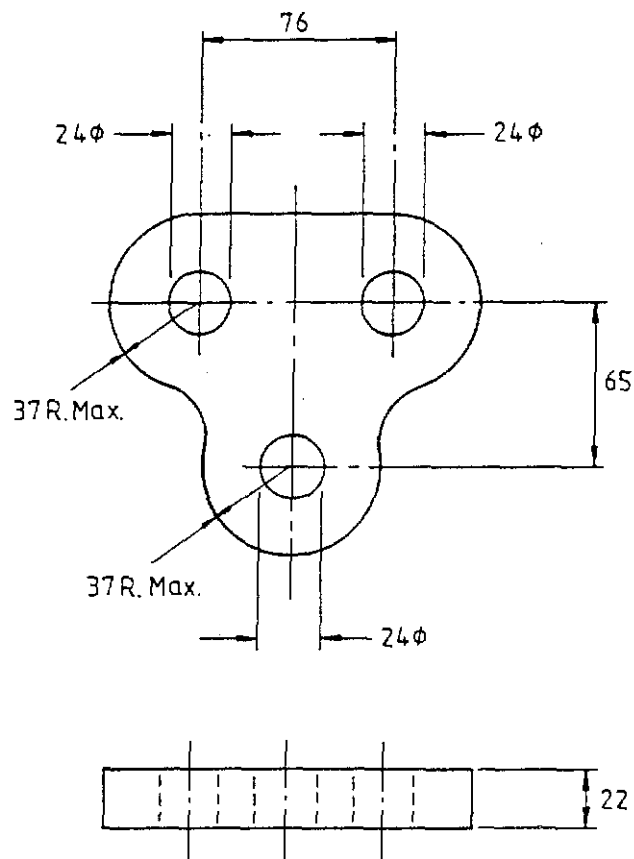
NOTE :-

To B.S. 3288 Part 1 & 2 Fig. 13 Ref. 42/30

SCALE	NTS
DATE	17-10-85
DRAWN	RAC
TRACED	CAR
CHECKED	
APPROVED	W.D.H.

BALL ENDED EYE LINK
MINIMUM FAILING LOAD 190kN

ISSUE	MIDLANDS ELECTRICITY BOA
	ORG. No.
	M 431



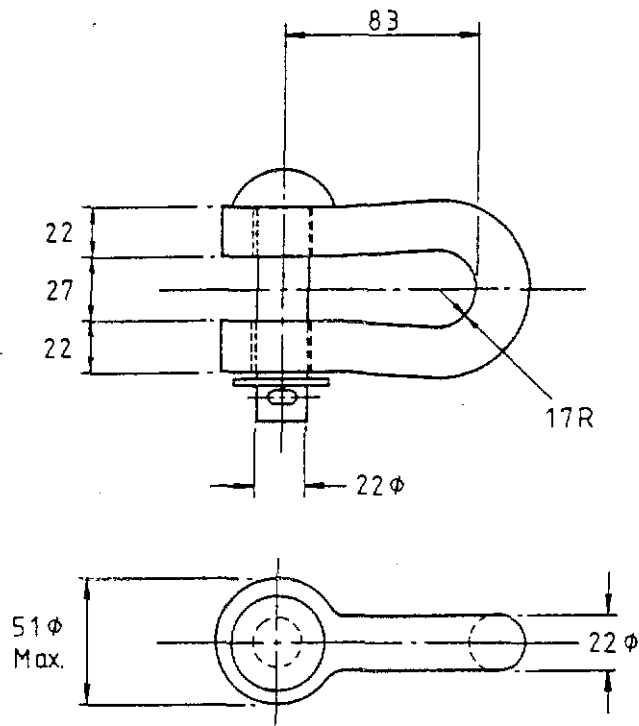
NOTE :-

To B.S. 3288 Part 1 & 2 Fig. 45 Ref. 42/21

SCALE	N.T.S.
DATE	17.10.85
DRAWN	R.A.C.
TRACED	C.A.R.
CHECKED	
APPROVED	<i>W.D.H.</i>

LANDING PLATE
MINIMUM FAILING LOAD 190kN

ISSUE	MIDLANDS ELECTRICITY BOARD
	ORG. No.
	M 432



TYPE	B.S. Ref.	PIN or BOLT SIZE
M433/1	42/29	PIN - 22φ × 95
M433/2	42/29A	BOLT - M22 × 115

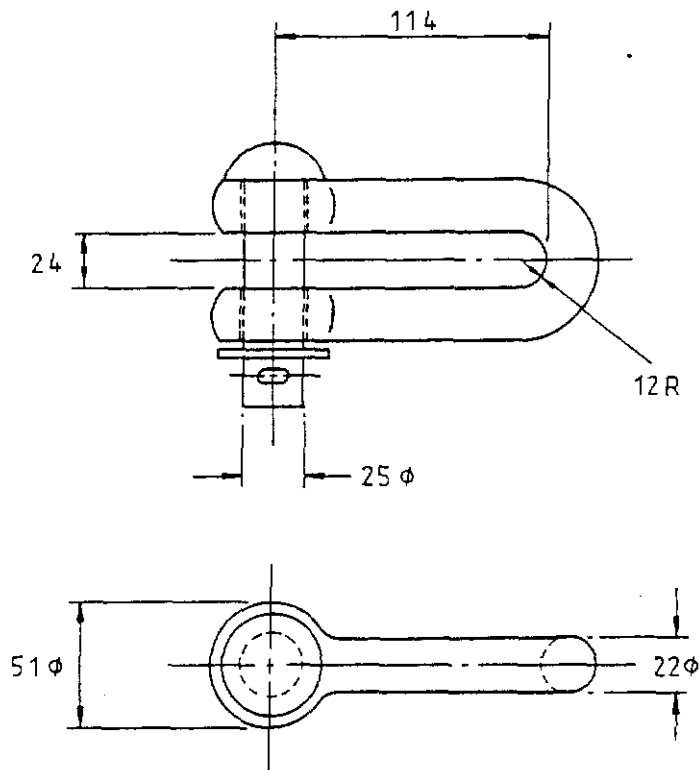
NOTE :-

To B.S. 3288 Part 1 & 2 Fig. 7 Ref. 42/29 (With Pin) 42/29A (With Bolt)

SCALE	NTS
DATE	17-10-85
DRAWN	R.A.C.
TRACED	C.A.R.
CHECKED	
APPROVED	WPH

SHACKLE
MINIMUM FAILING LOAD 190kN

ISSUE	MIDLANDS ELECTRICITY BOARD
	ORG. No.
	M433

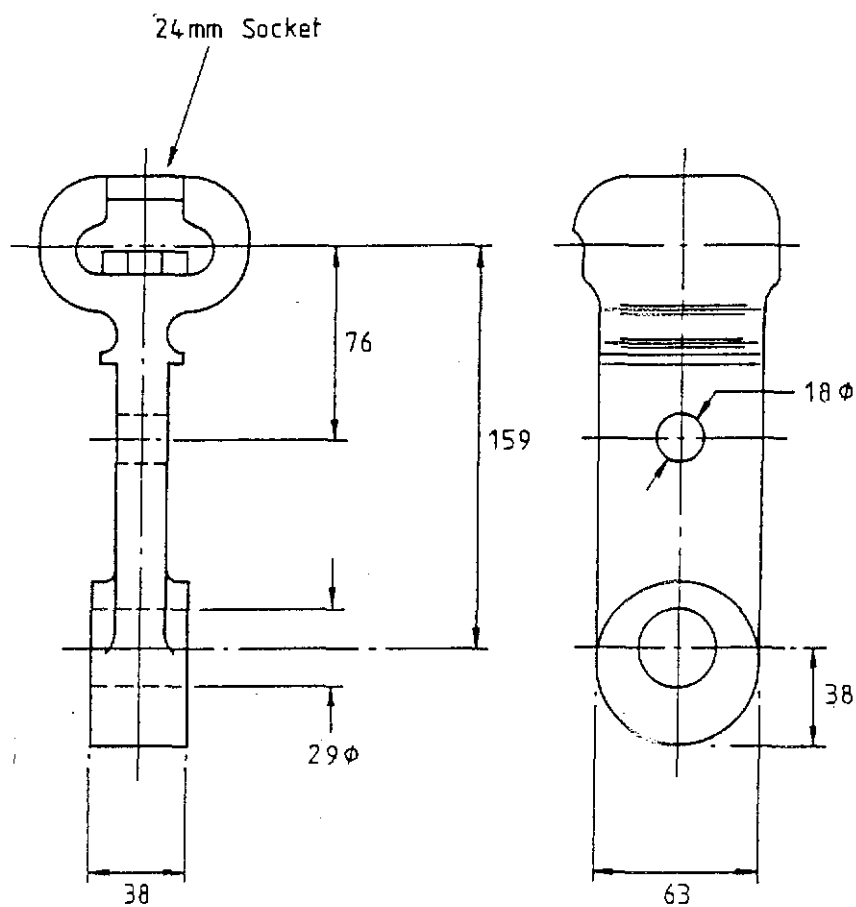


NOTE :-

Manufactured in accordance with B.S.3288 Part 1 & 2
To Dimensions shown.

Complete with 25x90 Clevis Pin, Split Pin and Washer.

SCALE	NTS.	TOWER SHACKLE 400 mm² A.C.S.R. CONDUCTORS MINIMUM FAILING LOAD 190 kN	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M434
DATE	17.10.85			
DRAWN	R.A.C.			
TRACED	C.A.R.			
CHECKED				
APPROVED	WAP			



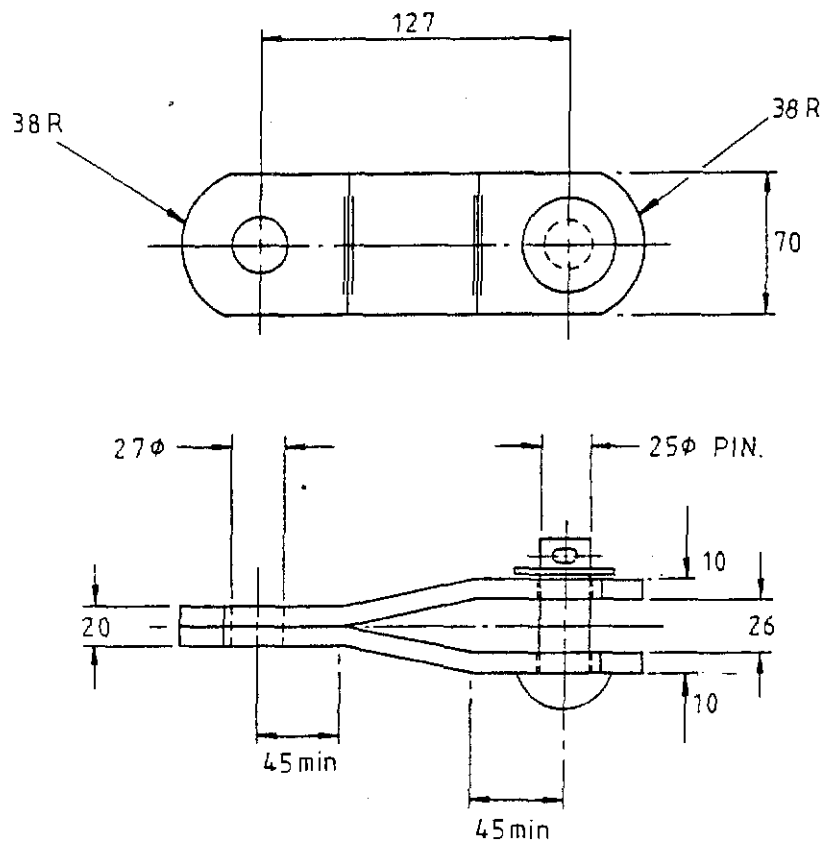
NOTE :-

To B.S.3288 Part 1 & 2 Fig. 25 Ref. 42/36

SCALE	NTS
DATE	17.10.85
DRAWN	R.A.C.
TRACED	C.A.R.
CHECKED	
APPROVED	W.A.H.

SOCKET TONGUE
MINIMUM FAILING LOAD 190kN

ISSUE	MIDLANDS ELECTRICITY BOARD
	ORG. No.
	M435

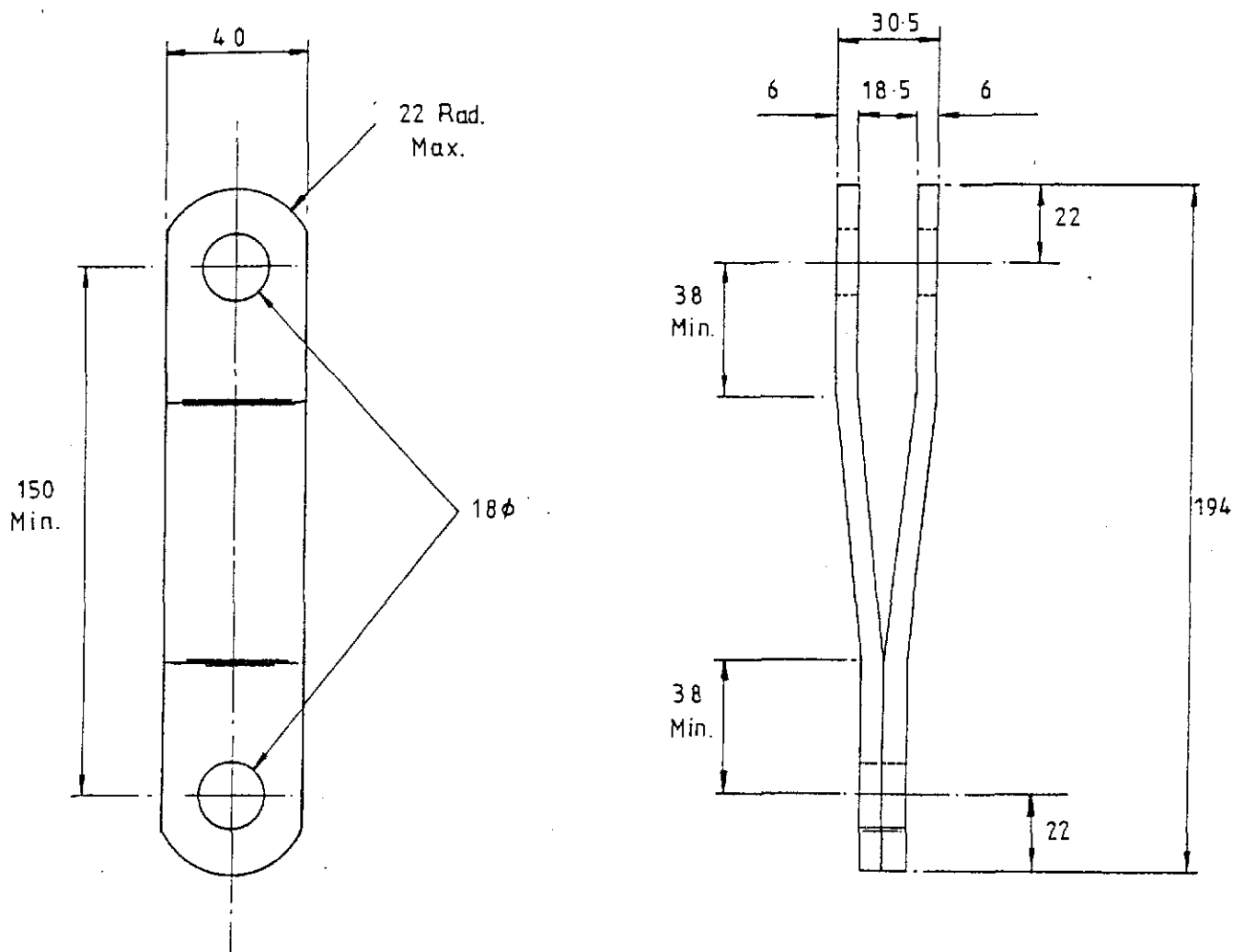


NOTE :-

To B.S. 3288 Part 1 & 2 Fig. 43 Ref. 42/87

Complete with 25x70 Clevis Pin, Split Pin and Washer.

SCALE	N.T.S.	CRANKED LINKS MINIMUM FAILING LOAD 190kN	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M436
DATE	17-10-85			
DRAWN	R.A.C.			
TRACED	C.A.R.			
CHECKED				
APPROVED	W.H.H.			



MINIMUM FAILING LOAD 70 kN

TO BS. 3288 REF. 1587

MAXIMUM WORKING LOAD 2268 kgf (5000 lbf.)

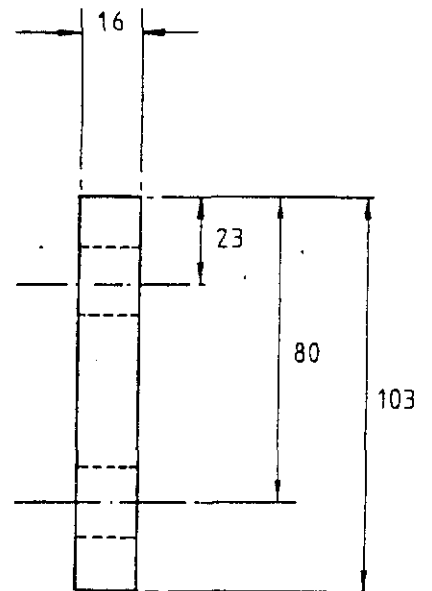
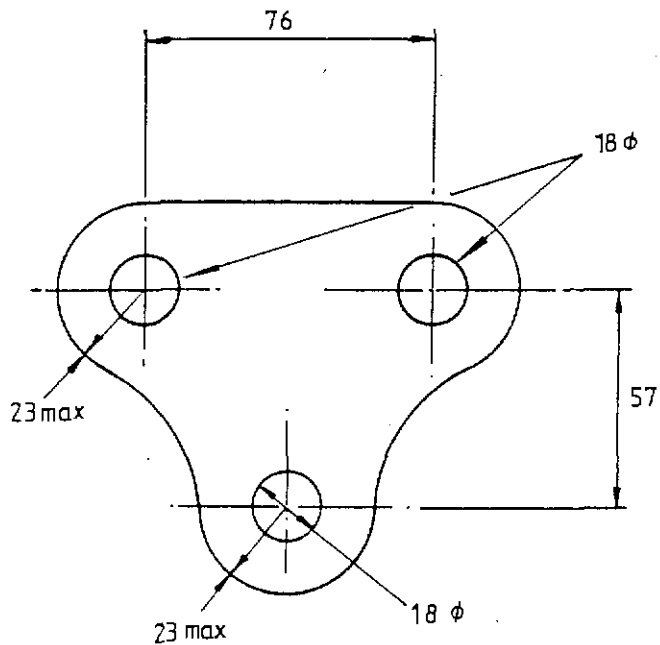
Supplied with 16φ×50 Pin, washer and split pin.

M 443/1

SCALE	1:20
DATE	7.1.86
DRAWN	RAC
TRACED	SJH
CHECKED	
APPROVED	WDH

CRANKED LINKS
MFL 70kN.

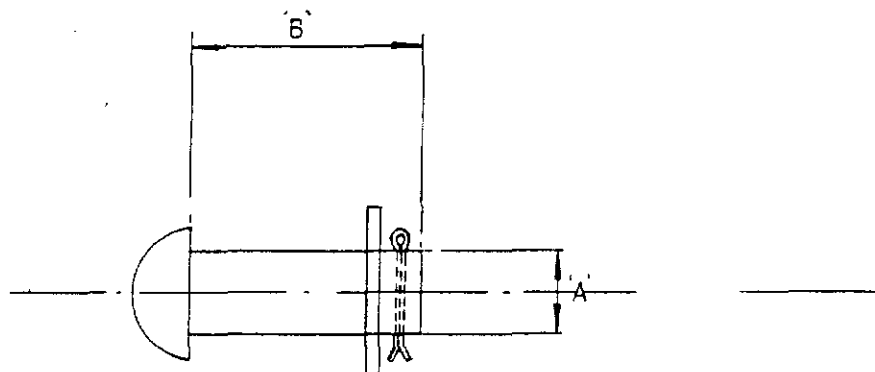
ISSUE	MIDLANDS ELECTRICITY BOARD
	DRG. No.
	M.443 (MF)



MINIMUM FAILING LOAD 70 KN
TO BS 3288 REF. 15 / 21
MAXIMUM WORKING LOAD 2268 Kgf. (5000 lbf)

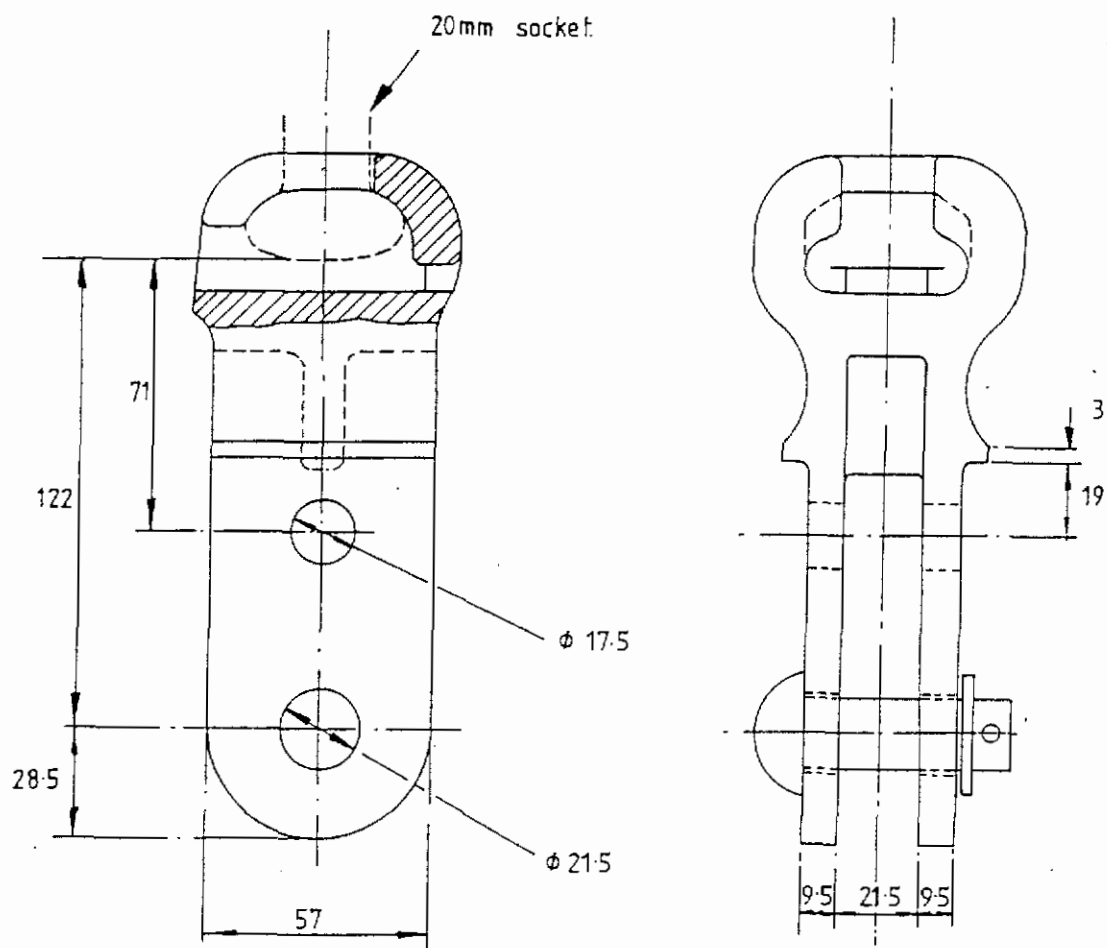
M 444/1

SCALE	1:20	LANDING PLATE MFL 70 KN	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	17.1.86			DRG. No.
DRAWN	S.J.H.			M.444 (MF)
TRACED				
CHECKED				
APPROVED	W.L.P.			
C. D. W. Ltd. 3297		E.S.I. STANDARD		ESI DRG N°



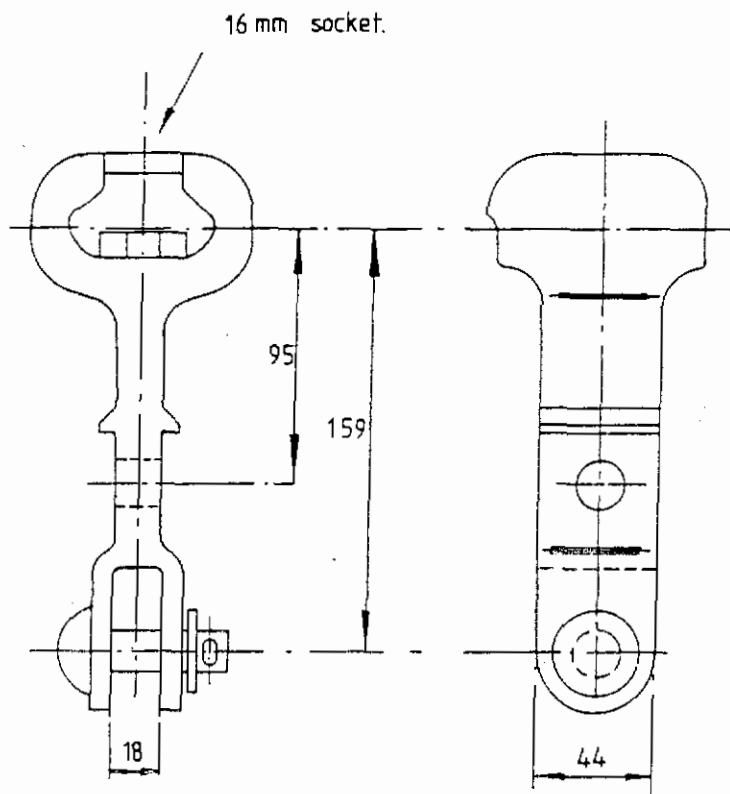
TYPE	USE	'A' ϕ	'B'	WASHER	PIN.
509/1	190kN Insulator fittings.	22	60		6.3 ϕ
509/2	190kN Insulator fittings	22	70		6.3 ϕ
509/3	70mm ² -175mm ² ACSR D. End	1(25.4)	4 1/4 (108)		
509/4	400mm ² ACSR Dead End	1 1/8 (28.6)	5 (127)		
509/5	70 kN Insulator fittings.	16	55		5.0 ϕ
509/6	" " " "	16	60		5.0 ϕ

SCALE	NTS	INSULATOR FITTINGS / CONDUCTOR TERMINATION CLEVIS PINS.	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M. 509. ESI. DRG. No.
DATE	16.11.82			
DRAWN	RAC			
TRACED				
CHECKED				
APPROVED	WDAH	ESI STANDARD		



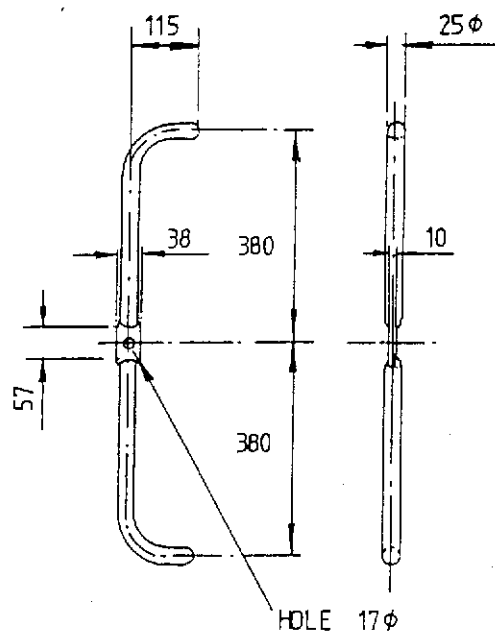
ME. REF	C.E.G.B. CC N°	UNIDARE OHLF REF
510	4323450	28/31

SCALE	1-2	SOCKET CLEVIS MINIMUM FAILING LOAD 125kN (12700 kgf)	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M. 510 E.S.I. DRG. N°
DATE	28-7-82			
DRAWN	R.A.L.			
TRACED				
CHECKED				
APPROVED	W.P.	E.S.I. STANDARD		



ME. REF	C.E.G.B. CC N°	UNIDARE OHLF REF
513.	4323474	0-554.

SCALE	N.T.S.	SOCKET CLEVIS MINIMUM FAILING LOAD 70 kN (7140 kgf)	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	29-7-82			
DRAWN	R.A.C.			DRG. N°
TRACED				M. 513
CHECKED				
APPROVED	W.L.H.			
E.S.I. STANDARD			E.S.I. DRG. N°	



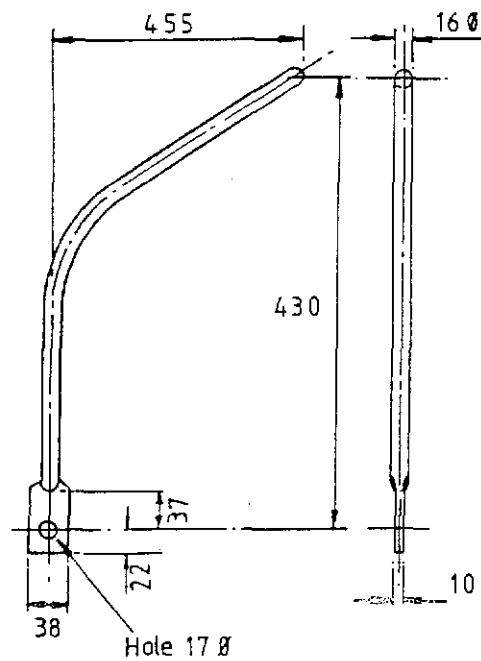
ME. REF	C.E.G.B C.C. N°	UNIDARE OHL F	REF
550	4323005	4SC	

MATERIAL

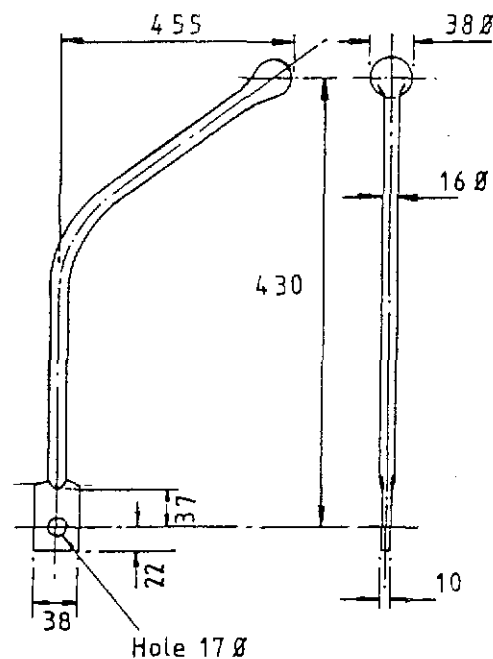
Mild Steel
Galvanised to
BS. 729

SCALE	N.T.S.	INSULATOR ARCING HORN 132 kV SUSPENSION LIVE END TYPE 4SC.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	26-7-82			
DRAWN	R.A.C.			DRG. No.
TRACED				M. 550
CHECKED				
APPROVED		E.S.I. STANDARD		E.S.I. DRG. N°

M. 551/1



M. 551/5



NOTE:-

For Fixing Bolts
See Drg. No M437

MATERIAL

Mild Steel Galvanised
to B.S.729

M. REF	CEG.B. cc No	UNIDARE OHLF. REF
551/1	4323016	4 LA
551/5	4323020	7 LA

SCALE	—
DATE	27.7.82
DRAWN	RAC
TRACED	S.J.H.
CHECKED	
APPROVED	

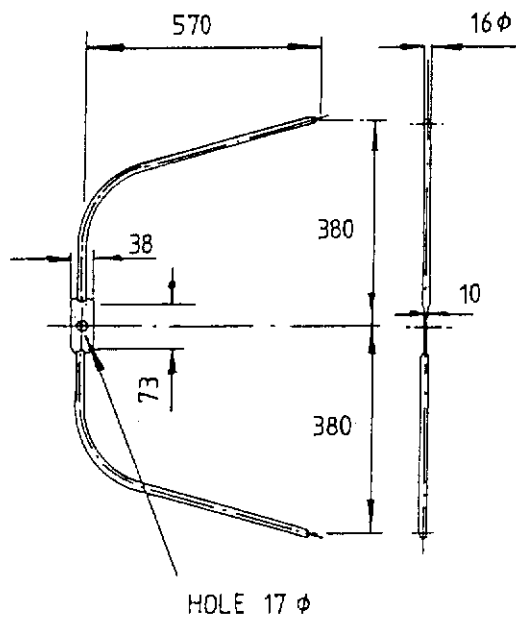
INSULATOR ARCING HORN
132 K.V. LOW DUTY EARTH END
TYPES 4LA & 7LA

E.S.I. STANDARD

ISSUE	
A	
M 551/5 Modified. 25-9-85 GHR	

MIDLANDS ELECTRICITY BOARD	
DRG. No.	
M. 551.	
E.S.I. DRG. No.	



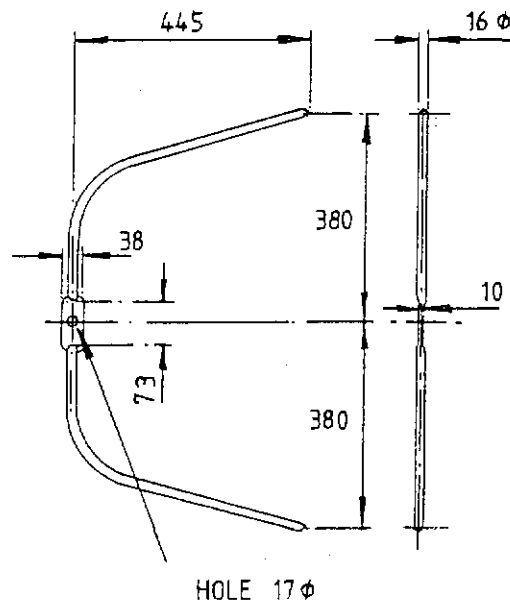


ME REF	CEGB C.C. N°	UNIDARE OHL F REF
552	43 23 017	4 S A.

MATERIAL

Mild Steel
Galvanised to
B.S. 729

SCALE	N.T.S.	INSULATOR ARCING HORN 132KV SUSPENSION EARTH END TYPE 4SA FOR 39" GAP	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	26-7-82			
DRAWN	R.A.C.			DRG. No
TRACED				M. 552.
CHECKED				
APPROVED		E.S.I. STANDARD		E.S.I. DRG. N°

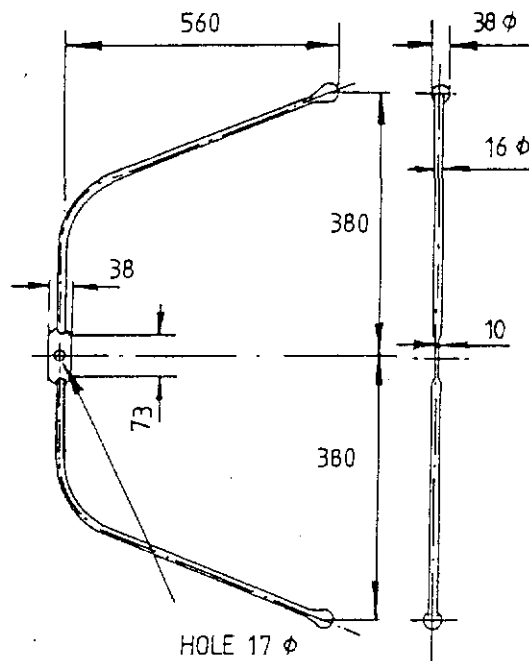


ME. REF	CEGB CC N°	UNIDARE	OH L F	REF
553	4323018		4 S B	

MATERIAL

Mild Steel
Galvanised to
BS 729

SCALE	N.T.S.	INSULATOR ARCING HORN 132kV SUSPENSION EARTH END TYPE 4SB FOR 44" GAP	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	27-7-82			DRG. No.
DRAWN	R.A.C.			M. 553
TRACED				
CHECKED				
APPROVED		E.S.I. STANDARD		E.S.I. DRG. N°

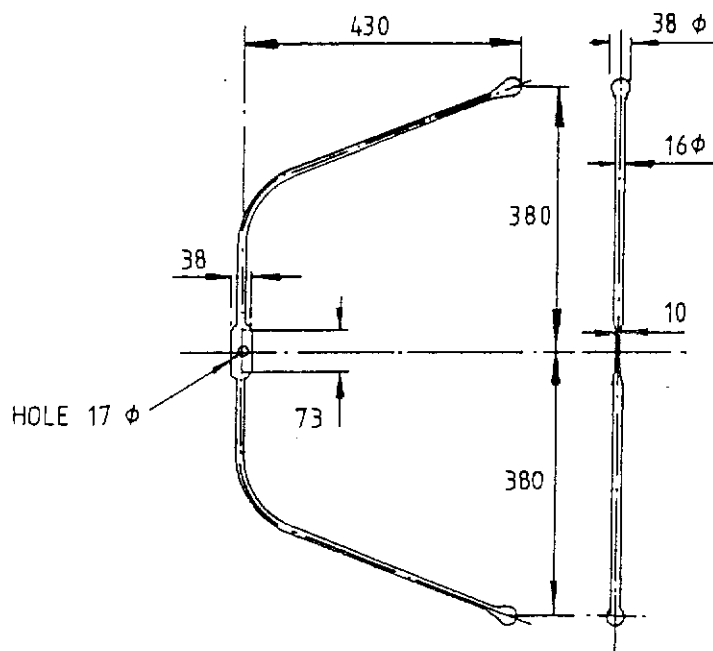


ME. REF	CEGB CC N°	UNIDARE	OHLE	REF.
554	43 23 021		7SA	

MATERIAL

Mild Steel
Galvanised to
BS 729

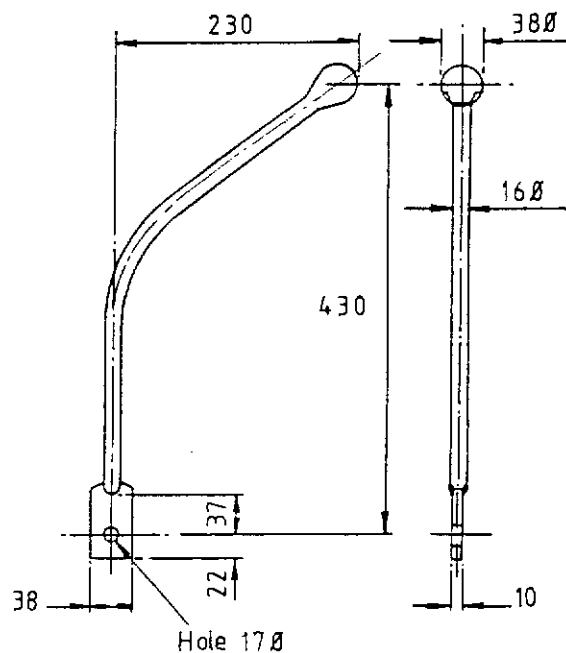
SCALE	N.T.S.	INSULATOR ARCING HORN 132 kV SUSPENSION EARTH END TYPE 7SA FOR 39" GAP	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	27-7-82			DRG. No.
DRAWN	R.A.C.			M. 554
TRACED				
CHECKED				
APPROVED		E.S.I. STANDARD		E.S.I. DRG. N°



ME REF	CEGB CC N°	UNIDARE	O.H.L.F	REF
555	4323022		7 S B.	

MATERIAL
MILD STEEL
GALVANISED TO
BS 729

SCALE	N.T.S.	INSULATOR ARCING HORN 132kV SUSPENSION EARTH END TYPE 7SB FOR 44" GAP	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	27-7-82			
DRAWN	R.A.C.			DRG. N°.
TRACED				M. 555
CHECKED				
APPROVED	<i>W. J.</i>	E.S.I. STANDARD		E.S.I. DRG. N°

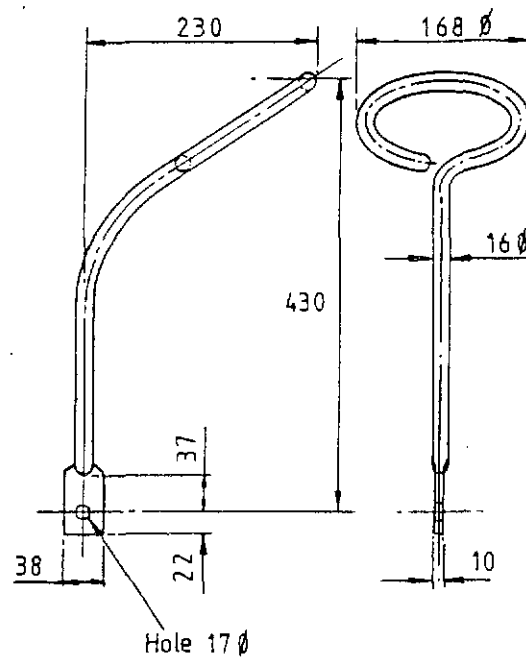


M.E. REF	C.E.G.B. cc No.	UNIDARE OHLF REF.
556	4323028	4 TC.

MATERIAL

Mild Steel Galvanised to B.S. 729

SCALE	—	INSULATOR ARCING HORN 132 k.V. TENSION LIVE END TYPE 4 TC	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	26.7.82			DRG. No
DRAWN	R.A.C.			M. 556
TRACED	S.J.H.			
CHECKED				
APPROVED		E.S.I. STANDARD		E.S.I. DRG. No

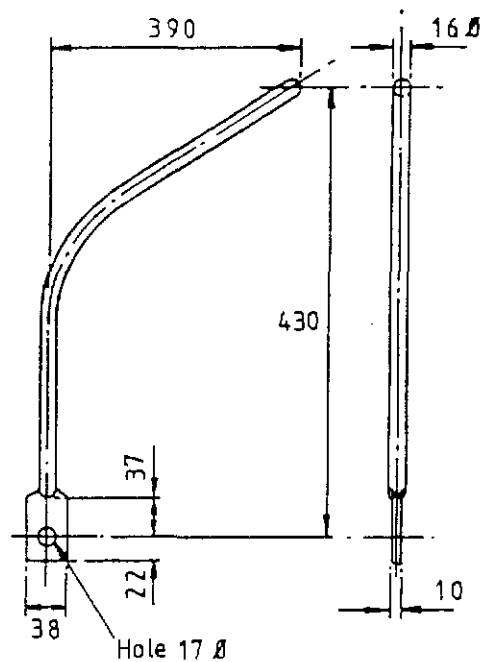


M.E. REF.	C.E.G.B. cc. No.	UNIDARE OHLF REF.
557	4323 031	7 TC

MATERIAL

Mild Steel Galvanised to B.S. 729.

SCALE	—	INSULATOR ARCING HORN 132k.V. TENSION LIVE END TYPE 7 T.C.	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. N° M. 557. E.S.I. DRG. N°
DATE	26.7.82			
DRAWN	RAC			
TRACED	S.J.H.			
CHECKED				
APPROVED		E.S.I. STANDARD		

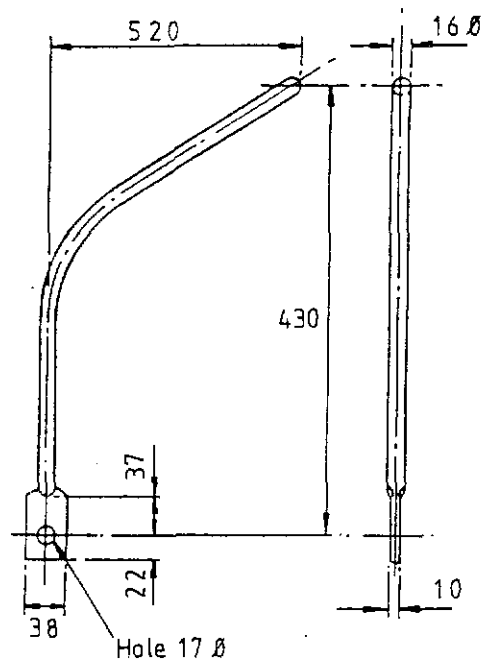


M.E. REF.	CEGB. CC. N°	UNIDARE OHLF REF.
558	4323043	4TB

MATERIAL

Mild Steel Galvanised to B.S.729

SCALE	—	INSULATOR ARCING HORN 132 k.V. TENSION EARTH END TYPE 4TB FOR 44" GAP	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	28.7.82			
DRAWN	R.A.C.			DRG. No. M.558.
TRACED	S.J.H.			
CHECKED				E.S.I. DRG. N°
APPROVED	<i>[Signature]</i>			
E.S.I. STANDARD				

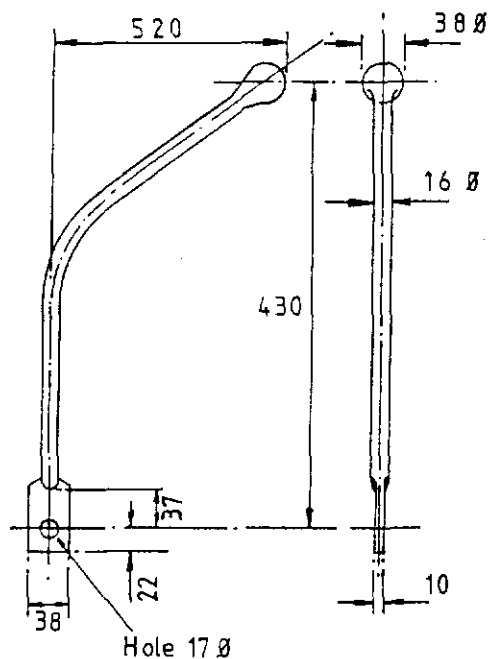


M.E. REF.	CEGB. C.C. N°	UNIDARE OHLF REF.
559	4323042	4TA

MATERIAL

Mild Steel Galvanised to B.S. 729

SCALE	—	INSULATOR ARCING HORN 132 k.V. TENSION EARTH END TYPE 4TA FOR 39" GAP	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M. 559 E.S.I. DRG. N°
DATE	28.7.82			
DRAWN	R.A.C.			
TRACED	S.J.H.			
CHECKED				
APPROVED				
E.S.I. STANDARD				



NOTE:-

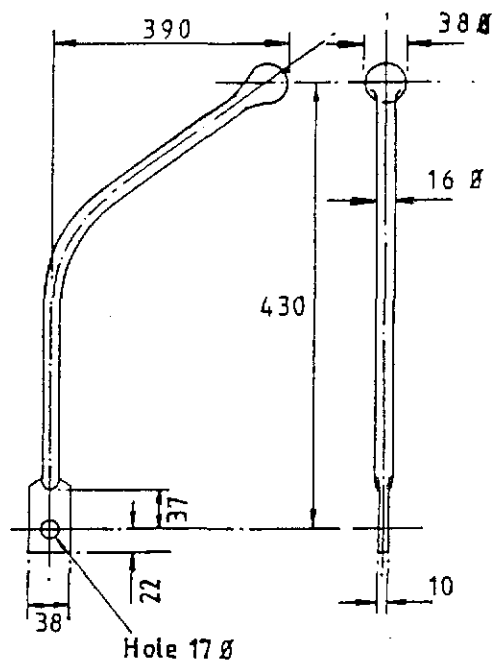
For Fixing Bolts
See Drg. M437

M. REF	CEGB. cc. N°	UNIDARE O.H./L.F. REF.
560	4323047	7TA

MATERIAL

Mild Steel Galvanised to B.S.729.

SCALE	—	INSULATOR ARCING HORN 132 kV. TENSION EARTH END TYPE 7TA FOR 39" GAP	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	28-7-82		A	
DRAWN	RAC		Fixing corrected.	
TRACED	S.J.H.		25-9-85 GHR	
CHECKED				
APPROVED	W.D.H.	E.S.I. STANDARD		DRG. No M. 560
				E.S.I. DRG. N°



NOTE :-

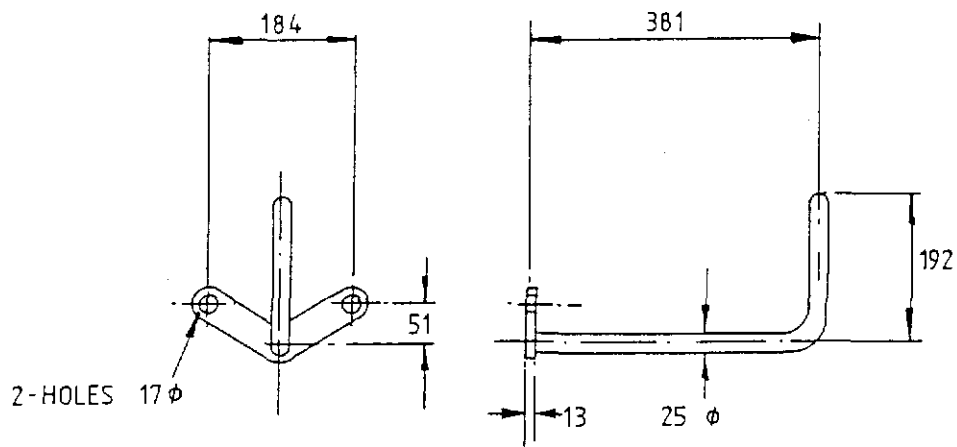
For Fixing Bolts
See Drg. M437.

M. REF	CEGB cc N°	UNIDARE O.H / LF REF.
561	4323048	7TB

MATERIAL

Mild Steel Galvanised to B.S.729.

SCALE	—	INSULATOR ARCING HORN 132 kV. TENSION EARTH END TYPE 7TB FOR 44" GAP	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M. 561 E.S.I. DRG. N°
DATE	28-7-82		A	
DRAWN	RAC		Fixing corrected.	
TRACED	S.J.H.		25-9-85 GHR	
CHECKED				
APPROVED	W.A.H.			
E.S.I. STANDARD				

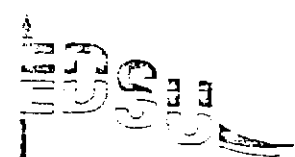


2 - OFF PER SET

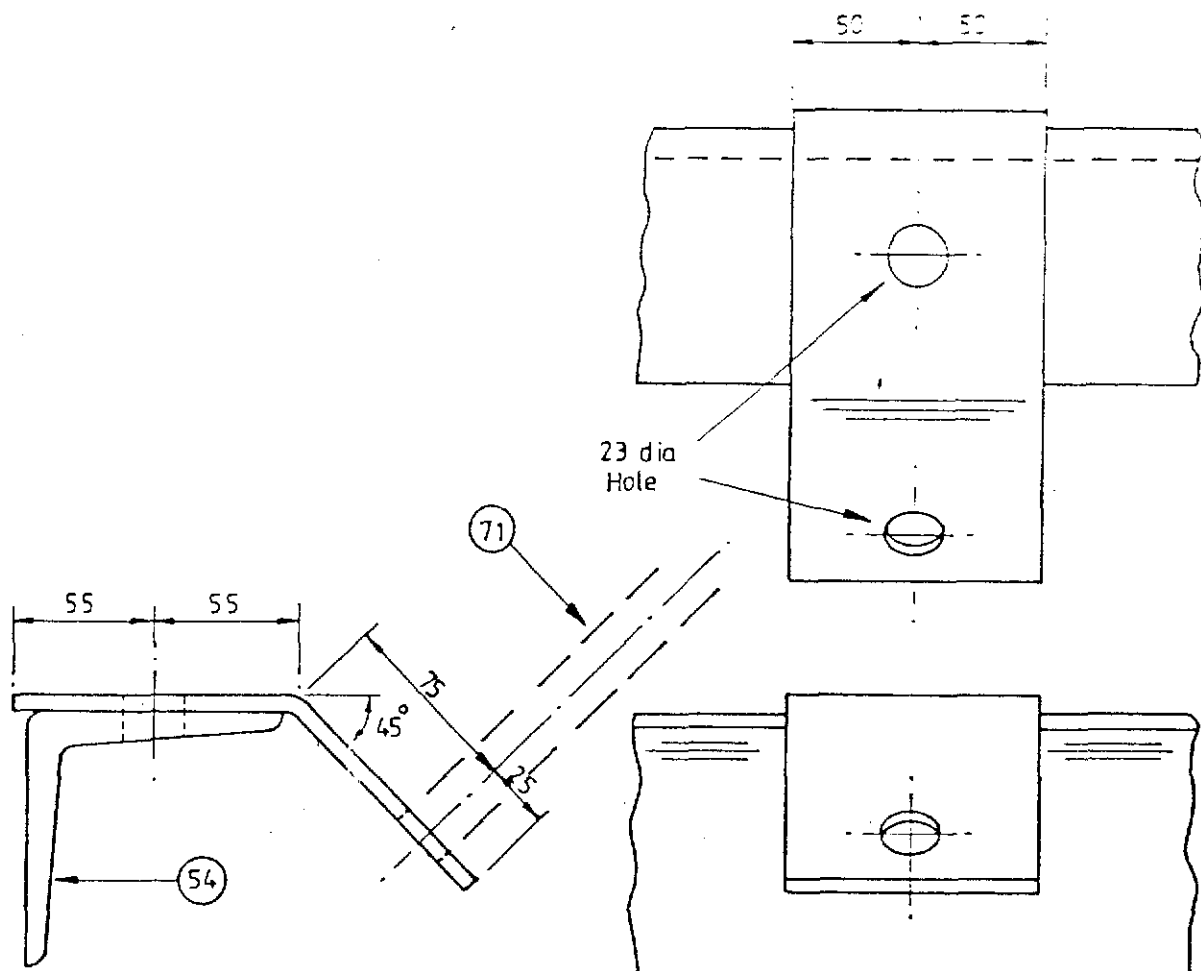
ME REF	CEGB CC N°	UNIDARE OHLF REF
562	432 3002	3/2/132 SC

MATERIAL
MILD STEEL
GALVANISED TO
BS 729.

SCALE	N.T.S.	INSULATOR ARCING HORN 132kV SUSPENSION LIVE END TYPE 3/2/132 SC.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	27-7-82			
DRAWN	R.A.C.			DRG No.
TRACED				M. 562
CHECKED				
APPROVED	<i>[Signature]</i>			
E.S.I. STANDARD				E.S.I. DRG. N°

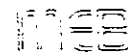


 MIDLANDS ELECTRICITY PLC



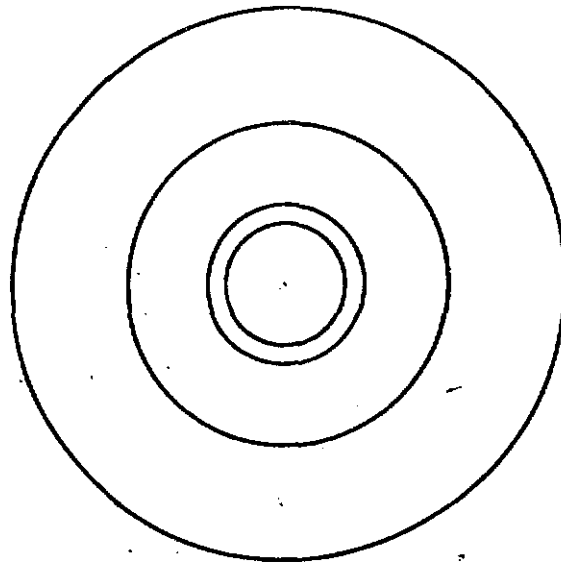
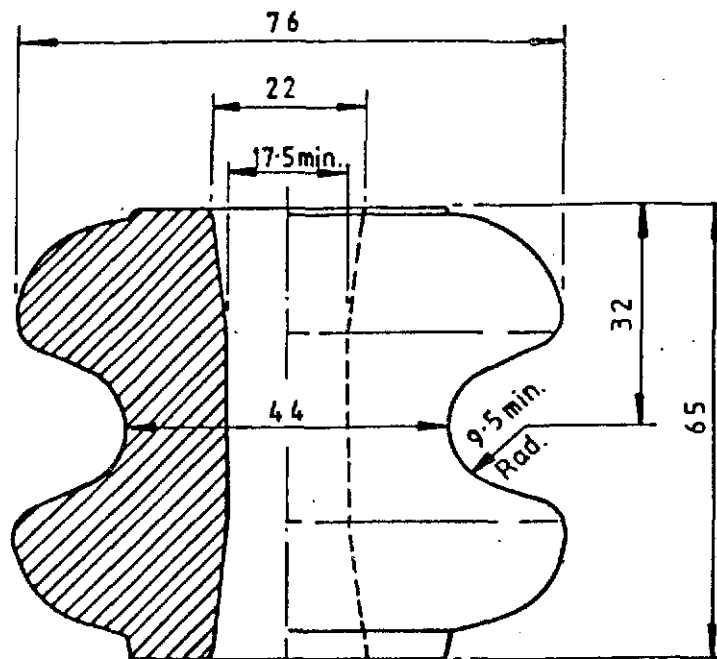
MATERIAL:- 100 x 6 flat Mild Steel

FINISH:- Galv- to BS 729 after cutting, drilling and bending

DRAWN BY DATE	TITLE INSULATOR BRACKET	SCALE	—	 Midlands Electricity plc FIG No M569
		DATE	11-6-93	
		DRAWN	A J B	
		TRACED	S G T	
		CHECKED		
		APPROVED		

SECTION 16

MIN NUMBER	M	DESCRIPTION
3450130	13	INSULATOR - L.V. REEL TYPE
3450140	14	D' IRON & PIN
3450220	22	TIE ROD FOR STRUT POLES
3452900	290	EXTENSION STRAP FOR HOUSE SERVICE BRACKETS
3453730	373	A.B.C - STAND-OFF CLEAT
3453740	374	A.B.C. - ANCHOR CLAMP
3453800	380	A.B.C COMPRESION TYPE END SEAL
3453810	381	A.B.C COMPRESION JOINT INSULATED
3453820	382	A.B.C COPPER LUG (INSULATED BI-METAL)
3453830	383	A.B.C. SERVICE ANCHOR CLAMP
3453840	384	A.B.C. INSULATOR PIERCING CONNECTOR
3453850	385	A.B.C. - SUPPORT HOOK
3453860	386	A.B.C. - EXTENDED SUPPORT HOOK
3453870	387	A.B.C. - NUT HOOK
3453880	388	A.B.C. - SUSPENSION STRAP
3453890	389	A.B.C. - LIVE POT END
3453910	391	A.B.C - SUSPENSION CLAMP
3453920	392	ASSEMBLY A.B.C. SERVICE TERMINATION BOX
3453930	393	A.B.C. INSULATING SLEEVE FOR TENSION & NON TENSION JOINTS



SCALE	N.T.S.
DATE	1.6.81
DRAWN	
TRACED	S.J.H.
CHECKED	
APPROVED	

INSULATOR - L.V. REEL TYPE

E.S.I. STANDARD 43.93

ISSUE	A
Min. added to 17.5 dims	
3.9.92 A.J.B	

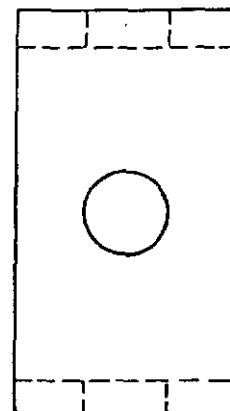
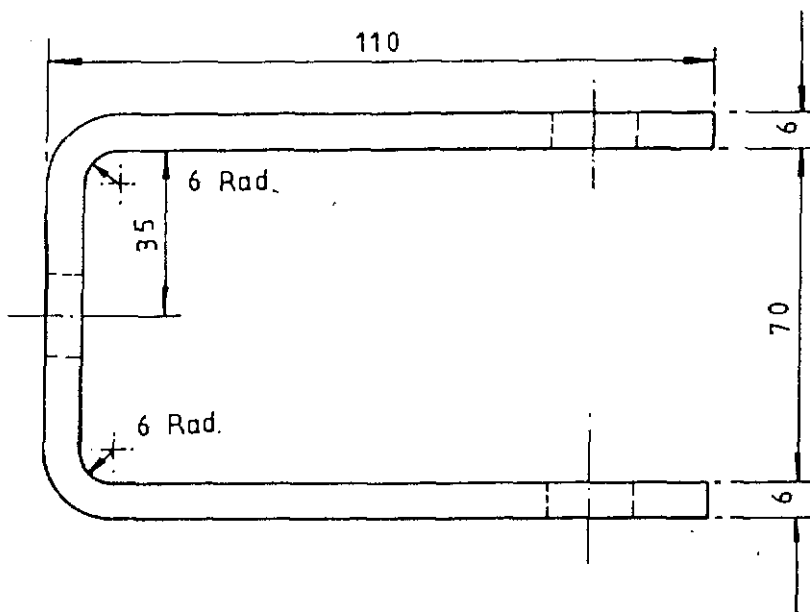
MIDLANDS
ELECTRICITY BOARD

DRG. No.

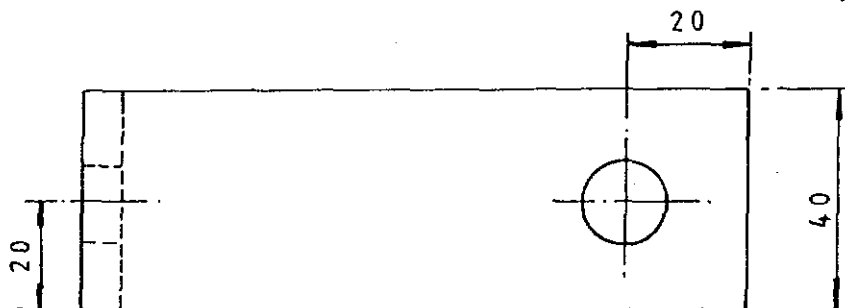
M. 13.

E.S.I. Drg. N° 43.9304

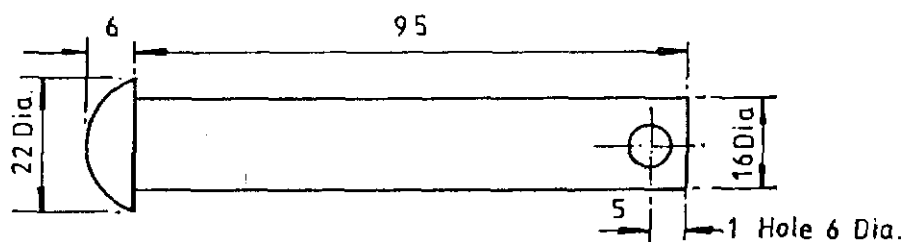
MF



3 Holes 18 Dia.



BRACKET



PIN

MATERIAL

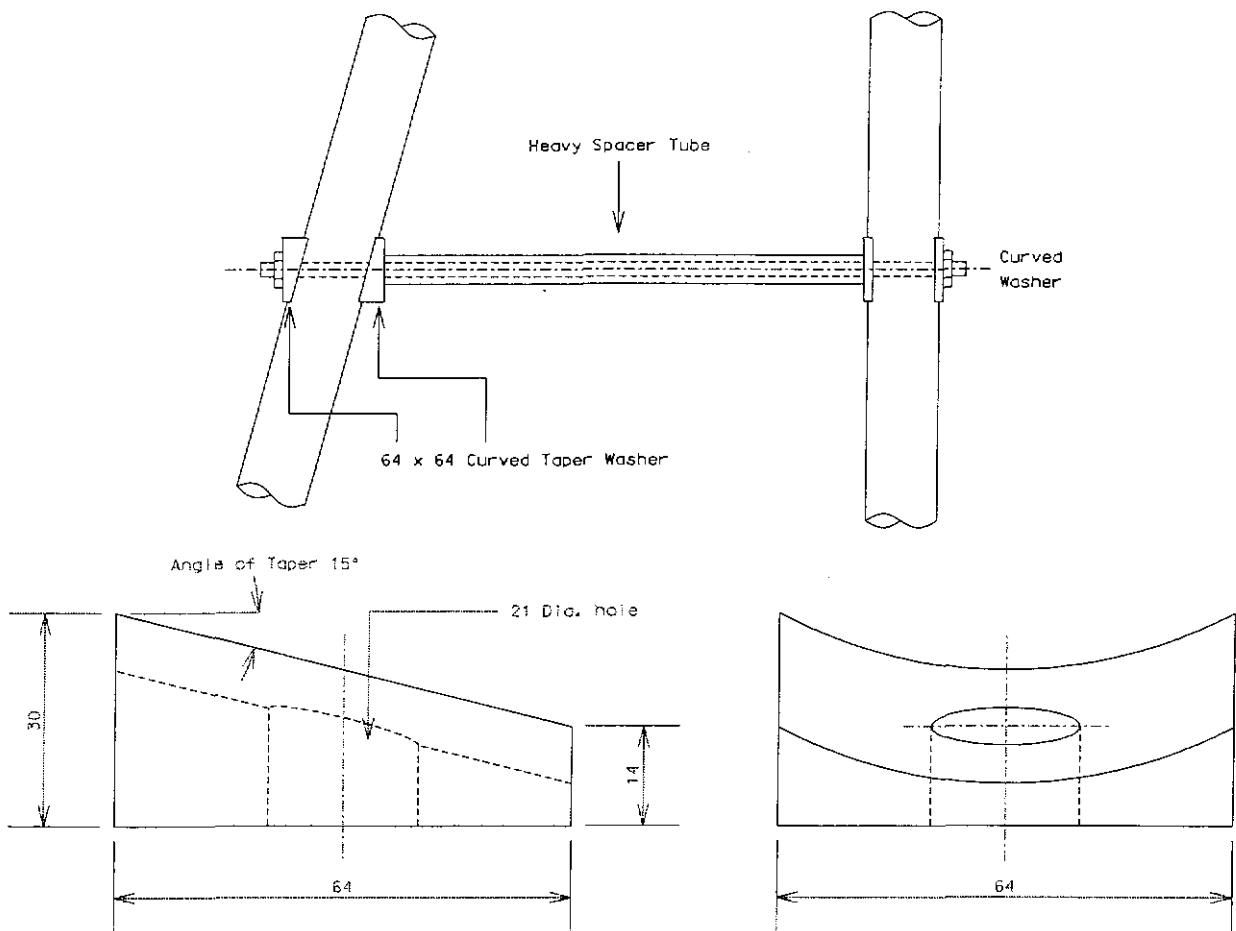
Mild Steel to B.S.4 20/33 ton quality

FINISH

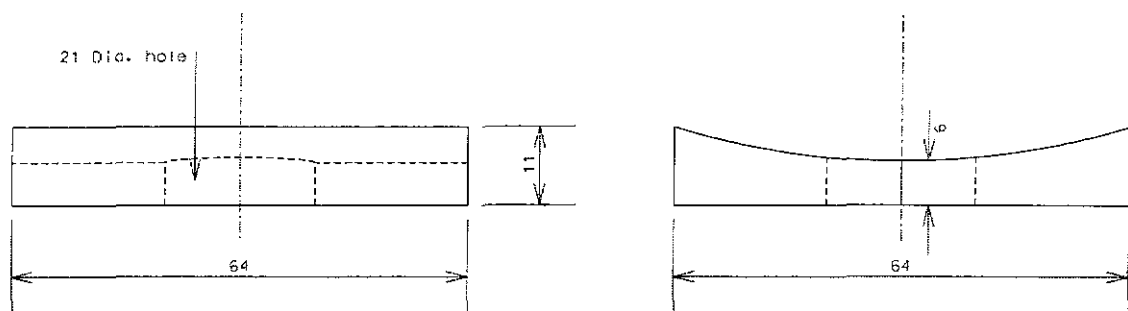
Galvanised to B.S. 729 after cutting, forming and drilling

Pin to be supplied with 5 dia. non-ferrous
or stainless steel split pin

SCALE	N.T.S.	'O' IRON AND PIN.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	1.6.81		A	
DRAWN			Matl. & Finish added 9-9-92 A.J.B	DRG. No. M.14.
TRACED	S.J.H.			
CHECKED				
APPROVED				
E.S.I. STANDARD. STD.43-95		E.S.I. Drg. No. 439511		



DETAIL 64 X 64 CURVED TAPER WASHER



DETAIL 64 X 64 CURVED WASHER

FINISH

Galv. to B.S. 729 Screw threads to be
Brush Galv.

MATERIAL

19 dia. x 1.5 24 Tie Bolt threaded 114
each end C/W. 2 Nuts.
25 x 914 B.S. Wrought Pipe.
2 - Curved Washers.
2 - Curved Taper Washers.

SCALE	1-1
DATE	5/9/00
DRAWN	ADD
TRACED	
CHECKED	28/9/00
APPROVED	KD

TIE ROD FOR STRUT POLES

ISSUE	DATE

Produced by
Output Services

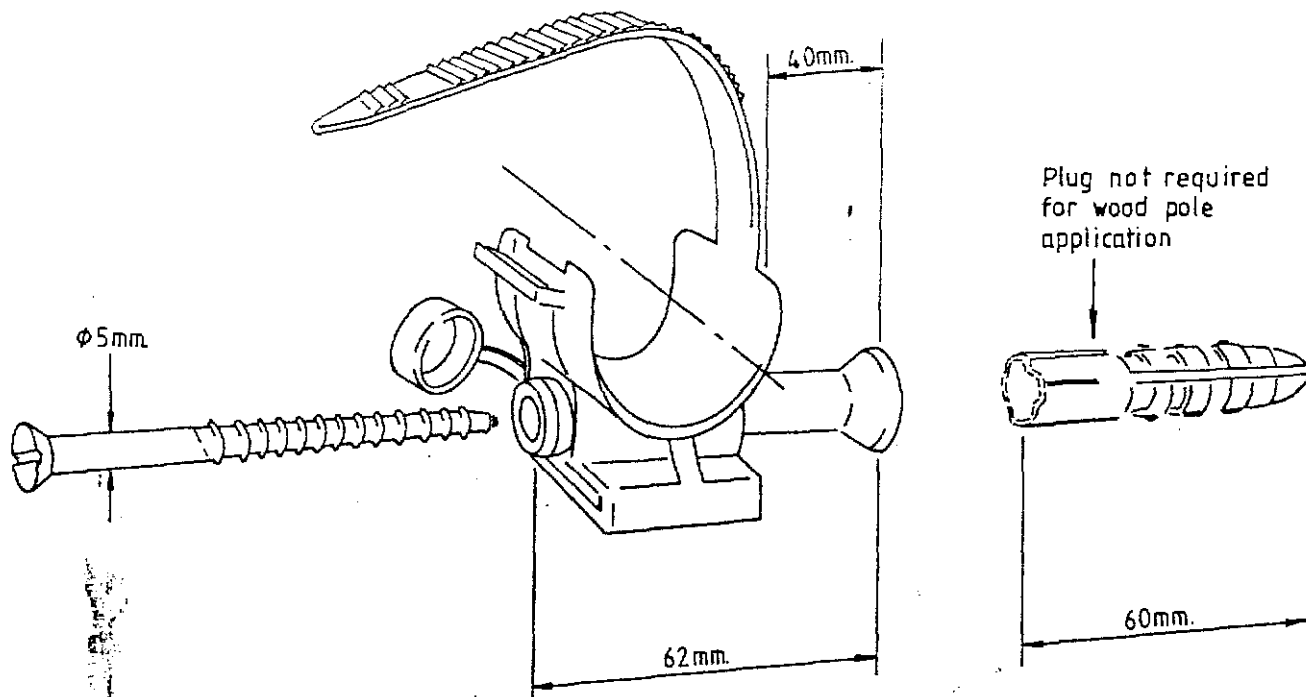
GPU
POWER UK

DRG. No.
M22

DRG. No.

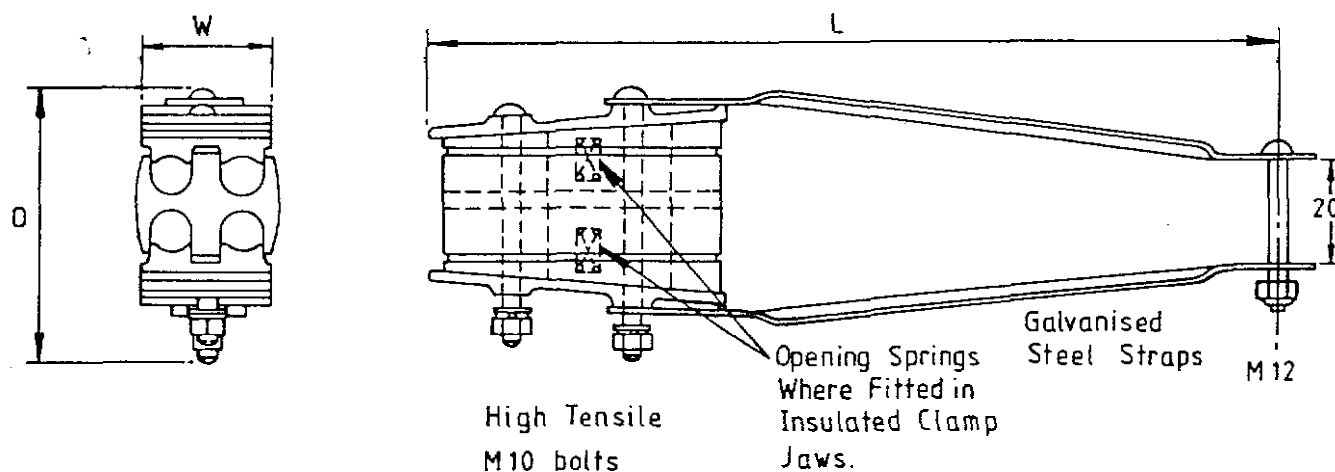
M22

A4



M.E.B. REF.	BICC REF.	CONDUCTOR mm ²
373/0	ACM5095	Up to - 4 x 95

SCALE	NTS.	A.B.C. - STAND-OFF CLEAT	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M373 (MF)
DATE	9-12-83			
DRAWN				
TRACED	HPS.			
CHECKED	BN			
APPROVED				
C. D. W Ltd. 3297		E.S.I. STANDARD	E.S.I. DRG. No.	

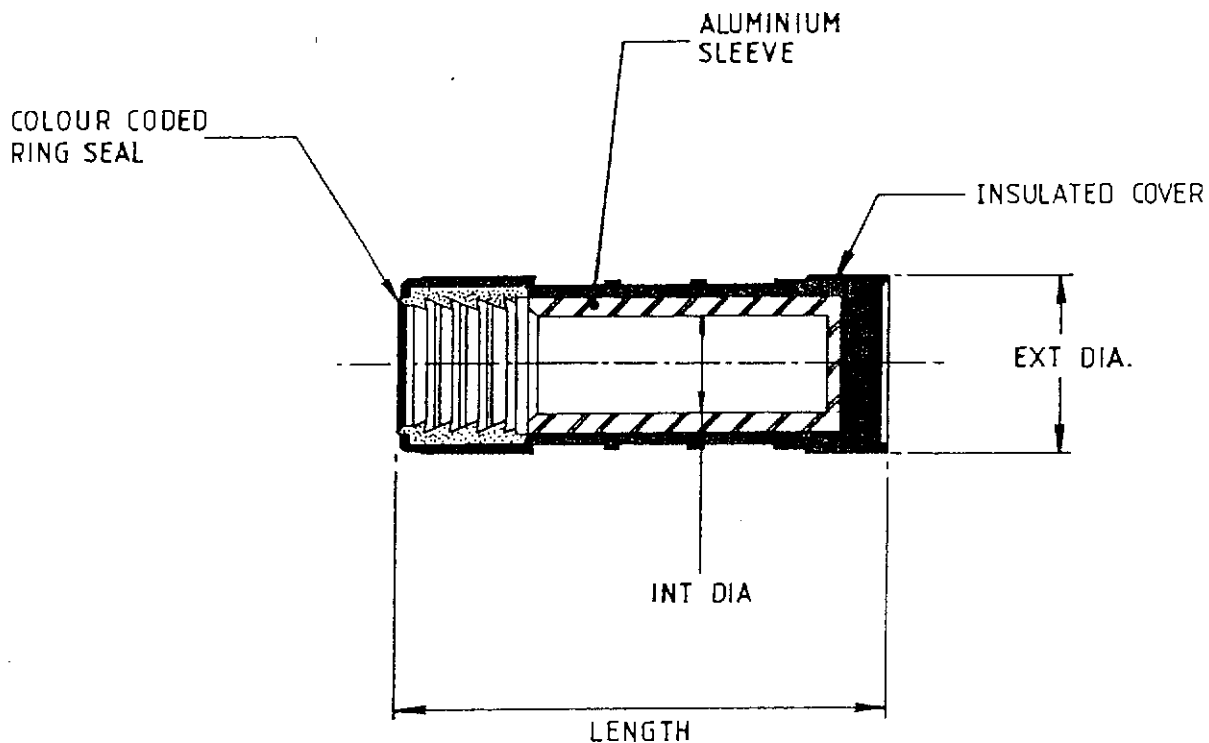


MEB REF.	CONDUCTOR SIZE (mm ²)	MANUFACTURER REF.	DIMENSIONS (mm)			WEIGHT (kg)	TORQUE (Nm)
			L	D	W		
374/0	4 core 95	BICC AAC5095	420	95	50	1.65	50
		MOSDORFER 5012/2	310	115	52	1.20	40
	4 core 50	MALICO PA 35 95 CB	310	112	52	1.30	60
		SICAME PA11	330	115	46	1.18	50
374/1	2 core 95	BICC AAC25095	420	50	50	1.50	30
	2 core 50						

Tightening Torque to be marked on Clamp

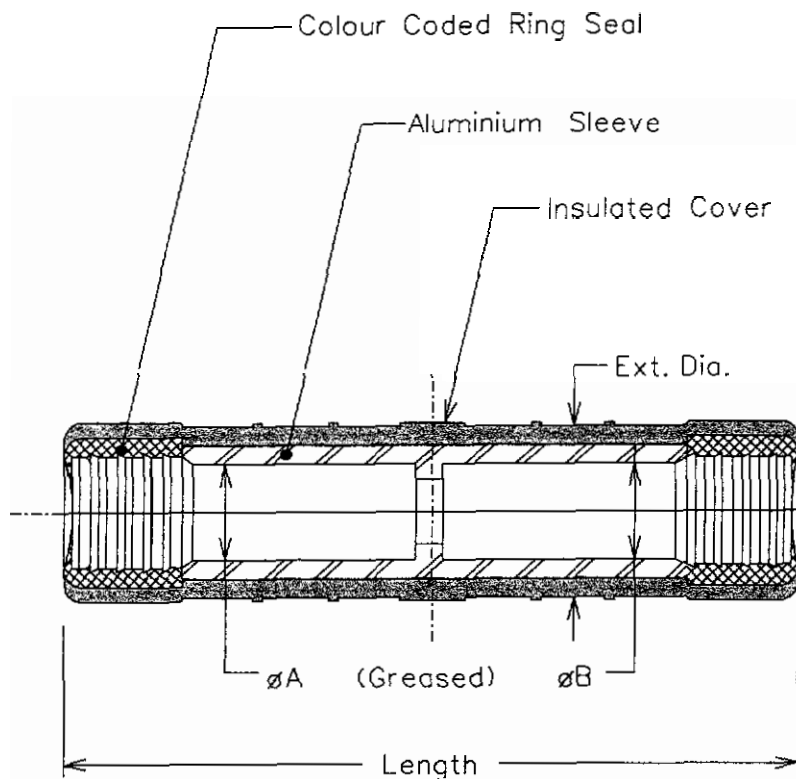
MF

ADDED BY A.J.B. DATE 1-3-93 Table Revised.	TITLE ABC - ANCHOR CLAMP	SCALE	—	 Midlands Electricity plc DRG No M374
		DATE	27-6-89	
		DRAWN	R.A.C.	
		TRACED	P.J.W.	
		CHECKED	<i>R.C.</i>	
AMDO15828E	ESI STANDARD 43-14	APPROVED		ESI ORG. N°



MEB Ref.	Conductor Size (mm ²)	Colour Code	Manufacturer Ref.	Dimensions (mm)			Die / No. of Compressions			
				Int Dia	Ext Dia	Length	Y35/G10	No.	G5	No.
380/1	120	Pink					UE 215	2		
380/2	95	Grey					UE173	2	WE173	3
380/3	50	Yellow					UE 173	2	WE173	3
380/4	35	Red					UE173	2	WE173	3

ADDED BY DATE	TITLE A B C COMPRESSION TYPE END SEAL	SCALE	—	 Midlands Electricity plc DRG No. M380
		DATE	20-3-95	
		DRAWN	R.C.F.	
		TRACED	P.J.W.	
		CHECKED	<i>RCF</i>	
		APPROVED		



MEB Ref	Conductor (mm ²)		Colour Code		Type	Manufacturer		Dimensions (mm)		Die/No. of Compressions			
	A	B	A	B			Ref	Length	Ext ϕ	Y35/G10	No	G5	No
381/1	120	120	Pink	Pink	Full Tension	SICAME	MJPT120	106	25	UE 215		N/A	
381/2	120	95	Pink	Grey	Non Tension	SICAME	MJPT120/95	106	25	UE 215		N/A	
381/3	120	50	Pink	Yellow	Non Tension	SICAME	MJPT120/50	106	25	UE 215		N/A	
381/4	95	95	Grey	Grey	Full Tension	SICAME	MJPT95/(21)	106	25	UE 215		N/A	
381/6	95	50	Grey	Yellow	Non Tension	SICAME	MJPT95/50	106	20	UE 173		WE173	
381/7	50	50	Yellow	Yellow	Full Tension	SICAME	MJPT50	106	20	UE 173		WE173	
381/8	50	35	Yellow	Red	Non Tension	SICAME	MJPT50/35	106	20	UE 173		WE173	
381/9	35	35	Red	Red	Full Tension	SICAME	MJPT35	106	20	UE 173		WE173	

SCALE	
DRAWN	09.05.94
TRACED	R.C.F
CHECKED	P.J.W
APPROVED	R.C.F

ABC COMPRESSION JOINT INSULATED

ISSUE	DATE
A	05.08.02
REDRAWN ON CAD MINOR MODS.	
MJB/KD	



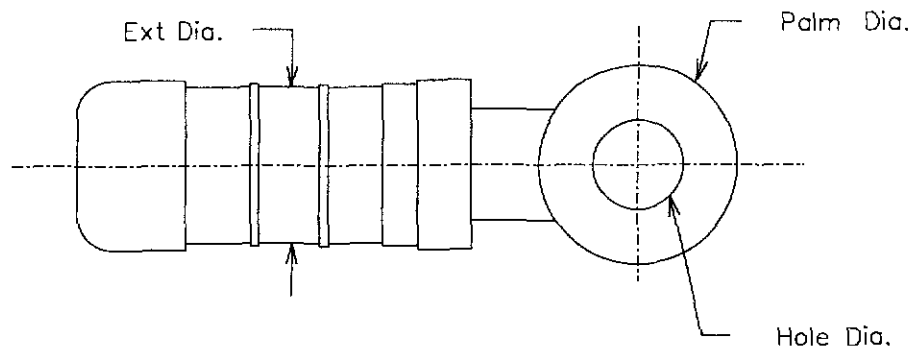
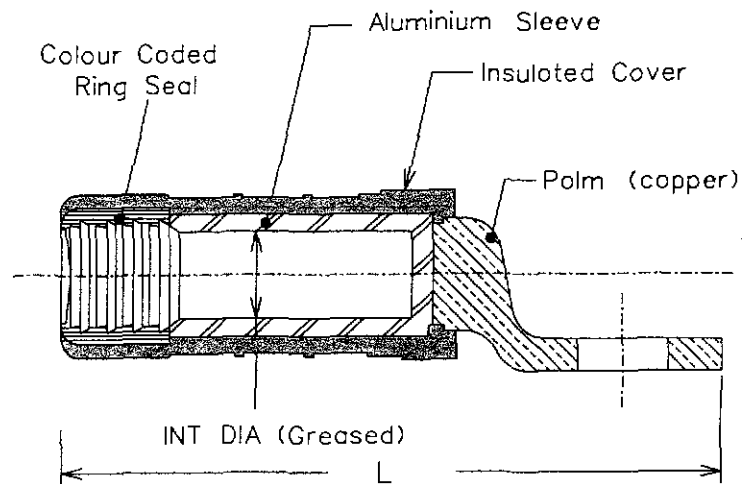
Aquila

ORG. No.

M381

ORG. No.

M381



MEB Ref	Conductor Size (mm)	Colour Code	Manufacturer		Dimensions (mm)		Hole Dia.	Palm Dia.	Stud Size	Die/No of Compressions			
			Ref		Ext Dia	Length				Y35/G10	No	G5	No
382/1	120	Pink	SICAME	CPTAU 120	25	104	12.8	30.5	M12	UE 215			
382/2	95	Grey	SICAME	CPTAU 95	20	94	12.8	25.5	M12	UE 173		WE173	
382/3	50	Yellow	SICAME	CPTAU 50	20	94	12.8	25.5	M12	UE 173		WE173	
382/4	35	Red	SICAME	CPTAU 35	20	94	12.8	25.5	M12	UE 173		WE173	
382/6	120	Pink	SICAME	CPTAU 120 B21	25	114	21	37.5	M20	UE 215			
382/7	95	Grey	SICAME	CPTAU 95 B21	25	114	21	37.5	M20	UE 215			
382/8	50	Yellow	SICAME	CPTAU 50 B21	25		21		M20	UE 215			

SCALE	
DATE	18.04.94
DRAWN	R.C.F.
TRACED	C.E.
CHECKED	
APPROVED	

ABC COPPER LUG (INSULATED BI-METAL)

ISSUE	DATE
A	07.08.02
REDRAWN ON CAD MINOR MODS.	
MJB/XD	



Aquila

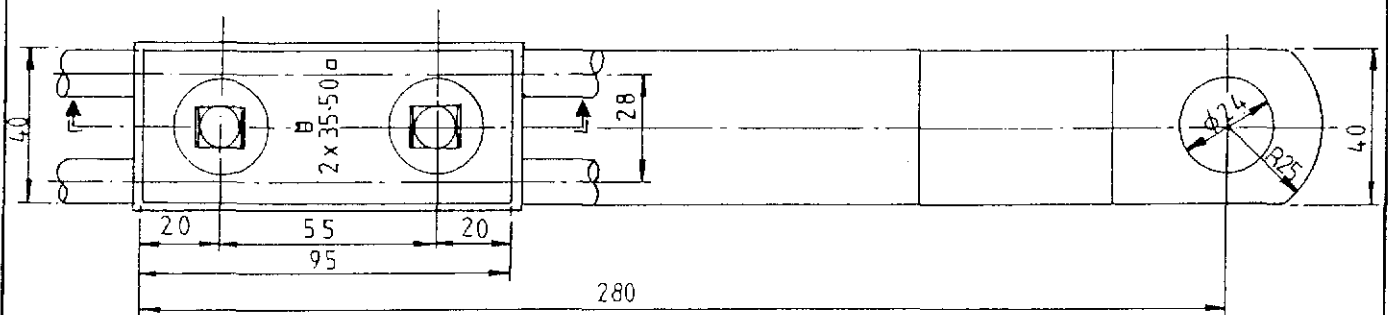
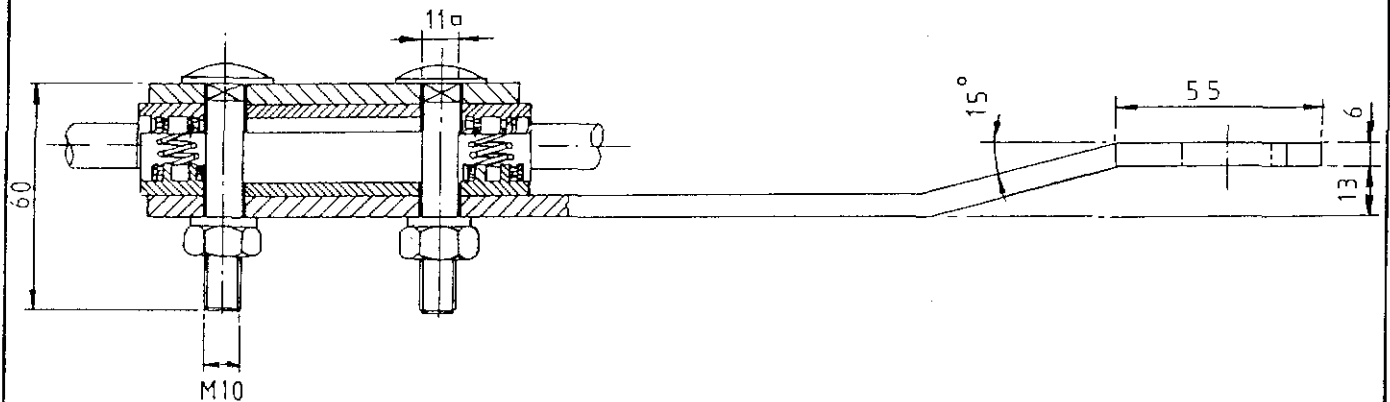
DRG. No.

M382

DRG. No.

M382

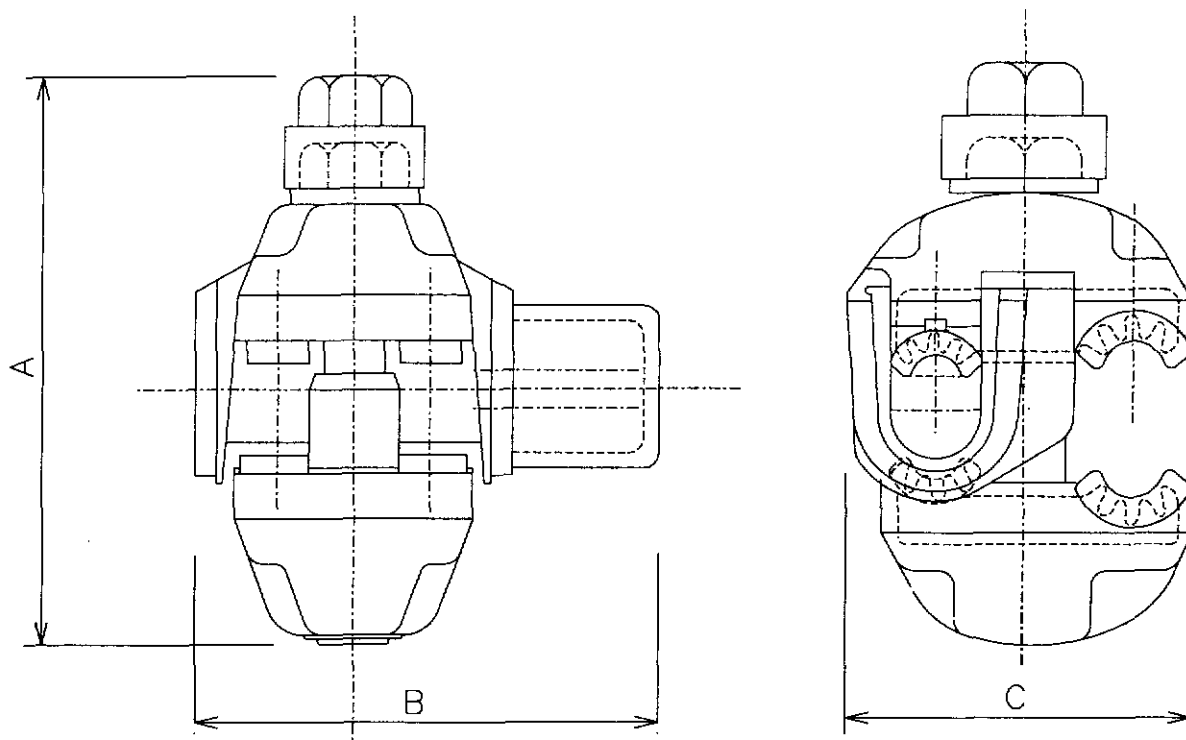
A4



M.E.B. REF	CABLE SIZE mm. ²	MOSDORFER REF	TORQUE
M383/1	2 x 35	5001/10	46 Nm

Tightening Torque to be marked on Clamp

B ADDED BY AJB DATE: 4.8.93 Torque Added.	TITLE	SCALE	-	 Midlands Electricity plc M383
	A.B.C SERVICE ANCHOR CLAMP	DATE	1.7.91	
		DRAWN	A.J.B	
		TRACED	S.J.H.	
		CHECKED	Ref	
	APPROVED			



MEB REF.	CONDUCTOR SIZE (mm ²)		MANUFACTURERS REFERENCE	DIMENSIONS (mm)			WEIGHT (kg)	BOLT SIZE	SPANNER SIZE
	MAIN	TAP		A	B	C			
384/0	35-95 ABC	35-95 ABC 50-100 AAC PVC 35-70 ABC	SICAME TTD 20 IF	90	61	57	0.20	M8	13mm
384/1	25-95 ABC	25-35 ABC 25 AAC PVC 14-35 Cu PVC 6-35 Copper Concentric Phase 25-35 Hybrid Concentric Phase	SICAME TTD 15 IF	77	54	43	0.10	M8	13mm
384/2	35-120	35-120	TTD 27 IF	90	61	60	0.20	M8	13mm

Notes :-

- 1) Refer to Method O.H.L. 16 for Fitting Instructions.
- 2) The Tap Conductor must never be under Tension.
- 3) Use on Single Insulated Conductors only.

SCALE	26.06.89
DRAWN	A.J.B.
TRACED	P.J.W.
CHECKED	R.C.F.
APPROVED	R.C.F.

ABC INSULATED PIERCING CONNECTOR

ISSUE	DATE
C	08.08.02
REDRAWN ON CAD MINOR MODS.	
M.J.B/K.D	



Aquila

DRG. No.

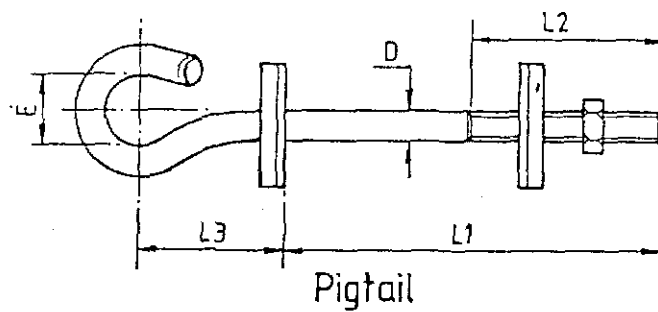
M384

DRG. No.

M384

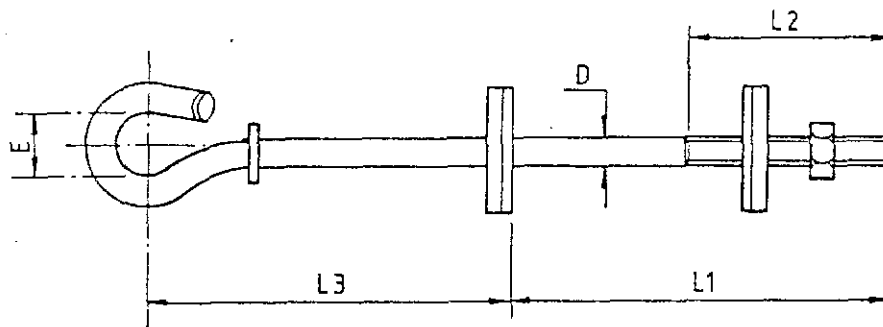
A4

Produced by
Output Services



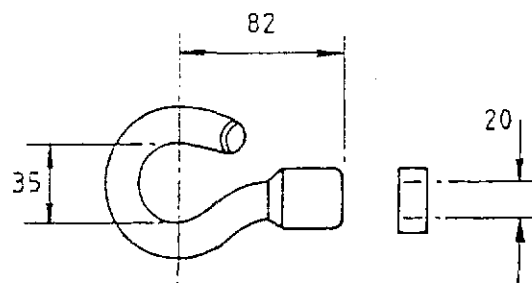
M.E.B. Ref.	BICC Ref.	Type	Dimensions (mm)					Weight kg.
			E	L1	L2	L3	D	
385 / 0	APH 20020	Pigtail	35	200	125	80	20	1.4
385 / 1	APH 24020		35	240	125	80	20	1.5
385 / 2	APH 32020		35	320	125	80	20	1.8

SCALE	—	ABC — SUPPORT HOOK.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	6.5.87			DRG. No. M 385 MF
DRAWN	RAC			
TRACED	PAO			
CHECKED				
APPROVED	BNJ			
MARQUE 7637/85 (7/21)		E.S.I. STANDARD 43-14	E.S.I. DRG. N°	



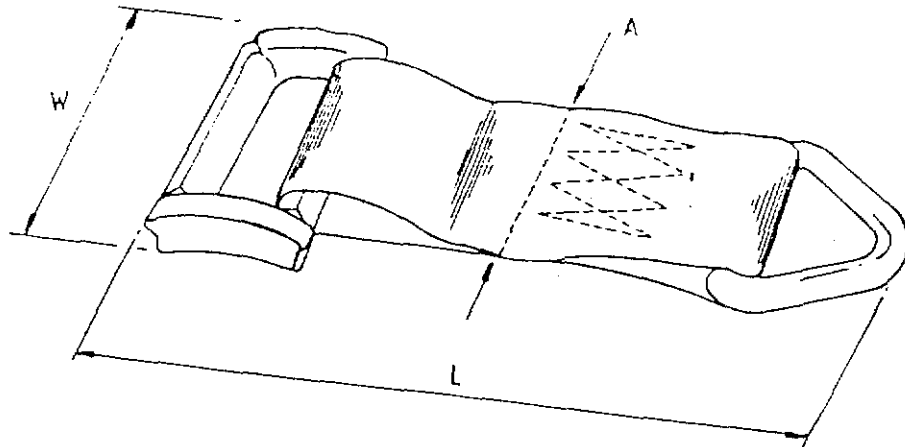
MEB Ref.	BICC Ref.	Type	Dimensions (mm)					Weight kg.
			E	L1	L2	L3	D	
386 / 0	APEH24020	Pigtail	35	240	125	215	20	2.0
386 / 1	APEH32020		35	320	125	215	20	2.2

SCALE	—	ABC EXTENDED SUPPORT HOOK.	ISSUE	MIDLANDS ELECTRICITY BOARD DRG. No. M386 (MF)
DATE	6.5.87			
DRAWN	RAC			
TRACED	PAO			
CHECKED	<i>[Signature]</i>			
APPROVED	<i>[Signature]</i>			
MARQUE 7637/85 (7/21)		E.S.I. STANDARD 43-14	E.S.I. DRG. N°	



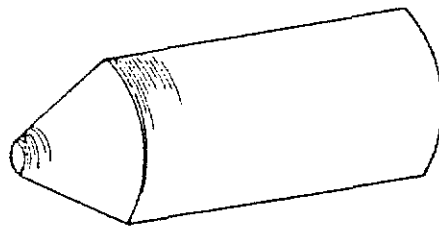
M.E.B. REF.	B.I.C.C. REF.	TYPE
387 / 0	APNH20	PIGTAIL

SCALE	NTS	ABC-NUT HOOK	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	8-5-87			
DRAWN	RAC			DRG. No. M387 (MF)
TRACED	G.H.R.			
CHECKED	Z			
APPROVED	BN			
MARQUE 7637/85 (7/21)		E.S.I. STANDARD 43-14	E.S.I. DRG. N°	

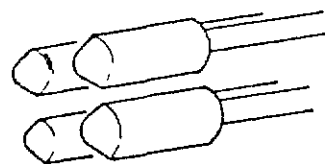


M.E.B. REF.	CABLE SIZE m.m. ²	DIMENSIONS m.m.			BICC REF.	MAXIMUM WORKING LOAD k.N
		L	W	A		
38810	2 x 35	250	80	50	ASS95	3.0

SCALE	—	A.B.C. SUSPENSION STRAP	ISSUE	MEB
DATE	28-6-89		A	
DRAWN	RAC		Cable size change 25-6-91 AJB	ORG No M388 <i>MEB</i>
TRACED	S.J.H.			
CHECKED	<i>Bos</i>			
APPROVED		E.S.I. STANDARD	E.S.I. ORG No	



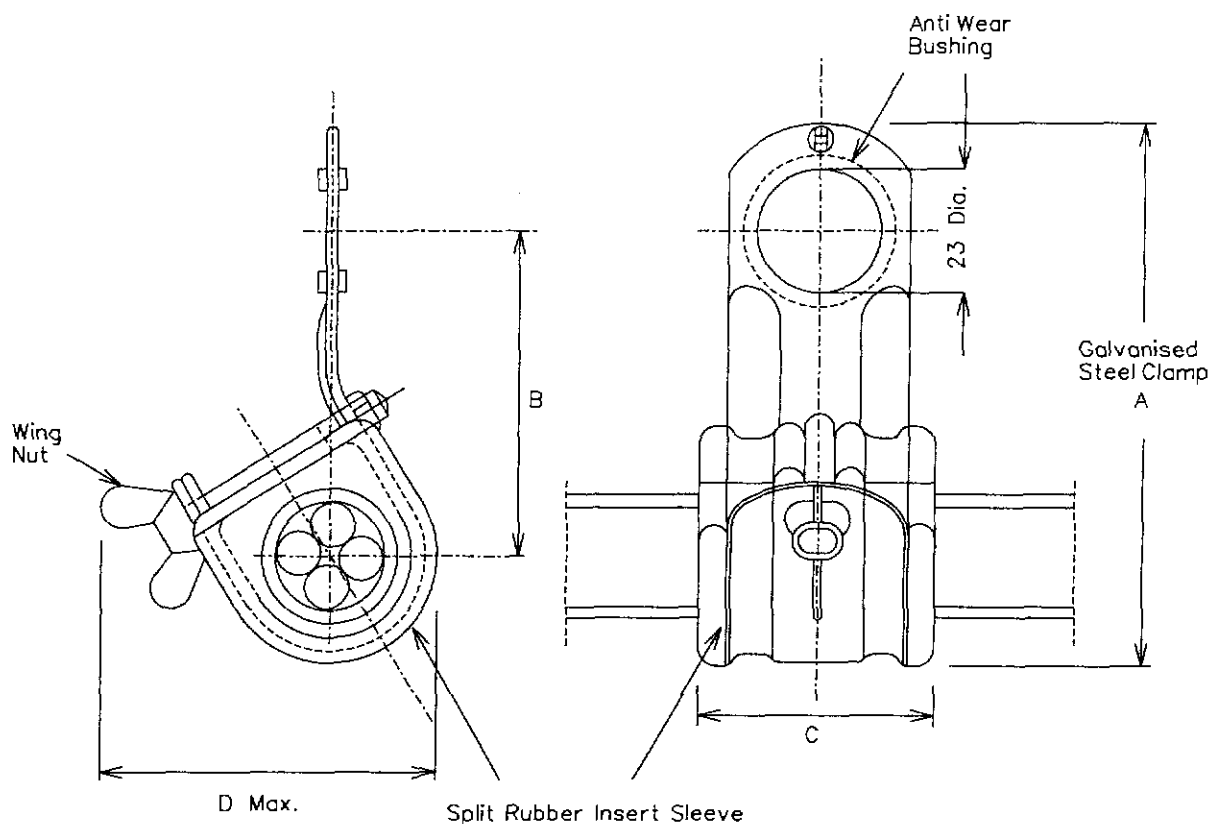
Outer Cap.



Heat Shrink Caps.

MEB Ref.	RAYCHEM Kit Ref.	CONDUCTOR Size (mm ²).
389/0	EPKE 0064	50-120

SCALE	---	ABC - LIVE POT END.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	6.5.87			DRG. No.
DRAWN	RAC			M389 MF
TRACED	PAQ			
CHECKED	RJ			
APPROVED				
AARQUE 7637/85 (7/21)		E.S.I. STANDARD	E.S.I. DRG. No.	



Notes :-Clamp is suitable for intermediate or intermediate angle position up to 30° .
Clamp supplied complete with insert.

AQUILA REF.	CONDUCTOR SIZE (mm ²)	MANUFACTURERS REFERENCE	DIMENSIONS (mm)				WEIGHT (kg)
			A	B	C	D	
391/0	4 x 95 ABC	SICAME PS 40 PM	159	95	64	114	0.40
391/1	2 x 95 ABC 4 x 50 ABC	SICAME PS 30 PM	159	95	64	114	0.43
391/2	2 x 50 ABC	SICAME PS 27/ 2PM	159	95	64	114	0.44
391/3	4 x 120 ABC	SICAME PS 43PM	159	95	64	114	

SCALE	-
	17.09.93
DRAWN	A.J.B.
TRACED	S.J.H.
CHECKED	R.C.F.
APPROVED	R.C.F.

ABC - SUSPENSION CLAMP

ISSUE	DATE
C	14.08.02
REDRAWN ON CAD	
M.J.B/K.D.	



Aquila

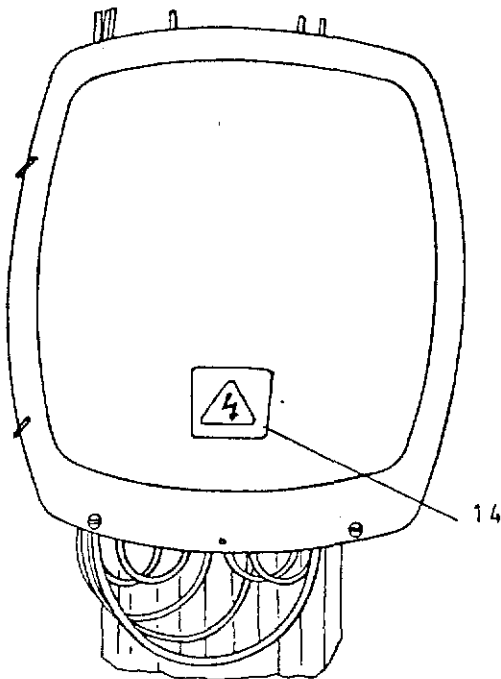
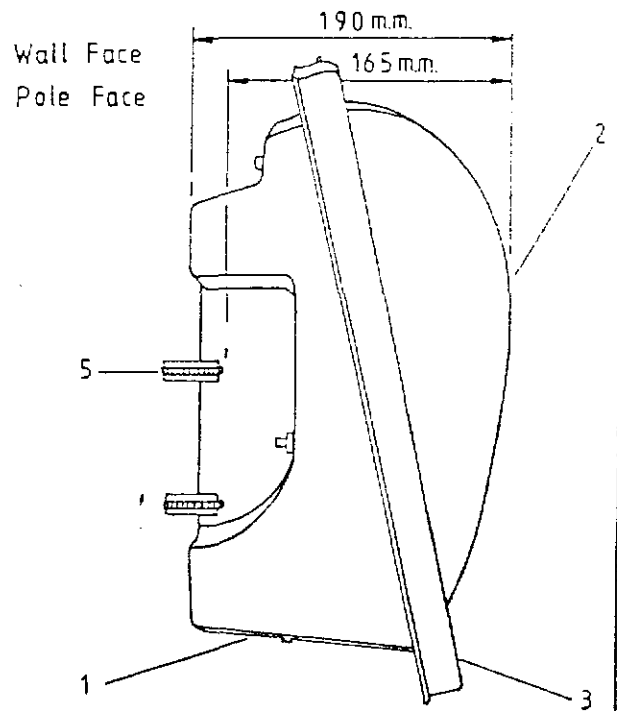
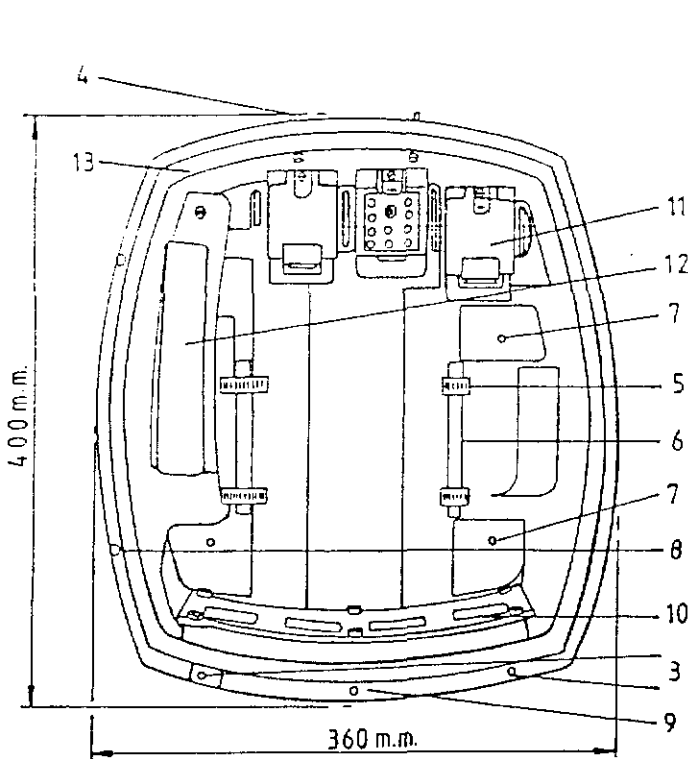
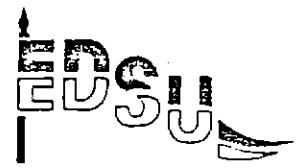
DRG. No.

M391

DRG. No.

M391

A4



- 1 Base
- 2 Cover
- 3 Captive Cover Fixings
- 4 Cover Locators
- 5 Pole Fixing Straps
- 6 Reinforcing Rods
- 7 Wall Mounting Fixing
- 8 Cover Retaining Holes
- 9 Holes for Padlock
- 11 Insulated Phase Connectors (3)
- 12 Shrouded Neutral Block
- 13 Weatherproof Seal
- 14 Warning Label

MEB REF N°	BOWTHORPE REF N°	N° OF SERVICES
392 / 0	ASDB-1	6 x 35 mm ²

SCALE	—
DATE	30.6.89
DRAWN	R.A.C.
TRACED	S.J.H.
CHECKED	
APPROVED	

ASSEMBLY A.B.C. SERVICE
TERMINATION BOX

ISSUE

MEB

DRG. No.

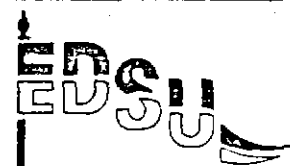
M 392

(MF)

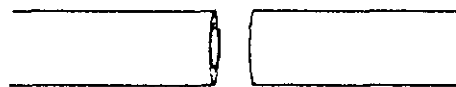
E.S.I. STANDARD

ES1. DRG. N°

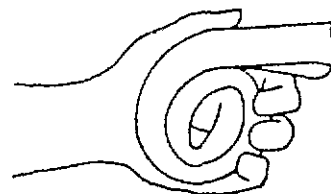
UDOG 2955



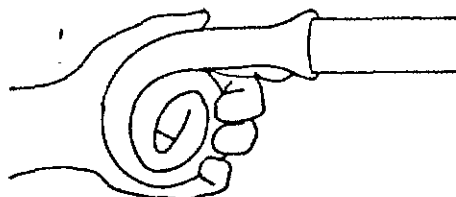
1. Clean insulation of cable ends to be spliced with solvent wipe provided in kit.



2. Roll Rayvolve into a snail shape as shown squeezing lubricant toward the outside layer.



3. Push end of Rayvolve onto cable end and roll back onto insulation.



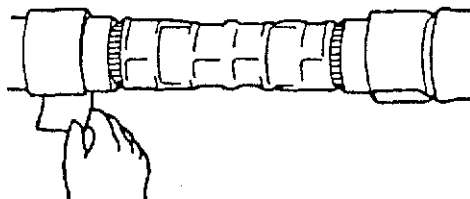
4. Remove insulation from cable ends to one-half connector length + 1/4".



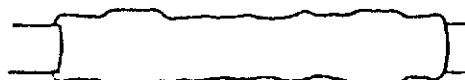
5. Connect cable ends with appropriate connector and connector manufacturer's recommended installation tool.



6. Apply a wrap of sealant around the cable insulation at each side of connector, area as shown.



7. Position Rayvolve over splice with sealant beneath each end.



M.E.B. REF.	RAYCHEM CATALOG KIT N°	APPLICATION O.D. RANGE mm	JOINT LENGTH mm	A B C SIZES mm
393/0	RVS - 12	12.2 - 22.1	240	50, 70 & 95

SCALE	—
DATE	22-6-88
DRAWN	
TRACED	P.J.N.
CHECKED	<i>Ros</i>
APPROVED	

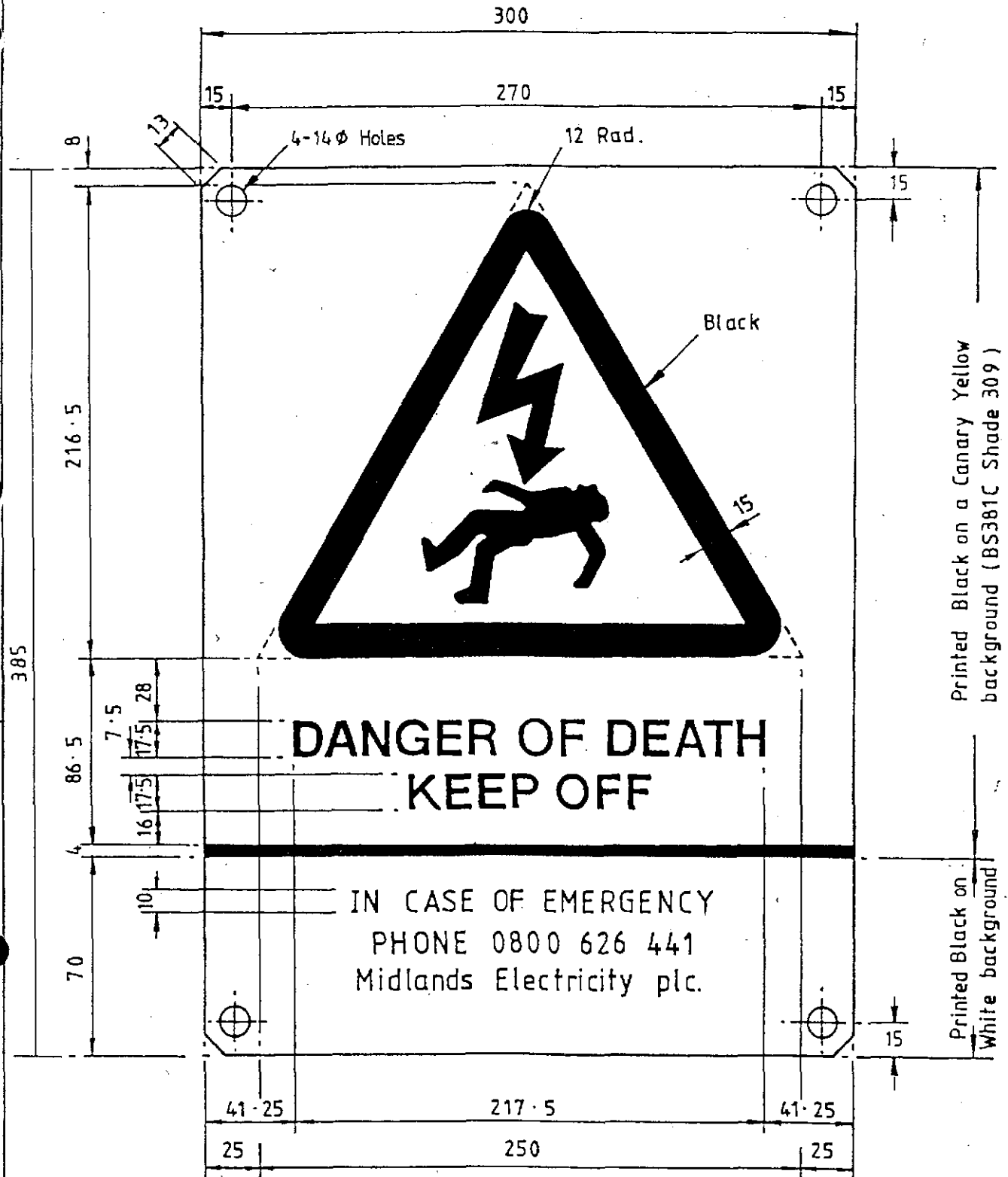
A.B.C INSULATING SLEEVE
FOR TENSION & NON TENSION JOINTS

ISSUE	
	MEB
	DRG. No. M. 393 (MF)

SECTION 17

MIN NUMBER	M	DESCRIPTION
3451390	139	POLE - CIRCUIT IDENTIFICATION PLATE (SYMBOL TYPE)
3452760	276	POLE NUMBER AND NAMEPLATE
3452811	281/1	SAFETY SIGN TRANSMISION TOWERS
3453690	369	FISHING WARNING & PROHIBITION NOTICES
3454621	462/1	SAFETY SIGN
3454900	490	TOWER SAFETY SIGN BRACKET
3455800	580	POLE NUMBERING SYSTENFITTING RAILS
3455810	581	DIGITS FOR POLE NUMBERING SYSTEM

NO DEVIATION FROM THIS SPECIFICATION PERMITTED WITHOUT WRITTEN CONSENT FROM THE ENGINEERING DEPT.



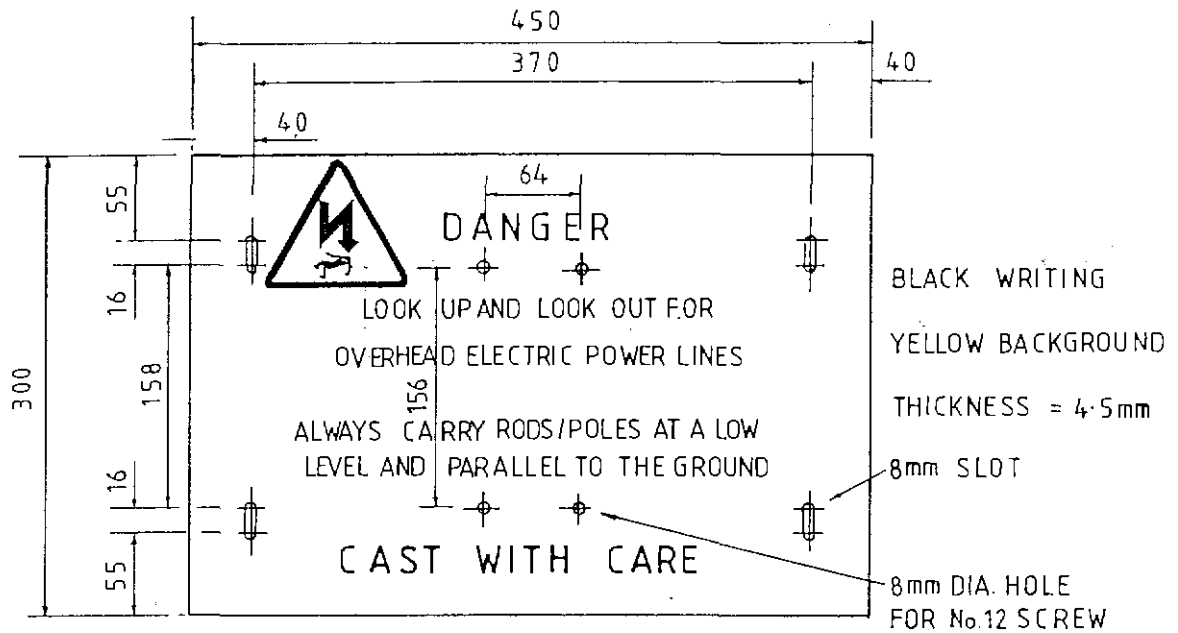
MATERIAL 3mm White rigid A.B.S.
plastic printed black & Yellow on
white & over laminated with 80mic
clear P.V.C.
SIZE 385mm x 300mm

DIVISIONS Shropshire & Hereford,
Tipton, North Staffs
South Staffs

A	ADDED BY: A.J.B. DATE: 25-8-92	Hole dia. change dims added.	TITLE SAFETY SIGN FOR OVERHEAD LINE TRANSMISSION TOWERS	SCALE	—	MEB Midlands Electricity plc DRG. No. M281/1
				DATE	13-11-90	
				DRAWN		
				TRACED	C.E.	
				CHECKED	alb	
				APPROVED		

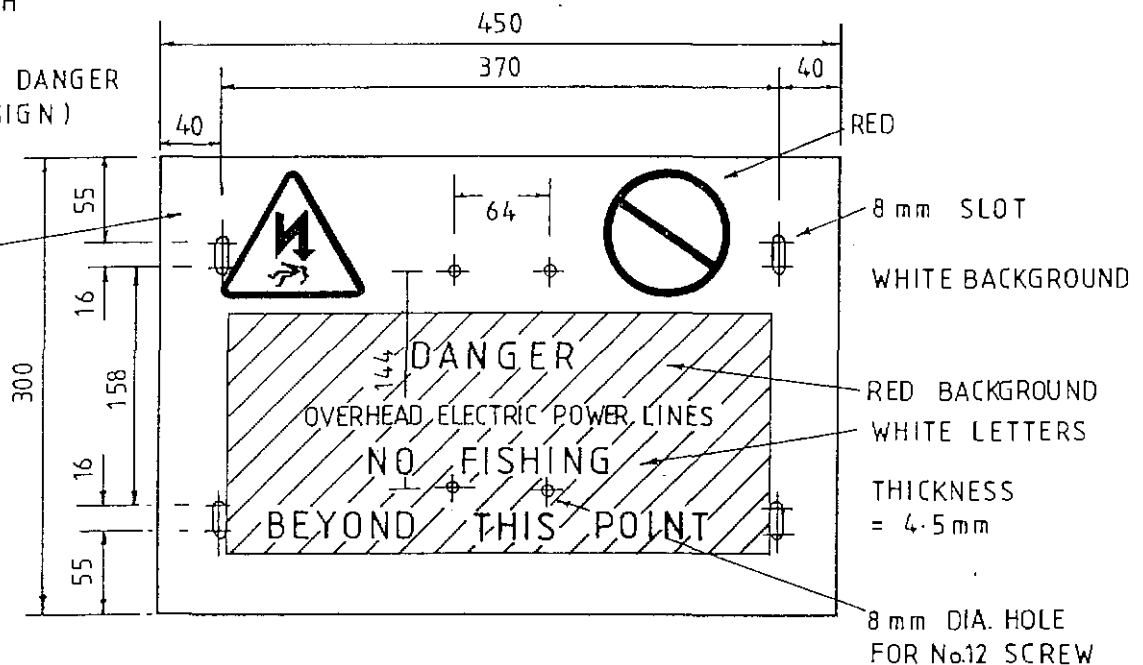
(MF)

SIGN -369/1 WARNING NOTICE

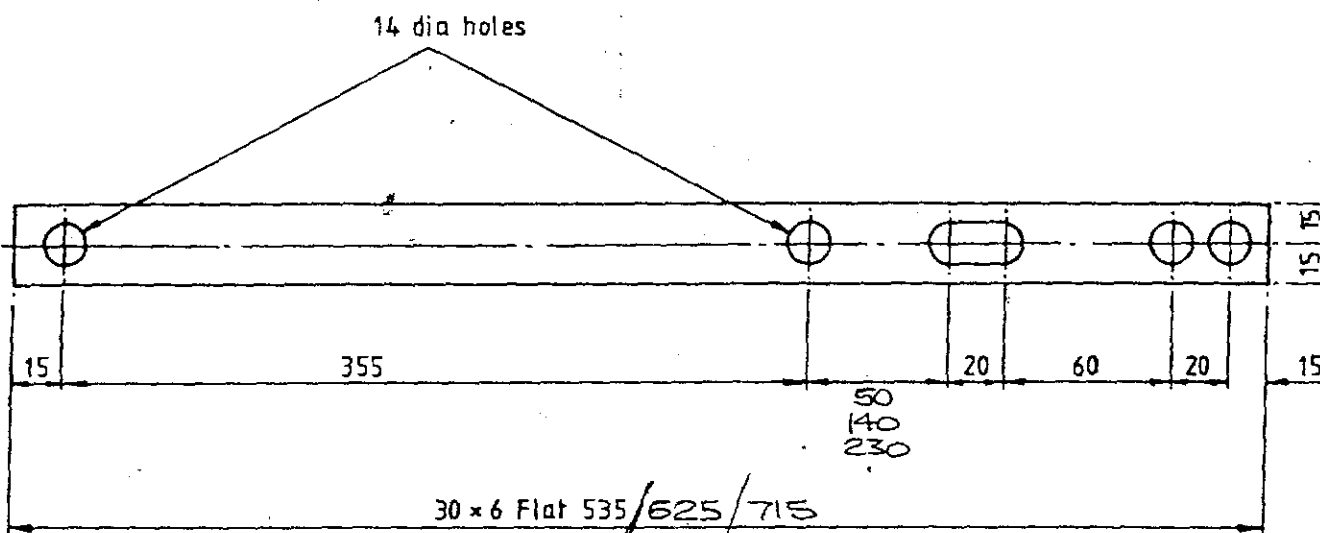


SIGN - 369/2 PROHIBITION NOTICE

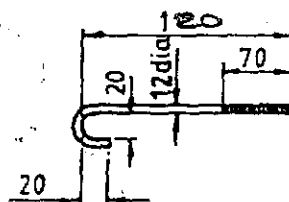
YELLOW WITH
BLACK
(STANDARD DANGER
OF DEATH SIGN)



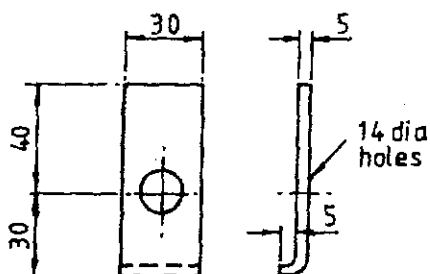
ADDED BY DATE	TITLE FISHING WARNING & PROHIBITION NOTICES	SCALE	—	MEB Midlands Electricity plc. ORG No M 369
		DATE	22-7-96	
		DRAWN	R.C.F.	
		TRACED	E.L.W.	
		CHECKED	RCF	
		APPROVED	RCF	



DETAIL OF SUPPORTING STRAP



DETAIL OF J BOLT



DETAIL OF FIXING CLIP

NOTE

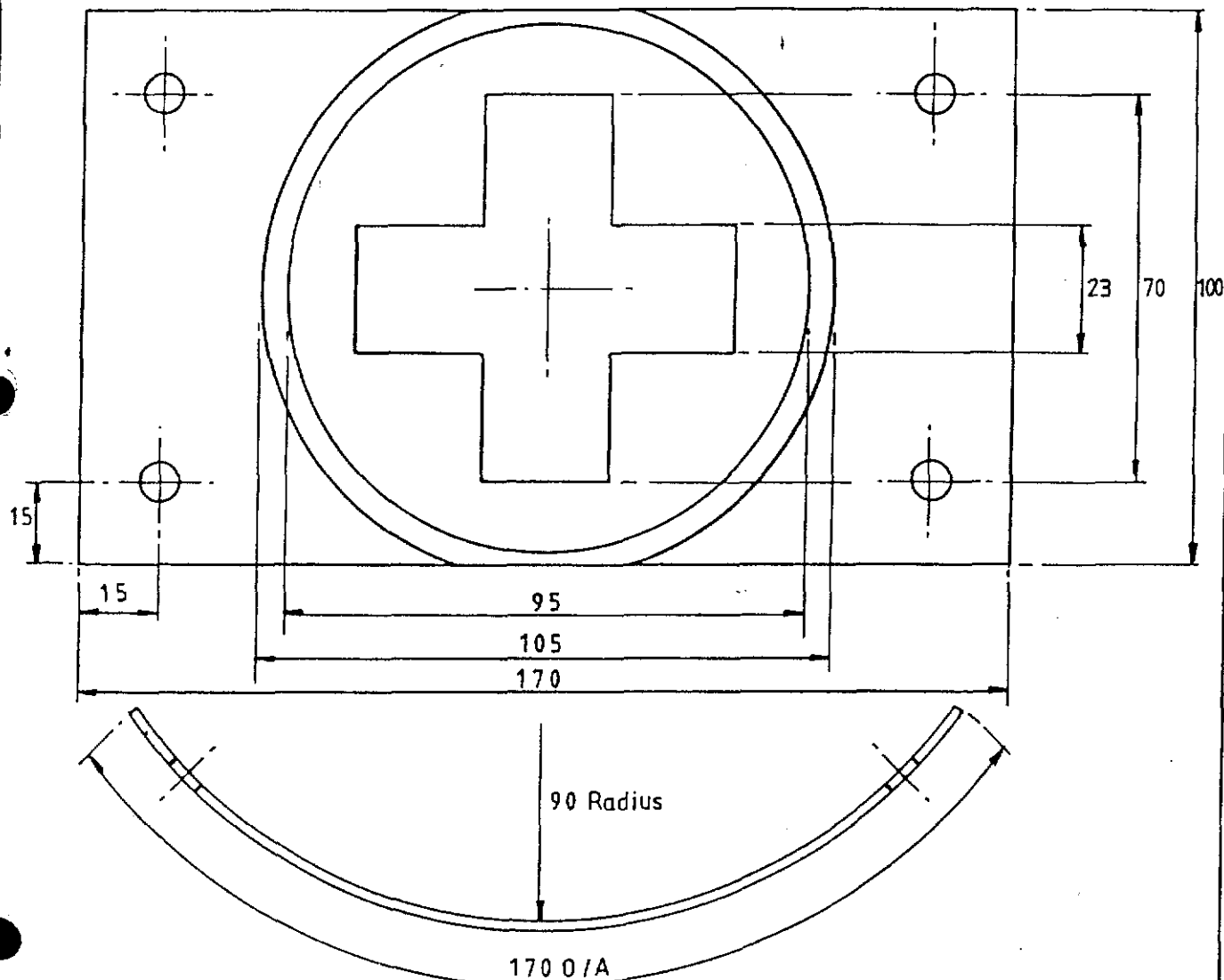
All steel grade 43A BS4360
Galvanised to BS729 PT1

MATERIALS PER SET

- 2 off - Supporting straps (see detail for appropriate length)
- 2 off - J Bolts complete with washer & nut
- 2 off - Fixing clips complete with M12 Bolts 35long with washer & nut
- 4 off - Plate fixing M12 Bolts 35long with 2 Fibre washers, Flat washer & nut

ADDED DATE A.Dims REVISED	TITLE	SCALE	—	MEB Midlands Electricity plc M 490
	TOWER SAFETY SIGN BRACKET	DATE	27-8-92	
		DRAWN	D.H.	
		TRACED	G.S.	
		CHECKED		
		APPROVED	W.D.	

No deviation from this Specification permitted without written consent from the Engineering Dept.



MATERIAL

1.5 mm. Thick White A.S.A. Plastic
printed and over laminated on both
sides with 80mic. clear PVC.
for resistance to creosote
7mm. dia holes drilled in each corner.

FIXING

Fixed with 4 - 75 mm x No 14 (5 mm.)
Galvanised Drive Screws.

COLOUR

1. White Background.
2. Outer Symbol Border to be same colour as Symbol.
3. Symbol type and colours to be specified on order (Typical Symbol shown).
4. Colours to be C.E.G.B. STD. 062908.

ADDED BY DATE	TITLE POLE - CIRCUIT IDENTIFICATION PLATE (SYMBOL TYPE)	SCALE	---	 Midlands Electricity plc ORG No M.139
		DATE	29-4-93	
		DRAWN	A. J. B.	
		TRACED	S. J. H.	
		CHECKED	<i>Ref</i>	
		APPROVED		

CURVED NUMBER PLATE

M.E. 276/0

10 Letters/Numbers/1 Line

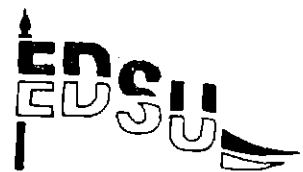
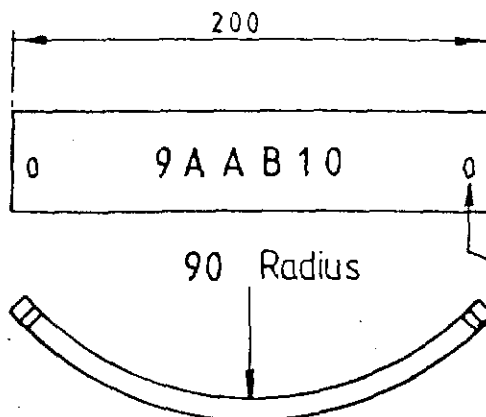
Depth 30mm.

NOTES

Fixing M.276/0, 1, 2, 3, 8 & 9
plates fitted by 2 - 62mm. N°14
Drive screws.

MATERIAL & LETTERING

1.5mm thick White A.B.S. Plastic
printed 10mm Black letters/Numerals
on White background & both sides to be
lacquered to resist Creosote attack



CURVED NAME PLATES

NOTES

See above.

M.E. 276/1

16 Letters/Numbers/1 Line

Depth 30mm

M.E. 276/2

32 Letters/Numbers/2 Lines

Depth 50mm.

M.E. 276/3

48 Letters/Numbers/3 Lines

Depth 70mm.

M.E. 276/8

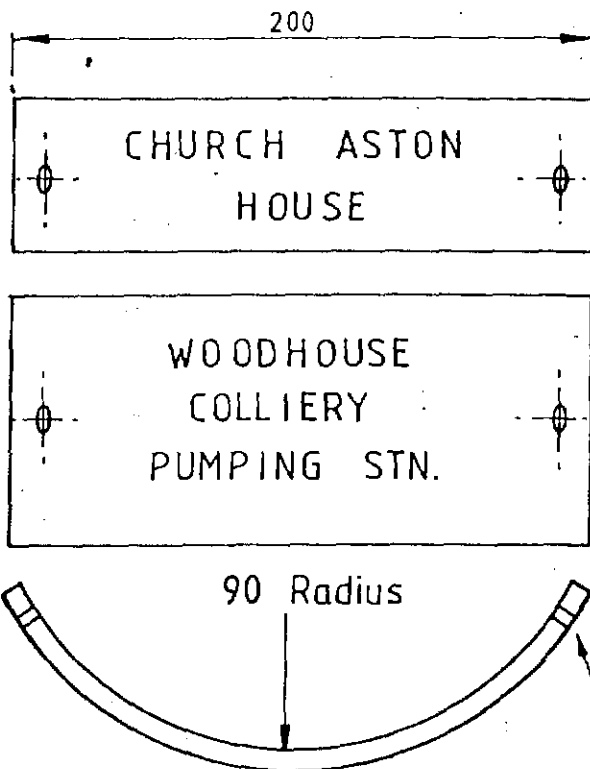
64 Letters/Numbers/4 Lines

Depth 90mm.

M.E. 276/9

80 Letters/Numbers/5 Lines

Depth 110mm.



FLAT NUMBER / NAME PLATES

NOTES See above.

M.E. 276/4

18 Letters/Numbers/2 Lines

Size 150mm x 50mm

M.E. 276/5

28 Letters/Numbers/2 Lines

Size 200mm x 50mm

M.E. 276/6

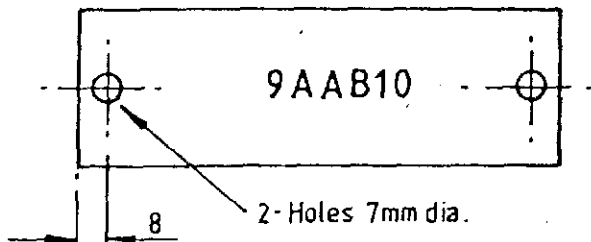
42 Letters/Numbers/3 Lines

Size 200mm x 63mm

M.E. 276/7

36 Letters/Numbers/2 Lines

Size 250mm x 50mm



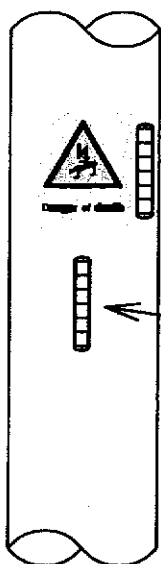
ADDED BY P.J.W.
DATE 11-5-92
MODIFIED

TITLE

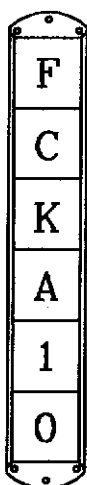
POLE NUMBER AND NAMEPLATE

SCALE	N.T.S.
DATE	25-9-91
DRAWN	R.A.C.
TRACED	G.S.
CHECKED	ABB
APPROVED	

MEB Midlands Electricity plc
DWG No M. 276



ACCEPTABLE
FIXING
POSITIONS
(numbers must
be fixed above
and not obscured
by barbed wire)



DIGITS
MIN 345581-2-3-4

DIGITS TO READ
TOP TO BOTTOM
"FCKA10"

43

MIN	LENGTH	No DIGITS
3455801	312	12
3455802	178	6

FIX USING A MINIMUM OF
4 - 3.35 X 40MM
GALVANISED CLOUT NAILS.
MIN-8500517.

ENDS TO BE FLATTENED
USING PLIERS TO
SECURE NUMBER DIGITS

SCALE	1:1
DATE	6/12/00
DRAWN	ADD
TRACED	
CHECKED	RSG
APPROVED	9.01.01

POLE NUMBERING SYSTEM FIXING RAILS

ISSUE	DATE
A	05.01.01
MINOR MODS.	
D.B. / R.C.	

Produced by
Output Services
GPU
POWER UK
DRG. No.
M580

DRG. No.

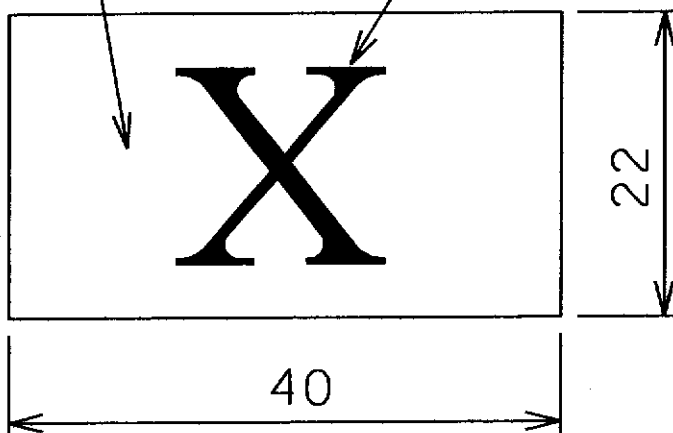
M580

Digits	Min. Numbers
0	345581/0
1	/1
2	/2
3	/3
4	/4
5	/5
6	/6
7	/7
8	/8
A	345582/0
B	/1
C	/2
D	/3
E	/4
F	/5
G	/6
H	/7
I	/8
J	/9
K	345583/0
L	/1
M	/2
N	/3
O	/4
P	/5
Q	/6
R	/7
S	/8
T	/9
U	345584/0
V	/1
W	/2
X	/3
Y	/4
Z	/5
-	/6
/	/7

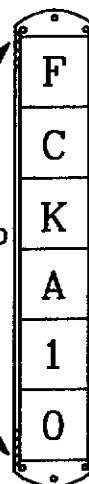
"DIGITS TO BE SUPPLIED
IN PACKS OF 100"

ALUMINIUM

BLACK EMBOSSED



ENDS FLATTENED TO
SECURE DIGITS



FIX USING A MINIMUM
OF 4 - 3.35 X 40MM
GALVANISED CLOUT NAILS.
MIN - 8500517

SCALE	-
DATE	6/12/00
DRAWN	ADD
TRACED	
CHECKED	RSG
APPROVED	9.01.01

DIGITS FOR POLE NUMBERING SYSTEM

ISSUE	DATE
A	6/12/00
MINOR MODS	
DB/RC	

Produced by
Output Services

GPU
POWER UK

ORG. No.
M581-2-3-4

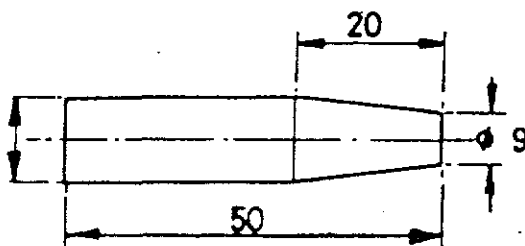
ORG. No. M581-2-3-4

A4

SECTION 18

[illegible]

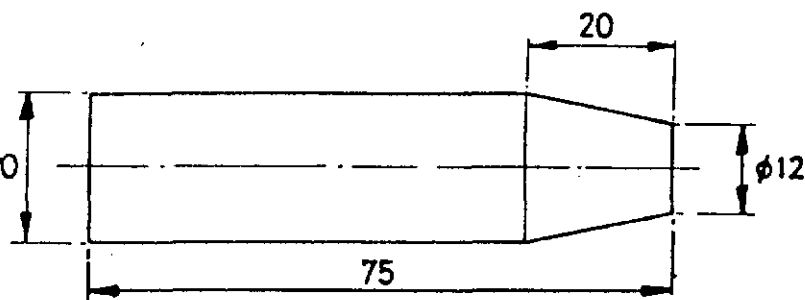
M425/1

 ϕ 12

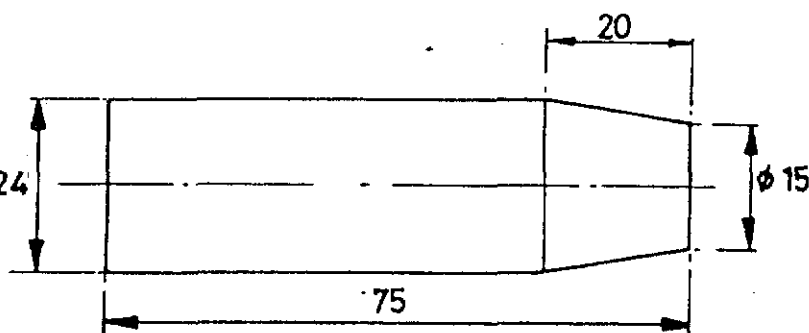
Gen Tolerance.

 $\phi \pm 1 \text{ mm.}$ $l \pm 5 \text{ mm.}$

M425/2

 ϕ 20

M425/3

 ϕ 24

APPLICATION:—

M425/1 For plugging holes made with the
Mattson size N° 3 increment borer.

M425/2 For plugging unused L.V. pole holes.
($11\frac{1}{16}$ " or 18 mm dia.)

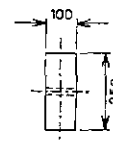
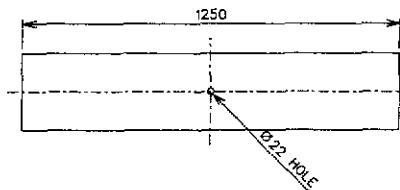
M425/3 For plugging unused H.V. pole holes.
($13\frac{1}{16}$ " or 22 mm dia.)

MATERIAL:—

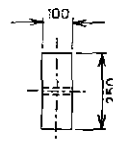
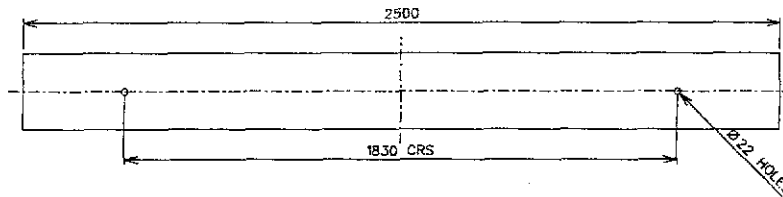
Hardwood (eg Beech) straight grained & free
of knots pressure creosoted as B.S.913.

SCALE	FULL	WOOD POLE PLUGS.	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	10-4-79		A	
DRAWN	R.A.C.		425/2 corrected to ϕ 12	ORG. No. M 425 
TRACED			10-11-92 AJB	
CHECKED				
APPROVED				
2044G		ESI STANDARD 43-88		ESI DRG. N° 

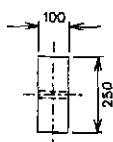
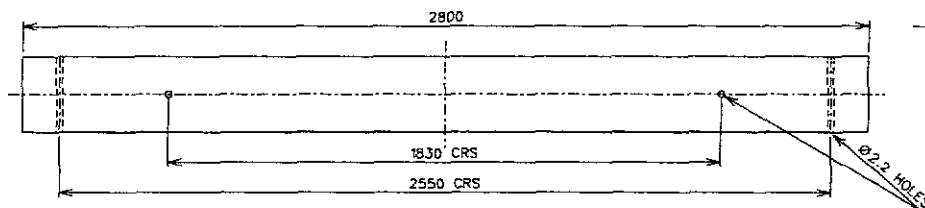
TYPE M.441/1
BOG SHOE
STABILISING BLOCK



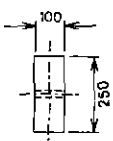
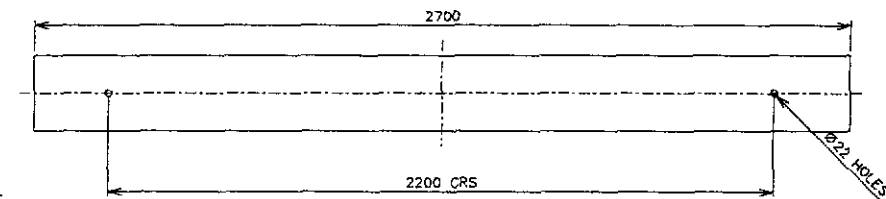
TYPE M.441/2
6' H-POLE
LOWER BAULK
NO BOG SHOE



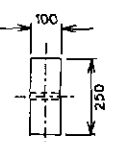
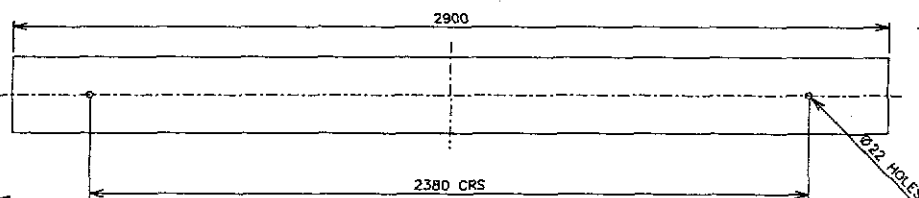
TYPE M.441/3
6ft H-POLE
LOWER BAULK
BOG SHOE



TYPE M.441/4
6ft H-POLE
≤ 14m
DIAGONAL BRACE



TYPE M.441/4
6ft H-POLE
> 14m
DIAGONAL BRACE



NOTE: ALL BLOCKS TREATED AS E.S.I. Std. 43-88 ISSUE 3

SCALE	1/25
DRAWN	29.10.85
TRACED	RAC
CHECKED	CAR
APPROVED	-
	WOK

FOUNDATION BLOCKS
FOR
SINGLE & H. POLES

ISSUE	DATE
B	24.01.03
Definitions	
added.	
JN/ADD	



Aquila

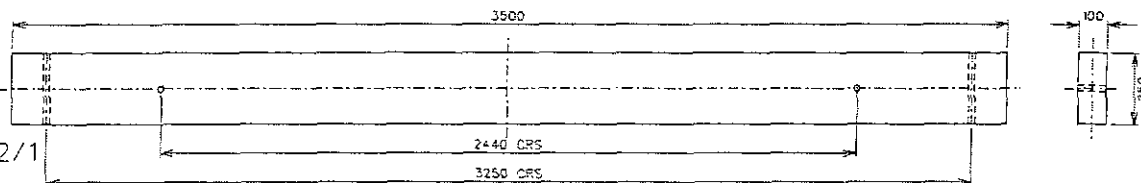
DRG. No.

M 441

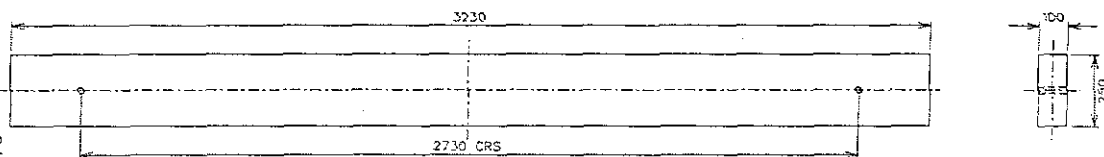
DRG. No.

M 441

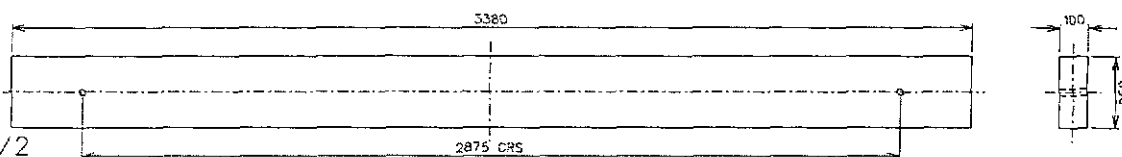
TYPE M.442/1
8ft H-POLE
LOWER BAULK
BOG SHOE



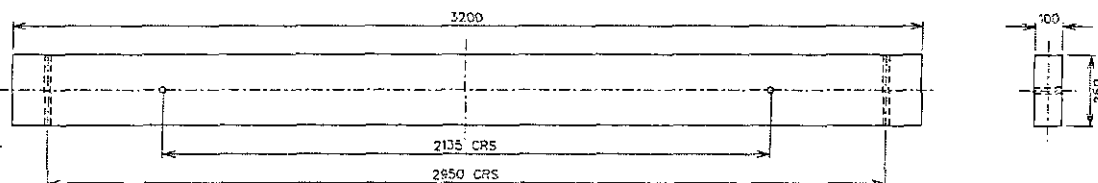
TYPE M.442/2
8ft H-POLE
≤ 14m
DIAGONAL BRACE



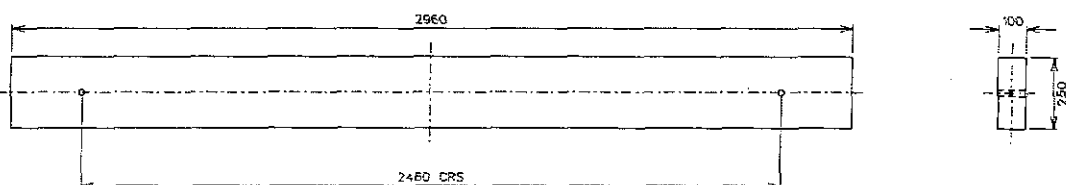
TYPE M.442/2
8ft H-POLE
> 14m
DIAGONAL BRACE



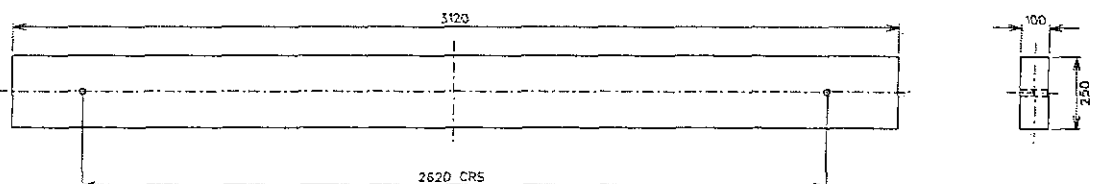
TYPE M.442/4
7ft H-POLE
LOWER BAULK
BOG SHOE



TYPE M.442/5
7ft H-POLE
≤ 14m
DIAGONAL BRACE



TYPE M.442/6
7ft H-POLE
> 14m
DIAGONAL BRACE

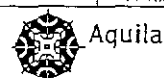


NOTE: ALL BLOCKS TREATED AS E.S.I. Std. 43-88 ISSUE 3

SCALE	1/25
DATE	29.10.85
DRAWN	RAC
TRACED	CAR
CHECKED	-
APPROVED	WOK

FOUNDATION BLOCKS
FOR
SINGLE & H. POLES

ISSUE	DATE
A	27.01.03
Definitions	
added.	
JN/ADD	



Produced by
Primary CAD

DRG. No.

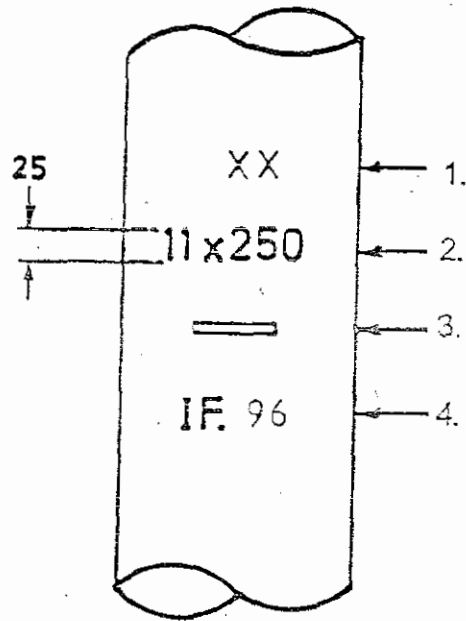
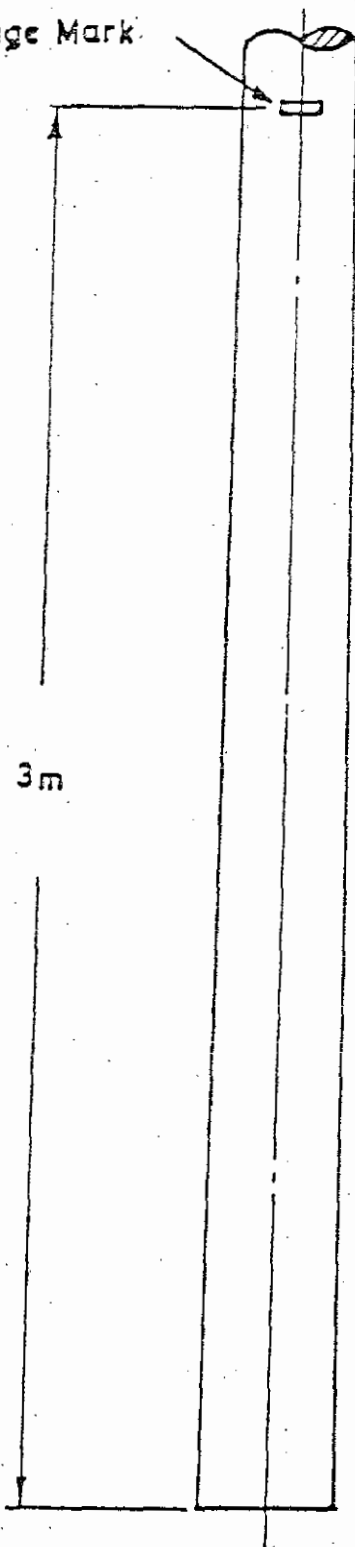
M 442

DRG. No.

M 442

A4

Gouge Mark



1. Initial of supplier
2. Length and diameter at 1.5m
3. Gouge mark min 50mm long
5 mm wide x 3 mm deep
3m from pole butt
4. Imported Fir and year of
impregnation (1996)
5. Overall length of pole and diameter
at 1.5m to be stamped on butt
6. Lettering to be min 25mm high,
Gouged min 5mm wide, 3mm deep
7. Marking and gouging to be on the
same face of pole as pole top scarfing

TITLE

STANDARD POLE MARKINGS & GOUGING

SCALE

DATE

DRAWN

TRACED

CHECKED

APPROVED

6-8-96

G.P.

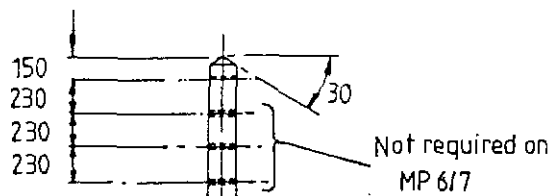
N.R.W.

MEB

Midlands Electricity plc

DRG No

M 450



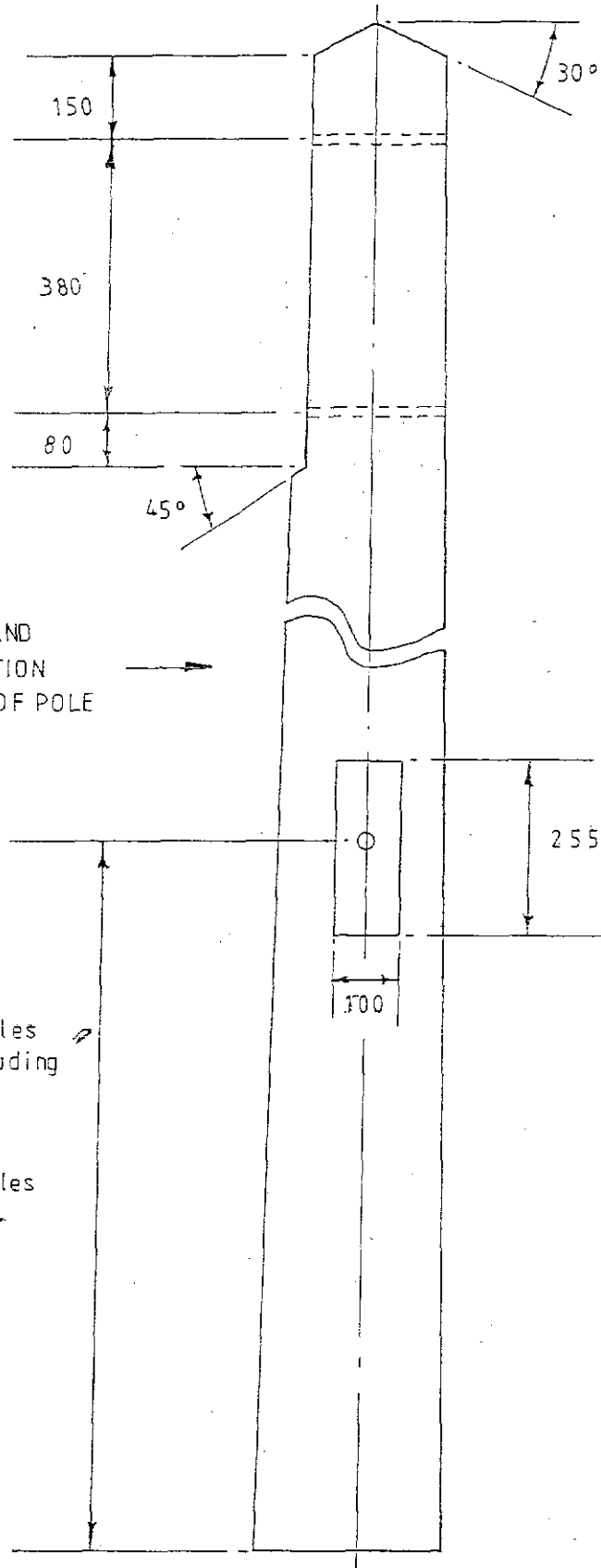
A

NOTE:— ALL POLE DRILLINGS
18 ϕ

MP 6/1
MP 6/2
MP 6/3
MP 6/4
MP 6/6/3
MP 6/6/4
MP 6/7

Note: Treated as E.S.I. Std. 43 - 88 Issue 3

SCALE	1-50	POLE DRILLINGS MP TYPES AS SHOWN	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	17-11-86		A	
DRAWN	R.A.C.		ESI issue 3	
TRACED				
CHECKED				
APPROVED	<i>G.D.H.</i>		10-11-92 A.J.B	
			DRG. No.	M 450/1



SCARFING ON
ONE SIDE ONLY
GIVING MIN. 50mm
WIDE FLAT FACE

1. ALL DRILLINGS TO BE
22 mm ϕ
2. POLES TREATED TO
CURRENT EATS 43-88
PARTS 1&2
3. DIMENSIONS IN MM.

GOUGE MARK AND
POLE INFORMATION
ON THIS FACE OF POLE

SCARFING FOR
FOUNDATION BAULK
255 x 100 FLAT FACE

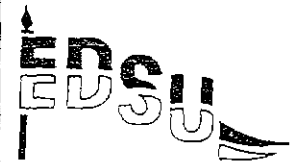
1-000 for poles
upto & including
10.5m

1-300 for poles
11.0m & over

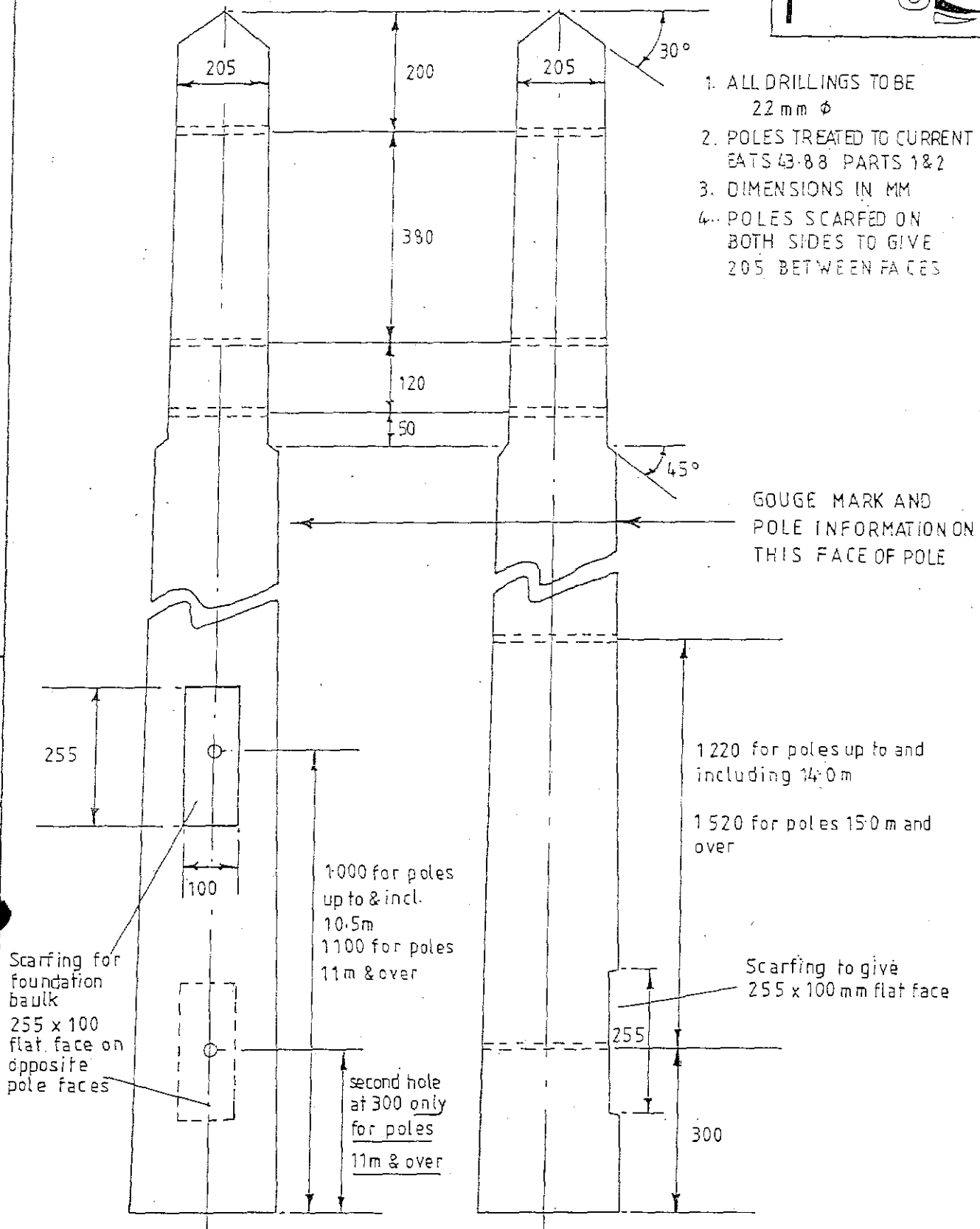
B A U D I O B Y E L W D A T E 7 - 8 - 96 I S S U E B A U G 96 - R E P L A C E S I S S U E A	TITLE STANDARD POLE DRILLINGS AND SCARFING MEDIUM CLASS POLES	SCALE	—	MEB Midlands Electricity plc DRG No M45012
		DATE	6-8-96	
		DRAWN	G.P.	
		TRACED	ELW	
		CHECKED		
		APPROVED		

SINGLE POLES

MATCHED H POLES

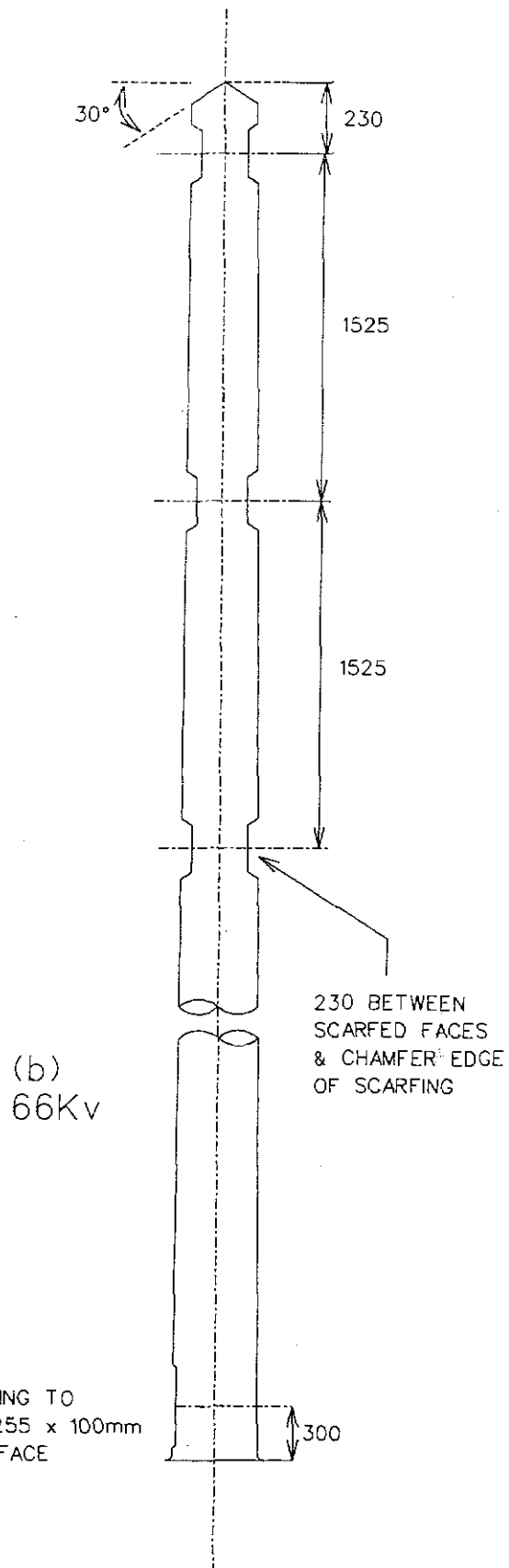
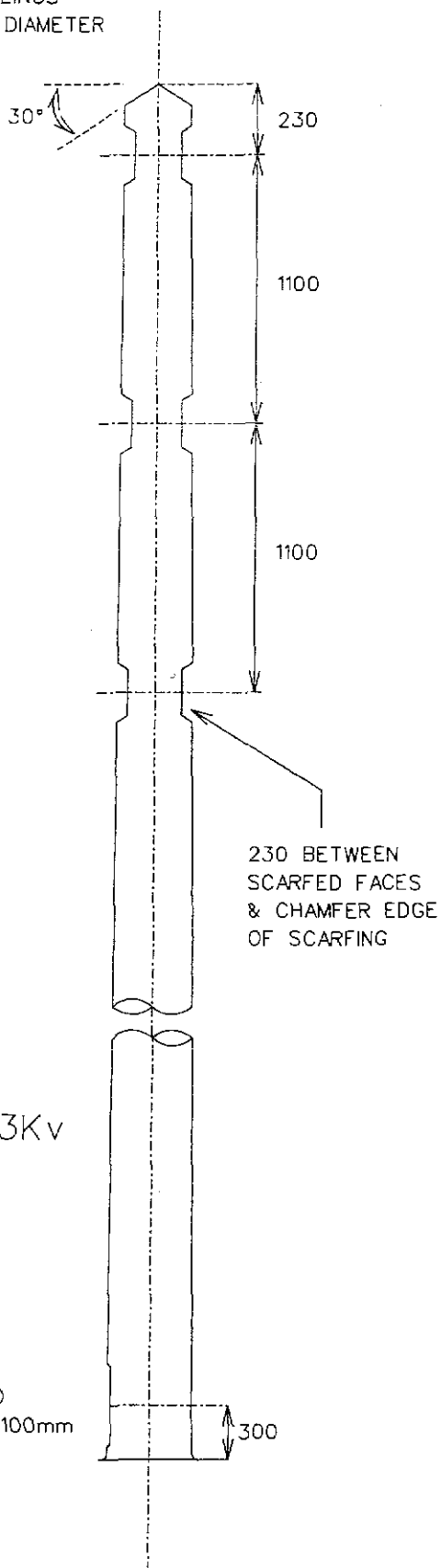


1. ALL DRILLINGS TO BE 22 mm ϕ
2. POLES TREATED TO CURRENT EATS 43-88 PARTS 1&2
3. DIMENSIONS IN MM
4. POLES SCARFED ON BOTH SIDES TO GIVE 205 BETWEEN FACES



ISSUE A DATE 7.8.96 ISSUE B DATE 6.8.96 ISSUE C DATE 1.8.96 ISSUE D DATE 1.8.96 ISSUE E DATE 1.8.96 ISSUE F DATE 1.8.96 ISSUE G DATE 1.8.96 ISSUE H DATE 1.8.96 ISSUE I DATE 1.8.96 ISSUE J DATE 1.8.96 ISSUE K DATE 1.8.96 ISSUE L DATE 1.8.96 ISSUE M DATE 1.8.96 ISSUE N DATE 1.8.96 ISSUE O DATE 1.8.96 ISSUE P DATE 1.8.96 ISSUE Q DATE 1.8.96 ISSUE R DATE 1.8.96 ISSUE S DATE 1.8.96 ISSUE T DATE 1.8.96 ISSUE U DATE 1.8.96 ISSUE V DATE 1.8.96 ISSUE W DATE 1.8.96 ISSUE X DATE 1.8.96 ISSUE Y DATE 1.8.96 ISSUE Z DATE 1.8.96	TITLE STANDARD POLE DRILLINGS AND SCARFING STOUT CLASS POLES (INCL. EXTRA STOUT)	SCALE DATE 6.8.96 DRAWN G.P. TRACED E.L.W. CHECKED APPROVED	MEB Midlands Electricity plc ORG No. M 450/3
---	---	---	--

ALL POLE DRILLINGS
TO BE 22mm DIAMETER



ALL DIMENSIONS IN mm EXCEPT WHERE OTHER WISE STATED

SCALE	-
DATE	22.01.03
DRAWN	MJB
TRACED	-
CHECKED	-
APPROVED	-

DUAL CIRCUIT POLE
DRILLING & SCARFINGS

ISSUE	DATE



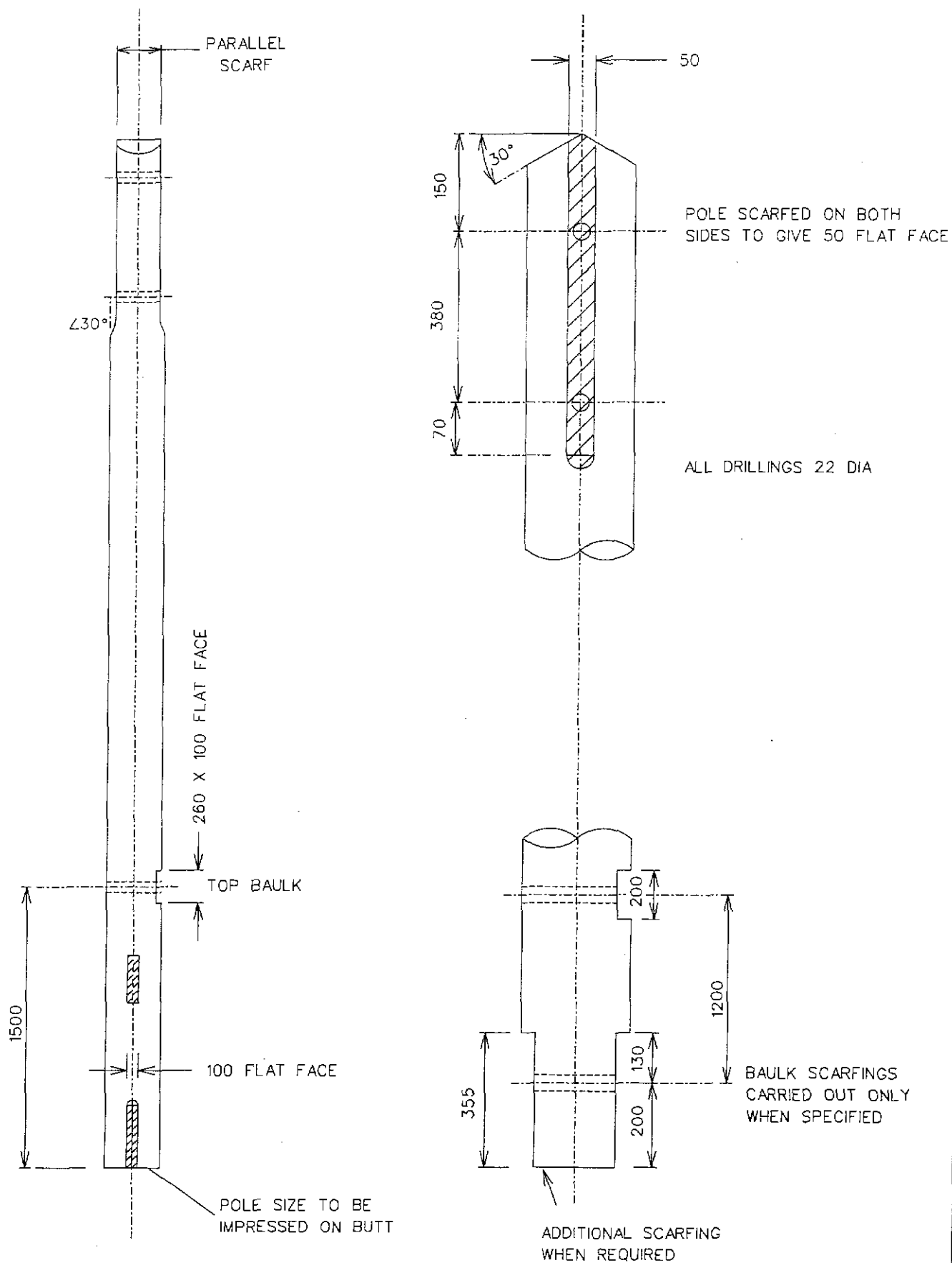
Aquila

DRG. No.

M450-4

DRG. No.

M450-4



ALL DIMENSIONS ARE IN mm EXCEPT WHERE OTHERWISE STATED

SCALE	-
DATE	20.01.03
DRAWN	MJB
TRACED	-
CHECKED	-
APPROVED	-

TRIDENT H-POLE DRILLINGS & SCARFINGS INTERMEDIATE SUPPORTS

ISSUE DATE

DATE



Aquila

DRG. No.

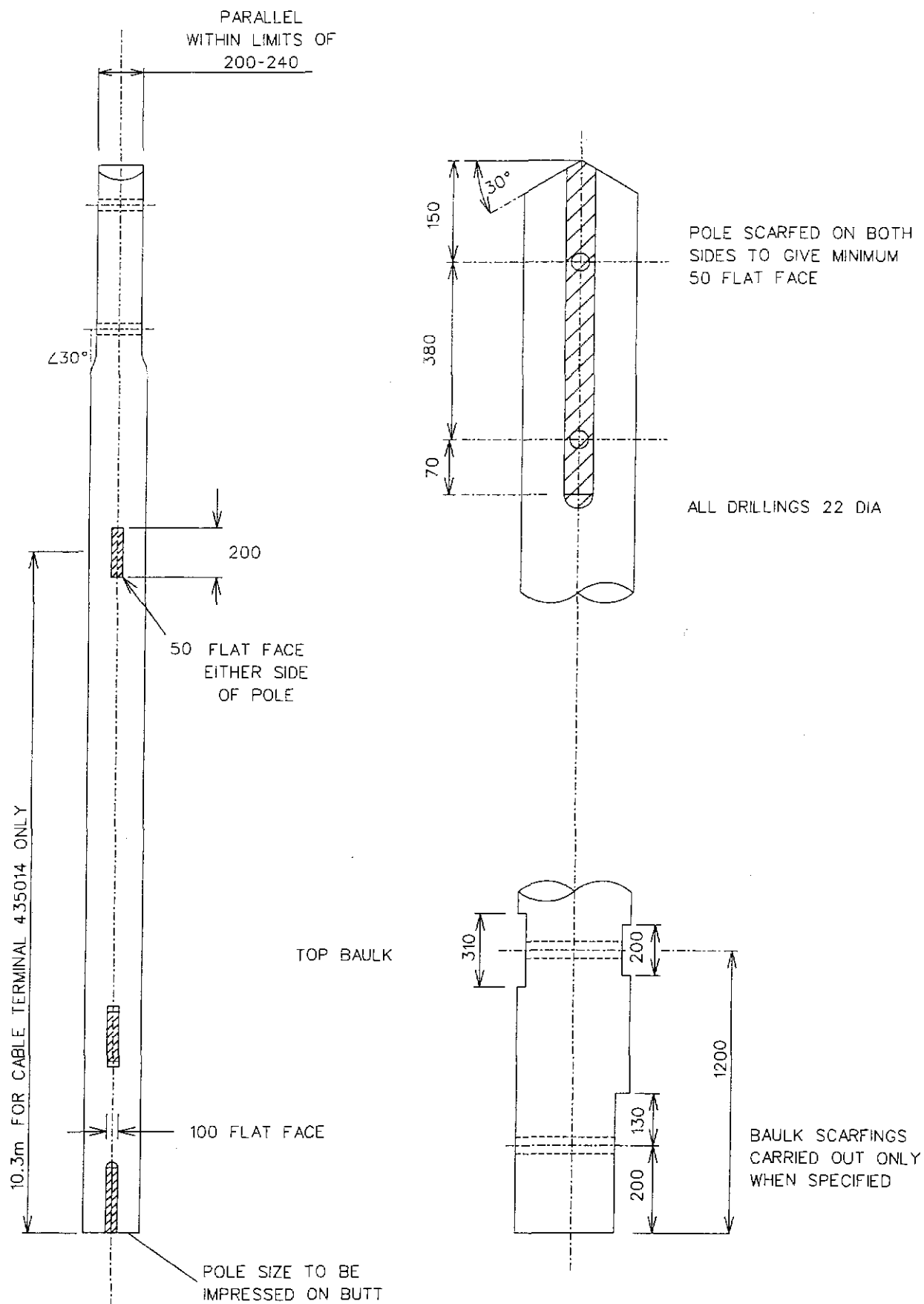
M450-5

DRG. No.

M450-5

A4

Produced by
Output Services



ALL DIMENSIONS ARE IN mm EXCEPT WHERE OTHERWISE STATED

SCALE	-
DATE	20.01.03
DRAWN	MJB
TRACED	-
CHECKED	-
APPROVED	-

TRIDENT H-POLE DRILLINGS & SCARFINGS
INTERMEDIATE SUPPORTS
TERMINAL + SECTION SUPPORTS

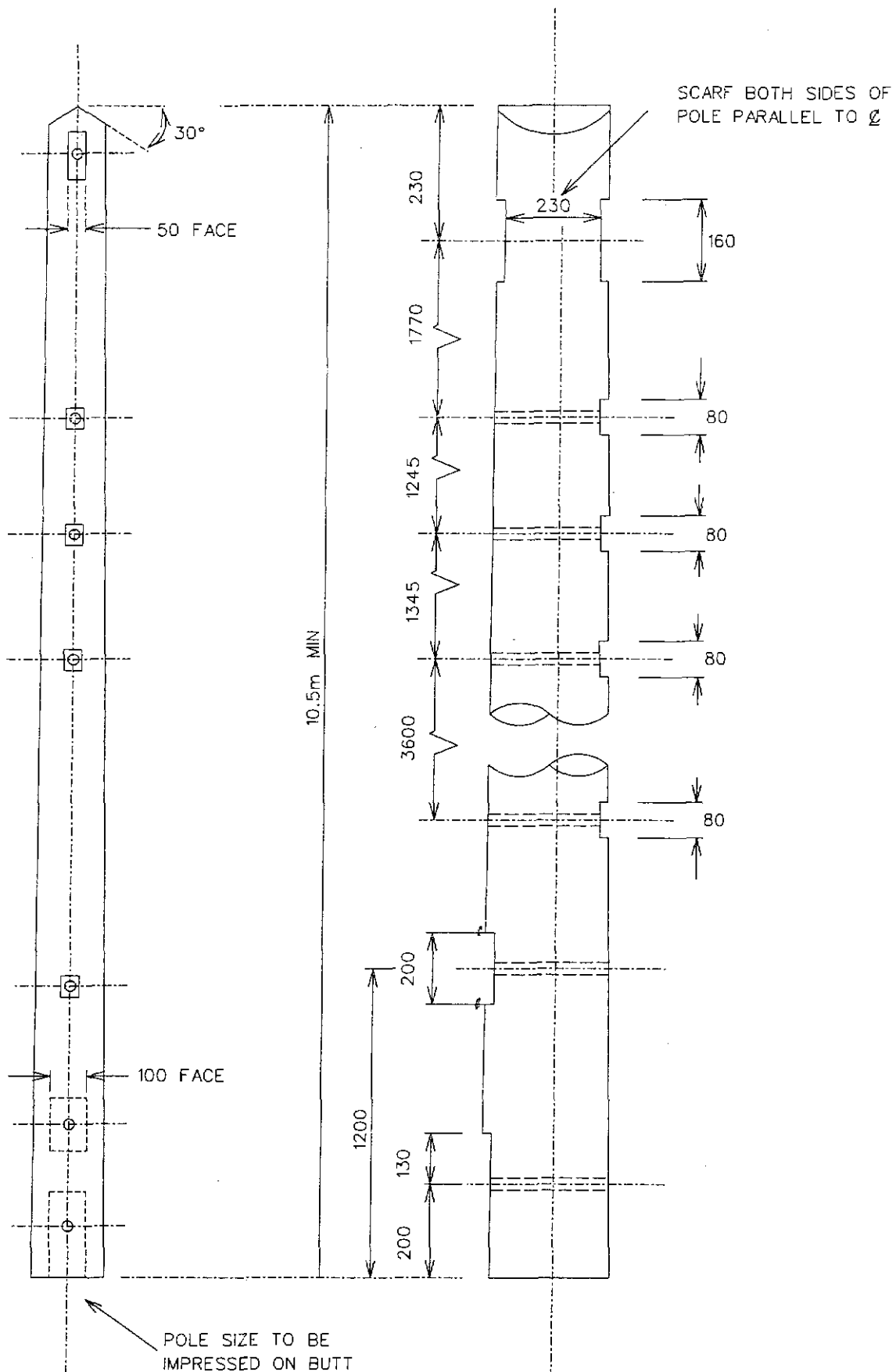
ISSUE	DATE

Produced by
Output Services

 Aquila

DRG. No.
M450-6

DRG. No. M450-6



ALL DIMENSIONS 22 DIA ALL DIMENSIONS IN mm EXCEPT WHERE OTHER WISE STATED

SCALE	-
DATE	22.01.03
DRAWN	MJB
TRACED	-
CHECKED	-
APPROVED	-

TRIDENT AUXILIARY POLE DRILLINGS & SCARFINGS

ISSUE DATE



Aquila

DRG. No.

M450-7

DRG. No.

M450-7

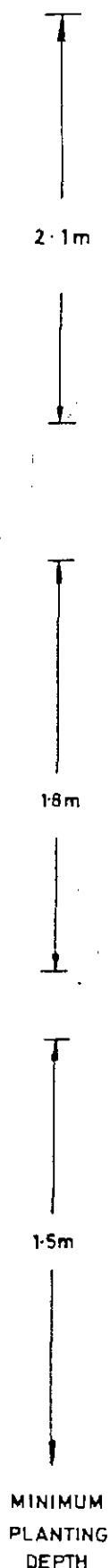
O/H LINE WOOD POLES
DIMENSIONS AND SRN NUMBERS.

USE	H.V. & LV.		L.V. ONLY			
SPECIES PRESERVATIVE Mean Bending Strength Mean M.O.F. E.	Imported Scots Pine Creosote 53.8 N/mm ² 10480 N/mm ²		Home Grown Corsican Pine C.C.A. 53.8 N/mm ² 11318 N/mm ²		Home Grown Norway & Sitka Spruce C.C.A. Norway 452 N/mm ² Sitka 441 N/mm ² " 9641 N/mm ² " 8908 N/mm ²	
LENGTH	DIA AT 1.5M	MIN	DIA AT 1.5M	MIN	DIA AT 1.5M	MIN
M	mm	Nº	mm	Nº	mm	Nº
<u>LIGHT POLES</u>						
8	170	3441080	170	3446080	185	3448080
8.5	180	3441085	180	3446085	195	3448085
9	180	3441090	180	3446090	195	3448090
9.5	185	3441095	185	3446095	200	3448095
10	185	3441100	185	3446100	200	3448100
10.5	190	3441105	190	3446015	205	3448105
11	195	3441110	195	3446110	210	3448110
<u>MEDIUM POLES.</u>						
8.5	215	3442085	215	3447085	230	3449085
9	220	3442090	220	3447090	235	3449090
9.5	225	3442095	225	3447095	240	3449095
10	230	3442100	230	3447100	245	3449100
10.5	235	3442105	235	3447105	250	3449105
11	240	3442110	240	3447110	255	3449110
11.5	245	3442115	245	3447115	260	3449115
12	250	3442120	250	3447120	265	3449120
13	260	3442130	-	-	-	-
14	275	3442140	-	-	-	-
15	290	3442150	-	-	-	-
16	305	3442160	-	-	-	-
17	320	3442170	-	-	-	-
18	330	3442180	-	-	-	-
<u>STOUT POLES.</u>						
9	275	3443090				
9.5	280	3443095				
10	285	3443100				
10.5	290	3443105				
11	295	3443110				
11.5	300	3443115				
12	305	3443120				
13	320	3443130				
14	335	3443140				
15	350	3443150				
16	365	3443160				
17	375	3443170				
18	390	3443180				
<u>STRUT POLES</u>						
9.5		3444095				
10.5		3444105				
12.0		3444120				
13.0		3444130				
14.0		3444140				
15.0		3444150				
16.0		3444160				

SCALE	N.T.S.
DATE	5-1-87
DRAWN	R.A.L.
TRACED	
CHECKED	
APPROVED	

OVERHEAD LINE WOOD POLES
STANDARD RANGE - DIMENSIONS AND
MATERIAL IDENTIFICATION: NUMBER (MIN).

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
		ORG. No
		M 451/1



METRES	FEET (METRES)
18	60 (18.29)
m.	ft.
17	55 (16.76)
16	
15	50 (15.24)
14	
13	45 (13.71)
12	
11.5	42 (12.80)
11	
10.5	40 (12.19)
10	
9.5	38 (11.58)
9	
8.5	36 (10.97)
8	
	34 (10.36)
	32 (9.75)
	30 (9.14)
	28 (8.53)
	26 (7.92)

OVERALL POLE LENGTHS

mm	ins (mm)
415	16 1/4" (413)
	16" (406)
	15 3/4" (400)
	15 1/2" (394)
390	15 1/4" (388)
380	15" (382)
375	14 3/4" (375)
	14 1/2" (368)
365	14 1/4" (362)
360	14" (356)
350	13 3/4" (349)
	13 1/2" (343)
	13 1/4" (336)
335	13" (330)
330	
325	12 3/4" (324)
320	12 1/2" (317)
	12 1/4" (311)
305	12" (305)
300	11 3/4" (298)
295	11 1/2" (292)
290	
285	11 1/4" (286)
280	11" (279)
275	
270	10 3/4" (273)
265	10 1/2" (267)
260	10 1/4" (260)
	10" (254)
250	9 3/4" (248)
245	
240	9 1/2" (241)
235	9 1/4" (235)
230	9" (229)
225	8 3/4" (222)
220	
215	8 1/2" (216)
210	8 1/4" (209)
	8" (203)
200	7 3/4" (197)
195	
190	7 1/2" (190)
185	7 1/4" (184)
180	7" (178)

POLE DIAMETER
AT 1.5m & 5ft FROM BUTT

SUPERSEDES DRG MD-75

SCALE	
DATE	15.5.70.
DRAWN	T.R.P.
TRACED	
CHECKED	
APPROVED	

OVERHEAD LINE WOOD POLES SIZE CONVERSION CHART.



ISSUE	MIDLANDS ELECTRICITY BOARD
	DRG. No. M. 451/2

CLASS	LENGTH (m)	TOP DIA.		MIN. DIA. at 1-5m (mm)	PLANTING DEPTH (m)	CANTILEVER		STRUTS		Approx pole weight (tonnes)
		min. (mm)	max. (mm)			Load at 0-6m from top (kN)	1mm Deflection (N)	Effective Dia. (mm)	Crippling Load (kN)	
Light	6	125	150	150	1.2	4.39	9.3	135.6	60.3	.09
	7	"	"	160	"	4.31	6.0	139.2	45.9	.11
	8	"	"	170	1.5	4.40	4.8	142.1	39.7	.13
	8.5	"	"	180	"	4.81	4.5	145.7	37.8	.14
	9	"	"	180	"	4.46	3.6	145.6	32.8	.16
	9.5	"	160	185	"	4.52	3.1	147.3	30.2	.18
	10	"	"	185	"	4.25	2.8	147.1	26.6	.19
	10.5	"	"	190	"	4.32	2.5	148.9	24.9	.21
	11	"	"	195	1.8	4.40	2.3	149.8	24.5	.23
	11.5	"	165	200	"	4.48	2.1	151.5	23.0	.25
	12	"	"	200	"	4.26	1.7	151.5	20.8	.26
	13	130	170	210	"	4.46	1.6	158.1	20.5	.30
Medium	8.5	150	180	215	1.5	8.20	9.1	174.5	77.8	.2
	9	"	"	220	"	8.15	7.8	176.2	70.4	.22
	9.5	"	"	225	"	8.13	6.8	177.9	64.3	.23
	10	"	185	230	"	8.15	6.4	179.5	59.1	.25
	10.5	"	190	235	"	8.16	5.7	181.2	54.7	.27
	11	"	"	240	1.8	8.18	5.0	181.9	53.2	.29
	11.5	"	"	245	"	8.21	4.5	183.6	49.6	.32
	12	"	"	250	"	8.23	4.1	185.3	46.6	.35
	13	160	200	260	"	8.46	3.7	195.1	47.5	.40
	14	"	205	275	"	8.98	3.3	200.2	44.4	.46
	15	165	210	290	2.1	9.60	3.3	207.6	45.9	.54
	16	170	215	305	"	10.23	3.1	216.0	46.3	.62
	17	180	220	320	"	11.07	3.0	227.6	49.7	.70
	18	"	"	330	"	11.14	2.7	231.0	46.3	.78
Stout	20	"	230	360	"	11.98	2.5	241.0	43.3	.95
	22	190	240	380	2.4	12.63	2.3	253.4	44.1	1.12
	8.5	190	240	265	1.5	15.35	21.5	218.3	190.4	.29
	9	"	"	275	"	15.91	19.2	221.8	176.8	.32
	9.5	"	245	280	"	15.66	16.4	223.4	160.0	.35
	10	"	250	285	"	15.52	15.5	225.1	146.1	.39
	10.5	"	250	290	"	15.36	13.4	226.7	134.0	.42
	11	"	"	295	1.8	15.24	11.8	227.3	129.5	.46
	11.5	"	"	300	"	15.15	10.5	228.9	120.0	.49
	12	"	"	305	"	15.08	9.4	230.6	111.7	.53
	13	195	255	320	"	15.73	8.3	238.9	106.7	.59
	14	"	260	335	"	16.24	7.3	244.0	97.8	.66
	15	"	265	350	2.1	16.68	6.8	247.9	93.2	.75
	16	200	"	365	"	17.35	6.2	256.2	91.6	.83
	17	"	"	375	"	17.20	5.4	259.5	84.0	.93
	18	"	"	390	"	17.57	5.0	264.5	79.6	1.03
	20	"	"	415	"	17.74	4.2	272.9	71.1	1.24
	22	"	"	435	2.4	17.49	3.6	278.4	64.3	1.45
	24	"	"	470	"	18.36	3.3	290.1	62.4	1.67

SCALE: _____
DATE: 24-2-78
DRAWN: W.D.H.
TRACED: C.E.
CHECKED: *gmb*
APPROVED: _____

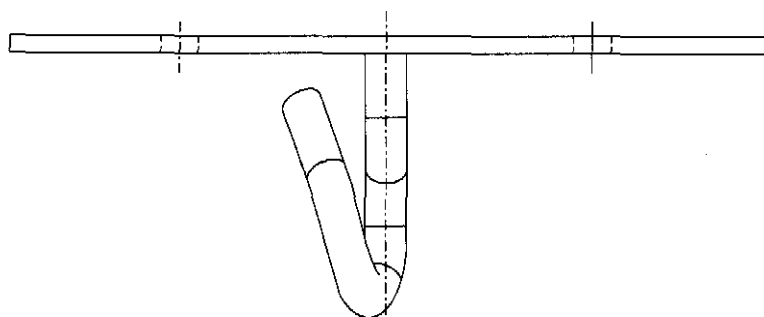
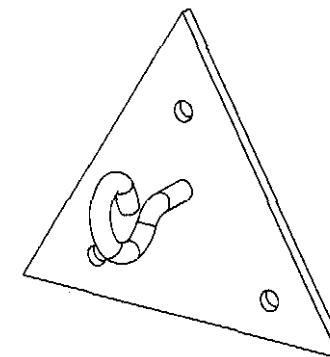
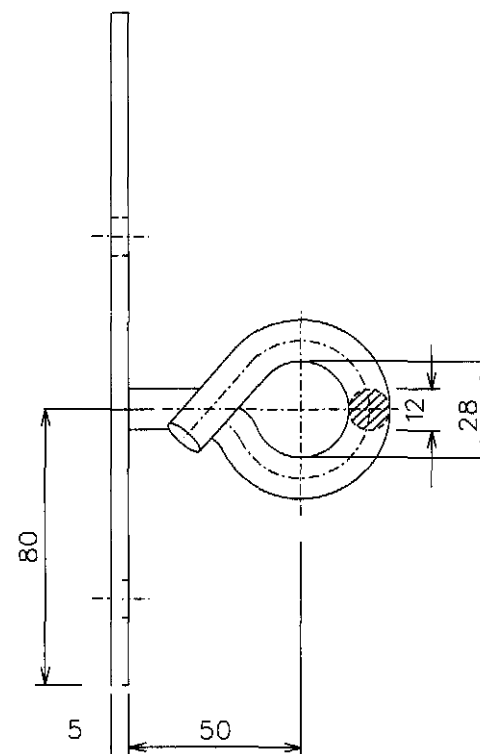
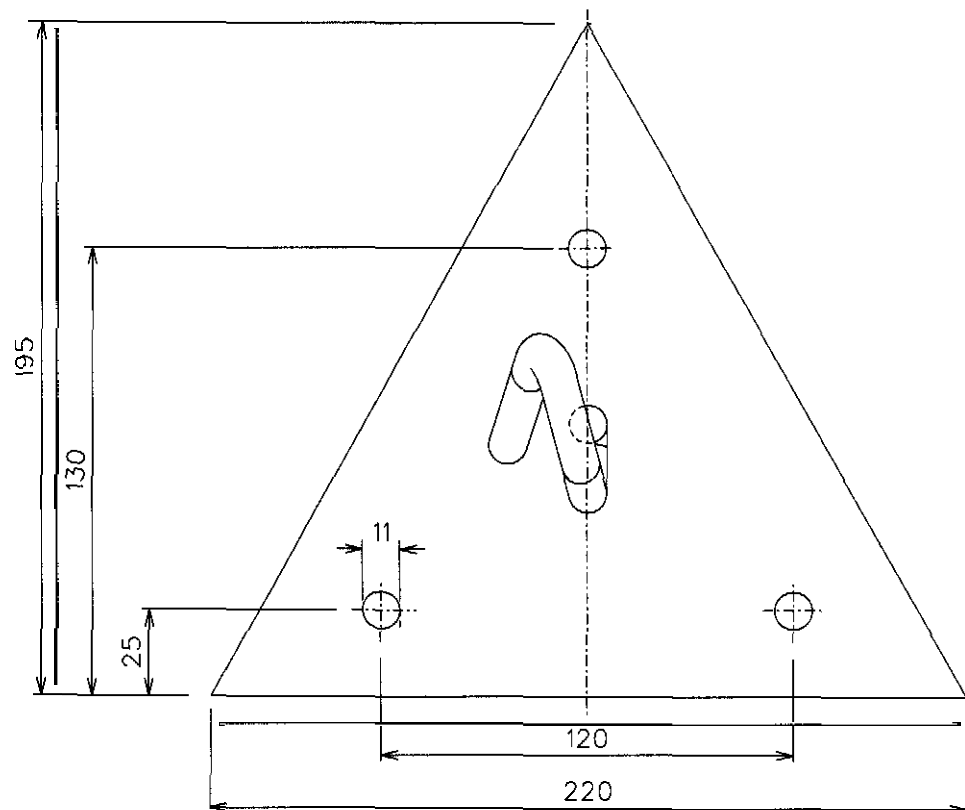
METRIC WOOD POLES (Scots Pine)
(To B.S. 1990 PART 2:1971)

ISSUE: _____
MIDLANDS
ELECTRICITY BOARD
DRG. No
M 451/3

SUPERSEDES DRG MD 60.

SECTION 19

MIN NUMBER	M	DESCRIPTION
3450370	37	HOUSE SERVICE WALL BRACKET (SHORT)
3450380	38	HOUSE SERVICE WALL BRACKET ABC PIG TAIL
3450390	39	HOUSE SERVICE WALL BRACKET (LONG)
3450430	43	HOUSE SERVICE CORNER BRACKET (DIAGONAL)
3450440	44	EAVES EXTENSION BRACKET
3450520	52	COACH SCREW SERVICE INSULATOR
3452900	290	EXTENSION STRAPS FOR HOUSE SERVICE BRACKETS
3452910	291	SERVICES ENTRY TUBE
3452990	299	INSOLE 'CABLE TIE'
3454520	452	UNDER EAVES SERVICE BREECHES JOINT BOX CONCENTRIC CABLES



Hot dip galvanized steel
Weight: 1.0 kg

Produced by
Primary CAD



Aquila

DRG. No.

M38

SCALE	1:22
DATE	20.10.03
DRAWN	M.J.B
TRACED	
CHECKED	17.11.03
APPROVED	KD

A.B.C Pigtail wall bracket

ISSUE	DATE	ISSUE	DATE	ISSUE	DATE



SCALE	1-1
DATE	17/9/01
DRAWN	ADD
TRACED	-
CHECKED	08.10.01
APPROVED	K.D.

ISSUE	DATE
-------	------

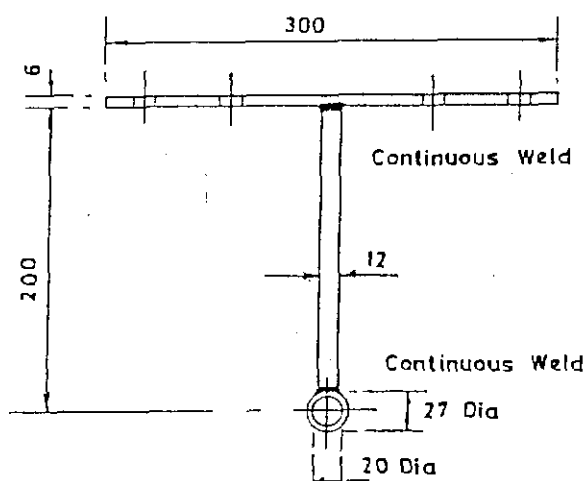
GPU
POWER UK

ORG. No.

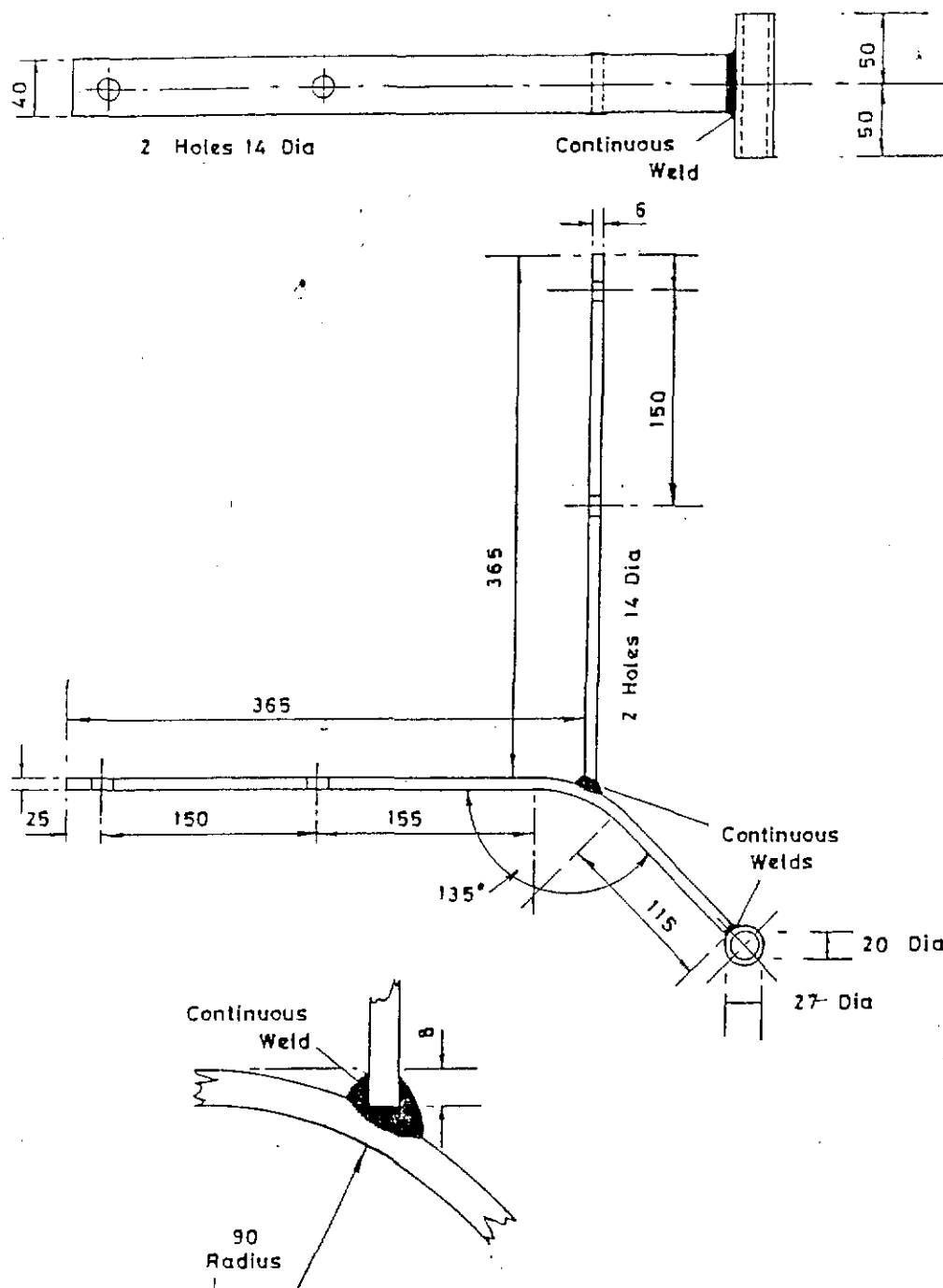
M37

DRG. No.	DRG. Description	DRG. Weight	DRG. Rate	DRG. Total
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2	DRG. 2 Description	1.0	100.00	100.00
3	DRG. 3 Description	1.0	100.00	100.00
4	DRG. 4 Description	1.0	100.00	100.00
5	DRG. 5 Description	1.0	100.00	100.00
6	DRG. 6 Description	1.0	100.00	100.00
7	DRG. 7 Description	1.0	100.00	100.00
8	DRG. 8 Description	1.0	100.00	100.00
9	DRG. 9 Description	1.0	100.00	100.00
10	DRG. 10 Description	1.0	100.00	100.00
11	DRG. 11 Description	1.0	100.00	100.00
12	DRG. 12 Description	1.0	100.00	100.00
13	DRG. 13 Description	1.0	100.00	100.00
14	DRG. 14 Description	1.0	100.00	100.00
15	DRG. 15 Description	1.0	100.00	100.00
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17	DRG. 17 Description	1.0	100.00	100.00
18	DRG. 18 Description	1.0	100.00	100.00
19	DRG. 19 Description	1.0	100.00	100.00
20	DRG. 20 Description	1.0	100.00	100.00
21	DRG. 21 Description	1.0	100.00	100.00
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61	DRG. 61 Description	1.0	100.00	100.00
62	DRG. 62 Description	1.0	100.00	100.00
63	DRG. 63 Description	1.0	100.00	100.00
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66	DRG. 66 Description	1.0	100.00	100.00
67	DRG			

MS

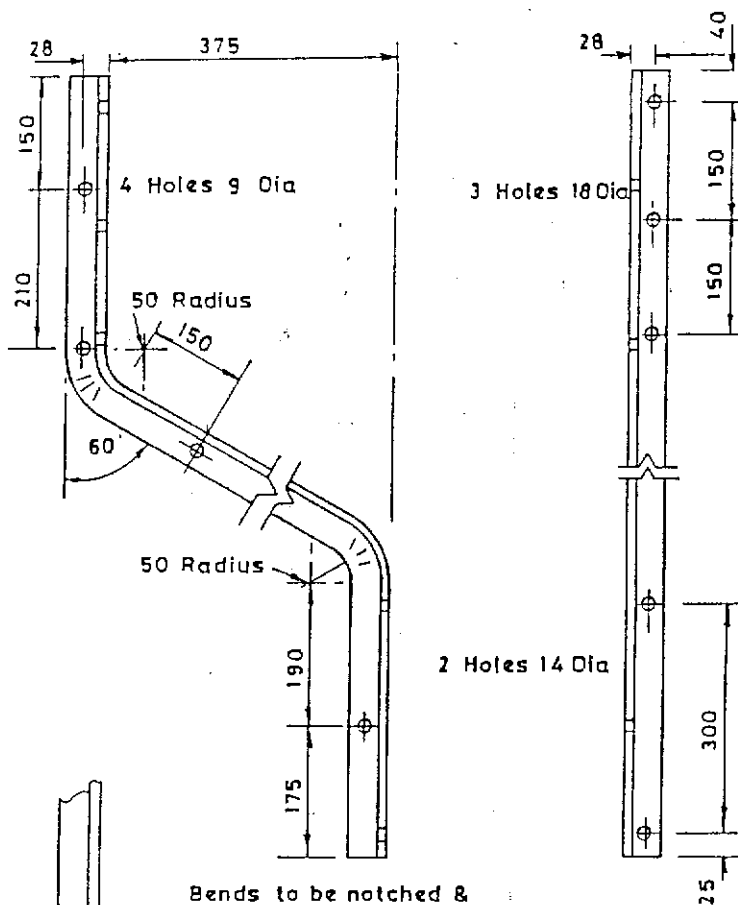


SCALE	1 - 5	HOUSE SERVICE WALL BRACKET (LONG)	ISSUE	MIDLANDS
DATE				ELECTRICITY BOARD
DRAWN	R.A.C.			DRG. No.
TRACED				M 39
CHECKED				
APPROVED				
		ES.I. STANDARD	ES.I. DRG No 439509	



ALL DIMENSIONS IN mm

SCALE	1 - 5	HOUSE SERVICE CORNER BRACKET (DIAGONAL)	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE				ORG. No.
DRAWN	R.A.C.			M. 43.
TRACED				
CHECKED				
APPROVED				
ESI STANDARD 43-95			ESI ORG No 439510	

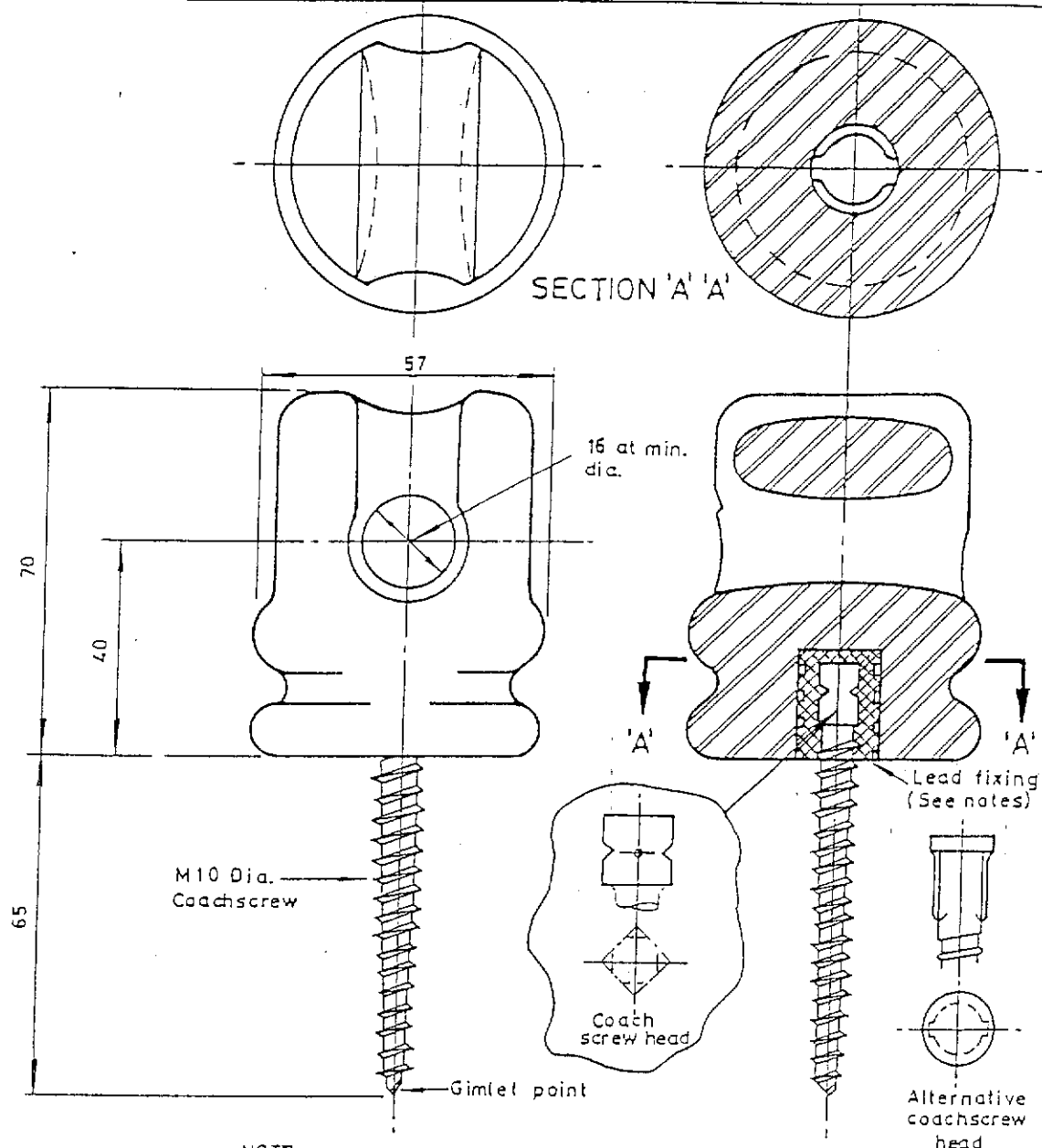


Bends to be notched & welded or cut & welded.

Alternative method of fabrication.

50 x 50 x 6 B.S.E.A.
ALL DIMENSIONS IN mm

SCALE	1-10	EAVES EXTENSION - BRACKET	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE				
DRAWN	R.A.C.			ORG. No.
TRACED				M. 44.
CHECKED				
APPROVED		E.S.I. STANDARD 43-95		E.S.I. ORG N° 439511



NOTE

For wood fixing, drill 6.4 dia. pilot hole.

MATERIAL Dark brown glazed porcelain

METAL FIXING SCREWS

The screw to be of suitable steel to BS 1494 part 2, clause 7.2, and to be completely galvanised to BS729 after fabrication, the shape of the head portion is left to manufacturer provided the completed service insulator will comply with the mechanical tests as specified below. The exposed portion to be a clean cut "wood type" screw, throughout its length, with gimlet point. The screw to be fixed into the porcelain by pouring molten lead alloy into the the pre-heated porcelain.

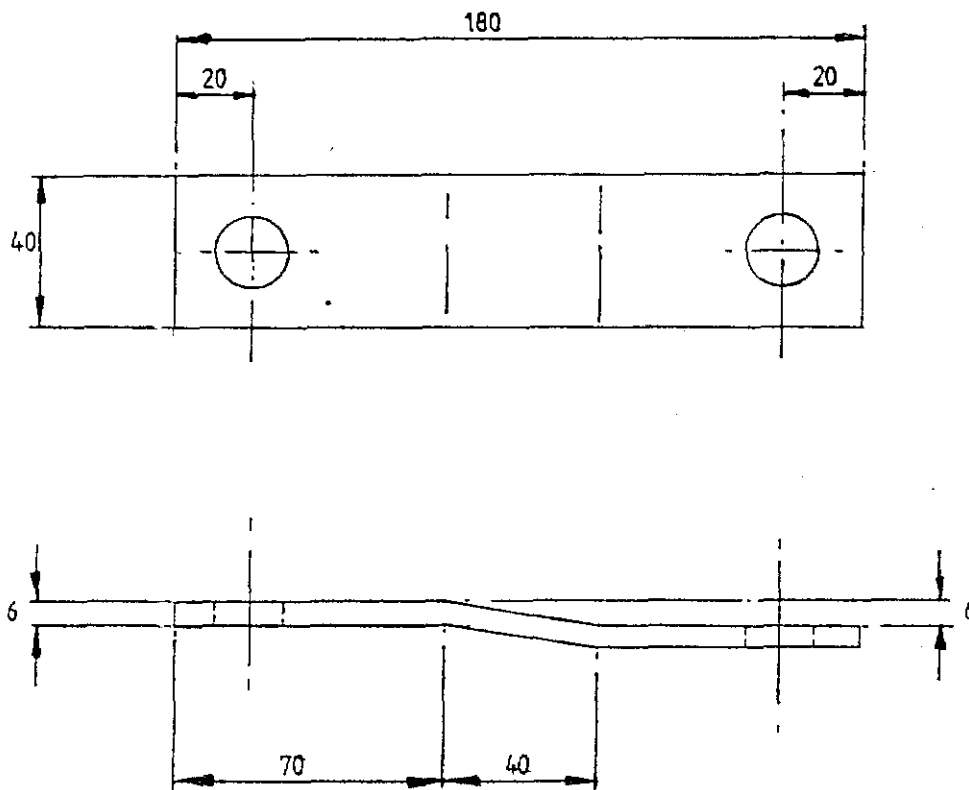
TESTS MECHANICAL TEST

Minimum failing load on complete insulator mounted as in service

Direct pull in line with screw - 4kN per 10secs

Lateral pull at right angles - 2.5kN " "

SCALE	-	COACH SCREW SERVICE INSULATOR	ISSUE -	ISSUE -	MIDLANDS ELECTRICITY BOARD
DATE	26.10.82				ORG. No.
DRAWN	A.R.B.				M.52. 
CHECKED	L.F.M.				
APPROVED					
			E.S.I. STANDARD 43/93	E.S.I. ORG.Nº 439305	



SCALE	1-20
DATE	17-11-82
DRAWN	RAC
TRACED	
CHECKED	
APPROVED	#

EXTENSION STRAPS FOR HOUSE SERVICE BRACKETS

ESI. STANDARD 43-95

ISSUE

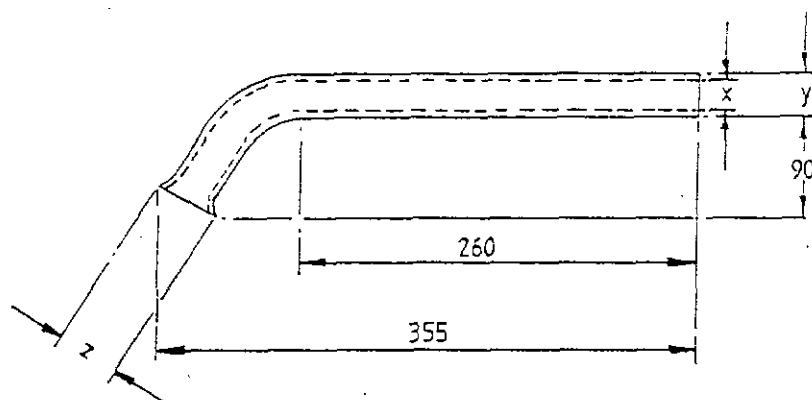
MIDLANDS
ELECTRICITY BOARD

DRG. No.

M. 290

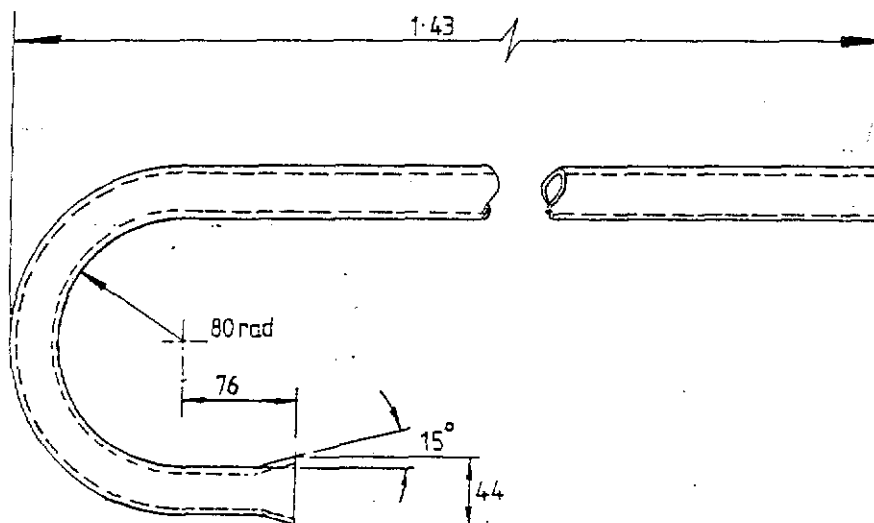
ESI Drg N° 439504

MAE



MATERIAL: —
HIGH IMPACT RIGID
PVC. COLOUR - BLACK.

	X	Y	Z
M291/1	25	32	35
M291/2	38	45	48



M. 291/5 (Preformed from a 1.7m L)
35.05 O/D. & 31.75 I/D.

MATERIAL
SEMI-RIGID PVC.
TUBING.

SCALE	1:5
DATE	26.10.82
DRAWN	R.A.C.
TRACED	
CHECKED	
APPROVED	

SERVICE ENTRY TUBE.

E.S.I. STANDARD

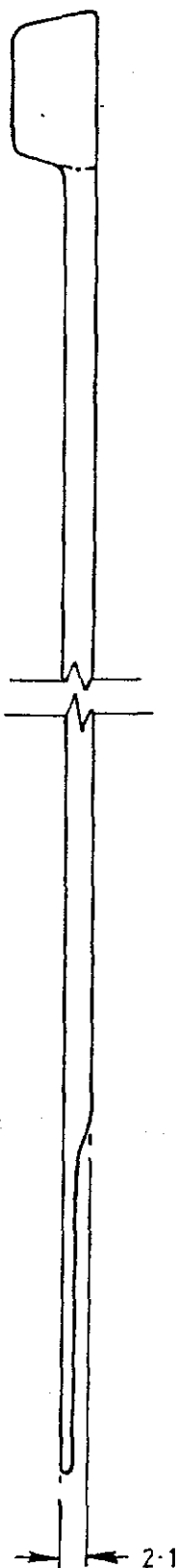
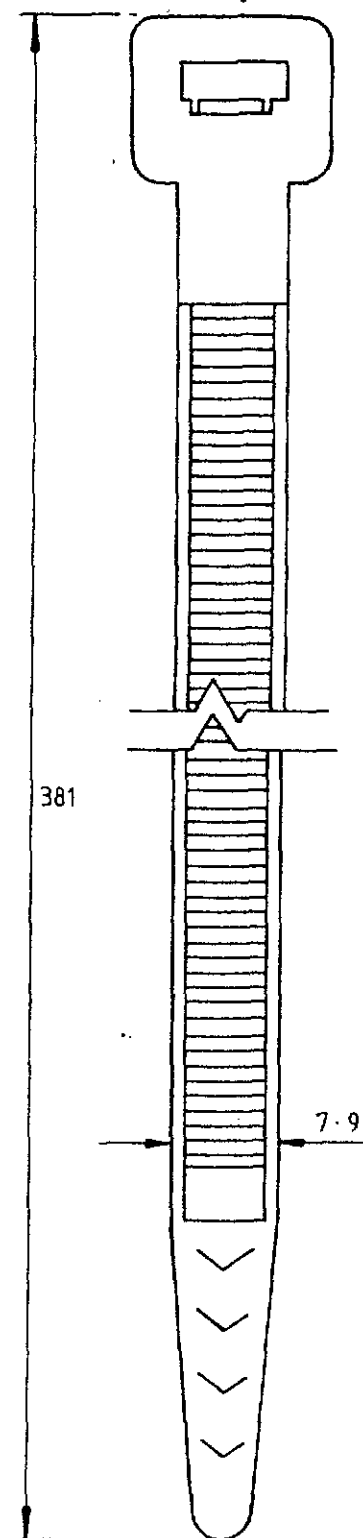
ISSUE

MIDLANDS
ELECTRICITY BOARD

DRG. No.

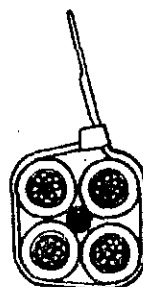
M. 291.

E.S.I. DRG. N°



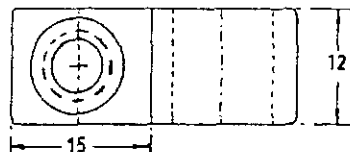
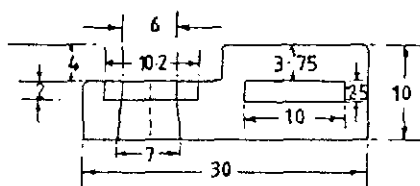
Cut off remainder
of strap to suit

Strap securing $2 \times 35 \text{ mm}^2$ Pvc/Pvc single core
copper conductor & $1 \times 25 \text{ mm}^2$ Al/Alloy PVC
covered.



Cut off remainder
of Strap to suit

Strap Securing $4 \times 35 \text{ mm}^2$ Pvc/Pvc single core
copper conductor & $1 \times 25 \text{ mm}^2$ Al/Alloy PVC
covered.



M299/1

Tie base to be used
with cable tie for
attaching service cables
to buildings.

FINISH - BLACK

SCALE	-
DATE	3.8.81
DRAWN	RAC
TRACED	S.J.H.
CHECKED	
APPROVED	

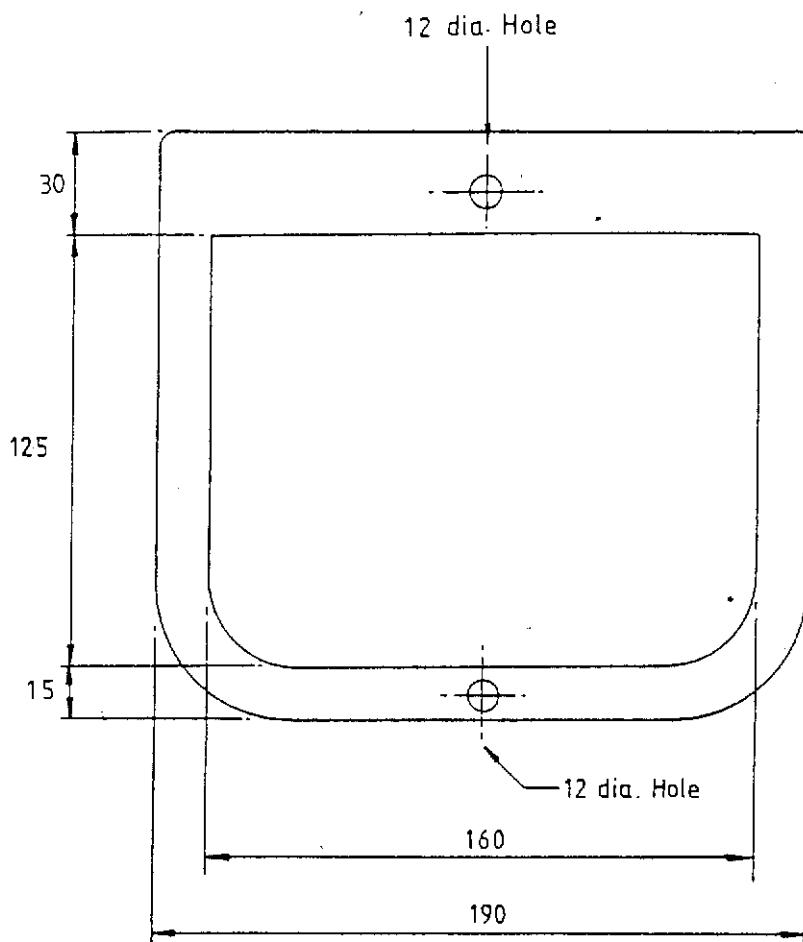
INSULOK 'CABLE' TIE

E.S.I. STANDARD

ISSUE	MIDLANDS ELECTRICITY BOARD
B	DRG. No.
TIE BASE ADDED.	M.299
1.12.89	ES.I. Drg. N°

MF

MF



Material:- 1.5 m.m. thick P.V.C. Sheet

Manufacturer:- Berger Elastomers,
Portland Road,
Newcastle upon Tyne,
NE2:1BL.

Manufacturer's
Ref. No:- UE1

Supplied complete with Resin Pack as a Kit.

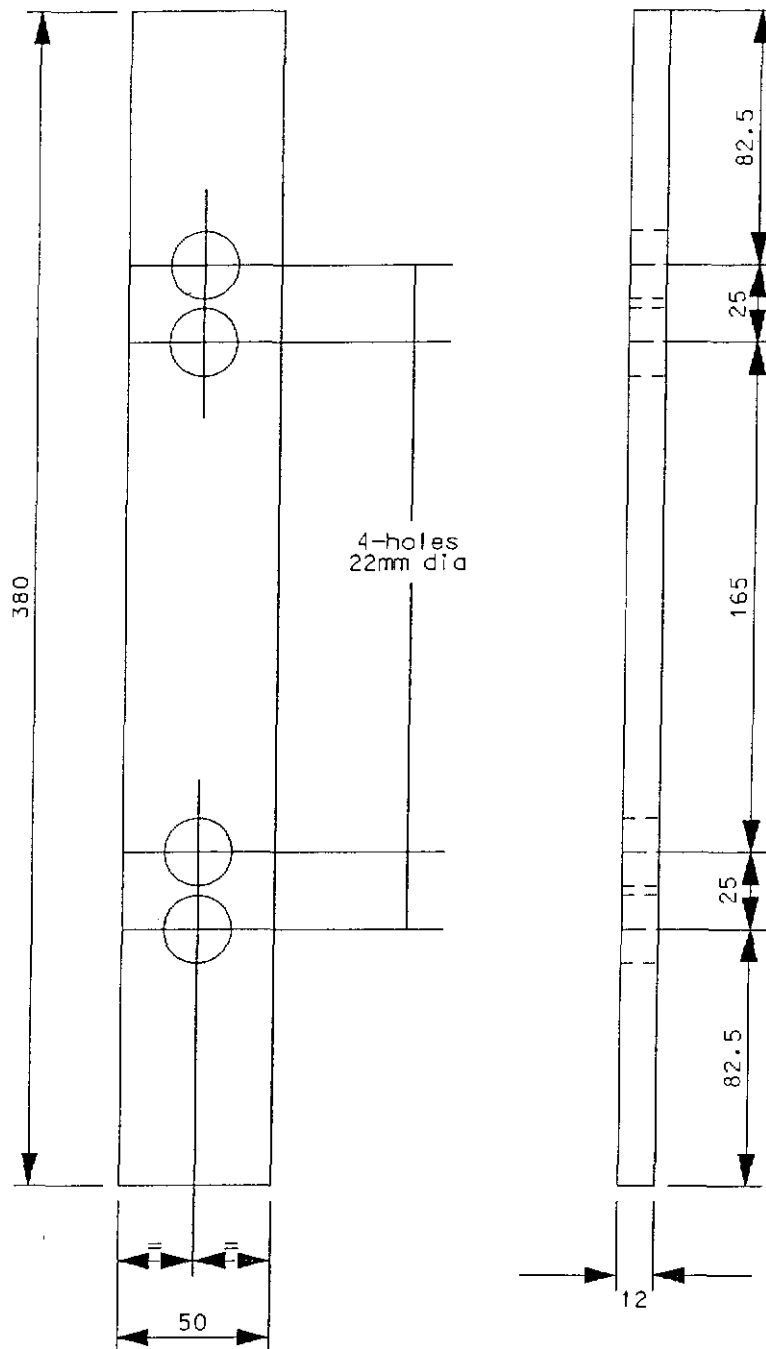
SCALE	—
DATE	21-1-87
DRAWN	AJ.B.
TRACED	S.H.
CHECKED	LP
APPROVED	LF

UNDER EAVES
SERVICE BREECHES JOINT BOX
CONCENTRIC CABLES

ISSUE	MIDLANDS ELECTRICITY BOARD
	DRG. No.
	M.452. LF

SECTION 20

MIN NUMBER	M	DESCRIPTION
3450020	2	TRANSFORMER FIXING STRAP
3450540	54	PILOT CROSSARM
3450751	75/1	TX PLATFORM GUSSET PLATE
3450920	92	TRANSFORMER PLATFORM SINGLE POLE
3451554	155/4	LUCY M.P.C.400 POLE MOUNTED L.V. FUSE UNIT
3452120	212	PILOT BAR ASSEMBLY
3452470	247	CABLE CLEAT



FOR METHOD OF USE SEE M.D.9

Material:- 380mm x 50mm x 12mm M.S. with
 2 - 150 x M20 bolts threaded for 50mm
 each c/w nut & 2 washers (1 spring & 1 flat)
 2 OFF PER TRANSFORMER

Finish :- Galv to BS 729 after
 cutting and drilling
 bolt threads to be
 brush galv.

SCALE	
DATE	14.7.00
DRAWN	J.S.
TRACED	
CHECKED	
APPROVED	

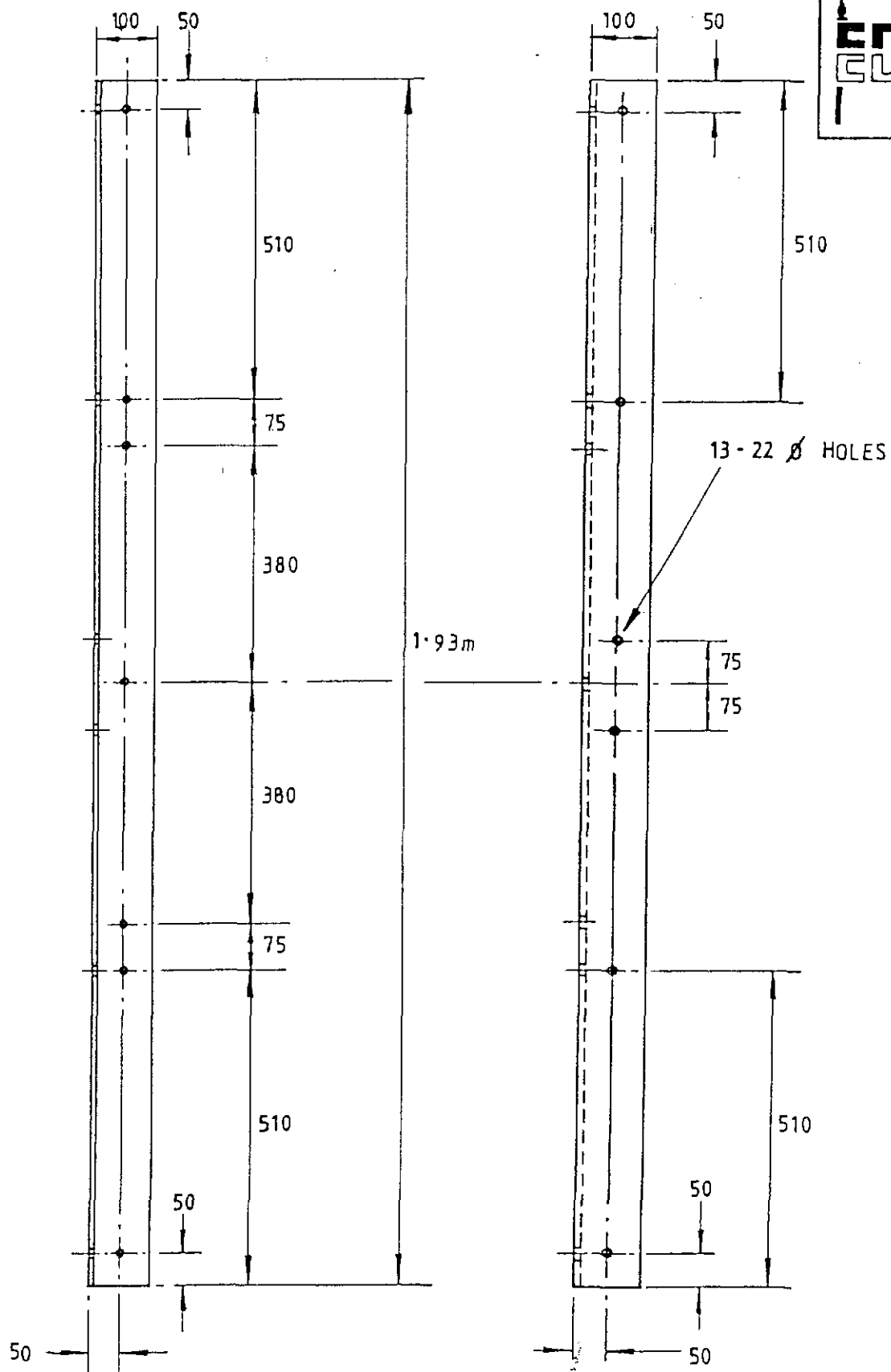
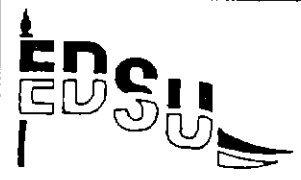
TRANSFORMER FIXING STRAP

ISSUE	DATE

Produced by Output Services
GPU POWER UK
DRG. No. M2

DRG. No.

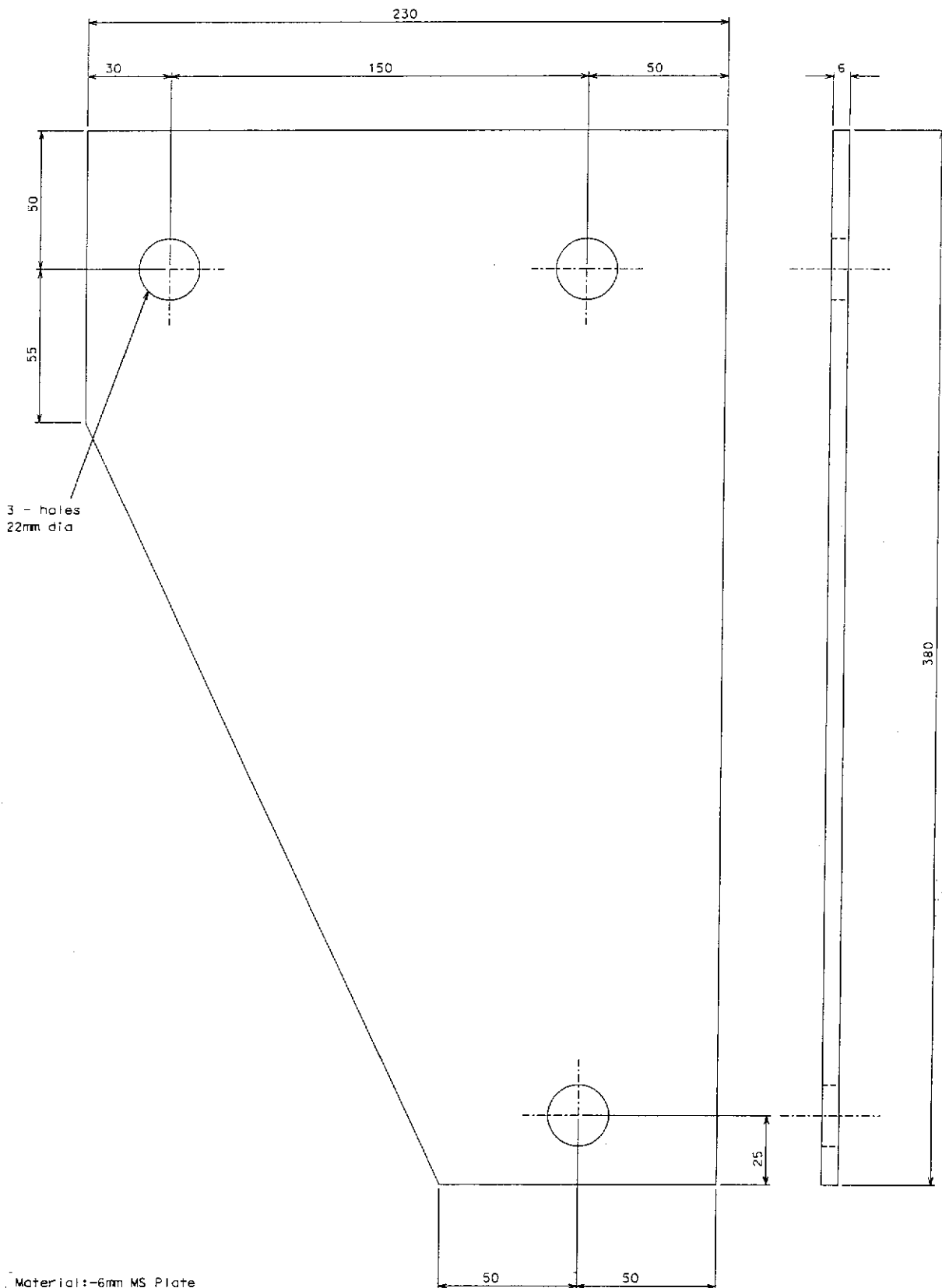
M2



FINISH:
Galv. to BS 729 after cutting
& drilling.

MATERIAL:
100 x 100 x 10 Angle to BS 4

DRAWN BY DATE	TITLE PILOT CROSSARM	SCALE	—	MEB Midlands Electricity plc ORG No M54
		DATE		
		DRAWN	A. J. B.	
		TRACED	C. E.	
		CHECKED		
		APPROVED	W. D. H.	



Material:-6mm MS Plate
28/33 ton quality
Finish:- Galv to BS729 after
cutting and drilling

SCALE	-
DATE	5.9.00
DRAWN	A.J.E.
TRACED	
CHECKED	27.9.00
APPROVED	KD

TX PLATFORM GUSSET PLATE

ISSUE	DATE

DRG. No.



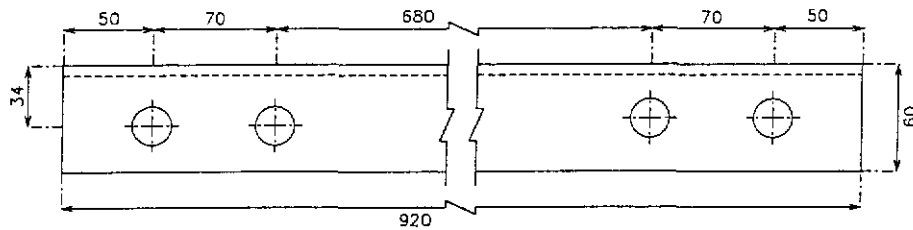
M75/1

Produced by
Output Services

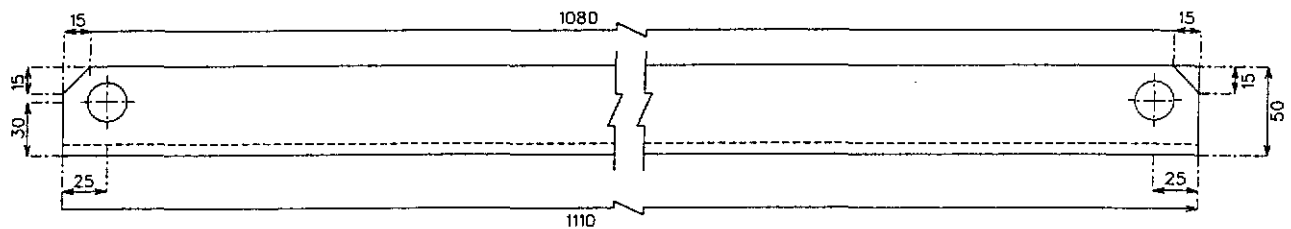
DRG. No.

M75/1

A4



ITEM 1 60x60x6 BSEA



ITEM 2 50x50x6 BSEA

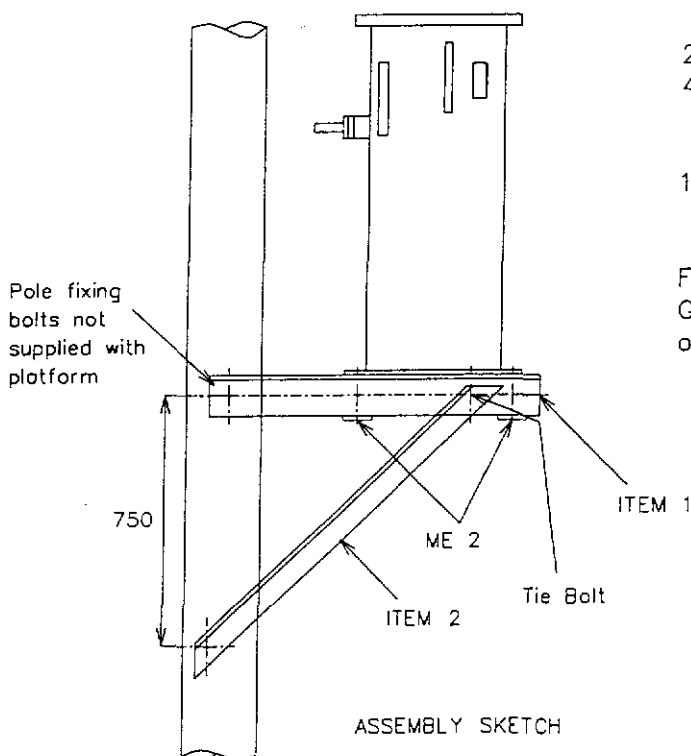
Holes 22 Dia.

MATERIAL

- 2- 60x60x6
MS Angle to BS 4.
- 2- 50x50x6
MS Angle to BS4.
- 2- Fixing Straps ME 2
- 4- M20x130mm HRH Bolts
screwed 50mm, each C/W 2 flat & 1
grover washers.
- 1- M20x450 Tie Bolt
100mm threaded end.
+ 4 nuts

FINISH

Galv to BS 729 after cutting
and drilling



ASSEMBLY SKETCH

Unless otherwise stated all dimensions are in mm.

SCALE	-
DATE	8.5.00
DRAWN	ADD
TRACED	
CHECKED	
APPROVED	KPD/14.8.00

TRANSFORMER PLATFORM SINGLE POLE

ISSUE	DATE
B	24.07.01

Modification

Produced by
Output Services



DRG. No.

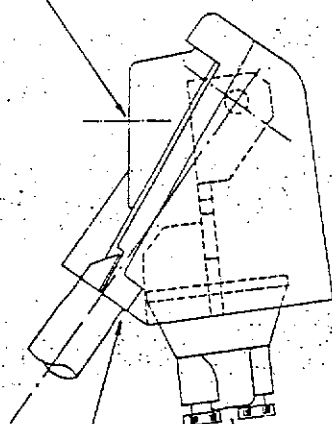
M92

DRG. No.

M92

A4

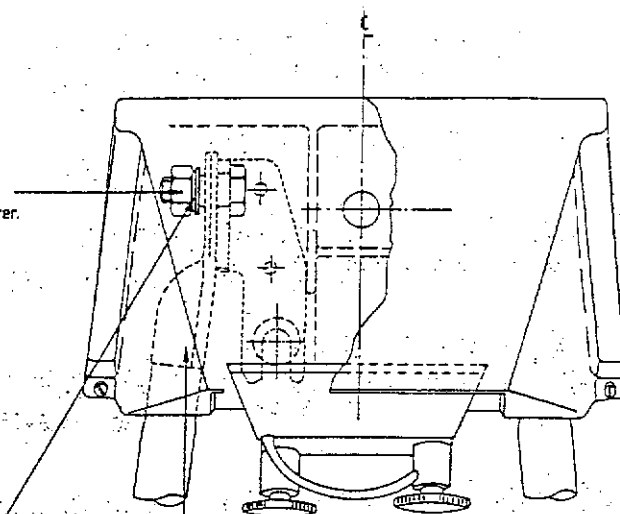
To be fixed with $\frac{1}{2}$ " (M12) dia. x 4" (101.6) long
coach screw to ME 46/5 (can be supplied by manufacturer)
c/w flat steel washer & waxed felt washer to ME 16.



For underground cable entry
arrangement see note *, below.

Fuse Carrier.

M12 x 35 Bolt c/w nut
(cad' plated steel)
supplied by manufacturer.



Stepped washer and Locking
washer supplied by manufacturer.

Compression Lug, see table for size.

LUCY M.P.C. 400 FUSE UNIT (Max Rating 400A Copper)

Copper P.V.C. Insulated / Sheath. to ME 231/1					
Cable Size	Compression Lug		Die Ref.		Number of Crimps
	BICC Burndy Ref.	MEB Ref.	Y35	MD 6-8	
35mm ²	Y.C.A. 2C5 - 62	237/2	See drg. N° ME 237		
70mm ²	Y.C.A. 258 - 1	237/3			
95mm ²	Y.C.A. 275 - 1	237/6			
120mm ²	Y.C.A. 291 - 1	237/7			

Aluminium P.V.C. Insulated / Sheath					
Cable Size	Compression Lug		Die Ref.		Number of Crimps
	BICC Burndy Ref.	MEB Ref.	Y35	MD 6-8	
50mm ²			See drg. N° ME 285		
95mm ²					
150mm ²					

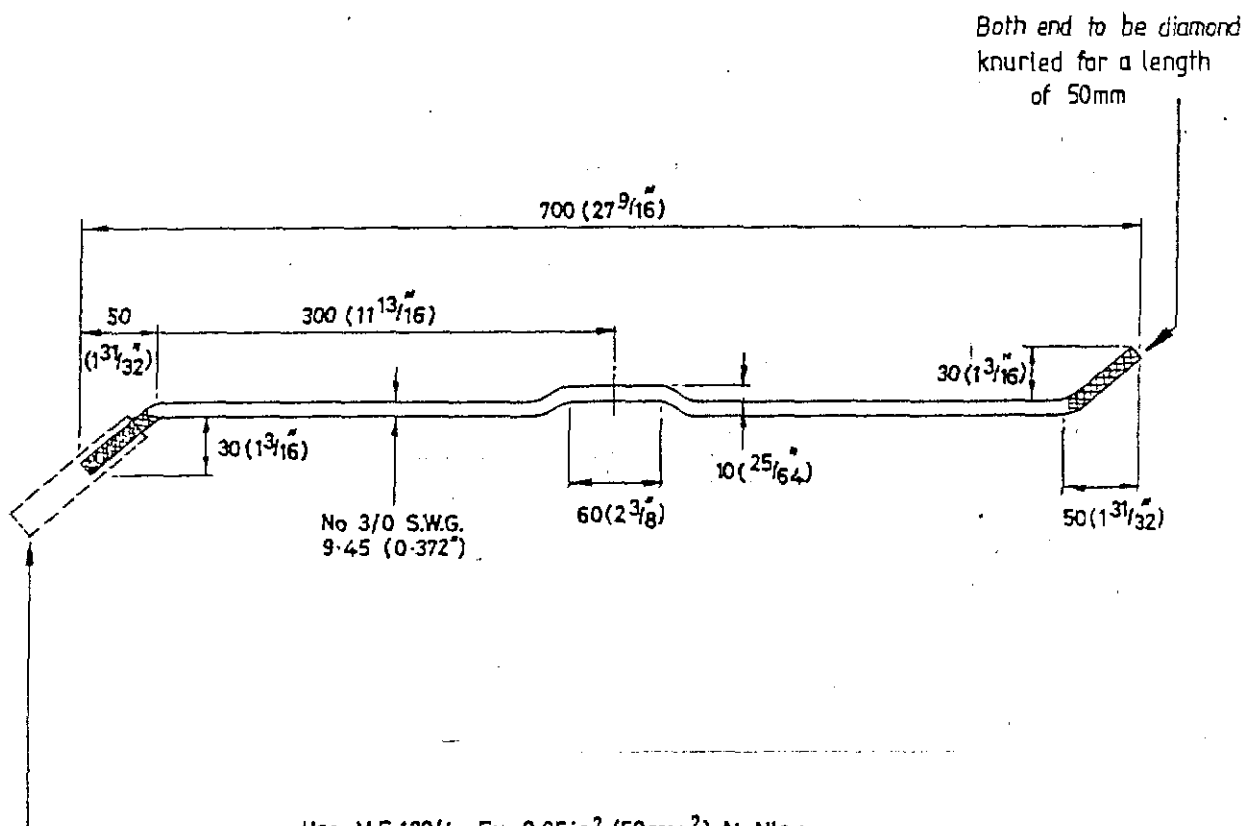
* Note:- For underground cable entry see
Cable Jointing Instruction Manual
Drg. N° MU/JA3/15-11.

SCALE	1:2
DATE	30.9.1980
DRAWN	A.C.T.
TRACED	C.A.R.
CHECKED	
APPROVED	

LUCY M.P.C. 400.
POLE MOUNTED LV FUSE UNIT.

ISSUE	ISSUE

MIDLANDS ELECTRICITY BOARD	
DWG. N°	ME 155/4. (H.C.)



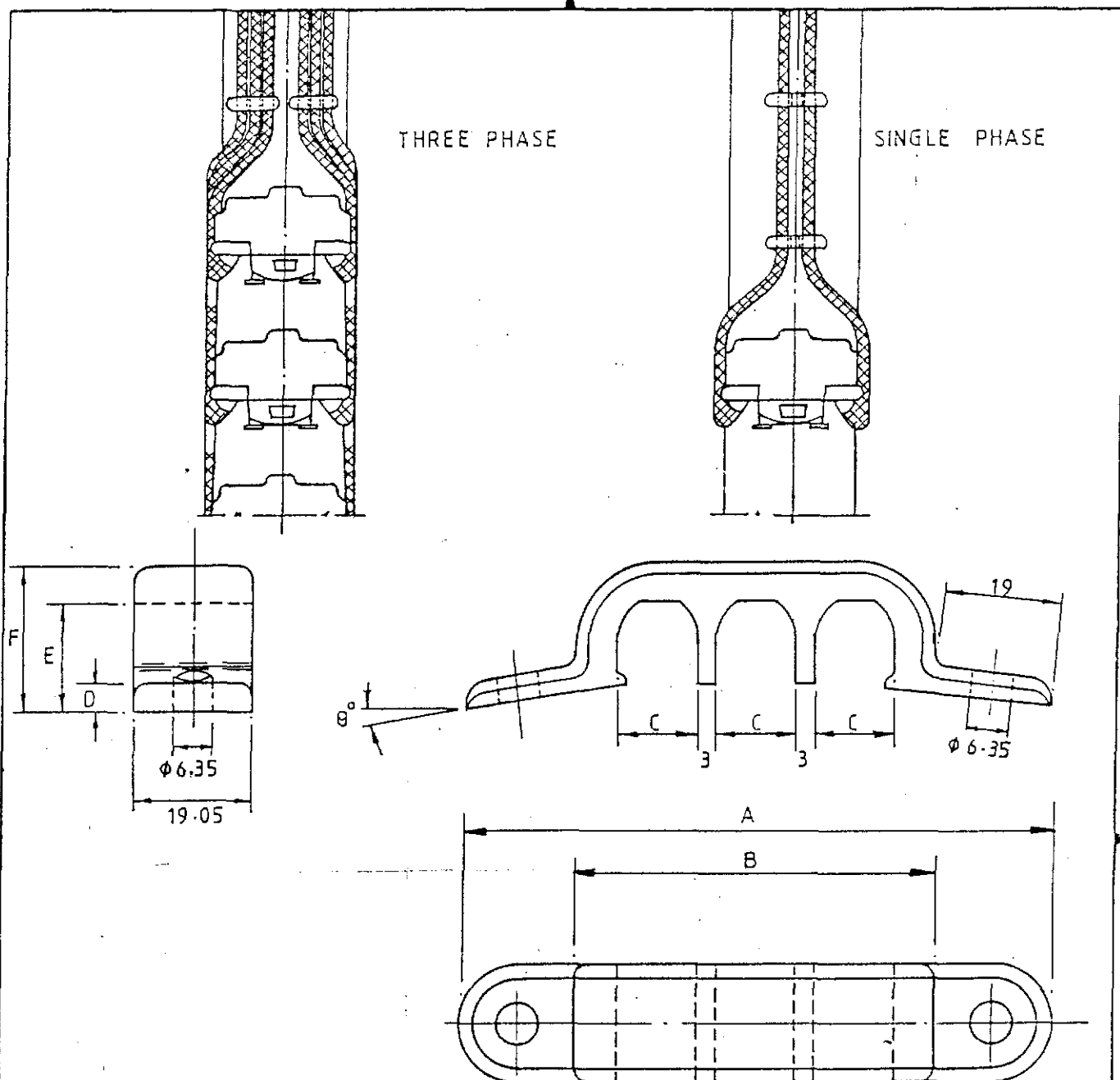
Use M.E.120/4 For 0.05in² (50mm²) Al. Alloy.
 M.E.120/5 For 0.05in² (50mm²) Al. Flexible.
 M.E.135/2 For 0.05in² (35mm²) Cu. Flexible.

TYPE	MATERIAL
212/1	Al. Alloy Rod - 9.5mm Dia (3/0 S.W.G) HE. 30 WD - 730mm Long.
212/2	HC Copper Rod 9.5mm Dia (3/0 S.W.G) - 730mm Long

SCALE	1/5
DATE	20.3.70
DRAWN	W.D.H.
TRACED	
CHECKED	
APPROVED	

PILOT BAR ASSEMBLY

ISSUE	MIDLANDS ELECTRICITY BOARD
H	
Type 212/1/2	DRG. No.
Material added R.A.C.	M.E. 212



Type No	PVC/PVC Cable Size mm ² (Cu)	A	B	C	D	E	F	θ°
247/1	35	93.6	55.5	13.2	3.96	17.48	22.2	8
247/2	70	101.2	63.09	16.5	3.96	19.71	24.48	8
247/3	95	111.3	73.22	19.1	4.75	23.88	28.63	10
247/4	120	118.7	80.62	21.750	4.75	26.34	31.09	10

MATERIAL : BLACK POLYPROPYLENE

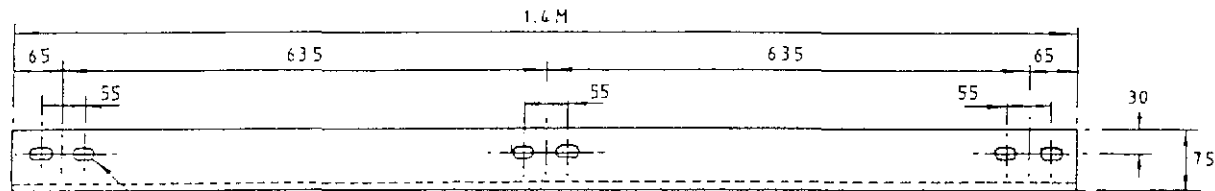
NOTE : CLEAT FIXED BY 2- 6-35

GALVE. DRIVE SCREWS 63 LONG (BS 1494).

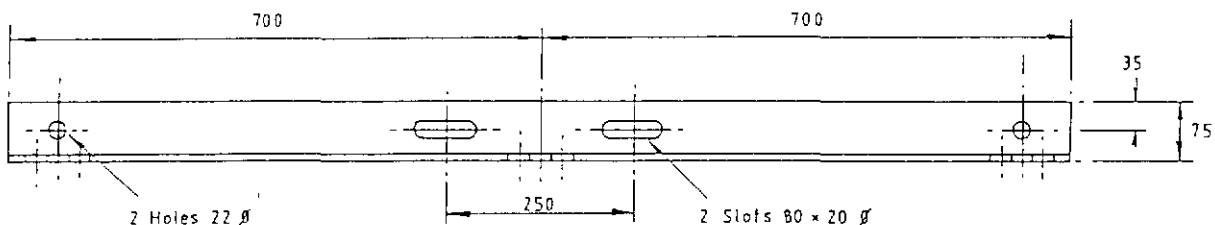
SCALE	N.T.S.	CABLE CLEAT	ISSUE	MIDLANDS ELECTRICITY BOARD
DATE	1-11-82			DRG. No.
DRAWN	R.A.C.			M.247.
TRACED	SGT			
CHECKED				
APPROVED		E.S.I. STANDARD		ESI DRG. N°

SECTION 21

MIN NUMBER	M	DESCRIPTION
3450721	72/1	SUPPORTING BRACKET FOR FUSES ON SINGLE POLES
3450722	72/2	STAND - OFF SUPPORT BRACKET FOR FUSES ON SINGLE POLES
3451360	136	66kV SURGE DIVERter SUPPORTING BRACKET
3452340	234	33kV SURGE DIVERter SUPPORTING BRACKET
3452342	234/2	33kV SURGE ARRESTER BASE PLATE HEC 36
3452820	282	BUSBAR & CONNECTORS FOR 33kV MOUNTED TERMINATION
3453421	342/1	BOWTHORPE BMP 33kV SURGE ARRESTER TYPE HEC 36
3454601	460/1	11kV SURGE ARRESTER
3454602	460/2	24kV SURGE ARRESTER

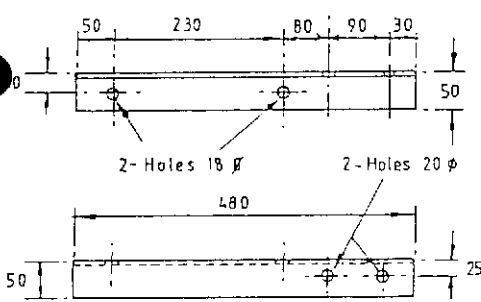


6 Slots 20 x 14 ø



2 Holes 22 ø

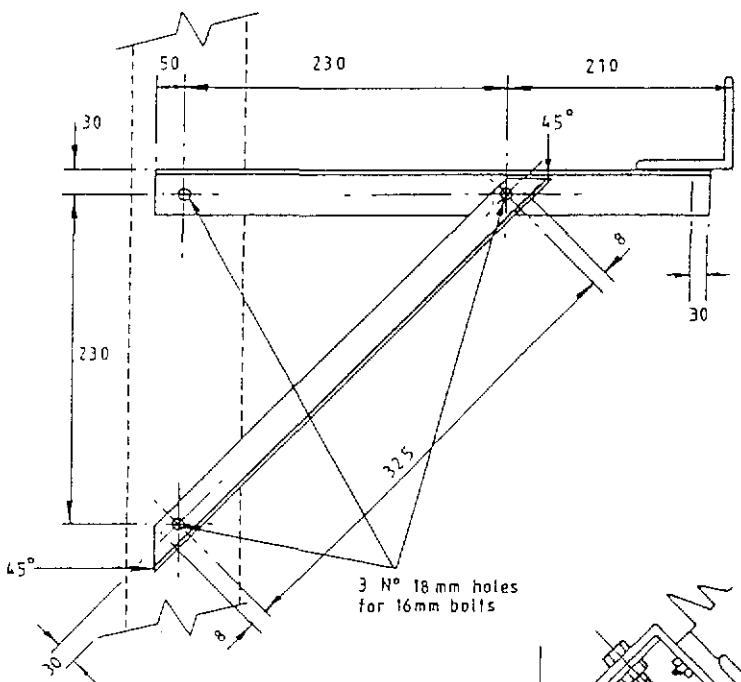
2 Slots 80 x 20 ø



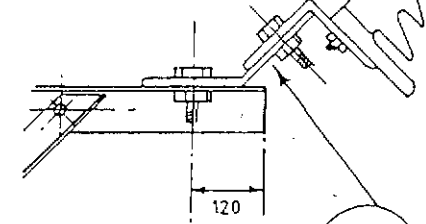
2 Holes 18 ø

2 Holes 20 ø

ITEM 'A' 1 off Left Hand
1 off Right Hand



Alternative arrangement for Surge Arresters



ITEM 'B' 2 off

MATERIAL

- 1 OFF x 1.4m x 75 x 75 x 8 Angle.
- 2 OFF x { 480 x 50 x 50 x 6 Angle
441 x 50 x 50 x 6 Angle
- M.S. to BS 4 2B/33ton quality
- 2 x M16 x 60 Bolts C/W nut & flat washer.
- 2 x M18 x 60 Bolts C/W nut & flat washer

FINISH

Galvanised to BS 729 after welding, cutting and drilling.

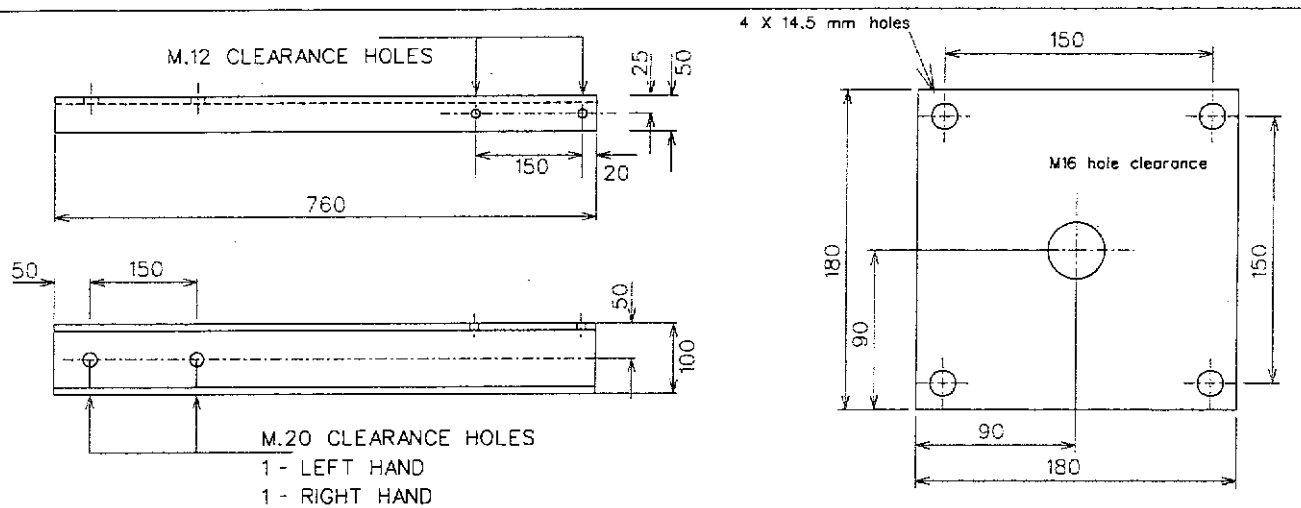
569/1

STAND - OFF SUPPORT BRACKET FOR FUSES ON SINGLE POLES

SCALE	
DATE	
DRAWN	
TRACED	
CHECKED	
APPROVED	

MEB
Midlands Electricity plc

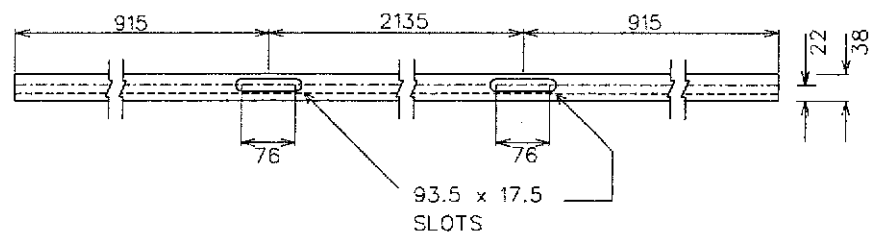
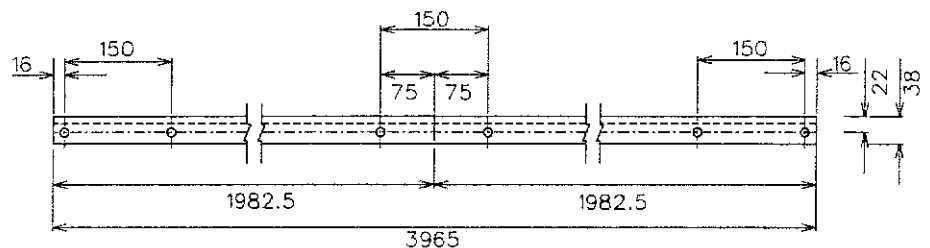
ARC No
M72/2 MF



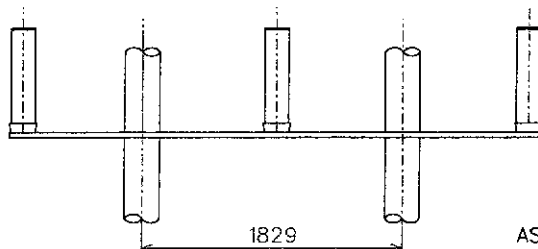
1:20

BASE PLATE
M.234/2 30FF

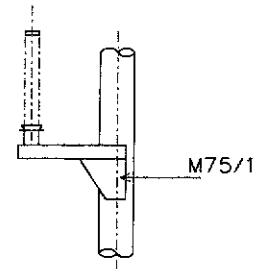
1:2



1:50



ASSEMBLY DETAIL



MATERIALS

ME.234/1

3965 x 76 x 38 6.7kg/M. MS CHANNEL TO B.S. 4 - 20FF

760 x 100 x 50 10.4kg/M. M.S. CHANNEL TO B.S. 4 - 20FF

180 x 180 x 10 M.S. PLATE - 3M 234/2

GUSSET PLATE M.S. - 2 OFF x M75/1

12 - M12 x 50

4 - M16 x 40

2 - M20 x 40

BOLTS EACH C/W NUT AND 1 FLAT WASHER

FINISH

GALVANISED TO B.S. 729 AFTER CUTTING AND DRILLING

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE STATED

SCALE	-
DATE	01.03.73
DRAWN	WE
TRACED	CE
CHECKED	-
APPROVED	-

33kV SURGE DIVERTER
SUPPORTING BRACKET

ISSUE	DATE
E	29.07.02
REVISIONS	
MODIFICATIONS	
METRIC MEASUREMENTS	
ADDED	
Q.C.	



Aquila

DRG. No.

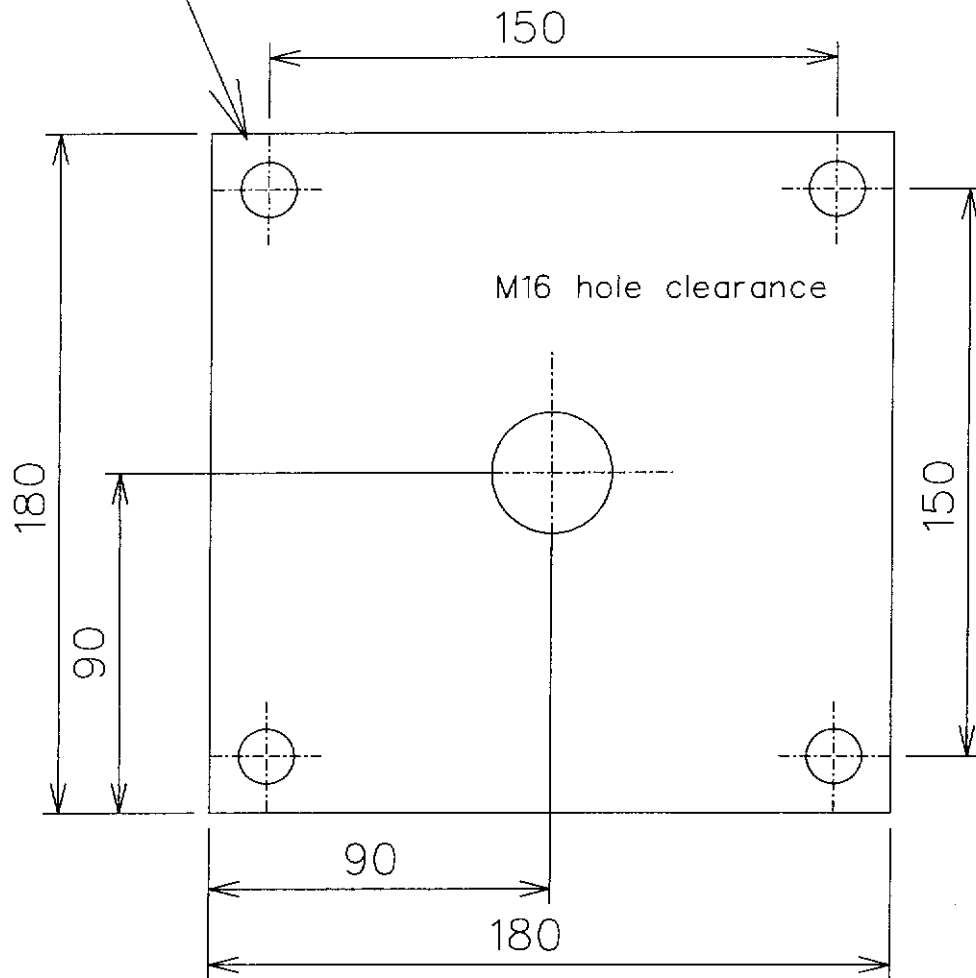
M234

DRG. No.

M234

A4

4 X 14.5 mm holes



MATERIAL

Mild Steel Plate 10mm thick.

FINISH

Galv. to B.S. 729
after cutting and drilling.

All dimensions are in millimeters unless otherwise stated.

SCALE	1-2
DATE	25/07/02
DRAWN	ADD
TRACED	
CHECKED	31/07/02
APPROVED	KD

33kV Surge Arrester
Base Plate For HEC 36

ISSUE DATE

DRG. No.



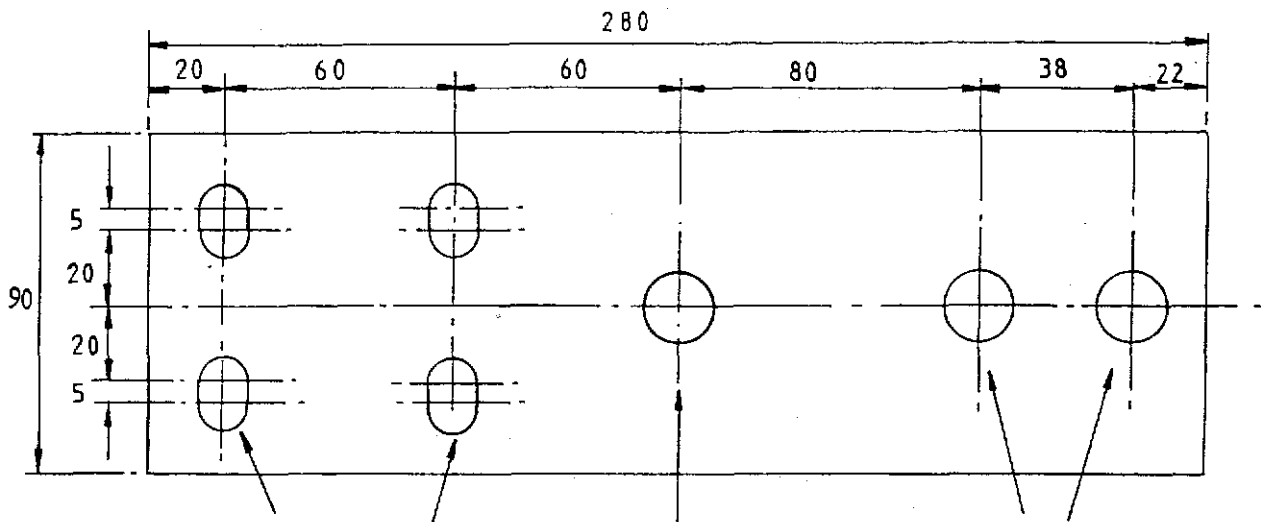
Aquila

M234/2

Produced by
Output Services

DRG. No.

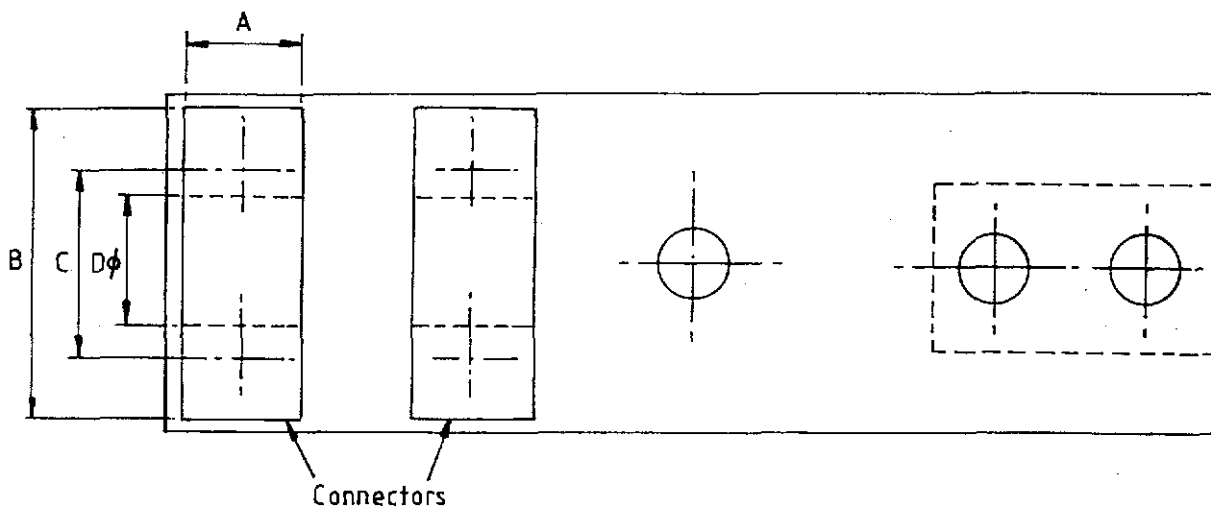
M234/2



4 Slots 14 mm. x 19 mm. 17.5 mm. dia. Hole

2 Holes 18 mm. dia.

BUS BAR - 10 mm. thick Electro-finned C.U.

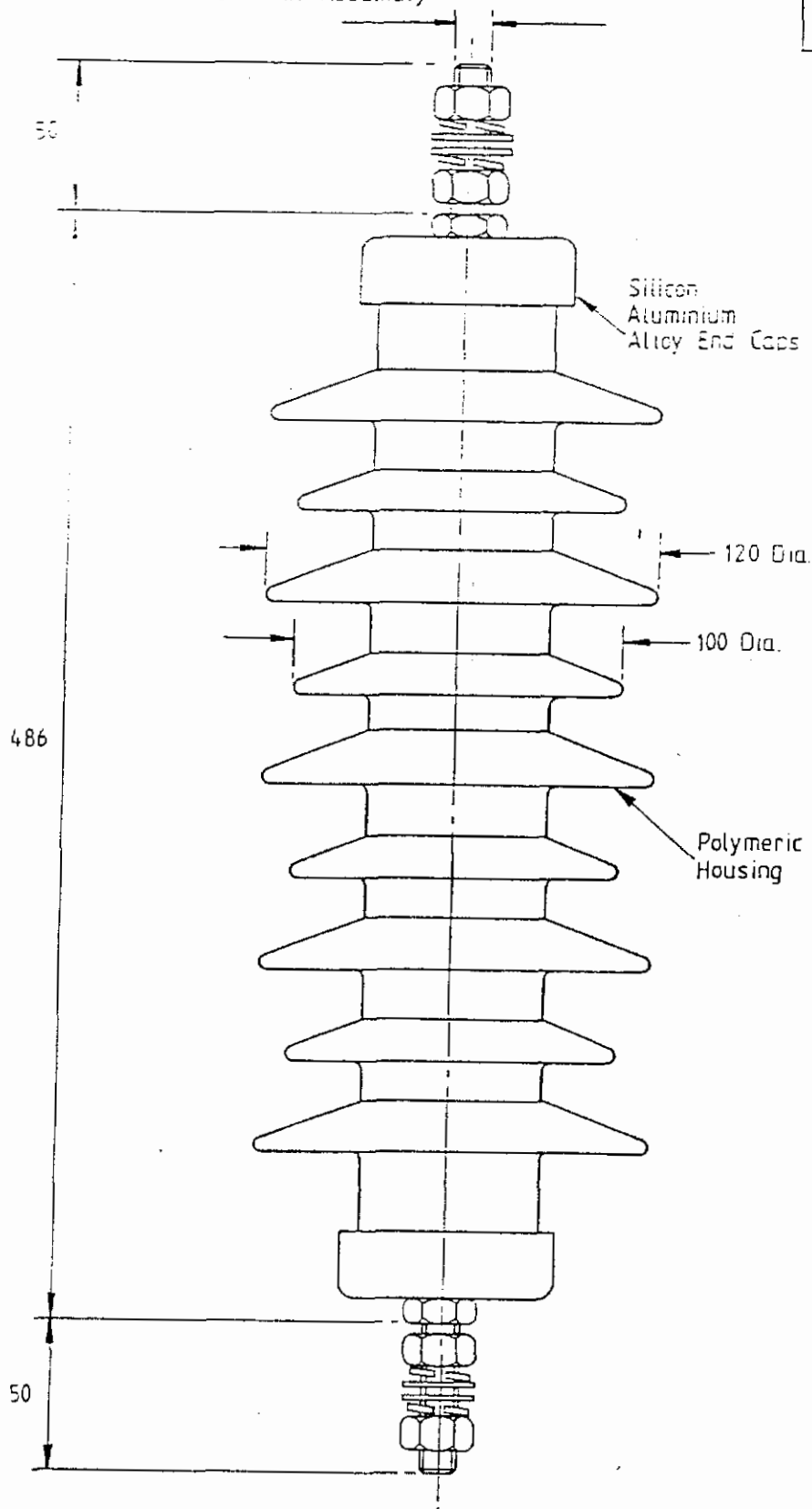
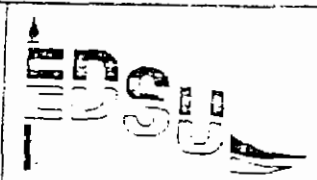


M.E.B. REF.	CONNECTOR REF.	A	B	C	D	CONDUCTOR SIZE & TYPE
282/1	C25	25.4	73	41	25	200 AAAC / 175 ACSR
282/2	C35	32	82.5	50	35	400 ACSR

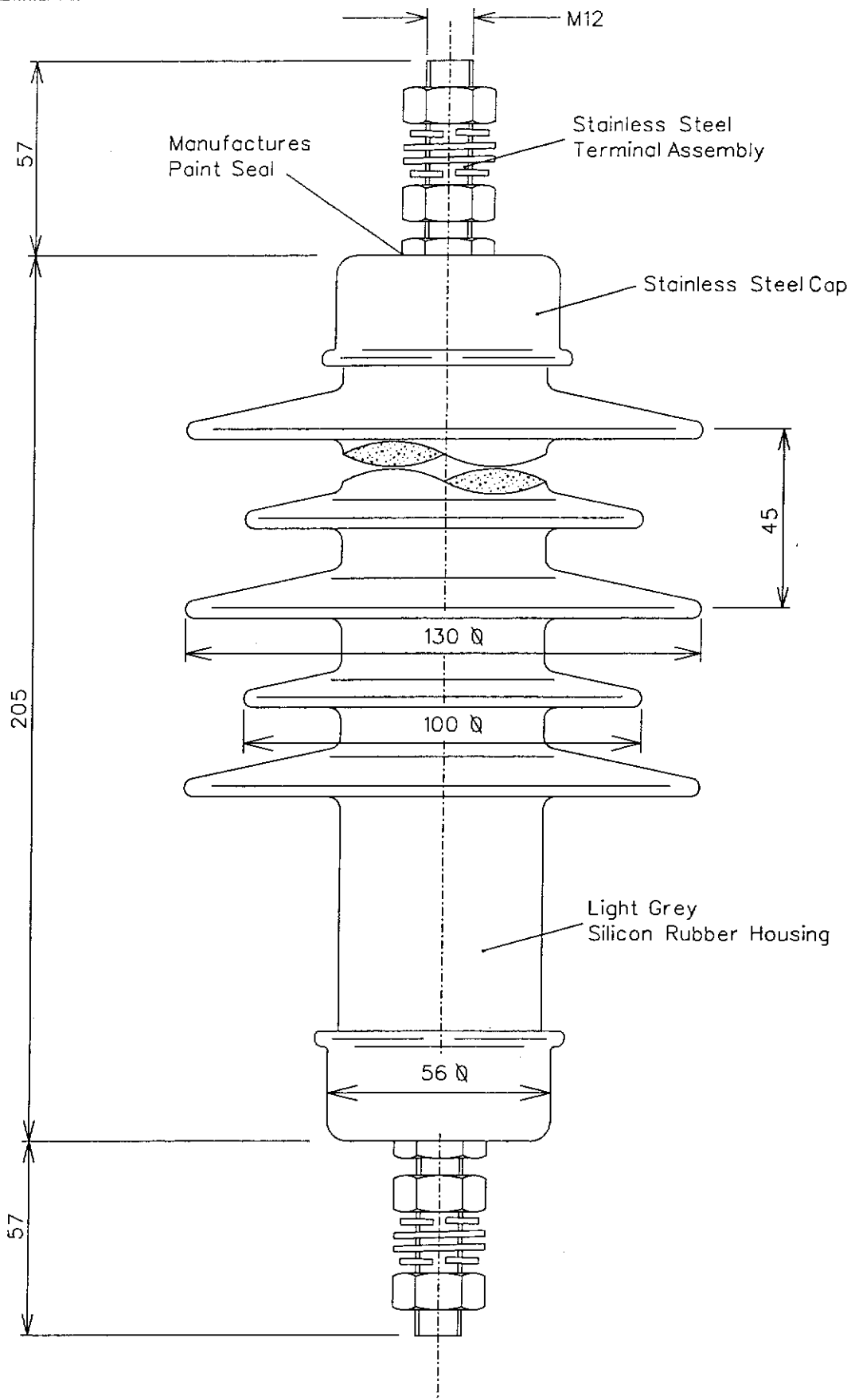
Supplied with 2-M16 x 45 H.T. Bolts, Plain Washers, Lock Washers and Nuts. Galv. to B.S. 729.

ADDED BY DATE	TITLE BUSBAR AND CONNECTORS FOR 33kV. POLE MOUNTED TERMINATION	SCALE	—	 Midlands Electricity plc DRG No M 282 
		DATE	25.2.93	
		DRAWN	A.J.B.	
		TRACED	S.J.H.	
		CHECKED	W.A.H.	
		APPROVED		

M16-2 Stainless
Steel Terminal Assembly



ADD'D BY DATE	TITLE BOWTHORPE EMP 33kV SURGE ARRESTER TYPE HEC 36	SCALE	—	 Midlands Electricity plc 11/88 M342/1
		DATE	4-10-91	
		DRAWN	A.J.B.	
		TRACED	P.J.W.	
		CHECKED		
		APPROVED		



HSRASS 12 L1E1 MO 9.6KV $\sim$ 10KA 12kV/100kA HIGH CURRENT DUTY

Produced by
Output Services

SCALE	-
DATE	22.02.02
DRAWN	MJB
TRACED	-
CHECKED	25.02.02
APPROVED	K.D.

11kV - SURGE ARRESTER

ISSUE	DATE
A	01.07.02
Minor mods.	
ADD/KD	



Aquila

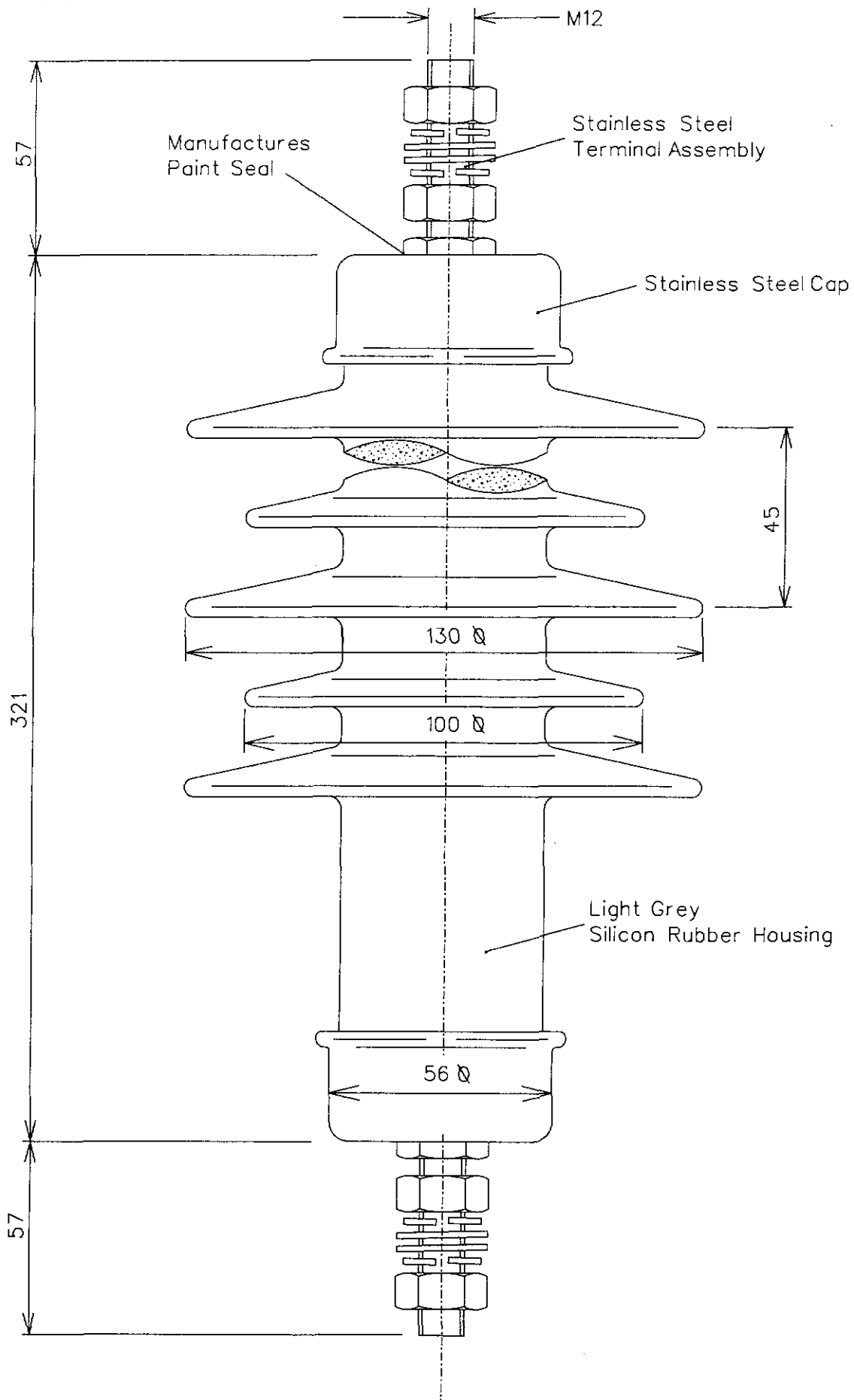
DRG. No.

M460-1

DRG. No.

M460-1

A4



HSRB 24 ~ 10KA 24kV/100KA HIGH CURRENT DUTY

SCALE	-
DATE	09.04.03
DRAWN	ADD
TRACED	-
CHECKED	-
APPROVED	K.D.

24kV - SURGE ARRESTER

ISSUE DATE

Produced by
Primary CAD



Aquila

DRG. No.

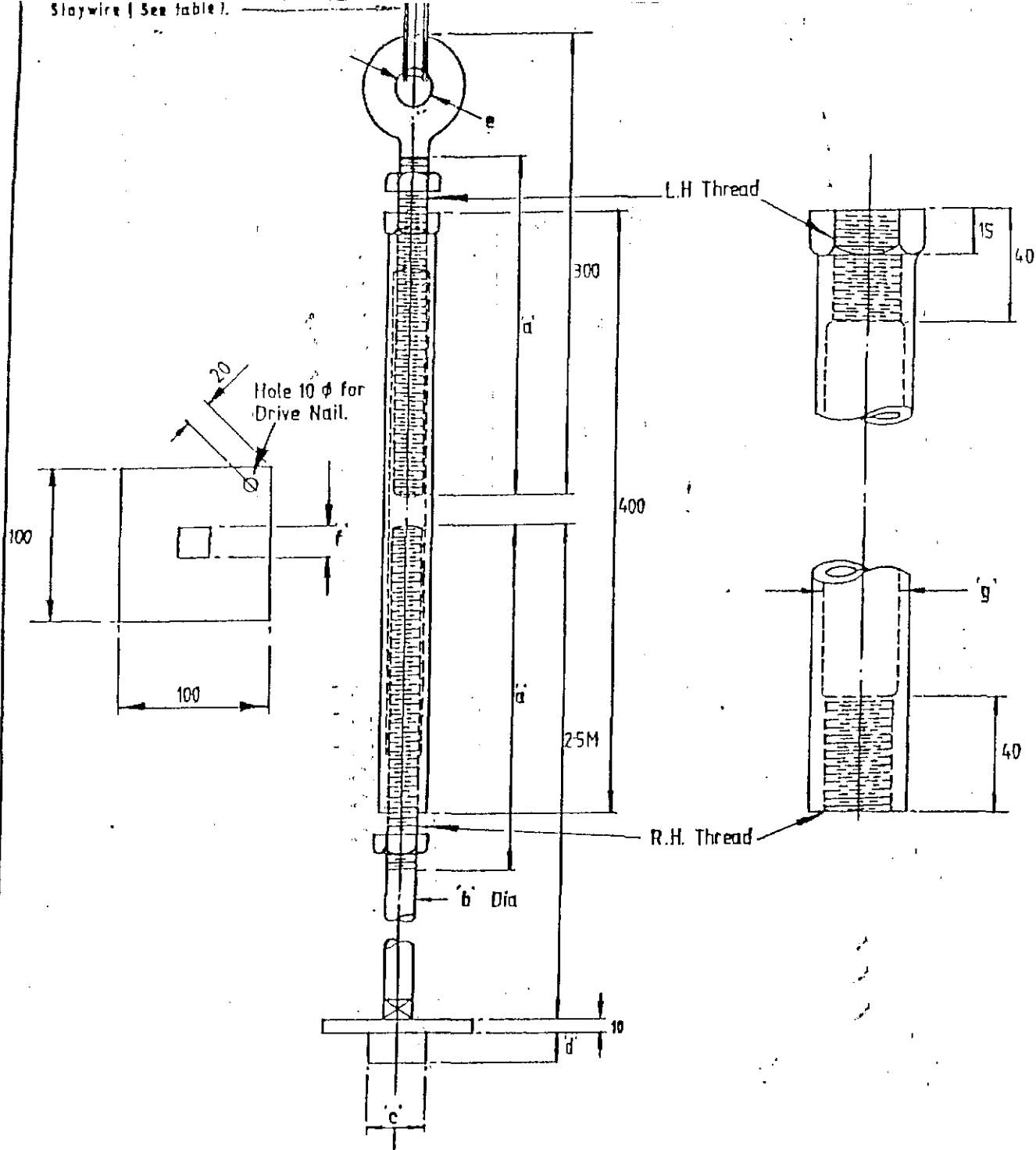
M460-2

DRG. No.

M460-2

SECTION 22

MIN NUMBER	M	DESCRIPTION
3450170	17	TUBULAR STAY ROD
3451150	115	STAY TERMINATING THIMBLE
3451190	119	STAY STRAP
3452700	* 270	PREFORMED STAY GRIPS
3452710	* 271	PREFORMED GRIPS
3453050	305	STAY INSULATORS FOR LV,11kv & 33kv LINES
3454090	409	DUCKBILL STAY ANCHOR (2.2 TONNE)
3454100	410	WOOD STAY BLOCKS
3454450	445	STAY INSULATORS FOR 66kv LINES

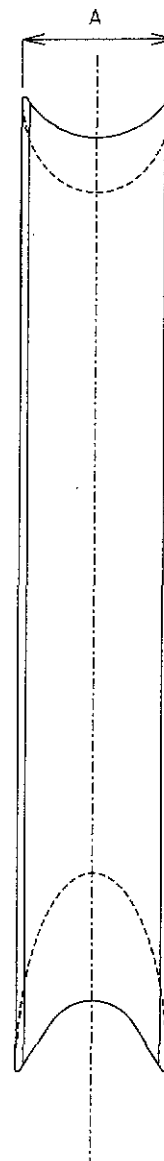
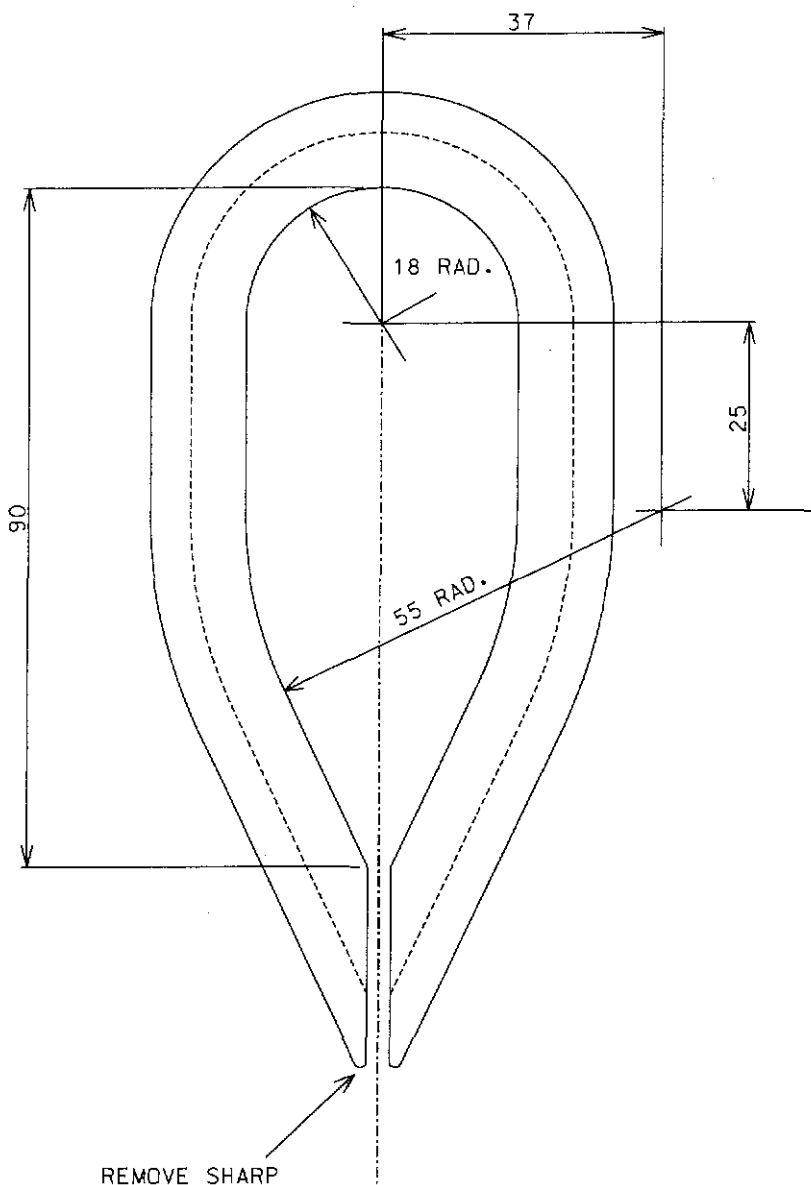


REF Nº	a	b	c	d	e	f	g	For use with Stay Wire	Min failing Load of rod.
M 17/2	230	22	40	20	30	25	25	19/3-55mm	110kN

SCALE	DATE	TUBULAR STAY ROD				ISSUE	MIDLANDS ELECTRICITY BOARD		
3:6:81						A	DRG. No.		
DRAWN						M17/3 added	M.17.		
TRACED	S.J.H.					R.A.C 3/88			
CHECKED									
APPROVED		E.S.I. STANDARD 43-91					E.S.I. Drg. N°439101.		

(ME)

MF



REMOVE SHARP
POINTS BY
GRINDING

See Drg.M461
for conductor
terminating thimble

REF. No.	A	FOR USE WITH STAY WIRE	THIMBLE TYPE
M115/1	20	7/3.25mm 7/4. 0mm	CLOSED CONDITION
M115/3	20	7/3.25mm 7/4.00mm	OPEN CONDITION 25mm GAP

SCALE	FULL
DATE	28.05.81
DRAWN	R.A.C.
TRACED	C.A.R.
CHECKED	R.C.F.
APPROVED	R.C.F.

STAY TERMINATING THIMBLE

ISSUE	DATE
C	18.12.01

TEXT REMOVED.

D.B.



DRG. No.

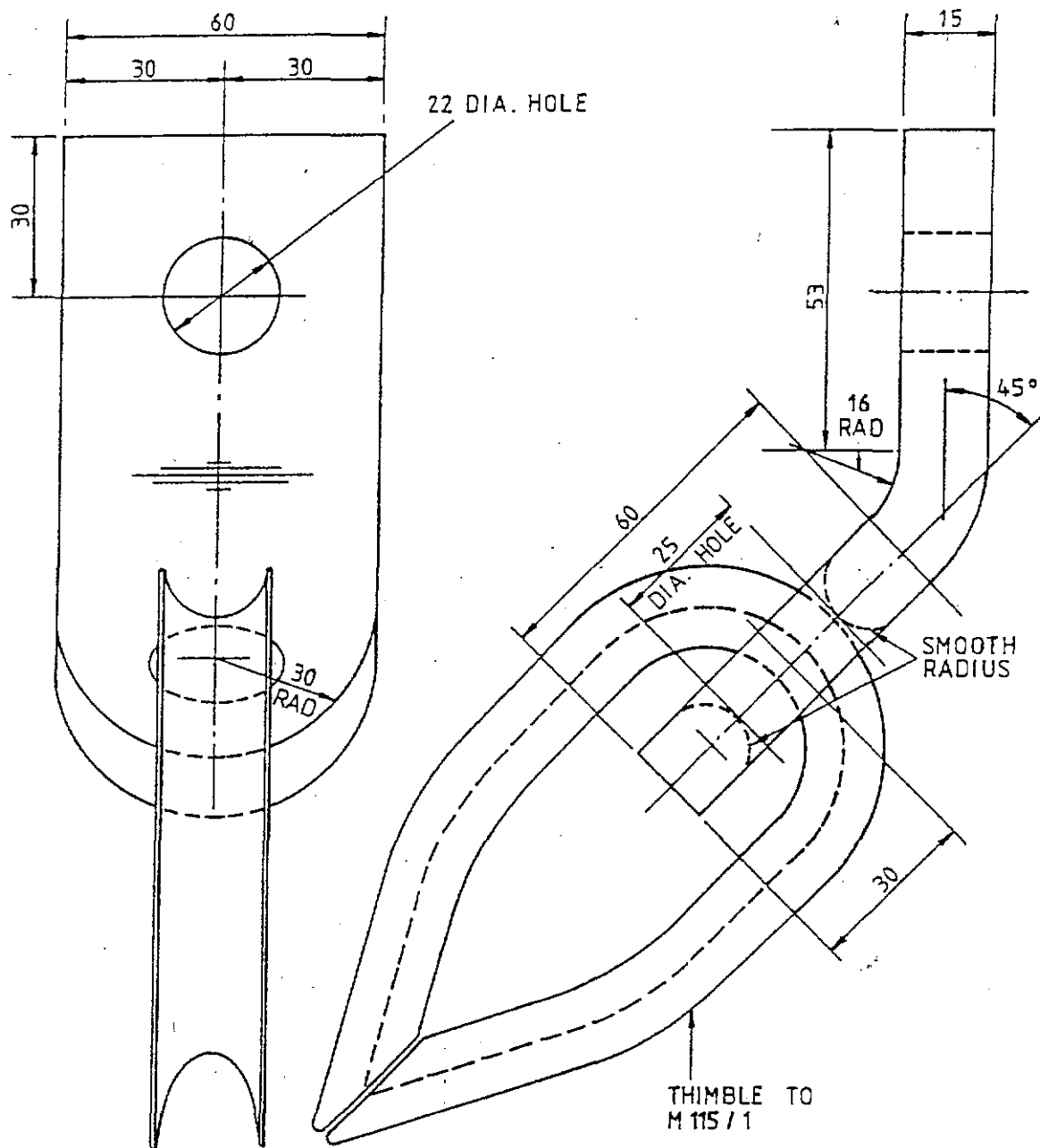
M115

Produced by
Output Services

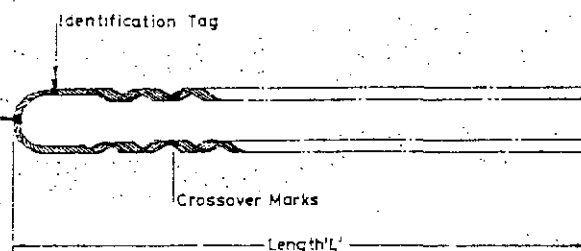
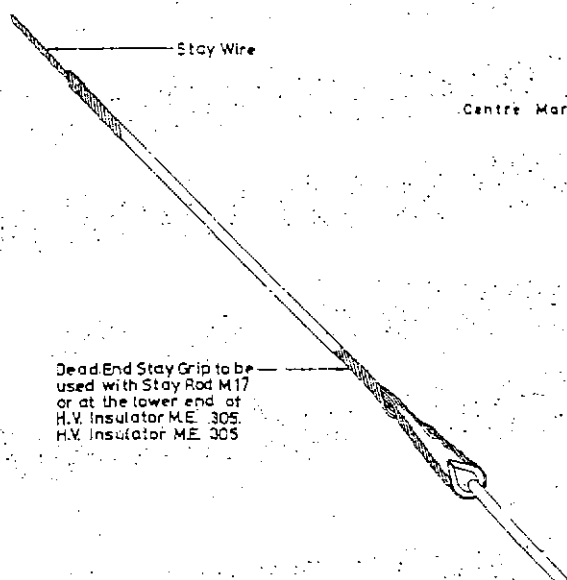
DRG. No.

M115

A4



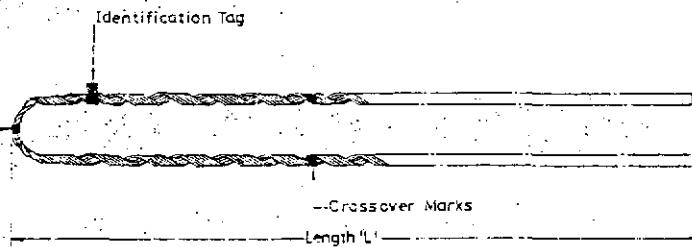
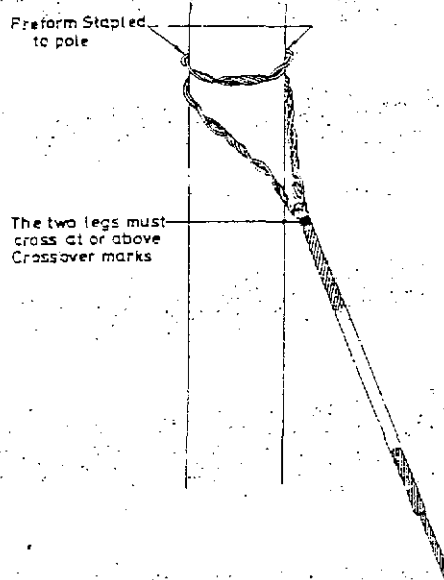
SCALE	N.T.S.	STAY STRAP	ISSUE	MIDLANDS
DATE	1-6-81			ELECTRICITY BOARD
DRAWN	R.A.C.			ORG. No. M.119
TRACED	C.A.R.			
CHECKED				
APPROVED				
E.S.I. STANDARD 43-91		E.S.I. Drg. No 439110		



DEAD END
GUY GRIPS

DEAD END STAY GRIPS				
ME. N°	PL.P. CATALOGUE N°	STAY CAPACITY	LENGTH 'L'	RATED HOLDING STRENGTH lbs.(kg)
270/1	DE 1303432	4/8 (4/4.0)	35" (914.4)	17,200 (7802)
270/2	DE 1303438	7/8 (7/4.0)	40" (1012.7)	24,900 (11290)
270/3	DE 1303452	19/10	51" (129.5)	

270/4



POLE TOP
GUY GRIPS

POLE TOP STAY GRIPS				
ME. N°	PL.P. CATALOGUE N°	STAY CAPACITY	LENGTH 'L'	RATED HOLDING STRENGTH lbs.(kg)
270/4	PT 1303445	4/8 (4/4.0)	54" (1.37m)	6,396 (2902)
270/5	PT 1303450	7/8 (7/4.0)	69" (1.75m)	9,995 (4534)
270/6	N/A	19/10		
POLE TOP STAY GRIPS (FOR P.T. DIA'S OVER 9")				
270/7	PT 1303450	4/8 (4/4.0)	68" (1.73m)	6,396 (2902)
270/8	PT 1303451	7/8 (7/4.0)	79" (2.0m)	9,995 (4534)
270/9	N/A	19/10		

As H.S. 57/1

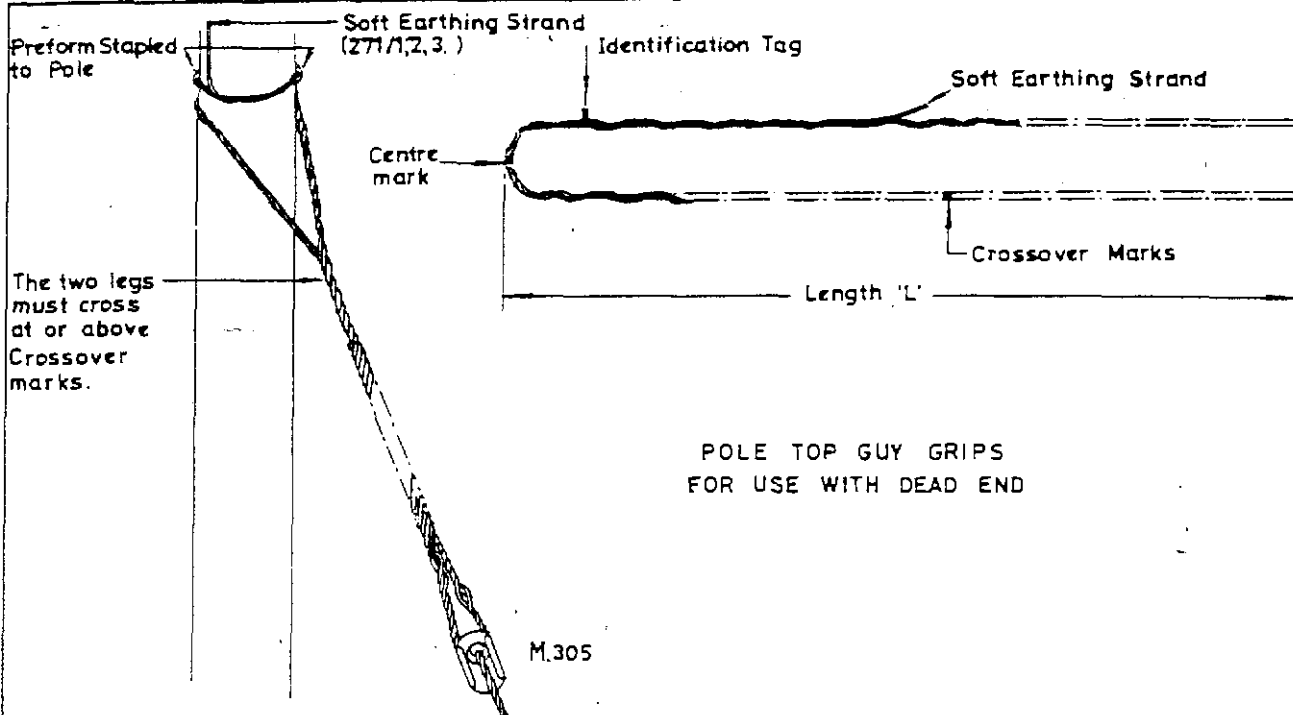
SCOTT	NTS
DRUM	WOMBAR.
CEL	LFC
CH. FAS	CH.
CH. FAS	CH.

PREFORMED STAY GRIPS

ISSUE	PL.P.
G	H
Cal. N° change.	Insulator N°
21-10 77 RAC	6-3-75 RAC

MIDLANDS
ELECTRICITY BOARD

M.E. 270



POLE TOP STAY GRIPS FOR CONNECTION TO HV. INSULATOR (WITH EARTH STRAND)					
ME. No.	STAY CAPACITY	P.L.P. CATALOGUE No.	LENGTH 'L'	RATED HOLDING STRENGTH lbs (kg)	MATCHING DEAD END
271/1	4/8 ()	PT 1303448	68" (1.77m)	14,119 (6404)	DE 1303432
271/2	7/8 ()	PT1303447	78" (1.98m)	22,064 (10010)	DE 1303438
271/3	19/10 ()	NOT AVAILABLE USE 1303447			DE 1303438
POLE TOP STAY GRIPS FOR CONNECTION TO LV. INSULATOR (To provide 6ft Insulator distance)					
271/4	4/8 ()	PT1303456	92" (2.33m)	14119 (6404)	DE 1303432
271/5	7/8 ()	PT1303455	90" (2.28m)	22064 (10010)	DE 1303438
271/6	19/10 ()	N/A			



- ME.271/0 PREFORMED DEAD END FOR 35mm² Cu CONCENTRIC SERVICE CABLE MANUFACTURERS CAT N° CND 5732100-R
- ME.271/7 - PREFORMED DEAD END FOR 16mm² Cu. CONCENTRIC SERVICE CABLE MANUFACTURERS CAT.No. CNDE 5732129.
- ME.271/8 - PREFORMED DEAD END FOR 25 mm² Cu. CONCENTRIC SERVICE CABLE MANUFACTURERS CAT.N° CNDE 5732155.

ME 271/9

NOTE:— Comprises of 3 sections



33kV Insulator Link Assembly.

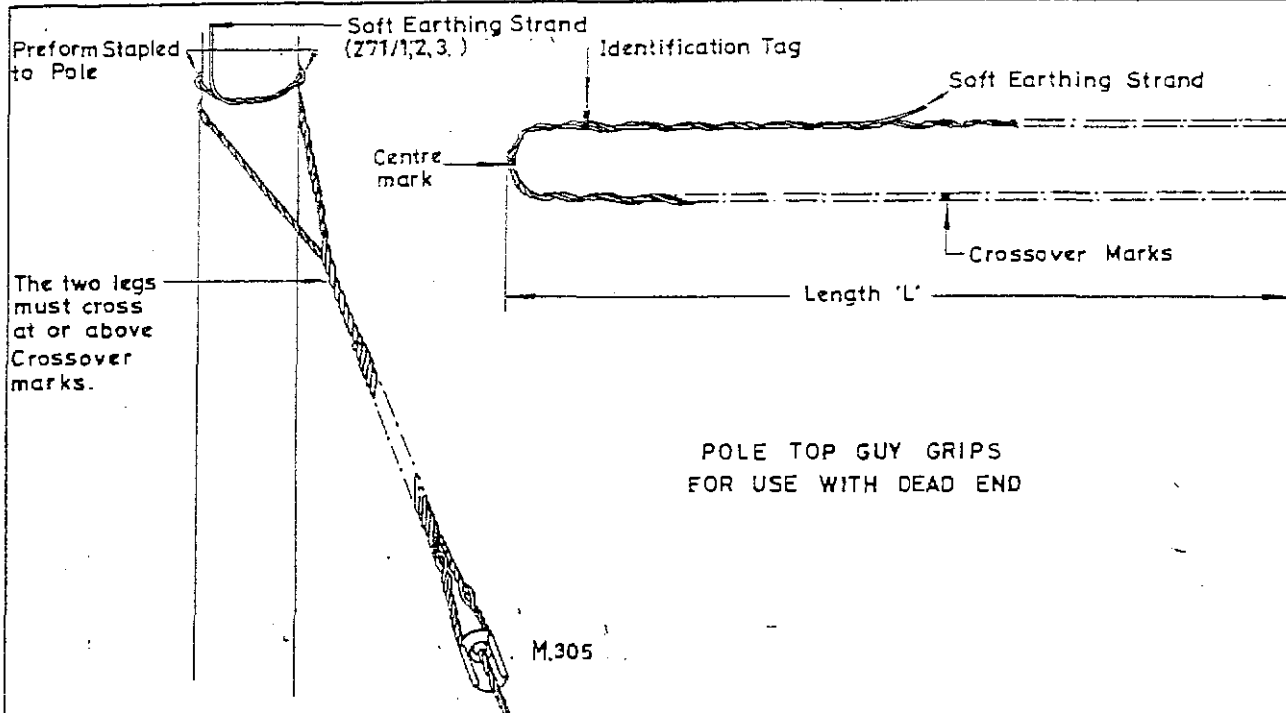
CAT N° 5715275

As H.S.57/2

SCALE	
DATE	2.3.73
DRAWN	F.A.H.
TRACED	A.C.
CHECKED	
APPROVED	

PREFORMED GRIPS

ISSUE	ISSUE	MIDLANDS ELECTRICITY BOARD
	F	
33kV Ass M305 1 was M305 & M 271/0 added RAC. 4-3-87		DRG. No. M.E. 271



POLE TOP GUY GRIPS
FOR USE WITH DEAD END

POLE TOP STAY GRIPS FOR CONNECTION TO HV. INSULATOR (WITH EARTH STRAND)					
ME No.	STAY CAPACITY	P.L.P. CATALOGUE No.	LENGTH 'L'	RATED HOLDING STRENGTH lbs (kg)	MATCHING DEAD END
271/1	4/8 ()	PT 1303448	68" (1.77m)	14,119 (6404)	DE 1303432
271/2	7/8 ()	PT1303447	78" (1.98m)	22,064 (10010)	DE 1303438
271/3	19/10 ()	NOT AVAILABLE USE 1303447			DE 1303438
POLE TOP STAY GRIPS FOR CONNECTION TO LV. INSULATOR (To provide 5ft Insulator distance)					
271/4	4/8 ()	PT1303456	92" (2.33m)	14119 (6404)	DE 1303432
271/5	7/8 ()	PT1303455	90" (2.28m)	22064 (10010)	DE 1303438
271/6	19/10 ()	N/A			



ME.271/7 - PREFORMED DEAD END FOR 16 mm² Cu. CONCENTRIC SERVICE CABLE. MANUFACTURERS CAT.No. CNDE 5732129.

ME.271/8 - PREFORMED DEAD END FOR 25 mm² Cu. CONCENTRIC SERVICE CABLE. MANUFACTURERS CAT.No. CNDE 5732155.

ME 271/9

NOTE: — Comprises of 3 sections



33kV Insulator Link Assembly.

CAT N° 5715275

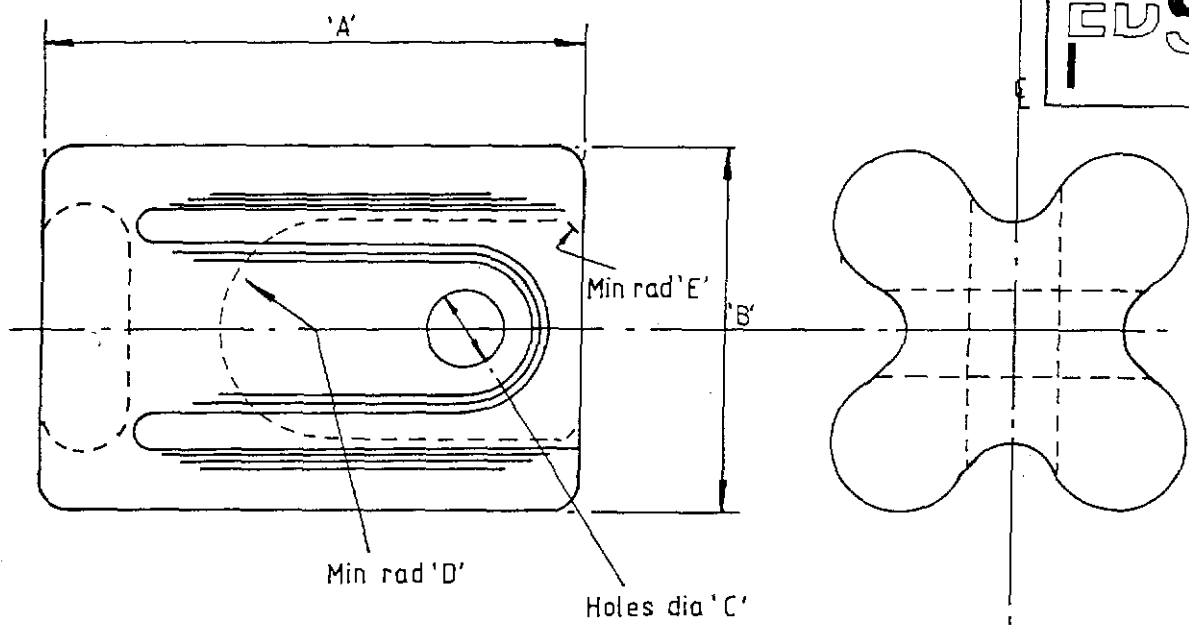
As H.S.57/2

SCALE	
DATE	2.3.73
DRAWN	F.A.H.
TRACED	A.C.
CHECKED	<i>[Signature]</i>
APPROVED	<i>[Signature]</i>

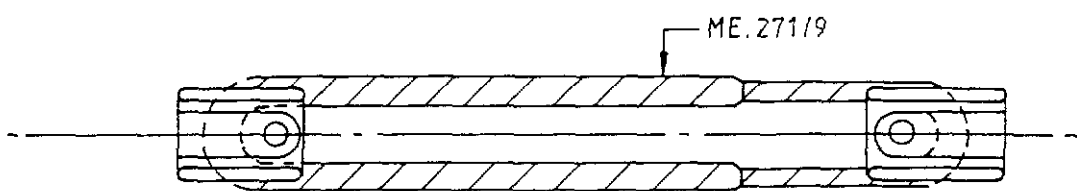
PREFORMED GRIPS

ISSUE	ISSUE
EMP	D.
Note added & Cat N° change 5-3-79 RAC	Cat N° change Item 271/8 added.

MIDLANDS ELECTRICITY BOARD
DRG. No.
ME.271



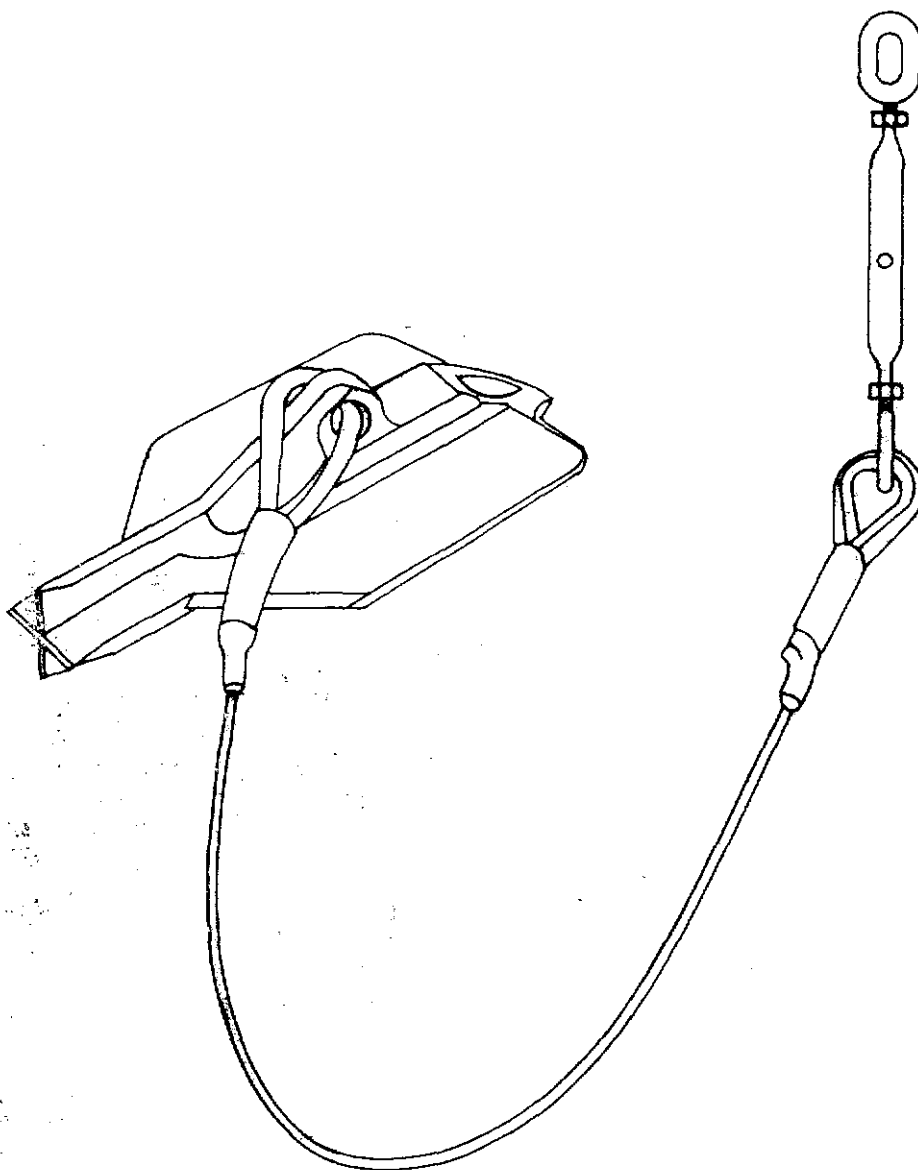
MIN No	305 / 0	305 / 1
ESI 43-91 Type No	1	2
System Voltage	LV & 11kV	33kV
Minimum failing load	110kN	110kN
Minimum creepage distance	50mm	95mm
Power frequency flashover voltage (dry)	38 kV	50 kV
Power frequency flashover voltage (wet)	16kV	25kV
Impulse withstand voltage	50kV	75kV
Minimum dimension A	115mm	220mm
Minimum dimension B	75mm	120mm
Minimum dimension C	25mm	28mm
Minimum dimension D	25mm	40mm
Minimum dimension E	6mm	20mm
B.S. Reference	B.S. 137	
Finish	Brown Glazed	



33k.V. STAY INSULATOR ASSEMBLY
FOR UNEARTHED STRUCTURES SHOWING
PREFORMED LINK

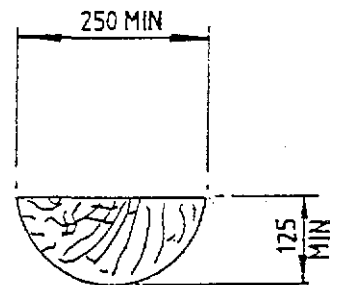
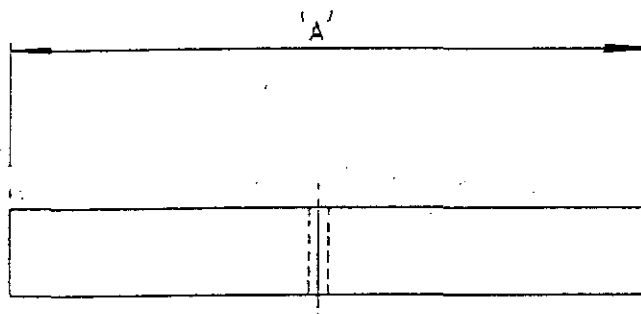
NOTE
Set comprises
of three items.

B DATE 9-8-93 Mod's to table.	TITLE STAY INSULATORS FOR LV, 11kV & 33kV LINES E.S.I. STANDARD. 43-91	SCALE	N.T.S.	MEB Midlands Electricity plc Drg No M. 305. E.S.I. Drg No 439107
		DATE	27-7-81	
		DRAWN	R.A.C.	
		TRACED	G.S.	
		CHECKED	<i>Ref</i>	
	APPROVED	<i>Ref</i>		

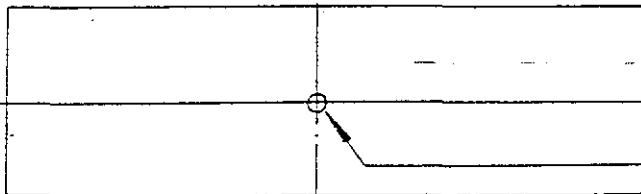


198-DBUTC 2.5 - 198 Duckbill Anchor connected with
2.5m of 10mm polypropylene impregnated
tendon, terminated with M20 x 230
turnbuckle (including locknuts). Fitted
with continuity bonds.

ADDED BY DATE	TITLE DUCKBILL STAY ANCHOR (2.2 TONNE)	SCALE		 Midlands Electricity plc DRG No M409 
		DATE	15.1.93	
		DRAWN	A.J.B	
		TRACED	S.G.T.	
		CHECKED	RC7	
		APPROVED		



or



26 DIA HOLE

REF Nº	A	FOR USE WITH STAY WIRE
M 410/1	850	7/4.0 mm
M 410/2	1300	19/3.55 mm

SCALE	1-10
DATE	19-1-82
DRAWN	R.A.C.
TRACED	
CHECKED	
APPROVED	#

WOOD STAY BLOCKS.

ISSUE

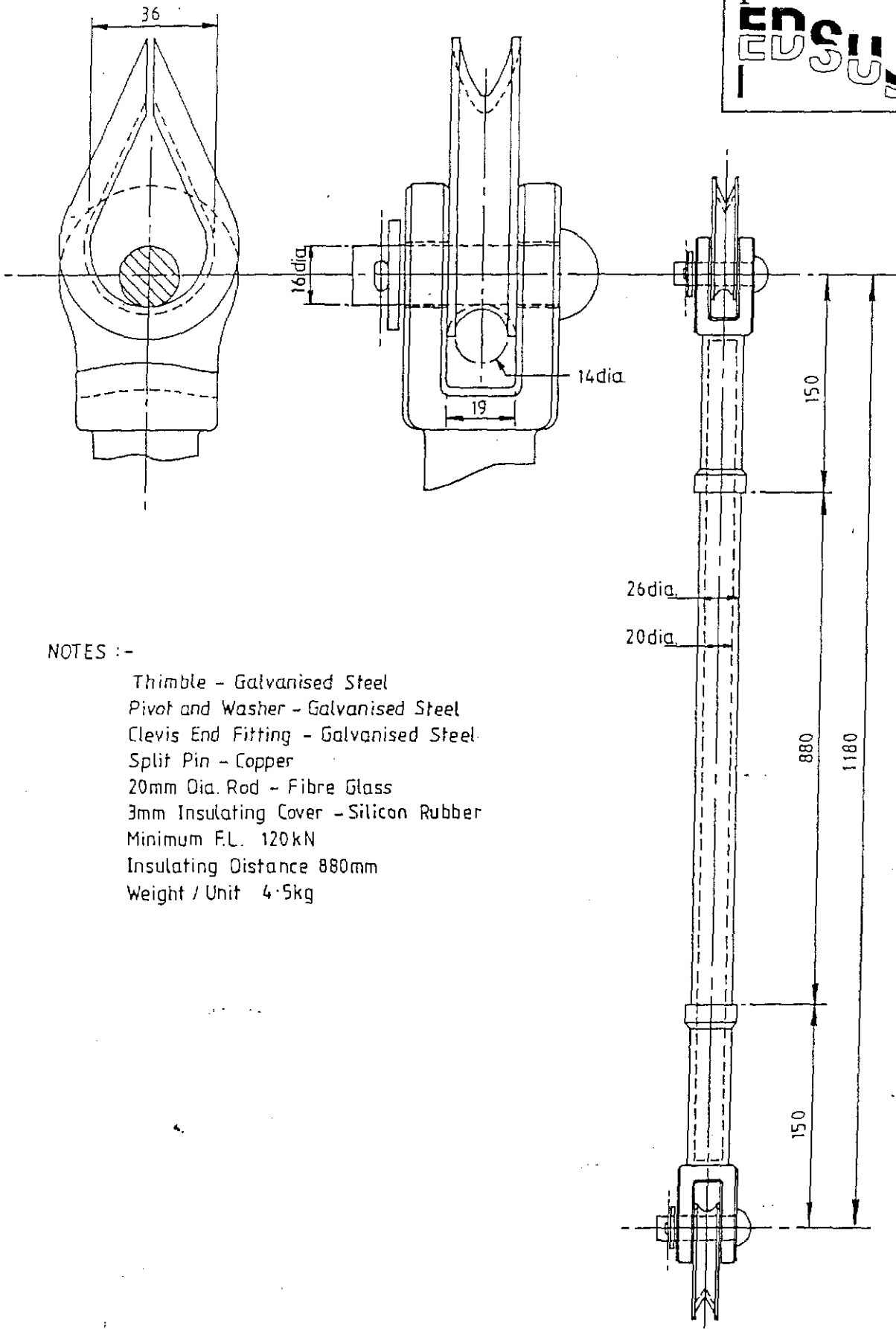
MIDLANDS
ELECTRICITY BOARD

DRG. No.

M 410

ESI STANDARD 43-91

ESI Drg N° 439103



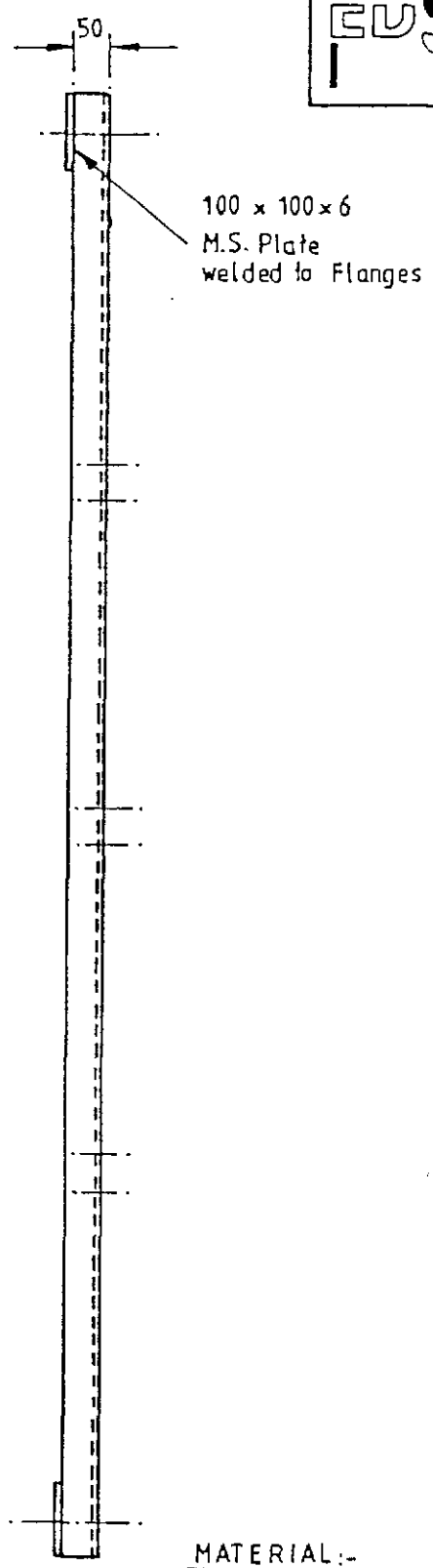
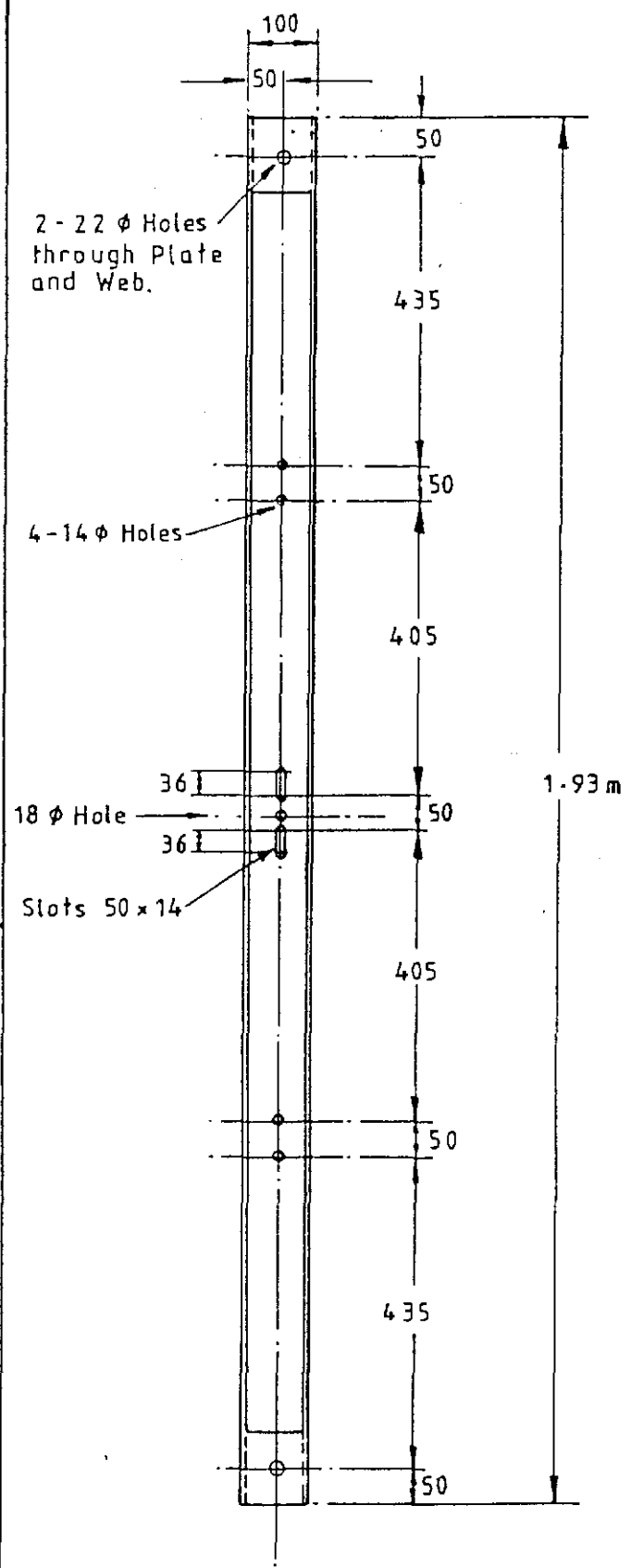
NOTES :-

Thimble - Galvanised Steel
 Pivot and Washer - Galvanised Steel
 Clevis End Fitting - Galvanised Steel
 Split Pin - Copper
 20mm Dia. Rod - Fibre Glass
 3mm Insulating Cover - Silicon Rubber
 Minimum F.L. 120kN
 Insulating Distance 880mm
 Weight / Unit 4.5kg

A DATE 14-10-93 Change of Manufacturer	TITLE STAY INSULATOR - 66KV	SCALE	—	MEB Midlands Electricity plc DAG No M445
		DATE	14-10-93	
		DRAWN	A. J. B.	
		TRACED	P. J. W.	
		CHECKED	Ref	
		APPROVED		

SECTION 23

MIN NUMBER	M	DESCRIPTION
3450530	53	CROSSARM - 'H' POLE FUSE & CABLE SUPPORT
3450910	91	OUTRIGGER A.C.D BRACKET
3452452	245/2	POST INS 11kV SWITCH & FUSE GEAR
3453440	344	11kV POLE MOUNTED EXPULSION FUSE UNIT BRUSH TYPE A3463
3453450	345	11kV POLE MOUNTED EXPULSION FUSE UNIT TYPE DJP (GEC)
3454590	459	ASSEMBLY OF SINGLE POLE ANGLE BRACKET TYPE A3468 (HAWKER)
3455050	505	UNIVERSAL EXPULSION FUSE ELEMENTS
3457150	715	CROSSARM MOUNTING STRAP FOR SPRITE ENCLOSURE
3457160	716	MOUNTING BRACKET FOR COOPER KYLE VCS SWITCH

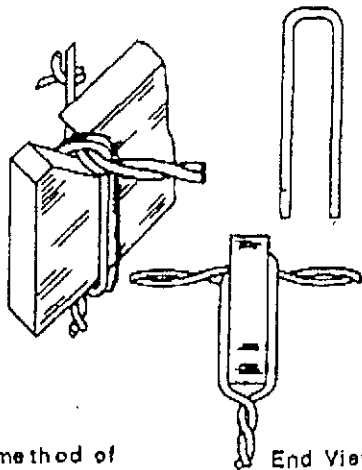


MATERIAL:-
100x50x6 M S Channel to BS 4

FINISH
Galv. to BS 729 after cutting drilling and welding.

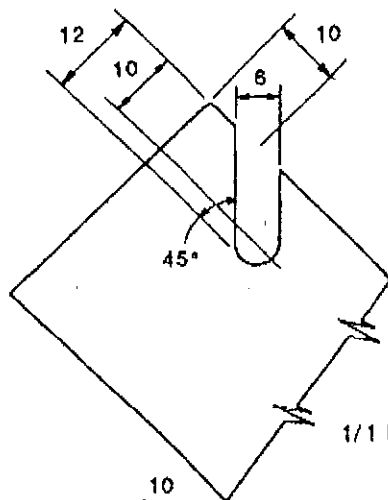
ADD'D BY DATE	TITLE CROSSARM - H POLE FUSE AND CABLE SUPPORT	SCALE		MEB Midlands Electricity plc M. 53. MF MF
		DATE	4.8.92	
		DRAWN	A.J.B.	
		TRACED	S.G.T.	
		CHECKED		
		APPROVED	W.D.M	

Preformed No. 12 S.W.G.
Galvanised wire

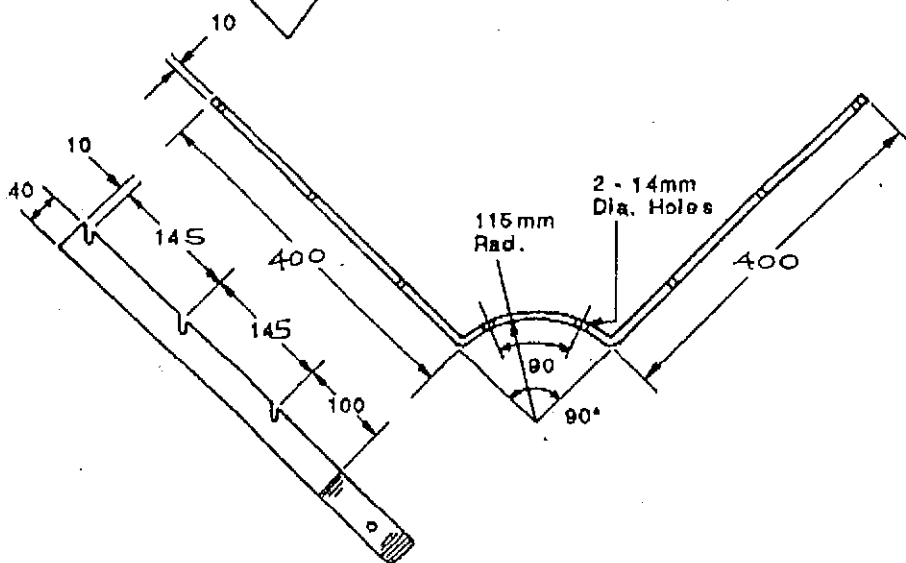


Typical method of
fixing barbed wire
to outrigger bracket

End View



1/1 Detail of Slot



MATERIAL

40 x 10 M.S.
Flat to B.S. 4

2 - Off per set.
o/w 4 - 12 x 75
Galv. coach screws.

FINISH

Galv. to B.S. 729
after cutting and drilling.

All dimensions are in millimetres unless otherwise stated.

TITLE

OUTRIGGER AND
BRACKET.

SCALE

DATE

DRAWN

TRACED

CHECKED

APPROVED

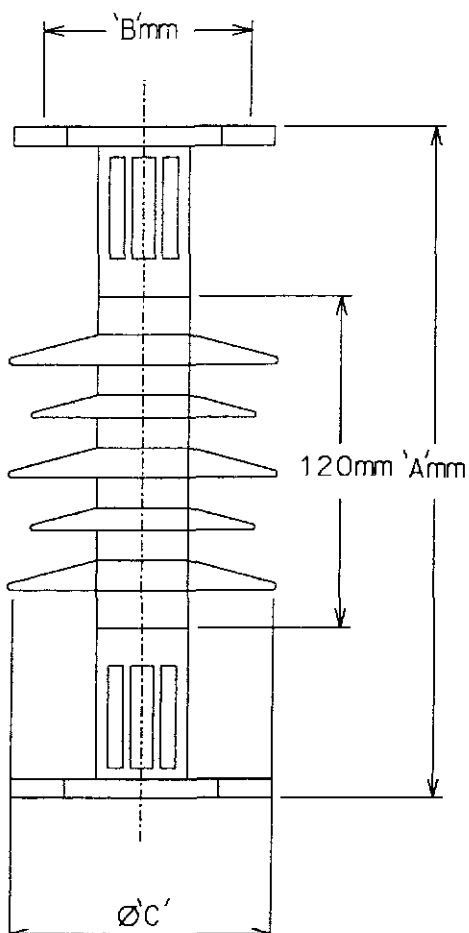
MEB

Midlands Electricity plc

ORG No.

M. 91

ADDED BY
DATE



ELECTRICAL DATA		
TEST	56145-27	SPECIFICATION
CREEPAGE (mm)	398	IEC 815
TAUT STRING (mm)	170	
IMPULSE FLASHOVER (DRY) (KVp)	120	IEC 383: 1983 BS137: 1982
IMPULSE WITHSTAND (DRY) (KVp)	100	
AC 1min FLASHOVER (DRY) (KV)	90	
AC 1min WITHSTAND (DRY) (KV)	80	
AC 1min FLASHOVER (WET) (KV)	60	
AC 1min WITHSTAND (WET) (KV)	50	

MECHANICAL DATA		
TEST	56145-27	SPECIFICATION
CANTILEVER DEFLECTION AT 1kN. LOAD (mm)	5	W T HENLEY
MINIMUM TENSILE FAILING LOAD (kN)	<50	
MINIMUM FAILING TORQUE (NM)	<90	

ROUTINE PRODUCTION TESTS			
COMPONENT	TEST	56141-10	SPECIFICATION
MOULDED SUBSTRATE	WATER IMMERSION/ ELECTRICAL	MIN. 1%	BB. QA/P.1-A 1988
FINAL ASSEMBLY	MECHANICAL	1%	0.5xSML-1min. 1.2xSML-1min. PULL TO FAIL

INSULATOR MARKING:

1. PRODUCT CODE.
2. MAUFACTURE'S IDENTITY
3. YEAR/WEEK NUMBER IDENTITY

PRODUCT CODE	SYSTEM VOLTAGE	WEIGHT (kg)	STAND-OFF LENGTH 'A'	FIXING CENTRE 'B'	SHED DIAM. 'C'
56145-27	11kv	1.20	230mm	76 mm	Ø 108mm

Suitable for re-placering porcelain Insulators in similar applications with 76mm cts end-fittings

SCALE	
TE	12.02.03
DRAWN	M. J. B
TRACED	-
CHECKED	KD
APPROVED	13.02.03

COMPOSITE POST INSULATOR FOR
11KV SWITCH AND FUSE GEAR

ISSUE	DATE



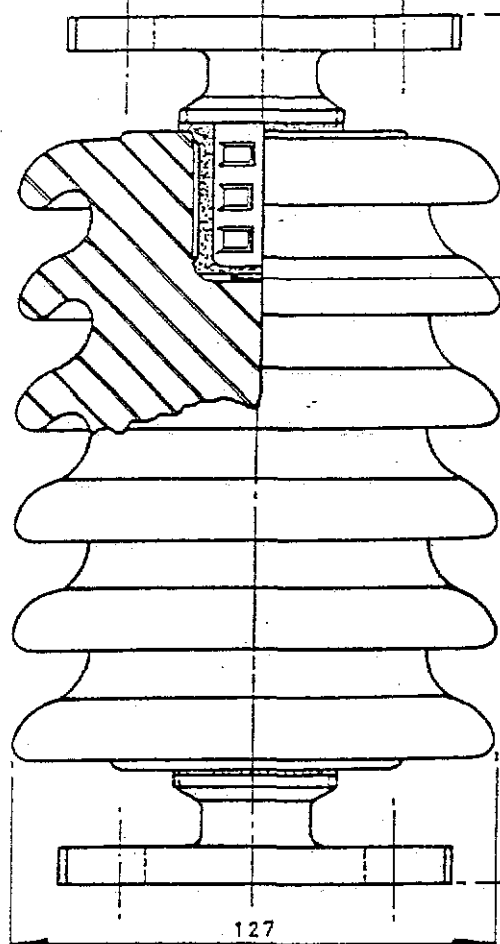
Aquila

DRG. No.

M245

DRG. No.

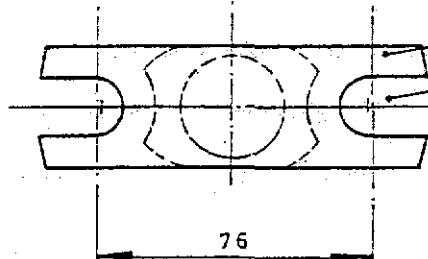
M245



Resilient pad 29 x 1.6
(2 off).

230

127



M. C. I. Galvanised inserts to BS.729

Slots for 12mm. dia bolts.

76

COUPLING

WEIGHT/UNIT 5.0 Kg.

MINIMUM FAILING LOAD (CANT) 363 Kg.

MAXIMUM WORKING LOAD (CANT)

CREPAGE LENGTH - SURFACE 381

" " PROTECTED 111

ELECTROSTATIC CAPACITANCE

FLASHOVER VOLTAGE

D.F.O. NORMAL 85 kV.

" 1 MIN

W.F.O. NORMAL 54 kV.

" 1 MIN

I.F.O. 50% 1/50 μ s +

" " " " "

IMPULSE WITHSTAND 50 Hz 125 kV.

PUNCTURE VOLTAGE 50 Hz

B.S. REFERENCE

B.S.3297

345245/2

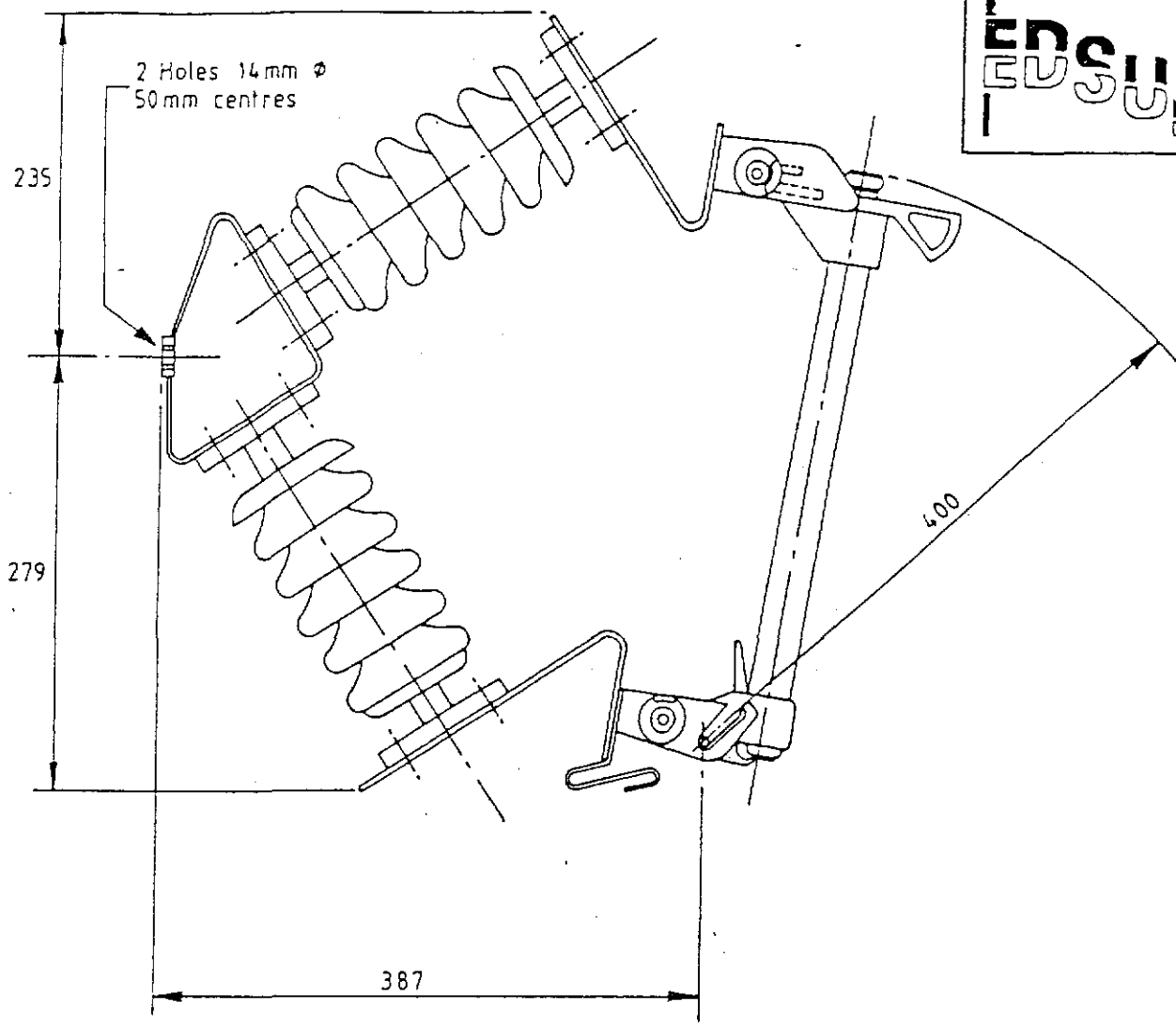
CHARACTERISTIC TABLE FOR INSULATORS MINIMUM REQUIREMENTS.

SCALE	1 : 2
DATE	25-10-82
DRAWN	A. R. B.
TRACED	
CHECKED	
APPROVED	#

POST TYPE INSULATORS FOR
11 kV. SWITCH AND FUSE GEAR

E.S.I STANDARD.

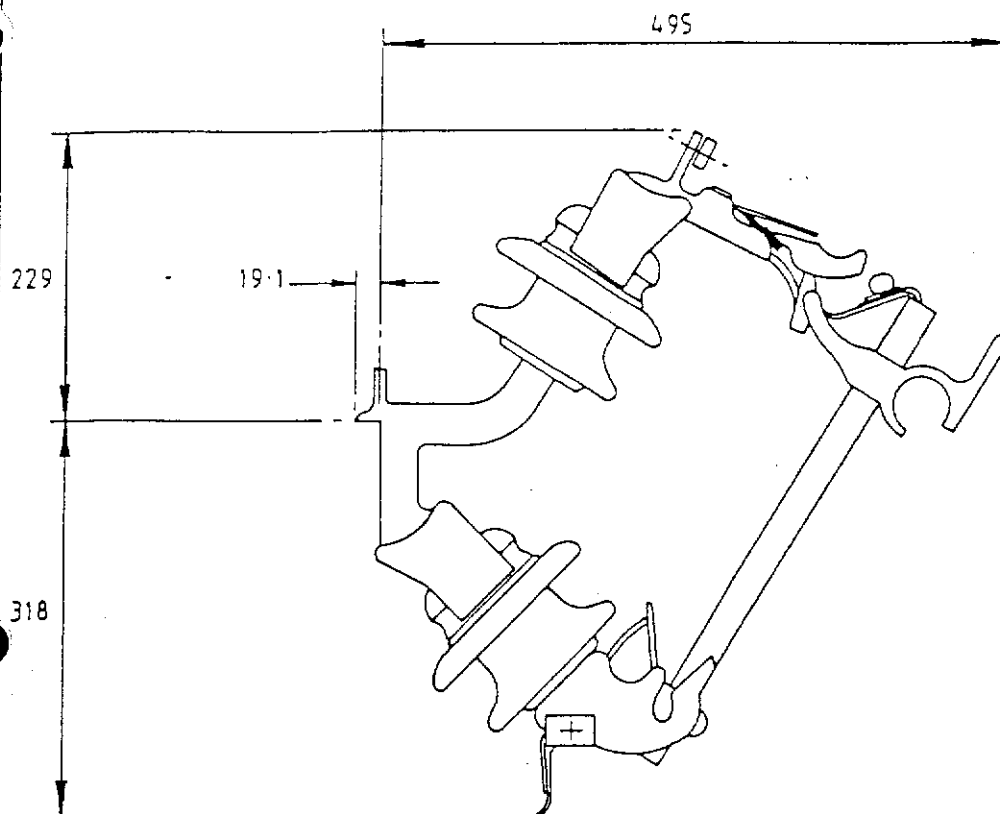
ISSUE	MIDLANDS ELECTRICITY BOARD
DRG. No.	M.245.
E.S.I. DRG. No.	



MIN N°	ITEM	MANUF. REF.
345344/1	UNIT C/W HOLDER	A 3463/PR50
345344/5	200A SOLID LINK	A3463/200A LINK
345344/6	FUSE ELEMENT HOLDER	A3463/FUSE HOLDER

- NOTES:-
1. This unit has been superseded by A3468/PR50 type to Drg. M459
 2. Universal Expulsion Fuse Elements see M505
 3. Replacement Insulator as M245

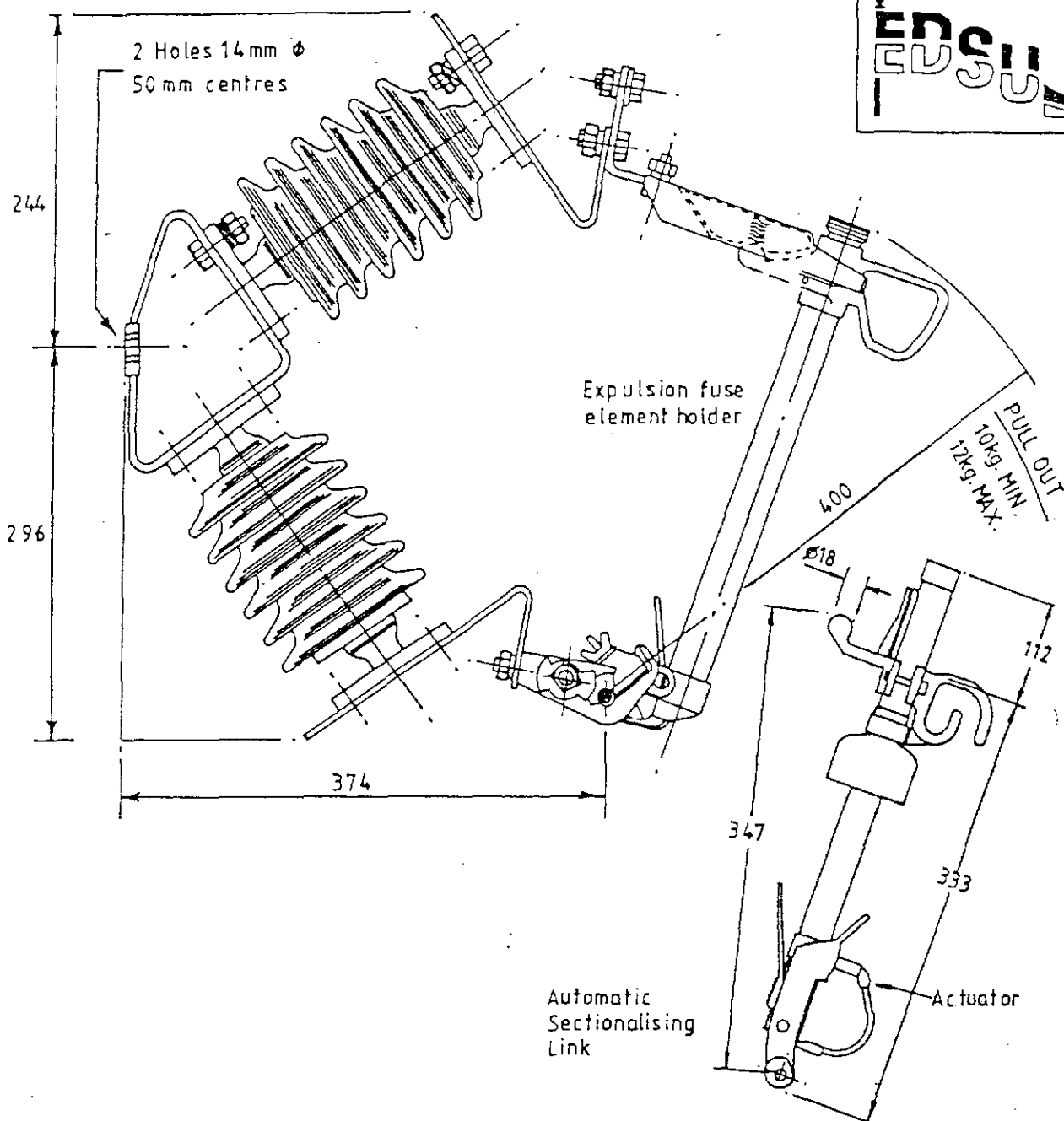
C 40000 BY A.J.B. DATE 6-5-92 Redrawn & Table revised MF D 40000 BY A.J.B. DATE 2-8-94 Mod to note 2.	TITLE 11kV POLE MOUNTED EXPULSION FUSE UNIT (BRUSH) TYPE A3463	SCALE	N.T.S.	M344	M344	MF
		DATE	12-1-83			
		DRAWN	R.A.C.			
		TRACED	P.J.W.			
		CHECKED	Ref			
APPROVED						



MIN N°.	ITEM	MANUF. REF.
345345/1	UNIT C/W HOLDER	11-22kV SWITCH (DJP)
345345/5	200A SOLID LINK	11kV-SOLID-LINK
345345/6	FUSE ELEMENT HOLDER	11kV-CAR-SYM5756

Not approved for new construction use 3454591
For Universal Expulsion Fuse Elements see M505

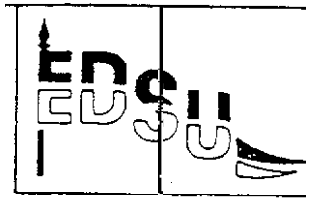
B ADDD BY AJR DATE 6-5-92 Redrawn & table revised C ADDD BY A.J.B. DATE 2-8-94 Table rationalised.	TITLE 11kV POLE MOUNTED EXPULSION FUSE UNIT TYPE 'DJP' (GEC)	SCALE	N.T.S.	MEB Midlands Electricity plc DRG No M34S
		DATE	13-1-83	
		DRAWN	R.A.C.	
		TRACED	P.J.W.	
		CHECKED	<i>[Signature]</i>	
		APPROVED	<i>[Signature]</i>	



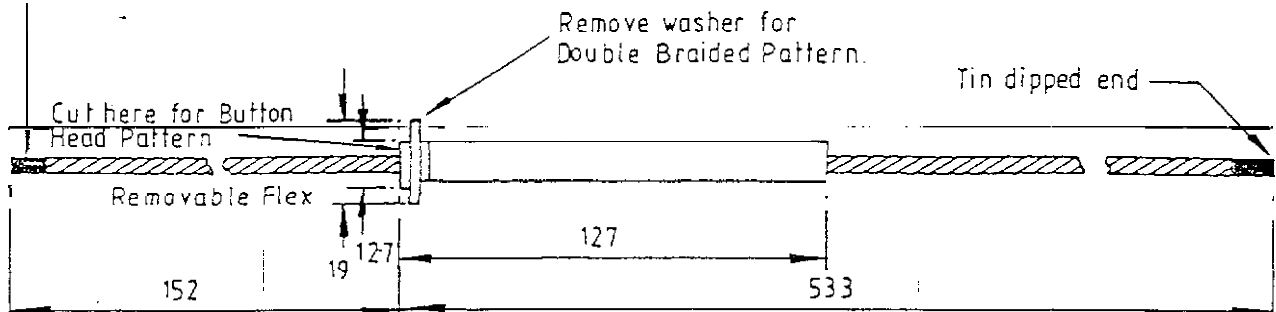
MIN. N°	ITEM	MANUF. REF.
345459/2	MOUNTING	A 3468 / PR50
345459/3	AUTO SECTIONALISING ACTUATOR	E 2906
345459/5	100AMP SOLID LINK	15 SEC 0 BR3
345459/6	EXPULSION FUSE ELEMENT HOLDER	2A 3724 / 10
345459/7	AUTO SECTIONALISING LINK 100AMP 3 SHOT	15 SEC 100 BR3-3
345459/8	" " " 50AMP 3 SHOT	15 SEC 50 BR3-3
345459/9	" " " 50AMP 2 SHOT	15 SEC 50 BR3-2

- 1 EXPULSION FUSE ELEMENTS SEE M505
- 2 AUTO SECTIONALISING LINKS ARE SUPPLIED COMPLETE WITH ACTUATOR
3. REPLACEMENT INSULATOR AS M245

ADD BY A.J.B. DATE 10-8-92 Redrawn & table revised C DATED BY A.J.B. DATE 3-8-94 Drg & table mods	TITLE	SCALE	N.T.S.	 Midlands Electricity plc M 459 MF
	ASSEMBLY OF SINGLE POLE ANGLE BRACKET	DATE	6-8-92	
	TYPE A3468 EXPULSION FUSE SWITCH	DRAWN	A.J.B.	
	(HAWKER) WITH 3" P.C.D. INSULATORS	TRACED	E.L.W.	
		CHECKED	RCJ	
		APPROVED		



Tin dipped end



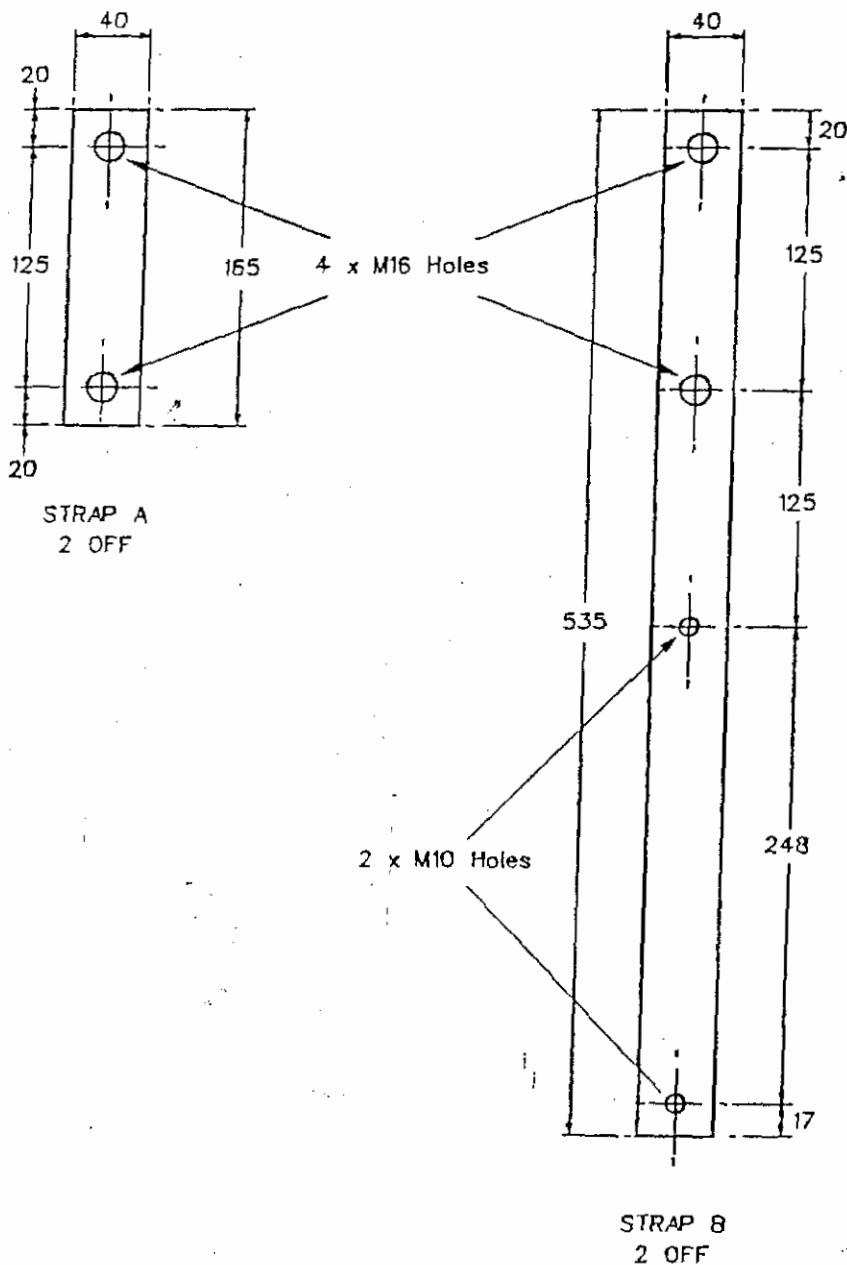
MIN N°	ITEM	MANUF. REF.
345505/1	3A Slow Blow Fuse Element	15T003UNI
345505/2	5A Slow Blow Fuse Element	15T005UNI
345505/3	10A Slow Blow Fuse Element	15T010UNI
345505/4	20A Slow Blow Fuse Element	15T020UNI
345505/5	30A Slow Blow Fuse Element	15T030UNI
345505/6	50A Slow Blow Fuse Element	15T050UNI
345505/7	65A Slow Blow Fuse Element	15T065UNI
345506/1	3A Fast Blow Fuse Element	15K003UNI
345506/2	5A Fast Blow Fuse Element	15K005UNI
345506/3	10A Fast Blow Fuse Element	15K010UNI
345506/4	20A Fast Blow Fuse Element	15K020UNI
345506/5	30A Fast Blow Fuse Element	15K030UNI
345506/6	50A Fast Blow Fuse Element	15K050UNI
345506/7	65A Fast Blow Fuse Element	15K065UNI

NOTE: 1 For Fuse Mountings and Element Holders see Drawings M344, M345 and M459

2 INSTALLATION-

- Either a) Cut Flex at position indicated on drawing for Button Head Pattern
(Brush and Hawker Type Fuse Switches)
or b) Remove Washer for Double Braided Pattern
(GEC Type Fuse Switches)

ADD'D BY Date	TITLE	SCALE	 Midlands Electricity plc M505
	UNIVERSAL	DATE 25.7.94	
	EXPULSION FUSE ELEMENTS	DRAWN A.J.B.	
	FOR ALL	TRACED C.E.	
	TYPES OF EXPULSION FUSE SWITCHES	CHECKED	
		APPROVED	



MATERIAL THICKNESS 8mm MILD STEEL
 GALVANISED TO BS729 AFTER CUTTING AND DRILLING
 SUPPLY TWO STRAP A's AND TWO STRAP B's
 COMPLETE WITH 4 - M16 x 90 BOLT, NUT AND 2 WASHERS (GALVANISED)
 AND 4 - M10 x 30 BOLT, NUT AND 2 SPRING WASHERS (STAINLESS)

SCALE	
DATE	3-11-97
DRAWN	R.C.F.
TRACED	C.E.
CHECKED	
APPROVED	

CROSSARM MOUNTING STRAP
 FOR SPRITE ENCLOSURE

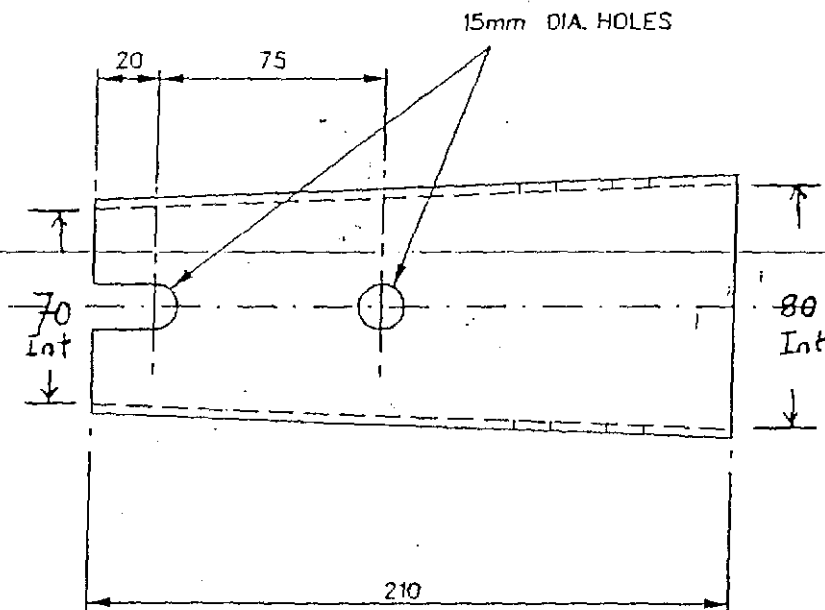
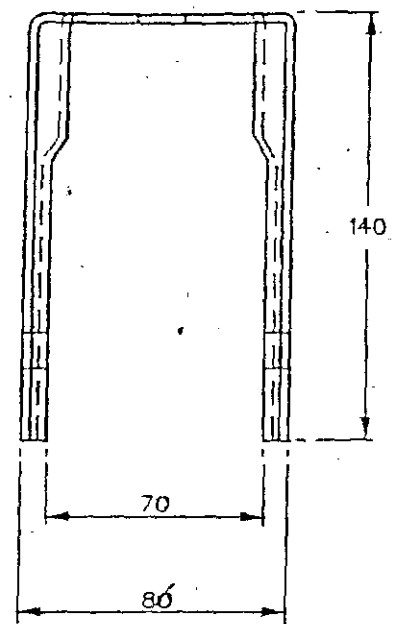
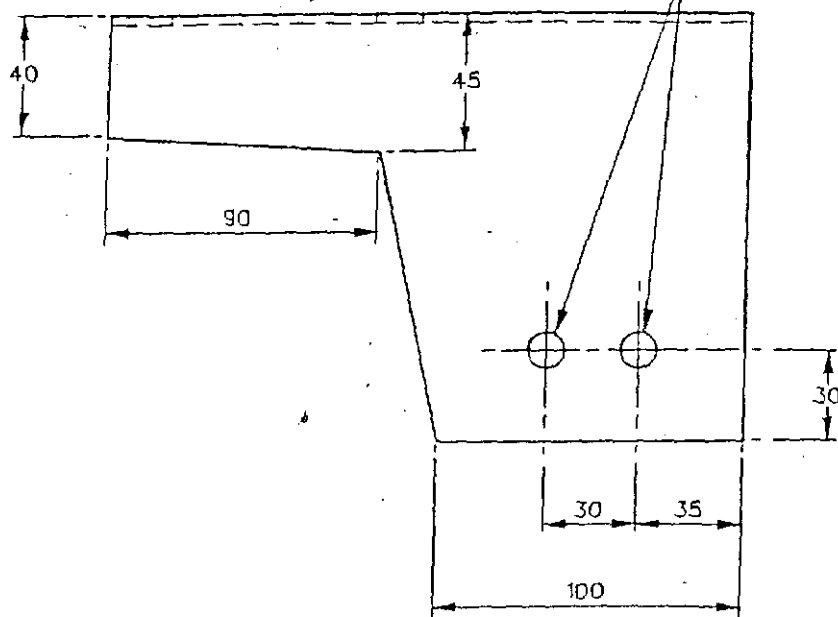
C.A.D. PRODUCED

ISSUE

MEB
 Manda Electric plc

DRG. No.

OHME 715



MATERIAL THICKNESS 3mm MILD STEEL
GALVANISED TO BS729 AFTER CUTTING, DRILLING AND BENDING

ALL DIMENSIONS IN MILLIMETRES

C.A.D. PRODUCED

SCALE	
DATE	4-11-97
DRAWN	R.C.F.
TRACED	C.E.
CHECKED	
APPROVED	

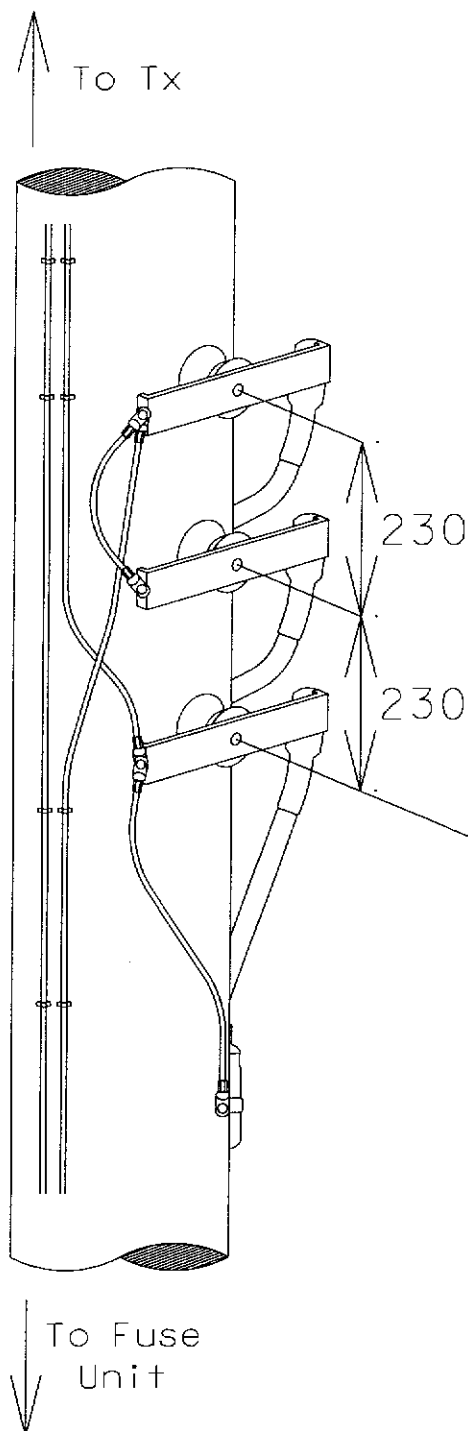
MOUNTING BRACKET
FOR COOPER KYLE VCS SWITCH

ISSUE

meb
Mildsteel Engineering & Fabrication Ltd.

ORG. No.

OHME 716



NOTE:-
Pictorial Details
of Fittings are Typical
only, as these vary for
each manufacturer

For jointing Instructions
CJI 2.4 WAVECON

Dimensions in millimetres

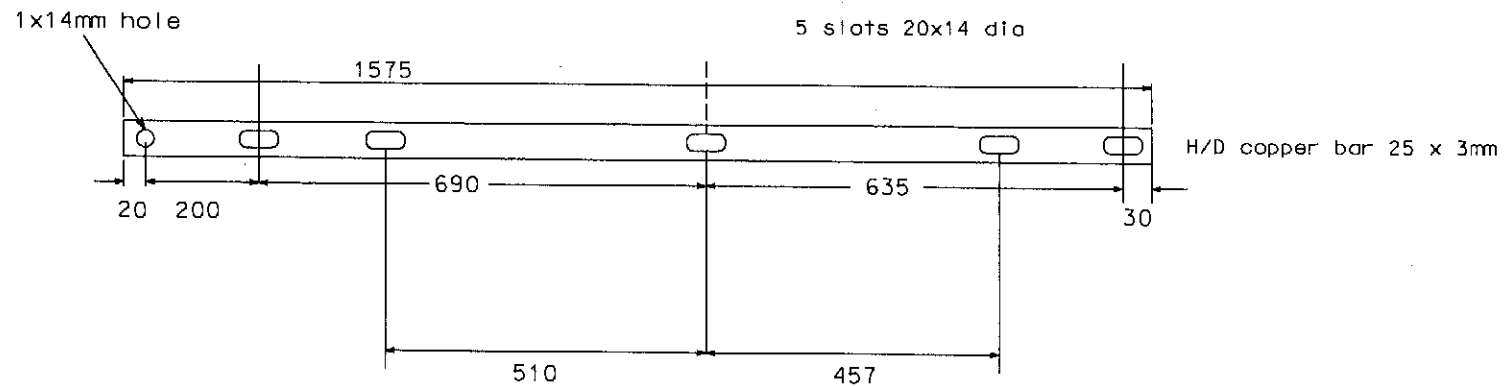
SCALE	1-1
DATE	27/09/00
DRAWN	ADD
TRACED	
CHECKED	KD
APPROVED	29/09/03

(A B C) TERMINAL SUPPORT TO
UNDERGROUND BUNCHED 3Ø RY-BN CABLE

ISSUE	DATE

Produced by Primary CAD	
 Aquila	
DRG. No.	
FIG 30	

H/D Copper Bar for use with M72 & M72/1.& ME74



SCALE	-
DATE	20.08.01
DRAWN	D.B.
TRACED	-
CHECKED	20.08.01
APPROVED	K.D.

H/D Copper Earth Bar for use with 11kv
Surge Arresters and M72, M72/1 & ME 74

ISSUE	DATE	ISSUE	DATE	ISSUE	DATE
				A	21.08.01
				Title	
				Amended	

Produced by
Output Services

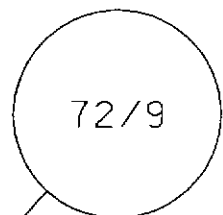
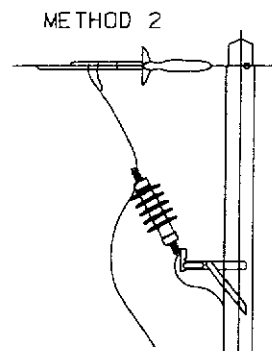
GPU
POWER UK

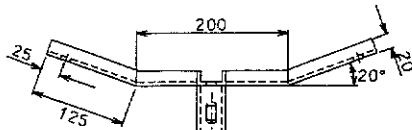
DRG. No.
M 72/9

DRG. No.

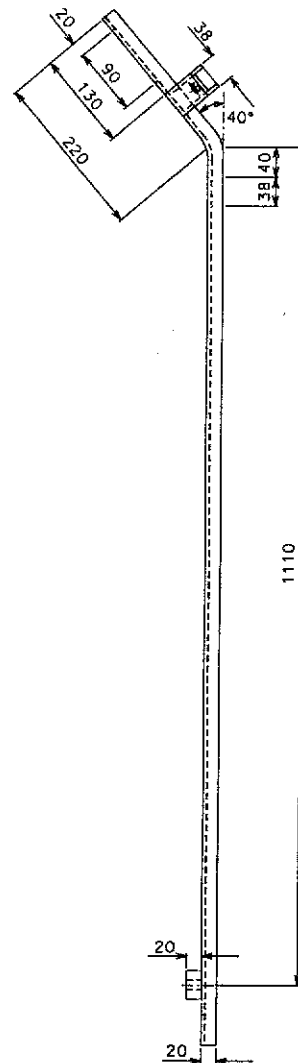
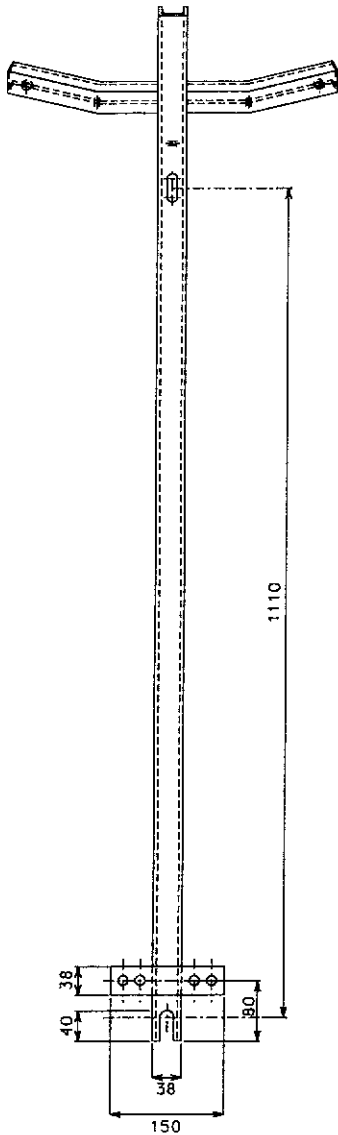
M 72/9

A4





MATERIAL: 1410 x 38 x 20 Galv. Steel Channel
 450 x 38 x 20 Galv. Steel Channel
 150 x 38 x 20 Galv. Steel Channel
 Slots for fixing Bolts 18 dia.
 Holes and Slot for mounting Insulators 13 dia.



MIN 3361517

HQ 7/7 A4 107

Produced by
Output Services

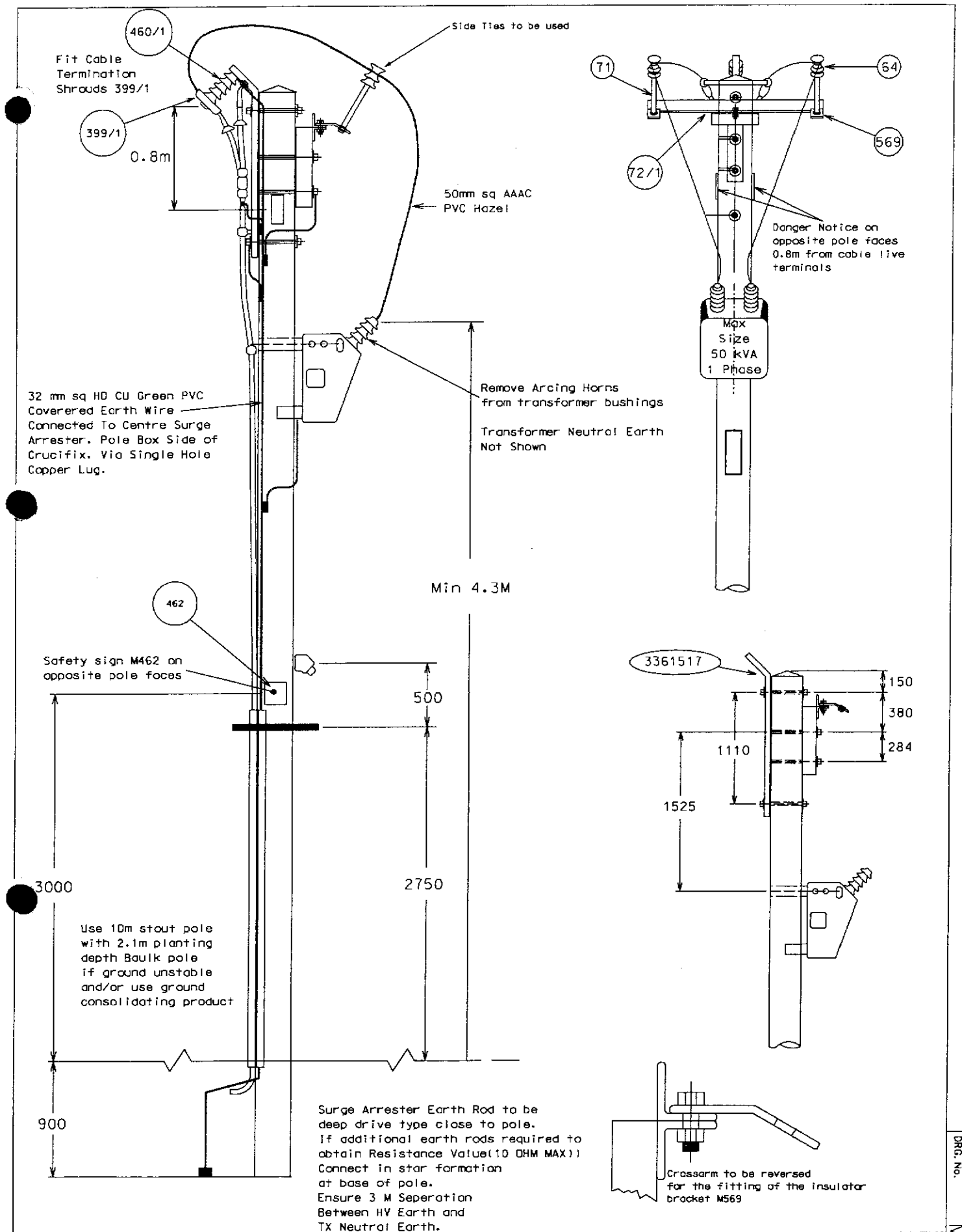
SCALE	-
DATE	12/12/00
DRAWN	ADD
TRACED	
CHECKED	28.12.00
APPROVED	D.S.S.

INSULATOR SUPPORT BRACKET FOR
11KV POLE MOUNTED TERMINATION

ISSUE	DATE
A	17.10.01
MIN NUMBER ADDED.	
D.B.	

GPU POWER UK
DRG. No. UGRD/190/001

DRG. No. UGRD/190/001
A4

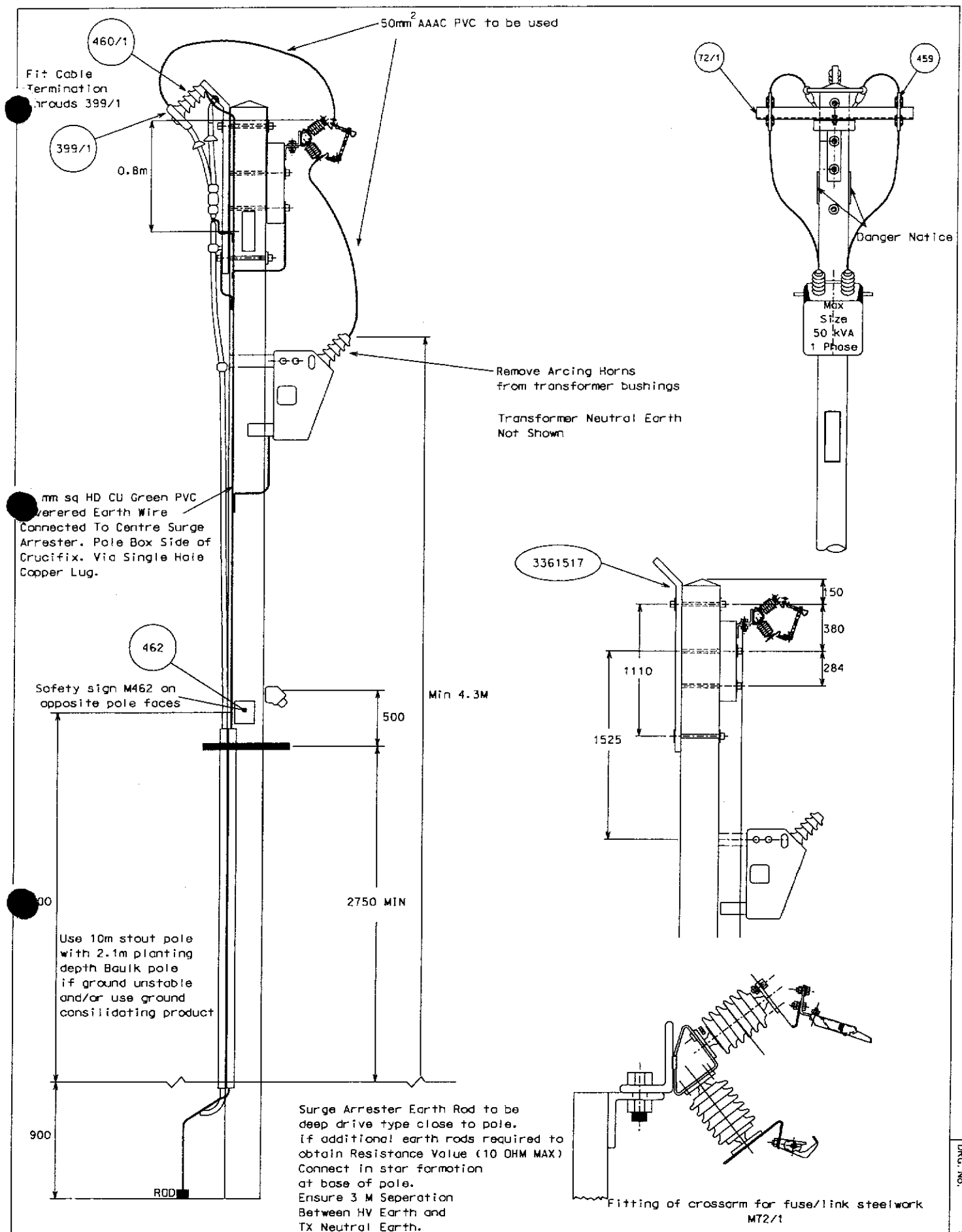


SCALE	1:40
DATE	11.10.01
DRAWN	D.B.
TRACED	
CHECKED	K.D.
APPROVED	17.10.01

MS2/29-11kV 50mm² AAAC Construction
Single Pole Transformer mounting
with HV U/G cable

ISSUE	DATE

Produced by Output Services	
DRG. No.	MS2-29



SCALE	1:40
DATE	11.10.01
DRAWN	D.B.
TRACED	-
CHECKED	17.10.01
APPROVED	K.D.

MS2/29a-11kV 50mm² AAAC Construction
Single Pole Transformer mounting
with HV U/G Cable and Fuses/Links

ISSUE DATE

DRG. No.

GPU
POWER UK

MS2/29a

Produced by
Output Services

DRG. No.

MS2/29a

A4

(11.0 metre pole shown)

Surge Arrester
Down lead Connection to
Surge Arresters via
copper tape

Surge Arrester
Separation 10mm
on Outgoing side,
455mm on Incoming
side

70mm² Cu PVC Surge
Arrester Earth to be
bonded to Recloser and
VT tanks

Surge Arrester Earth Rod
to be Deep Drive type
close to Pole. If additional
Earth Rods required to
obtain resistance Value
(10 Ohms max.) connect
in Star formation at
base of Pole

Radio Aerial and RTU
box bolted to Pole at
lowest possible
position at least 600mm
below Recloser mounting
bracket

Safety Sign M462
on each leg on
opposite faces

Voltage Transformer cable
(4 core 4mm² SWA)
Min bending radius 300mm
Cleat at 600mm intervals
using M275/3

Danger Plate ME66
on opposite pole faces

Multicore cable
Min bending radius
200mm
Cleat at 600mm
intervals using
M275/3

Separation from
Main Earth
Down Lead
150mm min

All H.V Steelwork
Recloser tank, V.T.
Control box to be
bonded to 32mm²
Cu PVC HV Earth
Down Lead

Main Earth to
be Deep Drive
type close to Pole
If additional Earth
Rods required to
obtain Resistance
Value (10 Ohms max.)

run separate
earth rods
Maintain
separation from
Surge arrester
earth

Arrester Earth wire
connection 900mm min
below Ground level

Earth wire
500 min
below ground level

C.A.D. PRODUCED

SCALE	N.T.S.
DATE	07.04.99
DRAWN	
TRACED	L.V.L.B.
CHECKED	N.B.W.
APPROVED	R.C.F.

33KV HEAVY CONSTRUCTION
'H' POLE SECTION SUPPORT
WITH NULEC RECLOSER

ISSUE	ISSUE

DRG. No.
MS4A/65

